

WINEMAKING

WINEMAKING

FROM GRAPE GROWING TO MARKETPLACE

SECOND EDITION

Richard P. Vine, PhD

Purdue University
West Lafayette, Indiana

Ellen M. Harkness, MS

Purdue University
West Lafayette, Indiana

and

Sally J. Linton

Purdue University
West Lafayette, Indiana

Springer Science+Business Media, LLC

Library of Congress Cataloging-in-Publication Data

Vine, Richard P.

Winemaking: from grape growing to marketplace/Richard P. Vine, Ellen M. Harkness, and Sally J. Linton.—2nd ed.

p. cm.

Includes bibliographical references and index.

ISBN 978-1-4613-5215-0 ISBN 978-1-4615-0733-8 (eBook)

DOI 10.1007/978-1-4615-0733-8

I. Wine and wine making. I. Harkness, E. M. (Ellen M). II. Linton, Sally J. III. Title.

TP548 .V4844 2002

663'.2—dc21

2002024122

© 2002 Springer Science+Business Media New York

Originally published by Kluwer Academic / Plenum Publishers, New York in 2002

Softcover reprint of the hardcover 2nd edition 2002

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PREFACE

Over the past several decades, consumer interest in the fine vintage wines produced by small “boutique” vintners across the United States has grown to rival that of many European estates. This attention continues to intensify, especially for the truly good wines that are reasonably priced. Consumers are, however, unforgiving—especially wine enthusiasts. Second-class wines do not succeed just because a vintner is new. The methods and controls essential to vintners in the production and marketing of top-grade wines have advanced. This second edition of *Winemaking* has updated and, in some cases, completely revised the material associated with these disciplines.

Fine wine is much like other art forms, as it is the infinite variability of factors pertaining to the subject that renders it so complex—and able to attract buyer’s attention. Hundreds of different vine varieties are cultivated around the world, and no doubt an even greater number of fruit and berry cultivars. And with the addition of such factors as *terroir* (soil and climate attributes) changing every vintage season, varied vineyard cultivation and harvesting techniques, advancing production technology, dynamic markets, and overall operational philosophy, one can easily understand the enormous breadth and depth of variation that exists. This diversity generates an unimaginable number of different wine possibilities. One need only visit a fine wine shop in a large city to see hundreds of different

wines for sale, yet these are still only a very small percentage of what the entire world vinifies every vintage. Since the first edition of *Winemaking*, the world of commercial wine has gotten not only better but bigger.

In larger wineries one can usually find viticulturists (grape-growing technologists), food chemists, enologists (winemaking technologists), recordkeepers, marketing specialists, and other well-trained professionals who respond and supervise within their respective fields of expertise. For the small estate vintner, the comprehension and responsibility of wine production, administration, and marketing usually weigh heavily, often entirely, on the owner and his or her family. The preface to the first edition of *Winemaking* stated that it was not necessary for the small winery vintner to have advanced degrees in the physical and business sciences. While this remains true, the second edition addresses the reality of a bigger and better wine industry placing greater demands upon start-up vintners. A solid academic background generally helps in carrying out the increasing administrative, production, and marketing functions necessary for managing a successful new winery.

One should fully understand the integral involvement of the Bureau of Alcohol, Tobacco, and Firearms (ATF) in the permits and regulations that must be complied with in operating a winery. Part 240 of Title 27 of the Code of Federal Regulations (CFR) is a source often quoted in this book to outline the application, documentation, and reporting forms and procedures necessary for vintners in the start-up and operation of a sound, legal, business. Further, each state has supplemental requirements with which vintners must be in compliance.

The second edition of *Winemaking* remains in design a foundation text on grape growing, wine making, and wine marketing from which both would-be and established vintners can advance to more comprehensive texts. *Winemaking* will also continue to serve independent wine wholesalers, distributors, retailers, and restaurateurs as a reference from which to understand the rationale of methods employed by estate vintners. Advanced amateur wine makers will also find this book a handy guide in their cellars. Similarly, wine connoisseurs can find value in its promotion of a deeper understanding of wine craftsmanship. Overall, this book should bridge the ever-widening gap between the art and science in the entire world of wine.

Everyone having more than a casual interest in wine should own a copy of the second edition of *Winemaking*. Its low price can be repaid many times over by its wealth of easy-to-read, updated information for beginners trying to understand the ways that wine is made by experts.

ACKNOWLEDGMENTS

The authors would like to thank Purdue University for its generous support in the preparation of this book. In particular, Dr. Victor Lechtenberg, Dean of Agriculture, and Dr. Randy Woodson, Associate Dean of Agriculture, have given of their time, energies and resources to help realize this project.

Dr. Phil Nelson, Purdue Department Head of Food Science has also given us sound advice, along with comprehensive assistance, without which this work would not have been possible. Dr. Richard Linton was a tireless reader and critic in the advancement of this second edition. As with almost every book authored in an academic community, much is owed to all the enthusiastic students who contribute far more than is generally thought about. Thank goodness for their boundless energy.

We are grateful to the Indiana Wine Grape Council for its many contributions toward creating *Winemaking: From Vineyard to Marketplace*. Council President Bill Oliver, Jr., previous Council President Jim Butler, and all members of the Council provided both material and inspirational influences. Special gratitude goes to Indiana vintner Dave Gahimer, who redesigned the cellar record forms presented throughout this book.

Similarly, the Indiana Wine Guild, from which the initial need for this book was first identified, must be recognized. Special thanks go to Bill Oliver, Sr., and Ben

Sparks for their leadership in the rebirth of commercial winegrowing in midwestern America. Indiana Wine Guild President David Lundstrom continues this guidance and has been an essential resource for us.

Most of all, we would like to thank our families and friends, who excused us from considerable personal quality time to create this manuscript.

To all of you, we are deeply grateful.

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INTRODUCTION

Man made wine long before the dawn of civilization. Paleontologists infer that crude fruit wines were familiar to early human beings more than 100,000 years ago.

According to the Bible (*Genesis* 9:20–21), Noah landed his ark on Mount Ararat across the northern border of Turkey and grew grapes for wine. Documents affirm that pre-Islamic Arabs drank wine in early Mesopotamia. Hammurabi wrote strict laws for wine commerce in the 18th century B.C. The great Egyptian pharaohs enjoyed a variety of wines grown from irrigated vineyards along the Nile, and Homer wrote of fine wines made in ancient Greece. In July 1996, the Associated Press reported that King Herod's 2,000-year old wine jug was discovered near Jerusalem.

Several centuries later, during the rise of the Roman Empire, wine was introduced to Europe, but it was Christianity that developed the wine industry as we know it today. When Christ offered the cup at the Last Supper, He willed that wine be the everlasting symbol of His blood—an event that was the motivation for monks to establish vineyards wherever they settled in the Middle Ages and the Renaissance. It continued in the New World: the development of the first wines in New York and California, among other states, was influenced by the Church.

Wine continues to replace fouled water supplies, as it has throughout the history of western civilization. For thousands of years wine was our most important medicine and our only anesthetic. The “magic” of wine fascinated the ancients:

that if one simply crushed grapes, wine would literally make itself. It wasn't until Pasteur's discoveries in 1860 that the phenomenon of fermentation, whereby microscopic one-celled plants, called yeasts, convert simple sugars into ethyl alcohol and carbon dioxide gas, could be explained.

In many foreign countries the attitude toward winemaking is much the same as that toward preserving fruits and vegetables from the garden. There is little concern about alcohol content, since alcohol serves as a natural wine preservative, enhances digestion, and adds to carbohydrates in the diet.

The wine industry is vast: British wine expert, Hugh Johnson, states that one person in every 100 around the world is employed in a vineyard, in a winery, or in the marketing of wine. Wine is, however, far from global. Some nations find difficulty in cultivating the vine because of harsh environmental conditions, and other countries are populated with cultural groups that restrict wine drinking or are prohibitionist.

Grapevines grow abundantly in the United States. In nearly every state one type of vine or another can be grown. Of the 50-odd species and thousands of varieties known, a greater selection of vine types can be grown in America than in any other country.

According to the Wine Institute, Wine consumption in France and Italy surpasses 16 gallons per person per year, which far outpaces the average of just 2 gallons annually per capita across the United States. Winegrowing in the United States is regulated nationally by the U.S. Bureau of Alcohol, Tobacco, and Firearms (ATF) and by individual state Alcohol Beverage Control (ABC) commissions.

Consistent winemaking success requires a systematic approach, and the design here is to help organize thoughts and materials into an efficient step-by-step methodology. The methodology is fundamental but still offers plenty of room for variations that may better suit individual tastes. While one should expect good results from following the recommendations made in this book, it cannot be considered as any form of assurance or guarantee that resulting products will be acceptable or superior.

The text is written in the "language" of the wine industry so that, as one becomes proficient, the commercial wine literature will serve to expand upon the same fundamental precepts.

Richard P. Vine, PhD
Ellen M. Harkness, MS
Sally J. Linton, BS

CHAPTER 1

HISTORY OF WINE IN AMERICA

In comparison to the 6,000 years of wine-growing history in Europe, the 400-year span of American vines and wines seems minuscule. But it should be remembered that most of the major development in the Old World took place during these same four centuries. Few modern-day European wine producers can boast continuous ownership of estate vineyards and cellars since before the 17th century.

The first wines in America were made more than 500 years ago by the Papago Indians, who fermented the juice of the saguaro cactus growing in what is now Arizona and New Mexico. Cactus wine is still made there, but demand for it falls far short of wine made from grapes.

This profusion of vines is undoubtedly what Leif Erikson found when he first visited the North American continent in the year 1000. The following passage comes from *The Discovery of America in the Tenth Century*, written several centuries ago by Charles C. Prasta:

A delightful country it seemed, full of game, and birds of beautiful plumage; and when they went ashore, they could not resist the temptation to explore it. When they returned, after several hours, Tyrker alone was missing. After waiting some time for his return, Leif, with twelve of his men, went in search of him. But they had not gone far, when they met him, laden down with grapes. Upon their inquiry, where he had stayed so long,

he answered: "I did not go far, when I found the trees all covered with grapes; and as I was born in the country, whose hills are covered with vineyards, it seemed so much like home to me, that I stayed a while and gathered them." . . . And Leif gave name to the country, and called it Vinland, or Wineland.

An enchanting story, and convincing, too, as wild grapevines are still prolific along the coastline of eastern America.

EASTERN AMERICA

A ship's log kept by navigator Giovanni da Verrazano dated in 1524 made note of vines in what is now North Carolina—and the possibility of making wine from them.

French Huguenots settled in Florida during the mid-1500s and made America's first wines from some of the native grapes found there. This fruit was from the species *Vitis rotundifolia*, known today as the Muscadine varieties. These New World grapes must have seemed strange to the Huguenot wine masters, since Muscadines ripen their fruit by individual berries in small clumps rather than in large clusters. Further, the flavor of Muscadines is very exotic when compared to the more subtle nuances found in the Old World *Vitis vinifera* grapes.

In 1565 Sir John Hawkins, a British admiral, reported upon winemaking from the native Florida grapes during his visits there. It seems likely that the Florida wines met with little favor among the early settlers.

Further north, Captain John Smith wrote of his observations in Virginia during the early 17th century:

Of vines great abundance in many parts that climbe the toppes of highest trees in some places, but these beare but few grapes.

There is another sort of grape neere as great as a Cherry, they [probably Indian natives] call Messamins, they be fatte, and juyce thicke. Neither doth the taste so well please when they are made in wine.

Beauchamp Plantagenet, in a London-published text that described the colonies, made these observations on the native grapes: "Thoulouse Muscat, Sweet Scented, Great Fox and Thick Grape; the first two, after five months, being boiled and salted and well fined [clarified], make a strong red Xeres [sherry]: the third, a light claret; the fourth, a white grape which creeps on the land, makes a pure, gold colored wine." The "Great Fox" that Plantagenet mentioned was probably a variety of *Vitis labrusca*, a species that bears its heavily scented fruit in moderately sized clusters. Even in modern times, *foxy* is a rather loose organoleptic term used to describe the aroma and flavor typical of the *labrusca* species—the Catawba and Concord being prime examples.

Even further north, more native vines were discovered in the early 17th century by the Frenchman Samuel de Champlain as he explored the islands in the St. Lawrence River. One of these islands was so overgrown with vines, probably *Vitis labrusca*, that Champlain named it Bacchus Isle.

The colonists often built their community inn or tavern near their church. Between services the worshippers would gather around the tavern's log fire, enjoying food, drink, and good fellowship. Vineyards were mostly private plantings, and the wines were homemade.

Disappointment with the native American vines and the unfamiliar wines they made no doubt prompted the plantings of European *Vitis vinifera* varieties in the Colonies during the 1600s. The London Company of Virginia may have made the first attempt at growing Old World grapes in the New World. Lord Delaware brought French viticulturists and vines to establish Virginia vineyards in the 1620s. The vines died, however, and colonists blamed the failure on the French vigneron's mishandling of the project.

As the 17th century progressed, hope continued that the European vines could be successfully grown in the New World. Viticulture was encouraged through economic incentives and even required by law in some places. In New York, wine could be made and sold without taxation. Virginia promulgated a law whereby "all workers upon corne and tobacco shall plant five vynes . . . upon penaltie to forfeit one barrell of corne."

Governor John Winthrop of Massachusetts planted vines on Governor's Island, and in Maryland Lord Baltimore established vineyards. William Penn had a vineyard plot near Philadelphia where he tried to grow classic European vines. One by one, all the Old World plantings died. In Georgia, at Savannah, the enthusiastic Abraham de Lyon, from Portugal, planted a large-scale *vinifera* vineyard in 1730. Despite encouragement throughout the community, his vines died.

King Charles II ordered vines to be planted in Rhode Island in what turned out to be yet another failed attempt to grow the classic Old World *Vitis vinifera*. John Mason offered to trade the British monarch all of what is now New Hampshire in return for 300 tons of French wine, but Charles II rejected the proposal.

The major inspiration for the frustrated hopes of early Colonial winegrowing pioneers was, of course, the profusion of wild vines; if the native vines grew so luxuriantly, why shouldn't the cultivated European varieties do all the better?

The failures were, of course, not the fault of anyone—rather, the then unknown *Phylloxera vastatrix* root louse, plus a lack of hardiness against the colder winter temperatures. These adverse conditions thwarted eastern American winegrowing for more than two centuries.

Thomas Jefferson maintained a very keen interest in American vineyards but continued to import wine from Europe, where, as ambassador to France, he was well known by his peers for his appreciation of wines. He walked the entire breadth of France, visiting Burgundy, Bordeaux, and other winegrowing regions. As U.S.

Secretary of State, Jefferson chose the wines for President Washington. When Jefferson became president, he selected and paid for his wines personally, keeping careful inventory records. He also assisted in stocking the presidential cellars of John Adams, James Madison, and James Monroe.

The first commercial cultivation of native American *Vitis labrusca* vines is credited to John Alexander, a Pennsylvanian whose viticultural research in the latter 1700s brought forth the aptly named Alexander grape. The Alexander became famous for its heavy production of fruit and resistance to disease and cold winters. The wine from Alexander, however, still fell short of the prized vines from mother Europe.

Jean Jacques Dufour left his native Vevey, in Switzerland, during the latter 1700s, eventually settling in southeastern Indiana along the Ohio River. By 1804 Dufour's first vines were planted near the settlement village of Vevay, near Madison, in what is now Switzerland County. Others followed his lead, and the region soon became known as the "Little Rheinland," owing to the ascending riverbank terraces upon which hundreds of acres of the Alexander vine were planted. Indiana became one of the top 10 states in volume of commercial wine production.

The Alexander grape variety interested the likes of Thomas Jefferson and Henry Clay, who also provided encouragement for Dufour to write *The American Vine Dresser's Guide*, a grape grower's manual first published in 1826. In 1809, following some of his own unsuccessful *Vitis vinifera* experiments at Monticello, Thomas Jefferson stated that "it will be well to push the culture of this grape [Alexander] without losing time and effort in the search of foreign vines which it will take centuries to adapt of our soil and climate." But Jefferson was not the only early hopeful winegrower with good foresight. William Penn was also of the opinion that the proper course was to move toward cultivating native *labrusca* vines.

John Adlum, from Georgetown, D.C., introduced the *Vitis labrusca* variety Catawba, named for the Catawba River in North Carolina, where the vine was discovered in the wild. The comparatively delicate fruit flavor of Catawba brought it quickly to the attention of winegrowers. Without modesty, Adlum informed the world that his providing the world with the Catawba grape had given a greater service to America than if he "had paid the national debt."

Thomas Jefferson never lost his optimism that native American winegrowing would eventually be achieved on a commercial scale—despite his fondness for the fine wines from Europe. He foresaw in 1808 that "we could in the United States make as great a variety of wines as are made in Europe, not exactly of the same kinds, but doubtless as good." Later, he stated that

wine being among the earliest luxuries in which we indulge ourselves, it is desirable it should be made here and we have every soil, aspect & climate of the best of wine countries . . . these South West mountains, having a S.E. aspect, and abundance of lean & meager spots of stony & red soil, without sand, resembling extremely the Cote of Burgundy from Chambertin to Montrachet, where the famous wines of Burgundy are made.



Fig. 1–1 Thomas Jefferson's Wine Cellar at Monticello.
 Courtesy: *Vinifera Wine Grower's Association*.

Jefferson kept a Garden Book at Monticello from 1766 to 1826 in which he kept the data of his vine plantings and experiments. Among his notes was recorded the planting of “30 plants of vines from Burgundy and Champagne with roots, 30 plants of vines of Bordeaux with roots.” Several years following was another entry reporting the planting of several famous Italian vines, which included Trebbiano and Aleatico. The Garden Book does not, however, show any significant success at the Monticello vineyard in testing any of the European selections. Jefferson's wine cellar at Monticello is illustrated in Figure 1–1.

The first commercial American winery may have been the Pennsylvania Vine Company, situated near Philadelphia and founded by Pierre Legaux in 1793. Legaux was a Frenchman determined to succeed in cultivating French vines, but like all previous attempts to grow the vaunted *vinifera*, this was yet another failure.

Located in Washingtonville, New York, about 50 miles northwest of New York City, the Brotherhood Winery has caves that are reminiscent of many in Europe. Founded in 1816 by a French immigrant shoemaker, Jean Jacques, Brotherhood remains the oldest operating winery in the United States.

Nicholas Longworth, a real estate magnate, built a majestic winery near Cincinnati, extending the “Little Rheinland” approximately 50 miles northeastward along the Ohio River from Vevay, Indiana. Longworth was so highly impressed by Adlum's Catawba grape that he planted a large acreage of the variety near his winery in the early 1820s. Longworth made the first American “*Champagne*,”

labeling his product with the honest name of “Sparkling Catawba.” Longworth took great pride in his wine, declaring that it was even superior to the sparkling wine from Champagne in France.

Henry Wadsworth Longfellow, a noted wine critic and connoisseur, was intrigued by the lofty claims of Nicholas Longworth and paid a visit to the Ohio winery in the mid-1850s to taste Longworth’s sparkling wine. Impressed and convinced, Longfellow wrote a poem in 1854, entitled “Ode to Catawba Wine,” which proclaims:

Very good in its way
Is the Verzenay
Or the Sillery soft and creamy;
But Catawba wine
Has a taste more divine,
More dulcet, delicious, and dreamy.

Longworth’s wine must have been truly remarkable, as Longfellow’s testimonial ranking it superior to the French Champagne from Verzenay is very high praise.

The Church grew very wealthy from winegrowing during the Renaissance in Europe. Christianity was influential in New World winegrowing too—this time from Protestant denominations existing in eastern America. Elijah Fay, A Baptist deacon, planted the first vineyard in western New York State, in 1818.

The vineyards of central New York State are nestled along the banks of the Finger Lakes (see Figure 1–2). Local legend has it that when God finished making the Heavens and Earth, he laid *His* hand upon that particular site to bless His creation, thereby leaving the great handlike impression that identifies the beautiful Finger Lakes map. Geologists point out, however, that the glens and hills of the Finger Lakes region resulted from the clawing effects of the Ice Age.

St. James Episcopal Church still stands today as a landmark in Hammondsport, New York. This village, situated at the southern tip of Lake Keuka, the “thumb” of the Finger Lakes, is where the New York State wine industry began. Father William Bostwick propagated cuttings of Catawba and Isabella, another native *labrusca* variety, into roots and planted them behind the church in 1829 to make wine for his Anglican Eucharist.

The success of Bostwick’s work became well known and attracted the interest of several prospective commercial vintners. In 1860 Charles Champlin, heading an investment consortium, founded the Pleasant Valley Wine Company just south of Hammondsport in Rheims, New York. The original cave was lined with stone walls more than 6 feet thick—a remarkable cellar that was to become the first federally licensed winery in America.

Champlin’s obsession was to make the finest “*Champagne*” in the New World—certainly to share the fame generated in Ohio 30 years earlier. He hired Jules Masson, a French wine master, to ensure that the very finest New York State



Fig. 1–2 Finger Lakes Vineyards in Upstate New York.

“Champagne” was made. In 1871 a tasting of some of the first wine from Pleasant Valley was held at the Parker House in Boston. Among the prestigious critics was Marshall Wilder, then president of the American Horticultural Society, who proclaimed that Champlin’s sparkling wine was “great” wine from the new “western” world. The famous “Great Western” name was thus born and became the trademark for the Pleasant Valley Wine Company.

The early 1860s marked the beginning of a promising era for American winegrowing. The success of Champlin and Longworth, among others, awakened new commercial winegrowing interests in the East and prompted considerable vineyard expansion—in turn bringing about a fierce competition in the marketplace for the consumer’s scarce dollars. In no fewer than 20 states and territories, significant vineyard and winery investments were taking place. Commercial American wines were existing, if not thriving. Mary Todd Lincoln was the first to serve American-grown wines in the White House.

New market potential for barrels, casks, and vats enticed cooper Walter Taylor, descendant of a sailing ship captain, to set up a cooperage shop on Bully Hill, at the foot of which the village of Hammondsport is situated. Taylor’s farm was in an area known for large stands of white oak timber. High on Bully Hill, it overlooked miles and miles of the young, but dynamic, New York State wine industry. Taylor also dabbled in winemaking, but his expertise lay in the crafting of fine cooperage, much of which was in service at the local wineries under credit agreements.

The wine production capacity of the Finger Lakes grew faster than the market's propensity to consume, and the competition proved to be too much for many of the financially weaker vintners. As this worsened, Walter Taylor was ultimately forced to act upon his delinquent accounts receivable, often taking bulk wine inventories for payment. Taylor and his three sons, Fred, Clarence, and Greyton, promptly found themselves deep in the wine business. During the ensuing 50 years the Taylor family parlayed this somewhat impromptu start into one of the single largest wineries of its time.

During the mid-1800s the Catawba and Isabella grape varieties found their way to Missouri and were adapted to yield large bounties of fruit. This success attracted widespread plantings there much as in New York.

Many of the Missouri growers were inexperienced with grapes, however, and did not recognize infestations of the dreaded Black Rot fungus. The attack of this disease throughout the heartland of America caused discouragement even among dedicated believers. A large number of the Old Swiss growers in the "Little Rheinland" of Indiana moved to northern Ohio, where they joined German immigrants in establishing vineyards near Sandusky and out on the Bass Islands in Lake Erie.

Ephraim Bull, from Concord, Massachusetts, introduced eastern vinegrowers to the Concord variety. It came from the long labors of Bull's research with vine hybrids and was released in 1854 after 7 years of testing. The Concord was bred from disease-resistant native *Vitis labrusca* varieties and, despite the then astronomical cost of \$5 per vine, was of interest to the sparse wine industry remaining in Missouri.

Along with another new disease-resistant and high-producing variety called Norton's Virginia, Concord rekindled enthusiasm in Missouri. By the mid-1860s vineyards had expanded to the largest acreage ever, making Missouri the leading grape-producing state in the Union. It was during this time that the first vine rootstock materials were taken from American vineyards to combat the *Phylloxera* root louse in Europe. Nurserymen Bush and Meissner of St. Louis published their initial catalogue in 1869, and by 1876 a French translation was being used by European winegrowers for ordering the rootstocks necessary for the great grafting project in the Old World. Even today, classic vines in France and other European countries are still grafted upon American roots.

The finest years for Missouri viticulture proved to be the 1860s as production again grew to far oversupply demand. The New York City newspaper editor Horace Greeley advised, "Go west, young man, go west." And people were doing exactly that: the Gold Rush, the land grants, and the transcontinental railroad were among many fascinations for opportunity in the West. Viticulture Professor Hussmann moved from Missouri to California, where he proclaimed that "this [California] was the true home of the grape." In 1870 California took the lead as the largest winegrowing state in America—a position that the Golden State has never since relinquished.

WESTERN AMERICA

Winegrowing in the western states was pioneered, as in Europe and in eastern America, by the Church. The “island” of California was first recognized by Pope Alexander VI in 1493. Spain made expeditions to explore the California coastline in 1542 and 1602. The conquistadors found new frontiers to explore along the western shores of the Americas, and vineyards were established in their wake.

Though the European *Vitis vinifera* vines did not succeed in the eastern states, they flourished in the western New World. In fact, vines grew so well that in 1595 the Spanish king decreed that new vineyard plantings be terminated in the provinces. The edict was intended, of course, to maintain sales of Spanish wine to the West, and it remained in effect for nearly 150 years. This decree offers an interesting historical parallel to a similar abatement made by Rome some 1,500 years earlier to curtain vineyard plantings in its colonial European provinces.

Despite the order from Madrid, Franciscan missions continued planting vines in the Americas. The first vineyard in the Baja, or lower California, was established at Mission Xavier by Father Juan Ugarte in the late 1690s.

Gaspar de Portola, explorer-captain of the Eighth Company of Dragoons, arrived in Mexico from Spain in 1767. The following year he was ordered by General Jose de Galvez to establish missionary settlements in Alta (upper) California to prohibit Russian fur trappers and British explorers from taking over the Spanish territorial claims.

De Portola embarked upon his mission project in March 1769 with a small force of soldiers and a devout missionary priest, Padre Junipero Serra (Figure 1–3). It was Serra’s responsibility to teach the native American Indians the ways of Christianity in order to subdue any pagan violence that might be encountered. The procession arrived in California in July of the same year, and Serra raised the Cross that marked the first mission, called San Diego. Since Padre Serra knew he would need wine, he brought vines and planted the very first vineyard in California.

More missions were established northward, each about one hard day’s horseback ride apart, along what became known as El Camino Real. By 1771, the vineyard at the San Gabriel Mission had become a large plot, called the *vina madre*, or “mother vineyard,” because of its superior soil and weather conditions. By the end of the 18th century Serra had carried his vines to five outposts in southern California: San Diego, San Juan Capistrano, San Buenaventura, San Gabriel, and Santa Barbara. His Mission vines flourished in the California sun, and the wines they produced, though perhaps not of the finest quality, were abundant. In all, 12 were missions constructed as El Camino Real progressed to San Luis Obispo, Santa Cruz, and San Jose and yet further north—the last begun at Sonoma in 1823 by Father Jose Altimira.

As merchant ships began to stop more frequently at ports along the Alta California coast, the missions of El Camino Real came increasingly into contact



Fig. 1–3 Serra's Personal Quarters in Mission San Diego.

with the outside world. This, in turn, brought about increased demand for wine there during the early 1800s.

This growth was short-lived, however, as the Mexican revolt against Spain resulted in the curtailment of all agricultural endeavors by the missions. When the situation worsened, some of the early Church vineyards in California were abandoned, and others were destroyed by the enraged monks themselves. Consequently, while the birth of California wine must be credited to the Church, the huge development of the California wine industry was made by the commercial laity.

Joseph Chapman planted the first commercial vineyard in California on the Los Angeles pueblo land in 1824. Chapman was originally an easterner who had gained experience in viticulture by working at Junipero Serra's Santa Ynez and San Gabriel missions.

Junipero Serra's Mission vines proved to be hardy, vigorous, and exceptionally well adapted to California. Some specimens eventually grew to become truly magnificent plants. The U.S. Department of Agriculture reported that a single vine, the Carpinteria vine, had a trunk nearly 3 feet in diameter and once yielded a crop of 8 tons of grapes! According to Ruth Teiser in her book *Winemaking in California*, the Carpinteria vine was dug up and sent off to the 1876 World Fair Exposition in St. Louis.

With the concentrated efforts of the monks, winegrowing had developed faster in southern than northern California. With the Mexican loss of California to the

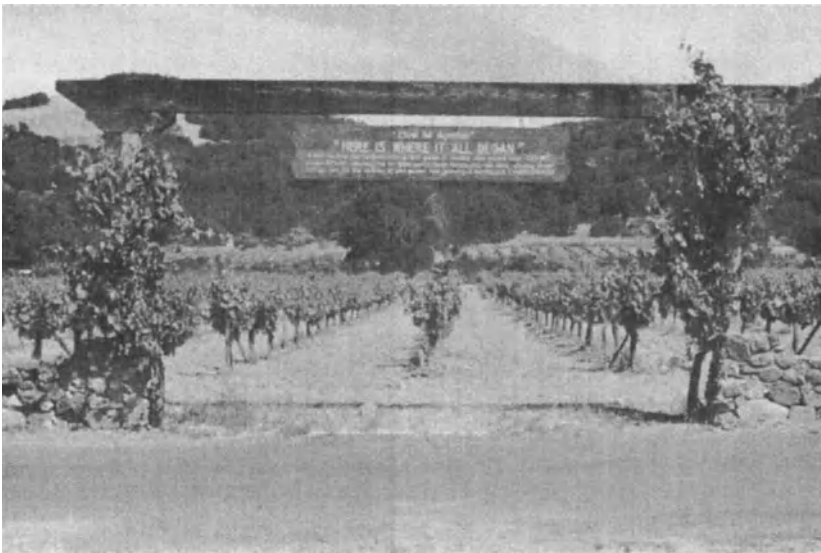
United States in 1846, and the Gold Rush of 1849, that situation was quickly reversed. Many European winegrowers emigrated to the San Francisco area, seeking refuge from political oppression in Europe and a mother lode of gold. California welcomed this new populace and encouraged land investments for vineyards. The State Agricultural Society promised that “capital put into vineyards would bring greater returns than when outlaid in fluming rivers for golden treasures.”

The face of California winegrowing, indeed, in all of America, would be changed forever by Agoston Haraszthy, an exiled Hungarian count. Haraszthy initially emigrated to Wisconsin in the early 1800s, planting vines and building a beautiful stone winery near what is now the village of Prairie du Sac. His vines died, from *Phylloxera* or the Wisconsin winters or both. In any event, about 1 year before the Gold Rush, Haraszthy moved to the Sonoma Valley, near the last mission of El Camino Real.

Count Haraszthy’s optimism over the possibilities of quality winegrowing became infectious in northern California. He developed the Buena Vista Vinicultural Society and was determined that the wine industry there should be built upon a sound foundation. His speeches and articles prescribed that only the finest varieties of *Vitis vinifera* should be planted in only the choicest land and that only the latest viticultural methods should be used (Figure 1–4).

In 1861 Governor Downey delegated Haraszthy to make a trip to Europe to select the choicest *vinifera* vine stocks there and bring back cuttings from them to

Fig. 1–4 Original Haraszthy Vineyard in Sonoma, California.



California. About 300 different varieties were shipped, but difficulties in handling and labeling the plant materials resulted in some of the varieties' becoming either mislabeled or lost altogether. It took years to untangle the puzzle of identifying varieties. Some of them, such as the now famous Zinfandel, have yet to be traced with absolute surety.

The estate and winery at Buena Vista were lavish, their cost eventually leading to Haraszthy's bankruptcy. With further financial support denied, Haraszthy left California quite mysteriously. His classic Old World vines remained and have since become a perpetual monument to his dedication.

Ten miles to the east of Sonoma is the Napa Valley, felt by many to be the single finest winegrowing district in America. A number of choice European varieties were planted there in the 1860s, some via Haraszthy's Sonoma project and some from other sources for Old World *vinifera* vines.

The reputation of the wines being grown in the Napa Valley spread rapidly. Robert Louis Stevenson commented in the 1880s that the Napa locale was "where the soil sublimated under sun and stars to something finer, and the wine is bottled poetry." Stevenson went on with his praises, comparing Napa wines with those from the great growths of Bordeaux and Burgundy, namely Chateau Lafite and the Clos de Vougeot. The future seemed very bright indeed, but Sonoma and Napa vines were soon attacked by the *Phylloxera* root louse in much the same manner as in Europe.

Phylloxera was checked in the latter 1800's through the viticultural research work of Eugene Waldemar Hilgard and the ex-Missourian Hussmann. Hilgard, first professor at the grape and wine laboratory at the University of California, had studied winegrowing in Michigan and Mississippi and had published in *Hilgardia* magazine.

With the root louse seemingly out of the way, a general opinion emerged that the potential of the California wine industry was limitless. As in New York and Missouri earlier in the century, wine production in California was developed far in excess of demand.

This situation was aggravated further by get-rich-quick vintners who marketed cheap, poor-quality wines. Their schemes crashed in the late 1860s, bringing about a depressed market for California wines. Industry leaders learned that more technical and economic knowledge was required before California wines could become the American standard. Professor Hilgard wrote in 1879:

As the depression was, beyond doubt, attributable chiefly to the hasty putting upon the market of immature and indifferently made wines, so the return of prosperity has been, in great measure, the result of steady improvement in the quality of the wines marketed—such improvement being partly due to the introduction of grape varieties better adapted than the Mission grape to the production of wines suited to the taste of wine-drinking nations.

Two colossal personalities led the recovery of California winegrowing in the 1880s—George Hearst and Leland Stanford. Hearst made a fortune in mining, became a U.S. Senator, and helped launch the monumental publishing career of his son, William Randolph Hearst. George Hearst purchased a 400-acre tract in Sonoma County, but *Phylloxera* promptly killed his vines. Undaunted, Hearst ordered new grafted vines, replanted, and ultimately attained production capacity of nearly 250,000 gallons of wine each year. Stanford, a railroad magnate and patron of Stanford University, built a winery amid the 350 acres of vineyard he owned near San Jose. By 1888 his output exceeded 1 million gallons.

Because of projects such as these, California vineyards once again expanded to the point of overproduction. A subsequent recession was even worse than that of the previous decade. Grapes were sold for less than \$10 per ton and wines for less than 8 cents per gallon.

To make matters worse, another mysterious disease was found in vineyards, principally in the Los Angeles vicinity. Vines rather suddenly became defoliated and died. This condition, which bewildered plant pathologists for decades, was finally identified as a bacterium introduced to vines by a leafhopper insect. It was called Pierce's disease, and to date no cure has been found. Pierce's disease is generally found in subtropical climates and was probably involved in many of the *vinifera* failures experienced in the South.

But most winegrowers forged ahead with little concern for economic and biological threats, perhaps encouraged by French winegrowers who admitted that California could grow Old World grapes well enough to be capable of entering serious competition with the wines of Europe. American vintners were invited by the French to enter their wines in competition at the Paris World Exposition of 1889. The results stunned the international wine scene: U.S. vintners winning 42 medals awarded by French wine judges! Among the California winners were Beringer, Krug, and Schramsberg. Stone Hill was a Missouri medalist, as was Great Western from New York State.

NATIONAL PROHIBITION

It had been building for some 80 years—a disaster worse than any disease or depression that had yet been known to American vintners. It was a reflection of profound and perhaps excessively rapid change in the fabric of American life. Though wine had been the symbol of the Church throughout the entire history of Western civilization, the Church became, ironically, the primary influence used to destroy American winegrowing.

The end of the Civil War did not end America's differences. Dynamic advances in industrial technology, a huge influx of immigrants coupled with a

population explosion, increased urbanization, lingering influences from the Victorian era, and Protestant sectarianism were all an aftermath that contributed to the idea that alcohol, in any form or measure, was sinful and a detriment to human welfare.

It formally began with the 1846 passage of Prohibition in Maine; Massachusetts, Rhode Island, and Vermont followed suit in 1852. More notable, however, was the 1880 enactment of Prohibition in Kansas. The force behind this action was the infamous Carrie Nation, who assumed, at one time or another, the roles of judge, jury, and executioner in taking action against any facility that might have had anything to do with an alcoholic beverage (Figure 1–5). Her disregard for private property and human rights spread quickly. As a result, much of our early American wine history was lost as books, diaries, and journals were burned by the *Prohibitionists*.

The first signal of national Prohibition came with the “Dry Movement.” Proponents demanded that any mention of wine be struck from textbooks and from the U.S. Pharmacopoeia. The “Drys” published books claiming that the word *wine* in the Bible was really grape juice, while at the same time advocating the ban of classical Greek and Roman literature that mentioned wine. By the time World War I broke out in Europe, a total of 33 states had voted to go dry. By 1919, the Drys had succeeded in gaining wartime Prohibition. The 18th Amendment to the U.S. Constitution, in conjunction with the Volstead Act, brought Prohibition across all of America in 1920.

Vintners who had foreseen the reality of Prohibition taking shape geared up their production and marketing forces for medicinal, sacramental, and salted cooking wines, which remained legal under the Volstead Act. Some wine masters went into the grape juice business, but many were forced to close their doors altogether.

Consumers who were determined to keep drinking wine found ways around Prohibition. Fake churches and synagogues were founded for the purpose of dispensing legal “sacramental” wines. Wine was often prescribed as “medication” for all sorts of human conditions.

Of all the schemes, perhaps the most blatant was the sale of an array of grape juice products with a “pill” of dehydrated dormant yeast attached for home winemakers. The complete package was usually labeled with a printed warning not to add the pill “because if you do, this will commence fermenting and will turn into wine, which would be illegal.”

Home winemaking soon became legal, with the head of each American household allowed to make up to 200 gallons per year for personal and family consumption. Some growers converted their vineyards from wine grapes to juice grapes. Concord growers knew that their grapes made high-quality grape juice. In fact, the founder of Welch’s grape juice company was a devout Prohibitionist.

The owners of Stone Hill Wine Cellars at Hermann, Missouri, had the novel idea of remodeling their cellars into caves for mushroom farming. The Glen Winery in

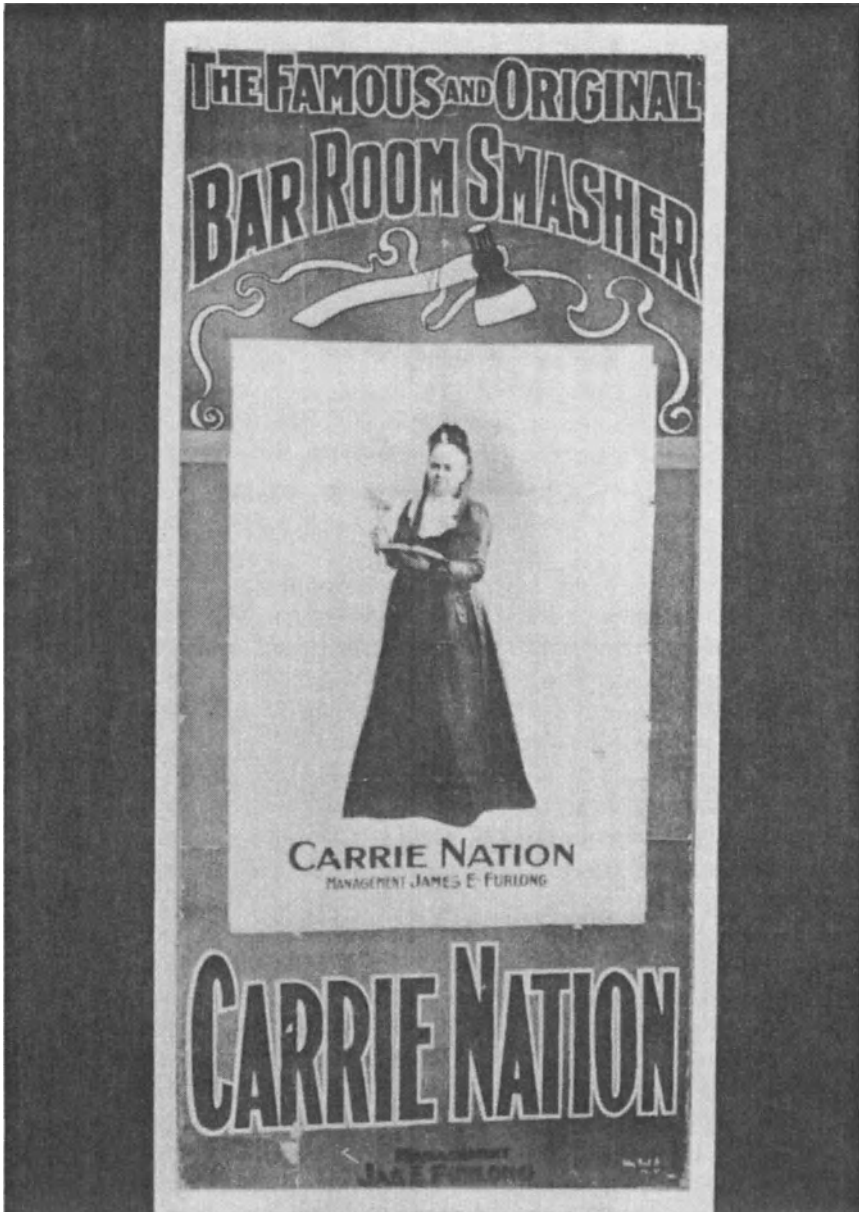


Fig. 1-5 Carrie Nation Ready for Battle.
Courtesy: The Kansas State Historical Society.

Hammondsport, New York, was transformed into a storehouse for Glenn Curtiss airplane parts.

A number of vintners desperately tried to continue their operations in secret, illegally, becoming “bootleggers.” For the most part, though, such efforts were ultimately discovered and destroyed by U.S. Treasury officers, or “T-Men,” who, standing amid the rubble of a winery just destroyed, would strike triumphant poses for news photographers.

Ironically, most of the bootleggers who prospered were criminals who had access to the gangster network of underground labor and resources that could circumvent or corrupt the enforcement efforts of the T-Men. It was a conflict in which the names Al Capone and Eliot Ness would be etched forever.

California vineyard land actually grew during Prohibition, totaling more than 640,000 acres. Most of the grapes grown were of the variety Alicante Bouchet, which resisted spoilage in storage and transport. More important was that this variety is so dark and dense that 1 gallon of crushed grapes could be ameliorated with another gallon or more of water and sugar. The resulting wine, often referred to as *Dago Red*, gave the bootleggers a beverage they could market cheaply to lower income people.

Despite the tenacity of the T-Men, the nation was well supplied with alcoholic beverages in countless ways, with perhaps the memorable “speakeasy” private clubs being the most colorful. Bootlegger greed eventually made Prohibition wine expensive—and made the gangsters rich. Desperate alcoholics who couldn’t afford the contraband prices resorted to drinking rubbing alcohol, aftershave lotions, colognes, and the like, which were made from butanol, methanol, and other toxic alcohols.

It didn’t take long for most people in America to realize that the “great experiment” of Prohibition was a huge mistake. Many attempts were made for the reversal of the Volstead Act, as well as some proposals for modification of the Prohibitionist regulations, but such efforts were fragmented by egoistic disunity. Thomas Jefferson’s statement, a century earlier, rang vividly appropriate: “No nation is drunken where wine is cheap.”

Finally, recognizing that Prohibition was destroying America, the nation commenced to rally around the demand for national Repeal. The impetus was, of course, not from the underworld, for they had built an empire with *Dago Red* and bathtub gin.

Nor did the Repeal movement originate with the legal vintners. They were, except for the remaining medicinal and sacramental wine producers, long since out of business. The support for Repeal came from a force perhaps best described as a national conscience. Early in December 1933, Repeal took effect, bringing to an end 14 years of devastation to the entire country, and particularly the American wine industry.

AMERICAN WINE IN THE 20TH CENTURY AND THE NEW MILLENNIUM

Repeal came in the midst of the Great Depression, which was a major hurdle for commercial winegrowing recovery. Consequently, few winegrowers in 1933 believed that American vines and wines could ever be resurrected to become a viable industry again, much less ever offer serious competition to the wines of Europe, which had made great advances during the American Prohibition.

Apart from the Italian Americans who continued to drink Dago Red, the new generation of young Americans maturing during Prohibition found that dry table wines were an unfamiliar “sour” foreign-type of drink. In addition, many states voted to remain Prohibitionist, especially in the Deep South; Mississippi remained so until 1966. Other states voted in rigorous tax laws and government control, while still others heavily restricted marketing activities.

Perhaps the only positive thing born out of Prohibition was the need to start over again from scratch. The Prohibition years effectively ended many of the traditional vineyards, wineries, and practices that had been detrimental to the advancement of the wine art. With the slate swept clean, there was an opportunity to formulate a truly American wine industry rather than a poor imitation of the European one. In any event, American winegrowers, presented with the ruins of abandoned vineyards, decayed cellars, and broken lives, began their work anew.

Unfortunately, a plethora of articles and books written by self-proclaimed American wine experts disseminated wine fallacy and snobbery. The notions that only certain wines should accompany specific food entrees—and then at precise serving temperatures, and in specific types of stemware, and other nonsense—confused and intimidated potential wine consumers.

Nevertheless, the U.S. wine industry slowly re-emerged. Franklin Delano Roosevelt saw to it that American wines were once again served in the White House. The late Leon Adams, a young San Francisco news reporter and wine enthusiast, was a key figure in founding the Wine Institute. Adams (Figure 1–6) spent his life as a wine educator.

During the late 1930s several foresighted pioneers were to make a permanent mark on the American wine industry—each with a very different philosophy of winegrowing. Philip Wagner was a Baltimore newspaper editor and an amateur winemaker. After visiting Europe Wagner had become deeply interested in some of the little-known French-American hybrid vines. These had been developed as an alternative to save Europe from the *Phylloxera* root louse. The idea had been to breed new cultivars of grapes, from parentages of *vinifera*, with *labrusca*, *riparia*, *rupestris*, and other species, endeavoring to combine the high wine quality of the Old World vines with the disease resistance, cold resistance and high productivity



Fig. 1–6 The Late Leon Adams, Cofounder of the Wine Institute.

Courtesy: Vineyard and Winery Management.

of the native American selections. From thousands of crosses, several hundred promising new hybrids were developed, each identified by number codes instead of names. Those that captured Wagner's attention were imported and propagated in his Maryland nursery.

Wagner befriended many winegrowing people in the East, particularly in New York State. Greyton Taylor, youngest son of the Taylor Wine Company founder, planted experimental plots of Wagner's imported hybrids. The successful ones had their number codes changed to marketable names: the best whites now known as Seyval Blanc, Vidal Blanc, and Vignoles, and the reds as Chambourcin, Chelois, and Marechal Foch.

The French-American hybrids were instantly rejected by *vinifera* purists and *labrusca* traditionalists. It took more than 50 years to perfect the viticultural and vinification techniques necessary to result in competitive wines. Today, they compete very well indeed. In 1992 a Missouri Vignoles was determined by more than 30 premier wine judges at the New World Wine Competition in California to be the "Champion White Table Wine," besting more than 700 competing whites of every variety. Hybrid grape wines continue to win awards in prestigious competitions.

Nevertheless, considerable controversy still surrounds the French-American cultivars. According to some, the hybrids are a viable alternative to native *labrusca* grapes and good competition for the Old World *vinifera*. To others, the hybrids are

bastard children not worth planting on even the most marginal of vineyard land. Some still think that only wines from Europe are worthy of serious consideration.

Another prominent post-Prohibition figure was Charles Fournier (Figure 1–7), who emigrated in 1934 from the Champagne region in France to Gold Seal Vineyards in Hammondsport. With 200 years of failure in the eastern United States, Fournier was fully forewarned that *Vitis vinifera* would not survive the disease and harsh winters of upstate New York, but he experimented anyway.

A decade or so later, Fournier met Dr. Konstantin Frank, a Russian winegrower refugee. Frank convinced Fournier that it was possible to grow Old World vines in colder climates by employing highly selective cultivation techniques learned in mother Russia. Fournier hired Frank to work in the Gold Seal vineyards, and they experimented together with the tender *vinifera*. During the late 1950s they succeeded in harvesting the first commercial quantities of Old World grapes grown in eastern America. While devastating winters still bring caution and fear among eastern winegrowers, the development of Old World wines continues to advance upon the breakthroughs by Frank and Fournier.

Brothers Ernest and Julio Gallo had their roots in Asti, in the Piedmont region of Italy. The Gallos first grew grapes at Antioch, California, where their parents had immigrated from Italy via South America. During Prohibition the Gallo family shipped grapes for home winemaking to Chicago and New York. Instead of the Alicante Bouschet variety, however, they marketed the higher quality Zinfandel.

Following repeal the Gallos became interested in the commercial wine business and leased a warehouse in Modesto from the Santa Fe Railroad in which to commence their initial operations. Demand for their bulk wines grew rapidly, and by World War II Gallo wines were also competing for consumers in the retail marketplace. The marketing genius of Ernest and the production expertise of Julio combined in a management team of unequalled drive and success. In less than 40 years Gallo became the largest single winery in the world. Its success heralded the greatest growth in the U.S. wine industry to date.

THE AMERICAN WINE BOOMS

The first major U.S. consumer rush to buy wine began during the late 1940s, perhaps nurtured by returning American servicemen who brought home a newly acquired appreciation for wine from Europe. Subsequently, a burgeoning commercial airline industry gave business and pleasure travelers an entirely new exposure to European wine consumption.

The quest for wine technology was again rekindled. The American Society of Enologists was founded in 1950 at the University of California at Davis, where wine researchers followed up the early work of Hilgard.



Fig. 1-7 The Late Charles Fournier.

Courtesy: Vineyard and Winery Management Magazine.

Funding was made available in other states for commencing or renewing viticulture and enology programs of study. As both European and American wine masters conducted intensive research programs, the traditions of Old World methodology began to give way to innovations based upon New World technology.

The forward steps were by no means confined to production. There were many new ideas for wine marketing too. New and more efficient distribution channels opened up better access to retail-shelf space and restaurant wine lists. People began to tour the wineries, learning the jargon and sampling wines. Vintners printed sales promotion pamphlets replete with wine cookery recipes and down-to-earth wine-serving tips. Slowly the veils of wine snobbery and intimidation commenced to fall.

By far the most important contribution of the 1950s was a simple commonsense idea expanded upon by Frank Schoonmaker, a devoted and highly respected American wine authority. Schoonmaker's concept was that the labels of American-made wines should carry the name of the grape variety rather than continuing to borrow the traditional European geographic names. A wine made from Pinot Noir grapes in a style similar to that of French Burgundy would no longer be labeled "California Burgundy" but rather "California Pinot Noir." The great German *vinifera* vine varieties Johannisberg Riesling and Sylvaner were much more honest and descriptive terms on labels than, say, "California Rhine Wine."

The conversion proved to be rather easy in California, especially in the Napa, Sonoma, and other North Coast districts, where many of the classic Old World varieties had been grown since the Haraszthy influence a century earlier. In the East, however, conversion to varietal labeling proved to be more difficult. The native American grape varieties, such as Catawba and Concord, were not internationally known, and the French-American hybrid cultivars were not yet widely planted. Consequently, many eastern vintners retained the practice of using European geographic names, such as "New York State Burgundy," "Ohio Rhine Wine," and "Michigan Chablis," among many others. While some of these wines were tasty, the practice was nonetheless false labeling, as native and hybrid grapes cannot be vinified to *vinifera* flavor. Consequently, commercial eastern winegrowing in the East became suspect to consumers during the 1950s.

Clever eastern vintners devised many efforts to conserve their market share. One innovation was "pop wines"—simple table wines flavored with synthetic essences and given hip names. A mixture of "Champagne" and "Sparkling Burgundy" made a bubbly called Cold Duck, which was an instant sensation. A traditional Spanish drink called Sangria, a mixture of fruit and light red wine, achieved sales in the millions of cases. An even greater success was a rush for fruit wines from California, particularly Gallo's Boone's Farm brand. The success of these created a second wine boom that inspired both European and Californian competition, creating yet another glut of production.

The third wine boom in the United States gathered momentum during the latter 1970s and early 1980s via a totally different operational philosophy: the

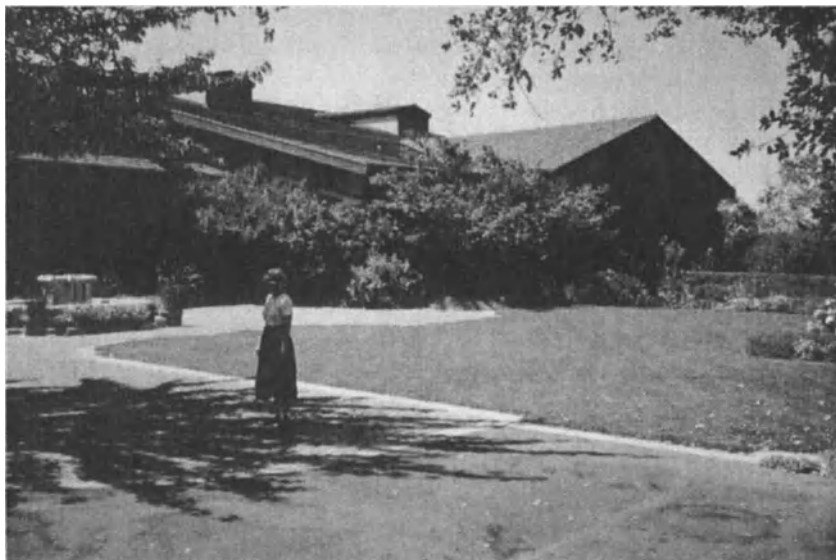


Fig. 1–8 Silverado Vineyards Winery: The Walt Disney Estate.

establishment of more than 1,000 small “boutique” wineries. The typical estate (illustrated in Figure 1–8) was a few acres of prime vineyard land planted to several select *vinifera* varieties that had as its center a state-of-the-art production facility housed in an architectural masterpiece. While most of this development occurred in the North Coast counties of California, boutique wineries emerged in most every state except the colder extreme upper central region of America. These new vintners prided themselves on individual wine character and style, but they were unified by the common expression of producing wines of high quality—wines that could challenge the classical Old World vineyards.

Small wine contests became comprehensively structured wine competitions. In the early 1990s U.S. wine consumption for the first time surpassed all distilled spirits combined. New consumer wine periodicals keep Americans abreast of news from the vineyards. Chapters of national and international wine societies and brotherhoods continue to grow.

Much of this success is credited to the technological advances in winegrowing across the United States. As previously mentioned, the University of California at Davis has pioneered viticulture and enology research, but other institutions have contributed significantly too. Among these are the Universities of Arkansas, Florida, Missouri, and Pennsylvania, along with the California State University at Fresno, Cornell University, Michigan State, New Mexico State, North

Carolina State, Ohio State, Oregon State, Purdue, Texas A&M, Virginia Tech, and Washington State.

Out of all this has come a national interest in fine wine such as America had never seen before. It is common to find large metropolitan wholesale distributors carrying portfolios of several thousand different selections for retail wine shops and restaurant wine lists.

Recent statistics published by the Wine Institute indicate that annual wine production of approximately 500 million gallons now ranks the United States fifth in wine production among nations, behind France (2 billion gallons), Italy, Spain, and Argentina, respectively.

CHAPTER 2

VITICULTURE (GRAPE GROWING)

Viticulture is the part of horticulture that deals solely with the growing of *Vitis*, the vine. A professional grape grower is a viticulturist.

More than 20 million acres of vines are cultivated in vineyards around the world, with more than 50 different species identified. The most widely grown species for wine is *Vitis vinifera*, or the “Old World” vine native to Europe. *Vinifera* is also the principal species for wine in California, Oregon, Washington, and other states having moderate climates.

Some species, such as *Vitis labrusca*, are native to the United States, so they are more disease resistant and endure colder climates than the *vinifera*. *Labrusca* is found in all of the midwestern and northeastern states, as well as the lower reaches of Canada. Hybrid cultivars are also found in almost every winegrowing region of the world.

An extra level of satisfaction can be gained in growing one’s own grapes for winemaking. Wines grown from vineyards owned by commercial vintners are often labeled “estate bottled” (Figure 2–1).

It may be the fifth or sixth year before a full vintage of grape production is taken. All things considered, it is common to find these years of vineyard establishment costs totaling more than \$7,000 per acre—not including the cost of land.



Fig. 2–1 Northern california estate vineyard and winery. Photo by authors.

Good places to commence the search for information relating to growing grapes and/or finding sources for grapes are the Cooperative Extension Service and regional winegrowing associations.

The grapevine is a woody perennial that can live for many years, generally several decades of productive growth, with proper care. Grape flowers and fruit are borne only on new shoots, which arise from dormant buds formed on the previous season's growth. When a shoot matures and drops its leaves, it is known as a *cane*. It is from canes that the next year's fruit wood is selected at pruning time.

Grape buds are classified as compound buds because they contain a primary bud and one to four separate, smaller buds. When growth starts in the spring, the primary (center) bud breaks dormancy and produces the fruiting shoot. Low midwinter temperatures may kill primary buds, and spring frosts may occasionally

kill the tender primary shoots during early stages of growth. If this happens, one of the smaller buds usually develops into a shoot. These shoots are usually less vigorous and produce fewer, smaller fruit clusters. However, as these shoots grow, new compound buds are formed that will provide the potential for a full crop the following season.

Flower clusters develop on shoots at nodes opposite a leaf. At nodes where fruit clusters are not formed, a tendril will develop. The number of clusters that develop per shoot is mostly a genetically controlled characteristic of the cultivar, with little influence by environment. The number of berries per cluster, however, is greatly influenced by the health and vigor of the vine, the environmental conditions during flower bloom, and other factors. Proper care of the grapevine is essential in meeting full fruiting quality and quantity potential.

SITE SELECTION

Climate

The major limiting factors to grape production in most temperate-climate locales are cold-temperature injury from winter cold and spring frosts, along with diseases encouraged by hot, humid weather and frequent rainfall during the growing season. Selection of a site with a desirable climate helps reduce the occurrence of these problems. The best sites for grapes are those with full sun exposure, mild winter temperatures, freedom from frost, and good soil drainage. Cultivar selection is obviously essential in determining the suitability of a given site.

Two important considerations in site selection are minimum winter temperatures and freedom from frost. Cultivar selection is often dictated by the minimum temperature expected for an area. Very hardy cultivars are capable of withstanding -15°F with little injury, while tender cultivars will suffer significant injury at temperatures slightly below 0°F . Varietal selection, preconditioning of the vine, the vine's state of acclimation, and the rate and extent of temperature change can dramatically affect the amount of cold injury sustained.

Freedom from spring and autumn frost events is another important characteristic of vineyard sites. Frost-free sites are usually areas with gentle slopes that are elevated above surrounding areas. Cold air drains from higher sites to lower levels, reducing the risk of damaging frost at the elevated sites. Higher elevations also have better air drainage throughout the growing season, which promotes rapid drying of foliage following dew or rain, thereby greatly reducing the incidence of disease organisms.

The length of the growing season is another important factor for grape production. It is determined by the dates of first and last frosts of the year. Length of the growing season (frost-free days) ranges from 150 to 160 days in cooler regions to

190 to 200 or more days in warmer locales. A frost-free period of at least 150 days is required for the earliest ripening cultivars. The length of the growing season is often extended within a region on sites near large bodies of water, which temper drastic changes in temperatures.

Soils

Grapevines are adapted to a wide range of soil types but perform best when they have healthy, well-developed root systems. Soil conditions favorable to root growth include good aeration, loose texture, moderate fertility, and good surface and internal drainage. Proper soil drainage cannot be overemphasized as a necessity for successful vine culture. Soils that are consistently wet during the growing season due to an impervious subsoil, high water table, or other drainage problem should be avoided. Root growth in poorly drained soils is usually limited to the top 2 feet or less, whereas in deep, well-drained soils roots may penetrate 10 feet or more. When root growth is restricted because of poor drainage, vine growth and fruit yields are generally low, and plant survival is limited to only several years.

THE VINE

The leaves, stems, and roots of the grapevine classify it biologically as a higher form plant. Such plants are able to use sunlight to make sugars from carbon dioxide and water through the process of photosynthesis. Some of these sugars are used in the metabolic processes of the plant's life system. Another portion of these precious sugar compounds is stored in the grapevine's roots until *veraison*, a French term referring to that time when grapes commence to ripen and sugars are transported from roots to berries.

Taxonomically, the grapevine is classified as follows: group Spermatophyta, division Tracheophyta, subdivision Pteropsida, class Angiosperma, subclass Dicotyledoneae, order Ramnales, family Vitaceae (Ampelidaceae), genus *Vitis*, subgenera *Euvitis* and *Muscadinia*.

There are more than 50 species of *Vitis* known, and many of these are indistinguishable to all but highly skilled viticulturists. Origins of the vine are largely limited to the Northern Hemisphere, and the North American continent is particularly abundant in native species.

Vitis is a deciduous plant that climbs by grasping supporting objects with outgrowths of very special leaf-type organs called *tendrils*. In the wild, vines may reach the tops of tall trees, but in culturing of the vine, the natural growth is controlled by annual pruning. Typically, vines in vineyards are trained onto post-and-wire trellis supports designed to optimize both the quality and the quantity of the plant (Figure 2–2). Exposure of grape leaves to sunlight increases the production of grape

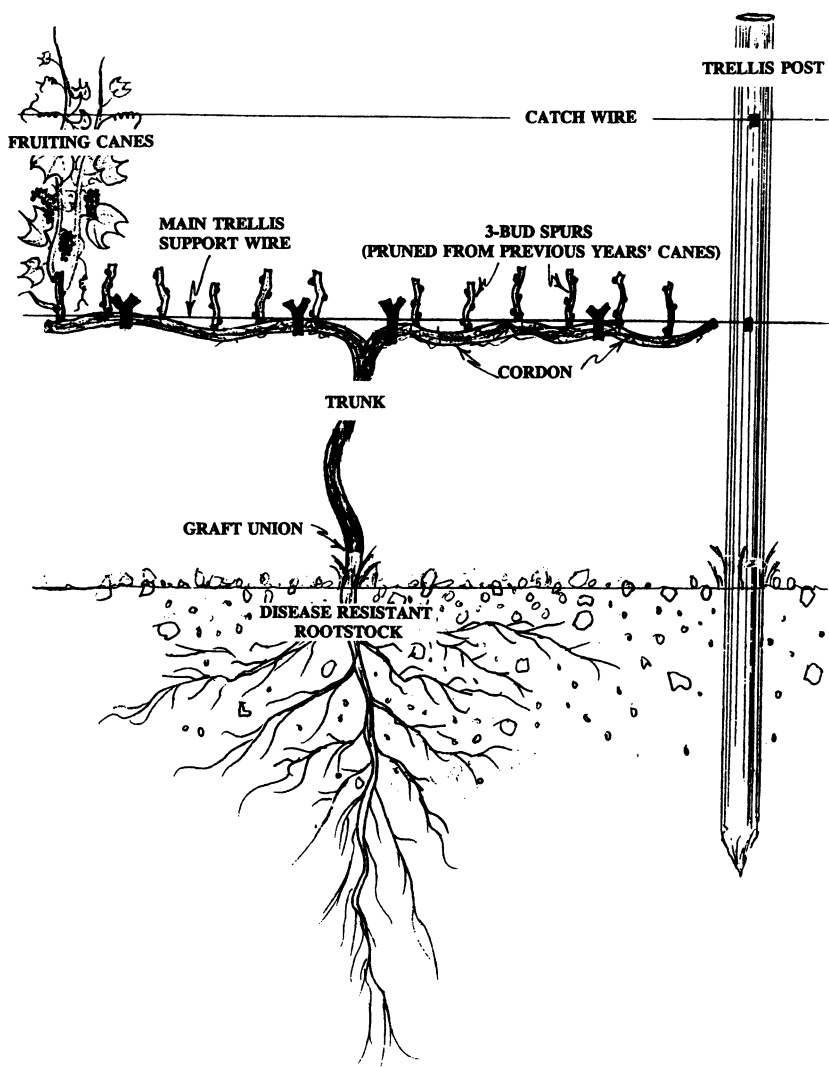
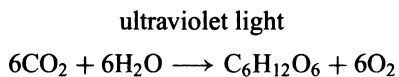


Fig. 2-2 Diagram of vine and trellis.

sugars by allowing photosynthesis to transform carbon dioxide, which eventually will result in grapes of varying levels of sweetness.

The following is a chemical equation for photosynthesis:



Carbon dioxide is the substrate from which the reaction is fueled, catalyzed by the ultraviolet rays in sunshine, with sugar as the product and oxygen as a byproduct.

Traditionally, grape quality is closely linked to sweetness, and grape growers monitor sweetness levels carefully. Severe pruning is often practiced for smaller crops of higher quality, with lesser pruning generally associated with larger quantity, lower quality fruit.

As a rule, two buds emerge from the *abscission point* of each leaf on the vine. These buds are covered by a single cap, or *caliptra*, and in many varieties a third, or *tertiary*, bud is also found, but it rarely sprouts. Unless damaged by spring frost or some other malady, the primary buds will sprout to produce new shoots, which will develop leaves and fruit. The flowers of the vine emerge on developing shoots. At the perimeter of the flower, several male *stamens* produce pollen, which, through wind, insect activity, or other mechanical means, finds its way to the central female *pistil*, where fertilization takes place and formation of grape berries commences.

Shoots mature from a supple green vegetative state to a hard woody cane, which is pruned according to varied viticultural motives and designs. When primary buds or shoots are damaged, secondary buds will grow to produce a grape crop—almost always of lesser quality and quantity than would have been achieved by the primaries.

In any event, all fruit results from the current season's growth. Bunches of grapes may be composed of fewer than 10 berries each for some of the more obscure varieties and more than 300 berries for some of the most prolific types. Cluster shapes range from long, cylindrical forms (with or without *shoulders*) to shorter conical structures (Figure 2–3). The grape is morphologically a true berry, since it is a simple fruit that has a pulpy *pericarp* consisting of four skin layers: the epidermis, hypodermis, outer wall, and inner wall (Figure 2–4).

Generally, grapes are native to temperate zones, primarily between 40° and 50° south latitude. The grapevine can, however, adapt to some rather difficult climate and soil conditions. In reproduction, vines often form *clones*, which are offspring with slight variations in character responding to natural selection processes. As compared to the classic type, some clones may be more resistant to cold winter temperatures, others may require a shorter growing season, still others produce heavier crops, and so forth. Some varieties have been catalogued with hundreds of different clones, from which selections are made to best meet the vintner's constraints and designs.

Young vines are commercially propagated asexually, usually from cuttings or graftings grown in a nursery for a year or so. A newly planted vineyard may be allowed to produce a token amount of grapes during its third year of growth, with other clusters pinched off shortly after bloom.

The subgenus *Euvinis* includes all the world's grapevines that produce grapes in clusters—some 50-odd species. The most important species of *Euvinis* is *Vitis vinifera*—vines that originated in Asia Minor and Europe.

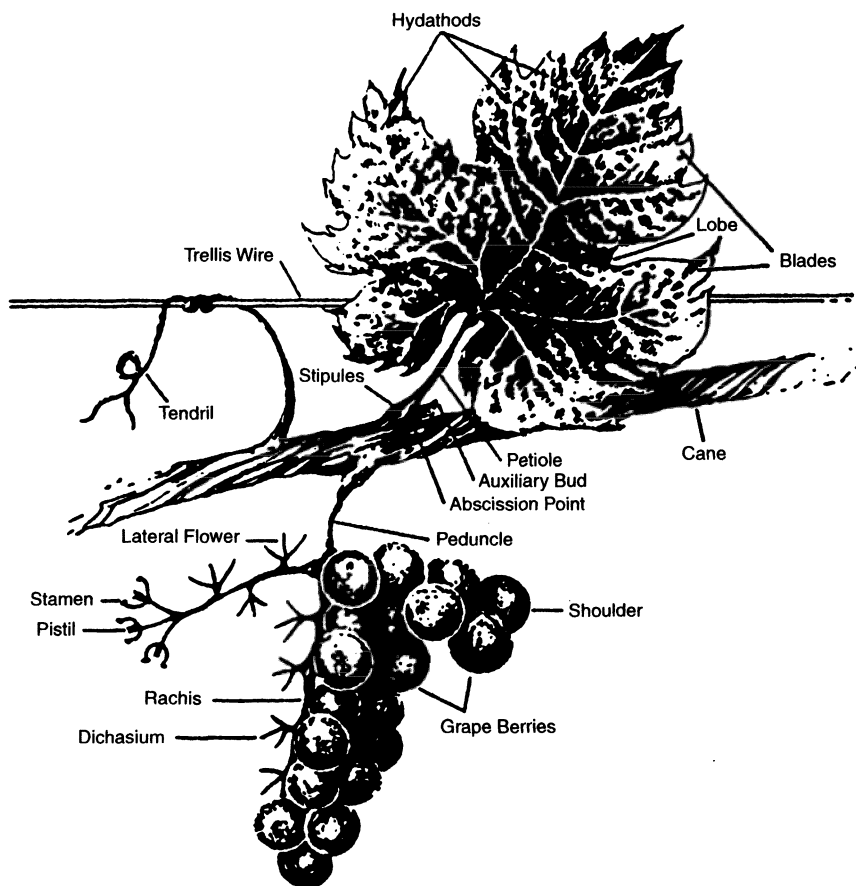
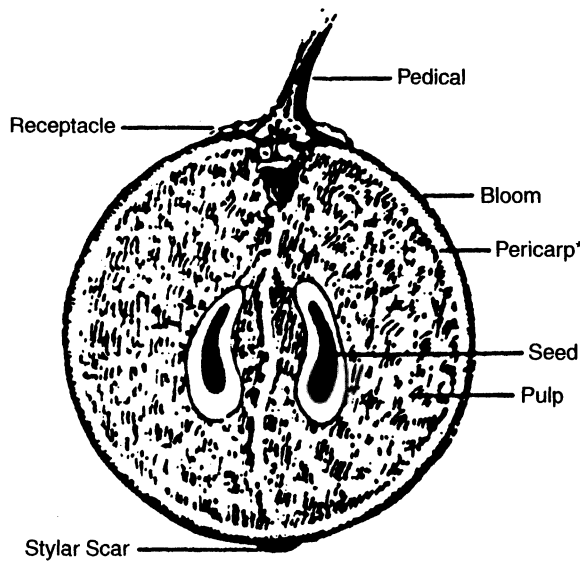


Fig. 2-3 Diagram of a grape cluster.

CULTIVAR SELECTION

Selection of the proper vine varieties and cultivars is obviously fundamental to the product line that will become an estate winery's product portfolio. It is equally fundamental to successful grape growing. Because the desired cultivars may not be ideal for the vineyard site selected, choices may be difficult to make.

Because wine enthusiasts are sensitive to the pedigree of vine varieties upon wine labels, some estate vintners accept high risk in planting high-profile varieties on sites that are marginal. It is recommended that supplemental cultivars be considered to make alternative wines and reduce that risk. Cultivar selection should be based upon the intended use of the fruit and its versatility, along with, of course, relative cold-hardiness and disease resistance.



*Pericarp – consists of epidermis, hypodermis, outer wall and inner wall

Fig. 2–4 Diagram of a grape berry.

VITIS VINIFERA

Hundreds of varieties of *Vitis vinifera* are cultivated around the world today as the “Old World” vine. More than 90% of the wine grown globally is from this genus and species.

Major *Vitis vinifera* White Varieties

Chardonnay (shar-doe-NAY)

This classic variety is native to France but is grown around the world. It ripens late midseason to small, compact, cone-shaped clusters of light green berries (Figure 2–5). Chardonnay produces sparse crops—the very best quality from struggling vines in marginal soils and climates. It is famous for complex apple-citrus, honey-olive flavors, from the noble vineyards in the Chablis, Meursault, and Montrachet districts in Burgundy. Chardonnay is also the principal variety grown in the Cote de Blancs district of Champagne.

Sauvignon Blanc (so-veen-YOHN BLAWN)

The white Sauvignon is another classic variety native to France and also very popular in many other countries. The name is derived from the French *sauvage* or “wild.”

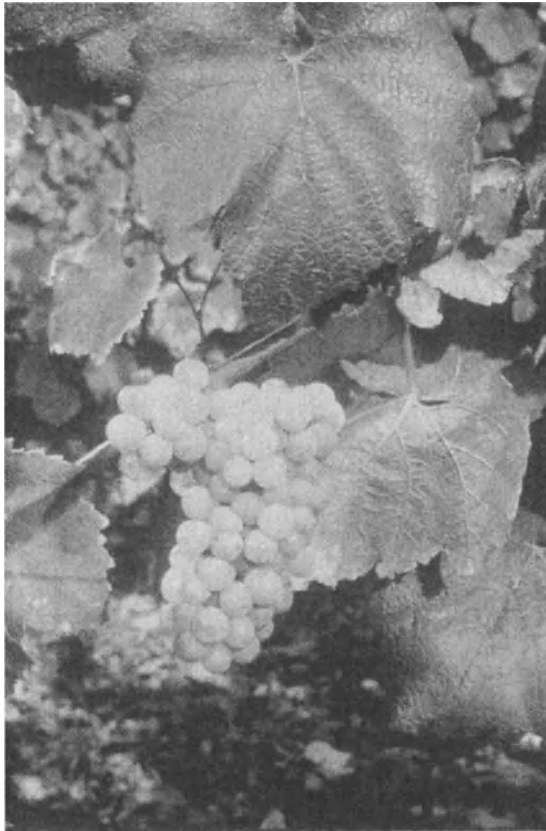


Fig. 2-5 Chardonnay cluster.

Courtesy: Food and wines from France.

It is a vigorous vine with long, loosely clustered yellowish-green berries maturing early midseason. From the Pouilly-Fume vineyards in the upper reaches of the Loire Valley region, Sauvignon Blanc wines are typically intense in a blossom-melon aroma, or “fume” in flavor character—a style sometimes labeled as “Fume Blanc” in the New World. In Bordeaux, New Zealand and in much of northern California, the wine has a pale greenish tint and bold grassy-pineapple-grapefruit flavors.

Johannisberg Riesling (yo-HANNES-berg REEZ-ling)

This vine is native to Germany and is the principal variety grown there for that country’s very best wines. The fruit takes the form of short cylindrical clusters of yellowish-gray berries dotted with russet spots. The prized Riesling, sometimes labeled as “White Riesling” or just “Riesling” in America and other countries, can

result in a golden wine of intense floral-apricot flavor. It is a variety increasingly planted in the Great Lakes and northeastern America, as well as in Oregon—less so in California.

Semillon (SEM-mee-yohn)

This variety is most closely identified with the famed sweet wines made from purposely over-ripened grapes in the Sauternes district of Bordeaux. Clusters of Semillon are comparatively small, with tightly compact yellowish-golden berries. While Sauternes has a rich golden-amber hue and profound honey-peach flavor, dry (not sweet) wine made from Semillon has a shy yellow color with a delicate pear-melon character. It is finding growing popularity in the warmer climes of California and Washington State.

Gewurztraminer (gah-VOORTS-trah-meener)

Native to northern Italy, Gewurztraminer is generally considered the major variety of the Alsace winegrowing region situated in northeastern France. It yields moderately heavy crops of small cylindrical bunches tightly packed with small berries curiously yellowish-pink in color. The German *gewurz*, or “spicy,” refers to its flavor—a rather ginger-cinnamon aroma. The acreage planted to Gewurztraminer has been slowly on the rise in the cooler climate locales of Oregon, the Great Lakes, and the northeastern states.

Muscat de Frontignan (MOOSkat dah frawn-teen-YAWN)

Muscat de Frontignan is one of many Muscat grape varieties that are prized for their bold, powerful fig-guava flavor. It is widely cultivated in southern France, and a close relative, Moscato, is grown in northern Italy for the famed Asti Spumante sparkling wines. Muscat de Frontignan produces elliptically shaped berries upon compact medium-sized clusters. Only token vineyards of this variety are cultivated in America, although this and other varieties of Muscat are increasingly planted for wines that are blended to enhance the fruit aroma in other white wines.

Pinot Blanc (PEE-noh BLAWN)

This variety is closely identified with the southern districts in the Burgundy region of France. Pinot Blanc gets its name from its “pine” cone cluster shape, packed tightly with small green berries. Its wine is very shy in flavor, recalling very light nuances of fig, melon, and raisins. Despite this, it is gaining a little popularity among vintners in California.

Sylvaner (sil-VAHN-ner)

Sylvaner (sometimes spelled *Silvaner*) is native to Germany and can generally be found in the warmer, less rigorous climates of the southern winegrowing districts there. It matures to small, cylindrical, compact clusters of spherical yellow-green berries. Sylvaner generally yields heavier crops, but less flavor intensity, than its

sibling, the noble Johannisberg Riesling. The Alsace region in France cultivates many Sylvaner vineyards. Interest in the variety has never been widespread in the United States.

Pinot Grigio

This variety from northern Italy is the same as Rulander in Germany and Pinot Gris in the Balkan countries. It produces light but distinctively fruity white table wines and is rapidly growing in popularity.

Major *Vitis vinifera* Red Varieties

Cabernet Sauvignon (cab-bair-NAY so-veen-YOHN)

The quintessential red Sauvignon is perhaps the most famous grape in the world. There are diverse opinions as to its origin, but it remains the principal variety grown for the renowned chateaux estates of Bordeaux in France. The vine is vigorous and makes long, loose, conical clusters of small spherical blue-black berries (Figure 2–6). The deeply lobed leaves of Cabernet Sauvignon are often used as the models for viniculturally inspired artwork. The wine is very dark ruby red with an aroma of green bell pepper and cedar and flavor of black currants. Cabernet Sauvignon is widely grown in California and Washington State, with several other states adding to this acreage.

Merlot (MAIR-loh)

Next to Cabernet Sauvignon, Merlot is the most important grape variety grown in Bordeaux. Merlot clusters are long and conical with compact medium-sized, spherical, blue-violet berries. Wine from Merlot is typically a bit lighter in color density and texture than Cabernet Sauvignon; it has similar flavor components but is markedly more fruitlike in overall character. In recent years the demand for Merlot has achieved remarkable growth, and a commensurate acreage of new vineyards has been planted in California, Washington State, and other milder climate states.

Pinot Noir (PEEN-oh nwah)

This is the regal variety of the magnificent “Slope of Gold” known as the grand “Cote d’Or” of northern Burgundy in east central France. It is called *Spatburgunder* in Germany. Pinot Noir yields its best quality fruit in cooler climates and poorer soils epitomized in Burgundy. It yields small blue berries tightly packed on cone-shaped clusters. Red wine from Pinot is very light in hue and density, with a distinctive coffeelike aroma and brambleberry flavor. Some districts in California and Oregon are building fine reputations for the great Pinot, as are certain locales in upper New York and other eastern states.



Fig. 2–6 Cabernet Sauvignon cluster.
Courtesy: Food and wines from France.

Gamay (gam-AAY)

Gamay is the premier grape variety of the famous Beaujolais district of the Burgundy wine region in France. Confusingly, the variety Gamay Beaujolais is neither Gamay or Beaujolais but rather a clone of Pinot Noir. Gamay is a vigorous vine producing abundant medium-sized conical, compact clusters of large, slightly elliptical blue berries. Young Gamay wine has a medium color density with a regal purplish hue, a black-cherry aroma, and a plum flavor. Comparatively few U.S. vineyards are planted to Gamay.

Sangiovese (san-gee-oh-VAAYZ-ah)

This is the revered vine of the Tuscany region in north central Italy. A clone called Sangiovese is grown for the famous Chianti wines of Tuscany. Sangiovese requires

a long warm growing season to produce its best fruit. Clusters are moderately large oval black berries filled with an abundance of large seeds. Young Sangiovese wine is very dark, with a rich scarlet-ruby hue and flavors recalling truffles and blackberries. While acreages of this variety planted in America are still very small, interest seems to be increasing among northern California vintners.

Nebbiolo (neb-BYO-loh)

Along with Sangiovese, Nebbiolo is the other aristocrat among red wine grapes in Italy. The grand vineyards of Barolo and Barbaresco in the Piedmont region are testimony to the famous Nebbiolo. It ripens late to long, conical clusters that are often shouldered. Medium-sized berries are spherical and a “foggy” (*nebbia* means “foggy” in Italian) gray-blue. Young Nebbiolo wine is dense in purplish-garnet color and is typified by an aroma of violets with prunelike flavors. There are several commercial Nebbiolo vineyards in the United States, but interest grows slowly.

Zinfandel (ZIN-fun-dell)

A variety closely identified with California vineyards. Controversy continues as to the origin of Zinfandel, with the most widely accepted notion being that it is perhaps the very same vine as the Primitivo di Gioia in southern Italy. Zinfandel is a productive vine yielding large cylindrical clusters, tightly packed and often heavily shouldered. Berries are spherical and have a small brown apical scar. Young Zinfandel red wine has rich plum-berry flavors. The popular “White Zinfandel” is only partially fermented in contact with berry skins so that a faint “blush” color is extracted.

Syrah (sear-RAH)

Syrah is thought by some to have been brought to the Rhone Valley in southern France from Asia Minor by returning Crusaders more than 700 years ago. It was called Shiraz then, and that name is still used in Australia and other countries. Syrah ripens rather early, with moderate production of compact clusters of slightly oval-shaped berries. It is often confused with Petite Sirah, a totally different variety. Syrah has a dense blue-black hue with plum and truffle/tar-like aroma, but its great intensity of color and flavor is its real character.

Cabernet Franc (cab-bair-NAY FRAHNK)

Cabernet Franc is another of the grand Bordeaux varieties still used primarily for blending but now finding an increasing popularity as an individual, particularly in cooler climates across the United States and Australia. Cabernet Franc berries are a bit larger, with less density in color, acidity, and tannin than those of its Sauvignon cousin. The aroma of Cabernet Franc is distinctively tobacco leaf with herbal and grassy nuances. The young wine is characterized by a fresh but complex earthy-berry flavor.

Mourvedre (moor-VEHD-rah)

This vine is perhaps the same as, if not a close relative of, the Spanish Mataro vine. Cultivated primarily for blending in the Rhone Valley of France, Mourvedre has a new and increasing popularity as a varietal red table wine in the United States. Clusters of Mourvedre are rather large, often double-shouldered, with spherical dark blue berries.

Petite Sirah

This northern California vine is the same as Duriff in the Rhone Valley of France. It makes dense, dark, full-bodied, often tannic red table wine.

VITIS LABRUSCA

This species is the so-called “fox” grape. Some people contend that the character of *labrusca* varieties is related to one or another of the odors associated with foxes. This takes a bit of imaginative stretch in the mind of almost anyone who has ever been close to these animals. More plausible legends relate to foxes being attracted by the intense fruit flavor—a flavor often associated with “grape” soft drinks, candies, and so forth.

Major *Vitis labrusca* White Wine Varieties*Catawba (kah-TAW-bah)*

Catawba is a vine that was found growing in the wild near the Catawba River in North Carolina circa 1802. Catawba was found to be disease resistant, productive, and hardy. It was widely planted in southwestern Ohio during the middle 1800s and made America’s first commercial sparkling wine. It ripens late, with medium-sized, conical bunches of large, pink-skinned, spherical berries. The Catawba remains heavily planted in eastern U.S. vineyards, but interest is gradually diminishing.

Delaware

Delaware County, Ohio, is the origin of this native American variety, which was discovered during the early 1800s. It is widely cultivated in eastern U.S. vineyards and remains primarily because of a once huge demand for “New York State Champagne,” most of which has long since been replaced by California sparklers. Delaware ripens rather early to small, cylindrical clusters of tiny, spherical, pink-skinned berries (Figure 2–7). In the hands of a talented wine master, grapes from this variety can make wines that compare favorably with wine types found in Germany and Austria.

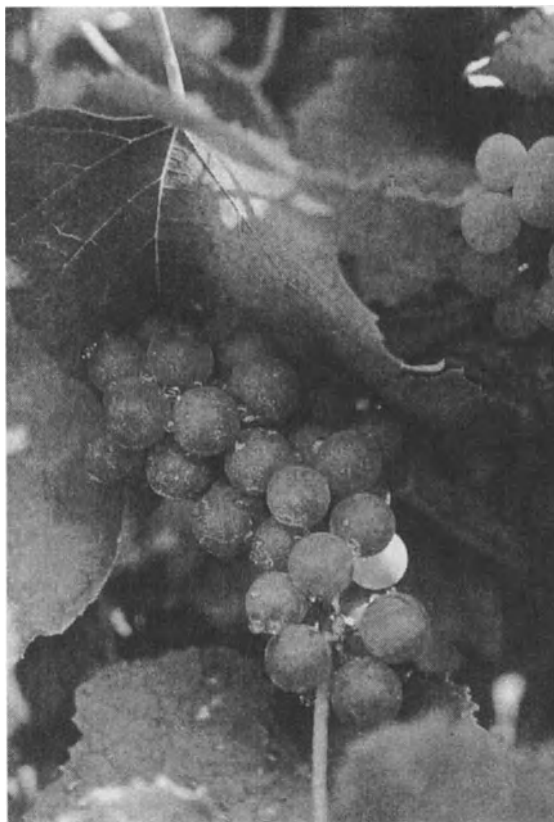


Fig. 2-7 Delaware cluster.

Niagara

This vine was developed by researchers in Niagara County, New York, during the 1860s. In preferred locales Niagara is a vigorous and productive vine, and its rich, heavily scented flavor makes it the standard bearer for rich, fruity native American white wines. Most vineyards of Niagara today are harvested for fresh table grapes and for white grape juice production.

Major *Vitis labrusca* Red Wine Varieties

Concord

Concord is the quintessential *labrusca* grape. A cultivar developed in Concord, Massachusetts, during the 1840s, it is better known for its juice, jelly, and table grape production than for wine. It is one of the most widely grown cultivar in the



Fig. 2-8 Concord cluster.

United States. Concord yields large crops of large, spherical, blue-black berries loosely arranged in moderate-size clusters (Figure 2-8).

Ives

There are several accounts as to the origin of Ives, but there can be no question about its immense color and flavor. It was first planted along the banks of the Ohio River during the 1850s and to a greater extent in upstate New York later. The grand New York State “Port” wines, very popular after repeal of Prohibition through the 1960s, were dependent upon Ives. It is a rather temperamental vine, not as productive as Concord or as cold and disease resistant—which is why it has long since been replaced with specially selected hybrid grape cultivars.

VITIS RIPARIA

This species is referred to as the “post-oak” or “frost” grape by some viticulturists. It is widely adapted to almost all of temperate North America but is mostly found east of the Rocky Mountains from Canada to the Gulf Coast. Because of its hardiness in cold-weather climates, it is often used for grafting as a rootstock or in grape breeding by researchers.

VITIS ROTUNDIFOLIA

The subgenus *Muscadinia* is more commonly known in America's Deep South as Muscadine. It generally has small, loose clusters of large berries typified by a thick skin and dense pulp consistency and literally bursting with jammy-fruit flavors. The bronze-skinned varieties are collectively referred to as Scuppernongs and the reds as Muscadines. Most are actually hybrid cultivars in the *Vitis rotundifolia* species, with the most commercially important of these being Carlos, Magnolia, Noble, and Tarheel.

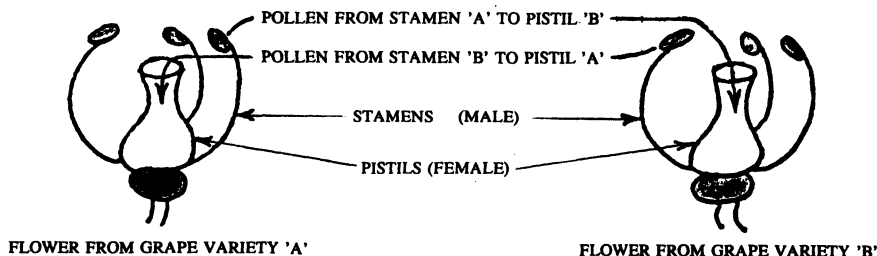
HYBRIDS AND GRAFTING

Natural variations in grapevine and/or fruit morphology result in what are termed *varieties*. Some varieties have certain mutations apart from a morphological change that are differentiated by the term *clone*. Varieties and clones are both the result of natural selection.

Hybrids, however, are another matter totally: humans, instead of nature, are involved in the breeding of grapevines. The process is one whereby pollen is gathered from the male stamens of selected grape flowers and used to fertilize female pistils on other specific grape flowers (Figure 2-9). In the resulting *meiosis*, a genealogical combination takes place in seeds that reflects the attributes of each parent. Each new vine resulting from hybridization is called a *cultivar*—a term derived from “cultivated variety.”

The rationale for hybridization is to improve resistance to disease and adverse climate/soil conditions so as to increase yields and advance fruit quality. Unfortunately, it is common to find negative characteristics associated with dominant genes. History indicates that it may take hundreds of crosses to find just one cultivar worthy of commercial cultivation. University breeding programs continue, but state-of-the-art research looks to the advent of biotechnology, which may serve to make classic varieties increasingly resistant to adverse climates and disease, with greater productivity of better quality fruit.

Fig. 2-9 Cross-pollination.



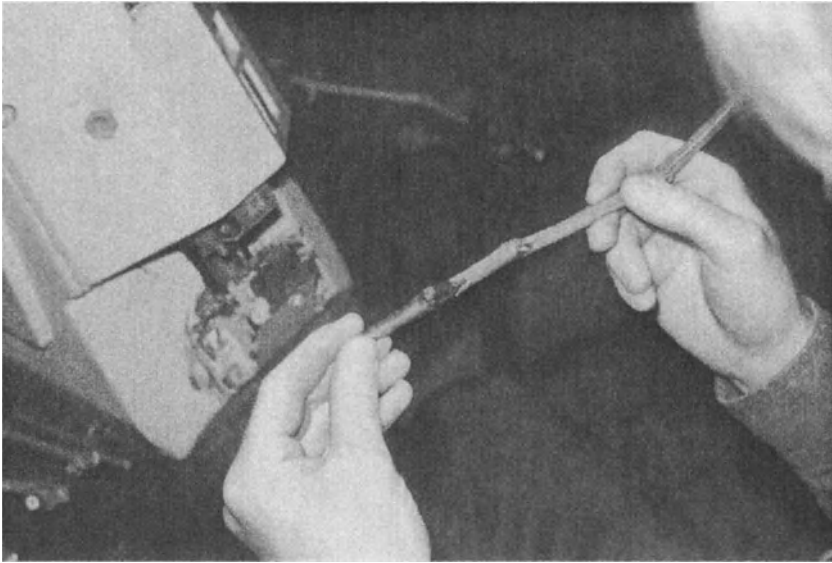


Fig. 2–10 Bench grafting.

Hybridizing should not be confused with *grafting*. In avoiding negative soil conditions, viticulturists will often graft one variety upon another. Great care must be taken to ensure that the *xylem* and *phloem* sap-flow tissue matches perfectly between the *scion* (upper fruit-bearing variety) and the *rootstock* (root-support variety). Vines can be “bench-grafted” (Figure 2–10) and propagated asexually in a nursery or “top-grafted” to change a mature vine from one variety to another in a vineyard.

French-American Hybrids

The term *French-American hybrid* evolved from efforts to find vines that were resistant to the great *Phylloxera* root louse blight that ravaged most of the world wine-growing community during the latter 19th century. French *vinifera* vines, well known for fine wine quality, were bred with disease-resistant *labrusca* and *riparia* American vines.

Major French-American Hybrid White Wine Cultivars

Seyval Blanc (SAY-vahl BLAWN)

This cultivar is considered by many to be one of the finest of the French-American hybrids. It ripens midseason to large, compact, conical bunches of slightly elliptical

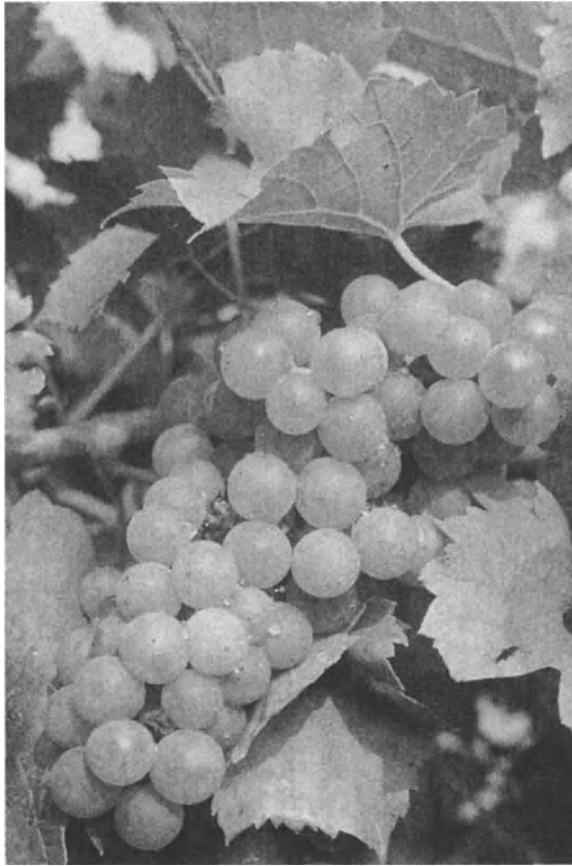


Fig. 2-11 Seyval blanc cluster.

greenish-yellow berries (Figure 2-11). Well-made Seyval Blanc can compete favorably with some dry white wines made from Chardonnay and Pinot Blanc.

Vidal Blanc (VEE-doll BLAWN)

In some environs this cultivar matures fruit with a rather distinctive Teutonic character. Vidal Blanc has very long cylindrical clusters (almost always shouldered) tightly packed with small, greenish-white berries with Riesling-like russet spots. It ripens late and is often left on the vine for “late harvest,” a natural dehydration process that concentrates sugars and flavors. Late-harvest Vidal Blanc can compare to similar wines made from Sylvaner.

Vignoles (VEEN-yole)

A Missouri-grown Vignoles shocked the world in 1992 when it was selected as the finest white table wine in a prestigious international wine competition held in California. Vignoles is a shy producer with tiny yellowish-white berries tightly packed on rather small clusters. Its flavor has a delicate flower-blossom character.

Major French-American Hybrid Red Wine Cultivars*Chambourcin (SHAM-burr-sahn)*

Chambourcin is a cultivar that has garnered considerable attention among eastern U.S. winegrowers during the past decade or so and may now be the most widely planted red hybrid cultivar. It usually yields good crops of small blue berries loosely clustered on long bunches that ripen rather late during the vintage season (Figure 2–12). The young wine has a moderate garnet hue and abundant berry-cherry flavors.

Chancellor

Chancellor is known for its rich plum-cedar aroma. If frequent mildew attacks can be controlled in the vineyard, Chancellor can be a bountiful producer. Clusters are long and loose, with medium-sized berries that ripen midseason. Chancellor wine is dense in ruby-violet color, heavy bodied, and possibly quite astringent unless tannins are tamed in oak.

Marechal Foch (MARR-shawl FOHSH)

This is a cultivar that ripens early, with tiny berries tightly compact on small clusters. From early-harvest fruit, young Foch wine can exhibit a moderate crimson hue and delicious cranberry-currant flavors.

VINEYARD ESTABLISHMENT**Cost and Value**

Establishment and operating costs can vary significantly within a region due to differences in cost for land, labor, machinery, and materials. Costs will also be affected by the vineyard site, grape cultivar, vine spacing, training system, pest management program, and other cultural practices selected (Figure 2–13).

The major costs during vineyard establishment are land, land preparation, labor, grapevines, trellis materials, pest management materials, and opportunity cost (interest paid on debt or return lost on capital). Additionally, irrigation (if needed), animal fencing, and major land preparation such as clearing, leveling, rock removal,



Fig. 2-12 Chambourcin cluster.

and drainage tiling can greatly increase initial costs. The added expenses of land improvement, however, will usually result in earlier and more consistent production of superior fruit.

Operating costs vary in much the same way as establishment costs—according to vineyard site, grape cultivar, vine spacing, training system, pest management program, and other factors. In particular, new growers should carefully consider the advantages and disadvantages of site selection and cultivar selection because of the profound impact these decisions have on the profitability of the planting.

Knowing the establishment and production costs is still not enough to make a sound decision about entering into grape production. Potential marketability and profitability must also be considered. Some varieties of grapes are more marketable,

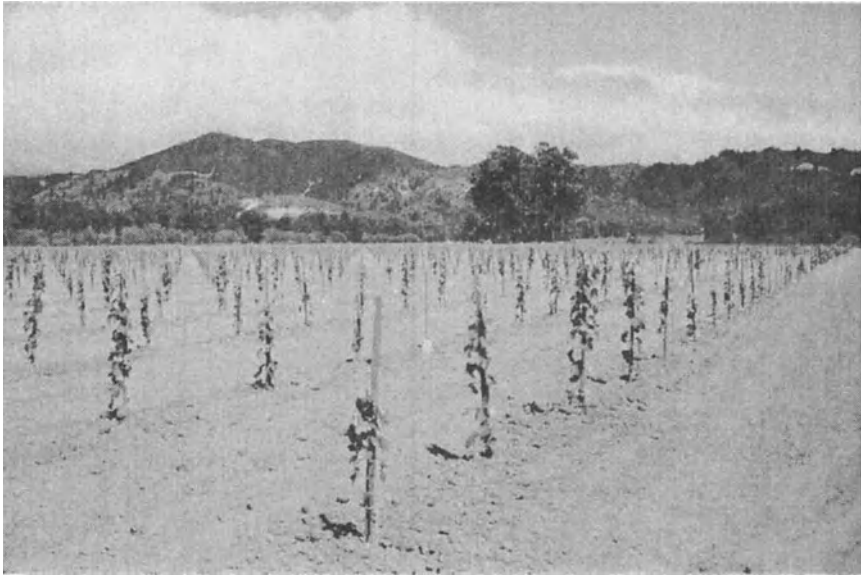


Fig. 2-13 Young vineyard planting.

both as fruit and as value-added wine, than others and are thus more valuable. For some vintners, appellation (location expressed on a wine label) of production is also an important characteristic for some vintners that can greatly increase grape value. Yield is a function of site, soils, vine variety, and vineyard management, as well as the highly unpredictable effects of weather—often referred to collectively as the quality of a vintage. Grape values and yields obviously have a combined direct effect upon profitability.

Table 2-1 is provided to illustrate typical grape prices and their impact of various sizes of wine production units.

Table 2-1 Value of grapes in wine at various prices per ton.

	\$500/Ton	\$600/Ton	\$700/Ton	\$800/Ton	\$900/Ton
\$/bottle	0.74	0.89	1.04	1.19	1.33
\$/gallon	3.70	4.44	5.19	5.93	6.67
\$/case	8.93	10.71	12.50	14.29	16.07

Note: Value base on 150 gallons of juice per ton minus 10% loss in processing for a total volume of 135 gallons per ton of grapes.

Each additional \$100 per ton paid for grapes adds \$0.15 to the cost per bottle of wine.

Start-Up

Labor Costs

Labor represents 30% or more of the cost of vineyard establishment, depending upon the amount of mechanization available, management skills, and quality of the labor pool. It is highly recommended that prospective grape growers seek discussions with experienced growers and state university Cooperative Extension Service personnel for advice and assistance.

Equipment Costs

Equipment ownership costs depend upon the particular circumstances of each vineyard operation. A diversified fruit farm that has all the necessary equipment available should have relatively low equipment costs for vineyard establishment. However, a grower who is starting from scratch will experience comparatively high capital requirements for a truck, tractor, sprayer (Figure 2–14), and other major equipment needs.

Major factors to consider in calculating equipment costs are initial cost, salvage value, years of life, annual use and maintenance, repair costs, insurance, interest, and operating expenses such as fuel and lubricants. Because accounting for the total cost for each piece of equipment is usually allocated over units of production, it may be difficult to justify starting a commercial vineyard of less than 5 acres. Sometimes it is possible for growers to find economies in one form or another of equipment sharing. Table 2–2 is provided to illustrate typical equipment costs that a prospective grower may expect in starting a vineyard.

Variable Costs

Variable costs are those that change directly with increases or decreases in acreage. Examples are trellis posts and wires, pesticides, fertilizer, wages, machinery fuel and lubricants, and even the grapevines themselves. Enterprising growers can reduce vineyard establishment costs by identifying alternative sources, such as making their own trellis posts; seeking assistance from family and friends to reduce wage costs; and propagating their own young vines. When a vineyard has been fully established and initial costs are recovered, labor costs will generally represent about 60% of the annual cost of producing grapes. In larger operations particularly those large enough to justify a machine harvester, this percentage reduces significantly.

Table 2–3 illustrates typical equipment variable costs that a grower may expect in establishing a vineyard.

Fixed Costs

Fixed costs are often referred to as *overhead* and are those that do not change directly with increases or decreases in acreage. Overhead includes machinery depreciation, opportunity cost, insurance, and taxes. In diversified farms or ranches,



Fig. 2-14 Vineyard tractor and sprayer.
Courtesy: Wine East magazine.

total overhead should, of course, be appropriately allocated to the vineyard operation.

Budget

A 7-year budget is provided to illustrate typical costs that a prospective grower may expect in starting a vineyard (Tables 2-4 through 2-10).

Table 2-2 Equipment purchase cost, operational cost, and anticipated annual hourly use.

Machine	Initial cost	Cost/hour	Hours/year			
			1	2	3	≥4
Tractor, 35–50 hp	\$20,000	\$4.05	85.2	14.0	2.17	26.7
Pickup truck						
(cost/mile & miles/yr)	18,500	0.29	500.0	500.0	500.0	500.0
Herbicide sprayer, 50 gal	2,125	1.27	2.0	2.0	1.0	1.0
Post driver	1,500	1.20	11.2			
Airblast sprayer, 100 gal	6,000	2.82	3.0	4.0	5.7	5.7
Mower/brush chopper, 6 ft	1,700	0.63	2.0	2.0	2.0	2.0
Fertilizer & seed spreader	1,800	1.44	0.5			
Auger, 12 in.	1,400	1.12	47.0			
Wagon/trailer, 5-ft × 12-ft flatbed	1,575	0.58	15.0		2.9	7.0
Total machinery purchase expense:	\$54,600					

Machinery capital recovery:

Machinery investment financed at 8% for 7 years: $\$54,000 \times 0.1921^a = \$10,489$.

Annual cost per acre (10-acre vineyard): $\$10,489/10 \text{ acres} = \$1,049$.

^a0.1921 is the capital recovery factor for the financing terms specified above.

Table 2-3 Vineyard equipment variable costs per hour of operation and per acre.

Machine	Cost/hour	Cost/acre for first 8 years			
		1	2	3	4–8
Tractor, 35–50 hp	\$4.05	\$345.1	\$56.7	\$87.9	\$108.1
Pickup truck (cost/mile)	0.29	145.0	145.0	145.0	145.0
Herbicide sprayer, 50 gal	1.27	2.5	2.5	1.3	1.3
Post driver	1.20	13.4			
Airblast sprayer, 100 gal	2.82	8.5	11.3	16.1	16.1
Mower/brush chopper, 6 ft	0.63	1.3	1.3	1.3	1.3
Fertilizer & seed spreader	1.44	0.7			
Auger, 12 in.	1.12	52.6			
Wagon/trailer, 5-ft × 12-ft flatbed	0.58	8.7		2.9	7.0
Total machinery cash expense		\$577.8	\$216.8	\$254.4	\$278.7

Table 2-4 Cost of vineyard establishment per acre, year 1.

Operation	Hours/ acre	Unit cost	Units/ acre	Cost/ acre
Site Preparation				
Liming—3 tons/acre		\$16.00	3	\$48
Plow & disk—labor	3.0	7.50		23
Sow cover crop—labor	1.0	7.50		8
Cover crop seed		0.68	50	34
Vineyard Layout				
Mark vine & post locations	5.0	7.50		38
Planting				
Vines		2.00	545	1090
Auger vine holes (two men @ 3 min/hole)	54.5	7.50		409
Planting (2 min/vine)	18.2	7.50		136
Trellising Materials				
3-in. × 8-ft. CCA-treated line posts		3.50	187	654
6-in. × 9-ft. CCA-treated end post		10.00	22	220
12.5 ga. HT wire (4000 ft per roll)		60.00	2.5	150
End-post anchors (5.8-in. × 4-ft helical end)		4.00	22	88
In-line ratchet wire strainers for cordon wires		2.00	11	22
Wirewise wire strainers for trunk support wires		1.50	11	17
Staples (28 lbs.) and wire crimping sleeves (44)				31
Trellising Labor				
Distribute & drive line posts (2 men @ 3 min/post)	18.7	7.50		140
Auger and set end posts	6.0	7.50		45
Mark & drill end posts for wirevises	3.0	7.50		23
Mark post for wires	2.0	7.50		15
Install end-post anchors	5.0	7.50		38
String wire, staple to posts, & tighten	15.0	7.50		113
Weed Control				
Postemergent application prior to planting		44.04	0.17	7
Preemergent application after planting		64.00	0.33	21
Herbicide application labor	2.0	7.50		15
Mowing row middles (6 times)	4.0	7.50		30
Hoing/weeding labor (2 times)	30.0	7.50		225
Fertilization				
Ammonium nitrate (0.25 lbs. per vine)		0.14	136	19
Hand application of fertilizer	2.0	7.50		15

(continued)

Table 2-4 (continued)

Operation	Hours/ acre	Unit cost	Units/ acre	Cost/ acre
Canopy Management (training)				
Bamboo/wood stakes for trunk support		\$0.25	545	\$136
Shoot thinning & tying to stakes (twice)	20.0	7.50		150
Tie materials				5
Flower cluster removal	2.0			15
Disease and Insect Control				
Spray materials				109
Spray labor (6 sprays @ 0.5 h/spray)	3.0	7.50		23
Machinery				
Cash operating expenses only (See Table 2-2)				578
Operating Interest				
Interest charged on yearly cash expenses for $\frac{1}{2}$ yr		$\frac{1}{2}$ yr @ 8.00%	4690	\$188
Annual Cash Expenses—Year 1				\$4,878

General assumptions made in developing this budget include:

1. Costs are based on a 10-acre vineyard planted on a good to excellent site to own-rooted (ungrafted) vines adapted to the region.
2. Vines are planted 8 feet apart in rows spaced 10 feet apart for a total of 545 vines per acre.
3. Row length is approximately 400 feet. Shorter rows would cost slightly more, longer rows slightly less.
4. The trellis is a standard two-wire system for either cordon training or cane pruning. All trellis posts are CCA-treated timber. End posts are 6 inches by 9 feet, while line posts are 3 inches by 8 feet and are spaced 24 feet apart.
5. The vineyard floor is managed as grass row middles with herbicide strips under the row.
6. Labor cost is calculated at \$7.50 per hour. Time required to complete the tasks will vary, and the times listed are considered averages.
7. All machinery and equipment expenses reflect *operating costs only*: no charges have been included for ownership items such as insurance, taxes, or depreciation/capital recovery.
8. Fixed costs for ownership of land are not included in this budget.
9. Accumulated costs are financed at 8% annual percentage rate. Interest on annual operating costs is computed at 8% APR for 6 months.

Table 2-5 Cost of vineyard establishment per acre, year 2.

Operation	Hours/ acre	Unit Cost	Units/ acre	Cost/ acre
Dormant Pruning				
Pruning & tying canes for trunks	10.0	\$7.50		\$75
Weed Control				
Preemergent application in spring		64.00	0.33	21
Spot treatment with postemergent		44.04	0.08	4
Total herbicide application labor	2.0	7.50		15
Mowing row middles (6 times)	4.0	7.50		30
Fertilization				
Ammonium nitrate (0.5 lbs. per vine)		0.14	250	35
Hand application of fertilizer	2.0	7.50		15
Replanting				
Plants (2% of initial planting)		2.00	11	22
Replanting labor	1.0	7.50		8
Canopy Management (training)				
Shoot thinning & tying to stakes (twice)	30.0	7.50		225
Flower cluster removal	8.0	7.50		60
Tie materials				5
Disease and Insect Control				
Spray materials				200
Spray labor (10 sprays @ 0.5 hrs/spray)	5.0	7.50		38
Machinery				
Cash operating expenses only (See Table 2-2)				217
Operating Interest		$\frac{1}{2}$ yr @		
Interest charged on yearly cash expenses for $\frac{1}{2}$ yr		8.00%	970	39
Interest on Year 1 Accrued				
Cash Expense		8.00%	4878	390
Annual Cash Expenses—Year 2				\$1,399
Total Accumulated Cash Expense (Years 1 and 2)				\$6,277

Table 2-7 Cost of vineyard establishment per acre, year 4.

Operation	Hours/ acre	Unit cost	Units/ acre	Cost/ acre
Dormant Pruning				
Spur pruning and brush pulling	20.0	\$7.50		\$150
Cordon training/tying	5.0	7.50		38
Weed Control				
Preemergent application in spring ^a				
(oryzalin)		64.00	0.33	21
(simazine)		2.85	1.0	3
Spot treatment with postemergence herbicide	2.0	44.04	0.08	4
Total herbicide application labor	4.0	7.50		15
Mowing row middles (6 mowings)		7.50		30
Fertilization^b				
Ammonium nitrate (0.5 lbs. per vine)		0.14	250	35
Spreading fertilizer	0.5	7.50		4
Leaf petiole sampling	0.3	7.50		2
Tissue analysis				20
Canopy Management (training)				
Shoot and cluster thinning	15.0	7.50		113
Shoot positioning	10.0	7.50		75
Tie materials				5
Disease and Insect Control				
Spray materials				287
Spray labor (12 sprays @ 0.5 h. per spray)	6.0	7.50		45
Harvest Costs^c				
Picking costs (\$1.25/25-lb. lug for 5 tons/acre)		1.25	400	500
Machinery				
Cash operating expenses only (See Table 2-2)				279
Operating Interest				
Interest charged on yearly cash expenses for 1/2 yr		8.00%	1626	65
Interest on Year 2 Accrued Cash Expense				
		8.00%	7883	631
Annual Cash Expense—Year 4				\$2,322
Total Accumulated Cash Expense Years 1-4				\$10,205

(continued)

Table 2-7 (continued)

Operation	Hours/ acre	Unit cost	Units/ acre	Cost/ acre
Income^d		\$500	5	\$2,500
Net Investment at End of Year 4				\$7,705

^aOryzalin and simazine may be used separately or in combination from year 3 on. Costs shown are for combination.

^bThe need for a particular nutrient fertilizer is best determined through visual assessment of vines and by routine leaf petiole sampling for nutrient analysis. Fertilizer costs will vary depending on the particular needs of the vines. A single tissue sample may suffice for an entire 10-acre planting.

^cHarvest costs of \$1.25 per 25-lb. lug, or \$100 per ton. Expected yield in year 4 and beyond is 5 tons/acre.

^dIncome based on harvest of 5 tons of grapes and a price of \$500/ton.

The cost of vineyard establishment and operation at various prices per ton for 5 tons per acre is shown in Table 2-11 for a 24-year period. Based upon this budget, the costs of establishing an acre of wine grape vineyard may be expected to range from \$7,000 to \$10,000—and far more in marginal vineyard sites.

It is essential that vine variety selections for a prospective vineyard be considered from several major viewpoints. The wine from some varieties has far more consumer demand than others. In most metropolitan markets, varietal wines grown

Table 2-8 Cost of vineyard operation per acre, year 5.

Operation	Unit cost	Units/ acre	Cost/ acre
Management Expense —(Similar to Year 4, including interest on yearly cash expenses for 1/2 yr)			\$1,691
Interest on Accrued Investment	8.00%	7,705	616
Annual Cash Expense—Year 5			2,307
Total Accumulated Cash Expense			\$10,012
Harvest Income 5 ton/acre yield	\$500	5	\$2,500
Net Investment at End of Year 5			\$7,512

Table 2-9 Cost of vineyard operation per acre, year 6.

Operation	Unit cost	Units/acre	Cost/acre
Management Expense Similar to Year 4			\$1,691
Interest on Accrued Investment	8.00%	7,512	601
Annual Cash Expense—Year 6			\$2,292
Total Accumulated Cash Expense			\$9,804
Harvest Income			
5 ton/acre yield	\$500	5	\$2,500
Net Investment at End of Year 6			\$7,304

from *Vitis vinifera* are in greatest demand, and some prime French-American hybrid cultivars are preferred. In more suburban and countryside markets, wines made from the native *Vitis labrusca* varieties and fruit wines may be more popular.

These consumer demands must, of course, be related to the perceived adaptability of a given site for the desired selection of vine variety types. A major concern is potential damage from extremes of hot and cold temperatures. Some varieties, especially in the *Vitis vinifera* genus and species, are so tender that they will frequently suffer bud and/or trunk damage during bitter winter temperatures. Measures such as hilling up in fall to protect the graft union and taking out in spring are very labor intensive and require special equipment, and they still may fail to

Table 2-10 Cost of vineyard operation per acre, year 7.

Operation	Unit cost	Units/acre	Cost/acre
Management Expense Similar to Year 4			\$1,691
Interest on Accrued Investment Minus Year 6			
Crop Sale	8.00%	7,304	584
Annual Cash Expense—Year 7			2,275
Total Accumulated Cash Expense			9,579
Harvest Income			
5 ton/acre yield	\$500	5	\$2,500
Net Investment at End of Year 7			\$7,079

Note: Cost analysis can be continued in this matter each year.

Table 2-11 Cost of vineyard establishment and operation at various prices per ton at 5 tons per acre yields.

Year	Total accumulated cash expense			
	\$500/ton	\$600/ton	\$700/ton	\$800/ton
1	\$4878	\$4878	\$4878	\$4878
2	6277	6277	6277	6277
3	7883	7629	7379	7129
4	7705	6930	6160	5390
5	7512	6175	4844	3512
6	7304	5360	3423	1484
7	7079	4480	1888	(706) 1st net
8	6836	3529	230	(2309) yearly net
9	6574	2502	(1561) 1st net	
10	6291	1393	(1809) yearly net	
11	5985	195		
12	5655	(1098) 1st net		
13	5298	(1309) yearly net		
14	4913			
15	4497			
16	4048			
17	3563			
18	3039			
19	2473			
20	1862			
21	1202			
22	489			
23	(281) 1st net			
24	(809) yearly net			

Note: Yearly cost of production from year 4 and on is \$1691/acre at 5 tons/acre. Increased profits from grapes sold at higher prices greatly reduces the number of years before establishment costs are recovered.

protect from a particularly severe winter. Untimely spring frosts can damage fruiting shoots on any vine.

There are considerations for each cultivar's individual susceptibility to diseases, pests, yields, and resulting fruit quality. Further, some vine types require more care and handling than others, such as shoot positioning, cluster thinning (green harvest), and leaf removal.

The additional costs anticipated for such risks must be figured into the cost of production to determine grape values. Years of reduced or no crops should also be calculated into the cost of production so that the value for grapes harvested during more successful years can accurately reflect the overall profitability on a sustainable basis.

Site Preparation

Thorough preparation of the planting site is essential to minimize weed problems and to increase the organic matter content of the soil. This should be done 1 year before planting. Problem weeds should be controlled with herbicides before soil preparation. Green manure crops such as rye add organic matter and help reduce weed problems.

Soil testing will provide information on fertility and liming needs for the first-year vineyard. If the soil pH is below 5.5, apply agricultural limestone to raise the pH to a more desirable level, in the 5.5 to 6.5 range. Lime applications should be made the year before planting, and the limestone should be incorporated deeply into the soil. Other nutrients, especially potassium, should be applied during site preparation if the soil test shows a deficiency.

Avoid poorly drained soils for grape production. If the soil has only fair internal drainage and drain tile is not installed, it may be advantageous to subsoil (chisel-plow) to a depth of 24 inches and establish a ridge upon which the vines will be planted by mounding the soil from the edges of the row to the middle. The ridge should be about 10 inches high and 2 feet wide. This will provide improved surface and internal drainage, as well as allow deeper rooting of the vines. It also will prevent ponding of water at the base of vines, which can lead to disease problems.

Vineyard Layout

Grapevines are generally planted in straight rows running north-south for maximum sunlight interception. However, sites on sloping land may require contour planting or straight rows across the slope, regardless of compass direction. In some areas it may be advantageous to orient rows parallel to the prevailing wind to aid in disease control. Row spacing depends upon the training system and the equipment to be used in the vineyard. Spacing of 9 to 10 feet between rows is commonly used but may be increased to accommodate larger equipment or on sites with steep slopes. Spacing between vines within the rows is usually 8 feet but may be closer for cultivars with low vigor. At maturity the vines should completely fill the trellis space without competing with each other for sunlight (Figure 2–15).

Planting

Planting should take place in early spring, after the danger of hard freezing temperatures has passed. Vines are usually purchased as dormant bare-root plants and should be stored in a cool, moist place, preferably cold storage, until planted. Soak the roots in water for 24 hours before planting to prevent the root system from drying out during the handling and planting process. Prune off broken and damaged roots and shorten excessively long roots for convenience when planting.

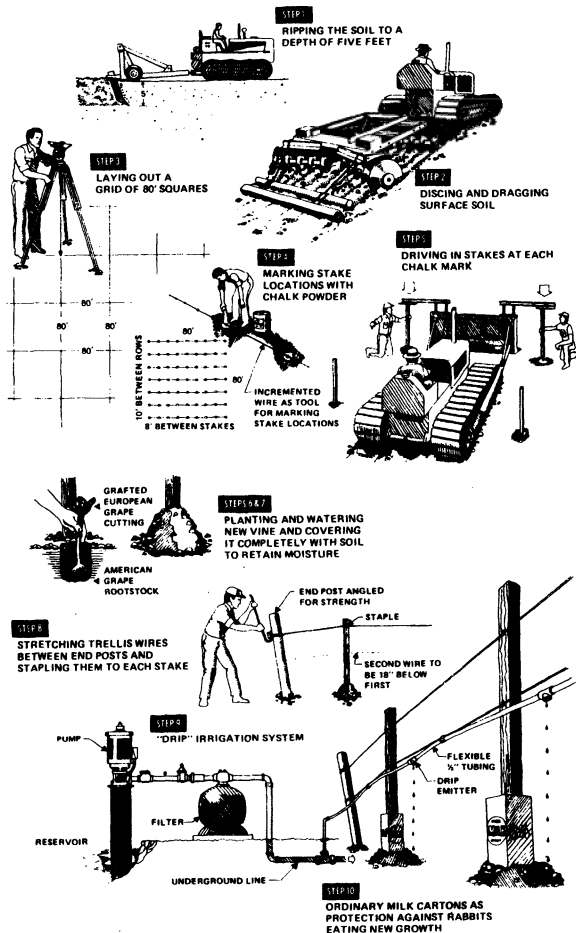


Fig. 2-15 Establishing a vineyard.

Courtesy: Sebastiani vineyards.

The planting hole should be large enough to accommodate the root system with ease. Excessive pruning of the root system is not advised. However, it is better to prune a few roots than to stuff the roots into a small planting hole. Spread the roots, cover with soil, and tamp well.

If soil moisture is low, water as needed until the plants have developed a root system large enough to support themselves during dry periods. Own-rooted (ungrafted) vines should be set to a depth where the lowest shoot of the dormant plant is just above the soil level. For grafted vines, the graft union should be situated at least 2 to 3 inches above the soil level to prevent scion rooting.

Vine Training

It is important to train vines properly during the first few years of growth to establish a vine form that will be easy to manage. After planting, but before growth begins, the top of the dormant plant should be pruned back to a single cane with 2 to 5 buds. After growth starts, all but the best two to four shoots should be removed. One of more of these shoots will become the trunks. Support should be provided for new shoots to keep them off the ground. This will greatly reduce disease problems and provide full sun exposure for maximum growth. The trellis should be established soon after planting to provide this support. String can be tied from a side shoot of the vine to the wires and the new shoots wrapped around the string. Never tie around the main trunk of the plant because the trunk will expand rapidly during the first growing season and can be girdled by the string. A stake can be driven next to each plant and the shoots tied to the stake instead of the trellis wire.

Trellising

Grapevines require support for ease of management and the heavy weight of ripening fruit. This can vary from a simple wire trellis to an elaborate arbor. Vine support should hold the vine up where it can be managed and cared for efficiently. The trellis should also expose a greater portion of the foliage to full sunlight, which promotes the production of highly fruitful buds. Trellising design features, such as the number and placement of wires, are dictated by the type of training system used. All training systems require a strong trellis assembly to support the weight of fruit and vines.

Training Systems

Grapevines are trellis-trained by several dozen major configurations. The most efficient methods provide a well-spaced, even distribution of fruiting wood along the trellis and promote full sun exposure for clusters and basal nodes. However, the selection of a training system usually is made after careful consideration of climate history, soil profile, and vigor of the vine variety.

Cane training systems are devised by selecting superior canes from the previous season of growth and arranging them in a manner that serves best.

An Umbrella Kniffin (Figure 2–16) may be employed for larger crops from vigorous vines in fertile soils situated in moderately cool environs. A Keuka Low Renewal system (Figure 2–17) is typically used for grafted vines and in colder locales. The short trunk conserves vine metabolism generated for woody tissues and allows for snow insulation or burying during winter if needed. Fewer, shorter canes control crop size to optimize vine vigor.

Cordon training systems function by extending trunks along the trellis wires, with canes and fruit developing from buds left on short “spurs” after pruning.

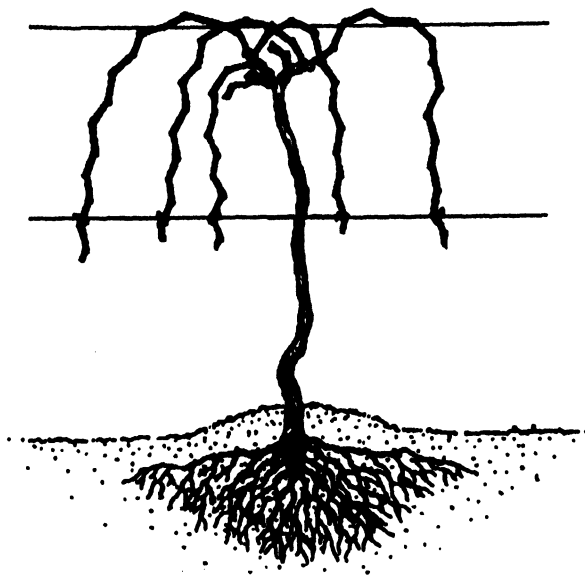
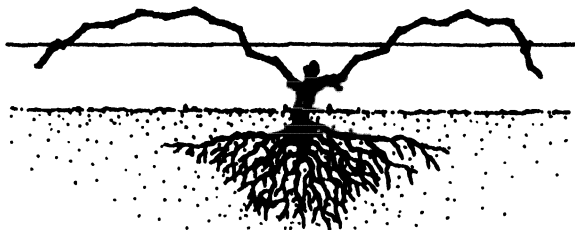


Fig. 2-16 Umbrella Kniffin cane training system.

The Bilateral Cordon (Figure 2-18) is often the choice in moderate climates and average soils. Vigorous vines cultivated in richer soils may do best upon a Geneva Double Curtain (Figure 2-19), whereby the trellis is divided to provide additional exposure to sunlight for enhanced photosynthesis.

The price tag on grape harvesters (Figure 2-20) ranges from \$50,000 to more than double that, depending upon whether the device is pulled by a tractor or self-powered. That amount can be doubled again by the extent of ancillary equipment needed, such as vineyard trailers, bins, heavy-duty trucks, and winery receiving machinery. Consequently, it is rare that estate vintners can find economies justifying individually owned and operated machine harvesting equipment. On the other hand, there may be an opportunity to share such investments with other estate vintners in the area or to hire custom machine harvesting services. In any case,

Fig. 2-17 Keuka Low Renewal cane training system.



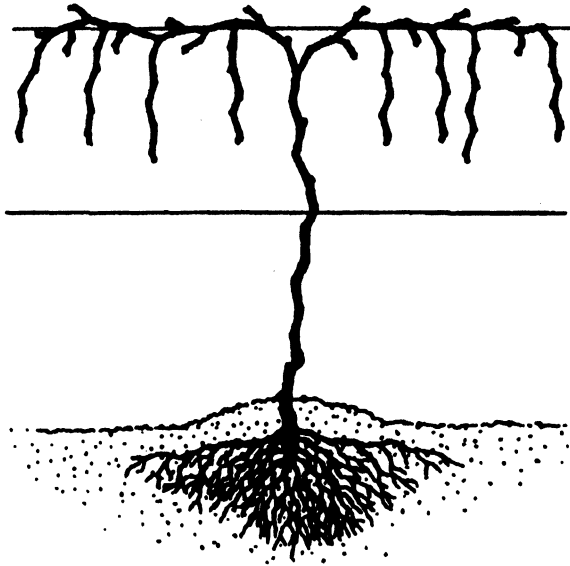


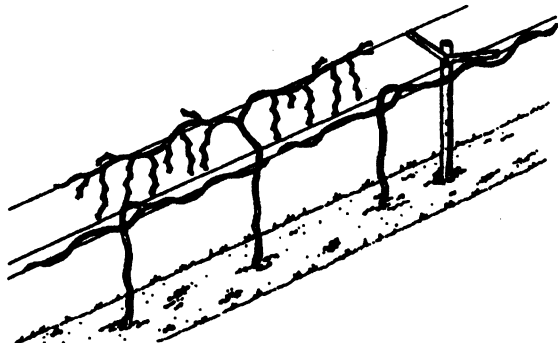
Fig. 2-18 Bilateral Cordon training system.

harvesting by machine requires trellis systems specially designed to accommodate the type of device that will be employed.

Weed Control

It is especially important to provide good weed control during the first several years of vineyard establishment. Young vines do not compete well with grasses and weeds for soil nutrients, moisture, and sunlight. Grasses and weeds can also harbor disease organisms that can harm developing vines. Plastic and biodegradable mulches, as

Fig. 2-19 Geneva Double Curtain cordon training system.



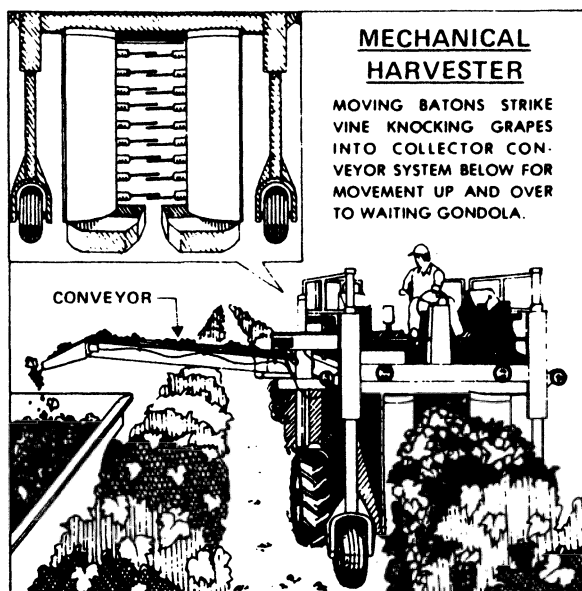


Fig. 2-20 Machine harvesting.

Courtesy: Sebastiani vineyards.

well as organic compost mulches, can be used to control weed growth. Mechanical weed control is effective as long as devices do not penetrate the soil deep enough to destroy vine feeder roots. For larger plantings, the use of herbicides may be necessary. Consult the Cooperative Extension Service's small fruit spray guides for information relating to suitable herbicides.

Herbicide Injury

Injury from 2,4-D herbicide is frequently confused with disease or insect injury. Most cultivated grapes are highly susceptible to injury by this herbicide and related compounds. These herbicides are used in a variety of forms on many different crops and noncrops. Because of their extreme volatility, many 2,4-D compounds can injure vines at considerable distances from the point of application. The use of these compounds in field crop production, especially reduced tillage situations, may limit the production of grapes in certain areas.

Vintners should diplomatically inform the local community, especially neighboring farmers, that grapevines will be cultivated in the area and that 2,4-D or related compounds are a threat to this endeavor. Requests should be made that they use only low-volatile formulations and make applications with equipment designed to reduce drift. Vines are most sensitive to this type of chemical injury during the early part of the growing season when shoots and leaves are expanding rapidly.

Symptoms of 2,4-D are easily diagnosed. The youngest terminal growth at the time of exposure will be stunted and misshapen and will have thick veins that tend to run parallel to each other rather than branching as in normal leaves. Growth will be stunted for several weeks following injury. Severely injured vines may die or not recover for several years.

VINEYARD MANAGEMENT

Pruning

Pruning is the most important cultural practice in the management of grapes. It is performed to select superior potential fruiting wood, to maintain vine shape and form for the best sun and wind exposure, and to regulate the number of buds retained per vine.

Grapevines require annual pruning to remain productive and manageable. It can be done anytime during the dormant season but is generally best delayed until late winter or early spring, as pruned vines are more susceptible to cold injury. In addition, delayed pruning allows for adjustments in the number of retained buds in the event of winter injury. It is thus essential to assess bud survival before final pruning.

An average mature grapevine will have 200 to 300 buds developed during the previous growing season. If all of these were retained, the result would be a large crop that would not ripen properly, reduced vine vigor, and poor cane maturation for succeeding years of production. To avoid this situation, researchers have developed a method of pruning to balance the fruit productivity and vegetative growth that will give maximum yields without reducing vine vigor or cane maturity. This procedure is commonly referred to as *balanced pruning*. The number of buds retained is balanced to match the situation in which the vine grows—its soil, its climate, and its vigor. All of this is termed *vine size*, which is measured by the weight of canes taken from vines during pruning.

To balance-prune a grapevine, estimate the vine size, then prune the vine, leaving enough extra buds to provide a margin of error, usually 70 to 100 total buds. Next, weigh the 1-year-old cane prunings with a small spring scale (Figures 2–21 through 2–24). Then use the pruning formula in Table 2–12 to determine the number of buds to retain per vine.

For example, a moderately vigorous vine cultivated in a cool climate having been pruned to a total of 3 pounds of 1-year-old prunings would be afforded 20 buds for the first pound of pruning wood and 10 buds for each of the additional 2 pounds. This is a total of 40 fruiting buds left, from which the upcoming crop of grapes would be born. Note that even if there were 4 or 5 pounds of prunings, the maximum total number of buds recommended for a vine in this situation would still be 40. After determining the appropriate number of buds to retain, prune the extra buds off, taking care to space the fruiting buds evenly along the trellis. Once several vines in a given vineyard are measured, experienced pruners will be able to closely

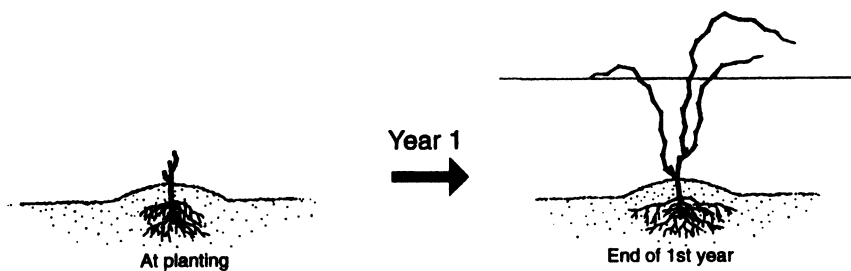


Fig. 2-21 Typical pruning and growth during first year.

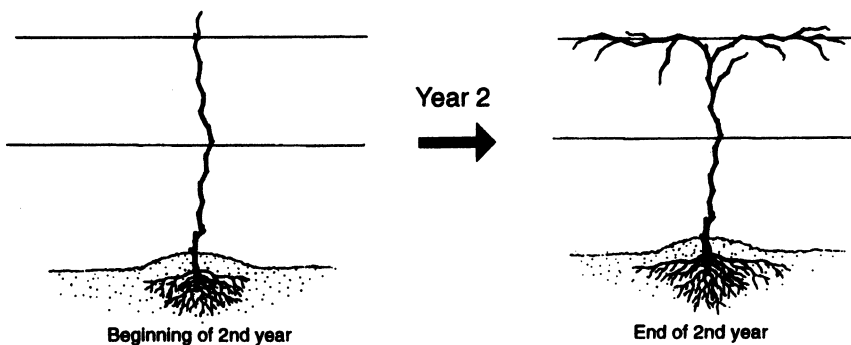


Fig. 2-22 Typical pruning and growth during second year.

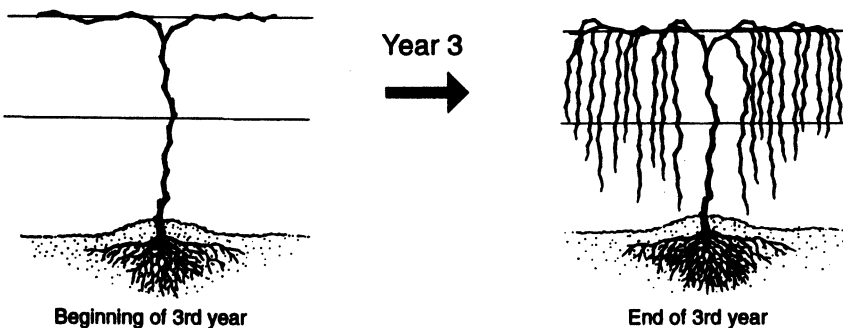


Fig. 2-23 Typical pruning and growth during third year.

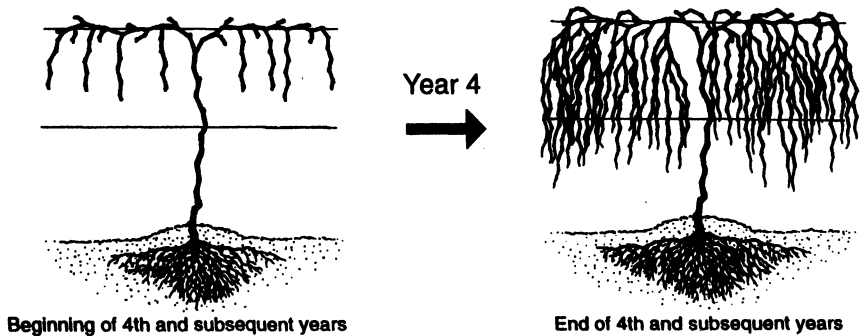


Fig. 2-24 Typical pruning and growth during fourth and mature years.

estimate vine size so as to dispense with the arduous task of weighing every vine. Pruning formulas developed for many cultivars have been devised and published by the Cooperative Extension Service in some states.

However, balanced pruning alone will not adequately balance the crop load with vegetative growth. Cluster thinning and/or shoot removal may be required on some cultivars, especially certain selections of hybrids that are highly productive. Cluster thinning typically involves the removal of all but the basal cluster, which is usually the largest. Shoot removal is performed by removing shoots that occur from secondary buds and primary buds from wood older than 1 year.

Pest Control

Diseases

Control of disease is necessary to maintain healthy leaves and protect the fruit from various types of deterioration. Most diseases of grapes are caused by fungi that thrive in hot, humid conditions.

The most common cluster and leaf diseases are black rot, bunch rot, downy mildew, and powdery mildew (Figures 2-25 through 2-28). Many factors affect

Table 2-12 Fundamental pruning formula for grapevines.

Vine condition	No. of buds left for first pound of prunings	No. of buds left for each additional pound	Max. total buds
Low vigor and/or cold climate	10	5	20
Medium vigor and/or cool climate	20	10	40
High vigor and/or moderate climate	30	20	70



Fig. 2-25 Black Rot.

Courtesy: Mississippi State University.



Fig. 2-26 Bunch Rot.

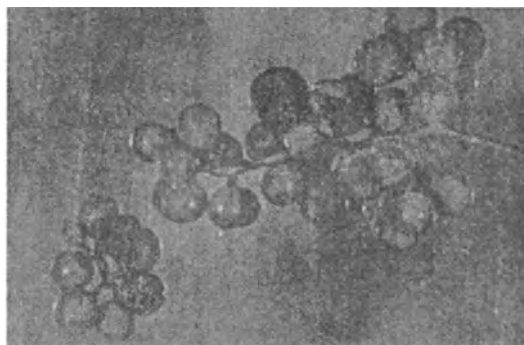


Fig. 2-27 Downy Mildew.

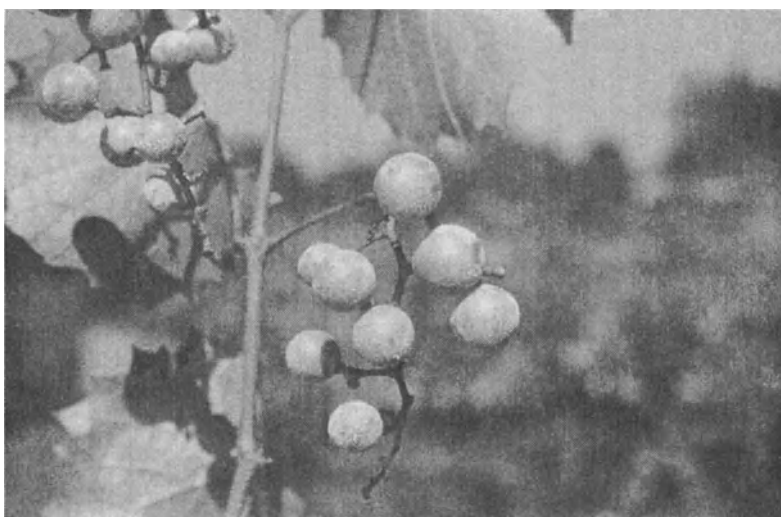


Fig. 2-28 Powdery mildew.
Courtesy: Mississippi State University.



Fig. 2–29 Pierce's Disease.

the amount of disease that can occur, such as site selection, vine vigor and susceptibility, canopy density, and other factors. Most cultivars are prone to one or another disease infection. A few cultivars are only slightly susceptible to the major diseases and can be cultivated with little or no pesticide use. Normally, however, some pesticide use is necessary to produce top-quality fruit.

Pierce's Disease (See Fig. 2–29) is a rather “arteriosclerosis-like” terminal condition vectored by the Sharpshooter Leafhopper. This insect introduces Pierce's Disease bacteria into vines as it sucks sap from succulent shoot tips. The bacteria multiply upon xylem walls, causing eventual total restriction of sap flow, and vine mortality. The adult insects emerge each year during summer and feed on grape foliage. The damage can be severe if uncontrolled, especially on young or weak vines.

Crown Gall (See Fig. 2–30) is usually caused by freeze or mechanical injury whereby lesions resulting in the vine trunks allow for *Agrobacterium tumefaciens* to enter the xylem vessels. Growth of this bacteria forms large galls which kill the vine. The bacterium itself does not cause disease but vectors a plasmid, a circular piece of DNA, that enters the plant cells, becomes incorporated into the plant genome and causes the cells to divide abnormally, leading to the development of tumors or galls.

Responsible pesticide use requires a knowledge of pest identity and biology. Growers should frequently inspect their vineyards carefully and seek help in



Fig. 2–30 Crown Gall.

identifying problems. Most fungicides registered for use on vines are effective against only certain diseases. Correct identification of disease problems is necessary for effective and responsible disease control. Consult state Cooperative Extension Service small fruit spray guides for information relating to suitable fungicides and pesticides.

Insects

Relatively few insect pests are found in grapevine cultivation. One of the most common is the Japanese beetle. The adult insects emerge each year during summer and feed on grape foliage. The damage can be severe if uncontrolled, especially on young or weak vines.



Fig. 2-31 Phylloxera.

Another important insect pest is the grape flea beetle. These insects emerge early in the spring and feed on developing grape shoots. If left uncontrolled, these pests can completely destroy all primary buds and cause heavy losses in production.

One of the most infamous of viticultural insects is *Phylloxera vastatrix* (Figure 2-31), the root louse that devastated millions of acres of vineyards in Europe during the latter 1800s and that recently invaded vineyards in northern California.

Insecticide control of destructive pests on vines should be made only if the infestation is present and the amount of damage is severe enough to warrant the cost of chemical control.

CHAPTER 3

WINE MICROBIOLOGY

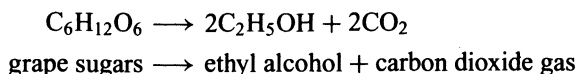
Microbiologically speaking, unknown wine is much safer than unknown water. Of the thousands of microorganisms now classified, relatively few have been found growing in wine. This is due to the nearly antiseptic combination of alcohol and acids that occurs in wine. Most microscopic organisms taken from contaminated water actually die if introduced into wine. Human pathogenic organisms cannot survive in the environment of wine. It is because of this phenomenon that wine has promoted human health throughout Western civilization.

This chapter will consider the major microscopic organisms that are involved in wine. It should be understood that the entire realm of wine microbiology embraces more than 100 yeast, bacteria, and mold organisms. The procedures for microbiological laboratory analysis are found in Appendix B.

FERMENTATION

Fermentation is a natural process in which the grape sugars (mostly glucose and fructose) are transformed into ethyl alcohol and carbon dioxide gas. In 1810 the French chemist and physicist Joseph-Louis Gay-Lussac described this process in

the following chemical equation:



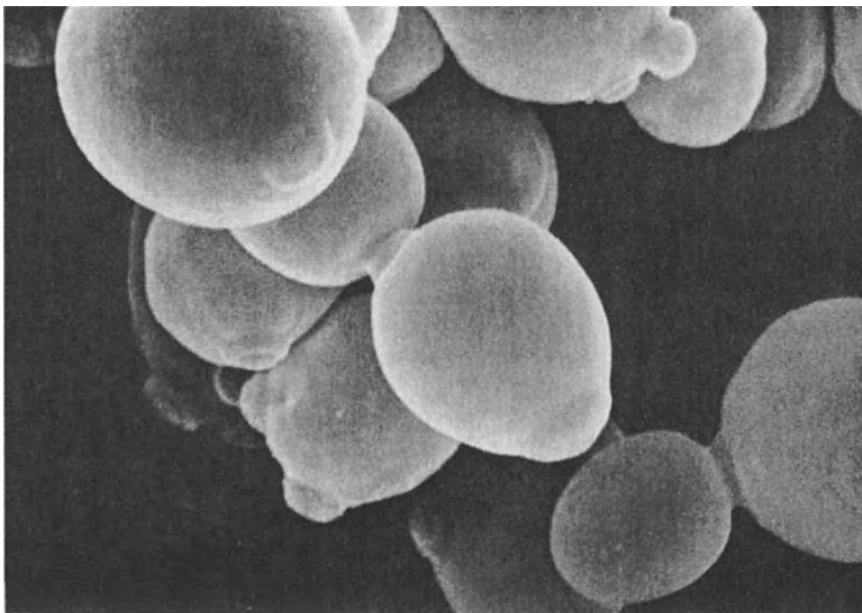
Today we know that wine fermentation is carried out by certain enzymes (protein compounds produced by yeast cells) that act as catalysts in facilitating the sugar to alcohol/carbon dioxide conversion. Students of the fermentation process learn that it very closely parallels the glycolysis cycle, which is elementary to modern biochemistry, and that heat energy is also a product of the transformation. Thus, the original Gay-Lussac equation is modified as follows:



Note that this reaction is the reverse of that given in the Chapter 2 discussion related to photosynthesis in grapevine leaves. Thus, photosynthesis and fermentation constitute a natural cycle.

It was Theodor Schwann who first discovered that this transformation was caused by a “sugar fungus” (i.e., *Saccharomyces* organism; see Figure 3–1). Louis Pasteur published a paper in 1857 that revealed an observation of a similar

Fig. 3–1 Electron Microscope Photo of *Saccharomyces cerevisiae*.
Courtesy: Michael Sullivan, SEM, Mississippi State University.



microscopic organism that was responsible for wine fermentation. Before this, fermentation was thought to have been due to a phenomenon known as “spontaneous generation,” loosely related to various forms of “magic.” The Greeks called fermentation *zestos*, or “boiling,” from which the term *yeast* was fashioned. In this way, the discipline of microbiology was born in France, largely in response to Pasteur’s curiosity about Gay-Lussac’s equation.

YEASTS

Yeasts are egg-shaped cells about 1/300,000 of an inch in length. Because they lack stems, leaves, roots, and chlorophyll, yeasts are referred to as lower-form plants (Figure 3–2). Under optimal conditions in a nutrient-rich medium, they can reproduce once every half hour or so—a rate that can generate a huge population in a relatively short period of time. Due to increasingly sophisticated methods for characterizing microorganisms at the genetic level, the taxonomy for microorganisms is constantly changing.

The current taxonomic classification of a popular commercial fermentative yeast is as follows: phylum Thallophyta, subphylum Fungi, class Eumycetes, subclass Ascomycetes, order Endomycetales, family Saccharomycetaceae, subfamily Saccharomycoidae, genus *Saccharomyces*, species *cerevisiae*, strain Prisse de Mousse.

Native Yeasts

There are millions of natural (wild, native or indigenous) yeast cells in the *cutin* of a grape berry. The cutin, often called the *bloom*, is a waxy coating on grape skins. Studies of the makeup of these native yeast populations on sound, mature grapes from vineyards worldwide show remarkably similar patterns. Aseptically prepared grape musts will have 10^3 to 10^5 yeast cfu (colony-forming units) per milliliter. This population is usually dominated by the genera *Kloeckera* (Figure 3–3) and *Haseniaspora* (50–75%), and, at lesser levels, *Candida* (Figure 3–4), *Cryptococcus*, *Rhodotorula*, *Pichia*, *Kluyveromyces*, and *Hansenula*. The desired fermentative species, *Saccharomyces cerevisiae* (Figure 3–5), occurs at extremely low levels (<0.1 – 0.001%).

Fermentations initiated by native yeasts are characterized by a changing progression of yeast species, each of which becomes dominant and then diminishes as the fermentation medium changes, especially the concentration of ethanol. If the alcohol-tolerant species *Saccharomyces cerevisiae* does not develop, fermentation may stop, leaving the resulting wine residually sweet and unbalanced. Additional hazards exist with spoilage yeast strains that can foul the wine with a vinegar (acetic acid) or paint thinner (ethyl acetate) flavor. Despite the risks, some winemakers still

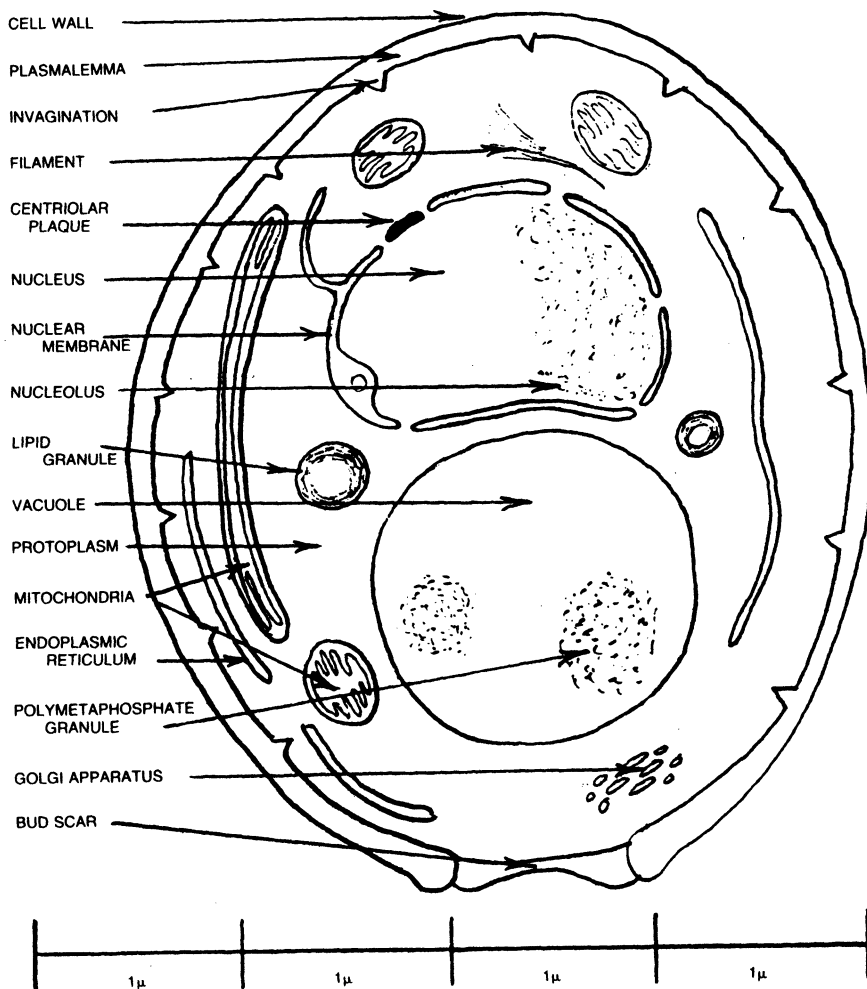


Fig. 3-2 Morphology of a Yeast Cell.

prefer to employ the native yeast population, attempting to develop unique flavor profiles in their wines. Chances for success depend on sound grapes, properly sanitized winery equipment, and, above all, a trained, dedicated, and observant wine-maker.

Yeast Spoilage

In the presence of air, several of the fermentative yeast species (*Candida*, *Metcschnikowia*, *Pichia*, *Hansenula*, *Saccharomyces beticus* (flor yeast), and

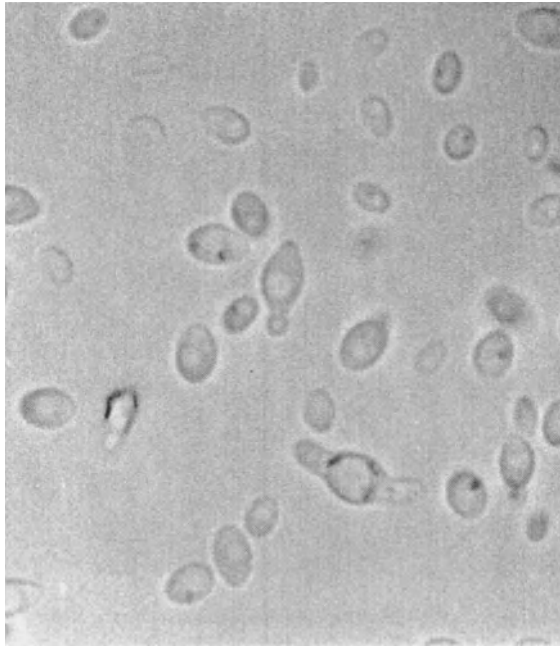


Fig. 3–3 Light Microscope Illustration of *Kloeckera*.
 Courtesy: A. Dumont, Lallemant, Inc.

Brettanomyces) are capable of producing chalky, wrinkled or blistered surface films on wine. Over time, dirty, musty off-flavors can develop due to the production of acetaldehyde, acetic acid, and esters of acetic acid such as ethyl acetate. Surface yeasts metabolize alcohol, glycerol, residual sugars, and malic acid, causing significant changes in wine balance and flavor. Contamination of bulk wines by film yeasts is extremely difficult to deal with, except by filtration and bottling of the infected wine. Many of the strains are very tolerant of sulfur dioxide. Maintaining storage temperatures at 45–55°F, as well as carefully managing the tank headspace with inert gases such as N₂ or carbon dioxide, can greatly inhibit surface growth.

Certain spoilage characteristics attributed to the genus *Brettanomyces* (Figure 3–6) are so ubiquitous to red wines produced in older wood cooperage and poorly sanitized environments that they debatably have become acceptable flavor characteristics at low concentrations. *Brettanomyces* can produce excessive volatile acidity and volatile phenolic compounds such as 4-ethyl guaiacol and 4-ethyl phenol, adding flavors described as spicy, medicinal, plastic, smoky, or horsy (barnyard). “Brett” can grow as a surface film; however, it is found more often as a slowly growing fermentative yeast, metabolizing low levels of residual sugars and causing

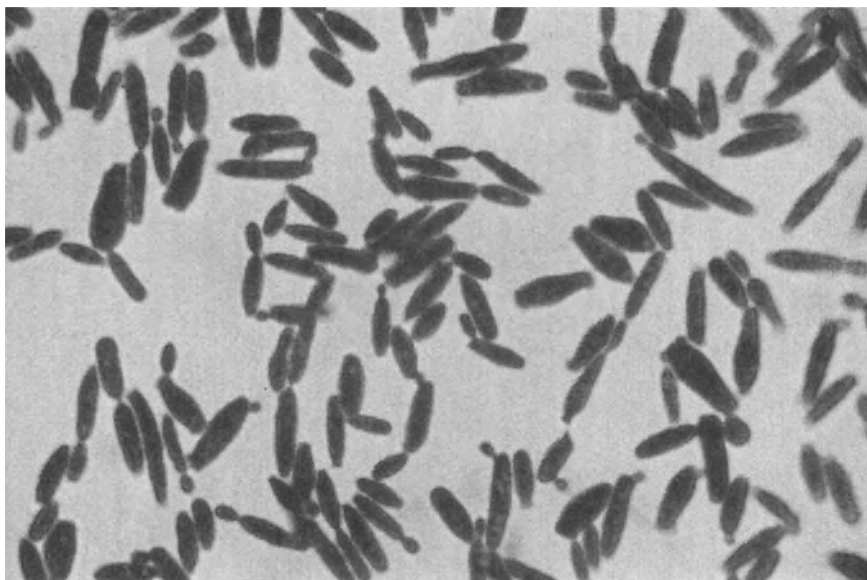


Fig. 3-4 Light Microscope Illustration of *Candida*.

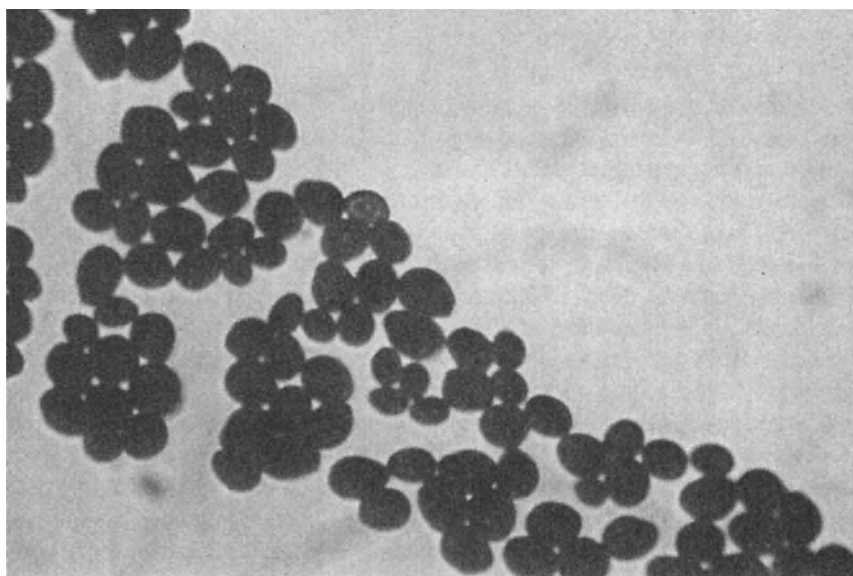


Fig. 3-5 Light Microscope Illustration of *Saccharomyces cerevisiae*.

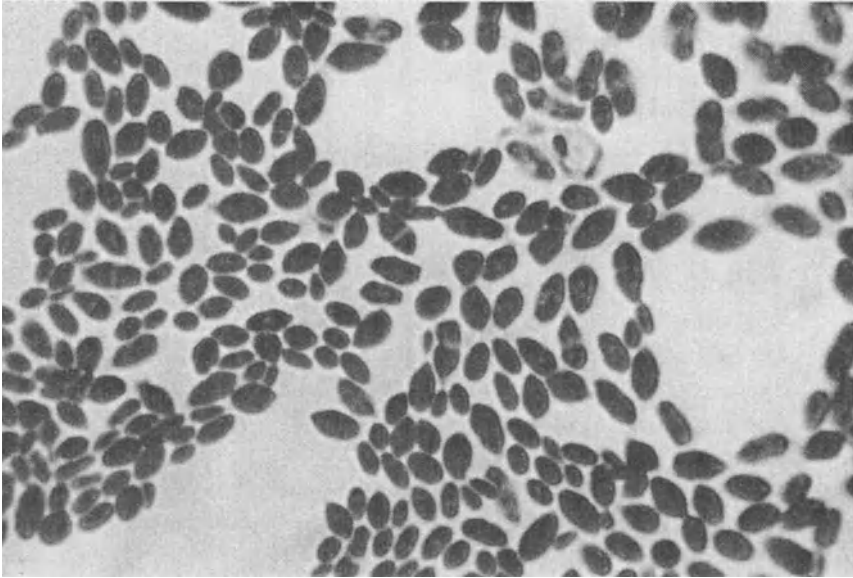


Fig. 3–6 Light Microscope Illustration of *Brettanomyces* sp.

haze and sediment and gradual deterioration of the wine. It is sensitive to sulfur dioxide at 0.8 ppm molecular. Microscopically, many cells of *Brettanomyces* will appear lemon shaped, but identification is based on its ability to grow slowly on media containing 100 ppm of cycloheximide.

Zygosaccharomyces bailii is an especially troublesome spoilage yeast, as it has unique tolerance to many inhibitive wine properties such as high concentrations of alcohol (>15%) and low pH (<2.0) and to wine preservatives. *Z. bailii* is not inhibited by 3.0 ppm of sulfur dioxide or 800 ppm of sorbic acid and is capable of growing in juice concentrates, a property that encourages its introduction in large numbers into otherwise sanitary wineries. It metabolizes sugar and malic acid, causing cloudiness, sediments, and increased acetic and succinic acid concentrations. Control of this organism depends upon filtration and rigorous high-temperature sanitation methods.

Yeast Identification

Identification of yeast at the genus and species level is a complex process, beyond the bounds of most wineries. However, the commercial winemaker can make presumptive judgments based on microscopic examination of the cells, knowledge of the history of the fermentation, and chemical and sensory evaluation of

the spoiled wine. Kenneth Fugelsang (1997) discusses detailed procedures for the identification of yeasts in his book *Wine Microbiology*.

Commercial Yeasts

Saccharomyces cerevisiae is the species within which the cultured wine fermentation yeasts are found, with many strains available for commercial vintners. All commercial strains have the following basic characteristics:

1. They ferment 25% to 30% of sugar to dryness.
2. They tolerate 50 to 100 ppm of free sulfur dioxide.
3. They produce minimum amounts of acetic acid, hydrogen sulfide, or sulfur dioxide.
4. They tolerate cool (50°F) to very warm (100°F) fermentation temperatures.
5. They tolerate low pH musts (<3.0).

Many commercial yeast strains have special properties allowing the winemaker to make small adjustments to the flavor or physical characteristics of the wine. Strains are available that can metabolize malic acid, increase body and mouth-feel, improve color stability, enhance volatile terpenes and overall fruitiness of the wine, produce “killer toxins” to inhibit wild yeasts, or even give the wine certain regional characteristics. Winemakers should be aware that these characteristics are usually quite subtle and may be dependent on certain fermentation temperatures or must pH. Normal oak exposure, malolactic fermentation, or poor winemaking practices, such as allowing the wines to oxidize, can mask any changes that the designer yeast strains may have imparted. New winemakers would be well advised to work with less expensive, general-purpose strains such as *Prise de Mousse* produced by Red Star and Lalvin (EC1118), Lalvin K1V, or Gist-brocades Fermivin.

Optimizing Must Conditions

Yeasts need adequate amounts of oxygen to replicate and develop alcohol-tolerant cell membranes rich in steroids and fatty acid components. During the introduction of yeast to the must, it is beneficial to pump over or stir vigorously to incorporate sufficient oxygen as well as to ensure that the desired yeasts are well distributed.

Replication also requires nitrogen in the form of amino acids (free α -amino acids, FAN) and ammonia for the production of yeast proteins. Although both are available in normal musts, many factors contribute to inadequate levels for optimal yeast replication. Free α -amino acid-level deficiencies in juice may be adjusted with diammonium phosphate (DAP) or proprietary formulations such as Yeastex or Fermaid, which contain mainly DAP. It is well established that stuck fermentations caused by limitations in nitrogen are irreversible. Thus, it is advised that DAP (200 mg/L) be added to the must before fermentation as a precautionary

treatment, with further additions of 100 mg/L added if hydrogen sulfide odors develop. In the United States, additions of DAP up to 950 mg/L are allowable, but in the European Common Market, 300 mg/L is the maximum addition allowed. Proprietary yeast nutrients may be added as a part of the total nitrogen addition. Excessive additions of DAP can give a salty flavor to wine as well as increase the potential for ethyl carbamate.

Preparation of Starter Cultures

Pure yeast cultures may be purchased in liquid form, or agar slants, but are most often sold as dehydrated pellets. When purchased from a reliable source, the cultures remain active for a year or more if stored under refrigeration. Although fermentation may occur if the dry yeast is simply sprinkled on the surface of the must, careful rehydration of the dried culture is a major deterrent to sluggish and stuck fermentations.

Recommended rehydration procedures will optimize yeast activity:

- Stir 1 g of dried yeast per gallon of juice or must to be inoculated into 10 times the total weight of water warmed to 102°F. Do not use juice for rehydration.
- Hold the mixture at 98 to 102°F for at least 15 minutes but no longer than 30.
- Mix the prepared starter culture into the juice or must that is within 10°F of the temperature of the culture. If the must is cold, add small amounts of the cold must to the culture over 2 to 3 minutes before adding the cooled mixture back to the bulk juice or must.

Stuck Fermentations

In spite of diligent precautions to provide an optimum environment for fermentation and well-adapted starter cultures, fermenting wines occasionally become sluggish or completely “stuck,” resulting in undesired residual sugar. The longer a wine stands without completing fermentation and going on to racking, sulfiting, fining, and cold stabilization, the more oxidation and acetic acid production will occur. Therefore, it is imperative that wines be monitored daily to provide the winemaker with information to facilitate reinitiation of problem fermentations.

Slow initiation of fermentation is often due to excessive free sulfur dioxide (especially in the case of low-pH musts), extremely high Brix, low temperatures, or unsatisfactory starter culture. Reinoculation is usually the best way to get the fermentation started, adapting the starter culture slowly with small additions of must in situations involving cold temperatures or high initial Brix.

When apparently normal fermentations slow down or stop abruptly, factors such as nutrient depletion and temperature shock, especially high temperatures with high ethanol concentrations, may be involved. Dr. Linda Bisson of the University

of California at Davis offered the following regimen based on her experiences and current industry practices:

1. Rack stuck wine, add 30 mg/L of sulfur dioxide, and bring to 60 to 65°F. Add 0.5 g per gallon of yeast hulls.
2. For every 250 gallons of stuck wine, prepare an inoculation medium of 4 gallons of water, 6 pounds of grape juice concentrate (or a 2:1 mix of fresh juice and water), and 30 g of DAP.
3. Rehydrate 2.2 pounds of dry yeast (Prise de Mousse strain) in 3 gallons of 102°F water for 15 minutes (this is an inoculation rate of about 4 g per gallon of stuck wine).
4. Add medium from step 2 to rehydrated yeast over a period of 30 minutes.
5. Monitor fermentation until half of the sugar has been used (a few hours).
6. Double the volume of the starter by adding an equal volume of stuck wine. Aerate well and monitor until half of the sugar is gone. Repeat this step two times (you should have about 40 gallons). The entire process may take 12 hours to 3 days.
7. Add adapted starter to remaining stuck wine, along with 20 g of dissolved DAP, and keep temperature at 60 to 65°F. It is important to pump over or stir the fermentation (without air) once a day to keep the yeast from settling out. The starter culture must not be allowed to go drier than the stuck fermentation; ideally it should be 2 to 3°Brix higher at the time of the final addition.

A microbiologically stable dry table wine typically has less than 0.2% residual sugar. When fermentations stick, leaving wine with higher levels of fermentable sugars, the winemaker must consider methods for stabilizing the wine, such as sterile filtration or the addition of sorbic acid, methods usually considered unacceptable for premium dry red table wines. If even small amounts of residual sugar are unacceptable for organoleptic or stability issues, the tedious and frustrating process of reinitiating stuck fermentations is worth the effort.

BACTERIA

In winemaking, two groups of bacteria dominate: lactic acid bacteria (LAB) and acetic acid bacteria. On very rare occasions involving high-pH wines, haze and sediment formation have been caused by the genus *Bacillus*, a spore-producing, gram-positive bacteria.

Lactic Acid Bacteria

As the name suggests, LAB are able to transform malic acid to lactic acid along with carbon dioxide gas. Three genera have been isolated from wines: *Lactobacillus*, *Pediococcus*, and *Leuconostoc*. They are gram-positive, rod-shaped or coccoid organisms that can be isolated from the grape cutin as well as from winery equipment, especially barrels, contaminated by “infected” wine. All are nutritionally

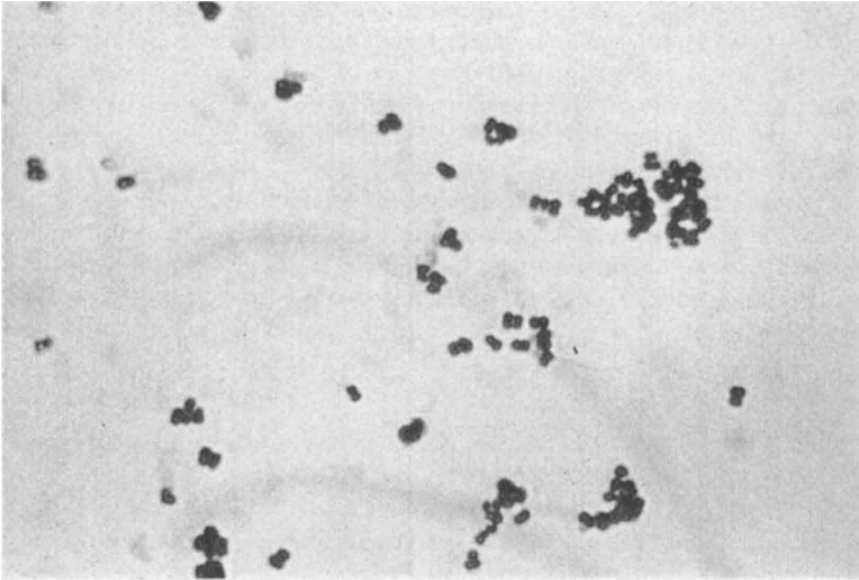


Fig. 3-7 Light Microscope Illustration of *Pediococcus cerevisiae*.

fastidious, requiring complex nutritional elements that may be supplied by the fruit components of the wine and dying yeast cells. LAB can utilize sugar, but they use different pathways and have different end products. Homofermentative species convert glucose to lactic acid only, while heterofermentative species yield lactic acid, ethanol, acetic acid, and carbon dioxide.

Pediococcus species can be identified readily by microscopic examination. These cocci exhibit characteristic tetrad formation resulting from cell division in two planes (Figure 3-7). They are homofermentative organisms, generally intolerant of wine pH less than 3.5. When they develop as a significant population in high-pH wines, pediococci have been associated with bitterness due to the formation of acrolein and with the formation of extracellular dextrans, referred to as “ropiness.” Wines with this condition have increased viscosity and “oiliness.”

Lactobacillus species may be homofermentative or heterofermentative. They appear in a range of forms, from coccobacilli (nearly round) to rods to long filamentous chains (Figure 3-8). Like the *Pediococcus* species, they are inhibited by low-pH wines but can cause serious sensory problems in wines with greater than 3.5 pH. Objectionable odors resembling urine-soaked mouse litter and sauerkraut have been attributed to *Lactobacillus* infections of high-pH wines.

Leuconostoc oenos is the only *Leuconostoc* species capable of growing in wine. It is an extremely slow-growing coccus that occurs in short to very long pearl-like chains (Figure 3-9). Because it can tolerate wine pH as low as 3.0, in the presence



Fig. 3-8 Light Microscope Illustration of *Lactobacillus brevis*.

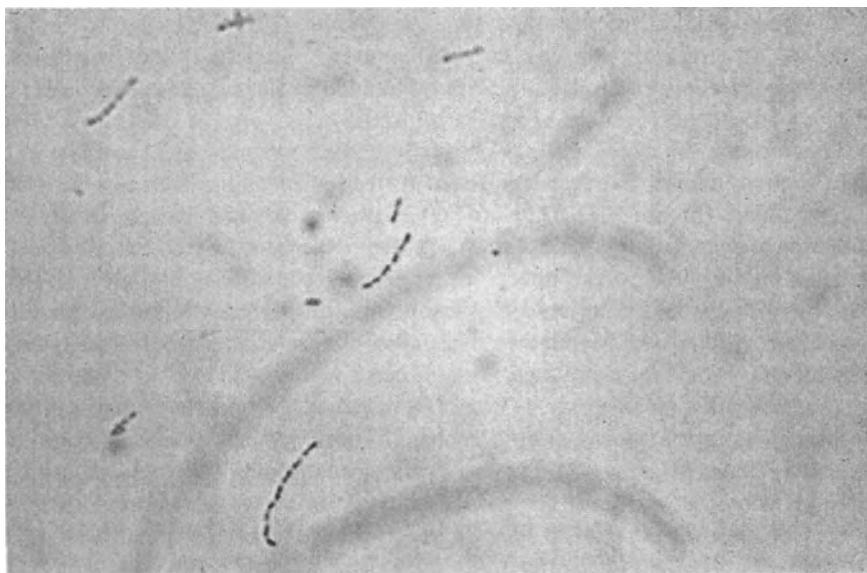


Fig. 3-9 Light Microscope Illustration of *Leuconostoc oenos*.

of 12% or higher alcohol, and can complete the conversion of malic to lactic acid without sensory deterioration of the wine, *Leuconostoc oenos* is the only LAB currently available in commercial starter cultures, although a few *Lactobacillus* and *Pediococcus* species have been used with varying success. In uninoculated malolactic fermentations, *Leuconostoc* usually becomes the dominant organism due to its ability to tolerate low pH. It is heterofermentative, causing a small increase in acetic acid but rarely enough to affect the perception of volatile acidity. *Leuconostoc* (but probably not *Lactobacillus* or *Pediococcus*) can metabolize sorbic acid, the short-chain fatty acid used to inhibit yeast refermentation of sweetened wines. The resulting compound, 2-ethoxyhexa-3,5-diene, has a pungent aroma resembling crushed geranium leaves.

Malolactic Fermentation

Although the exact pathway and biological reason for this transformation are still being studied, it is known that the malolactic enzyme converts dicarboxylic (two acid groups) malic acid to monocarboxylic (one acid group) lactic acid, with the release of carbon dioxide and a reduction in the total titratable acidity of the wine.

Malic Acid CO_2 *Lactic Acid*

$\text{HOOC}-\text{CH}_2-\text{CH}_3$

$\text{CHOH}-\text{COOH}$ $\text{CHOH}-\text{COOH}$

Although malolactic fermentation has been known to occur in wines since Pasteur's time, it is now considered common practice to encourage the process in most dry red wines and many whites. There are three major benefits in winemaking: deacidification, flavor modification, and microbial stability.

The deacidification impact depends upon the concentration of malic acid in the wine. Usually a decrease of 0.1% to 0.3% total titratable acidity is observed, with an upward pH shift of 0.2 to 0.4 units.

Both yeast and LAB metabolize citric acid, producing diacetyl, acetoin, and acetic acid. The buttery flavor of diacetyl can be quite pronounced when the growth of lactic acid bacteria occurs after yeast fermentation, especially in white wines lacking strong spicy or floral character. Lactic acid has a cheesy flavor and, when present in sufficient quantities, may affect the overall character of the wine. In some cases, reduced intensity of vegetative flavors has been noted, while fruitiness may be diminished or enhanced depending on the grape variety involved. Increased wine body and the length of aftertaste are often recognized as benefits of malolactic fermentation.

When a yeast fermentation has completed to dryness, malic and citric acid are the only components left in the wine in any significant amount that can be

metabolized anerobically. If these compounds have been removed during malolactic fermentation, the resulting wine may be considered microbiologically stable in the bottle, and the use of sulfur dioxide may be reduced (0.5 ppm molecular sulfur dioxide at bottling should be made available to prevent early aging of the wine due to oxidation). *Brettanomyces* spoilage has been associated with trace amounts of residual sugar. Wines destined to be bottled unfiltered and at reduced sulfur dioxide levels should be evaluated for the presence of *Brettanomyces spp.*

Commercial Malolactic Fermentation Cultures

Several excellent sources of commercial cultures are available. Since bacteria are more sensitive to drying procedures and storage conditions, it is especially important to purchase fresh cultures every year and to store them frozen. Although packages may be partially used and refrozen, some loss of cell viability usually occurs. Some cultures have been developed that can be added directly to wine; others must be rehydrated according to manufacturers' directions. When used as directed, the inoculums will provide about 10^6 viable bacteria per milliliter of wine. Unless some aspect of the wine is inhibitory, very little cellular replication will be needed to initiate the conversion of malic acid.

Strains of *Leuconostoc oenos* have been selected to give different flavor profiles as well as increased tolerance for low-pH wines. The effect on wine flavor is very subtle in most cases and may be influenced more by the wine character (high malic acid) or time of malolactic fermentation (during alcoholic fermentation, during barrel aging, etc.) than by the *Leuconostoc* strain chosen.

Encouraging Malolactic Fermentation

Successful completion of malolactic fermentation depends upon the ability of the dominant LAB species to reach a population level of 10^6 to 10^7 viable cells per milliliter of wine. If an uninoculated malolactic fermentation is desired, the following conditions should allow indigenous *Leuconostoc oenos* to attain the required population level and complete the transformation of malic acid:

- pH greater than 3.3
- free sulfur dioxide less than 10 to 5 ppm and bound sulfur dioxide less than 50 ppm (no sulfur dioxide added before or after fermentation)
- alcohol less than 12%
- wine temperature 65 to 75°F

Commercial cultures also thrive under these optimum conditions, but they may be able to overcome less than favorable environments and will most likely be necessary if sulfur dioxide was added to the must. At pH lower than 3.2, replication of LAB is very slow, and the necessary population may never develop during an acceptable

time between the end of fermentation and bottling. In this situation, the inoculation of a commercial malolactic culture may be the only option.

As the pH increases, especially above 3.5, indigenous LAB growth and malolactic fermentation will progress more rapidly and often will reach completion within a few days of the end of alcoholic fermentation. High pH also encourages growth of undesirable LAB and *Acetobacter*. When a high-pH wine shows no evidence of malolactic fermentation (by chromatogram) at the time of first racking of a white wine, or pressing of a red wine, inoculation of a commercial culture will ensure that malolactic fermentation progresses rapidly before the wine can be damaged by spoilage organisms.

Monitoring Malolactic Fermentation

The evolution of carbon dioxide, increase in wine pH, and decrease in total titratable acidity all accompany malolactic fermentation, but they are also indicative of other events in the life of a wine. Evaluation of malolactic fermentation activity should be done by paper chromatography, a simple and dependable procedure, described in detail in Appendix B of this book. Inexpensive kits are available from many commercial wine laboratories and wine supply companies, or wine samples may be sent to laboratories for accurate analysis of malic and lactic acid concentrations.

Preventing Malolactic Fermentation

Diligent use of sulfur dioxide remains the most reliable technique for controlling unwanted microbial growth in wine. Free sulfur dioxide in excess of 0.8 ppm molecular at bottling will be sufficient to maintain wine microbial stability. Several other chemical options are now available and are discussed later in this chapter.

Sterile filtration can also provide microbial stability. The extremely small size of *Leuconostoc oenos* dictates the use of filter media having an absolute pore size rating of 0.45 μ . This can be reliably provided only with membrane filter media assembled in a manner that allows for bubble point testing (see manufacturer's directions). Unless the filter, housing, and all tubing, connections, bottling line spouts, and so forth, downstream from the filter have been sanitized, expensive filters are futile. Wines passed through 0.45- μ filter membranes will lose some color, body, and even flavor, especially apparent in red wines. Sterile filtration of premium red wines is usually less desirable than maintaining a safe level of free sulfur dioxide.

Acetic Acid Bacteria

The microbes most feared by wine masters are the vinegar bacteria—more properly known as *Acetobacter* (Figure 3–10) or acetic acid bacteria. Two genera of bacteria are identified with acetic acid spoilage of wine, *Gluconobacter* and *Acetobacter*. They are both gram-negative, aerobic (requiring oxygen),

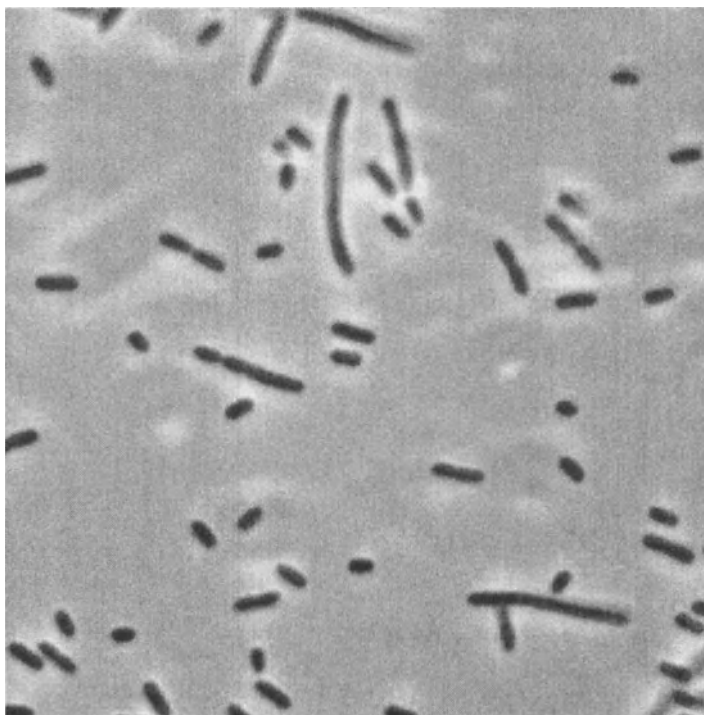


Fig. 3–10 Light Microscope Illustration of *Acetobacter*.

Courtesy: Neil Brown, Vinquiry, Inc. (New photograph).

pleomorphic (many-shaped) bacteria, capable of producing acetic acid from ethanol. The potential of these organisms to damage wine is a consideration throughout the winemaking process: (1) in the vineyard when grapes are damaged by disease, birds, or insects; (2) during the fermentation if yeast activity is interrupted and carbon dioxide formation is insufficient to protect the wine from exposure to air; or (3) during bulk wine storage.

Acetic acid is the major volatile acid in wine (others include formic, butyric, and propionic). The expression *volatile acidity* (VA) refers to the concentration all of these acids. A small amount of acetic acid, 0.2 to 0.4 g/100 mL, is produced by yeast during normal fermentation of glucose. Acetic acid bacteria are capable of producing large amounts of acetic acid by oxidizing ethanol. The extent of the conversion of ethanol depends upon the amount of aeration and exposure time. Acetic acid bacteria also produce significant amounts of acetaldehyde and ethyl acetate (nail polish remover), which cause even more damage to wine sensory character. Acetaldehyde binds free sulfur dioxide, requiring the use of large additions of sulfur dioxide to achieve stability.

In the United States, the legal limit of volatile acidity is 0.12% (1.2 g/L) in white wines and 0.14% (1.4 g/L) in red wines. Dessert wines made from grapes with 28 or more degrees Brix may have up to 0.15% VA in white wines and 0.17% VA in red wines. Natural fruit wines (containing no added brandy or alcohol) may have up to 0.14% VA.

Monitoring Vinegar Spoilage

By the time a classic, slimy vinegar film has developed on the wine surface, volatile acidity levels may be above the legal limit. Determination of volatile acidity can be accomplished in the winery by use of the Cash still (Appendix B). Wines in storage should be checked regularly. If the VA level appears to be increasing, steps to stabilize the wine must be taken immediately.

Controlling Vinegar Spoilage

Acetic acid bacteria are part of the indigenous microbial population of grapes. When fruit spoilage occurs in the vineyard or during shipment to the winery, incoming populations of vinegar bacteria can be immense. Damage will occur rapidly unless great care is taken to ensure a rapid fermentation and unless the wine is quickly secured in an oxygen-free environment immediately after completion. Large numbers of acetic acid bacteria exist in the winery wherever wine or juice is in contact with equipment or surfaces. Fruit flies, attracted to the smell of vinegar, carry the bacteria and infect everything they touch. Without diligent winery cleaning and sanitation as well as elimination of fruit fly populations, vinegar spoilage will be a recurring problem.

Maintaining all wines in an environment free of oxygen inhibits the bacterial conversion of ethanol to acetic acid. During bulk storage in barrels or partially filled tanks, several options are available to reduce acetic acid bacteria growth and activity. Sulfur dioxide acts both as an antimicrobial agent and as a scavenger of oxygen. A level in excess of 0.8 ppm molecular sulfur dioxide should be maintained during bulk storage. Regular applications of inactive gases such as carbon dioxide or N₂ also help to protect wines during bulk storage.

Winemakers who decide to encourage a wine to encourage the conversion of a wine to vinegar should move the infected tank to another spot, completely outside any wine production area. An old oak barrel, half to three-quarters filled, is an excellent vinegar generator. The bung hole can be covered with cheesecloth to control fruit flies while allowing air to enter. Acetic acid bacteria may form a thin, slimy surface film to a solid rubbery pad that grows thicker, settles to the bottom, and reforms slowly at the surface. The conversion yields approximately 1 part acetic acid from 2 parts of ethanol and may take several months to a year or more to complete in this manner. Commercial vinegar is usually 4% to 5% acetic acid.

MOLDS

Viticulturists may have to contend with such molds as black rot, downy mildew, powdery mildew, and others in the vineyard from year to year. Winemakers contend with a different set of molds, all of which affect wine indirectly.

Alcohol content and lack of oxygen prevent molds from growing directly in wine. Poor sanitation and humid conditions can lead to the growth of molds on equipment and storage vessels, especially older wood tanks and oak barrels. Wine exposed to these contaminated surfaces will absorb moldy odors and flavors that seriously deteriorate its quality.

Corks that have been stored in damp places or within porous containers previously contaminated with mold and *Streptomyces*, a moldlike soil microorganism, can develop an intense musty smell ("cork taint"), which permeates and contaminates wines. Compounds associated with this problem are 2,4,6 trichloroanisol, guaiacol, and geosmin and may be introduced into unprocessed corkwood during stockpiling.

With unexpected rains or improper handling in the vineyard, berries may become infected with fruit mold pathogens capable of shriveling berries to hard, black raisins or turning entire clusters into slimy decay. *Penicillium expansum* (Figure 3-11), not to be confused with the *Penicillium notatum* of medicinal fame, emerges as a white powder-like dust on grape berries and matures to an ominous bluish-green mold. Other molds such as *Aspergillus* and *Mucor* species produce similar fruit degradation in the vineyard. Wines produced from moldy grapes may exhibit unpleasant musty flavors and will suffer from serious oxidative problems due to the presence of laccase enzyme from the mold. Some molds utilize grape sugars and tartaric acid and produce gluconic and acetic acids. The grape grower must understand that poor vineyard practices that lead to moldy grapes will cost dearly in resulting wine quality and thus reduce the value of the harvest. The winemaker who is forced to deal with moldy grapes should take the time to sort out as much of the spoiled fruit as possible and make every effort to have a rapid, controlled fermentation, stabilization, and bottling.

The famous "noble mold," or, more properly, *Botrytis cinerea*, is an exception. When desired and when grown in optimal conditions, this fuzzy gray mold permeates grape skins and allows internal water to evaporate. The resulting berries are shriveled with concentrated tannins, acids, and sugars. *Botrytis cinerea* contributes slightly musty, honeyed flavors that have become acceptable characteristics of the style. Such grapes make the renowned French *pourriture noble* ("noble mold") wines of Sauternes in Bordeaux and Edelfaule wines in Germany. Many similar "late-harvest" wines of both types are also grown in the United States. Connoisseurs consider noble mold wines some of the most exquisite white wines made. The loss of natural grape water and the delicate techniques of making such wines successfully render them an expensive delicacy.

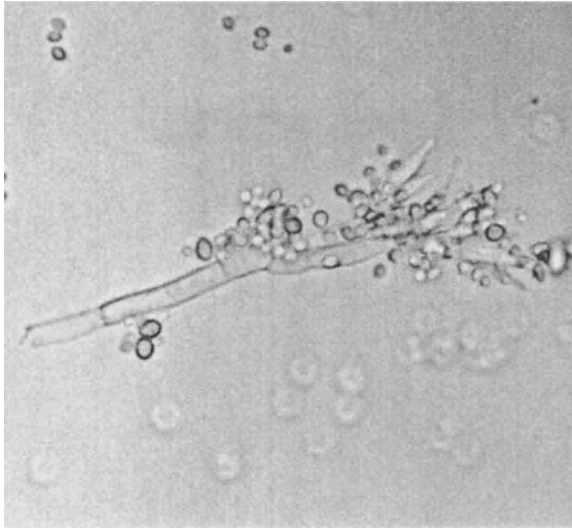


Fig. 3–11 Light Microscope Illustration of *Penicillium*.
Courtesy: A. Dumont, Lallemant, Inc.

THE MICROBIOLOGICAL WORKPLACE

Commercial winemaking requires vintners to become involved in handling microbes. Yeasts and malolactic bacteria need to be properly stored, rehydrated, and sometimes expanded for successful fermentations. Problem microbes need to be detected and overcome. The investment need not be large. Fully adequate microbiological facilities can be achieved in a winery for \$3,000, including a good used microscope about the cost of six premium oak barrels.

The ideal location for a microbiology laboratory is a small, well-lighted room, with washable, nonporous walls, ceiling, and floor. It should have a countertop, sink with hot and cold running water, and several electrical outlets.

Alternatively, microbiological tasks can be performed in a multipurpose area, such as the analysis lab, if the area can be isolated from fermentation vessels and winery operations. A “hood” or small self-contained workplace will protect the work area from air currents and dust. Used chemical hoods are not suitable, as they are generally designed for rapid air circulation in the wrong direction for microbiological work and are virtually impossible to sanitize.

Building a small effective hood made from plywood, Plexiglas, and hinges can be very cost-effective. The plywood back and adjoining sides are cut on a bevel and hinged together to collapse for storage (see Figure 3–12). The unit is covered

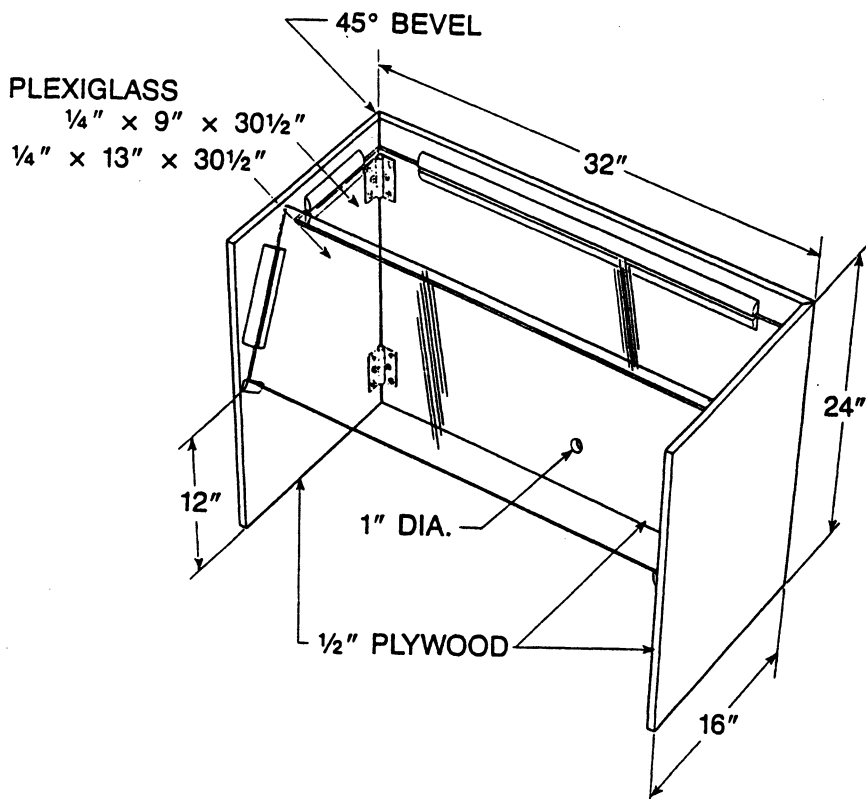


Fig. 3-12 Microbiology Hood.

with a primer and several coats of high-gloss washable enamel paint. Removable Plexiglas panels are mounted between railing strips.

A small Styrofoam incubator (Figure 3-13) will provide a continuous temperature to grow yeasts and bacteria rapidly. The temperature is controlled by a variable control switch connected to a light bulb in a socket assembly. The bulb is inserted through the side of the Styrofoam ice chest. This unit is inexpensive to construct and can maintain a constant temperature of 70 to 75°F with a 5 to 10 watt bulb.

EQUIPMENT

A technician with a fundamental knowledge of aseptic technique and rudimentary skills with light microscopy and slide preparation should be able to perform most of the basic tasks involved in presumptive identification of winery microorganisms.

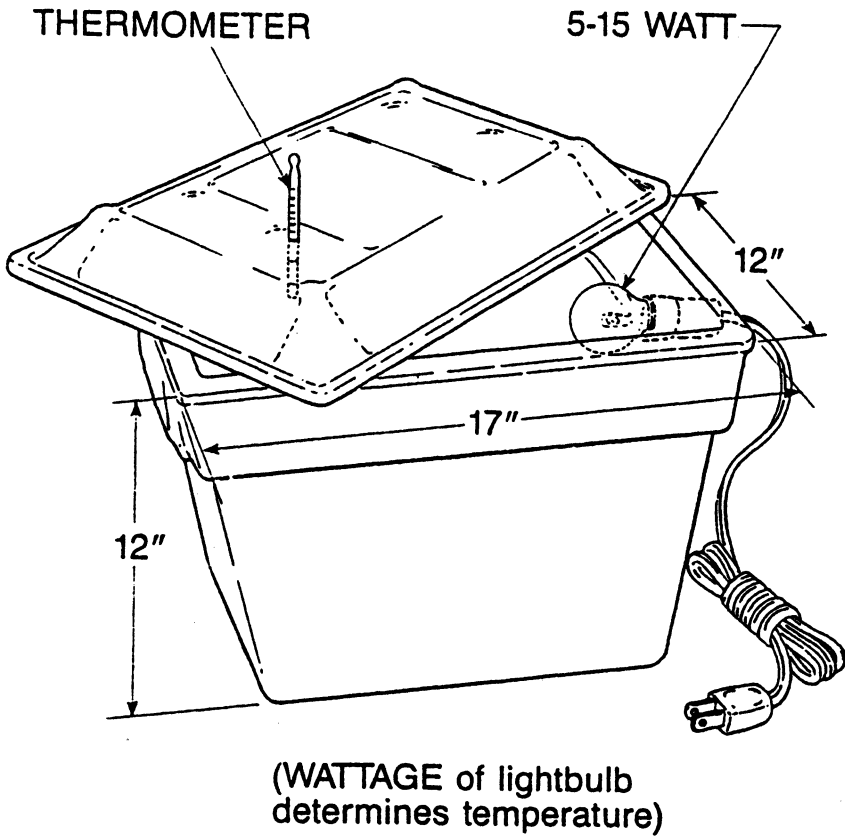


Fig. 3-13 Styrofoam Incubator.

Detailed instructions for use of the light microscope and slide preparation and staining are given in Appendix B. Also in Appendix B, instructions are specified for performing evaluations of winery sanitation and of microbiological contamination in beverages. The following list contains equipment needed to set up a small effective microbiological laboratory:

- *Alcohol Burner (Glass Lamp)*. The primary purpose of this is to create a flame to disinfect equipment, dry microscope slides, and sterilize loops.
- *Aspirator Filter Pump*. This device is powered by the velocity of water passing through a cold-water faucet, producing the vacuum needed to pull wine samples through a filter assembly.
- *Side-Arm Vacuum Flask (1,000 mL)*. A side-arm vacuum flask supports the filtration unit and collects filtered wine drawn through by vacuum. Size 8 rubber stoppers with a single hole are needed to connect the filtration unit to the vacuum flask. Heavy-duty black rubber tubing connects the aspirator to the vacuum flask.

- *Yeast and Mold Swab Test Kits.* These are used for detecting microbial contamination of winery equipment, tanks, corks, and so forth. Kits may be purchased from Millipore Company (see Appendix A).
- *Paper Chromatography Kits.* Detection of malolactic fermentation is made using these kits. Alternatively, a paper chromatogram can be made following the methodology outlined in Appendix B.
- *Microscope.* The cost of a microscope capable of 1,000X magnification is not within every vintner's budget. Often, laboratories in universities, high schools, hospitals, and other sources have surplus used microscopes for sale at prices far below new models. From time to time, some of the suppliers listed in Appendix A offer preowned as well as new microscopes. A good used microscope can cost several hundred dollars. Depending on features, a new microscope can cost several thousands of dollars. It is important to check all features and lenses by viewing a prepared bacterial slide before paying for a used microscope.
- *Microscope Accessories.* These include glass slides, cover slips, lens paper, inoculating loops, immersion oil, Gram Stain kit, and methylene blue stain.

FACTORS AFFECTING MICROBIAL GROWTH

As with all living things, the ability of microbes to thrive is directly dependent on their environment.

- *Moisture* is essential to cell growth and activity. In extremely dry environs, actively growing cells will eventually die or go into a dormant state if they have that capability. As the moisture content of the environment increases, cellular activity also increases. In solutions where there is a very high concentration of dissolved solids, and thus a commensurately lower moisture level, living cells cannot utilize the low moisture levels and are unable to multiply. For this reason, jellies, concentrates (above 68°Brix), honey, and brine solutions are resistant to microbial spoilage due to their "low water activity."
- *Temperature* is a controlling factor for microbial activity. Optimum growth rates for wine-related organisms are in the range of 60 to 85°F. However, some yeasts are capable of fermenting slowly at temperatures below 30°F. Most microbial growth ceases at temperatures above 100°F, and viable cells (actively growing cells) are killed if exposed to temperatures above 160°F for a few minutes.
- All cells require *nutrients*, but each microbe genus and species may have different specific requirements. Wine provides carbohydrates in the form of sugars, ethanol, and organic acids. Amino acids and ammonia are available in grapes as sources of nitrogen. Juices and musts contain vitamins and other complex factors required for the growth of microorganisms.
- *pH*, a measure of acid strength, is a major factor controlling microbiological activity. Low pH is one of the main reasons why so few microbes will grow in wine. Yeasts and some LAB will tolerate pH as low as 2.8. Wine organisms are increasingly active

as pH increases to 4.0. Because wines experience serious nonmicrobial problems at pH higher than 4.0, a high pH is not a factor in inhibiting microbial growth in juice, must, or wine.

MICROBIAL SPOILAGE

Juice and Must

The primary problem facing fresh juice processors, in both bulk storage and packaged final products, is yeast fermentation. According to U.S. Food and Drug Administration (FDA) regulations, a level of 0.5% (by volume) or greater of ethyl alcohol exceeds permitted concentrations for juice labels. A secondary problem is bacterial spoilage producing increased levels of acetic acid. If the surface of the juice is exposed to air, large dark greenish pads of wrinkled mold growth may develop, floating like lily pads on a pond. They will eventually give a musty character to the juice.

The presence of gas in juice is evidence of yeast spoilage. As this spoilage progresses, sour odors and flavors develop. Bulk juice should be monitored for ethyl alcohol concentration. Juice samples should be cultured on appropriate media (Appendix B) to determine the numbers of viable yeast cells, along with an observation under a microscope to confirm the presence of budding yeast cells.

Juice Concentrates

When juice has been concentrated to at least 68°Brix, sufficient water has been eliminated to achieve resistance to spoilage due to low water activity. Concentrates between 60 and 68°Brix should be stored at 30 to 32°F, whereas lower concentrations of juice should be stored at 23 to 25°F. The condition of concentrates should be inspected regularly; storage for more than several months is not recommended. One problem that can develop when concentrates are moved in and out of refrigerated storage is the condensation of moisture on the inner surfaces of the containers. This moisture can serve to dilute the concentrate surface to provide higher levels of water activity, thereby creating a fermentable condition for yeasts or mold growth. Yeast activity is accompanied by carbon dioxide gas and reduced Brix levels.

Mold growth occurs as small fuzzy islands developing on the surface of the concentrate or on the sides of the container, along with a musty odor. These colonies may become several inches in diameter over time.

Zygosaccharomyces bailii has been reported growing in concentrates and is an especially troubling contaminant because of its resistance to wine preservatives, as discussed earlier. For this reason, all concentrates should be evaluated for yeast activity. Wines sweetened with concentrates should be sterile filtered before bottling.

Bulk Wine

Although dry wines fermented up to at least 11% alcohol are relatively resistant to microbial growth, they are still susceptible to spoilage by the growth of a few undesired organisms. In all cases, the higher the wine pH, the greater the likelihood that microbial growth will occur, the faster it will occur, and the greater the intensity of the resulting spoilage.

Bulk wines should be stored with 30 to 50 ppm of free sulfur dioxide in completely filled tanks, maintained as cool as possible. Flushing headspaces with inert gases such as carbon dioxide or N₂ helps for short-term storage but is risky for longer periods unless controlled by sophisticated gas metering equipment.

The very first sign of microbial activity in bulk wine (or juice) is a reduction in the free sulfur dioxide. An excellent way to monitor wines in storage is to maintain records of the free sulfur dioxide on a weekly basis, also checking for any visual changes such as surface films. If the level of free sulfur dioxide suddenly decreases, the wine should be evaluated for yeast or bacterial growth (Appendix B) and analyzed for volatile acidity, and steps should be taken immediately to remedy any problems.

Barrel aging of bulk wines provides a perfect opportunity for *Brettanomyces* yeast growth. It is often slow to develop and may go undetected until the wine is severely damaged with unpleasant “barnyard” odors and flavors. *Brettanomyces* can be controlled by free sulfur dioxide. Wines stored in barrels (especially *used* barrels) should be evaluated for free sulfur dioxide and sensory characteristics on a regular basis. Barrels that have had *Brettanomyces*, *Acetobacter*, or lactic acid bacterial infections are nearly impossible to sanitize and should not be used again for wines.

Acetic acid bacteria will become active in bulk tanks where headspace allows air contact with the wine or when frequent, turbulent rackings introduce air after fermentations have completed. Careful management of free sulfur dioxide and headspace management can minimize spoilage from acetic acid bacteria.

Bottled Wine

In a properly filled and sealed bottle, acetic acid bacteria are rapidly inhibited. The most common microbial problem in bottled wine is yeast refermentation of wines having residual sugar. Haziness and profuse gas in the wine after the cork is pulled are characteristic of yeast spoilage. *Brettanomyces* yeast are capable of growing slowly after bottling, rarely producing more than a fine sediment but causing significant deterioration of wine aroma and flavor. LAB can grow rapidly in corked bottles, producing hazy wine and small amounts of gas, usually seen as slowly rising fine bubbles coming steadily to the surface after the cork is pulled. Table 3–1 presents a summary of the causes of microbial spoilage.

Table 3-1 Determination of causes of microbial spoilage.

	Bacteria			
	Yeast	Acetic acid bacteria	Bacilli	Lactic acid bacteria
Visual appearance	Fine haze or precipitate or film; wine may be gassy	Gray film on surface	Fine haze or precipitate or silky, streaming cloud when shaken; wine may be dull	
Odor	Not characteristic	Vinegary (acetic acid and ethyl acetate)	—	Wine may be slightly gassy. Sauerkraut or diacetyl character. Less than 4–5 mμ.
Microscopic appearance	Greater than 4–5 mμ	Ellipsoidal or rods (involutionary forms may be present)	Less than 4–5 mμ; Rods.	Rods or cocci (like tangled mass of hair = <i>Lactobacillus trichodes</i>).
Growth on Basic medium ^a	+	+	+	+
Basic medium + cycloheximide ^b	0 or + ^c	0 or +	0 or +	(Ferments malate to lactate = malolactic bacteria) +
Catalase test ^d	—	+	+	—
Spore formation	0 or +	—	+	—
Calcium carbonate plates ^e	—	Clearing.	No clearing	No clearing.

^aBasic medium: The basic culture medium contains 2.0 g/100 ml of Bacto tryptone, 0.5 of Bacto peptone, 0.5 of Bacto yeast extract, 0.3 of glucose, 0.2 of lactose, 0.1 of liver extract (Wilson), 0.1 ml of 5% aqueous Tween 80, 100 ml of diluted and filtered tomato juice. To prepare tomato juice, dilute 4-fold with distilled water, filter through Whatman No.1 paper using a Büchner funnel and Super Cel filter aid. To prepare the basic medium dissolve solid ingredients in diluted tomato juice by heating (avoid scorching by frequent agitation of the flask). When cool, adjust pH to 5.5 with concentrated hydrochloric acid, add 2 g agar/100 ml, and autoclave 15 min at 15 psig.

^b“Acti-dione”: 1 ml containing 10 mg of cycloheximide is added to each 100 ml of medium.

^cGrowth is presumptive evidence of *Brettanomyces* or *Dekkera* yeast.

^dBasic medium with cycloheximide: Add a drop of 3% of fresh hydrogen peroxide to colonies. If gas is evolved the organism is catalase positive.

^eBasic medium with cycloheximide minus the glucose and lactose and containing 2% calcium carbonate and 3% ethanol (the ethanol is added after the auto-claving):

If acetic bacteria grow they will cause clearing of this cloudy medium around the colonies.

Source of data: Tanner and Vetsch (1956) and personal observations of authors.

Source: From *Amerine et al. 1980. Technology of Winemaking*. AVI Publishing Company, Westport, Conn.

WINE PRESERVATIVES

Wines may be treated with compounds that act as preservatives or as sterilants. Sterilants, used correctly, should kill 100% of all viable microorganisms contacted.

Preservatives can cause cell death but generally act to suppress microbial growth and are not uniformly effective against all organisms. Sulfur dioxide and sorbic acid are preservatives whose activity is dependent upon pH: they are more active as pH decreases. Both preservatives and sterilants are most effective against inactive, low-density populations of microbes; they are usually quite inadequate when added to vigorously growing populations.

Sulfur Dioxide

Sulfur dioxide continues to be one of the most important food preservatives available. Since it is both an antimicrobial and an antioxidant, its value as a wine preservative is unequalled. Unfortunately, it is also an irritant, especially for people with asthma, and it is closely regulated by the Bureau of Alcohol, Tobacco, and Firearms (ATF). The current legal limit for total sulfur dioxide remains at 350 ppm. Further discussion of the use of sulfur dioxide in winemaking may be found in Chapter 4. Although sulfur dioxide is very inhibitory for most bacteria, it has limited activity against fermentation yeasts and many spoilage yeasts, especially in low-alcohol or high-pH wines. Sulfur dioxide should be added before yeast inoculation or after active fermentation has ceased. While yeasts are actively metabolizing sugar, acetaldehyde is present in quantities that immediately link with free sulfur dioxide, resulting in high levels of bound and inactive sulfur dioxide.

Several methods for analyzing sulfur dioxide are available. The Ripper method (Appendix B) is fairly reliable for white wines but will give elevated results for red wines due to nonspecific reactions with red wine pigments. Rapid but very unreliable results are obtained with Titrettor test kits. They should be used only for monitoring purposes, and results should be compared occasionally to more reliable methods. The most accurate method available for wineries is the aeration/oxidation method (Appendix B). Glassware is available for about the same cost as Ripper equipment. Although the procedure takes a bit longer to complete, the accuracy of the results is well worth the extra effort.

Sorbic Acid

Wines with residual sugar can be stabilized at bottling by the addition of 200 ppm of sorbic acid. The maximum amount allowed by the ATF is 300 ppm. Pure sorbic acid is extremely insoluble in wine and must be added as the salt potassium sorbate. The addition of 1 g of potassium sorbate to a gallon of wine will give 200 ppm of sorbic acid. Sorbic acid (sorbate) is only effective in wines with very low yeast

populations (fewer than 100 viable cells per milliliter). Since it has a potential for flavor deterioration if the wine becomes oxidized, it should be added just prior to bottling. Wood barrels should not be used for prebottling mixing, since sorbic acid soaked into the wood can be metabolized by resident LAB and can taint any other wines introduced to the barrel.

When the salt is fresh and relatively odorless, it has a nearly imperceptible effect on wine flavor, especially aromatic, fruity flavored wines such as Concord or Muscat. This preservative is not recommended for wines that will have long-term storage, since slow oxidation may damage flavor characteristics. LAB and some yeasts can metabolize sorbic acid, producing powerful geranium leaf odors, as mentioned earlier in this chapter; therefore, wines treated with sorbate must also be treated with adequate sulfur dioxide.

Sorbic acid has been shown to work synergistically with alcohol. Wines having alcohol levels above 11% should be stable with 150 ppm of sorbate, especially if they are low-pH wines. Sorbic acid is very stable in wine, unlike sulfur dioxide, making analysis unnecessary unless the winemaker suspects an incorrect addition. Analysis must be done at a laboratory that has access to an ultraviolet spectrophotometer.

Benzoic Acid

Although benzoic acid is approved as a preservative in wine coolers and juices, it is not allowed for use in table wines. It is usually used in combination with sorbic acid at levels around 500 ppm (250 ppm of benzoic acid and 250 ppm of sorbic acid) to obtain maximum preservation with minimum flavor impact. The ATF allows a maximum addition of 1,000 ppm of benzoic acid, well above the sensory detection threshold. As with sorbic acid, adequate sulfur dioxide levels must be maintained.

Fumaric Acid

Fumaric acid is a strong organic acid, approved by the ATF for acidulating wine as well as inhibiting the growth of LAB. The maximum allowable addition is 3,000 ppm, but the level usually used is 300 to 400 ppm. Due to difficulties with dissolving fumaric acid in wine, its distinctive flavor characteristics, and reports that some species are resistant, it is rarely used as a wine preservative.

Lysozyme

Lysozyme is an enzyme found in mucosal secretions of humans and many animals. It is a wine preservative that exhibits specific activity against gram-positive bacteria, causing lysis of cell walls, and is used in white wines to control all species of LAB. The suggested use level is 250 to 500 ppm. It is newly approved by the ATF,

though not listed in the regulations at this time. Winemakers can apply and receive authorization to use up to 500 ppm of lysozyme in commercial wine following the protocol found in 27 CFR, Part 4, Section 24.250. Egg whites represent the major commercial source of lysozyme.

Lysozyme is ineffective against yeasts or acetic acid bacteria, but in dry white wines it can allow a significant reduction in sulfur dioxide levels without loss of protection against LAB spoilage. This enzyme is active any wine pH and is not affected by alcohol. It causes a rapid disintegration of gram-positive bacteria after addition, and it is stable in white wine—a factor that allows it to be effective during wine storage. Since lysozyme is a protein, it is precipitated by bentonite and red wine tannins.

Dimethyldicarbonate

Dimethyldicarbonate (DMDC) is a sterilant sold under the trade name Velcorin. The ATF allows its use in table wines at a level of 200 ppm, generally added at bottling. It destroys yeasts and bacteria within a few minutes, depending upon the temperature of the wine, and then hydrolyses to carbon dioxide and an insignificant amount of methanol within 1 to 5 hours.

Because it has rapid hydrolysis and is relatively insoluble, DMDC is usually introduced into the wine on its way to the bottling line by a metering pump that can atomize the material into the wine. Ideally, DMDC is still active at the time the wine enters the bottle, allowing for the destruction of any organisms that may have been picked up from the line, bottle, or closure.

Although it appears to have great value, especially in the preservation of wines with residual sugar, DMDC requires special handling precautions. Velcorin sterilant is a liquid that does not evaporate rapidly; however, in its undiluted liquid form it can be severely irritating to the skin and must be handled with the same care as dilute sodium hydroxide or hydrochloric acid. Its vapors are irritating to mucus membranes in the same way that those of sulfur dioxide and ammonia are, although it is much less volatile. The use of designated equipment is highly recommended for safe and effective wine treatment.

WINERY SANITATION

Evaluating the effects of winery sanitation procedures requires frequent careful visual and sensory examinations of the entire area. Sticky juice, grape stem and seed particles in the crusher-stemmer and drains, mold residue on walls, tanks, pumps, filters and hoses, and particles of cork and trash on the floors are visual signs of an ineffective sanitizing program. Foul odors from drains and from the inside of hoses and tanks are sure indications that better control is needed.

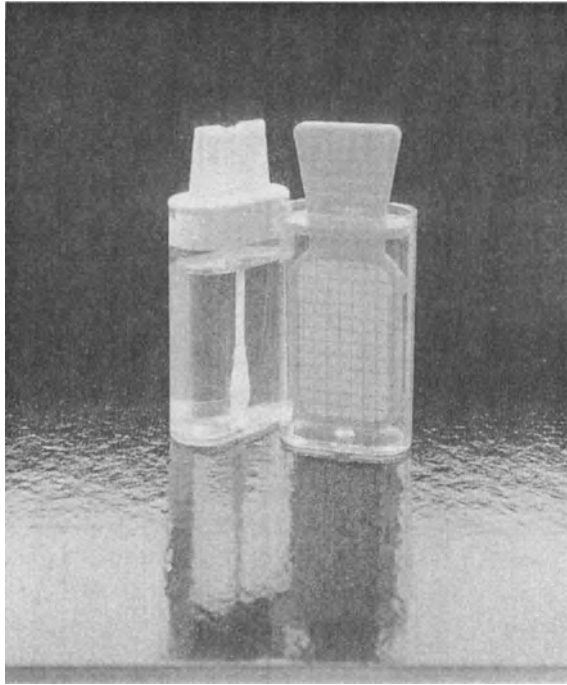


Fig. 3–14 Sampler and Swab Test Kit.
Courtesy: Millipore® Corporation, Bedford, MA.

Following thorough cleaning of all surfaces exposed to fresh juice, must, or wine, various methods of sanitation are available to the winemaker. Appendix B provides analytical procedures for the evaluation of winery sanitation procedures. Figure 3–14 illustrates a sanitation detection kit.

Heat Sanitation

Steam or hot water in excess of 180°F is the method of choice for complete sterilization in the winery because heat can penetrate sealed areas and is effective against all microorganisms harmful to wine. Added benefits include ease of use and absence of toxic and corrosive properties. It is critical that time and temperature be monitored carefully to ensure reliable results. With both steam and hot water, the surface to be sanitized should be maintained at 180°F for at least 20 minutes. All valves and ports should be open, with hot water or steam exiting from those points. Steam systems should be monitored at the furthest point from introduction of the steam, and condensate at that area should reach 180°F at least 20 minutes.

Equipment that has been disassembled for cleaning should be completely reassembled before final sanitation with heat. Plastic and rubber are damaged by extensive heat treatments and should be inspected often for cracks that may hide microbial contamination and prevent complete sanitation.

Ultraviolet Radiation

UV light is very effective against microorganisms. Unfortunately, due to its inability to penetrate even thin films, it has very limited usage in the winery.

Sulfur Dioxide

Sulfur dioxide in an acid solution ($\text{pH} < 3.3$) can be used effectively to *reduce* microbial populations. A water solution of 2 to 3 g of potassium metabisulfite and 12 g of citric acid per gallon (300–500 ppm of free sulfur dioxide) can be very valuable as a final rinse to well-cleaned tanks, hoses, filter units, and so forth. Since both sulfur dioxide and citric acid are compatible with wine, problems of chemical contamination are avoided. Irritating vapors, eventual corrosion of steel, and limited effectiveness against mold spores and fermentation yeast contamination are all concerns with dependency upon sulfur dioxide as the only sanitation option.

Ozone

Ozone (O_3) is an activated form of oxygen (O_2) that is naturally formed in the upper atmosphere. It is an extremely powerful oxidant, rapidly destroying yeast, molds, and bacteria before reverting to oxygen without any other byproducts. Neither pH nor temperature affects the activity of this sanitizer. Ozone generators are quite costly to purchase but very inexpensive to operate compared to hot-water sanitizing systems. Ozonated water prepared by dispersing ozone gas from the generator into a stream of water is especially effective in sanitizing barrels and bottling lines. It is less corrosive and toxic than chlorine-based sanitizers and does not produce any compounds that could be involved in sensory problems upon contact with wine.

Chlorine-Based Compounds

Hypochlorous acid (HOCl) is the active form of chlorine that may be found as household bleach. It acts as a powerful oxidant and sterilant in a pH range of 4 to 7. HOCl is active against all organisms and works rapidly, at a wide temperature range. Since it is inactivated by organic dirt, careful cleaning must precede any chlorine treatment. To sanitize surfaces, an active chlorine concentration of 100 to 200 ppm is required. Wood and other organic surfaces treated with chlorine-based sanitizers may be damaged by oxidation and will absorb the chlorine smell. Wine

contacting these surfaces may be tainted. Perhaps the best use of HOCl is in the sanitizing and deodorizing of drains and moldy floors and walls. Very effective buffered chlorine sanitizers are available to reduce the corrosion problems caused by simple HOCl solutions.

Chlorine-based sanitizers pose health risks and must be used with care. ***WARNING: CHLORINE SOLUTIONS MUST NEVER COME IN CONTACT WITH STRONG ACIDS OR BASIC [ALKALINE] COMPOUNDS, AS POISONOUS CHLORINE GAS WILL BE GENERATED.***

Iodine Compounds

Iodophores are commercial formulations containing iodine and nonionic wetting agents. Iodine, the active sanitizing component, is most active in the pH range of 4 to 5. The recommended level of iodine for surface sanitizing is 25 ppm. Iodine is active against all wine spoilage organisms. It is relatively nontoxic and has residual activity, retaining its antimicrobial properties for some time after the surfaces dry. Iodophore sanitizers may be used to sanitize bottling lines, hoses, and tanks if followed by extensive cold-water rinsing.

Quaternary Ammonium Compounds

Modern formulations of quaternary ammonium compounds (Quats) are effective against most wine spoilage organisms. Since they are heat stable, nontoxic, and noncorrosive and are not inactivated by organic dirt, they may be incorporated easily into winery cleaning operations. Like iodophores, Quats have residual activity long after the application and can prevent mold growth on walls and floors for several weeks. Great care must be taken to avoid contamination of wine with these chemical sanitizers. Although they are not toxic to humans at the recommended use levels, they certainly can cause undesirable aromas and flavors in the wines.

CHAPTER 4

ENOLOGY (WINEMAKING)

Sooner or later vintners will encounter the term *enology*, which is sometimes spelled in the British version as *oenology*. In either case, it is best simplified by its Greek origin of *oinos* (wine) and *logos* (logic). Wine logic is based upon the application of analysis, equipment, materials, and technology in designed methodologies to achieve predetermined results. Enology is constructed upon a gastronomic art form, heavily supported by the disciplines of biochemistry, microbiology, and mechanical engineering. A modern-day winemaker is often referred to as an *enologist*.

The U.S. federal regulatory body for wine, the Bureau of Alcohol, Tobacco, and Firearms (ATF), defines wine as “the product of the juice of sound, ripe grapes.” But there are also many famous wines made from unripened grapes and even more made from overripened grapes. More important, the ATF definition overlooks any stipulation for the all-important fermentation process necessary to generate “the product.”

Wine, as the word used in the wine industry, is defined as “the product of fermenting and processing grape juice or must.” Crushed grapes, generally with all or some portion of the stems removed, are known as *must*.

The ATF requires that commercial wines made from any other fruit than grapes must be qualified by fruit type on the label. Wine made from peaches, for example,

must be labeled as “peach wine,” wine made from blackberries as “blackberry wine,” wine made from plums as “plum wine,” and so forth. Wine made from any species or variety of grape can be commercially labeled with the word *wine* standing alone.

Wine is the result of fermenting and processing grapes or some other fruit—sometimes even vegetables. For virtually all white wines, the juice is pressed out prior to fermentation. Most reds are made by fermenting crushed grapes (must) before pressing so that the skins leach red pigments into the new wine. Pink wines, often called “blush wines” nowadays (and also known as “rosé wines”), are generally made by a very limited skin contact during fermentation and then are quickly pressed, so that only a small amount of color is extracted from the grape skins.

There are countless wine “recipes” and grandfathered methods that are treasured by some winemakers. These are encouraged but are not always easily justified by wine logic. The most consistent high-quality wines generally emerge from winemakers who use analysis and logic for making decisions about processing.

Grapes and other fruits vary in sugar, acidity, and flavor constituency each year, requiring adjustments that have a sound reasoning. Wines from previous vintages may have properties that need fine-tuned adjustments in later processing. The enological approach makes good sense for effective recordkeeping, too.

GRAPE AND WINE COMPONENTS

Because sugar and water have no flavor, only about 3% of the total wine composition accounts for flavor and color (see Table 4–1). In other words, the difference between the very finest and poorest of grapes and wines is determined by this small percentage of their constituency.

Table 4–1 shows a rather straightforward conversion of sugar to alcohol during fermentation. The increase of water percentage during fermentation is due to several biochemical pathways. The Embden-Meyerhof reaction diverts some of the sugars to the production of lactic acid, creating water as a byproduct. Yeasts also produce enzymes that convert pyruvic acid to carbon dioxide gas and water.

Pectic Enzymes

Pectins are generally classified as a group of polyuronides consisting of d-galacturonic acid units in long-chain polymers. The most important form of pectin in winemaking is pectinic acid, which is the operative component that reacts with sugar and fruit acids when heated and that forms the gel consistency as temperature is lowered. Pectin is added in fruit processing to make jelly. Arabans, dextrans, galactans, and other gumlike compounds formed during fermentation are also sometimes classified as pectin materials.

Table 4–1 Composition of grapes and natural table wine.

Component compound	Approx. % in grapes	Approx. % in wine
Water	75.0	86.0
Sugars (fructose, glucose, with minor levels of sucrose)	22.0	0.3
Alcohols (ethanol, with trace levels of terpenes, glycerol, and higher alcohols)	0.1	11.2
Organic Acids (tartaric, malic, with minor levels of lactic, succinic, oxalic, etc.)	0.9	0.6
Minerals (potassium, calcium, with minor levels of sodium, magnesium, iron, etc.)	0.5	0.5
Phenols (flavonoids such as color pigments, along with non-flavonoids, such as cinnamic acid and vanillin)	0.3	0.3
Nitrogenous compounds (protein, amino acids, humin, amides, ammonia, etc.)	0.2	0.1
Flavor compounds (esters such as ethyl caproate, ethyl butyrate, etc.)	Trace	Trace
TOTAL	99.0	99.0

Source: Adapted from Amerine, M.A., et al., *The Technology of Wine Making*, 4th ed., Westport CT: AVI, 1967, pp 111–112.

Pectinic acid can be an obstacle to obtaining optimal clarity and yield. Some natural depectinization occurs during fermentation, generally more in reds than in whites due to greater proportions of methyl esterase found in red grapes.

For many white wines, an appropriate treatment with pectic enzyme (pectinase) will serve to break down pectin, increasing free-run juice yield and ease in filtration, while reducing the potential for troublesome fining processes. In the case of blush wines, pectinase treatment at the crusher is essential, as it permits a more balanced extraction of tannin and pigment phenolics during the brief fermentation period on the skins. In most red grape varieties and cultivars, this provides a better ratio of phenolics, resulting in a more attractive pink hue as compared to dull tawny (brownish-orange) color values.

Sugars and Sweetness

Wine publications often use such terms as *reducing sugar*, *residual sweetness*, *invert sugar*, *dextrose*, *glucose*, *fructose*, *fermentable sugar*, and *nonfermentable sugar* in reference to this group of carbohydrates. This terminology and its proper context can sometimes be confusing and, more important, lead to mistakes.

Sugars are categorized in biochemistry as *saccharides*. A *disaccharide* such as sucrose yields two (hence the prefix *di-*) molecules of simple sugars when hydrolyzed. The broad definition of sugar *hydrolysis* is the decomposition of crystals in water. In enology, sucrose hydrolysis results from decomposing one “12-carbon” molecule ($C_{12}H_{22}O_{11}$) into two “6-carbon” molecules. These are often referred to as *hexoses*, expressed in chemical formula as $C_6H_{12}O_6$. Note that the recombination of two hexoses results in the addition of two atoms of hydrogen and one atom of oxygen, or H_2O , which is, of course, the water taken from hydrolysis. Because this cleavage of sucrose into two hexoses is made by invertase enzymes, the transformation is referred to as *inversion*. Sometimes we find liquid sugar being sold as *liquid invert*, which refers to the product composed of water in which 6-carbon sugars are dissolved, usually in a very heavy syruplike concentration.

Grapevine roots store sugars produced from photosynthesis in the sucrose form. As the fruit matures, the sucrose is typically inverted to a 1:1 ratio of glucose and fructose as it is translocated to the grape berries following veraison. Fully ripened grapes may contain up to 1% of uninverted sucrose, which is inverted later by yeast-synthesized invertase during fermentation.

Glucose and fructose are *monosaccharides*—the *mono-* indicating that the 6-carbon sugar molecules cannot be hydrolyzed to a simpler form. These are the sugars of greatest concentration in grapes and are essential substrates (source materials) for the fermentation process in winemaking. The literature often expresses these sugars as *d-glucose* and *d-fructose*, which are prefixes referring to the rotation angle variation with respect to the wavelength of light transmitted: *dextro* (d) the right, and *levo* (l) to the left. Dextrose and d-glucose are identical compounds and are often referred to as corn sugar, grape sugar, or honey sugar, while d-fructose is also known as levulose, or fruit sugar. Both glucose and fructose are chemically symbolized by the same $C_6H_{12}O_6$ structure, so both have a molecular weight of 180.09. The difference is that glucose has a melting point of 146°F and fructose is 105°F.

The definition of *reducing sugars* is less straightforward. Wines fermented to absolute dryness still contain between 0.1% and 0.2% of unfermented sugars. The most common reducing sugars are glucose and fructose, but the precise reason why they do not completely convert during fermentation remains unclear. About one-third of the reducing sugars can be attributed to various *polysaccharides*, which are complex forms of sugar that are unfermentable by yeasts but may be substrates in malolactic fermentation.

Many wines are designed to contain some specific amount of sugar for the wine to finish with some particular degree of sweetness. Traditionally, this *residual sweetness* results from natural grape hexoses. In modern times, the addition of cane or beet sugar, both of which are sucrose, is permitted. ATF regulations in 27 CFR, Part 4, Section 24.179, allow natural grape wines not more than 17% total solids (21% for natural fruit wines) by weight, provided the alcohol content

does not exceed 14% by volume. There are additional provisions for specially sweetened natural wines.

The language of sugar and dissolved solids in wine production is expressed in terms of *Brix*, *Balling*, and *extract*. Ripe grapes should arrive at the winery at about 22°Brix, which is roughly equivalent to percent total solids by weight. Native American *Vitis labrusca* grapes may achieve considerably less than 22°Brix. Late-harvest *Vitis vinifera* and French-American hybrid grapes may often reach more than 35°Brix. While nearly all dissolved solids in grape juice or must are sugar, there are also some solids attributed to acids, minerals, and phenols.

Balling is an indicator of wine density or weight on the human palate, often referred to in sensory terms as *mouth-feel*. An increase in dissolved solids will increase the weight of wine, as dissolved solids are heavier than water. As fermentation progresses, the increasing alcohol content decreases the weight of wine, as ethyl alcohol is lighter than water. The combination of solids and alcohol taken together is expressed in terms of Balling. A dry wine with virtually no dissolved solids would have a Balling of less than 0, which is the Balling of water.

Brix and *Balling* are terms often misunderstood and thus often misused, perhaps because they can both be measured using a simple hydrometer in analysis (see Appendix B). The difference between them is best exemplified by considering that it would be impossible to have a negative Brix.

Extract is virtually the same as *Brix* except that it refers to total dissolved solids in wine.

Late-Harvest Grapes

Despite the lack of wide market appeal, vintners are responding to the connoisseur niche by offering an increasing number of late-harvest wines in U.S. markets. Some of these wines are made from grapes that have been naturally raisinized by excessive sun exposure in the vineyard. Sluggish fermentation rates in higher Brix musts resulting from late-harvest grapes can be attributed to an increase in water activity, which is an osmotic force taking up water that places a dehydration strain called *zytorrhysis* on yeast cells. An increase from 20 to 50°Brix can reduce yeast cell water volume by about 50%. The need to minimize Brix levels, and thereby reduce the impact of this phenomenon, has made vintners focus more on identifying a rather precise amount of ethanol, residual sweetness, and acid balance to meet perceived consumer demand.

In conditions of moderate temperatures and higher humidity, *Botrytis cinerea*, the “noble mold,” can appear as fuzzy gray spots on grape skins during the harvest season. *Botrytis* spores bore through the skins and create a porosity from which berry water can evaporate. This serves to concentrate retained sugars, acidity, glycerol, and flavors. The infection process actually reduces some sugar and acidity (increases glycerol and mucic acid), but the net effect is a concentration. Botrytized

fruit flavors undergo change due to the formation of ethyl esters of hydroxy-, keto-, and dicarboxylic acids. Yeast enzyme synthesis and activity rates determine how these acids are metabolized into the ultimate ester flavor profile. Some natural fruit flavors, such as the aromatic monoterpenes, geraniol, and linalool, common to the Johannisberg Riesling, Gewurztraminer, and Muscat varieties, may be destroyed by *Botrytis* mold cells.

The mold can also form laccase, an enzyme capable of oxidizing important phenols, including anthocyanins. This accounts for the golden-brown colors in most botrytized wines. *Botrytis* infection can be quantified using a laccase assay available in laboratory kits. Samples are initially treated with polyvinylpyrrolidone (PVPP) to reduce polyphenol interference and then are mixed in a spectrophotometer cell with a syringaldazine-ethanol solution, along with a buffer. The change in absorbance is recorded over several minutes and calculated by formula to provide laccase activity measured in laccase units per milliliter.

The *Botrytis* fungus can reduce up to half of the protein, amino acid, and free ammonium nitrogen (FAN) constituency of grapes. Constrained FAN can induce deamination activity by yeasts upon protein and amino acid constituents, causing the development of unpleasant hydrocarbon (kerosenelike) flavors and other maladies of flavor. FAN is a more essential element in active yeast growth, and deficiencies can be supplemented with diammonium phosphate and other food-grade sources.

A more serious problem is a reduction of thiamine and other B complexes necessary for decarboxylation in the pyruvic acid cycle and the synthesis of various keto acids mentioned above. These vitamins can also be supplemented, often as components in concert with FAN in commercially available products. Yet another concern is the growing evidence that indicates that *Botrytis* mold can produce trace amounts of antibiotics that may be toxic to yeasts.

Botrytis infection produces polygalacturonase, which hydrolyzes pectins into polysaccharides. Mucic acid development can react with calcium to form slow-developing precipitates. Both are hurdles to effective fining, evidenced by the formation of haziness in the wines after bottling. Proper use of glucose oxidase enzymes followed by appropriate applications of kieselsol fining (described below) can provide additional help in overcoming problem clarifications.

Moderation of these perils and pitfalls remains best controlled by close monitoring of *Botrytis* development in the vineyard and processing techniques in the cellar.

Acids, Acidity, and pH

The principal acids involved in winemaking are tartaric, malic, lactic, and acetic. The sum total of these and other minor acids determines the amount of tartness a wine will deliver on the palate. While some people can detect hints of apple-like flavor in malic acid and cheeselike flavor in lactic acid, the aggregate flavor contribution by acids in wine is minuscule.

Total acidity (TA) is a measurement routinely made by analysis in winery laboratories to quantify tartness in juice, must, or wine. Levels of less than .500 g/100 mL are generally considered bland. TA levels exceeding .800 g/100mL are usually ranked as sharp. Sweetness levels tend to mask TA, and vice versa. The relativity of sweetness and TA in wines is the basis for determining gustatory balance in sensory evaluation.

The measurement of the active, or effective, acid “strength” in juice, must, or wine is expressed in terms of pH. The determination of pH is made by measuring the hydrogen ion concentration in a given solution. On a scale of 0 to 14, a pH of 7 would be neutral, as is pure water. From 7 to 0, the hydrogen ion concentration increases, and thus the solution becomes increasingly acidic. From 7 to 14, the hydroxyl ion concentration increases, and thus the solution become increasingly basic. These progressions are not linear but logarithmic. As a consequence, each gradient from 7 toward either 0 or 14 becomes more intense. For example, a change in pH from 3.40 to 3.30 is a major increase in effective acidity. This, however, is not nearly as large an increase logarithmically as the increase from 3.30 to 3.20.

The determination of pH is particularly important during the harvest season, as it can be used as a standard for grape ripeness. Tartaric acid is found in the highest concentration in grapes and is a relatively “strong” acid, resulting in lower pH ranges. As grapes ripen, the tartaric acid begins to diminish and the pH increases. Sufficient pH data collected from vineyards over several vintage seasons may justify setting standards from which to predetermine optimal harvest time.

Lactic acid is a product of malolactic fermentation, a bacterial fermentation in which malic acid is transformed to lactic acid, diacetyl, and carbon dioxide gas. This is usually considered beneficial to red wines and some whites, as the “buttery” flavor of the diacetyl adds tone and complexity to the overall flavor profile. Inoculations of cultured malolactic bacteria are commonplace in the wine industry.

Acetic acid is the vinegar flavor result of oxidized alcohol by an undesired bacterial fermentation. It almost always results from poor winemaking procedures.

Citric acid is found in grapes only in trace amounts but is an important additive used by winemakers. It is not added to unfermented juice or must, as it can slow fermentation by inhibiting enzymatic functions in the glycolysis pathway. Citric acid is active in metal chelation, aiding in reducing the probability of iron and/or copper haziness. Being a tricarboxylic acid in the Krebs cycle, it also has an overall stabilizing effect. Consequently, it is usually added to wines that are nearing the end of processing, just before bottling.

Analytical procedures for TA, volatile acidity (VA), and pH are found in Appendix B.

Oxygen and Oxidation

While musts that have not been treated with sulfur dioxide contain high levels of polyphenoloxidase and resultant pigmentation oxidation, the delay of sulfiting

ultimately results in wines containing reduced total phenolics and improved sensory quality. Appropriate additions of ascorbic acid can aid in minimizing persistent browning due to oxidation.

Free oxygen is rapidly utilized by cultured species of *Saccharomyces* wine yeasts (facultative anaerobes) at the outset of logarithmic growth, with full depletion of oxygen resulting in anaerobic fermentation. This condition results in the redistribution of certain lipid and sterol compounds essential for yeast membrane construction during cell division and for membrane mechanics during the glycolytic functions of fermentation. Some winemakers actually supplement oxygen by agitating white wine fermentations when activity commences to diminish, more often in combination with FAN supplements. Unfortunately, this is a treatment frequently overdone that results in an excessive level of oxygen.

Commercial strains of wine yeasts fully adapt to continuing fermentation without any available oxygen in the must. This anaerobic condition results in the redistribution of certain lipid and sterol compounds essential to yeast membrane construction during cell division, as well as membrane mechanics during the glycolysis functions that take place during fermentation. Under normal conditions with minimal oxygen present, these redistribution dynamics should cause no alarm or unusual difficulty.

With excessive oxygen present, however, the new wine is in jeopardy of color browning and flavor degradation. The greater a wine's buffering capacity, or its content of oxidizable compounds, the greater its aging potential. Wine aging is, thus, a controlled process of oxidation. For example, a young wine racked (decanted from lees sediment) promptly after fermentation to cool stainless-steel tanks and then clarified and coarse filtered would be much less susceptible to oxidation than a wine allowed to assimilate yeast and fruit pulp compounds by extended lees contact. Wood storage containers contribute nonflavonoid phenols that are oxidizable. Higher storage temperatures (more than 55°F) and higher pH levels are both conducive to higher oxidation rates.

Consequently, wines containing higher levels of yeast autolysis compounds (wine fermented *sur lies*, or treated with significant additions of yeast hulls) and/or containing an appropriate level of free sulfur dioxide are chemically equipped to bind with greater quantities of oxygen. The result is often measured in wines having less browning, with more complexity and structure in the flavor profile. Higher levels of phenols, as found in long-term macerated musts, also influence a higher buffering capacity and provide the major reason why red wines, particularly at lower pH levels, generally take far longer to age to maturity.

Minimizing oxygen exposure includes regularly "topping off" storage vessels to replace ullage. Cold-temperature short-term aging in stainless-steel tankage, nitrogen sparging, and adequate maintenance of free sulfur dioxide levels are essential for lower oxygen risk. Attention and care in racking, fining, filtering, and other transfer operations can reduce splashing and other mechanical sources of oxygen. Tanks should be sparged with carbon dioxide or nitrogen gas. Close monitoring

of temperatures and sulfur dioxide levels is also part of the quality assurance routine.

Sulfur Dioxide

The use of sulfur in wine production has several centuries of history, yet it remains one of the most controversial of wine treatments. It is also one of the most misunderstood, and its capabilities are often miscalculated.

The greatest variables exist in definition and communication. The term *sulfur* is used to describe the rhombic yellow sulfur solid employed in vineyard sprays and sulfur sticks and is the elemental form from which hydrogen sulfide and sulfur dioxide are derived.

Hydrogen sulfide (H_2S) has the odor of rotten eggs, which often results in wines made from grapes having been dusted with sulfur in the vineyard. There is evidence that excessively high inoculations with an overabundance of yeast cells can also contribute to hydrogen sulfide generation. Hydrogen sulfide can also arise from sluggish, nitrogen-deficient and/or extended low-temperature fermentations, or any other condition that may inhibit healthy yeast growth. Yet another source of hydrogen sulfide is from lees substrates, which is a common occurrence in poorly managed *sur lies* regimens.

Sulfur dioxide is a gas having a very prickly sharp pungency in the nose. Heavy doses of sulfur dioxide gas leave an unpleasant powdery residue on the palate.

This gaseous form of sulfur dioxide is generally used by larger wineries where greater quantities are required. In most smaller wineries sulfur dioxide is sourced from meta-bisulfite anions. This exists in two general forms. The bisulfite form, or *free sulfur dioxide*, has a rather “burnt match” odor and is the form that has the desired antimicrobial and antioxidant properties. The bound sulfur dioxide form includes bisulfites bound to aldehydes, pectic compounds, various phenol complexes, certain proteins, and other molecular sites. Bound sulfur dioxide has comparatively little distinguishable flavor. The free and bound forms combine to make *total sulfur dioxide*, which is currently regulated by the ATF to a limit of 350 mg/L or 350 ppm, as found in 27 CFR, Part 4, Section 4.22 (b)(1).

It remains rather commonplace for vintners to add 30 to 60 mg/L of free sulfur dioxide to grapes dumped in the crusher, or early on in the resulting must. The traditional rationale for sulfite additions at the crush is that this dosage inhibits or kills wild yeasts and bacteria attached to the bloom (waxy coating) of grape skins and also provides some measure of protection from oxidation. There is growing evidence to indicate that sulfur dioxide additions at the crusher may actually increase browning in resulting white wines.

While “killer” yeasts are not new to enology, their role in reducing or eliminating sulfur dioxide at the crusher is an application that is relatively recent. Killer yeasts are species that kill sensitive members of their own species and frequently those

of other species as well. They function by secreting a plasmid-coded protein toxin that binds with 1,6 beta D-glucan receptor components in the cell walls of sensitive strains. This toxin interacts directly with protein components of the cell membrane and, in turn, disrupts the normal state of cell activity. Recent research suggests that killer yeasts themselves are immune due to a precursor protein that functions as an inhibitor of toxin in its cell membrane metabolic processes.

Killer yeasts are particularly effective in reducing infection from *Brettanomyces* yeasts. This spoilage microorganism is most often identified with red wine spoilage in the form of acetic, isobutyric, and isovaleric acids, which emerge as "barnyard" flavors. *Brettanomyces* can resist mid-range dosages of free sulfur dioxide, is insensitive to sorbic acid, and may go unnoticed until growth has become widespread. Consequently, musts undosed with sulfur dioxide require immediate inoculation with killer yeast cultures to achieve maximal protection. Some enologists inoculate at the crusher hopper, taking advantage of heavy oxygen demand by the yeasts to lower the oxidation potential.

It has been reported that the killer system occurs in some natural yeast strains. Musts inoculated with a sensitive cultured yeast strain can be dominated by wild killer strain populations, causing stuck fermentations. The resulting wine can suffer from reduced ethanol yield, high VA, formation of hydrogen sulfide, and contaminant flavors caused by acetaldehyde, fusel oils, and lactic acid.

Vintners continue to evaluate a growing availability of killer yeast strains. Popular strains of killer yeasts are Champagne 111 and Montrachet 1107, the latter more popularly known as Prisse de Mousse. These and other strains of cultured killer yeasts are frequently used by progressive winemakers.

Conversely, some vintners prefer to allow natural yeasts to conduct the fermentation in order to achieve enhanced flavor complexity. This practice has some perils and pitfalls that should be avoided, particularly by new vintners.

Another argument against sulfur dioxide treatment at the crusher is that increased astringent phenols are extracted from the must. On the other hand, some traditional red wine vintners actually prefer this added dimension of phenolic complexity. Further, there is evidence that an absence of sulfur dioxide at the crusher permits yeast-inhibiting bacteria to flourish, which, in turn, contributes to stuck fermentations.

In the case of clean, freshly harvested grapes, it is becomingly increasingly common to find the initial dose of sulfur dioxide being made during the first racking. At this time, carbon dioxide has significantly diminished and sulfur dioxide is needed to achieve an appropriate degree of bacterial and oxidation protection. Little further complexity is usually desired in the case of most white and blush wines. Reds may require considerable malolactic bacterial activity and taming of harsh tannins via oxidation. Consequently, minimal sulfur dioxide treatment is essential.

The proper calculation of sulfur dioxide treatments depends heavily upon the pH of the wine being treated. The active portion of free sulfur dioxide is called

molecular sulfur dioxide and is much more active at lower pH levels. Table B-5 in Appendix B lists the approximate effective sulfur dioxide concentrations at different wine pH levels.

Wines with a pH of 3.50, for example, require about twice the sulfur dioxide addition to achieve the same preservative effect as wines with a pH of 3.20. For dry table wines at a pH of 3.25, it is typical to add 20 to 30 mg/L of free sulfur dioxide at the first racking and another addition just before bottling, raising the level up to 30 to 40 mg/L. On the other hand, some winemakers may have just cause to feel a need for upwards of 80 mg/L of free sulfur dioxide at the time of bottling high-pH wines, particularly those having a history embattled with spoilage organisms or other suspect problems. The question always begged for distressed wines is whether they should be bottled at all.

Table wines having residual fermentable sugar should be dosed to levels of free sulfur dioxide about 50% greater than for dry wines, in tandem with a sufficient treatment of sorbic acid, which has an ATF maximum of 300 mg/L, as provided in 27 CFR, Part 4, Section 24.246. Lower free sulfur dioxide levels result in wines that have greater fruit character and less harshness on the palate.

How potassium meta-sulfite (KMS) is added is also somewhat controversial. One rather archaic method is to dissolve KMS in a small amount of water first. Adding KMS directly to the wine allows the lower pH in the wine to disassociate sulfur dioxide gas, which conserves the amount of bisulfite ions, or free sulfur dioxide, lost to evaporation.

Phenols, Phenolics, and Polyphenols

As mentioned above, table wines are composed of about 85% water and 12% ethyl alcohol, leaving only about 3% to account for color, flavor, and body. A major group of compounds found in this small but essential portion of wine are *phenols*, sometimes referred to as *phenolic substances* and/or *tannins*. Technically, each is a different term. The chemistry of these structures is very complex and diverse, each having important relationships in winemaking.

Phenols include the fundamental form of phenic acid, phenylic acid, and oxybenzene. A phenol very common to wine is hydroxybenzene. Phenolics are more complex structures formed from primary phenols, as in ripening grapes and maturing wines. Tannins are condensed giant phenolic polymers that can be precipitated by proteins such as the mucoproteins on the human palate, accounting for the “puckery” astringent effect when high-tannin wines are tasted. Yet another family of phenolics are extracted from aging wines in barrels and vary according to the species of oak selected, the intensity of “toasting” given during construction, and the ratio of time/temperature employed in the cellar.

The entire family of phenols, phenolics, and tannins can be referred to as *polyphenols*. On average, about 65% of grape polyphenols are found in the seeds, while another 22% are in the stems, 12% in the skins, while only 1% is in the

pulp. The quality and quantity of polyphenols in grapes is derived from species and variety, vineyard locale, and cellar management.

Polyphenols are categorized into two major groups called *flavonoids* and *non-flavonoids*. Flavonoids are large-polymer molecules involved with wine color and tannins and with the highly publicized *resveratrol*, which is thought to be effective in reducing low-density cholesterol in the human body. Nonflavonoids are generally smaller molecules that are primarily associated with oak flavor extracts such as cinnamic acid and vanillic acid.

Color

The pigments in grape skins, which eventually determine wine color, are phenolic flavonoids called *anthocyanins*. These are a group of five closely related anthocyanidin compounds constructed in combination with sugar residues called *glucosides*. Monoglucosides have a single sugar residue, and diglucosides have two. Anthocyanidins decompose rather rapidly without a glucoside component. Consequently, the stability of wine color is largely dependent upon anthocyanin glucoside structure.

It is principally the monoglucoside form that is associated with wines made from *Vitis vinifera* grapes, generally more stable than the diglucoside form identified with the red Muscadine varieties belonging to the *Vitis rotundifolia* species. Varying density of purple hues is typical to the predominance of the monoglucoside anthocyanin *malvidin* in Cabernet Sauvignon, Merlot, and other *vinifera*. Some *Vitis labrusca* types, such as Concord, also possess a significant level of monoglucoside malvidin, but in concert with the other four monoglucoside anthocyanins—*cyanidin*, *delphinidin*, *peonidin*, and *petunidin*. Interspecific hybrid grapes such as the French-American cultivars each have a phenolic color profile inherited from their breeding lines.

Higher total acidity and lower pH values found in slightly immature grapes are generally associated with superior color hue and stability. These factors, however, may combine to create a specific color hue, and the intensity of that hue is primarily a function of viticultural response. Vines stressed in cooler climates and poorer soils can be expected to yield grape berries of smaller diameter. This leads to the famous “struggling vine” theory, which postulates that comparatively smaller berries resulting from challenging vineyard environs provide a greater ratio of skin surface area per unit of resulting wine and thus greater intensity of color and flavor.

Virtually all of the anthocyanin pigments are extracted from grape skins by the time a properly conducted on-skin fermentation has reached 5.0 Balling. No enhancement of color is gained by prolonged skin contact during maceration.

Color can be enhanced, however, by employing the *saignee* system of must concentration. This involves separating some portion of the free-run juice from the freshly crushed red grapes and processing it as white or blush wine. The remaining must is then either fermented by itself or added to other crushed grapes of the

same type to increase the availability for increased pigment extraction. Exacting control over the extent of extraction is necessary to avoid generating distorted flavor profiles and saturated concentrations of phenolics, as well as excessive astringency, bitterness, and eventual color precipitates.

A more simple method to achieve red wine color enhancement is by simple blending of a *teinturier*, a dense, inky wine made from Salvador, Colobel, and other heavily pigmented grape varieties. The wine extracted from the highest pressure can also be a ready source of supplemental pigmentation. Vintners differ in their approach to this: some feel that blending strays from the ideals of varietal purity, while others point out that many classic reds, such as Bordeaux and Rhones, are blends among different varieties anyway.

New products made possible by membrane separation processing techniques have resulted in the isolation of concentrated pigments in retentate form. While this may carry similar concerns of varietal purity, it does so on a far smaller scale, as comparatively little pigment is required to achieve favorable results.

Nitrogenous Compounds: Proteins

Amino acids, peptides, and proteins are all nitrogenous compounds derived from elemental nitrogen, as are amides, amines, and peptones. While the aggregate of these make up less than 1% of grape composition, they are nevertheless essential in making wine.

Yeasts have no known direct requirement for amino acids. However, fermentation requires amino acids as a catalyst in synthesizing nitrogen into the free ammonium state that is required by yeasts.

There are more than 25 different amino acids in most wines. The precise level of each depends upon grape variety, vineyard locale, and climate, among other factors. The amino acids can be condensed into peptides and proteins; this creates the common problem of protein instability in white table wines.

Proteins are complex nitrogenous compounds consisting of up to 20 different amino acids bound together by peptide linkages, forming polymer chains that can achieve molecular weights exceeding 150,000. Heavier, less soluble proteins are formed in higher temperature (over 75°F) fermentation and/or storage. Higher alcohol levels also aggravate protein solubility. The greater levels of phenolic compounds extracted in red table wines also precipitate protein complexes. To the extent that phenolics in reds may be excessive, proteins, in the form of egg white albumin, casein, and gelatin, are commonly used in finings.

Higher temperatures, higher alcohol levels, and greater concentrations of phenolics are not generally acceptable in delicate white table wine flavor profiles. Consequently, white wine fining regimens must be very carefully determined and carried out to achieve clarity at minimal expense of flavor loss. Typically, gelatin/kieselsol or bentonite, or Sparkolloid, is employed for white wine clarification.

MALOLACTIC FERMENTATION

The principal effect of malolactic fermentation is a reduction in total acidity, along with the development of a buttery flavor known as diacetyl. As a rule, malolactic fermentation is desired in more complex table wines but undesired in lighter types that express greater fruit flavor profiles. Most cultures of malolactic bacteria are found in the *Leuconostoc* and *Lactobacillus* species, with generally higher levels of diacetyl resulting from *Pediococcus* strains.

Along with pH levels higher than 3.25 and virtually no free sulfur dioxide content, cultured malolactic bacteria typically require supplemental B-complex vitamins to adequately conduct fermentation. Yeast cytoplasm extracts refined from autolysates (dead yeast cells) are a natural source of these substrates and are the rationale upon which the prolonged exposure to dead yeast cells following barrel fermentation of white wines (*sur lies*) is based. Similar substrates are extracted by extended skin contact time (maceration) for reds. Alternatively, ATF regulations found in 27 CFR, Part 4, Section 24.246, provide for additions of yeast cytoplasm “to facilitate fermentation of juice/wine” in amounts not exceeding 3 pounds per 1,000 gallons.

GRAPE FLAVORS

It is imperative that a winemaker be familiar with the nature of wine flavors. The literature often refers to chemical structures of flavor, and an increasing amount of casual discussion is focused on flavor chemistry.

One of the most advanced wine flavor sources is Jean Lenoir, who manufactures the Le Nez Du Vin (“The Nose of Wine”) aroma identification kits, which are highly recommended for the neophyte and even for more advanced enophiles (Figure 4–1).

In many white grape varieties we find what Lenoir calls *primary floral aromas*. Among these are acacia, grapefruit, honeysuckle, linden, peach, pear, rose, peony, and violets. These flavor values are most often found in grapes that have been harvested slightly before maturity. *Primary fruit aromas* form during full maturation.

Many of the primary floral and fruit aromas exist in the form of higher terpene alcohols such as citronellol, linalol, and geraniol. These are commonly found in Johannisberg Riesling, Gewurztraminer, Vidal Blanc, Vignoles, and most of the Muscat varieties. Other aromas are identified in the form of esters. Among these are the isoamyl acetate, associated with bananas; ethyl propionate, associated with apples; and the celebrated methyl anthranilate, found in the native American *Vitis labrusca* species, exemplified by the cultivars Niagara and Concord.

Primary fruit aromas in red grapes are often identified as cherry, black currant or cassis, strawberry, raspberry, and plum; these are often used as flavor descriptives in evaluation of wines made from Chambourcin, Gamay, and Pinot Noir. Many of

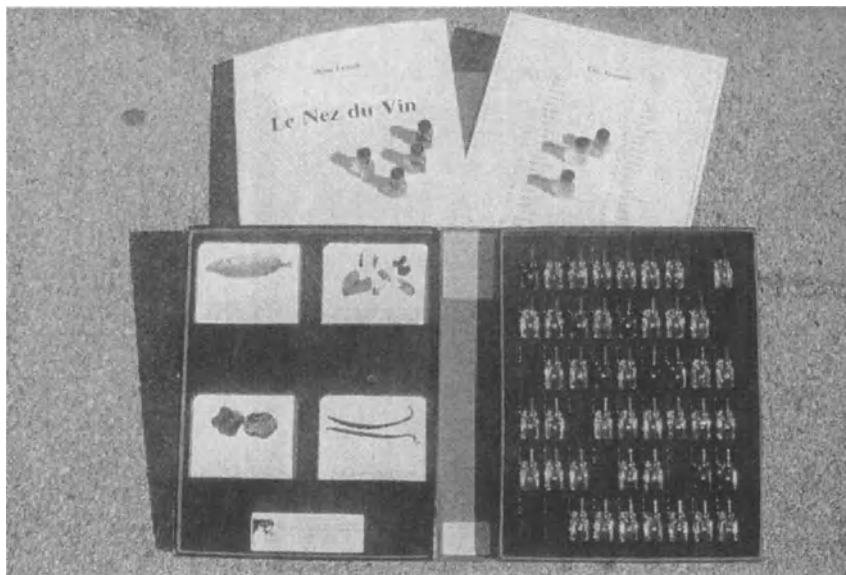


Fig. 4-1 Le Nez Du Vin. Photo by David Ferguson.

these are found in ester forms. Strawberry flavor is attributed principally to several butyrate compounds, and raspberry is identified with ethyl caproate.

Heavy floral and fruit aromas extracted from mature or slightly overripe grapes require particularly close attention to avoid oxidation and microbial infection. This involves culling out of bad clusters and berries (what the French call *triage*), fermentation at 50 to 55°F in sanitized stainless steel, rackings sparged with nitrogen, absolutely full tankage for storage, minimal finings, cold storage stabilization, and prompt bottling.

Another flavor group is called *primary vegetal aromas*. Good examples of these are anise, green pepper, tobacco leaf, and mint. This group is often structured in the form of carbonyl compounds, such as the methoxy-isobutyl-pyrazine that is the principal aroma component associated with bell peppers. These compounds can be very complex, such as the aroma associated with Cabernet Sauvignon, 2-methoxy-3-isobutylpyrazine. Vegetal aromas can also exist as phenols, such as cinnamic acid, found in grass and tobacco. This character is most often desired in very heavy red wines such as those traditionally made from Cabernet Sauvignon, Chancellor, and Merlot.

Yet another group, *primary wood aromas*, is exemplified by briar, cedar, hazelnut, resin, oak, and eucalyptus. These generally exist in the form of phenolic compounds such as vanillin, which is the aroma component associated with oak.

Wood aromas can also be carbonyl compounds such as benzaldehyde, the aroma associated with bitter almond.

As mentioned several times above, a vinegar flavor in wine is generally associated with the decomposition of ethyl alcohol to acetic acid by various species of spoilage bacteria. Ethyl acetate, described as a “paint-thinner” odor, is often associated with acetic acid formation. Ethyl acetate can, however, metabolize independently from action by certain species of yeast. Wine bacteria and yeasts are discussed in Chapter 3.

Also mentioned several times previously is an earthy character, sometimes described as “barnyard,” that is attributed to fermentation products resulting from the action of *Brettanomyces* yeast. Some wine judges call this condition simply as “Brett.” The occurrence of Brett in white wines is almost always considered a flaw, although judges often differ in opinion as to whether a little of this flavor contributes positively to the complexity in some reds.

Another set of spoilage flavors found in wine are due to one or another form of sulfur degradation. These conditions usually arise from elemental sulfur dusted on vines, essential in controlling various types of mildew and molds in vineyards. Sulfur is also an important natural constituent in the synthesis of essential proteins in grapevines. Whether dusted or natural, elemental sulfur on or in grapes can be transformed by both cultured and wild yeasts into hydrogen sulfide, or a “rotten eggs” character, in wine.

Sulfide compounds can be further degraded by bacterial action into mercaptan compounds that emerge in various types of complex compounds. At best, this is manifested in an “asparagus,” “cabbage,” or “green bean” flavor. As may be expected, this can sometimes be confused with primary vegetal flavors. At worst, mercaptan spoilage is expressed in a “rubbery” aroma, a foul “wet dog” smell, or, worst of all, a “skunky” character.

Moldy-musty flavors arise from either moldy grapes, moldy barrels, or some other exposure of the wine to a source of mold. Molds will not grow directly in wine due to its alcohol content.

Flavors that seem “cooked” or “brown apple” in character are often due to oxidation of wine resulting from extended exposure to air and/or higher temperatures (over 65°F) in processing and storage. These flavors can also be categorized with “prune” flavors that come from grapes that are over-ripened and/or have decayed. Some wines, such as Madeira, are purposely heavily oxidized to produce high levels of aldehyde compounds that result in a pronounced “nutty-caramel” flavor.

WINE BLENDING

History indicates that the blending of wines has gone on since ancient times. At first it may have been more wine “mixing” than blending: mixing wine with seawater,

honey, tree resins, and other additions to dilute spoiled wine so it didn't taste quite so bad. The salt in the water was also an aid in preservation.

In one way or another, most of the great wines in the world are the product of blending. Village-labeled wines are blended from small vineyards in Burgundy each vintage. Large Bordeaux estates blend wines from vineyard plot by vineyard plot and free-run with press-wine to achieve the perfect chateau style each year. In Champagne, blending is done on a large scale with wines from various vineyards from different villages, and sometimes across several vintages, in *assemblage* of the "house" cuvee.

There is no magic in blending. Successful blending, in its purest form, is synergistic, with the result being greater than the sum of its parts. Consider the following blending postulates:

1. Blending reduces the character of individual components.
2. Blending increases the complexity of the resulting product.
3. Blending a faulted wine with a sound wine results in a faulted wine.
4. Blending two stable wines can result in an unstable wine.
5. *Always* make a lab blend first.

Blend Expression Rationale

It is the creative ability of each individual winemaker that ultimately determines blending success. The form and character of each wine expresses the way the product has been conceived to parallel market demand. There are, of course, blends that are "monuments" to various egoist expressions. Often these motives are not shared by consumers, and so such wines may find difficulty in the marketplace.

Some fundamental blending parameters seem to find general acceptance among wine enthusiasts:

- Dry Whites
 1. greenish-straw hues
 2. lighter color densities
 3. lighter body
 4. crisp acidity
 5. moderate fruit flavor
 6. simplicity
- Sweet Whites
 1. golden-amber hues
 2. heavier color densities
 3. moderate-heavy body
 4. sweetness balanced with acidity
 5. complexity
- Blush
 1. pink (not orange) hues
 2. lighter color densities

3. lighter body
 4. delicate acidity and sweetness
 5. light fruit flavor
 6. simplicity
- Dry Reds
 1. crimson-tawny hues
 2. color density correlated with body
 3. tannins correlated with aging design
 4. moderate acidity
 5. broad complexity
 - Sweet Reds
 1. scarlet-purple hues
 2. dense color intensity
 3. heavy body
 4. sweetness balanced with acidity
 5. heavy fruit flavor
 6. simplicity

Key ATF Blending Regulations

Of major concern is that adequate supplemental records exist in the winery to permit ATF inspectors to readily account for the varieties, geographic origins, vintages, quantities, and other appropriate information relating to each blend made.

Blending grape wine with wines made from any other fruit or vegetable, and/or flavoring components, requires the application of Form ATF F 5120.29, Formula and Process for Wine, with ATF approval to make such special natural wines (Exhibit 4–1). Rules are found in 27 CFR, Part 4, Sections 24.195 to 24.198.

Key ATF Blending Regulations for Labeling

The following are regulations concerning grape type designations (27 CFR, Part 4, Section 4.23a):

- (a) *General.* The names of one or more grape varieties may be used as the type designation of a grape wine only if the wine is also labeled with an appellation of origin, as defined in Section 4.25a.
- (b) *One variety.* The name of a single grape variety may be used as the type designation if not less than 75% of the wine is derived from grapes of that variety, the entire 75% of which was grown in the labeled appellation of origin area.
- (c) *Exceptions.*
 - (1) Wine made from any *Vitis labrusca* [native American] variety (exclusive of hybrids with *Labrusca* parentage) may be labeled with the variety name if:
 - (i) Not less than 51% of the wine is derived from grapes of that variety;
 - (ii) the statement “contains not less than 51% (name of variety)” is shown on the brand label, back label or a separate strip label (except that this

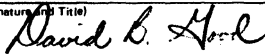
Exhibit 4-1 A Completed Form ATF F 5120.29.

FORM APPROVED: OMB NO. 1512-0059 (2/29/91)

DEPARTMENT OF THE TREASURY - BUREAU OF ALCOHOL, TOBACCO AND FIREARMS

FORMULA AND PROCESS FOR WINE

(See Instructions and Conditions on Reverse)

1. NAME, ADDRESS AND WINERY NUMBER OF PROPRIETOR		2. CLASS AND TYPE OF PRODUCT	
WONDERFUL WORLD WINERY 1234 PARADISE WAY PLEASANTVILLE, CA 99999 U.S. BONDED WINERY NO. 9999		OTHER THAN STANDARD CRANBERRY FLAVORED WINE	
		3. ALCOHOL CONTENT OF FINISHED PRODUCT (range may be shown)	
		11 - 13 PERCENT BY VOLUME	
		4. FORMULAS SUPERSEDED:	
		WINERY NUMBER	FORMULA NUMBER
5. FORMULA (List ingredients quantitatively. If more space is needed, use reverse or separate sheet)		6. PROCESS OF PRODUCTION (If more space is needed, use reverse or separate sheet)	
DRY WHITE WINE UNDER 14 PERCENT ALCOHOL BY VOLUME 80-90 GALS DRY RED WINE UNDER 14 PERCENT ALCOHOL BY VOLUME 10-20 GALS CANE OR BEET SUGAR 20-30 LBS XYZ CORP. CRANBERRY NATURAL FRUIT ESSENCE NO. AB-12 1-2 GALS OR ABC CORP. CRANBERRY NATURAL FRUIT ESSENCE NO. YZ-22 1-2 GALS POTASSIUM META-BISULFITE 1-3 OZ		THE FOLLOWING FORMULA IS SUBMITTED IN ACCORDANCE WITH 27 CFR, SUBPART J, SECTION 24.218: TO EACH 100-GALLON BLEND OF BASE WINE BLENDED IN MIXING TANK NO. 1 OR 2, THE SUGAR AND ESSENCES WILL BE ADDED AND THOROUGHLY MIXED. AFTER ANALYSIS, POTASSIUM META-BISULFITE WILL BE ADDED AS A PRESERVATIVE AND THE RESULTING WINE WILL BE ANALYZED AGAIN, THE WINE WILL THEN BE FILTERED AND BOTTLED.	
7. DATE	8. PROPRIETOR	8a. BY (Signature and Title)	
JAN 24, 1997	BETTY B. GOOD AND DAVID B. GOOD	 DAVID B. GOOD, WINEMAKER	

FOR ATF USE ONLY (Items 9, 10 and 11)

9.		LABELING (finished product only)	
☐ Approved subject to the provisions of 27 CFR _____. ☐ Natural and Artificial flavor, predominantly natural for ATF labeling (Contains not more than 0.1 percent top note material.) ☐ Contains more than 0.392 gram CO ₂ per 100 ml of wine.		☐ The designation of the product must include a truthful and adequate statement of composition such as: _____ ☐ The finished product label must indicate the use of: ☐ caramel ☐ certified color (other than FD&C Yellow #5) ☐ artificial flavor ☐ FD&C Yellow #5 ☐ _____	
10. DATE	11. APPROVED - ATF SPECIALIST FOR THE DIRECTOR, BUREAU OF ALCOHOL, TOBACCO AND FIREARMS	12. FORMULA NUMBER	

ATF F 5120.29 (6-90) REPLACES 698 - Supplemental which is obsolete.

Exhibit 4-1 (continued).**5. FORMULA (Continued)****6. PROCESS OF PRODUCTION (Continued)****INSTRUCTIONS****General Instructions**

The proprietor shall submit a separate ATF Form 5120.29 for each formula covering the production of special natural wine, agricultural wine, effervescent wine in which more than 10 percent of the volume is finishing dosage, essences and extracts produced on bonded wine premises, and other (than standard) wine except for distilling material and vinegar stock. The proprietor shall forward the completed Form 5120.29, in triplicate (Quintuplicate for applicable wine products produced in Puerto Rico), to the Product Compliance Branch, Bureau of Alcohol, Tobacco and Firearms, Washington, D.C., 20226. Production may commence upon the receipt by the proprietor of an approved formula on Form 5120.29.

Specific Instructions

Item 1. Enter the name, complete address and winery number of the applicant.

Item 2. The class and type shall conform to one of the class and type designations in the regulations issued under the Federal Alcohol Administration Act. Wine products containing less than 7 percent alcohol by volume shall be designated in accordance with the labeling regulations of the Federal Food, Drug and Cosmetic Act. Do not show a brand or fanciful name in item 2. Nonbeverage wine will be designated as such, since products are labeled under the Federal Food, Drug and Cosmetic Act.

Item 3. If the product is to be bottled at more than one alcohol content within the same tax rate, state the alcoholic content as a range to include all alcoholic contents at which the product will be bottled.

Item 4. Enter the winery number and formula number for each formula which is superseded.

Item 5.

(a) Specify the kind and quantity of each and every material or ingredient to be used in the formulation of a batch of the product, e.g., 100 gallons, 1000 gallons, etc.

(b) Identify flavoring or blending materials by the name of the flavor or blender, name of the flavor or blender manufacturer, the manufacturer's product number (if none, so indicate), drawback formula number (if none, so indicate), date of approval of the nonbeverage formula, alcohol content of the flavor or blender (if nonalcoholic, so indicate), a description of any coloring material contained in the flavor or blender.

Item 6. Show in sequence each step employed in producing the product including the step at which the specified materials will be added and the approximate period of time to complete production.

Item 12. Number the formulas in sequence commencing with the number "1." Nonbeverage formulas shall be prefaced with the symbol "NB."

Conditions for Granting Approval

This approval is granted under 27 CFR Part 24 and does not in any way provide an exemption from or a waiver of the provisions of the U.S. Food and Drug Administration regulations relating to the use of food and color additives in food products.

PAPERWORK REDUCTION ACT NOTICE

This request is in accordance with the Paperwork Reduction Act of 1980. The information collection is used by ATF to determine that wine is processed in an authorized manner and meets one of the standards of identity for wine. The information is required to obtain a benefit.

The estimated average burden associated with this collection of information is 2 hours per respondent or recordkeeper, depending on individual circumstances. Comments concerning the accuracy of this burden estimate and suggestions for reducing this burden should be directed to Reports Management Officer, Information Programs Branch, Room 7011, Bureau of Alcohol, Tobacco and Firearms, 1200 Pennsylvania Avenue, N.W., Washington D.C., 20226, and the Office of Management and Budget, Paperwork Reduction Project (1512-0059), Washington D.C., 20503.

ATF F 5120.29 (6-90)

- statement need not appear if 75% or more of the wine is derived from grapes of the named variety);
- (iii) the entire qualifying percentage of the named variety was grown in the labeled appellation of origin area.
- (2) Wine made from any variety of any species found by the Director upon appropriate application to be too strongly flavored at 75% minimum varietal content may be labeled with the varietal name if:
- (i) Not less than 51% of the wine is derived from grapes of that variety;
 - (ii) the statement “contains not less than 51% (name of variety)” is shown on the brand label, back label or a separate strip label (except that this statement need not appear if 75% or more of the wine is derived from grapes of the named variety);
 - (iii) the entire qualifying percentage of the named variety was grown in the labeled appellation of origin area.
- (a) *Two or three varieties.* The names of two or three grape varieties may be used as the type designation if:
- (1) All of the grapes used to make the wine are of the labeled varieties;
 - (2) the percentage of the wine derived from each variety is shown on the label (with a tolerance of plus or minus 2%);
 - (3) if labeled with a multicounty appellation, the percentage of the wine derived from each variety from each county is shown on the label; and
 - (4) if labeled with a multistate appellation, the percentage of the wine derived from each variety from each state is shown on the label.

The following are regulations concerning appellations of origin (27 CFR, Part 4, Section 4.25a):

(a) *Definition.*

- (1) *American Wine.* An American appellation of origin is:
 - (i) The United States;
 - (ii) a State;
 - (iii) two or no more than three States which are all contiguous;
 - (iv) a county (which must be identified with the word “county,” in the same size of type, and in letters as conspicuous as the name of the country);
 - (v) two or no more than three counties in the same State; or
 - (vi) a viticultural area (as defined in paragraph (e) of this section).

(b) *Qualification.*

- (1) *American wine.* An American wine is entitled to an appellation of origin other than a multicounty or multistate appellation or a viticultural area, if:
 - (i) At least 75% of the wine is derived from fruit or agricultural products grown in the appellation area indicated;
 - (ii) it has been fully finished (except for cellar treatment pursuant to Section 4.22(c), and blending which does not result in an alternation of class or type under Section 4.22(b)) in the United States, if labeled “American”; or, if labeled with a State appellation, within the labeled State or an adjacent

- State; or, if labeled with a county appellation, within the State in which the labeled county is located; and
- (iii) it conforms to the laws and regulations of the named appellation area governing the composition, method of manufacture, and designation of wines made in such place.
- (c) *Multicounty appellation.* An appellation of origin comprising two or no more than three counties in the same State may be used if all of the grapes were grown in the counties indicated, and the percentage of the wine derived from grapes grown in each county is shown on the label, with a tolerance of plus or minus 2%.
 - (d) *Multistate appellation.* An appellation of origin comprising two or no more than three States which are all contiguous may be used, if:
 - (1) All of the grapes were grown in the States indicated, and the percentage of the wine derived from grapes grown in each State is shown on the label, with a tolerance of plus or minus 2%;
 - (2) it has been fully finished (except for cellar treatment pursuant to Section 4.22(c), and blending which does not result in an alteration of class or type under Section 4.22 (b)) in one of the labeled appellation States;
 - (3) it conforms to the laws and regulations governing the composition, method of manufacture, and designation of wines in all of the States listed in the appellation.
 - (e) *Viticultural area*
 - (1) *Definition.*
 - (i) *American wine.* A delimited grape growing region distinguishable by geographical features, the boundaries of which have been recognized and defined in Part 9 [in this chapter of the ATF wine regulations]

The following are regulations concerning the term *estate bottled* (27 CFR, Part 4, Section 4.26):

- (a) *Conditions for use.* The term “Estate bottled” may be used by a bottling winery on a wine label only if the wine is labeled with a viticultural area appellation of origin and the bottling winery:
 - (1) Is located in the labeled viticultural area;
 - (2) grew all of the grapes used to make the wine on land owned or controlled by the winery within the boundaries of the labeled viticultural area;
 - (3) crushed the grapes, fermented the resulting must, and finished, aged, and bottled wine in a continuous process (the wine at no time having left the premises of the bottling winery).
- (b) *Special rule for cooperatives.* Grapes grown by members of a cooperative bottling winery are considered grown by the bottling winery.
- (c) *Definition of “Controlled.”* For purposes of the section, “Controlled by” refers to property on which the bottling winery has the legal right to perform, and does perform, all of the acts common to viticulture under the terms of a lease or similar agreement of at least 3 years duration.

The following are regulations concerning vintage wine (27 CFR, Part 4, Section 4.27):

- (a) *General.* Vintage wine is wine labeled with the year of harvest of the grapes and made in accordance with the standards prescribed in classes 1, 2, or 3 of Section 4.21. At least 95% of the wine must have been derived from grapes harvested in the labeled calendar year, and the wine must be labeled with an appellation of origin other than a country (which does not qualify for vintage labeling). The appellation shall be shown in direct conjunction with the designation required by Section 4.32 (a)(2), in lettering substantially as conspicuous as that designation. In no event may the quantity of wine removed from the producing winery, under labels bearing a vintage date, exceed the volume of vintage wine produced in that winery during the year indicated by the vintage date.

Meritage Blending

An association of California vintners, following a nationwide contest in search of an appropriate term for rather “Bordeaux-like” whites and reds, established *Meritage* as an approved ATF label designation in 1989 (Exhibit 4–2). The adoption of the category came with a set of rules that, loosely summarized, are as follows:

1. Membership in the Meritage Association is required to use the word *Meritage* on wine labels.
2. Meritage wines must be either white or red (no blush wines), and only grape varieties approved for classified Bordeaux commercial winegrowing may be used.
3. Wines labeled as Meritage must be the vintner’s finest in its type, often measured as the most expensive in the line.

Exhibit 4–2 Example of Commercial Meritage Label. Photo by David Ferguson.

MERRYVALE




MERITAGE
NAPA VALLEY

72% SAUVIGNON BLANC 28% SEMILLON

PRODUCED AND BOTTLED BY MERRYVALE VINEYARDS,
ST. HELENA, CALIFORNIA, USA, ALCOHOL 13.5% BY VOLUME

4. Meritage wines cannot also be varietal labeled, although appropriate variety percentages can be listed subordinately.
5. A limit of 25,000 cases of production is permitted for each type from each vintage.

The track record of Meritage wines indicates that a market niche has been created and is limited.

Proprietary Blending

While the Meritage rules above disallow the use of any grape varieties other than those of classic Bordeaux origin, vintners can and are encouraged to pursue the feasibility of producing and marketing *proprietary* wines. These are products that are identified by a name that, after ATF approval, becomes the exclusive property of the vintner.

Blending Cost Rationale

The typical purpose of blending is to provide an element missing in a base wine. Even more typically, the unit cost of a blend component may be much more than the unit cost of the base wine.

Consider the following simple calculation:

Color-deficient Chambourcin	950 gal @ \$5/gal =	\$4750
Dark red blending wine	<u>50 gal @ \$15/gal =</u>	<u>750</u>
Total:	1000 gal	\$5500

At first glance, the notion of paying \$15 or more per gallon for blending wine might cause a vintner trepidation. A closer look would reveal that the blend has increased the unit cost from \$5.00 to only \$5.50 per gallon, or just 10 cents per bottle. Of great virtue in this exercise is that the resulting wine may be much more attractive to consumers, perhaps a dollar or two, or more, per bottle. In this context, the blending investment of 10 cents per bottle seems very shrewd. Similarly, lackluster white and blush wines can be greatly enhanced by a comparatively small percentage blend addition of fruity-spicy Muscat Canelli or Muscat de Frontignan.

FINING

Young white wines are typically clouded with suspended proteins that require clarification. The extraction of phenolic compounds from grape skins usually neutralizes the proteins in red wines, leaving them clear but harshly astringent. Consequently, the clarification and softening processes in commercial winemaking are

known as *fining*. It is chemically complex, involving molecular interaction, bond formation, and both absorption and adsorption. The materials and techniques for fining are discussed in Chapter 8.

DETARTRATION

Most vintners achieve tartrate stability in new wines by the proper use of fining agents together with the immediate precipitation of potassium bitartrate crystals in cold storage.

The precipitation of unstable potassium bitartrate (KHT), or cream of tartar, is performed best in a controlled atmosphere of constant temperature, usually from 25 to 28°F for at least 4 weeks. This time can be cut in half by “seeding” KHT unstable wines with specially prepared tartar crystals that act as nuclei: focal points upon which new crystals are more readily formed.

Wines that have high calcium content (over 50 ppm) can be frustratingly difficult to stabilize. Even after months in a cold room storage, precipitates can form in bottled wines. This is why calcium carbonate treatments to reduce juice or wine acidity are avoided.

An alternative to this procedure is an *ion-exchange* process, much the same as what takes place in common water softeners, in which unstable potassium ions are exchanged for stable sodium ions on tartrate crystals. Major hurdles with ion-exchanging wine include a comparatively high cost for the equipment and resins required, quality control of metal content through expensive atomic absorption analyses, and the creation of a product that does not align well with low-sodium diets.

BARREL AGING*

Winemakers can find an abundance of literature relating to the composition of oak extractives and their effect on flavor of oak-aged wines. More than 200 volatile components of oak wood have been identified. Components of an oak barrel are illustrated in Figure 4-2.

American Oak (*Quercus alba*) typically makes the best barrels with regard to minimal seepage but also expresses minimal phenols. Limousin Oak (*Quercus robur*) generally offers the most potential extraction of desired phenols but also the most seepage.

Probably the most thoroughly studied group of volatile oak extractives is derived from oak lignin. These are complex organic compounds composed mainly of higher alcohols and make up a significant share of wood weight. After the oak timber is

*Adapted from Lincoln Henderson, *Barrels and Oak Aging of Wine*, Bluegrass Cooperage Company, Louisville, KY.

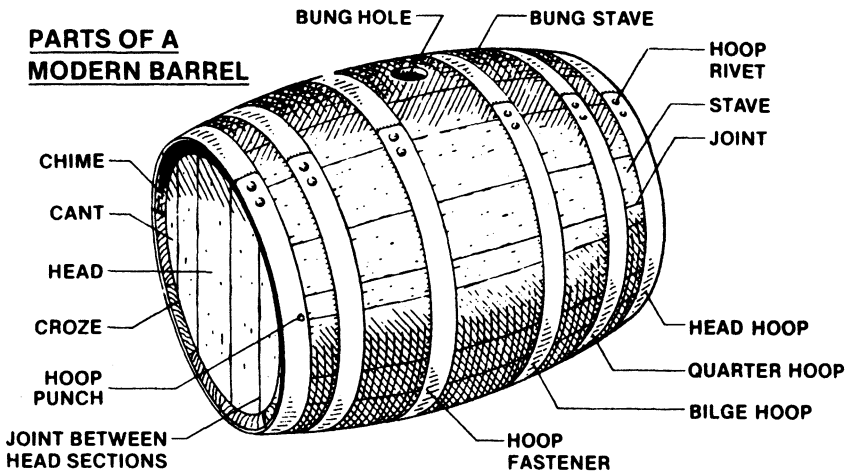


Fig. 4-2 Components of a wine barrel.

Courtesy: Sebastiani Vineyards.

harvested, lignin can be broken down by hydrolytic and microbiological reactions. Another major group of volatile oak wood components exist as a result of thermal degradation of the oak polysaccharides. These sugar-based compounds, such as maltol, which can impart a sweet toasty flavor, make up more than 50% of dry wood weight. The third major group of oak wood components are lactones, which impart coconut-like flavors but, when exposed to organic solvents found in wine, express the caramel-vanilla character that is desirable in wine to one degree or another. Other flavors are also attributed to oak barrel aging: bitter almond, leather, caramel, nutty, cedar, pencil shavings, cigar box, sap, clove, smoke, coffee, tar, dusty, and toast.

Barrel fermentation has become a popular technique employed by winemakers over the past decade. The general notion is that bitter phenolic compounds existing in the oak react with fermenting juice proteins, resulting in more oak flavor extraction and less astringent tannin extraction. The result is wines with slightly less fruit intensity and increased levels of higher alcohols.

Lees contact, or *sur lies*, also has a significant effect upon the complexity of barrel-fermented white wine. Greater frequency and duration of the *batonnage*, or lees stirring, exposes the wine to greater influence of enzymes created by yeast autolysis and the flavors generated from yeast protoplasm. This treatment requires daily surveillance to guard against the early development of acetic acid (vinegar), dirty-mustiness (wild yeasts), ethyl acetate (paint thinner), mercaptans (garlic-skunky), sulfides (rotten eggs, manure), and oxidative color changes. Lees provide a wealth of substrates for the growth of spoilage microorganisms.

Cost Implications

The fact that French Oak barrels cost upwards of \$600 each removes any question about the serious cost involved with barrel aging.

Essential elements in aging/storage equations include container cost or lease rate, maintenance costs, depreciation schedules, salvage value, capacity, and the measure of space/time requirements needed to achieve desired results in improved wine quality.

Consider two alternatives for storing 2,000 gallons of wine: a single 2,000-gallon stainless-steel tank (SST) and forty 50-gallon French oak barrels (FOBs).

The SST with single wall (no jacket), a manway door, and three valves (outlet, racking, and vent), on short legs, should cost about \$4,000. (See Figure 4–3 for the components of a wine tank.) This is a “ballpark” price: stainless steel is rather commoditylike, so prices fluctuate. It is typical to find vintners depreciating such tanks over 5 years, although accountants may advise otherwise for given operational strategies.

It must be remembered that this investment of \$4,000 over 5 years also has a cost, called *opportunity cost* by agricultural economists. If the \$4,000 was borrowed, the opportunity cost is in the form of paid interest. If the \$4,000 was taken from owned capital then the opportunity cost is in the form of lost interest. Opportunity cost calculated at 10% compounded over 5 years adds \$1,274 to the price, or a total outlay of \$5,274 for the tank. At 5 years of age, a salvage value of about \$2,000 can be expected if the tank remains in good condition. Subtracting this from the \$5,274 total outlay results in an actual cost of \$3,274. Dividing this figure by 2,000 gallons results in an actual cost of \$1.64 per gallon.

The same rationale can be used to calculate the cost of FOBs. Typically these vary from 50 to 60 gallons in capacity, but 50-gallon barrels will be used for this exercise.

Forty new 50-gallon FOBs will cost about \$600 each, or a total of \$24,000 for the 2,000 gallons of total capacity. Opportunity cost at the same rates over the same 5-year period amounts to \$7,644, or a total outlay of \$31,644. Retiring barrels after 5 years leaves a common salvage value of about \$150 apiece, or \$6,000 for the entire lot. Subtracting \$6,000 from the \$31,644 results in an actual cost of \$25,644, or \$12.82 per gallon.

The overall cost/value table can now be started:

	<u>\$/gal capacity</u>
SST	1.64
FOB	12.82

Tanks and barrels take up cellar space, which must be assigned a “rent” cost. Rent should be calculated as the total cost per square foot or per cubic foot of cellar space per year. Figure total fixed costs and variable costs appropriate to the cellar

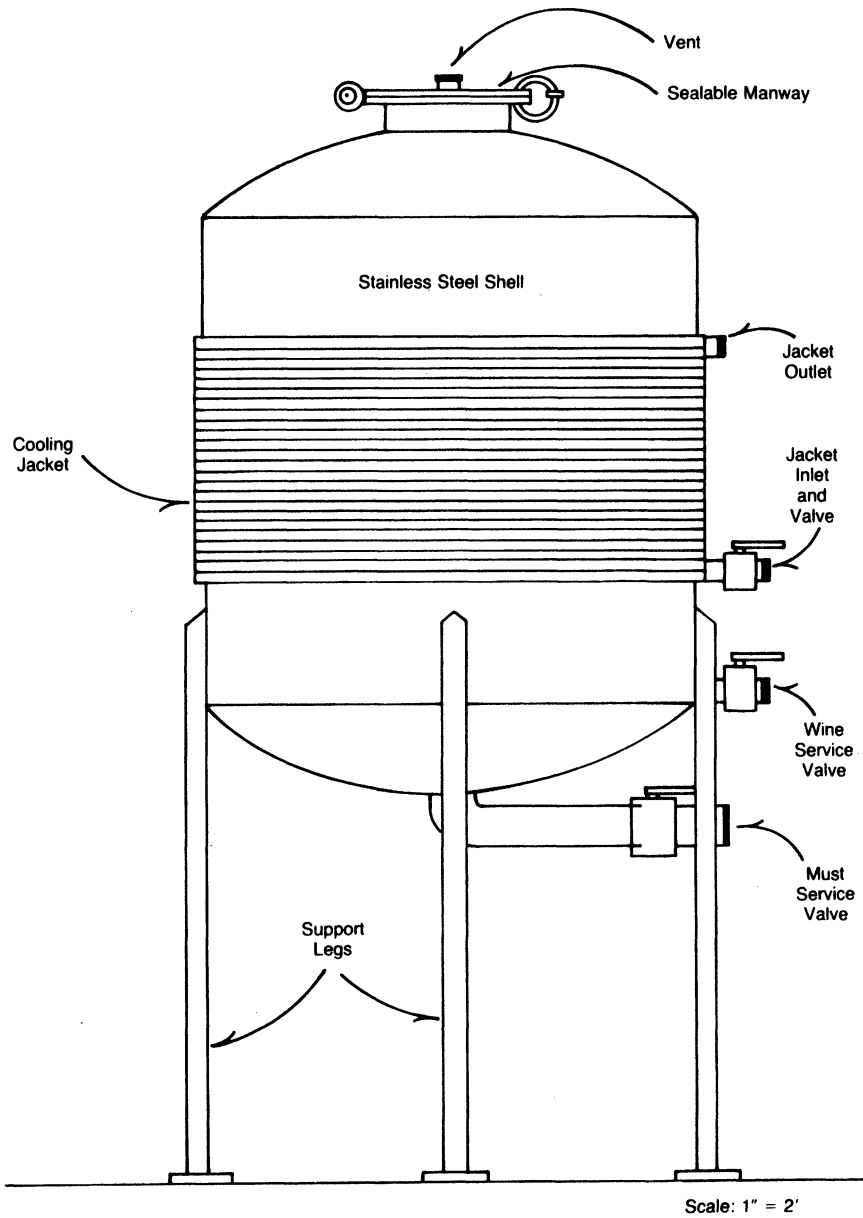


Fig. 4-3 Components of a wine tank.

in which tanks and barrels are to be housed, and then divide by the total footage. For this example, a rent of \$6.00 per square foot is assigned. The SST will occupy about 60 square feet, which will cost \$360 per year and \$1,800 for 5 years. Divided out, this becomes a per-gallon cost of \$.90. The 40 FOBs, stacked in tiers three barrels high, will occupy about 120 square feet and thus \$720 per year, totaling \$3,600 for 5 years. Divided out, this becomes a \$1.80 per-gallon cost.

	<u>\$/gal capacity</u>	<u>\$/gal rent</u>
SST	1.64	0.90
FOB	12.82	1.80

It takes a lot more time to clean and sanitize FOBs than SSTs. On the other hand, most vintners age their wine in barrels for a longer period of time. Estimate a need to clean and sanitize a SST about four times per year. Calculating one person-hour @ \$8 per hour, per treatment, results in a labor cost of \$32 per year, or \$160 for 5 years: \$.08 per gallon.

By the time barrels are received, cleaned, treated, and placed in service, cleaning and sanitizing treatments require about one person-hour per barrel per year. At the same \$8 per hour wage rate, this results in an annual barrel cleaning cost of \$320, or \$1,600 for 5 years: \$.80 per gallon.

The table is now expanded with these maintenance costs:

	<u>\$/gal capacity</u>	<u>\$/gal rent</u>	<u>\$/gal maintenance</u>	<u>\$/gal total</u>
SST	1.64	0.90	0.08	2.62
FOB	12.82	1.80	0.80	15.42

The common reaction to FOBs' costing more than five times what SSTs cost is negative: barrel aging is just too expensive. Obviously, the next step is to look at the ratio of cost/benefit achieved with this treatment.

Wines made and marketed in the "squeaky clean" manner often associated with stainless-steel aging can satisfy a market demand that is limited. Most blush wines are made with no barrel aging, and many fruitier whites and reds, particularly nouveau types and all wines made from native American *Vitis labrusca* varieties, are also better without wood complexity.

A common usage of new FOBs is for white wines such as Chardonnay or Seyval Blanc for the first year or two and then reds such as Cabernet Sauvignon or Chambourcin for the next 3 years. The benefit of FOBs is needed in a measure of cost so it can be compared with the treatment cost determined above.

Most of the popular "fighting varietal" (Chardonnay and Cabernet Sauvignon) wines are priced in the \$4 to \$6 range. As a rule, these are good everyday wines made with little, if any, benefit of barrel aging. This is a fiercely competitive market. With grapes valued at \$1,000 per ton, juice price alone is about \$1.15 per 750-mL

bottle. Common cellar processes add another \$.60 to \$.80 per bottle, and packaging increases the cost by another \$1 or so. It is difficult to find less than \$3 per bottle in total cost before marketing. Excise taxes, shipping, and wholesale markup result in yet another \$1 per bottle. Retail markup can easily bring the shelf price to more than \$5.

Consider the \$2.62 total 5-year, per-gallon cost figure for SSTs calculated above. A typical fighting varietal red wine process/aging regime would be 2 years, or 40% of this \$2.62, or \$1.50 per gallon. Dividing that by 5, as there are approximately five bottles in a gallon, we find \$.21 of cost attributed to SST containment.

By the same formula, the \$15.42 total 5-year per-gallon cost figure for FOBs over 2 years is \$6.16. Dividing that by 5 results in an FOB aging cost of \$1.23 per bottle. Subtracting \$.21 for the SST cost reveals that 2 years of FOB aging costs about \$1.02 per bottle more than the same 2 years in SST.

The remaining question for each wine is whether this added cost burden can be justified in terms of the final product being more salable in the marketplace. An initial response to this is that the barrel-aged product will need a packaging treatment that attracts attention to its barrel aging hierarchy: perhaps punted bottles, embossed labels, longer corks, or tin alloy capsules. Combined, this can easily add another \$.50 cost per bottle, and, added to the 2 years of FOB aging cost of \$1.02, the total added production cost becomes \$1.52 for each bottle of barrel-aged wine.

The fighting varietal production cost estimated at \$4 per bottle above included SST tankage cost. For the same wine to be aged in FOBs, the cost would advance by \$1.52, for a total cost of \$5.52 per bottle.

A secondary response to market acceptance is addressed to profitability. It is common to assign a 30% margin over cost for the vintner's profit. Thus,

$$\text{SST @ } \$4.00/\text{bottle} \times 1.30 \text{ profit} = \$5.20 \text{ winery price}$$

$$\text{FOB @ } \$5.52/\text{bottle} \times 1.30 \text{ profit} = \$7.18 \text{ winery price}$$

The addition of a wholesale channel cost at 25%, which includes FET, freight, and wholesale margin, is as follows:

$$\text{SST @ } \$5.20/\text{bottle} \times 1.25 \text{ wholesale} = \$6.50 \text{ to retailers}$$

$$\text{FOB @ } \$7.18/\text{bottle} \times 1.25 \text{ wholesale} = \$8.98 \text{ to retailers}$$

Finally, the addition of a retail markup of 35% results in the following consumer shelf price comparison:

$$\text{SST @ } \$6.50/\text{bottle} \times 1.35 \text{ retail} = \$8.78$$

$$\text{FOB @ } \$8.98/\text{bottle} \times 1.35 \text{ retail} = \$12.12$$

The question raised by the \$8.78 price is how the vintners can sell such wines at \$4 to \$6 and still make a profit. Virtually all of the answer is found in the size and scale

of larger operations, where hundreds of thousands of production gallons and rapid shelf movement can dilute costs to just fractions of what smaller vintners must pay.

Do consumers feel that FOB aging makes a wine worth \$3.34 more per bottle than a fighting varietal wine made in stainless steel? The many thousands of FOBs that exist in the cellars of successful vintners around the United States and the entire world of commercial winemaking attest to a positive response to that question. It is impossible for most small-scale vintners to compete with the large fighting varietal vintners, even when their wines are marketed exclusively at the winery. Consequently, the business of modest-size vintners has become largely a “boutique” operation: high-quality, barrel-aged wines that target classified European wines as direct competitors.

American Oak versus French Oak

There are numerous studies comparing various species of oak, such as the French and American oak barrels illustrated in Figure 4–4, and many more controversial debates on the topic. Prior to the 1970s, the “toasting” of oak barrels, which is a charring treatment, was employed throughout the Bourbon whiskey industry but was virtually unknown in wine production. Barrel aging for wines was made upon bare oak surfaces treated only with one or another chemical soaking and rinsing.

Today, with toasting regimes ranging for very light to very dark, there is a growing school of thought that the level of toast given an oak barrel has more flavor influence than the species of oak itself. As might be expected, this has brought even more research and debate to the topic.

The following are wood species and their generally accepted utility:

1. *Allier*: dry whites, light-medium toast; dry reds, medium-heavy toast
2. *American*: dry heavy-bodied reds, medium-heavy toast
3. *Limousin*: dry heavy-bodied reds, medium-heavy toast
4. *Nevers*: dry whites, light-medium toast; dry reds, medium-heavy toast
5. *Tronçais*: dry whites, light-medium toast; dry light-bodied reds, medium-heavy toast
6. *Vosges*: dry delicate whites, light-medium toast

Winemakers are encouraged to study and gather a database on oak aging in order to best fit the wine style desired. With new American Oak barrels sold at about one third the cost of new FOBs, the cost implications are obviously profound.

Conditioning*

Water quality is perhaps the single most important factor in maintaining sound cooperage. If heavily chlorinated water is used, a charcoal filter is strongly recommended. If well water, or water from other sources that is high in iron and elemental

* Adapted from *Barrel Care and Maintenance Manual*, Barrel Builders Inc., St. Helena, CA 94574.

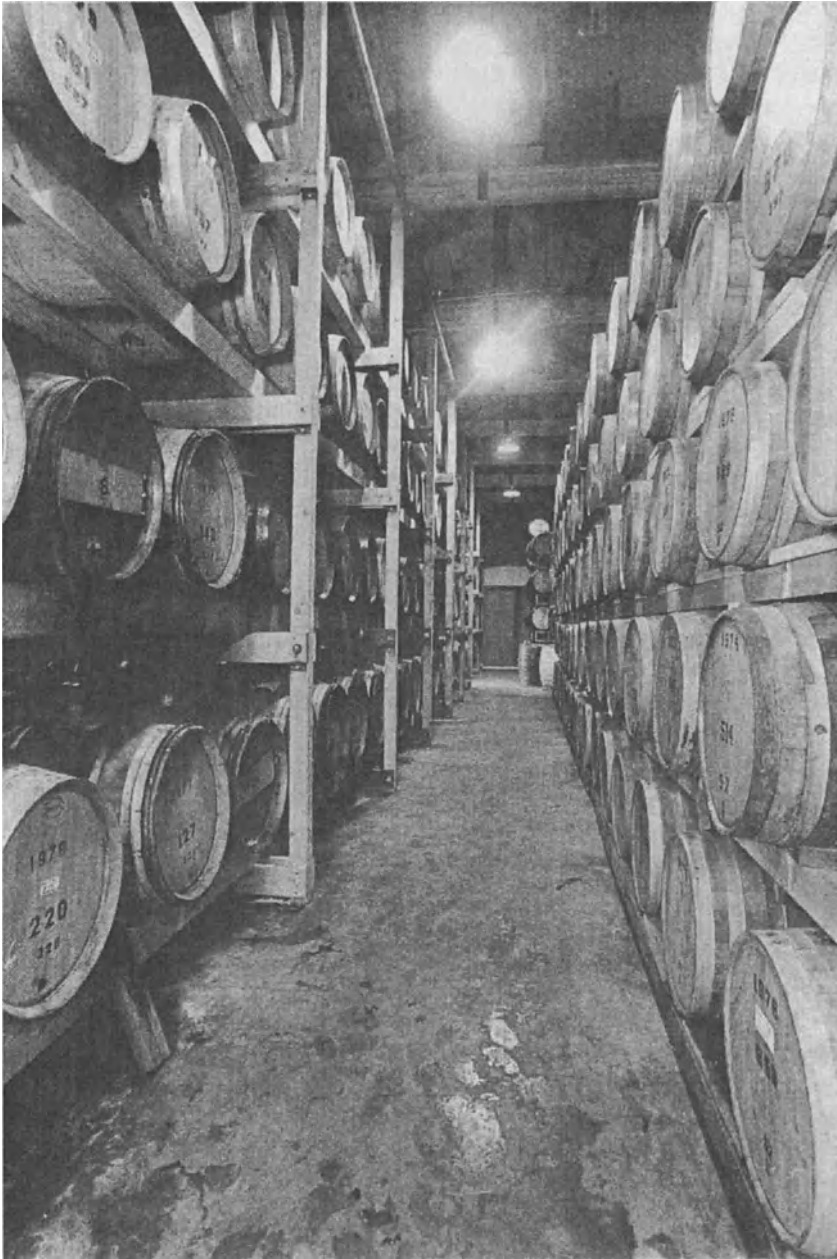


Fig. 4-4 French oak barrels racked on left row, American oak barrels stacked on right row.

Courtesy: Dry Creek Vineyard.

sulfur, is used, a potassium permanganate filter or other ion-exchange method that will remove these minerals is required. Residual iron can cause iron haze *casse* in wine, and residual sulfur can be a substrate for the formation of sulfides.

Wooden wine vessels are porous and can never be absolutely sterile. Conditioning treatments are employed to clean, detartrate, and discourage microbial growth.

When adding any treatment compound to a barrel, ensure that it is completely dissolved in water first. When adding both citric acid and potassium meta-bisulfite (KMS) to a barrel, ensure that these solutions are dissolved separately with the citric acid first. Citric acid lowers the pH of the water, which serves to make the sulfur dioxide emanating from the KMS more effective.

New Barrels

For obvious reasons it is recommended that a barrel be watertight before it is filled with wine. Before following the procedures below, be sure the hoops are snug. As a barrel swells, the hoops may be tightened a little at a time. If the barrel is excessively dry, a water mist sprayed inside and outside the barrel will conserve water and hasten the tightening. Boiling hot water should not be used to swell a barrel, as it may cause the inside of the stave to swell faster than the outside, possibly causing staves to crack.

The procedure is as follows:

1. Rinse the barrel to be sure it is free of dust and debris.
2. Fill the barrel halfway with cold water.
3. Add 90 g of citric acid to the water and stir until fully dissolved. Mixing may be performed by rocking barrels back and forth.
4. Add 45 g of KMS solution to the water and stir until fully dissolved.
5. Fill the barrel completely with water.

It usually takes from 3 to 5 days to completely soak up a barrel. Each barrel should be topped up daily with clean water. When barrels are completely soaked up, empty and rinse well several times before filling with wine.

Although a barrel may be tight when filled with water, it may seep when filled with wine. Wine has a lower specific gravity, a property that can allow it to leak through smaller pathways. It is best to introduce a fermenting must or a nonfiltered wine to the newly soaked barrel, as the higher solids content will help seal off these pathways. If the barrel does seep a little, keep it topped with wine, and wash off the outside to avoid microbe growth.

Maintenance of Empty Barrels—Short Term

Any chemical introduced to wooden barrels will penetrate the wood. Using hot water as a solution for these chemicals will increase the rate of this penetration. Citric acid additions function as neutralizers of residual chemicals in the wood.

Plain water can stagnate if left in a barrel for more than several days. If a barrel must be filled with water for more than 2 days, either empty the barrel, rinse, and refill with fresh water every 2 days thereafter, or add 180 g of citric acid and 90 g of KMS to the soaking water to help preserve it. This treatment is recommended only for the short term, several weeks maximum, since a holding solution will leach some of the oak flavor from the wood.

Maintenance of Empty Barrels—Long Term

Barrels left empty after wine or water storage will soon be invaded by bacteria and/or mold growth, which will eventually ruin them for any further wine usage. Consequently, it is important to create an atmosphere of sulfur dioxide gas inside empty barrels to discourage microbial growth.

The procedure is as follows:

1. Clean the barrel thoroughly and rinse it extensively several times with cold water. Drain dry; residual water standing in the barrel will combine with sulfur dioxide gas to form sulfurous acid, which will soak into the wood.
2. Hang a sulfur stick (1/2 stick for each 50-gallon barrel) from the bottom of the sulfur bung. Light the stick, ensure that it is burning properly, and insert it into the barrel with the bung left loosely in the bunghole. An alternative to sulfur stick burning is injection with sulfur dioxide gas from a pressurized canister for 3 to 5 seconds.
3. Allow the stick to burn at least 3 to 5 minutes. When it is completely burned, remove the sulfur bung and replace it with a regular bung driven into the bunghole tightly.
4. Store the barrel in a cool, dark, dry place for storage. Repeat treatment every 3 to 4 weeks to keep sulfur fresh.
5. If the barrel is to remain empty for a long period of time, it should be soaked overnight with fresh water prior to burning a sulfur stick to help keep it tight. If a barrel begins to dry and the staves shrink, air will enter the barrel, displacing the sulfur and creating an environment for destructive microbial activity.
6. When preparing the barrel for wine use, be sure it is tight and proceed with the new barrel treatment outlined above.

Soda Ash Treatment

Soda ash is a caustic chemical and a harsh treatment for distressed barrels. Its main function is to dissolve tartrate crystal deposits; it may also be used in attempts (usually not totally successful) to bleach out red wine color. Another major purpose of soda ash is to clean and leach out undesirable odors and flavors caused by microbial growth. Soda ash, in conjunction with chlorine treatment, is also used to sanitize barrels.

Treatment with soda ash will also leach out the oak flavor of a barrel, so it is not recommended for use on a new barrel. Soda ash should be dissolved in hot

water, 140 to 180°F. Protective clothing and handling with care are strongly recommended. Soda ash used in strength exceeding 1/4 ounce per gallon of water has burned oak staves.

The procedure is as follows:

1. Thoroughly clean and rinse the barrel, ensuring that it is tight with no leakage.
2. Fill the barrel halfway with hot water.
3. Add soda ash up to 1/4 ounce per gallon of water, and mix by rocking the barrel back and forth.
4. Fill the barrel completely with hot water. The bung may be placed in the bunghole loosely.
5. Allow to stand for 24 hours, then empty and rinse several times with cold water.
6. For severe cases of microbial spoilage, the barrel may then require supplemental chlorine treatment (outlined below). Otherwise, it should be neutralized with citric acid treatment (also outlined below).
7. Maintain barrel with the short-term treatment outlined above, and inspect carefully to ensure that undesirable odors and flavors have been removed. If not, repeat soda ash treatment. If so, thoroughly rinse several times with cold water, and immediately fill with wine.

Chlorine Treatment

This treatment should be employed only as a last resort to save microbe-infected barrels. Weak treatments of chlorine can form trichloroanisole (TCA), which can produce disagreeable “corky-musty” odors and flavors.

Chlorine treatments should be used in a strength of 200 ppm of available chlorine.

The procedure is as follows:

1. Thoroughly clean and rinse the barrel, ensuring that it is tight with no leakage.
2. Fill the barrel half with 200 ppm of chlorine cold-water solution, and place the bung loosely in the bunghole.
3. Allow to stand 48 hours, then empty and rinse thoroughly several times with cold water.

Citric Acid Treatment

A citric acid treatment is used to neutralize any residual chemicals in barrels and is essential to the maintenance of sound cooperage.

The procedure is as follows:

1. Rinse the barrel thoroughly several times with cold water.
2. Fill the barrel halfway with cold water.
3. Add 180 g of citric acid, and mix by rocking the barrel back and forth.

4. Fill the barrel completely with cold water.
5. Allow to stand 48 hours, then empty and rinse thoroughly several times with cold water.
6. Maintain the barrel with the short-term treatment outlined above, and inspect carefully to ensure undesirable odors and flavors have been removed. If not, repeat the soda ash treatment. If so, thoroughly rinse several times with cold water, and immediately fill with wine.

Filtration

In filtration, two important factors must be considered. First, of course, is the degree of clarity desired. Second, and arguably more important, is the amount of color and flavor lost.

All filtering operations remove some portion of color and flavor, justifying some red wine labels that prominently state “Unfiltered” as a proud indicator of higher quality. This is also used sometimes in combination with a disclaimer statement to excuse any haziness or precipitates that may have occurred in the bottle.

The advancement of food science and technology has given us many new processes and techniques readily applicable to producing higher quality wines with greater efficiency. One of the most meaningful developments in recent years is the ability to filter liquids through submicron-porosity media. In fine wine production, however, the application of this must focus upon adequate or optimal filtration to conserve precious color and flavor constituency.

Pressure and flow rate are important parameters of filtration performance. High pressure generally indicates improperly clarified wines causing difficult filtration with slow flow rates that result in excessive removal of color and flavor. Ideal wine filtration is characterized by a near-constant low pressure and moderate flow rates. Pressure surges can easily breach filter media, particularly delicate membranes.

Filtration efficiency depends upon three major elements of filter media: their properties, surface area, and porosity.

Primary filtration of very cloudy wines is typically performed with diatomaceous earth (DE), powders ranging in porosity from fine (just a few microns) to very coarse. Used to excess, this medium can cause considerable color and flavor loss. DE is most efficient when employed in a depth filter in which suspended wine particles are entrapped as powder slurries are “dosed” at specific intervals and collected upon screens inside the filter tank chamber. Some plate filters have frames available for DE filtrations, most often used for the filtration of lees (Figure 4–5). Wines that are obviously brilliant from good fermentation and clarification technique do not generally require a primary filtration.

Secondary, or “final,” filtrations in small wineries are usually performed with pads in frame filters, or cartridges in canister chambers, both of which are screen-type filtration devices. Pads and cartridges are nothing new, although the continued

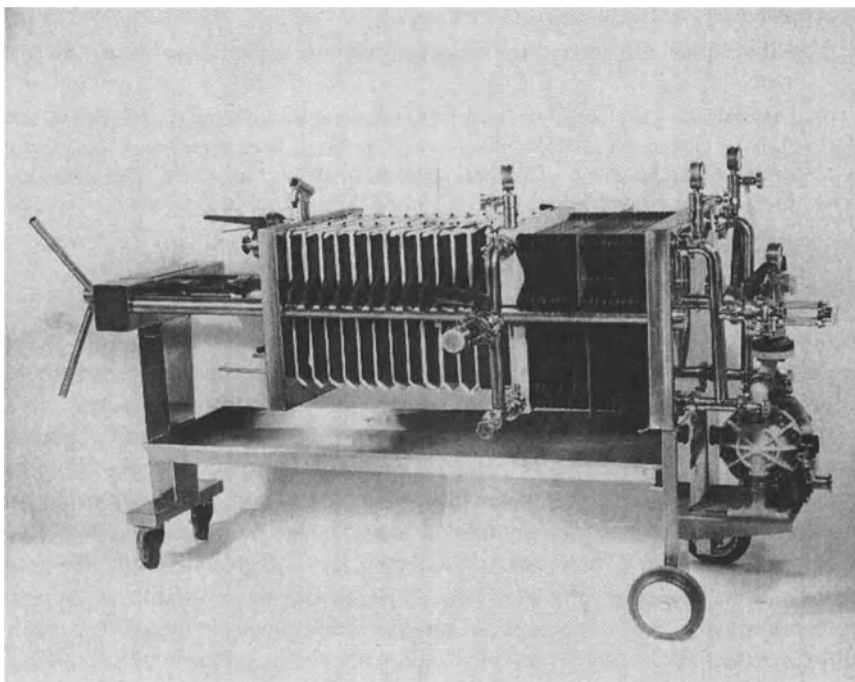


Fig. 4-5 Two-stage stainless-steel plate and frame filter with divider plate.
Courtesy: Prospero Equipment Company.

development of new plate and canister design, along with increasingly efficient pad and cartridge media, results in better economies of operation. State-of-the-art filters can conduct both primary and secondary filtration, either individually or, with a divider plate, simultaneously.

Commercial wine production on a smaller scale, with good quality control, can get along quite well with cartridge media, usually constructed from paper, glass fibers, cotton, hemp, and cellulose. Cartridges range from rather coarse ($40\text{-}\mu$ porosity) to fine (less than $1\text{-}\mu$). While the unit gallon cost of filtration with cartridges is quite high, it can be justified when compared to the much higher cost of investing in a frame filter apparatus.

The most common difficulty with pad and cartridge filtration is a surface area that is too small for the task at hand. Vintners often seek to use pads or cartridges with more filter time and higher pressure. This is a false economy, since the added time and energy are usually more costly than the medium and since the resulting wine quality will suffer undue color and flavor loss. A good rule of thumb is that the differential filter pressure should not exceed half of the recommended rate published by the medium manufacturer.

Microfiltration and Ultrafiltration

Microfiltration is the type of final filtration most often employed in the wine industry. It requires very low pressure, generally from 5 to 15 psi, and employs a semipermeable membrane for the removal of suspended wine particles at less than $.5\text{-}\mu$ porosity, well within nominal “sterile” filtration limits. Ultrafiltration uses a higher pressure range, from 10 to 150 psi, and employs less porous membranes that can separate high-weight molecules out of solution with what is actually a molecular “sieve.”

Microfilters are designed for a comparatively slow product flow across the membrane until an appropriate pore size allows the wine to pass through. This “cross-flow” technique reduces particle buildup on the membrane media, increasing throughput while reducing cleaning time. Continued improvement of both the filters and the membrane media, in tandem with a growing demand for microfiltration by the wine industry, has resulted in a considerable reduction in unit wine cost.

Unfiltered Wines

Experienced wine consumers actually expect older reds to have a sediment and ritualize the decanting procedure. It is probably in this market segment that most of the rationale and justification for producing unfiltered wines is found. Neophytes and occasional wine consumers may find some virtue in unfiltered wines as being somewhat ecologically friendly, and there may be an association to such wines being more “organic” or “natural.” There can, however, be no question that unfiltered red wines taste bigger and richer. Considerable color and flavor may be conserved with significantly fewer finings performed over a commensurately longer time. More and more vintners are returning to the time-honored method of egg white treatments for red wines. A light addition of tannic acid beforehand will help prevent the formation of phenolic degradation compounds. Despite the conventional wisdom of bypassing red wine cold stabilization, a few weeks under refrigeration can enhance clarity and eliminate the nasty, ugly tartrate crystalline formation on the cork. Acid balance can be regained with a small addition of citric acid, which, as a tricarboxylic acid, provides superb stability and gives the resulting wine a little lift to its fruit component.

PACKAGING

Having potential consumers visit a winery tasting room is, of course, a big advantage over having to depend solely upon the attention-getting of packaging on the shelves in an independent retail store. Vintners often make the mistake of paying too little attention to the importance of attractive packaging to maximize consumer

appeal. Dull, drab, unprofessional packaging takes the excitement out of even the finest of wines. Worse yet, some vintners get caught up in egoistic label schemes that consumers don't understand and/or find an identity.

Bottles

Consumer acceptance for nontraditional bottle shapes is questionable. There is plenty of evidence to show that packaging Cabernet Sauvignon in a Burgundy bottle or Pinot Noir in a Bordeaux bottle indicates that the vintner either doesn't know the difference or doesn't care, and either impression can only affect the vintner's image negatively.

One pathway to attention-getting bottles is via attractive colors. The traditional Burgundy shape in "dead-leaf" green continues to be very popular for U.S.-grown Chardonnay and Pinot Noir. Since the 1980s, this same bottle has been widely employed for fine quality Rhone varietals, Fume Blanc, Gamay, Seyval Blanc, Vignoles, Chambourcin, and Marechal Foch.

The traditional Bordeaux shape in colorless "flint" was widely adopted for Sauvignon Blanc, white Meritage, Vidal Blanc, and pink "blush" wines during the 1980s. The same shape in "Champagne" green has been used for Cabernet Sauvignon, Merlot, and, of course, the Bordeaux chateau wines for many years. A new "antique green" serves well, too, for red Meritage, Sangiovese, and other heavier reds. There is a slight tapered version of this that is also accepted for top-of-the-line wines.

Punted Burgundy and Bordeaux bottles, those with "push-up" indentations in the bottom, are another attribute of fine bottle quality and consumer appeal.

The tall, narrow, "hock" bottle shape is required to be green in the Mosel-Saar-Ruwer region of Germany, but German regulations require it to be amber in the Rheinland. In the Nahe region it is often blue. The extremely tall and narrow versions are not practical, as they are easy to tip over, don't stack on their sides, and are difficult to stand up on a shelf or in a refrigerator.

Most important is bottle quality. Some of the imported glass does not meet the standards set by U.S. bottle manufacturers, either in its composition or in its dimension tolerances. This is particularly important in the neck of the bottle, where a given cork diameter may be either too loose, causing leaks, or too tight, causing breakage. It is also important in the body of the bottle, where varied mold sizes can serve to significantly affect fill levels. See page 216 for various bottle shapes.

Closures

Many consider "screw-cap" types to be the perfect closure for wine bottles. Unfortunately, these are not acceptable to most fine wine buyers, and screw-caps are therefore generally relegated to lower priced wines.

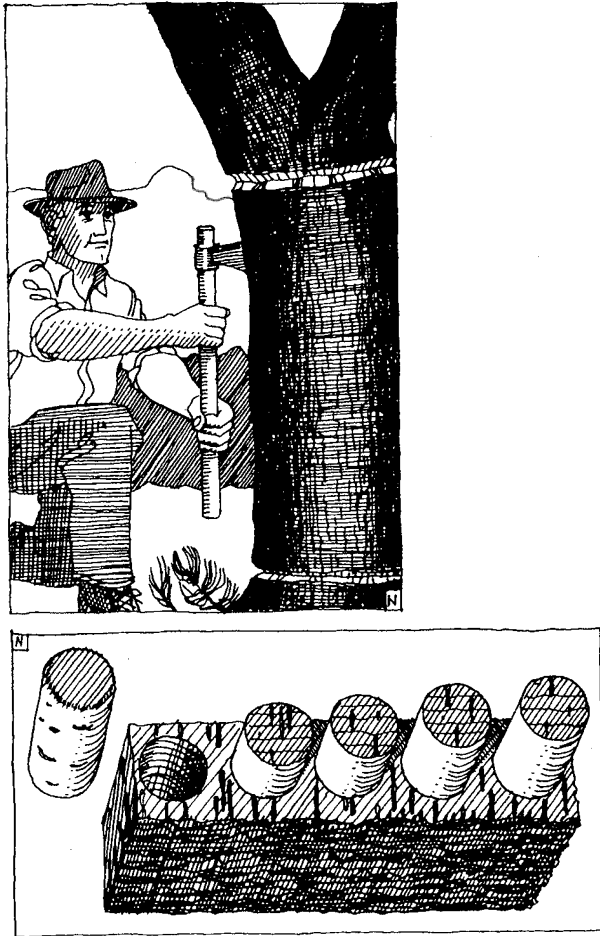


Fig. 4-6 Illustration of cork bark harvest and cork production.

Courtesy: Sebastiani Vineyards.

Natural cork (Figure 4-6) remains the closure of choice throughout most of the wine world, but the advancing technology of synthetic corks made of various ethylene vinyl acetate compounds is encouraging many vintners to take a closer look. Manufacturers say that most problems of difficult driving, and even more difficult pulling, are now ironed out. They advertise new technology, such as virtual immortality of their ethylene vinyl acetate co-polymer (EVA) formulation, improved lubricants, variable resiliencies, and even the possibility of scavenging headspace oxygen and sulfur dioxide. Considerable testing of new types continues with both vintner trials and university research. While only some vintners have totally

committed all production to synthetic corks, many are poised to move forward with the conversion in varying degrees. For the venturesome vintner, synthetic corks may be worth a try.

The key to cork selection is size: the proper diameter to ensure a good balance between ease in pulling and optimal seal. Perhaps the only thing more detrimental to a good packaging concept than an oversized cork virtually impossible to pull is a sticky bottle caused by an undersized leaking cork. Wines designed for long periods of bottle age, reds in particular, are usually bottled under longer corks.

Capsules

Traditionally, capsules have been a very important part of packaging, primarily for consumer appeal. Manufacturers have made much of the ability of capsules to “seal” bottles, but in fact most capsules can rather easily be removed and replaced. In the case of most hard poly types, it is simply a matter of pulling them off and putting them right back on again.

Up until the late 1980s, lead and lead-alloy capsules were the epitome of fine wine capsules: durable, efficient, rich in appearance, and expensive. With several controversial studies indicating the possibility of trace amounts of lead migrating from capsule into poured wine, lead was banned by the ATF.

Inexpensive capsules generally give the impression of cheap wine. Great advances have been made in the appearance of heat-shrink PVC film capsules, and they offer great resistance to tampering. Even more improvement has been achieved in replacing lead with tin-aluminum and aluminum poly laminate roll-on capsules.

As with corks, capsule diameter must be precisely fitted to the bottle neck to avoid folding, wrinkling, or tearing when applied.

The notion that it is better to have no capsule at all than a bad one is gaining momentum in the United States. The revolutionary “B-Cap” closure system features a custom-designed, drip-resistant, flange-lip bottle mouth. Corks are driven as usual, but instead of a capsule, a small decorative paper circle is applied to cover the top of the cork. The full length of the cork is usually perfectly visible, but some vintners prefer to partially mask the cork with a narrow garnish strip applied around the bottle neck. The B-Cap marketers advertise it as “functional, economical, attractive, user friendly, and environmentally correct.”

Label Rationale

Many vintners, particularly those who market chiefly to winery visitors and tourists, operate on the notion that they have a captive audience. This often takes the form of cheap, “in-house” labeling based on the “letting my wine speak for itself” syndrome. While the wine may indeed be very good, the question is whether it is salable. Is the label something that can inspire consumers’ pride in their cellar? Is the packaging something that serious wine imbibers want to present at their dinner

table with guests? Is the product something that qualifies as a nice gift? If the answer to any of these questions is in doubt, then results are likely to be a bottle or two bottles of “courtesy” sales as compared to full-case sales and repeat orders. Vintners should consider that the important thing is not what they think about their packaging but rather what the consumer’s impression is and whether the package sufficiently attracts consumers to buy it.

Another important part of label rationale is accoutrements, such as tags, stickers, and other supplemental attention-getters. There is no reason not to flaunt a prestigious award, medal, or critical accolade. It is common to find vintners taking advantage of medals by significantly increasing the price of medal-winning wines. This may sometimes slow movement of such wines, but, more important, it draws attention to vintner quality across the entire product line. Most important, the price increase generates additional profits, which may be needed to lower prices to make non-medal winners more attractive.

Label Shapes

While many label shapes in use radically depart from the traditional square or rectangle, prospective vintners should understand that odd shapes are a common cause of labeling problems and increased labeling cost.

Design and printing costs are generally higher in that odd shapes require custom dies for cutting. One way or another, the cost of these dies will be passed on to the vintner. Nontraditional shapes cause greater paper waste too—yet another cost borne by the vintner. Labeling machines usually require expensive custom parts fabricated to handle custom shapes.

Label Colors

There can be little question that color and its integration in wine packaging should be placed in the hands of proven professionals. This is expensive but absolutely essential to properly address each wine to its target market.

Color psychology seems to differ with each psychologist who feels qualified to write about it. Several references have been distilled down to this general set of consumer truisms:

- White: clean, winter, delicate
- Black: formal, death, night
- Brown: heavy, religion, obscure
- Red: hot, danger, stop
- Blue: cool, thin, distant
- Green: life, summer, proceed
- Yellow: floral, spring, fresh
- Orange: citrus, autumn, ripe
- Violet: royal, soft, expensive



Fig. 4–7 A selection of well-designed commercial wine labels. Photo by David Ferguson.

This brings up the notion of color combinations. Red and green are, of course, traditional decor for the holiday season, but that is hardly a combination of heat and summer. Adding white to the red and green brings to mind things Italian. The combinations are virtually endless and, as mentioned above, best dealt with by professionals.

A chief concern regarding color, and the entire realm of label design, is that of vintner ego expressed to a fault. Obviously, wine labels should be totally representative of their product and their producer, but there have been many wine failures because of overdone labels. Figure 4–7 illustrates some well-designed wine labels that have proven to be successful.

Perhaps even more damaging are underdone labels, with such a lack of character that there is insufficient consumer attraction to make a sale. This condition exists widely throughout the wine industry, principally from vintners who are ignorant to the importance of packaging in consumer appeal or who feel they are saving money by minimizing label cost. This is very poor economy. Vintners must understand that, no matter what the quality of their wine, it is the package that creates the desire for a prospective consumer to try it. There are gold medal-winning wines that go begging because of inept packaging. Conversely, there are mediocre wines that sell actively because of clever packaging.

Label Humor

Some wine labels are funny and are not meant to be, which is truly unfortunate. On the other hand, some vintners set out to generate label smile appeal.

While many wine marketing successes can be attributed to snappy label design, there is a danger that vintners can become typecast and that the winery may be regarded as flippant and insincere. Consumers may not take any wines in the line seriously, and a solid reputation may erode to novelty.

Label Experiments

A case can be made that every wine label is on a pathway of eventually becoming dated. Some wine labels, of course, have a shorter life span than others. Conversely, some labels in the marketplace have not markedly changed for decades and remain successful. Among these are classic European wines, such as some chateau labels that have not changed since before the turn of the century.

Few U.S. vintners can claim classic heritage. Thus, wine packaging, like any other consumer product, undergoes trends that are dynamic. It makes good sense to closely observe consumer reaction, both in the winery tasting room and in wine shops and restaurants. Obviously, positive appeal coupled with positive sales should not be of immediate concern. But when the proverbial “red flags” of consumer lack of interest appear, it behooves the vintner to seriously consider new label design.

Again, it makes sense to consult with packaging professionals to ascertain what possibilities for change may be recommended. A good packaging company or label-printing firm should have a design team available for the vintner at little or no cost. Their incentive is, of course, continued business with that customer.

It is impractical for most vintners to market-test new packaging concepts in stores and restaurants. The cost for small print runs and the complexity of obtaining ATF approval for each new label discourages this concept. On the other hand, mock-ups can be shown to interested visitors and tourists in the winery tasting room. Comments can be gathered over a sufficient period of time from which rational decisions can be made for the change of choice.

Before proceeding with label design, it is a good idea to integrate fully with the marketing program, as discussed in Chapter 13.

Key ATF Labeling Regulations

The following are ATF regulations concerning mandatory label information (27 CFR, Part 4, Section 4.32):

- (a) There shall be stated on the brand label:
 - (1) Brand name, in accordance with Section 4.33.
 - (2) Class, type, or other designation, in accordance with Section 4.34.
 - (3) Alcohol content, in accordance with Section 4.36.

- (b) There shall be stated on any label affixed to the container:
 - (1) Name and address, in accordance with Section 4.35.
 - (2) Net contents, in accordance with Section 4.37.

The following are regulations concerning brand names (27 CFR, Part 4, Section 4.33):

- (a) *General.* The product shall bear a brand name, except that if not sold under a brand name, then the name of the person required to appear on the brand label shall be deemed a brand name for the purpose of this part.
- (b) *Misleading brand names.* No label shall contain any brand name, which, standing alone or in association with other printed or graphic matter creates any impression or inference as to the age, origin, identity or other characteristics of the product unless the Director finds that such brand name, either when qualified by the word "brand" or when not so qualified, conveys no erroneous impressions as the age, origin, identity, or other characteristics of the product.

The following are regulations concerning wine class and type (27 CFR, Part 4, Section 4.34):

- (a) The class of the wine shall be stated in conformity with Subpart C of this part if the wine is defined therein, except that "table" ("light") and "dessert" wines need not be designated as such. In the case of still grape wine there may appear, in lieu of the class designation, any grape-type designation, semigenic geographic type designation, or geographic distinctive designation, to which the wine may be entitled. In the case of champagne, or crackling wines, the type designation "champagne" or "crackling wine" ("petillant wine," "Frizzante wine") may appear in lieu of the class designation "sparkling wine". In the case of wine which has a total solids content of more than 17 grams per 100 cubic centimeters the words "extra sweet," "specially sweetened," "specially sweet" or "sweetened with excess sugar" shall be stated as a part of the class and type designation. The last of these quoted phrases shall appear where required by Part 240 of this chapter, on wines sweetened with sugar in excess of the maximum quantities specified in such regulations. If the class of the wine is not defined in Subpart C, a truthful and adequate statement of composition shall appear upon the brand label of the product in lieu of a class designation. In addition to the mandatory designation for the wine, there may be stated a distinctive or fanciful name, or a designation in accordance with trade understanding. All parts of the designation of the wine, whether mandatory or optional, shall be in direct conjunction and in lettering substantially of the same size and kind.
- (b) An appellation of origin such as "American," "New York," "Napa Valley," or "Chilean," disclosing the true place of origin of the wine, shall appear in direct conjunction with and in lettering substantially as conspicuous as the class and type designation if:
 - (1) A grape type (varietal) designation is used under the provisions of Section 4.32a;

- (2) A semi-generic type designation is employed as the class and type designation of the wine pursuant to Section 4.24b;
- (3) A product name is qualified with the word “Brand” under the requirements of (j); or
- (4) The wine is labeled with the year of harvest of the grapes, and otherwise conforms with the provisions of Section 4.27. The appellation of origin for vintage wine shall be other than a country.

The following are regulations concerning name and address (27 CFR, Part 4, Section 4.35):

- (a) *American wine.* On labels of containers of American wine, there shall be stated the name of the bottler or packer and the place where bottled or packed (or until January 1, 1985, in lieu of such place, the principal place of business of the bottler or packer if in the same State where the wine was bottled or packed, and, if bottled or packed on bonded premises, the ATF registry number of the premises) immediately preceded by the words “bottled by” or “packed by” except that:
 - (1) If the bottler or packer is also the person who made not less than 75% of such wine by fermenting the must and clarifying the resulting wine, or if such person treated the wine in such manner as to change the class thereof, there may be stated, in lieu of the words “bottled by” or “packed by,” the words “produced and bottled by,” or “produced and packed by.”
 - (2) If the bottler or packer has also either made or treated the wine, otherwise than as described in paragraph (a)(1) of this section, there may be stated, in lieu of the words “Bottled by” or “Packed by,” the phrases “Blended and bottled (packed) by,” “Rectified and bottled (packed) by,” “Prepared and bottled (packed) by,” “Made and bottled (packed) by” as the case may be, or, in the case of imitation wine only, “Manufactured and bottled (packed) by.”
 - (3) In addition to the name of the bottler or packer and the place where bottled or packed (but not in lieu thereof) there may be stated the name and address of any other person for whom such wine is bottled or packed, immediately preceded by the words “Bottled for” or “Packed for” or “Distributed by” or other similar statement; or the name and principal place of business of the rectifier, blender, or maker, immediately preceded by the words “Rectified by,” “Blended by” or “Made by,” respectively, or, in the case of imitation wine only, “Manufactured by.”
- (b) *Form of address.* The “place” stated shall be the post office address (after December 31, 1984, the post office address shall be the address shown on the basic permit or other qualifying document of the premises at which the operations took place; and there shall be shown the address for each operation which is designated on the label. An example of such use would be “Produced at Gilroy, California, and bottled at San Mateo, California, by XYZ Winery”), except that the street address may be omitted. No additional places or addresses shall be stated for the same person unless
 - (1) such person is actively engaged in the conduct of an additional bona fide and actual alcoholic beverage business at such additional place or address, and (2) the

label also contains in direct conjunction therewith, appropriate descriptive material indicating the function occurring at such additional place or address in connection with the particular product.

- (c) *Trade or operating names.* The trade or operating name of any person appearing upon any label shall be identical with the name appearing on the basic permit or notice.

The following are regulations concerning alcoholic content (27 CFR, Part 4, Section 4.36):

- (a) Alcoholic content shall be stated in the case of wines containing more than 14% of alcohol by volume, and, in the case of wine containing 14% or less of alcohol by volume, either the type designation "table" wine ("light" wine) or the alcoholic content shall be stated. Any statement of alcoholic content shall be made as prescribed in paragraph (b) of this section.

The following are regulations concerning net contents (27 CFR, Part 4, Section 4.37):

- (a) *Statement of net contents.* The net contents of wine for which a standard of fill is prescribed in Section 4.73 shall be stated in the same manner and form as set forth in the standard of fill.

The following are regulations concerning general requirements (27 CFR, Part 4, Section 4.38):

- (a) *Legibility.* All labels shall be so designated that all the statements thereon required by Section 4.30 through 4.39 are readily legible under ordinary conditions, and all such statements shall be on a contrasting background.
- (b) *Size of type.*
 - (1) Containers of more than 187 millimeters. All mandatory information required on labels by this part, except the alcoholic content statement, shall be in script, type, or printing not smaller than 2 millimeters; except that if contained among other descriptive or explanatory information, the script, type, or printing of the mandatory information shall be of a size substantially more conspicuous than that of the descriptive or explanatory information.
- ...
- (f) *Additional information on labels.* Labels may contain information other than the mandatory label information required by Sections 4.30 through 4.39, if such information complies with the requirements of such sections and does not conflict with, nor in any manner qualify statements required by this part. In addition, information which is truthful, accurate, and specific, and which is neither disparaging nor misleading may appear on wine labels.

The following are regulations concerning prohibited practices (27 CFR, Part 4, Section 4.39):

- (a) *Statements on labels.* Containers of wine, or any label on such containers, or any individual covering, carton, or other wrapper of such container, or any written, printed, graphic, or other matter accompanying such container to the consumer shall not contain:
- (1) Any statement that is false or untrue in any particular, or that, irrespective of falsity, directly, or by ambiguity, omission, or inference, or by the addition of irrelevant, scientific or technical matter, tends to create a misleading impression.
 - (2) Any statement that is disparaging of a competitor's products.
 - (3) Any statement, design, device, or representation which is obscene or indecent.
 - (4) Any statement, design, device, or representation of or relating to analyses, standards, or tests, irrespective of falsity, which the Director finds to be likely to mislead the consumer.
 - (5) Any statement, design, device or representation of or relating to any guarantee, irrespective of falsity, which the Director finds to be likely to mislead the consumer. Money-back guarantees are not prohibited.
 - (6) A trade or brand name that is the name of any living individual of public prominence, or existing private or public organization, or is a name that is in simulation or is an abbreviation thereof, or any graphic, pictorial, or emblematic representation of any such individual or organization, if the use of such name or representation is likely falsely to lead the consumer to believe that the product has been endorsed, made, or used by, or produced for, or under the supervision of, or in accordance with the specifications of, such individual or organization; Provided, that this paragraph shall not apply to the use of the name of any person engaged in business as a producer, blender, rectifier, importer, wholesaler, retailer, bottler, or warehouseman of wine, nor to the use of any person of a trade or brand name that is the name of any living individual of public prominence or existing private or public organization, providing such trade or brand name was used by him or his predecessors in interest prior to August 29, 1935.
 - (7) Any statement, design, device, or representation (other than a statement of alcohol content in conformity with Section 4.36), which tends to create the impression that a wine:
 - (i) Contains distilled spirits;
 - (ii) Is comparable to a distilled spirit; or
 - (iii) Has intoxicating qualities.

However, if a statement of composition is required to appear as the designation of a product not defined in these regulations, such statement of composition may include a reference to the type of distilled spirits contained therein.
 - (8) Any coined word or name in the brand name or class and type designation which simulates, imitates, or which tends to create the impression that the wine so labeled is entitled to bear, any class, type, or permitted designation recognized by the regulations in this part unless such wine conforms to the requirements prescribed with respect to such designation and is in fact so designated on its labels.

- (b) *Statement of age.* No statement of age or representation relative to age (including words or devices in any brand name or mark) shall be made except (1) for vintage wine, in accordance with the provisions of Section 4.27; (2) references relating to methods of wine production involving storage or aging in accordance with Section 4.38(f); or (3) use of the word "old" as part of a brand name.

It should be noted that, providing no individual state regulation prohibits the practice, 27 CFR, Subpart C, 4.50, permits the use of vintage labels, without any appellation of origin, upon wines made from out-of-state fruit, provided that wine is sold exclusively within the state boundaries of the bottler.

The following are regulations concerning certificates of label approval (27 CFR, Part 4, Section 4.50):

- (a) Any bottler or packer of wine shall be exempt from the requirements of this section if upon application the bottler or packer shows to the satisfaction of the Director that the wine to be bottled or packed is not to be sold, offered for sale, or shipped or delivered for shipment, or otherwise introduced in interstate or foreign commerce. A "Certificate of Exemption from Label Approval under the Federal Alcohol Administration Act" (Form 1650) shall be issued by the Director upon application upon the form designation "Application for Certificate of Exemption from Label Approval under the Federal Alcohol Administration Act" (Form 1648) properly filled out and certified to by the applicant.
- (b) *Statement of bottling dates.* The statement of any bottling date shall not be deemed to be a representation relative to age, if such statement appears in lettering not greater than 8-point Gothic caps and in the following form: "Bottled in _" (inserting the year in which the wine was bottled).
- (c) *Simulation of Government stamps.*
 - (1) No labels shall be of such design as to resemble or simulate a stamp of the United States Government or any State or foreign government. No label, other than stamps authorized or required by the United States Government or any State or foreign government, shall state or indicate that the wine contained in the labeled container is produced, blended, bottled, packed, or sold under, or in accordance with, any municipal, State or Federal Government authorization, law, or regulation, unless such statement is required or specifically authorized by the laws or regulations of a foreign country. If the municipal, State, or Federal Government permit number is stated upon a label, it shall not be accompanied by any additional statement relating thereto.

And, any wine bottle on or after November 18, 1989, must include a statement on the label precisely as follows:

Government warning: (1) According to the Surgeon General, women should not drink alcoholic beverages during pregnancy because of the risk of birth defects. (2) Consumption of alcoholic beverages impairs your ability to drive a car or operate machinery, and may cause health problems.

WINEMAKING PERILS AND PITFALLS

Every winemaker will encounter wines that may fall short of expectations or may express faults that are disappointing. Some defects can be prevented if they are detected in time. The best way to avoid such problems is, of course, to ensure that equipment, materials, and methods are in proper order beforehand. Wines with serious problems are generally the products of grapes and/or procedures with equally serious problems.

Table 4-2 is provided to help identify some of the more common difficulties in finished wines. A comprehensive wine laboratory service should be employed to examine and advise upon all wines felt to be out of order.

Table 4-2 Common difficulties in finished wines.

Effect/Symptom	Cause/Comments
Visual Mode	
Browning	Exposure to air Low free sulfur dioxide level
Cloudy	Yeast fermentation Bacterial fermentation Insufficient peptic enzyme treatment Insufficient fining High iron level
Color deficiency	Overcropping of grapes Unripe grapes High free sulfur dioxide levels Amelioration dilution
Color instability	Grape rot High pH level Oxidative enzyme reaction High free sulfur dioxide level Bacterial fermentation
Effervescence	Yeast fermentation Bacterial fermentation
Floating solids	Cork dust Cork paraffin Equipment lubricants Insufficient sanitation
Haziness	High copper level Protein instability
Precipitate	Autolyzed yeast Breached filter media Potassium bitartrate crystals Calcium tartrate crystals High iron level Insufficient sanitation

(continued)

Table 4-2 (continued).

Effect/Symptom	Cause/Comments
Suspended solids	Filter fibers Mold from bottling equipment Case dust Insufficient sanitation
<i>Olfactory (Flavor) Mode</i>	
<i>Effect/Symptom</i>	<i>Cause/Comments</i>
Aroma deficiency	Overcropping of grapes Unripe grapes Amelioration dilution
Burn nasal passages	High alcohol
Burnt match	High sulfur dioxide level
Buttery	Malolactic fermentation
Cooked (burnt caramel)	High fermentation temperature Exposure to air
Hydrogen sulfide (rotten eggs)	Sulfur residues from vineyard Wild yeast fermentation Insufficient yeast nutrients Excessive lees contact Insufficient sanitation
Mercaptans (skunky)	Extended hydrogen sulfide condition Bacterial fermentation Insufficient sanitation
Moldy/musty	Moldy grapes Moldy barrel Decayed barrel Bacterial fermentation Decayed cork Insufficient sanitation
Oxidization (stale bread)	Wild yeast fermentation Exposure to air
Paint thinner	Low sulfur dioxide level Bacterial fermentation
Prune-like	Low sulfur dioxide level Dehydrated grapes High pH
Rubbery	High fermentation temperature
Vinegary	Sulfide compounds Bacterial fermentation Low sulfur dioxide level

Table 4-2 (continued).

Effect/Symptom	Cause/Comments
Wet wool	Sulfide compounds Wild yeast fermentation Bacterial fermentation Low sulfur dioxide level Insufficient sanitation
Woody	Insufficient new barrel treatment Excessive barrel storage
<i>Gustatory (Taste) Mode</i>	
<i>Effect/Symptom</i>	<i>Cause/Comments</i>
Astringency	Unripe grapes High phenol level in grapes Insufficient destemming
Bitter	High sulfur dioxide level at crusher Unripe grapes High phenol level in grapes Cracked grape seeds High sulfur dioxide level at crusher
Coffee-like	Sulfide compounds Insufficient new barrel treatment Excessive barrel storage Insufficient sanitation
Flat—insipid	Overripe grapes Overcropping High pH Amelioration dilution High sweetness level
Metallic	High copper level High iron level
Nutty—toasty	Oxidation Wild yeast fermentation Exposure to air Low sulfur dioxide level
Thin body	Overcropping Amelioration dilution
Vegetal	Sulfide compounds Bacterial fermentation Low sulfur dioxide level
Vinegary	Bacterial fermentation Low sulfur dioxide level
Woody	Insufficient new barrel treatment Excessive barrel storage

CHAPTER 5

WINE CLASSIFICATION

Wine experts generally classify wine into five categories, with the distinctions among the classes based primarily upon major differences in their manner of vinification.

TABLE WINES

The overwhelming majority of the wines produced in the world fall into the category of table wines. These range from the obscure and ordinary to the most expensive and celebrated classics. As the name suggests, table wines are designed for use at the table as a complement to good food. For this same reason, table wines are sometimes referred to as “dinner wines.”

There are white, blush (pink), and red table wines. These are the base wines needed to make every other wine type. While neophytes may prefer wines with a little sweetness, the overwhelming preference among veteran wine consumers is for dry wines. Sugar blunts crisp acidity and has a tendency to mask delicate wine flavors.

Two of the most common terms used in connection with table wine identification are *varietal* and *generic*. Varietal wines are labeled with the variety of grape or

fruit used in making the wine, such as “Chardonnay” or “Johannisberg Riesling.” Generic wines are named for a specific geography of origin, such as Chablis, part of Burgundy, one of only several locales in France where Chardonnay is permitted by regulation to be grown commercially. In America, we have the freedom to plant Chardonnay, or any other vine species, anywhere. In similar fashion, we often see wines labeled as “Rheingau” or “Rheinpfalz” from the German Rheinland, where Johannisberg Riesling is widely cultivated. The Riesling is also planted in many states across our country. ATF regulations permit the commercial use of a select list of European generic names by U.S. vintners, but, as may be expected, this is very controversial.

U.S. wines are, consequently, more commonly varietal, while European wines are typically generic. Table 5–1 shows some examples of generic wines, along with the varietals that are cultivated in those regions.

Among the most widely consumed American table wines are the “jug” wines that can be found in most markets. Unfortunately, these often have less than good reputations, owing to the old myth that “only expensive wines can be good.”

Jug wines are defined as any wine commercially marketed in a container of 1.5-L capacity or larger. The bag-in-a-box packages can also loosely be considered as jug wines. Most jug offerings are California generic and varietal table wines, with a few from eastern American and European vintners.

To some extent, jug wines epitomize the many advancements in wine production technology. Research has met consumer demands for freshness, flavor, and overall appeal with new techniques that allow better wines to be made more quickly at

Table 5–1 Generic wines from varietal grapes.

Generic	Varietal
White table wines	
Alsace (France)	Gewürtztraminer and Pinot Blanc
Chablis (Burgundy in France)	Chardonnay
Gavi (Piemonte in Italy)	Cortese
Graves (Bordeaux in France)	Sauvignon Blanc and Semillon
Rheingau (Germany)	Johannisberg Riesling
Orvieto (Umbria in Italy)	Trebbiano and Malvasia
Red table wines	
Barolo (Piemonte in Italy)	Nebbiolo
Beaujolais (Burgundy in France)	Gamay
Medoc (Bordeaux in France)	Cabernet Sauvignon and Merlot
Chianti (Tuscany in Italy)	Sangiovese
Cote d’Or (Burgundy in France)	Pinot Noir
Rhone (France)	Grenache and Syrah

less cost. To these specific ends, jug wines are primarily designed for immediate consumption and usually do not improve with further aging. One frequently hears that jug wines are “America’s answer to the ordinary wines of Europe.”

ATF rules require table wines to have an alcohol content not exceeding 14% by volume. Commercial table wines must remain “in bond” (federal excise tax [FET] unpaid) at the winery until the FET is paid to the ATF. Vintners producing less than 100,000 gallons per calendar year pay a rate of \$0.17 per gallon FET. Vintners producing more than 100,000 gallons annually are subjected to a sliding FET rate scaled up to a maximum of \$1.07 per gallon, which is paid by all vintners producing 250,000 gallons or more each year.

SPARKLING WINES

These are wines that “effervesce,” or bubble. The first sparkling wines may have been made in the mid-1600s by a French monk named Dom Perignon. He is also thought to have been the first to use plugs made from cork tree bark for bottle stoppers. Apparently the seal was so good that wine bottled in the winter, before it was completely finished fermenting, resumed fermenting in the bottle during springtime temperatures. This resulted in some buildup of carbon dioxide gas pressure inside the bottle during storage, and this was later released as “sparkling” bubbles when the cork was removed.

Nowadays there are people who still use the terms *Champagne* and *sparkling wine* interchangeably, but this is incorrect. Champagne is a sparkling wine-producing region in France and is thus a generic term. All Champagnes are sparkling wines, but not all sparkling wines are Champagne.

Needless to say, fermentation pressure in a table wine bottle would be very dangerous. Modern methodology for the traditional Champagne process, or *methode champenoise*, is by a highly controlled secondary fermentation of a low-alcohol table wine. Pressure may build up to 100 pounds of pressure per square inch in specially designed bottles.

In the Champagne region, the French are permitted by regulation to use only Chardonnay, Pinot Noir, and Pinot Meunier grapes—the latter two having blue-black skins, but white juice inside the berry. “Blanc de Blancs” on the label indicates that the white sparkling wine is made exclusively from white grapes, and Blanc de Noirs is a white made exclusively from black grapes.

Sparkling wine fermentation in tanks is a process called *Charmat*, or the “bulk process.” Due to EEC regulation, sparkling wine made in Germany must be labeled as *Sekt*, while in Italy it is referred to as *Spumante*, and in Spain as *Cava*. In the United States it can legally be referred to as “California Champagne,” or “Napa Valley Champagne,” or with some other ATF-approved preface for identification.

The driest of sparkling wines is generally referred to as *naturel*, while very dry is *brut* or *sec*—these being the most popular. *Demi-sec* is noticeably sweet and *doux* very sweet. One finds *dry* and *extra dry* on sparkling wine labels, too, and these are often somewhere between *sec* and *demi-sec* in sweetness level.

The ATF's FET rate on naturally fermented sparkling wines is \$3.40 per gallon for all producers, regardless of annual output. Artificially carbonated wines are taxed at a rate of \$2.40 per gallon.

Vintners are advised to make sparkling wines only after a high level of proficiency with table winemaking has been achieved. Sparkling wine fermentations require precise calculations and remain very dangerous even in the hands of the most experienced masters. A good place to start is by reading advanced texts on the subject. Seek out seminars and workshops presented by academic and industry groups. Visit other small wineries operated by people who are willing to illustrate the essentials in their sparkling wine operations. Best of all, endeavor to enlist the help of an experienced winemaker through the first batch or two.

DESSERT WINES

As the name implies, these are wines generally consumed with, or instead of, dessert courses. Credit is given to the medieval Moors occupying Spain and Portugal, despite their Islamic prohibition, for inventing the distillation process from which "wine spirits," or brandy, is made. But it was the British, in the 17th and 18th centuries, who perfected the addition of brandy to wines grown in Portugal that are now exemplified by sweet, rich Ports. Wines having undergone a brandy addition are sometimes referred to as "fortified." Sherry from Spain, Madeira from the isle of Madeira, and Marsala from Sicily are also good examples of fine dessert wines.

Dessert wines are usually made by the addition of grape brandy to a fermenting juice or must, less often to a completely fermented table wine. The brandy addition usually increases alcohol content up to 19% to 20% by volume, not to exceed 24% by ATF regulations found in 27 CFR, Part 4, Section 24.233. Dessert wine FET is \$1.57 per gallon for all commercial vintners.

The brandy used in wineries is "high-proof" wine spirits refined to more than 95% ethanol and, following approval by the ATF, is added in bond so as to avoid paying the comparatively expensive brandy FET.

Vintners are cautioned to investigate state winemaking regulations carefully, as some states prohibit high-proof brandy storage and additions.

APERITIF WINES

These are wines that are designed to serve as appetizers to prime the palate before a special meal. Dubonnet, Lillet, and Pernod are well-known proprietary names of

light and dark aperitifs in Europe. In the United States, aperitif wines are consumed largely as cocktail “mixers,” such as dry vermouth in martinis and sweet vermouth in Manhattans.

Aperitif wines are fortified with brandy, generally up to a level of 17% or 18% alcohol by volume, and carry the same \$1.57 FET as for dessert wines. ATF approval of winery applications is required prior to making these “special natural” wines, as they are sometimes called. Essences from various barks, herbs, peels, roots, and/or spices, combined in a special closely guarded, sometimes patented, recipe, are added to create a consistently distinctive wine.

The history of this category may date to the ancient Greeks, who added, among other things, honey, seawater, and wood resins to their wines. Coniferous resins are still added to Greek wines called *retsina*.

POP WINES

This is a wine type thought by some to be named after its popularity among young adults and ethnic groups. Others will testify that pop wines have a similarity to soda pop. Perhaps both sources apply. In any event, it is a category that has emerged just during the past several decades.

Commercially produced pop wines may be made at alcohol levels under 14% by volume, in which case they are subject to the table wine FET rate, or from 14% to 24% by volume, in which case they are subject to the dessert wine FET rate.

In short, *Pop* wines closely resemble aperitif wines, except that the added essences are more exotic, typically from pronounced fruit and/or berry flavors.

CHAPTER 6

WINERY DESIGN

Many people drawn toward commercial wine production have more rewards in mind than making a profit. A flair for the romantic life or some set of Bacchanalian notions leads them to the idea of starting their own winery. The prospect of realizing such a dream is almost always driven by eagerness: the excitement of awaiting the first set of drawings and the ensuing anxiety to get the structure operational.

This type of enthusiasm is as it should be, as long as such passions don't derail prudent judgment in working with winery design concepts. The point here is to resist the "add-ons syndrome," which can burden a workable plan with untimely cost.

An essential key in winery start-up is to remain as financially "liquid" as possible, with a tenacious attitude toward conserving capital. One old adage is that "to make a small fortune in the wine business one must start with a large fortune."

Most aspiring commercial vintners do not have large fortunes, and finding alternative ways of getting things properly accomplished with significant reductions or replacement of dollar outlays is absolutely imperative.

MANAGEMENT

The first order of new vintner management is the pursuit of exhaustive research and planning. This includes the market audit, business strategy, site selection, production design, financial support, and all the clever synergies that will eventually separate winners from losers. A careless approach is an invitation to disaster. New wineries can be overdone but never overplanned.

It is not the way of modern business to forgive poor management. Whether it can be blamed upon bad information, poor production quality, insufficient capital, superior competition, or some other excuse, once the final analysis of a failed business is made, it will always reflect upon the inability of management to overcome adversity.

Even with ideal plans in place, winery managers can expect that they will have to make changes to deal with unexpected events. This is called *contingency management* and is a particularly important factor in steering new wineries out of harm's way. There are vintners who will attest that their business is always in the contingency mode.

The gathering and processing of salient production, marketing, and administrative information are essential to effective decision making. Using the full benefit of one of the integrated computer software programs for wineries is the first step recommended in assuming this responsibility.

These programs start with accounting for grapes received at the winery and go on to a quick and easy recording of origin, weight, units, Brix, acidity, pH, lot number assignments, grower settlements, must and juice adjustments, and almost any other particular needs a vintner may wish to address. The system will then track fermentation, racking, materials and treatments, blending, fining, aging, filtration, and bottling, all with attendant analyses. Each lot of wine should be singled out to show its share of both variable and fixed costs. Tracking will continue for each wine through warehousing, sales and depletion, customer history, order processing, and printing of bills of lading and invoices, all the while keeping track of complex discount rates, promotional costs, sales incentives, and performance rates. Integration permits the creation of all required ATF reporting documents, as well as up-to-the-minute reports for cost accounting, accounts receivable, accounts payable, and ledgers. Computer software is available for tracking all aspects of production and for cost analyses and recordkeeping (Exhibits 6-1 and 6-2).

Good management will see to it that such systems are provided with all the inputs necessary to generate outputs that can go a long way toward effective decision making for any part of the winery business.

The standard expectations for managers of any business are tireless pursuit of tough choices, protection of assets, achievement of goals, exceptional foresightfulness, and inspiring leadership. This is classic entrepreneurship. Those who are interested in working from 8:00 to 5:00, 5 days per week, should consider doing something else.

Exhibit 6-1 Printout of computerized grape receiving and press record.
Courtesy: Agrisystems, Inc., Littleton, CO.

29-Sep-85
39

WORK ORDER

Press

From: z (not specified)
Via: UPS

For/By: (not specified)

Description Product Code	Container	Lot Number	Location	Add	Remove	Unit Wt Quality	Count/Unit Count/UnitWt	OK?
Grapes, Pinot Gris P. Gris Grapes...			Press	0	5.19	Tons	0 0	
Juice, Pinot Gris P. Gris Juice...	Tank, 136	lg1.....	Main Warehouse	802	0	gallons	0 0	

Press let lg1 to tank 136. No SO2 addition. chill to 45 - 50 deg. gas head space. hold 24 to 48 hrs. - before yeast - raise to before yeast

02-Oct-85
41

WORK ORDER

Rack

From: z (not specified)
Via:

For/By: (not specified)

Description Product Code	Container	Lot Number	Location	Add	Remove	Unit Wt Quality	Count/Unit Count/UnitWt	OK?
Juice, Pinot Gris P. Gris Juice...	Tank, 136	lg1.....	Main Warehouse	0	802	gallons	0 OK	
Juice, Pinot Gris P. Gris Juice...	Tank, 127	lg1.....	Main Warehouse	822	0	gallons	0 0	
Yeast Yeast P. Dm....		chem3.....	Main Warehouse	0	2.2	lbs	0 0	
Tartaric Acid Tartaric Acid....		chem2.....	Main Warehouse	0	4.9	lbs	0 0	

tank 136 - rack to tank 127 - add 2.2 lbs. p. dm yeast, 4.9 lbs tartaric acid

Exhibit 6-2 Printout of computerized inventory cost analysis.
Courtesy: Agrisystems, Inc., Littleton, CO.

Inventory Activities In Progress

15-Jul-96

Reference Product	Activity	Lot Name Container	Organization Location	Quantity	Supplier	Carrier	Ttl Costs	Unit Cost	Wt/Unit	Weight	Volume	Quality Saleable
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Friday, September 29, 1995

39	Press		(not specified)		z (not specified)							
					UPS							
Juice, Pinot Gris		tg1.....		+	802 gallon		\$2,595.00	\$3.24	/gallon	6416	802	
P. Gris.Juice....		Tank. 136	Main Warehouse	-	0		\$0.00	\$0.00		0	0	
		Total 39		+	802		\$2,595.00			6416	802	
Press for tg1 to tank 136. No												
SO2 addition. chill to 45 - 50												
deg. gas head space. hold 24												
to 48 hrs. - before yeast -												
raise to before yeast												
				-	0		\$0.00			0	0	
				Var			\$2,595.00			6416	802	
				+	802		\$2,595.00			6416		
				-	0		\$0.00			0		
				Var			\$2,595.00			6416	802	

FEASIBILITY AND FINANCE

The structure of the winery business is another essential element facing anyone considering operating a commercial winery.

Once the decision is made to move forward with a new winery project, it is a good idea to employ the services of both an attorney and an accountant. They should be able to provide answers to the many questions that surround whether such a venture is feasible or realistic. Further, they should explain all the hurdles, perils, and pitfalls that are involved with starting a business. This can be challenging, even frustrating, and is often where the whole idea of being a commercial vintner ends. If the new project has merit, these consultants can suggest the best way the business may be structured to serve an individual, a family, a partnership, or a group of investors.

A business plan should be written that explains the rationale for projecting how the winery shall succeed. Several computer software programs include good outlines for generating a sound business plan. This document should include a market audit in which perceived product demand is identified, tested, and measured. Product design parameters and production levels should create a supply that parallels the perceived demand. Marketing strategies should be outlined in detail. Pro forma income statements, cash-flow analyses, and balance sheets should be prepared, along with detailed personal financial statements and vita. While supporting statistics, demographics, and other specific testimony can be part of a good business plan, these should be simple and direct.

Table 6-1 shows a typical format for a pro forma income statement for a new small winery in its second year of operation.

The income statements in Table 6-1 project a poor performance as related to most new businesses. Nevertheless, reaching break-even in the second year is generally considered acceptable, as most new winery businesses, even under favorable operating conditions, fail to become profitable during the first several years.

Construction costs and operating expenses are the major needs for capital in a new winery. Unfortunately, they are also the most difficult for which to obtain credit. How loans may be arranged is heavily dependent upon the financial condition of those who will be principals in the business. The analysis of this situation is typically the point where financial negotiations begin.

Capital Investments

Most capital investments in new wineries are made by the operators themselves. Since few wineries make a profit during the first few years of operation, it is rare that external venture capital is raised for start-up. Venture capital investors justify risk by demanding large rates of return and major shares of ownership. It is much the same for initial public offerings, or IPOs. These devices are usually structured

Table 6-1 Pro forma income statement, second fiscal year.

XYZ WINERY	
Gross sales:	
1,000 cases (12 × 750-mL bottles) @ retail (\$100/case)	\$100,000
1,000 cases @ wholesale (\$50/case)	50,000
Gift shop sales @ retail	<u>25,000</u>
Total gross sales	\$175,000
Less:	
Transportation out (wholesale @ \$5/case)	5,000
State excise taxes @ \$0.05/gal.	240
Federal excise taxes	—
Returns and allowances @ 3% of gross sales	<u>5,250</u>
	10,490
Net sales:	\$164,510
Cost of goods sold:	
Grapes—32 tons @ \$800 (150 gal./ton net yield)	\$25,600
Cellar materials (KMS, enzymes, fining agents, etc.)	1,800
Packaging materials (bottles, labels, corks, etc.)	3,000
Gift shop merchandise	17,500
Transportation in	4,000
Real estate taxes	3,000
Insurance	2,400
Utilities	4,400
Depreciation (building, equipment, fixtures, etc.)	20,000
Repairs and maintenance	3,000
Direct labor (2,200 hours @ \$7/hour)	15,400
Fringe benefits @ 20% of labor	3,080
Miscellaneous manufacturing expense	<u>3,000</u>
Cost of goods sold	106,180
Gross profit	\$58,330
Marketing expense:	
Samples (50 cases @ \$40/case cost)	\$2,000
Pamphlets (10,000 @ \$0.35/each)	3,500
Travel	3,500
Public relations (tasting room materials)	3,500
Sales promotion (retailer activities)	2,500
Advertising	2,000
Direct labor (2,200 hours @ \$7/hour)	<u>15,400</u>

Table 6-1 (continued)

Fringe benefits @ 20% of labor	3,080
Miscellaneous marketing expense	2,000
Marketing expense	<u>\$37,480</u>
Marketing profit	\$20,850
General and administrative expense:	
Office supplies and materials	\$2,000
Computer supplies and materials	1,500
Telephone	2,400
Postage	600
Bonds, licenses and permits	2,000
Professional services	6,000
Travel	3,500
Miscellaneous general and administrative expense	2,000
General and administrative expense	<u>\$20,000</u>
Operating income	<u>\$850</u>

for large amounts of capital, usually not less than \$5 million, designed to expand existing wineries that have demonstrated potential for rapid growth. An IPO will require the services of an underwriter, will create a number of stockholders, and add considerable administrative responsibility to winery management.

Prospective vintners having personal means with which to fund all or part of a winery should seek the counsel of a professional financial expert. There may be advantages in seeking funding from one or more alternative sources to minimize personal risk, maximize equity control, and optimize liquidity.

Lending Institutions

Before making appointments with prospective lenders, be sure you have a copy of your credit report and financial statements so that you know where you stand. Review over and over the amount of capital necessary, and be prepared to fully justify each item for which the loan proceeds will be employed. Know the business plan and pro forma documents thoroughly, and have local personal contacts ready to pledge their support to the project. Ask your attorney and accountant to recommend lending institutions that they believe will be not only the most apt to approve the initial loan but also likely to work with the winery over the long term.

Conventional bank loans supported by substantial collateral are the most common vehicle for funding new and expanded winery projects. With good credit these

types of loans have the advantage of relatively low interest rates (at this writing) but the disadvantage of relatively short repayment terms.

A traditional problem in obtaining loans for commercial wine operations is that the prospective vintner has had limited winery management experience. Although the vintner may be an award-winning winemaker or an accomplished wine marketer, bankers often consider winery management another matter.

Bankers are placing an increasing emphasis upon cash flow financing, as compared to collateral lending, and focusing on the soundness of the business to repay instead of the value of the assets. Consequently, business plans that show dynamic revenues and retained cash liquidity are more apt to be financed. Another field of preparation should be in the development of contingency planning. Bankers will devise "worst-case" scenarios and expect alternative action plans to be in place.

One of the most successful pathways for small winery loans is via the Small Business Administration (SBA). This federal bureau does not loan money directly but does guarantee a major percentage of each loan made by cooperating banks. Typically, an SBA 7(A) loan will guarantee about 75%. This, of course, relieves banks of undue risk, especially with start-up winery operations. At this writing the SBA charges a sliding scale interest rate (on top of the bank's interest rate), starting with 3.0% for the first \$250,000, increasing to 3.5% for the next \$250,000, and so on. Repayment schedules for working capital loans are maximized at 7 years, for equipment loans at 10 years, and for real estate loans at 25 years. SBA 504 loans are designed for existing wineries to finance real estate and facilities expansion. SBA rates and rules change from time to time. Again, the advice and counsel of an attorney and accountant can help a great deal in procuring such loans.

Leasing

Many vintners choose to lease cooperage, tankage, equipment, and vehicles. Leasing companies purchase and own the items agreed upon, and the winery pays a monthly payment, in much the same way as personal auto and truck leasing operates. At the end of the lease, vintners are usually given the option of buying the items at a predetermined residual value, leasing the items again under a new residual, or turning the items over to the leasing company. The major advantage of leasing is the conservation of capital by smaller down payments, more flexibility in monthly payment options and the possibility of tax advantages. Leasing also provides an opportunity to obtain more advanced equipment at the end of the lease. The major disadvantages of leasing are the comparatively higher rates of interest and the possibility of being "locked in" with pieces of equipment that are troublesome.

Insurance

Prospective vintners must understand that winery operations embrace more than general liability. Beyond this are various forms of product liability that have increasingly become litigated. Negligence regarding the winery facility, such as poorly marked steps, loose railings on stairs, wet floors, or uncovered drains, can, of course be avoided with good management. It is, however, a mistake to assume that a new winery cannot be held legally responsible for events over which it has no control. Health problems, broken bottle cuts, and wine stains are damages for which hundreds of such suits are brought to court. Personal injury and property damage caused in automobile accidents by drivers who have left a winery tasting room have resulted in vintners losing their entire estate.

Consequently, membership in the American Vintners Association (AVA) should be considered. This organization is located at 1301 Pennsylvania Avenue NW, Suite 500, Washington, DC 20004. The AVA sponsors most forms of vineyard and winery insurance.

LOCATION

The ultimate effectiveness of administrative, production, and marketing functions is dependent upon winery location. Some of the key administrative parameters involve regulatory matters—federal, state, and sometimes local too.

In that wineries produce an alcoholic beverage, prospective vintners can expect, over and above other small businesses, an extra list of state and local concerns attached to every potential winery site. There are, of course, different beverage control regulations for each state, which must be carefully considered when choosing the winery location. These often include minimal distances from the highway serving a winery and restrictions regarding the distance between the winery and other buildings, such as churches and schools. Local codes and statutes are often a bit more political in nature and may involve permanent zoning variances, utility services, wastewater treatment, building permits, and highway access. Figure 6–1 illustrates an ideal new estate winery facility.

Along with all the legal site parameters, it is a good idea to have the blessing of all local officials before committing to any site. A good public relations program can go a long way to secure a warm welcome and operating assistance in a community. Being at odds with the local citizens can result in insurmountable hurdles. The general attitude that the community may have about a new winery in the area is important. A good locale can result in ongoing support that facilitates the winery's ultimate success. Persistence in the establishment of a facility within hostile territory can be very aggravating, frustrating, and, most importantly, an invitation to financial disaster.



Fig. 6-1 A newly constructed estate winery. Photo by authors.

The premiums for adequate insurance coverages, particularly fire insurance, are generally heavily influenced by winery location. Attracting quality employees is also easier when it is convenient for them to get to work.

Production parameters need careful scrutiny when one is choosing a winery site. A readily accessible supply of 220-volt, three-phase electric power is preferred, along with plenty of quality water. There must be an adequate method to readily dispose of effluents that satisfies both the ATF and ABC or local authorities. The winery should ideally be located near vineyards so that delivery time can be minimized during the harvest season. This reduces time in which fruit quality can deteriorate and allows for maximizing personnel deployment during the concentrated seasonal demands in both vineyard and winery. Ideally, the winery location should have good access to freight pickup and delivery. Trade services such as electricians, plumbers, mechanics, air-conditioning and refrigeration technicians, processing engineers, and other specialists should be close at hand.

Marketing is generally the management area in which new wineries struggle or fail. Most newly established vintners must depend upon immediate revenue to help stem the tide of capital outlay in building and establishing their winery. Retail sales at the winery are essential.

Locating the winery near a metropolitan area has the potential for several added market resources. Highway traffic to and from the city provides a high-volume source of commuters, business travelers, and, most important, tourists.

Promotional efforts at the winery, such as spring celebrations, summer concerts, harvest festivals, and holiday programs, have a much greater chance for success with a large populace to draw from within a reasonable distance. An urban concentration of wine retail outlets and restaurants enables vintners to operate, if state ABC regulations permit, more efficiently as their own wholesale channel. Urban proximity also facilitates access to professional marketing services such as label design and printing, brochure development and production, advertising creation and placement, sales promotion programming, and other resources necessary to optimize revenue generation.

Apart from these general suggestions, there are no rules for determining winery location. Sometimes prospective vintners already own property that is well known in the area, such as a restaurant or a farmer's market, where the addition of a winery may seem feasible.

On the other hand, there is risk that the vintner will exercise poor judgment, often swayed by personal bias, as to whether a location really is "well known." Forcing a justification for establishing the winery at a remote location is a common error in site selection. A farm situated back from a secondary road is rarely a good choice for a winery site. Conversely, a site situated within a mile or so of an intersection to a busy highway typically has a far better chance of attracting sufficient visitors from whom to achieve adequate retail sales.

Another major factor in site selection is lot size. All too often, prospective vintners provide inadequate room for parking, for outdoor events, and, most important, for long-term growth.

SCALE AND SIZE

Perhaps no other manufacturing business is as forgiving of the poor economies of small size as is wine production. The higher price categories of wines constitute a very small share of the total market but may be sufficient to make economic sense of many small winery projects.

Most aspiring vintners have a well-defined plan for what they are going to produce long before any pro forma business statements are drafted. If a thorough market study reveals that a new winery may successfully market 3,000 cases (7,200 gallons) of two white wine types during the first year of operation, then the winery structure and its equipment should be geared to produce that amount, plus an appropriate capacity for processing and short-term growth. This might, for example, consist of the following 10,150 gallons of tankage:

- six 1,000-gallon closed jacketed stainless-steel tanks
- six 500-gallon closed jacketed stainless-steel tanks
- two 300-gallon refrigerated stainless-steel mixing tanks
- ten 55-gallon stainless-steel drums

If, instead, the same amount of two red wine types is to be produced, winery space and capacity will need to be significantly greater, as red wine aging regimens generally require at least 1 additional year of inventory for barrel aging. Consequently, the amount might consist of the following 15,400 gallons of tankage:

- twelve 500-gallon stainless-steel tanks with dome manways or “floating” heads
- two 300-gallon refrigerated stainless-steel mixing tanks
- ten 55-gallon stainless-steel drums
- one hundred and fifty 55-gallon oak barrels

Usually both white and red table wines are included in the planned product portfolio. Producing a total of 3,000 cases over a line of, say, two whites and two reds provides more efficiency and flexibility in tankage. Consequently, one might find the following 12,275 gallons of capacity sufficient to fulfill combined short-term needs:

- four 1,000-gallon closed jacketed stainless steel tanks
- six 500-gallon stainless-steel tanks with dome manways or “floating” heads
- two 300-gallon refrigerated stainless-steel mixing tanks
- ten 55-gallon stainless-steel drums
- seventy-five 55-gallon oak barrels

Under this arrangement, the 500-gallon tanks can serve double duty as both red wine fermenters and white wine storage vessels. Each time another wine type is added to the product line, a corresponding demand is placed upon space and flexibility of winery capacity, both in tankage and in cased-goods storage.

Depending upon design motif, a new 15,000 gallon winery should require a minimum of 5,000 square feet and should have side walls 12 feet high, providing about 60,000 cubic feet of space. This is, of course, only a broad estimate. Tight design modes can greatly reduce this, and extensive tasting-retail areas can likewise increase the space needed.

DESIGN MOTIF

It is essential that winery design fit the personality of the vintner and the environment. Almost any chosen architectural flavor can be incorporated into an efficient floor plan. Style modes should all support a central theme. Visitors will be confused to find a winery with a Spanish-style motif producing wines with French names in a countryside that is predominantly German. Perhaps even worse is a lack of design flair—a facility that is dull and uninteresting. Function in design should, of course, attend the requirements for maximum product quality, but it is equally important that winery design be executed with good taste and create a pleasant

atmosphere. Overdone esthetic interpretations are often gauche and create negative visitor impressions.

CASE STUDIES

Remodeled Small Barn

The first example is a remodeled barn winery having a capacity of about 8,000 gallons with a “no frills” approach. It has an interior space of 1,971 square feet, along with another 432 square feet of outdoor concrete apron (Figure 6–2).

The apron and lower floor were poured with “three-bag mix” (a superior grade) concrete reinforced with steel mesh. A floor pitch of about 1% was made toward the central drain trough so that cleaning water could not collect and provide a medium for growth of microbes. Several coats of food grade concrete sealer were applied to protect the floor from corrosive fruit acids.

The apron is used for crushing, stemming, and pressing, with that equipment stored out of the way in an adjacent storage area during the balance of the year. Two refrigerated tanks once used for farm milk storage were overhauled and installed in this facility. One above the other, these tanks provide instant chilling for white and blush juice extracted from the press and chilling for water that can be circulated through fermenting tank jackets. The agitators allow for gentle mixing operations needed for bulk wine processing operations throughout the year.

The cold room uses a remodeled chilling device taken from a semitrailer refrigerated van. This can operate to maintain either 55°F for white wine fermentation during the vintage season or 26°F for tartrate stabilization. Seven custom-designed single-wall stainless-steel tanks were chosen to maximize initial capacity.

Five 250-gallon jacketed stainless-steel tanks with dome manways and heavy-duty agitators serve as red wine fermenters. The jackets serve for either heating or cooling fermenting red musts. This winery is in a cool climate and often requires cold musts to be heated to reach the ideal 85°F temperature required by the vintner. The agitators are not essential but came as a bonus on this set of used food-processing tanks. All five tanks are mounted upon legs sufficient for a gravity outlet feed into the press, which is wheeled in from the apron as needed.

The manual bottling and packaging setup is minimal in capacity and helps to save many thousands of dollars in more sophisticated machinery. Empty bottles are brought from another barn storage area when required.

The second floor has a ground-level entry opposite the apron and contains the laboratory-office, tasting bar, retail shop, and a single restroom. This was subsequently changed when state public facilities regulations were found to require two restrooms. Obviously, the two levels make for a rather arduous routine for tourists and transfer of cased goods for retail. See Figure 6–3 for the layout of the interior elevations.

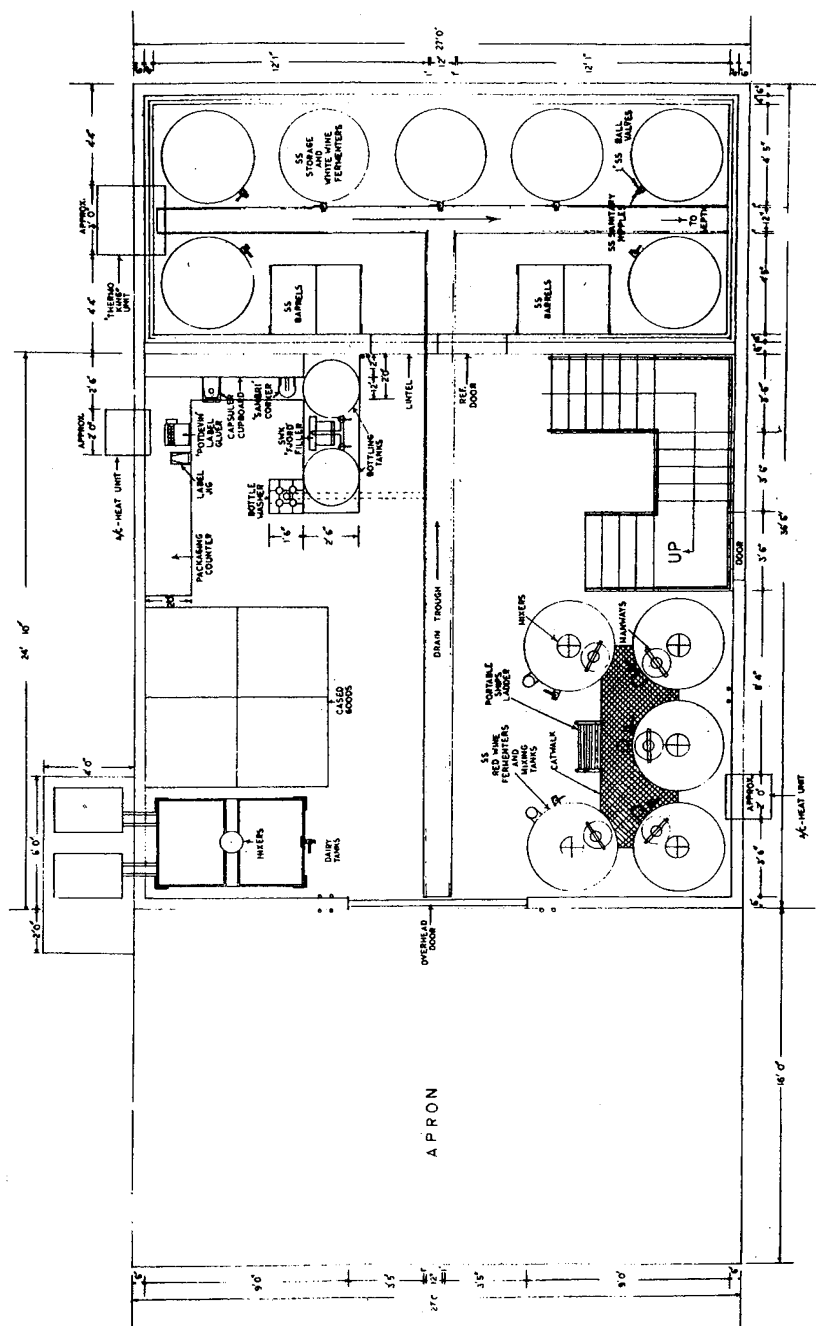


Fig. 6-2 Floor plan of remodeled barn winery.

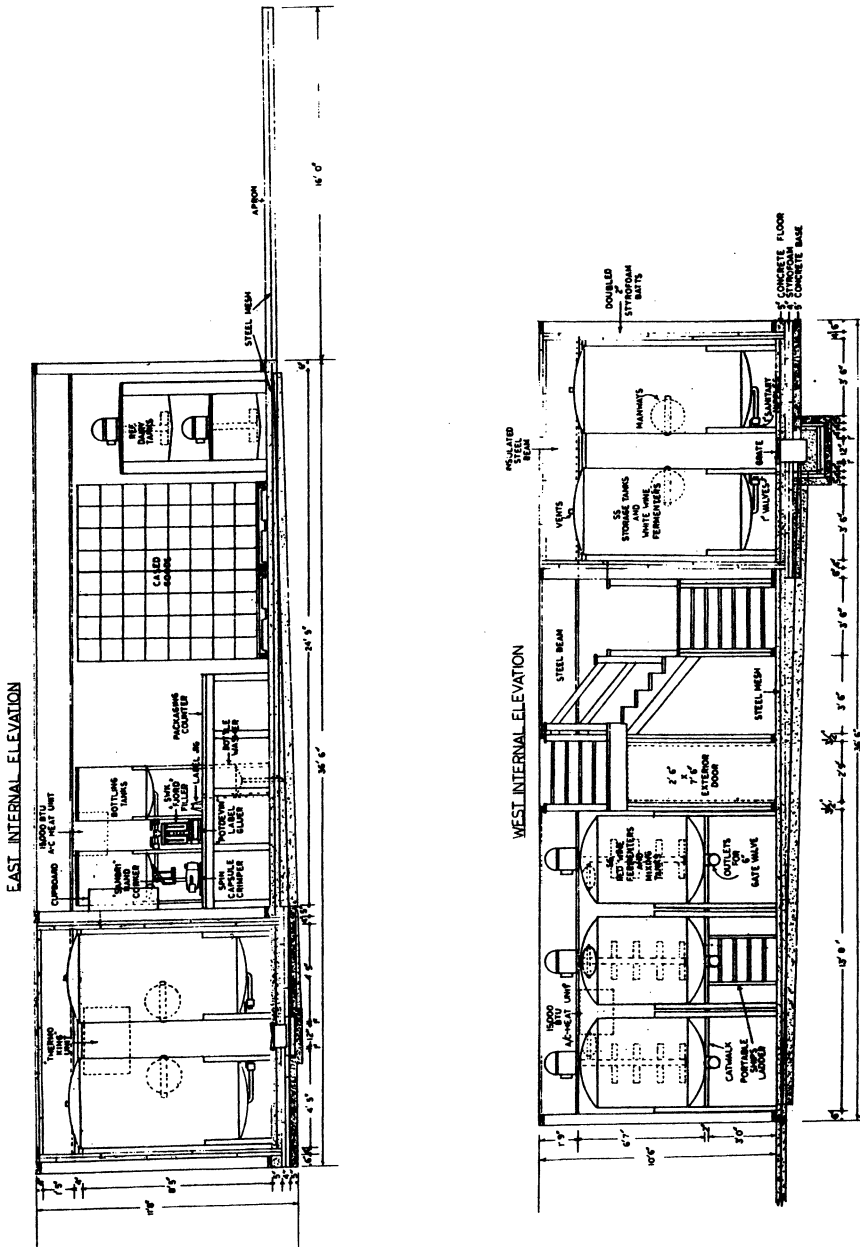


Fig. 6-3 Interior elevation of remodeled barn winery.

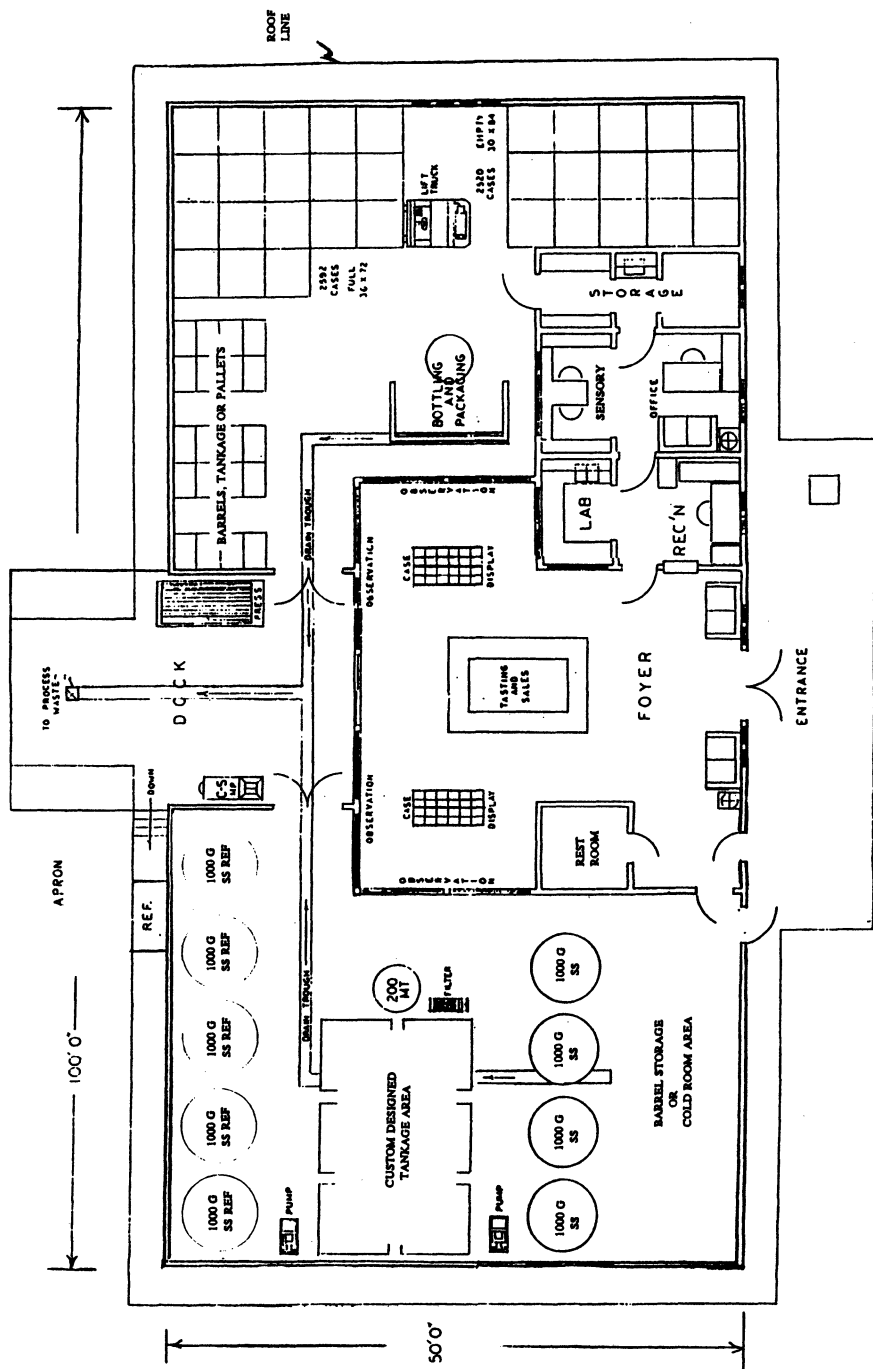


Fig. 6-4 Floor plan for a simple design winery.

Simple Custom Design

The basic format of this concept can be used for any size of winery and is expandable to most any reasonable degree. A design sketch provided previously illustrates a moderately small winery where absolute minimal costs in construction, maintenance, and operation were imperative. This division of production, with fermenting, processing, and bulk storage on one side and bottling, packaging, and cased-goods storage on the other, provides for virtually limitless expansion in these opposite directions.

This building was prefabricated insulated steel construction in a simple rectangle cleverly designed to appear complex and interesting. The floor plan (Figure 6-4) shows a glass wall-enclosed center; crushing/pressing, cellar operations, laboratory, and bottling/packaging can all be observed from inside the hospitality/sales area, eliminating the need for guided tours. The dock area also serves as a stage for outdoor presentations for various programs and festivals held each year.

Semiautomatic bottling and packaging machines permit production of more than 300 cases per day. Because the cost of such machines often places a severe financial burden on new winery ventures, it may be far less expensive to produce fewer cases daily with hand-operated machinery. This has the secondary benefit of displaying “handmade production” with part-time employees during peak visitor times, which can enhance production action and image.

This particular concept is designed for the production of several wine items, so smaller sized tanks were selected. For a smaller product portfolio, larger tanks would be needed.

The laboratory, reception area, and office area are clustered, as there is often a single person fulfilling these functions in a small winery, particularly during off-season times, when this person may also serve as the tasting and sales clerk.

CHAPTER 7

REQUIREMENTS, RESTRICTIONS, AND REGULATIONS

Wine production is regulated by an extensive set of statutes. It is highly recommended that anyone seriously considering starting a winery first obtain a complete set of both federal and state regulations. These should be read, understood, and kept readily available for future reference.

As mentioned above, the U.S. Bureau of Alcohol, Tobacco and Firearms (ATF) is the federal regulatory and enforcement authority for wine production in America. The ATF will not issue permits for commercial wine premises in the design phase. A winery must be operational, or nearly so, before an ATF inspector will visit the facility for inspection. Consequently, ensure that winery property, buildings, and major equipment are in place before applying to the ATF. It often takes several inspections before the permit is issued.

With land acquisition, winery design, construction, and ATF, ABC, and local permits, the decision to establish a winery may need to be made more than 1 year in advance of the first grape crush.

The ATF provides a booklet entitled *Information for New Wineries*, which can be obtained by writing to Department of the Treasury, Bureau of Alcohol, Tobacco, and Firearms, Wine and Beer Branch, Ariel Rios Federal Building, 1200 Pennsylvania Avenue NW, Washington, DC 20226. This booklet is very helpful

in outlining the necessary forms and procedures required to begin the lengthy task of ATF permit application.

It is essential to understand that ATF regulations are not always straightforward. Sometimes a single section can be interpreted in several different ways. Similarly, different ATF inspectors may provide different opinions on a given question. While this can be a source of confusion and frustration, it is recommended that all such topics be dealt with patiently—and actions fully documented in files that can be easily retrieved.

No two application processes are precisely the same. Each principal involved in a prospective winery may undergo a background check by ATF inspectors. Some of this investigation is focused on personal character and funding sources for the prospective commercial winery. Obviously, everyone has a different history and thus a different length of time needed to satisfy the ATF. Application process time will vary according to the way the winery will be structured—whether, for example, it will be a sole proprietorship, a partnership, or a closed corporation. The time will also vary according to the potential environmental impact of the wine production facility: water consumption, disposal systems, effluent quality and quantity, distances from public waterways, and air and noise pollution, among other factors.

These and other ATF concerns all help to determine a unique process and time frame for every winery permit application—and the time frame will be further influenced by the workload facing ATF inspectors at the time the application documents are submitted. It is poor judgment to anticipate that any given winery project will be simple. Some prospective vintners make more work out of attempts to circumvent the application process than the actual process requires.

APPLICATION FOR BASIC PERMIT

This application is made by using ATF Form 1637. It is primarily involved with identifying the applicant, the type of business in which the applicant desires to be engaged, and its location. Instructions for completing this form are provided on its reverse side.

APPLICATION TO ESTABLISH AND OPERATE WINE PREMISES

ATF Form 5120.25 must be filled out and approved to establish and operate a winery. This serves to disclose the name, address, and organizational format of the applicant, such as a sole proprietorship, a partnership, or a corporation. The work and rework necessary to create and compile all of the detailed supplemental

information required by the ATF often results in frustration for hopeful vintners. Formal documentation of partnerships and corporations ownership interests, articles and charters creating the organization, and specific resolutions authorizing the application to establish and operate a winery are often required as attachments. Names and addresses of principals involved in the prospective winery must also accompany this application form. Complete instructions for completing ATF Form 5120.25, as well as detailed requirements for the necessary attachments, are provided on the reverse side of this form.

ATF regulation 27 CFR, Section 24.111, requires that a description of the premises be attached to the application. Its language follows:

The application will include a description of each tract of land comprising wine premises. The description will be by directions and distances, in feet and inches (or hundredths of feet), with sufficient particularity to enable ready determination of the bounds of the wine premises. When required by the regional director (compliance), a diagram of the wine premises, drawn to scale, will be furnished. The description will clearly indicate any area of the wine premises to be used as bonded wine premises, used as taxpaid wine premises, or alternated for use as bonded wine premises and taxpaid wine premises. The means employed to afford security and protect the revenue will be described. If required by the regional director (compliance) to segregate operations within the premises, the manner by which the operations are segregated will be described. Each building on wine premises will be described as to size, construction, and use. Buildings on wine premises which will not be used for wine operations will be described only as to size and use. If the wine premises consist of a part of a building, the rooms or floors will be separately described. The activities conducted in the adjoining portions of the building and the means of ingress and egress from the wine premises will be described.

It should be clear from this passage that a diagram of the wine premises “may” be required by the regional director of the ATF. It is advised that a diagram be made beforehand to avoid potential delays during the application approval process. Thus, it is a good idea to contact the regional director in order to ascertain the format in which the diagram is to be rendered.

BONDS

The ATF is charged with the responsibility of protecting uncollected excise taxes on wines located upon bonded premises. The application process for a commercial winery permit seeks authorization to hold wines “in bond,” or excise tax unpaid. This places a burden upon the vintner to assure the ATF that all excise taxes are guaranteed payment, with a minimum bond of \$1,000. Unless a prospective vintner wants to make a monetary pledge covering the maximum production tax, a wine bond must be applied for with a bonding company, using ATF Form 5120.36. As

these are generally rather inexpensive for responsible citizens, it is a good idea to calculate the bond principal well over any perceived maximum tax exposure so that there can be no question that this is fully covered.

It is best to provide the bonding company extra copies of Form 5120.36, as any erasures or alterations to the bond form require written consent to those changes. Consequently, it makes good sense to ensure that the document is executed without error. Other important details are listed in the instructions on the reverse side of this form.

ENVIRONMENTAL IMPACT

New wineries are subject to the same concerns for the environment as any other business project. ATF administers this responsibility for compliance with two documents, ATF Forms 1740.1 and 1740.2.

The first of these, 1740.1, has two sides to complete and deals with general information relating to the composition and quantity of liquid and solid wastes, as well as noise pollution projected to be generated by the winery. It is recommended that the response to each question be calculated carefully and that those figures be kept at hand for ready reference when the ATF inspector visits the premises. It is also recommended that copies of local effluent, disposal, and noise codes be kept at hand to avoid delays in ensuring compliance.

The second form, 1740.2, has only one side to complete and supplements Form 1740.1 regarding how liquid wastes from the winery will be handled. The primary concern is the condition and manner in which effluents may enter public waterways. It is recommended that the plat diagram discussed with the application form above have a supplemental page indicating location, size, and type of winery wastewater treatment facilities, as well as accurate distances of any nearby public waterways.

Affidavits relating to potential winery impact may be gathered, if possible, from local environmental authorities and attached as addenda to ATF Forms 1740.1 and 1740.2 when they are submitted.

Wineries planned in or near a registered historic site should have documented approval from the state historic preservation authority before the application is made.

PERSONNEL QUESTIONNAIRE

The intent of the personnel questionnaire is to help ensure that people with unsuitable backgrounds are denied responsibility in commercial wine production. This is administered by ATF Form 5000.9, a three-page form that is generally

closely scrutinized by ATF officials. Approach this form with close attention to detail, including all zip codes, telephones, and other contact information, as inspections usually include interviews with some or all of the personal references listed. Prepare a fully detailed financial profile, including bank statements, security transactions, loan schedules, and any other pertinent documents that substantiate funding sources for the prospective winery.

Forms 5000.9 must be completed by all partners and any person who has, or will have, a significant financial interest in the winery.

SIGNATURE AUTHORITY

If the winery structure is in the form of a corporation, ATF Form 5100.1 provides for a list of winery officials who will be authorized to sign on behalf of the company. This document must be executed with the corporate seal.

For absentee owners and similar partnership or corporate situations, it may be necessary to provide authority for another person to sign official ATF documents. This can be ensured with a Power of Attorney via ATF Form 5000.8.

LABEL APPROVALS

Prospective vintners should not delay label design until after the winery application is approved. As all labels must be approved by ATF Form 5100.31, and can be subject to their own time-consuming processes, it is a good idea to move forward with artwork and other preliminary label design activity to obtain label approvals as soon as possible.

Membership in the American Vintners Association (AVA) provides assistance in processing wine label approvals. For full information on this service, write the AVA at 1301 Pennsylvania Avenue NW, Suite 500, Washington, DC, 20004.

FORMULA WINES

Prospective vintners may desire to make wines in which flavorings are added or some other additive that is not naturally part of the grape or fruit. Wine jargon sometimes refers to these as "special natural wines." Each of these wines requires a completed ATF Form 5120.29 to be submitted for approval. Instructions on the reverse side of the form are easy to follow. Since formula approval processing can also be a source of delay in getting a new winery operation, it is advised that the research and development in devising such products be finalized and readied for executing Form(s) 5120.29 as soon as practicable.

STATE ABC REGULATIONS

Each state has its own wine regulatory statutes. It is beyond the scope of this book to outline them all. However, it may serve prospective vintners to peruse some of the Indiana ATC rules for examples of how these are structured. These are, of course, valid only in the state of Indiana. Below are key sections of the Indiana Code relating to winery start-up and operation.

The following are regulations concerning the scope of a small winery permit (Section 7.1-3-12-5):

- (a) The holder of a small winery permit is entitled to manufacture table wine, to bottle wine, and to bottle table wine produced by his small winery. He is entitled to serve complimentary samples of his table wine on the licensed premises. He also is entitled to sell his table wine on the licensed premises to consumers either by the glass, or by the bottle, or both. The permittee also is entitled to sell table wine by the bottle or by the case to a person who is the holder of a permit to sell wine either wholesale or retail. The holder of a small winery permit is exempt from the provisions of IC 7.1-3-14 [Wine Retailers Permits]. The holder is entitled to advertise the name and address of any retailer or dealer who sells wine produced by his winery.
- (b) With the approval of the commission, a holder of a permit under this chapter may conduct business at a second location that is separate from the winery. At the second location, the holder of a permit may conduct any business that is authorized at the first location, except for the manufacturing or bottling of wine.

The following are regulations concerning location of premises in proximity to school or church (prohibitions and restrictions) (Section 7.1-3-21-11):

The commission shall not issue a permit for premises situated within two hundred feet (200') from a school or church if no permit has been issued for the premises under the provisions of Acts 1933, Chapter 80.

The following are regulations concerning the small winery permit (Section 7.1-4-1-25):

The annual license fee for a small winery permit is two hundred fifty dollars (\$250).

The following are regulations concerning the supplemental retailer's permit (Sunday sales permit) (Section 7.1-4-1-31.5):

The annual license fee for a supplemental retailers permit is thirty-three and one-third percent (33 1/3%) [\$83.33] of the amount of the annual license fee that the applicant paid, in the aggregate, for the retailer's permit that qualifies him to hold a supplemental retailer's permit.

The following are regulations concerning the rate of tax (Section 7.1-4-4-1):

An excise tax at the rate of forty-seven cents (\$0.47) a gallon is imposed upon the manufacture and sale or gift, or withdrawal for sale or gift, of wine within this state.

The following are regulations concerning beverages to which tax is applicable (Section 7.1-4-4-2):

The wine excise tax shall apply to wine that contains less than twenty-one percent (21%) of absolute alcohol reckoned by volume. The wine excise tax also shall apply to an alcoholic beverage that contains fifteen percent (15%), or less, of absolute alcohol reckoned by volume, mixed with either carbonated water or other potable ingredients, or both, by either the manufacturer or the bottler, or both of them, and sold in a container filled by the manufacturer or bottler, and which is suitable for immediate consumption directly from the original container. An alcoholic beverage that is subject to the wine excise tax shall not be also subject to the liquor excise tax.

The following are regulations concerning when employment of person 18 years or older on licensed premises is lawful (Section 7.1-5-7-13):

The provisions of IC 7.1-5-7-12 [prohibition of employing minors on licensed premises], shall not prohibit the employment of a person eighteen (18) years of age or older on or about licenses premises where alcoholic beverages are sold, furnished, or given away for consumption either on or off the licensed premises, for a purpose other than selling, furnishing, consuming or otherwise dealing in alcoholic beverages. Nor shall the provisions of IC 7.1-5-7-12, prohibit a person eighteen (18) years of age or older from ringing up a sale of alcoholic beverages in the course of his employment.

The following are regulations concerning employees between eighteen (18) and twenty-one (21) years of age (Section 905 IAC 1-15.31):

Sec. 1. (a) A minor eighteen (18) years of age or older employed at a licensed premises may ring up a sale of alcoholic beverages in the course of his employment, provided that at the time the sale is rung up, there is at least one (1) other employees on the licensed premises who is:

- (1) twenty-one (21) years of age or older; and
- (2) responsible for supervising said minor.

The following are regulations concerning times when sales are unlawful (Section 7.1-5-10-.1):

- (a) It is unlawful to sell alcoholic beverages at the following times:
 - (1) At a time other than that made lawful by the provisions of IC 7.1-3-1-14 [basically, Monday through Saturday 7 a.m. until 3 a.m.];
 - (2) On Christmas Day and until 7:00 o'clock in the morning, prevailing local time, the following day; and
 - (3) On primary election day and general election day from 3:00 o'clock in the morning, prevailing local time, until the voting polls are closed in the evening on these days.
- (b) During the time when the sale of alcoholic beverages is unlawful, no alcoholic beverages shall be sold, dispensed, given away, or otherwise disposed of on the licensed premises and the licensed premises shall remain closed to the extent that the nature of the business carried on the premises, as at a hotel or restaurant, permits.
- (c) It is unlawful to sell alcoholic beverages on New Years Day for off-premises consumption.

CHAPTER 8

GETTING STARTED

RECORDKEEPING

The distress of repeatedly making the same mistakes is perhaps only surpassed by the frustration of having made a good wine and forgetting how it was done. It is wise to keep a good price and expense history in the files, and essential to have up-to-date processing and inventory information to supplement records required by the Bureau of Alcohol, Tobacco, and Firearms (ATF). All of this obviously indicates that a functional recordkeeping system is essential in operating a successful winery business. A personal computer program, as discussed below and in Chapter 6, is ideal. It may, however, be good education and thrift to start out with manual card files and ledgers in order to fully experience all of the wine production, marketing, and accounting operations in detail before applying computerization. A flowchart for basic recordkeeping is presented in Figure 8-1.

It is recommended that every vintner carefully review 27 CFR Part 4, Subpart O, Records and Reports, which states that “a proprietor who conducts wine operations shall maintain wine transaction records and submit reports as required by this part” (Section 24.300).

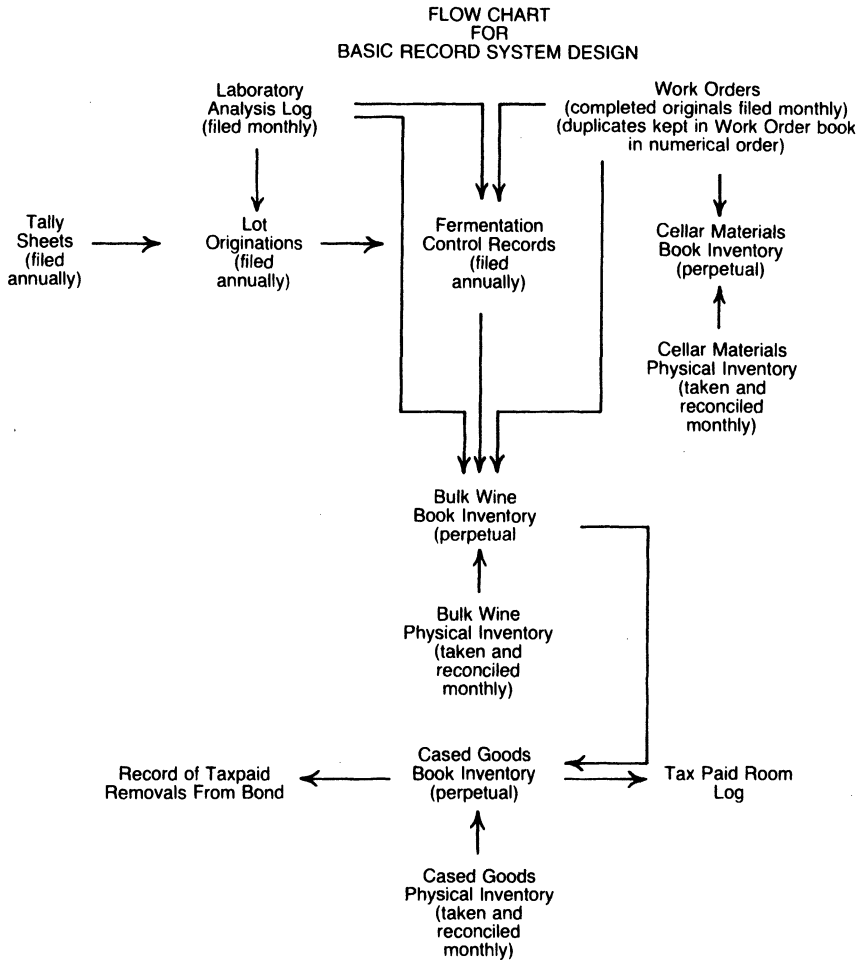


Fig. 8-1 Supplemental recordkeeping flow chart.

COMPUTERIZED RECORDS

Virtually every successful commercial wine producer has found the computer to be an essential tool. Cost savings are gained by optimizing efficiency in processing, minimizing inventories, and other important strategies. Some vintners have hired computer specialists to custom-design programs to fit their individual vineyard-winery-marketing-administration needs. These are, of course, very expensive and typically find their greatest value in large, comprehensive wineries.

Other winery operations have consulted with software companies such as those listed in Appendix A. Still others have cleverly adapted general software programs available in computer stores and catalogs.

An integrated computer program, especially designed for commercial wine operations, is best of all. Information essential to the ATF can be segregated and instantly copied on a printout as a worksheet, leaving the system and the vintner free to do other things during the process of generating routine report forms, as well as during an on-site ATF inspection. Computer systems can easily be used to generate valuable historical, trend, and other data essential to a progressive operation.

Whatever the manner of setting up a computer system of recordkeeping, it should be able to include as much of the detailed vineyard-to-market history of every wine as may be practicable.

Regulations in 27 CFR Part 4, Subpart O, stipulate the following:

- *Bulk still wine record.* "A proprietor who produces or receives still wine in bond, (including wine intended for use as distilling material or vinegar stock to which water has not yet been added) shall maintain records of transactions for bulk still wine" (Section 24.301).
- *Effervescent wine record.* "A proprietor who produces or receives sparkling wine or artificially carbonated wine in bond shall maintain records showing the transaction date and details of production, receipt, storage, removal, and any loss incurred" (Section 24.302).
- *Formula wine record.* "A proprietor who produces beverage formula wine shall maintain records showing by transaction date the details of production" (Section 24.303).
- *Chaptalization (Brix adjustment) and amelioration record.* "(a) *General.* A proprietor who chaptalizes juice or ameliorates juice or wine, or both, shall maintain a record of the operation and transaction date. Records will be maintained for each kind of wine produced (grape, fruit, or berry). No form of record is prescribed, but the record maintained will contain the information necessary to enable ATF officers to readily determine compliance" (Section 24.304).
- *Sweetening record.* "A proprietor who sweetens natural wine with sugar or juice (unconcentrated or concentrated) under the provisions of this part shall maintain a record of sweetening by transaction date" (Section 24.305).

27 CFR Part 4 goes on to include

- Distilling material or vinegar stock record (Section 24.306)
- Nonbeverage wine record (Section 24.307)
- Bottled or packed wine record (Section 24.308)
- Transfer in bond record (Section 24.309)
- Tax-paid removals from bond record (Section 24.310)
- Tax-paid wine record (Section 24.311)

- Unmerchtable wine returned to bond record (Section 24.312)
- Inventory record (Section 24.313)
- Label information record (Section 24.314)
- Materials received and used record (Section 24.315)
- Spirits record (Section 24.316)
- Sugar record (Section 24.317)
- Acid record (Section 24.318)
- Carbon dioxide record (Section 24.319)
- Chemical record (Section 24.320)
- Decolorizing material record (Section 24.321)
- Allied products record (Section 24.322)

One of the best ways to simplify recordkeeping is to use coded lot numbers. For example, the first two digits in a lot number could indicate the vintage year, and a third digit could reveal the source of the grapes, with 1 denoting estate-grown grapes, 2 denoting grapes purchased from XYZ Vineyards, and so forth. A fourth digit might note the processing progress of the lot, with 1 denoting fermenting juice or must and a new digit assigned for each processing step completed, ascending to 9, which could indicate wine ready for bottling. From this code, a lot numbered, say, 0118 Seyval Blanc would instantly inform the winemaker that this was vintage 2001 Seyval Blanc, from his own estate vineyard, which might need only filtration to be ready for bottling. A 0025 Chambourcin would be 2000 vintage grown at XYZ Vineyards, a red wine in some middle stage of processing.

It is essential that the precise source, variety, and weight have documentation. ATF inspections will require this information to trace production output and will require label statements on the resulting wine to be factual. A Receiving Tally Sheet should record information that can help to plan for more efficient time, labor, and material resource allocations for future grape-receiving activities. Exhibits 8–1 through 8–5 are five forms that vintners may reprint and use in their supplemental wine production recordkeeping. The first of these, the Laboratory Analysis Record (Exhibit 8–1), is a standard form for all winery analyses.

A Lot Origination Form (Exhibit 8–2) also helps to trace information that can enable one to plan for more effective time, labor, and material resource allocations in future grape-receiving activities. The third form, the Fermentation Control Record (Exhibit 8–3), documents the transition of grapes into fermenting must or juice. This is often called a wine “birth certificate.”

Once fermentation is completed, the juice or must has, of course, become wine and should be transferred to a perpetual Bulk Wine Record (Exhibit 8–4). At that time the Lot Origination Form is retired to a record file. The Bulk Wine Record card can be reprinted and used in a 5" × 7" card format for each lot of wine made. The reverse side of the card should serve to continue the record needed for each wine.

As each wine lot is bottled, it should be removed from the Bulk Wine Record to the Bottled Wine Record (Exhibit 8–5). If activity in the history of a wine goes

QUALITY CONTROL

Few vintners look forward to the tedious demands of an effective quality assurance program. But quality control must be carefully designed and faithfully performed. Wine enthusiasts have a keen sense of value. They will not excuse low-quality wine, and they will not buy it.

An adequate quality control program does not have to be capital intensive, and it does not require extensive training. The worst-case scenario is spending the time and money to set up a workable program and then not using it.

Ideally, the assurance that every wine will meet prescribed standards of quality should be a task totally separated from production. This has led most larger vintners to establish a department within the management framework that has an entirely autonomous quality assurance obligation and authority. Samples from every phase of production are gathered, analyzed, evaluated, approved, or disapproved. It is when a sample fails the scrutiny of testing that the quality control system really shows how it works as protection for the vintner. The wine problem has an opportunity to be corrected before it becomes a much larger problem in the marketplace.

A divided production/quality control authority can be criticized as impractical in small wineries. This should not, however, stop any vintner from employing the spirit and philosophy of such a system. Most small winery operators must assume several different roles, and the unbiased judgment of honest quality control is probably the most important of them all. In its simplest form, quality control can be a thorough sensory evaluation made outside the winery, perhaps at home, at a time and place that creates total separation from production.

The road to achieving this as routine may take some initiative, but it does not have to be difficult. It is a matter of identifying potential sources of problems and standards of acceptance, then designing timely analytical and testing procedures to measure and record the findings. The immediate interpretation of these findings thus becomes key to any action that may be necessary to steer wine production out of harm's way.

ANALYTICAL INSTRUMENTATION

Wine was made long before any testing was performed in a laboratory setting. In modern times, though, it makes sense to use affordable technology when it can take the guesswork out of making wine from expensive grapes.

pH Meter

Fruit acidity is one of the most important criteria in making wine. The pH test is, in simple terms, an indication of the "strength" of acidity in a juice or wine, and thus

a measure of how ripeness has progressed. More precisely, pH is a measurement of hydrogen ion concentration and the relative ability to neutralize basic (alkaline) hydroxyl ions. The pH test is expressed on a scale from 0 (the most acidic) to 14 (the most basic), with neutrality at pH 7.

In winemaking, the pH meter is frequently used during the harvest. While the taste buds of an experienced wine master may often be the final word for when grapes are to be harvested, the pH test often helps in pinpointing the optimal time. Litmus papers are inadequate for wine pH testing, as they are not precise enough to be of any value.

Most grapes are green at pH 2.70, and as ripeness progresses, pH will increase. Winemakers will often standardize pH at 2.90 to 3.10 for wines that are meant to have sharp, crisp acidity, particularly wines earmarked for secondary fermentation into sparkling wines.

Fresh, fruity white and blush wines are made from fruit in the 3.10 to 3.20 range. This is usually just slightly on the green side of ripeness. More important, this lower level of pH resists bacterial activity and allows for better efficiency in the use of sulfur dioxide as a preservative.

Heavy, complex red wines are frequently harvested in the 3.30 to 3.50 range. Higher pH ensures complete flavor and pigmentation development in the fruit and allows for malo-lactic bacterial activity, which has become commonplace in red wine processing.

It may take some convincing to spend \$300 to \$600 for a reliable pH meter and the several hours of time reading the instructions and practicing its proper operation. But other than convenient access to pH testing from another source, there are few good alternatives, and the prevention of just one lot of good fruit from being ruined more than justifies the expense.

Refer to Appendix B for the proper procedure in operating a pH meter.

Brix and Balling

These tests address the sugar content in grapes and wine and are the simplest of testing instruments essential for commercial winemaking.

A set of hydrometers, or “spindles,” with the following ranges are needed:

- –5.0 to +5.0 degrees
- 0.0 to 8.0 degrees
- 8.0 to 16.0 degrees
- 16.0 to 24.0 degrees

Hydrometers float at different levels depending upon the density of the liquid in which they are placed. In pure water, the –5.0 to +5.0 range hydrometer should float with a 0.0 reading.

The same hydrometers can be used during and after fermentation. While increasing the sugar levels makes the hydrometer float increasingly higher, increasing the alcohol levels has the opposite effect.

Consequently, the hydrometer floating in a wine sample is read as a Balling test and indicates relative viscosity—the “thinness” or “heaviness” in mouth-feel of the wine. A 12% alcohol wine, fermented completely dry, should have a Balling of -1.8 to -2.2 , depending upon non-sugar solids remaining in the wine. Note that this is a minus reading, which means that wine density, in this case, is actually less than water at 0.0 Balling. As will be seen below, the Balling test is the perfect tool for determining fermentation progress and controlling finished wine sweetness levels. A set of the four Brix-Balling hydrometers listed above should cost less than \$50.

A quicker and more convenient method of measuring Brix is by the use of a refractometer, although this is considerably more expensive than the hydrometer method. It operates by measuring the light refracted through a drop or two of grape juice on a prism. That light is directed to a scale that is read as one holds the refractometer up to a bright source of light. A refractometer is particularly good for testing grape sugars in the vineyards. This instrument cannot be used to measure Balling. Reliable Brix refractometers can be budgeted at about \$150 or so.

Every vintner will face the prospects of adjusting the amount of sugar in juice, must, and wine. Grape juice or must measuring at 23°Brix (roughly a percentage of sugar solids) will ferment to about 12.3% alcohol by volume. The calculation is made by multiplying the Brix reading from the hydrometer by .535, which is the amount of alcohol resulting from each degree of Brix. It is readily apparent that Brix is an essential test in calculating the precise amount of sugar addition, if any, to achieve the desired alcohol level from fermentation.

Methods for the operation of Brix/Balling hydrometers and refractometers are found in Appendix B. A table of total solids in solution for each degree of Brix is also provided in that section.

As in the wine cellar, sanitation in the laboratory is absolutely essential. After use, instruments should be thoroughly rinsed at least three times with distilled or deionized water and then placed upright in a storage rack to dry.

Residual Sugar

The Balling scale cannot be used to determine whether a “dry” wine is absolutely free of remaining fermentable sugar. It is important to know this before wine bottling to ensure that yeasts will not begin a secondary fermentation.

One simple and inexpensive way to analyze for residual sugar is by using Dextro-Chek tape. A piece of the tape is dipped into the wine sample, removed, and allowed to dry. Fermentable sugar, if any, will be revealed by a color change of the tape that is compared to a graduated scale included in the kit.

Total Acidity

Total acidity is a measurement of tartness. It is difficult to make definitive judgments upon how tart each lot of juice or must is by taste alone. Sweetness has a “masking” effect that blunts the sensory effect of tartness. Notice that the term *sourness* wasn’t used, as this has to do with wine spoilage, such as the formation of vinegar.

Most table wine total acidity measures in the range of .500 to .750 g/100 mL—from bland to tart, respectively. This numerical value can also be considered a rough expression of total acidity percentage.

The total cost for buret, buret stand, sodium hydroxide solution, and several 250-mL Erlenmeyer flasks should not exceed \$100.

Appendix B provides the proper method for total acidity analysis, as well as a table that can help in easily determining results.

Volatile Acidity

Most of the acids formed by spoilage bacteria are volatile and can be isolated by using a Cash volatile acid distillation apparatus. Commercial vintners rely on this instrument to monitor the extent of volatile acid formation in wine, and thus also the extent of spoilage bacteria activity. The ATF (27 CFR, Part 4, Section 4.21 (a)(1)) limits volatile acidity in commercial white wines to .120 g/100 mL, while red wines are set at a .140 g/100 mL maximum.

The Cash apparatus, complete with condenser and attending glassware, can exceed \$400. Novices are encouraged to seek some assistance in individual operational instruction before attempting to operate the device.

Appendix B provides the proper method of using a Cash apparatus in the determination of volatile acidity and a table that can help in easily determining results.

Sulfur Dioxide

Another essential test is that for sulfur dioxide (SO_2), the agent used for preserving wines from further yeast and/or bacterial activity. Dry table wines having lower pH levels can be protected with as little as 30 mg/L. Wines having residual fermentable sugar and higher pH levels may require more than 100 mg/L. The milligrams per liter expression is often given as parts per million, or ppm, which is virtually the same value.

The instrumentation and materials needed to perform the Ripper method used in this analysis cost less than \$100. The standardized iodine reagent, however, degrades rather quickly and should be kept refrigerated.

A better analysis for sulfur dioxide in wines is the aeration oxidation distillation method. The glassware and other materials needed to set up this apparatus are a bit more expensive, but the results are more precise.

Analytical procedures for determining sulfur dioxide are provided in Appendix B.

Sensory Evaluation

The investment in instrumentation for this analysis is simply a good set of durable wine glasses. A dozen or so restaurant-grade, all-purpose, “tulip-shaped” glasses are recommended. They should be identical so that wines can be compared next to each other. Good-quality wine glasses should not exceed \$5 each in cost. Finer grade glasses can exceed \$50 each.

Essential to meaningful sensory evaluation of wines is an experienced palate. Sensory education can be the single largest expense incurred in tooling up for wine analysis. One of the best ways to dilute this cost is finding ways to participate in tasting functions. Chapter meetings of the American Wine Society, Les Amis du Vin, and other wine groups usually include wine evaluations. Serving at wine competitions is another excellent way to develop expertise. Local wine retailers and restaurateurs may be sources for learning about promotional tasting programs. Groups of friends can certainly share on their own.

Many different wine evaluation scales are in use around the world. In the United States, two are used most frequently, the 20-point scale and the 100-point scale. Principles of sensory evaluation are outlined in Appendix B.

For consumers, wine judgment may be limited to a very narrow set of personal likes and dislikes. Commercial wine judging expertise, however, requires a mind-set that can separate personal preferences from actual flaws and virtues. See Appendix B for sensory wine analysis procedure.

Laboratory Refrigerator

Access to a refrigerator, including a freezer unit, is essential in the commercial wine laboratory. Sometimes reliable low-cost used units can be found. Most analytical solutions, indicators, and reagents require refrigerated storage. Dried yeast should be kept frozen to maximize viability and storage life. Yeast cultures on agar or in broth should be refrigerated.

WINERY EQUIPMENT

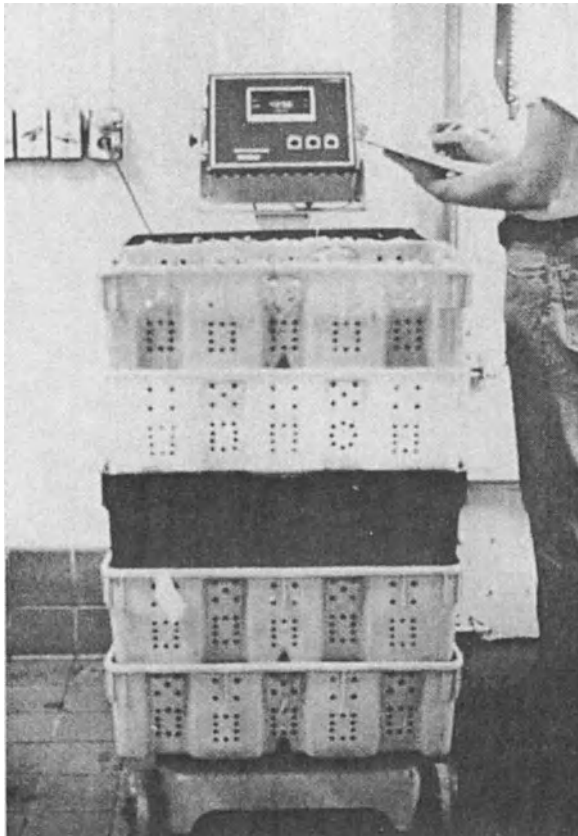
There is virtually no end to the amount of equipment that can be employed in the modern commercial wine facility. This part of winemaking can be approached with a number of rewarding philosophies. Some vintners pursue the challenge of spending as little as possible for equipment—of “making do” and being creative. Conversely, others enjoy the pride of building and equipping what amounts to a rather “high-tech” winery. Most beginning wine producers find themselves somewhere between these extremes.

Several equipment needs are essential to making wine. They can be relatively expensive. Some winemakers reduce this burden by sharing equipment costs with friendly competitors.

Scales

It is necessary to fully document the net weight of all grapes and other fruit received at the winery, whether from estate vineyards and orchards or from independent suppliers. ATF inspectors will require this information as a starting point from which to trace wine production histories. Some of the new digital scales, with 4' × 4' platforms and 5,000 pounds of capacity, can be found for less than \$3,000 (Figure 8-2).

Fig. 8-2 Electronic scales.



Crusher-Destemmer

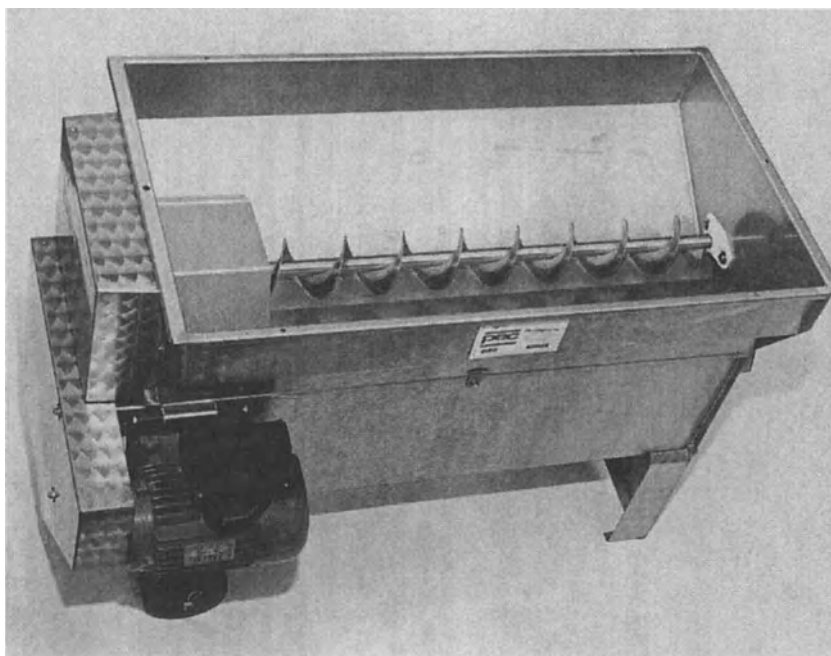
While “whole-cluster” pressing is preferred by some vintners, the vast majority of wine is made from grapes that have undergone a gentle cracking of the skins and pulp (not the seeds) sufficient for the pressing operation to be effective and efficient. Disintegrated seeds and stems will cause the resulting wine to be bitter and astringent due to excessive extraction of phenolic compounds.

Sometimes vintners are fortunate enough to find grower-suppliers who will sell fruit already crushed and destemmed. Be prepared to pay a premium for this service, however. The cost for a good-quality crusher-stemmer-must pump (Figure 8–3) should be budgeted at about \$2,000 for a 1-ton-per-hour unit. The heavy-duty usage of these machines for each vintage will wear the food-grade paint off steel hopper and shaft surfaces. These should be fully refinished immediately following the last crush each year. Larger capacity machines, fabricated in stainless steel, will cost much more, but are easier to maintain.

For some red wines, a portion of stems remaining in the must can be desirable. The best types of destemming devices are equipped with must pumps

Fig. 8–3 Crusher-stemmer-must pump.

Courtesy: Prospero Equipment Corporation.



permanently installed at the bottom of the receiving hopper. A 1-ton-per-hour crusher-stemmer-must pump generally costs upwards of \$3,000 and much more for higher capacity machines built from stainless steel.

Press

No other piece of equipment is more symbolic of the wine art than the press. A basic piston-type basket press having a quarter-ton must capacity generally costs less than \$1,500 but requires considerable labor to operate and are comparably inefficient.

Such presses operate by a plate, or *piston*, forced downward by the press head, which is turned down the screw shaft running vertically through the center of the basket. The juice or wine is extracted through the spaces left between the basket slats. Keep the shaft well lubricated with a food-grade vegetable grease. For another several hundred dollars of investment, the piston can be augmented with hydraulic pressure generated by a hand pump—well worth the price.

Better yet are vertical and horizontal pneumatic basket presses (Figure 8–4) in which heavy-duty rubber “balloons” are inflated inside to exert pressure upon the must against the press walls. These cost a bit more and are, again, well worth the higher price.

Best of all are tank presses, which function on the balloon principle but with gentle action within stainless-steel baskets and minimize extraction of pulp and seed particles (Figure 8–5). The smallest of these can exceed \$10,000 in cost which, usually, can be recovered over time through efficiency.

Pumps

Very small wineries can get along with centrifugal plastic-head pumps so small that they can be easily carried around the cellar. These cost less than \$300 each and usually have a maximum capacity of about 10 gallons per minute. Greater capacities and efficiencies are found in larger stainless-steel pumps mounted on rubber wheels. Depending upon size and options, these generally range from \$1,000 to \$5,000 each.

Hose

The hose needed for transferring crushed white grapes (must) from the crusher-stemmer-must pump to the press or for transferring red must to fermenters will need to be at least 3 inches in inside diameter (ID). This should be a good-quality food-grade hose 10% to 15% longer than the longest anticipated distance for must transfer. For fine-quality, 3-inch must hose, budget at least \$15 per foot.

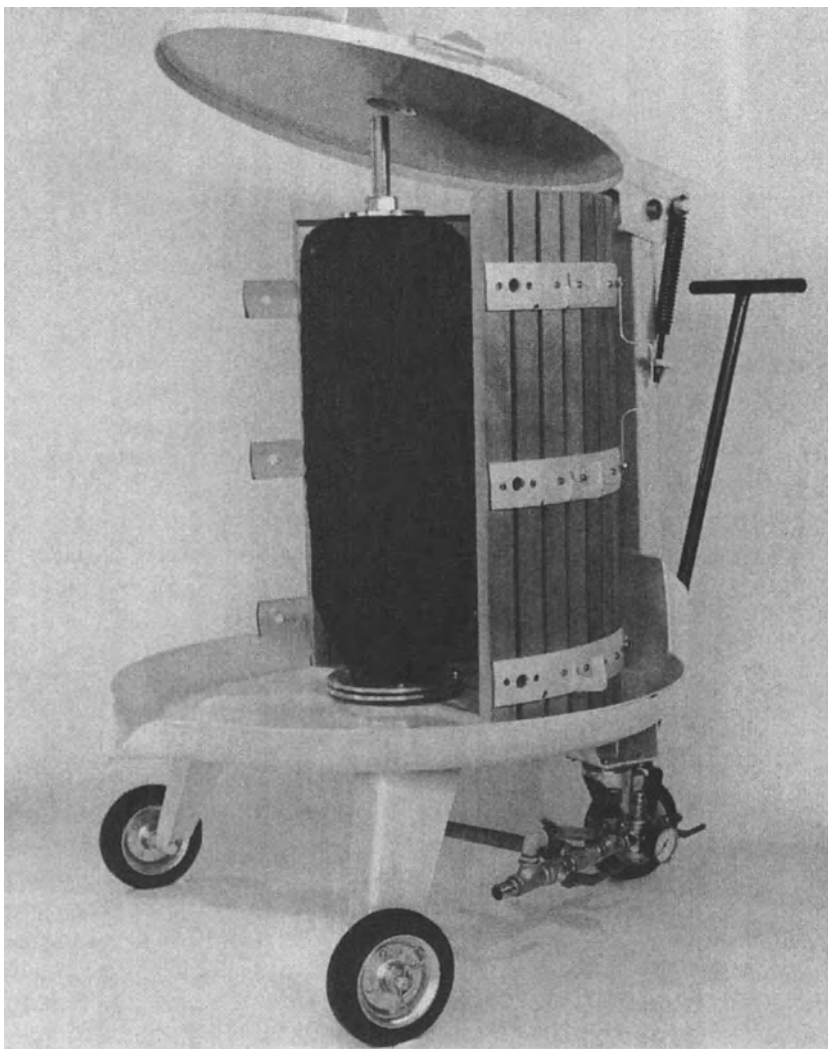


Fig. 8-4 Pneumatic basket press.
Courtesy: Prospero Equipment Corporation.

Transparent or translucent dairy-type hoses are recommended, as they are relatively inexpensive and durable if properly maintained. The ability to see liquid in movement through the hose has obvious advantages in operation.

The wine transfer hose recommended for the small winery has a 1½-inch ID. Rarely is there a need for less than 100 feet of such hose. Typically, two sections of



Fig. 8-5 Stainless-steel tank press.

Courtesy: KLR Machines.

about 6 feet are fitted as “sucker” hoses for the inlet of the pumps, and two longer sections are cut as “delivery” hoses for the outlet of the pumps. For fine-quality 1½-inch food-grade transparent transfer hose, budget at least \$8 per foot.

In all cases, female fittings should be made at the end of the hoses—not upon tanks, filters, or other units. The rationale for this is that the female fittings can better withstand hoses’ continually being dropped or hit upon hard surfaces. Male fittings, whether threaded or adapted, will soon dent and foul, wearing to the point where leakage is difficult to control. Figure 8-6 illustrates typical winery hoses and fittings. Clamp fittings are equally suitable.

Tanks, Barrels, and Other Bulk Wine Vessels

All sorts of containers are used for making wine. Of critical importance is the material from which they are made. The best are constructed from stainless steel (Figure 8-7), although glass carboys are often used for very small lots. These should be budgeted at about \$15 each.

Stainless-steel and glass containers can be cleaned effectively with sufficient warm-water rinsings to wash out any juice, wine, or solids that remain when emptied. This should be followed by the addition of 1/4 teaspoon of potassium

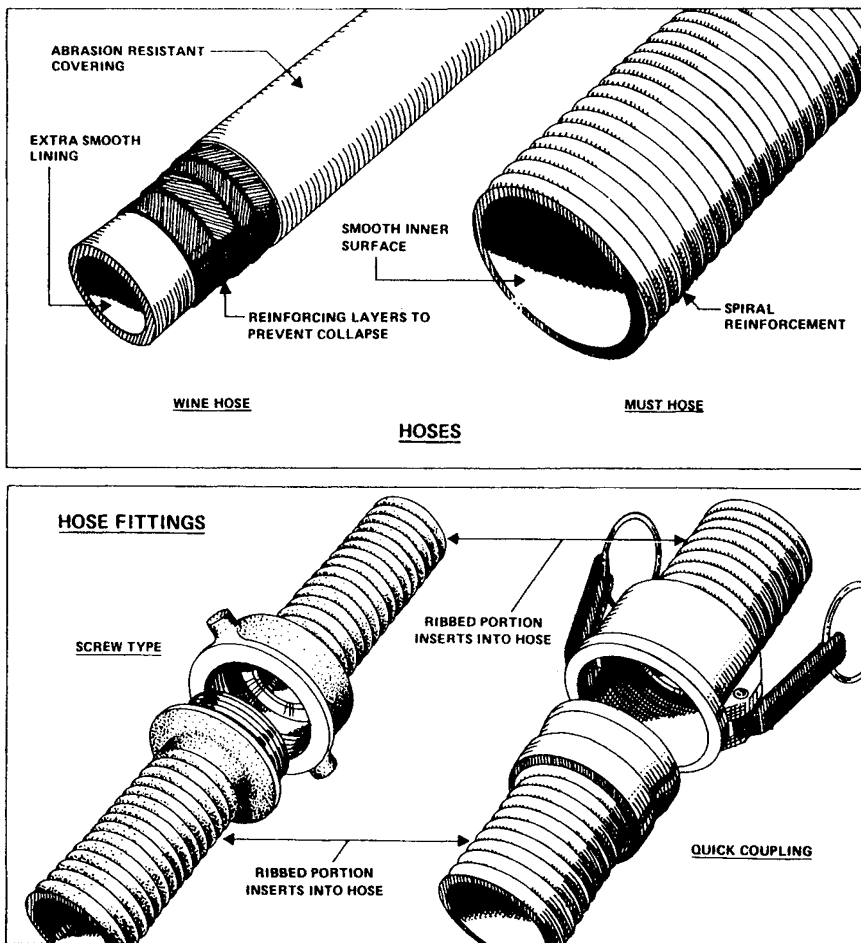


Fig. 8-6 Typical winery hoses and fittings.
Courtesy: Sebastiani Vineyards.

meta-bisulfite per gallon of rinse water. Containers should then be drained dry.

Polyethylene can be acceptable if used only for short-term processing needs, such as fermentation. Extended use has given rise to controversy over the development of “off” flavors in wines. Used poly drums (55-gallon capacity) can be found at very reasonable prices, sometimes at less than \$10 each.

While there is plenty of wine romance in using kegs and barrels in winemaking, they are also very difficult to employ properly. Wood is organic and can suffer



Fig. 8–7 Stainless-steel drums.

deterioration, even ruination, from spoilage microbes. Wooden vessels must be kept moist, as they will shrink from drying, creating leaks and losses. Wood aging essentials are discussed in Chapter 4.

Mild steel, iron, copper, aluminum, ceramics, and other materials are unacceptable. The danger is in the potential for extracting metals and minerals that can ruin wine clarity and flavor.

Containers for fermenting white wines should be closed so that the only opening is that upon which a *fermentation lock* (see Chapter 9) is attached. Red wine fermenters should have open tops of removable lids so that the must “cap” can be easily “punched” down several times each day.

It is recommended that the small winery facility be designed with containers that can be easily moved to optimize usage of space seasonally. A 5-gallon glass carboy weighs about 50 pounds when full, which is generally about the largest vessel one wants to consider moving without any mechanical assistance. Stainless-steel vessels from 100 to 1,000 gallons capacity should be mounted upon legs for ease in use and relocation around the winery.

Following fermentation, all wines should be processed and stored in vessels that are completely filled, a practice that keeps air from oxidizing wine. White wine fermenters can often serve double duty as good wine-processing and storage/aging containers.

Key ATF Tank Regulations

Tanks (27 CFR, Part 4, Section 24.167)

- (a) *General.* All tanks on wine premises used for wine operations or for other operations as are authorized in this part will be suitable for the intended purpose. Each tank used for wine operations will be located, constructed, and equipped as to permit ready examination and a means of accurately determining the contents. Any tank used for wine operations not enclosed within a building or room will be enclosed within a secure fence unless the premises where the tank is located are enclosed by a fence or wall, or all tank openings are equipped for locking and are locked when used for wine operations and there is no proprietor's representative on the wine premises, or the regional director (compliance) has approved some other adequate means of revenue protection. All open tanks will be under a roof or other suitable covering.

Identification of Tanks (27 CFR, Part 4, Section 24.168)

- (a) *General.* Each tank, barrel, puncheon, or similar bulk container used to ferment wine or used to process or store wine, spirits, or wine making materials will have the contents marked and will be marked as required by this section.
- (b) *Tank markings.* (See Figure 8–8.)
 - (1) Each tank will have a unique serial number.
 - (2) Each tank will be marked to show its current use, either by permanent markings or by removable signs of durable material; and
 - (3) If used to store wine made in accordance with a formula, the formula number will be marked or otherwise indicated on the tank.
- (c) *Puncheon and barrel markings.* Puncheons and barrels, or similar bulk containers over 100 gallons capacity, will be marked in the same manner as tanks. A permanent serial number need not be marked on puncheons and barrels, or similar bulk containers of less than 100 gallons capacity, used for storage, but the capacity will be permanently marked.

Measuring Devices and Testing Instruments (27 CFR, Part 4, Section 170)

- (a) *Measuring devices.* The regional director (compliance) may at any time require proprietors to provide at their own expense equipment for ascertaining the capacity and contents of tanks and other storage containers, and scales and measuring devices for weighing and measuring wine, spirits, volatile fruit-flavor concentrate, or materials received and used in the production of wine. Where winemaking materials or other materials used in the treatment of wine are used immediately upon receipt on wine premises, or received and stored on bonded wine premises in original sealed shipping containers with a stated capacity, the quantity shown on the commercial invoice or other document covering the shipment may be accepted by the proprietor and entered into records in lieu of measuring the materials upon receipt.

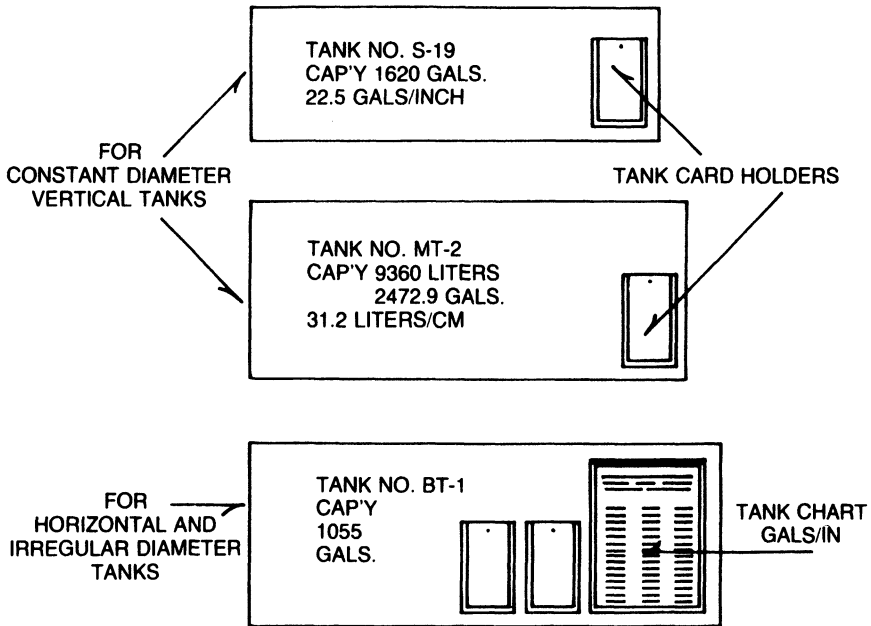


Fig. 8-8 Winery tank markings.

Fermentation Locks

These are devices that allow carbon dioxide gas to escape from fermenters but that keep air, dust, and other contaminants from entering the vessel (Figure 8-9). Air exposure can be devastating to wine—a source of oxygen required by vinegar bacteria and also of oxidative “browning.”

Perhaps the most effective fermentation locks are inexpensive devices in which water floats a small plastic cylinder inside a larger one. As carbon dioxide gas is generated, it is allowed to bubble out through the water. Empty sparkling wine bottles, with sloping shoulders, are often placed neck down in fermenter bungholes. Carbon dioxide gas pressure will build in the tank until sufficient to lift the bottle out but disallow air back in.

Refrigeration

Because much of the flavor in white wines remains in the skins after pressing, it is obvious that very careful controls should be placed upon conserving the flavor extracted in the resulting juice. Fermenting white grapes in contact with

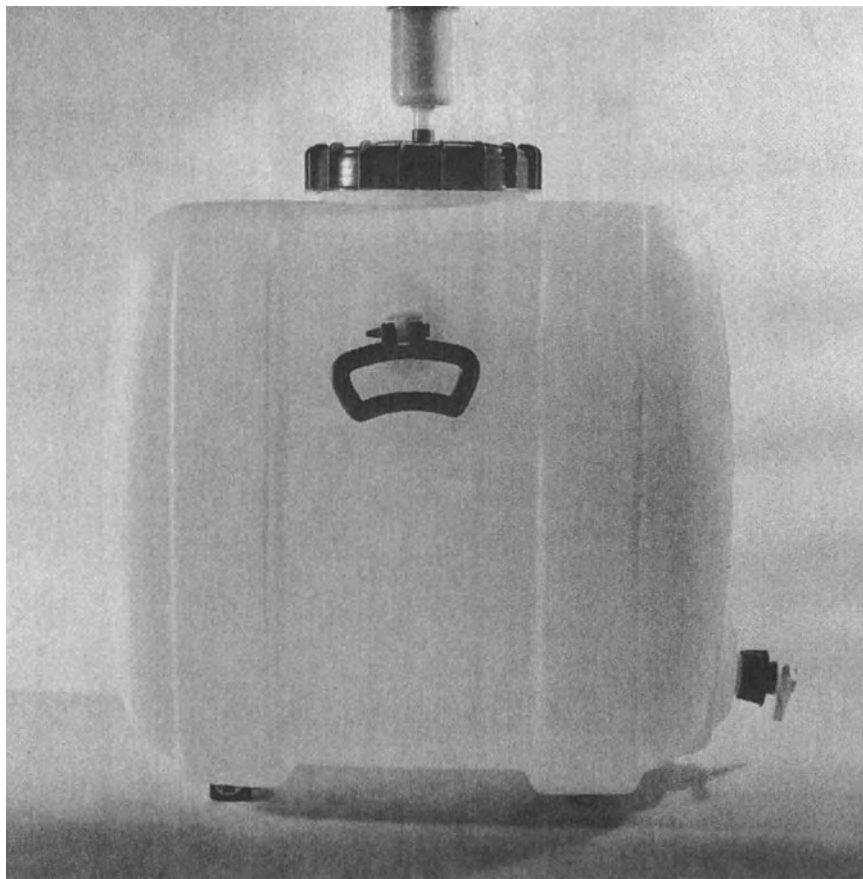


Fig. 8–9 Portable poly all-purpose tanks with fermentation lock.

Courtesy: Prospero Equipment Corporation.

the skins will serve to extract greater intensities of fruit flavor but also will leach out undesirable phenols. Consequently, vintners should equip their wineries with either a refrigerated room, specially designed, with adequate insulation, cooled with an appropriately sized refrigeration unit (Figure 8–10), or jacketed tanks in which fermenting white wines generating heat can be cooled in a custom-designed facility with remote refrigeration compressor and condenser.

Many new wineries are designed with jacketed stainless-steel tanks that can be used for fermenting, blending, processing, and storage (Figure 8–11). Chilled water or alternative refrigerants can be circulated through these jackets for pinpoint temperature control.

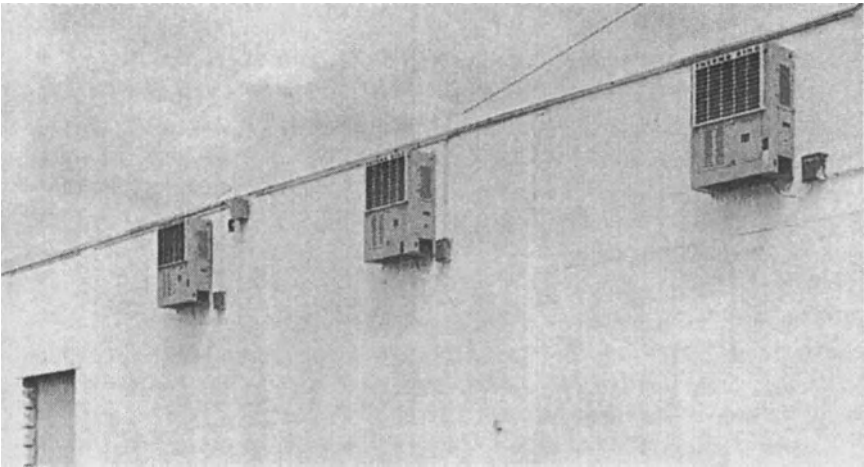


Fig. 8-10 Semi-truck refrigeration units adapted to winery cold room refrigeration.
Courtesy: Thermo King Corporation.



Fig. 8-11 Dimple-jacketed stainless-steel tanks.
Courtesy: Paul Mueller Company.

Filters

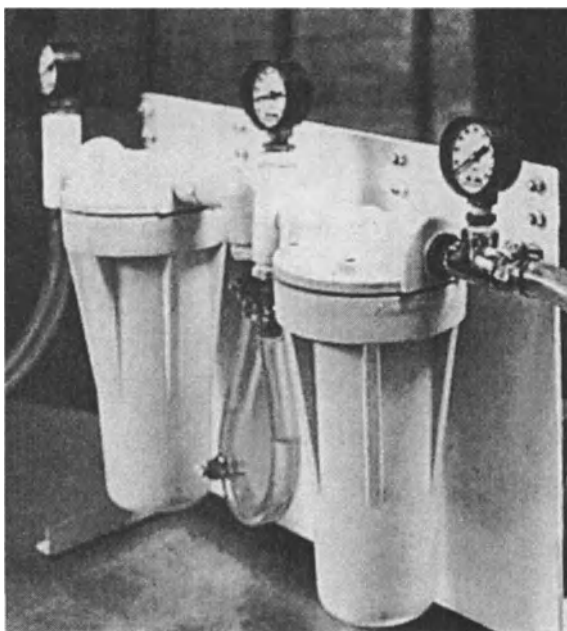
To have a “finished” look, commercial wines should be brilliantly clear. Ill-conceived “straining” operations often do more harm than good.

Vintners may choose a cartridge filter equipped with pump and hoses on a handy portable stand for filtrations of up to 100 gallons. The entire unit usually costs about \$600 with hard durable plastic cartridge housings and a bit more for stainless steel. This type of filter employs disposable cartridges and is easy to use. As a bonus, the pumps and hoses can also be detached for use in transferring wine from one container to another (Figure 8–12).

For larger quantity filtrations, a plate-and-frame filter using cellulose pads may be the best choice (Figure 8–13). These filters are priced according to the number of plates and the type of materials used in construction. A stainless-steel filter with ten 16" × 16" poly plates should be budgeted at about \$4,000. Stainless plates are better but are, of course, much more expensive.

Coarse filtration using diatomaceous earth (DE) in filters that are much akin to swimming pool “sand”-type filters serve to remove suspended particles in young wines long before they are naturally precipitated. This is particularly recommended in white and blush wines in which optimal fresh fruit character is desired. This type

Fig. 8–12 Cartridge filter.



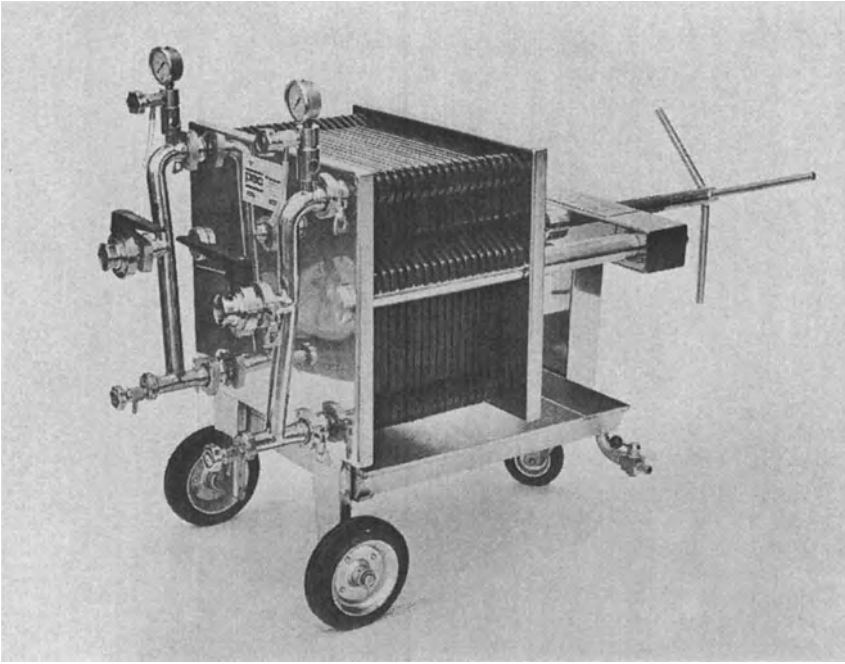


Fig. 8-13 Small winery plate-and-frame filter.
Courtesy: KLR Machines, Inc.

of filtration can be achieved by employing DE plates and frames in appropriate filter types.

There are also “dosing” type filters that are specially designed only for DE filtration.

Fillers

Manual four- and six-spout stainless-steel gravity fillers costing from \$600 to more than \$1,000 can suffice most small wineries well into the long term. These devices operate best with a constant supply of wine from a tank above that is regulated by a float-level inlet valve (Figure 8-14).

Corkers

Wines can, of course, be packaged in bottles and sealed with screw caps. Corks are expensive, but their value in maintaining a resilient seal, displacing headspace

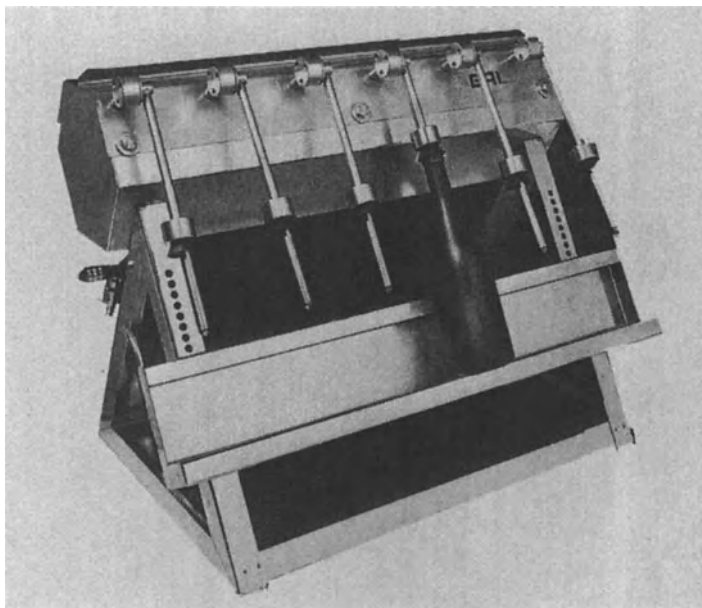


Fig. 8-14 Small winery manual filler.

Courtesy: Prospero Equipment Corporation.

air in bottles, and promoting an aesthetic appearance is far more marketable than caps.

A floor or bench-type corker (Figure 8-15) at less than \$200 can work adequately over the short term, but a semiautomatic corking machine ranging upwards from \$3,000 will be required in the growing winery.

Most corkers operate by a set of “jaws” in which the cork is squeezed. A piston then drives the cork down into a bottle securely positioned under the jaws.

Labelers

The start-up winery can do well in the short term by using a manual label-pasting machine, which can range in cost from \$400 to more than \$1,000. These devices spread glue evenly over the backs of paper labels, which are then applied directly to the bottles by hand. Various types of jigs can be devised to ensure that labels are placed at a uniform location on each bottle. If this uniformity is not closely adhered to, bottles upon shelves look uneven and create a poor product image.

It should be noted that haphazard label application (e.g., labels that are crooked or wrinkled or have lapped-over corners) illustrates to the consumer the vintner’s same lack of care and pride for what is inside the bottle.

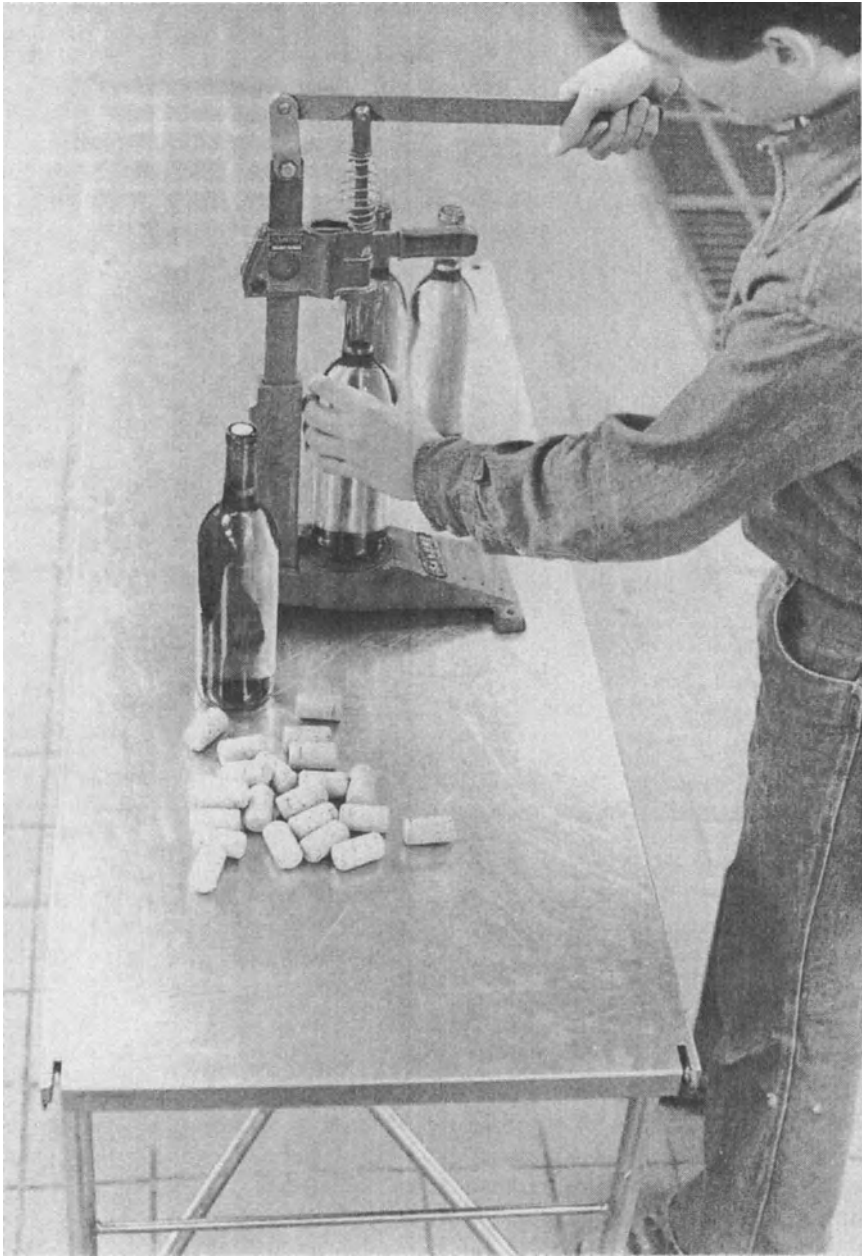


Fig. 8-15 Bench-model manual corking machine.

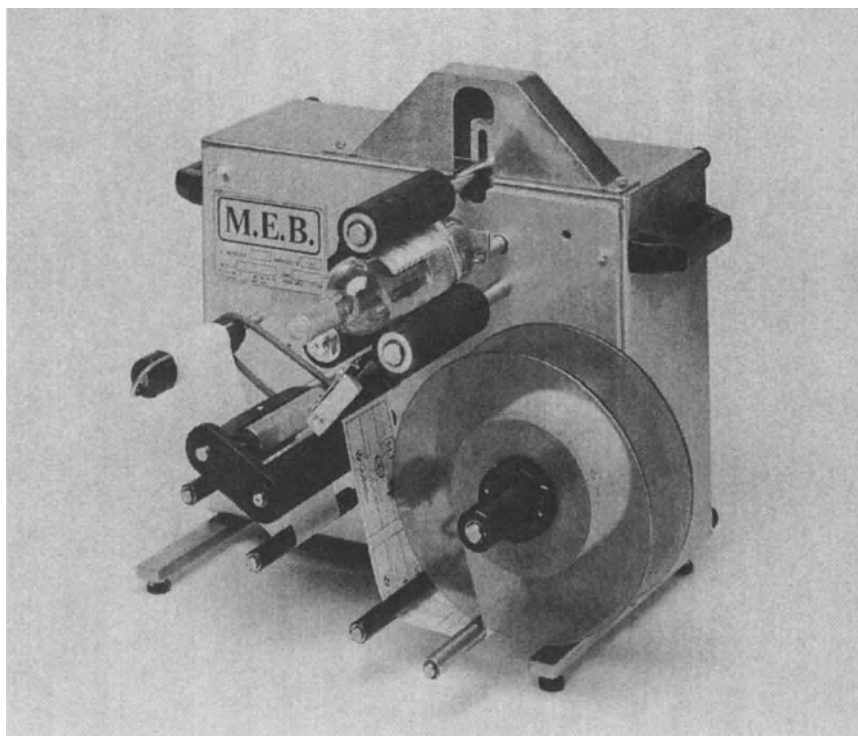


Fig. 8-16 Semiautomatic bottle labeler.

Courtesy: Prospero Equipment Corporation.

There are very few inexpensive semi-automatic labeling machines that operate dependably. Most successful vintners go directly from manual label pasting to an automated labeling machine (Figure 8-16), which may cost upwards of \$7,000.

MATERIALS

The philosophy of applying materials in commercial winemaking generally parallels that of investing in wine instruments and equipment. Sometimes equally good arguments can be made for the frugal, the fancy, and considerations in between. It would be difficult, however, to justify considerable expense for equipment while skimping on materials, and vice versa.

Try to minimize material inventories, with the ideal being to have on hand only what is needed for each season. Some materials, such as potassium meta-bisulfite,

lose strength over extended storage periods. Even in constant freezer storage, yeasts will lose viability. Inventory control is cost control.

This is another area in which cooperating producers can find economies in sharing cost burdens. Larger unit purchases often significantly reduce unit prices.

Potassium Meta-Bisulfite

Also known as KMS, this compound is an absolute necessity in small winery operations. Potassium meta-bisulfite is an inexpensive white powder that serves as both an antiseptic and an antioxidant. Sodium bisulfite is equally effective, but the potassium form is much more popular. With good acidity strength (pH 3.30 or below), 1 ounce of highly active KMS should result in about 30 ppm of “free” sulfur dioxide in 100 gallons of must, juice, or wine. KMS solutions are essential in cellar sanitation.

Be careful when using sulfites. The active ingredient is sulfur dioxide, which is toxic and corrosive. Because KMS loses strength rather quickly, especially in warmer storage conditions, it makes sense to buy KMS from sources that are sure to have a fresh supply. Keep in tightly sealed glass or poly containers—in cool, dry storage.

Pectic Enzymes

Grape cell walls contain several major types of pectins that can reduce press yield efficiency, as well as cause further difficulty in resulting wine clarification and filtration. In the making of jellies, pectin is often added as an aid in solidifying fruit juice. In the making of wine, the opposite problem exists, and commercial winemakers often employ pectic enzymes, sometimes referred to as *pectinase*, to musts before pressing. Due to high cost and the short shelf life of most pectic enzymes, most vintners buy only what is necessary for the upcoming vintage season.

Yeast

Be very careful in selecting a source for yeast. It is tempting to take cultures offered by friendly vintners having the best of intentions. The danger exists in the inadvertent transfer of spoilage organisms along with the yeasts. It is generally acceptable to freeze clean yeast cultures for use the following year. However, there are typically fewer problems and better results from fresh yeast supplies purchased each year just before the vintage season.

While much is currently published in consumer magazines about using natural yeasts, cultured wine yeasts (*Saccharomyces cerevisiae*) are recommended for beginning commercial winemakers. The Red Star Company produces cultured

strains such as Champagne, Montrachet, Premier Cuvee, and Premier Red, which are offered in dehydrated pellets, the most convenient and efficient form to use. There are more good choices available from other producers.

One heaping teaspoon of dry yeast pellets is intermittently stirred in a cup of warm (110°F) water for about 30 minutes before adding to a 3-gallon starter batch of must or juice. Each starter batch of 1 gallon should be sufficient to inoculate juice and must lots of 100 gallons.

It is highly recommended that the material provided in Chapter 3 be fully understood before making any final decision prior to yeast selection, preparation, or application.

Fining Agents

One of the better clarification agents used nowadays is a proprietary compound called Sparkolloid. Prepare in batches that can be easily handled. To each gallon of boiling water, slowly add up to 2 ounces of Sparkolloid while constantly stirring. One gallon of slurry is generally sufficient for 100 gallons of wine; divide or multiply as necessary. While it is still hot, slowly add it directly to the wine and gently stir. Again, be careful, as hot liquids can easily break cold glass. High grades of gelatin (over 100 bloom) can be used similarly.

Fine red wines are often fined with egg whites. The white of just two eggs, beaten to a froth (not stiff), is usually sufficient for a 50-gallon barrel of young red wine.

One of the most common fining agents is bentonite, a clay that swells enormously in water and has a huge adsorptive capacity. Preparation requires extreme care to avoid clotting. Prepare by very slowly (it clumps very easily) adding up to 2 ounces of bentonite per gallon of constantly stirred boiling water. The resulting suspension slurry can be added directly to the young wine. One gallon of slurry should be fully sufficient for 100 gallons of wine. Larger batches of slurry can be made up beforehand and stored cold for up to several weeks prior to use.

Make laboratory trials to determine the minimal amount of finings that can be employed, as most clarification agents reduce both color and flavor. Difficult wines may require increased dosages of finings. Sparkolloid, gelatin, and bentonite can be kept for several years if held in cool, dry storage.

Kieselsol has widely replaced bentonite in Europe and is increasingly doing so in the United States. First, a light gelatin treatment (gelatin readily dissolves in water) is made, and then Kieselsol can be added directly to the wine. It quickly coagulates to embrace any phenol degradation products having resulted from the gelatin fining made previously. The important advantage of Kieselsol over bentonite and Sparkolloid is that it will not noticeably reduce color and flavor unless they are already precariously unstable.

Isinglass is another protein compound preferred by some white wine vintners. Gelatin is often used to reduce astringent phenols, as is polyvinylpolypyrrolidone, or PVPP.

Ascorbic Acid

More commonly known as vitamin C, ascorbic acid is sometimes added in amounts of 3 to 5 ounces per ton of white grapes which have a high propensity to brown from oxidation. The chief use of ascorbic acid is as an additive to white wines just before bottling to add crispness to acidity, add freshness to overall character, and enhance resistance to browning.

Citric Acid

Citric acid is often added at a rate of 1 to 2 ounces per 100 gallons after wines have been fined and stabilized but before filtering. It is an organic acid that helps to further stabilize wines from forming precipitates. Citric acid solutions are also used in “neutralizing” filter media and in cellar sanitation.

Sorbic Acid

For wines finished with residual sugar, sorbic acid is highly recommended to help inhibit secondary fermentation. Sorbic acid is an effective inhibitor of yeasts (not bacteria or molds) at minimum levels of 200 to 300 mg/L. Wines having higher pH, lower total acidity, and weaker alcohol levels generally require a greater dosage. The ATF limit (27 CFR, Part 4, Section 24.246) is 1,000 mg/L for commercial wineries.

Bottles

Considerable history is involved in the shape of wine bottles. The original Roman shape was rather cylindrical, with an abrupt shoulder reduced down to a narrow neck. This was the principal influence on bottles as we know them today (Figure 8–17). As the Roman Empire progressed to the West, into Spain, Portugal, and Bordeaux in southern France, the bottle remained in this traditional shape. But in northern Europe, over several hundred years, the occupation would evolve a new tradition—a more gradual slope in the shoulders, as now expressed in the bottles from Burgundy, Champagne, and the Rheinland. Chapter 4 provides further background on the essentials of selecting wine bottles. Commercial prices range from \$4 to \$8 per case of twelve 750-mL wine bottles.

Table 8–1 indicates what is generally considered appropriate for some of the more popular varietal table wines.

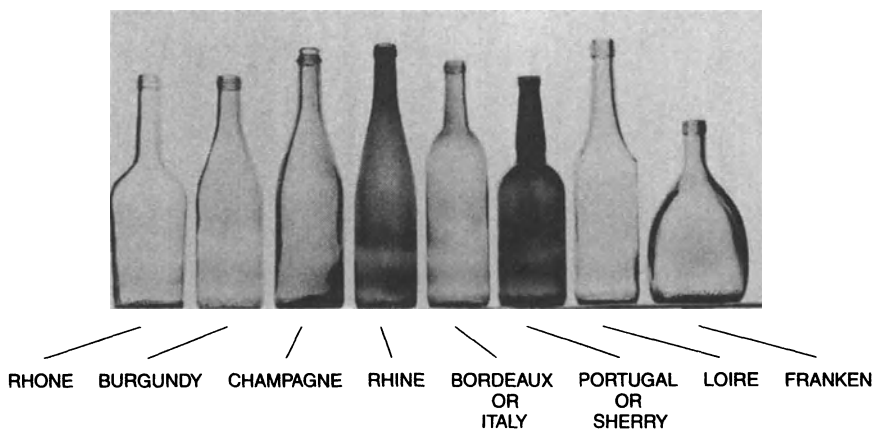


Fig. 8-17 Traditional bottle shapes.

The most common wine bottle size is 750 mL, or 3/4 L, which replaced the 1/5 gallon, or “fifth” (757 mL), when the ATF changed to standardized metric measurements. See 27 CFR, Part 4, Section 4.37, for ATF bottle contents regulations.

Following is a list of common wine bottle sizes and names:

- Split (or quarter-bottle): 187 mL
- Half-bottle: 375 mL
- Bottle: 750 mL
- Magnum (double bottle): 1.5 L
- Jeroboam: 3.0 L
- Imperial: 6.0 L

Splits are used extensively in the transportation and hotel industries but are usually difficult to justify for small winery production due to the high cost of bottling and packaging. In addition, 187-mL bottles have a relative short shelf life. Half-bottles are designed chiefly for restaurant use when only one or two people are dining. Magnums, or double bottles, are used primarily for festive occasions when eight or more guests participate. Some vintners fill a small percentage of production into half-bottles and magnums, with the major share of inventory going into the popular 750-mL size. Jeroboams and imperials are very expensive to buy and difficult to handle.

The use of soft drink bottles, spirits bottles, jugs, and the like is not recommended. They can be used successfully, but they create an impression of carelessness and questionable ethics.

New bottles are usually transported in paperboard cartons. Cartons not in use can easily be “broken down” and folded for storage. When needed they can quickly be restored with glue and/or durable reinforced packing tape.

Table 8-1 Suggested table wine bottles.

Bottle shape	White varieties and types
White Wines	
Bordeaux	Aurora Blanc Niagara Sauvignon Blanc
Burgundy	Catawba Cayuga Chardonnay Delaware Seyval Blanc
Rhine (Hock)	Gewürztraminer Johannisberg Riesling Missouri Riesling Vidal Blanc
Red Wines	
Bordeaux	Cabernet Sauvignon Chambourcin Chancellor Ives Marechal Foch Merlot Norton
Burgundy	Baco Noir Concord Gamay Isabella Pinot Noir

Most important is that bottles be clean before filling. All wine bottles, even new ones, need to be thoroughly cleaned. Used bottles should be scrubbed both inside and outside with a stiff-bristled brush and then rinsed at least three times with scalding hot water. If detergent is used, four or five rinsings are recommended.

Cleaning can be made easier by rinsing bottles as soon as they are emptied. After draining, place them upside down in a paperboard container and seal with tape. Cases of empty bottles can be stored in almost any place that is free from moisture, insects, and rodents and is out of harm's way.

Corks

Corks are made from the bark of the cork oak tree, *Quercus suber*. They can cost more than 15 cents apiece, especially those in the higher quality grades.

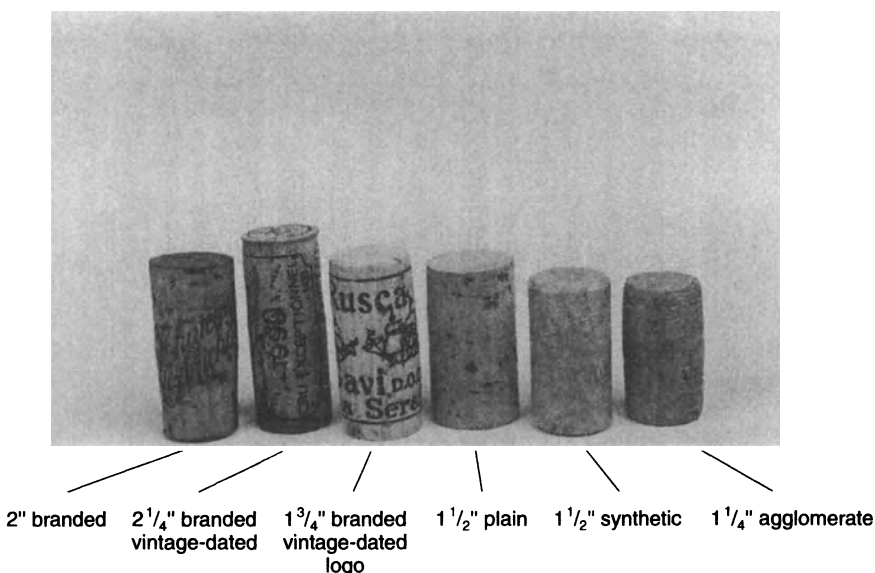


Fig. 8-18 Various cork types and sizes. Photo by David Ferguson.

Corks are available in different sizes and in different finishes, as well as different grades (Figure 8-18). Bottles are molded to a number of varying diameters, for which matching cork diameters are needed. Typical cork dimensions are 9 mm in diameter by 1.75 inches in length. Shorter corks can be used for wines that are expected to be in bottle for only a few months; longer corks are necessary for reds designed for several years of bottle aging.

The best corks—those that have minimal natural imperfections—have designations such as “Extra Fine Quality” or “Grade AAA.” They are chamfered (edges rounded off), de-dusted, coated with a thin coat of paraffin and silicone, and sterilized. These specifications add a few more pennies to cost, but added sealing integrity and fewer application problems are results well worth the expense. Figure on high-quality corks costing more than 15 cents apiece.

Cellukork® and SupremeCorq® plastic corks manufacturers seem to be surmounting the initial problems of difficult driving and even more difficult pulling. There are indications of a rapidly improving technology, such as the virtual immortality of their ethylene-vinyl acetate co-polymer (EVA) formulation, improved lubricants, variable resiliencies, and even the possibility of scavenging headspace oxygen and sulfur dioxide. With prices remaining significantly lower than those for natural corks, more and more vintners seem to be moving toward usage of these synthetic closures.

One of the newest developments is the “hybrid” cork, which is designed to combine the virtues of natural cork with the dependability of plastic. Natural cork and plastic are molded together to form a uniform closure that does not leak and protects against mold formation. See Chapter 4 for more information relating to corks and other closures.

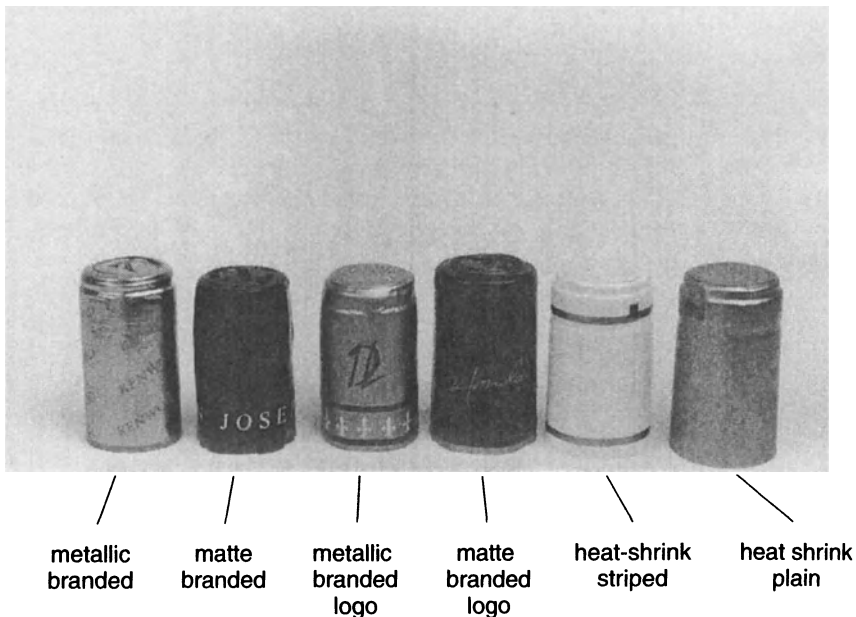
Capsules

Capsules do not add any type of pilfer-proof seal to a wine bottle. Their value is totally aesthetic but is nevertheless important. They provide a “finished” appeal to the final product and express the degree of pride and craftsmanship that a vintner has given a wine.

One popular type of capsule is shrunk on the bottle using a hot air gun similar to a hair dryer. Firm plastic capsules need no application machinery and are relatively inexpensive, although questionable in appearance when compared to the “roll-on” types, which provide a higher quality image. Various capsule types are illustrated in Figure 8–19.

Capsule types are offered in a number of colors and decor schemes. Prices range from less than a penny or two apiece in larger quantities to more than 5 cents

Fig. 8–19 Various capsule types. Photo by David Ferguson.



for high-grade custom capsules. See Chapter 4 for more information relating to capsules and bottles that are designed for no capsule application.

Labels

Labels disclose the winemaker, the wine grape variety or type, the vintage year, and other important information. The ATF has exacting rules relating to label statements and decor that are strictly enforced in the wine industry. Every commercial wine label and label change must be submitted for approval upon ATF Form 5100.31. Only the vintage date can be changed on an approved wine label without ATF authorization.

There is no single more important item in the wine business than the quality of the wine label. For beginning wineries especially, a case can be made for label quality being more essential than even the wine quality itself. The label is what gives potential consumers their first impression of the wine. The reader is encouraged to review the discussion on label design given in Chapter 4.

The application of glue to labels is too slow and messy without at least a label-pasting machine. Semiautomatic labeling machines can cost several thousand dollars. Straight-edge paper labels cost less and work best in these types of small winery labeling machines. Pressure-sensitive labels are not recommended in the start-up winery until experience can serve to deal with the difficulties of adjusting position and removal.

Cases

Bottles are sold by manufacturers by the case in a plain case with appropriate “contents” markings as standard procedure. Cases can be custom printed, but most start-up vintners opt for the plain cases. ATF regulations for identity can be met by simply applying the same labels to each end of the case as are on the bottles inside. Some vintners prefer to apply various types of stamps to identify what is in each case.

CHAPTER 9

WHITE TABLE WINES

Dry white table wines are easy to make but challenging to make consistently well. They require delicate handling to avoid oxidation (browning) and careful attention to protect and develop delicate flavors throughout formative processing.

GRAPE VARIETIES

Chardonnay, Chardonel, Pinot Gris, Sauvignon Blanc, and Seyval Blanc grape varieties make whites that are “vinous” in character (herbal and citrine flavors) and are recommended for dry white wines. Varieties such as Aurora Blanc, Cayuga, Gewurztraminer, Riesling, and Traminette are typically “floral” (honey and spicy flavors) and are recommended for both dry and semisweet wines. Varietals such as Catawba, Delaware, and Niagara are “fruity” (flavors of melons and raisins) and are recommended for both semisweet and sweet wines.

Whatever varieties are selected, they should obviously taste good and have the same type of flavor that is expected in the resulting wine. Harvest the first batches of white grapes from a new vineyard when they are still slightly green, which may be in pH range of 3.10 to 3.20. Adjust to earlier or later harvest, as may seem prudent, thereafter.

One variation on this theme is wines made from purposely overripened grapes, using processes that have been pioneered in Germany. The goal is to maximize sugar content and develop added floral and fruit flavor components in Riesling and other varieties that are cultivated there. Success depends upon high total acidity (TA) levels (over 0.7 g/100 mL) and an agreeable environment. As grapes mature past optimal ripeness, there is an increasing dependency on cold temperatures to inhibit the formation of molds. One mold, the celebrated *Botrytis cinerea*, or “noble mold,” actually aids in the flavor development process. This microorganism permeates grapes skins, allowing berry water to evaporate out and concentrating sugar and flavor components retained in the fruit. Less experienced commercial vintners interested in trying their hand at late-harvest wines should experiment with small lots beforehand.

INCREASING THE FRUIT FLAVOR INTENSITY OF WHITE WINE

The most important factor in extracting optimal fruit flavor is close monitoring of pH and flavors during fruit maturation. Grape flavors change during ripening. Slightly underripened grape flavors, generally in the pH range of about 3.10 to 3.20, may tend to be masked by high TA—something akin to the green-apple flavor found in many cool-climate Riesling wines. The same Riesling allowed to mature in warmer weather may reach a pH range of 3.30 to 3.40 or higher, in which the typical terpene flavors of apricot and peach are readily distinguishable. A higher pH encourages the growth of bacteria and wild yeasts. Consequently, preservation of fruit flavor will be dependent upon the winemaker’s ability to conduct a straightforward cool fermentation of clean juice in sanitized stainless-steel vessels.

Many vintners add pectolytic enzymes to musts held for up to several hours prior to pressing. While this can contribute significantly to fruit flavor extraction, it can also result in a partial disintegration of grape skins and phenol extraction. Consequently, reduced pressure is needed during the pressing operation.

Clarity cannot be a critical factor in the sensory evaluation of fermenting juice or must. During the peak of a normal fermentation, the yeast cell population may reach more than 1 million per cubic inch of liquid. Consequently, winemakers should expect a heavy degree of turbidity due to this heavy suspension of yeast cells.

Thus, one of the most important means of increasing fruit flavor potential is through reduction of solids resulting from crushing and pressing operations. The general idea is that fewer solids provide fewer substrates from which complex flavors develop and mask fruit flavor. In this case, ensure that crusher rollers are not cracking seeds and creating bits of skin and stems so as to minimize these sources for bitter phenols. On the other hand, some pulp solids are needed to

ensure that there is sufficient retained nitrogen and key vitamins necessary for yeasts to perform a complete fermentation.

Suspended and precipitated solids are generally composed of grape pulp, fragments of stems, skins, and seeds; sometimes colloidal materials are also imparted from the must under heavy pressure. If left in the juice, these gross lees (sediment) solids can release distasteful compounds formed by enzymatic breakdown and oxidation. Such solids are also related to colloidal formations, which may require difficult filtrations that diminish and/or degrade fresh fruit flavors. Ideally, pressed juice should be held in cool 50°F stainless-steel tanks for at least 48 hours before beginning the fermentation regimen. The juice should then be racked (separated) from the settled gross lees into another tank before inoculating with yeast.

Some white wines, such as *sur lies* Chardonnay and Seyval Blanc, are designed to increase flavor resulting from extended exposure to secondary grape and autolyzed yeast lees following fermentation, which is generally conducted in oak barrels. It is common to find *sur lies* terms ranging up to several months in duration, with a *batonnage* (stirring with a clean stainless-steel rod) several times per week. As would be expected, this practice is beset with potential microbiological and oxidative perils. An especially tenacious quality control program is required to ensure that this technique does not go awry.

Some vintners are finding more flavor by pressing some portion, as much as half, of their grapes in whole-cluster form. Champagne cuvee (table wines made before secondary fermentation) wines are totally pressed from whole clusters. Lower gallon-per-ton efficiency can be expected, although stems can serve as a press aid and a source of higher tannin content. It is important not to equate whole-cluster pressing with whole-cluster fermentation. While some “nouveau” red wines benefit from whole-cluster fermentation, in whites this can extract bitter leucoanthocyanins, which often oxidize to a yellow pigmentation and a “leathery” tannic flavor.

Whole-berry fermentations are also made, and these require a special type of destemming machine. Individual research is required to ascertain to what extent, if any, whole-berry fermentation may improve the final product. It is common to see whole, uncrushed berries survive more than 500 psi in the pomace of basket-type presses. Some vintners counter this by exerting excessive pressure, which can extract undesirable bitter seed and stem phenols.

Yet another method for increasing white wine fruit flavor intensity is the retention of some portion of unfermented juice in cold storage. This is sometimes referred to as a *muté* and is added to the wine later, but before clarification and stabilization. If the variety and source of the *muté* are different from those of the wine in which it is blended, care must be exercised that the character of the finished product is not distorted. There are specific Bureau of Alcohol, Tobacco, and Firearms (ATF) regulations pertaining to this in 27 CFR, Part 4, Section 4.23.

HARVESTING

Machine-harvested fruit will have broken skins and some juice already extracted along with bits and pieces of leaves, stems, and other foreign matter. Consequently, grapes harvested by hand into clean containers are preferred. Expect a yield of about 130 to 150 gallons of finished wine to result from each ton of grapes.

LATE HARVEST

Vintners may decide to produce a wine from grapes purposely harvested beyond ripeness so that the fruit will have the opportunity to dehydrate to some degree and thus concentrate sugar and flavor content. Beerenauslese wines from Germany are made this way. Late-harvest fruit upon which the gray mold *Botrytis cinerea*, or “noble mold,” has grown are further dehydrated to raisinlike berries exuding very sweet juice with exotic flavors. The famous Sauternes in Bordeaux, France, and Trockenbeerenauslesen in Germany, are made from such “botrytized” grapes. ATF regulations in 27 CFR, Part 4, Section 4.24, do not permit U.S. vintners to use the names *Sauternes*, *Beerenauslese*, *Trockenbeerenauslese*, and other such European late-harvest wine designations. Further, ATF Ruling 78–4 allows “the use of ‘Late Harvest,’ and certain other similar terms on labels of grape wine, provided the amount of sugar contained in the grapes at the time of harvest and the amount of residual sugar in the finished wine were accurately stated in the same kind of measure on a front or back label for the product.” And the ATF ruled in Industry Circular 82–3 that “wine that has been made from concentrate or that has been ameliorated, sweetened or fortified may not bear labels containing the terms ‘Late Harvest,’ ‘Late Picked,’ ‘Botrytis Infected,’ ‘Pourriture Noble,’ ‘Ice Wine,’ or variations thereof.” This is obviously very restrictive, and vintners considering producing such wines should fully understand the provisions of 27 CFR, Part 4, Section 4.39(a)(1), beforehand.

CRUSHING-DESTEMMING

Chill grapes, if necessary, as soon as possible after harvest. A temperature in the range of 45 to 55°F is ideal. If there is excessive “MOG” (material other than grapes), set up a moving belt triage line in which people are assigned the task of removing leaves, sticks, and other trash, along with unripe or spoiled berries. Record net weight, and rinse grapes (hand harvested only) with clean, cold water. In perfect-quality white grape musts, one should consider minimizing or eliminating sulfur dioxide additions, as these increase the extraction of bitter phenolic compounds from seeds and skins. However, if the fruit is subject to degradation from some

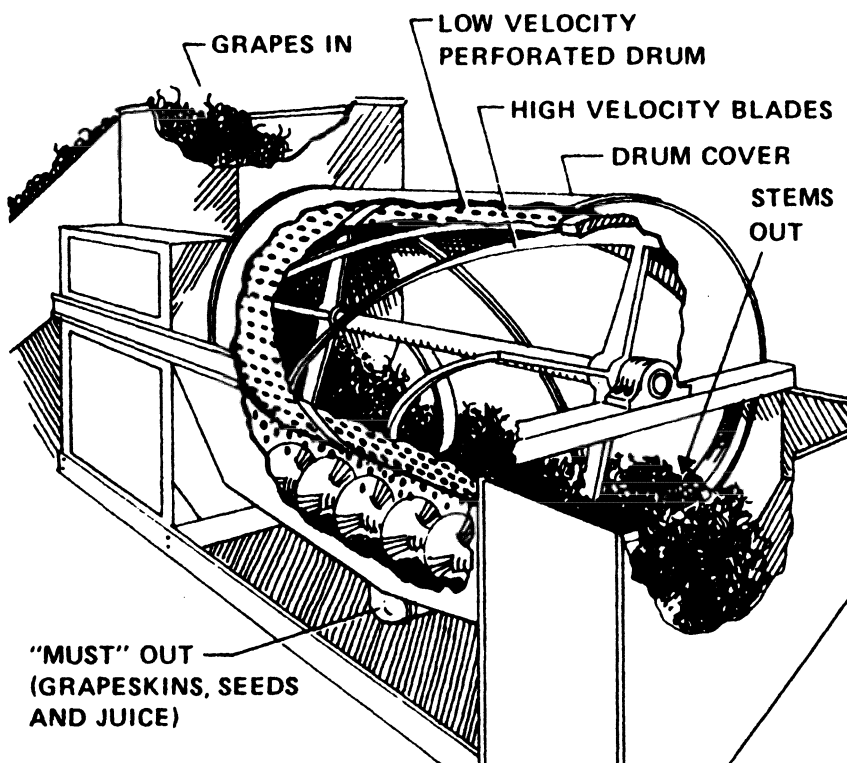


Fig. 9-1 A crushing-destemming operation.
Courtesy: Sebastiani Vineyards.

combination of delay after harvest, heat exposure, microbe infection, or pest invasion, it may be best to consider a sulfur dioxide application of from 40 to 60 ppm. However, as the pH of must increases past 3.20 or so, vintners should expect commensurately increasing problems from heat, microbes, and pests, since higher pH reduces the amount of molecular sulfur dioxide and consequently its effectiveness.

Figure 9-1 shows a crushing-destemming operation.

PECTIC ENZYMES

Winemakers generally add pectolytic enzymes at the crusher to aid in pressing efficiency and juice clarity. Further information on enzyme utility can be found in Chapter 4.

COLD SOAKING

A rather new twist on an Old World technique has developed into a technique that has proven successful for some white wine vintners. Many French and German white wines extract additional flavor compounds from grape skins by holding musts for up to a day prior to pressing. The danger of this rests with exposure to oxygen and spoilage organisms, along with the extraction of undesirable tannins. With adequate refrigeration this exposure can be greatly reduced. A typical procedure is to immediately refrigerate crushed grapes through a heat exchanger so as to reach a temperature from 40 to 45°F and hold them in a tank under a nitrogen or carbon dioxide cap for several hours, or until it is decided that the optimal amount of flavor has been extracted along with minimal tannins. Pressing then proceeds as outlined below.

PRESSING

One of the most common errors in pressing juice from grapes is that of overpressing—of attempting to gain efficiency by long, high-pressure regimens. While this succeeds in maximizing quantity, it is achieved at a costly reduction in quality. Free-run juice is the highest quality; the next-best juice fraction is extracted with gentle pressure. After that, the wine master should carefully evaluate the juice being extracted. Once the juice starts to clear, there is a good chance that undesired bitter and astringent phenols are being released from the skins and seeds.

The juice should be immediately transferred to cool temperature (45 to 55°F) for about 24 hours. During this time, undesired suspended solids from grape pulp and seeds will precipitate. Juice is then racked (decanted) from this lees (sediment) and transferred to a fermenting vessel.

Pomace (skins, pulp, and seeds remaining in the press) should be immediately removed from the winery, as it quickly attracts insects and other pests. A good option for handling pomace is to transport it to a well-designed and maintained compost facility.

BRIX (SUGAR) ADJUSTMENT

Brix in white grape juice typically measures between 14 and 23°. Native *Vitis labrusca* varieties will generally ripen at the lower end of this range, *Vitis vinifera* at the higher end, and the French-American hybrids at midrange. There are exceptions.

Each degree of Brix will convert to about .535% alcohol during fermentation. Consequently, to achieve 12% alcohol by volume, the white grape juice would require approximately 22.5°Brix. A natural Brix of 18.1°Brix would thus need to be adjusted with sufficient sugar to achieve a 4.4° increase in Brix.

The addition of sugar is called *chaptalization*. In some cases, grapes are excessively high in acidity, leading to the addition of both sugar and water—a process referred to as *amelioration*. Both of these adjustments have historically been controversial in fine wine circles.

Chaptalization and amelioration increase the resulting quantity of juice. Vintners must fully interpret the ATF regulations in this regard. They read as follows:

In producing natural grape wine from juice having a low sugar content, pure dry sugar or concentrated juice of the same kind of fruit may be added before or during fermentation to develop alcohol. In producing natural fruit wine from juice having a low sugar content, pure dry sugar, liquid sugar, or concentrated juice of the same kind of fruit may be added before or during fermentation to develop alcohol. The quantity of pure dry sugar, liquid sugar or concentrated juice added may not raise the original density of the juice above 25 degrees Brix. If grape juice or grape wine is ameliorated after chaptalization, the quantity of pure dry sugar added to juice for chaptalization will be included as ameliorating material. If fruit juice or fruit wine is ameliorated after chaptalization pure dry sugar added under this section is not considered as amelioration material. However, if liquid sugar is added to fruit juice, the volume of water contained in the liquid sugar will be included as ameliorating material. (27 CFR, Part 4, Section 24.177)

A simple mathematical formula can be used to calculate granular sugar additions:

$$\text{Juice gallons} \times \frac{(\text{desired Brix} - \text{existing Brix})}{(100 - \text{desired Brix})} = \text{pounds of sugar to add}$$

Consider the following example:

$$1,743 \text{ gallons} \times \frac{(22.5 - 18.1)}{(100 - 22.5)} = 99 \text{ pounds of sugar to add}$$

Sugar additions are, however, illegal in some states, including California, and deficiencies in Brix must be made with the addition of concentrates. Consider the following formula:

$$\text{Juice gallons} \times \frac{(\text{LSR} - \text{LSE})}{(\text{LSC} - \text{LSR})} = \text{gallons of concentrate to add}$$

Where

LSR = pounds of solids per gallon required

LSE = pounds of solids per gallon existing in the juice

LSC = pounds of solids per gallon existing in the concentrate

Consider the example of 1,743 gallon of white grape juice. It has 18.1°Brix, but in this case, grape concentrate at 68.0°Brix is to be added in a sufficient quantity to raise the entire blend to exactly 22.5°Brix. First find LSR, LSE, and LSC from the Brix Table provided in Appendix B.

$$1,743 \text{ gallons} \times \frac{(2.054 - 1.623)}{(7.586 - 2.054)} = 135.8 \text{ gallons of concentrate}$$

However, is common for native *Vitis labrusca* grapes to ripen with total acidities well beyond the 0.900 g/100 mL, which may be considered maximum tartness. Consequently, winemakers may opt to ameliorate with both sugar and water in order to dilute this acidity.

A well-designed Lot Origination Form (Exhibit 9–1) will permit vintners to accurately calculate sugar/water additions and simultaneously maintain a good record of increases in gallonage.

Total acidity (TA) in white grape juice should ideally measure between .65 and .75 g/100 mL, or, roughly, between .65% and .75% acidity. In the more delicate white varieties, such as Chardonnay and Seyval Blanc, it may be best to leave a slightly higher acidity unadjusted. Juice from the floral and fruity varieties can benefit by ameliorating high acidity with the judicious addition of clean, potable water.

A good rule of thumb is to optimize, not maximize, amelioration. Calculate water additions by first dividing the desired TA by the naturally existing TA. Second, use the *inverse* of the result as the percentage of water to add to the juice. The inverse is that number required in addition to an *obverse* decimal number needed to result in 1.00. The following is an example of 100 gallons of juice analyzed to contain .96 g/100 mL TA:

$$.85 \text{ (desired TA) divided by } .96 \text{ (existing TA)} = .89$$

$$\text{inverse of } .89 = .11 \text{ (percent water addition to juice)}$$

$$100 \text{ gallons of juice multiplied by } .11 = 11 \text{ gallons of water}$$

The inverse for a desired TA of .80 in the above example would be .17 (0.80 divided by .96 = .83). For a desired TA of .75, the inverse would be .22 (0.78 divided by .96 = .78), and so forth.

It is essential that winemakers fully understand the ATF rulings relating to amelioration, as set forth in 27 CFR, Part 4, Section 24.178 (note: below, is the conversion of the ATF's usage of "grams per liter" to "grams per 100 milliliters," as the latter is more commonly used throughout the U.S wine industry, although Europeans more often use the former):

Exhibit 9-1 White wine Lot Origination Form in service.

LOT ORIGATION			
DATE	<u>9-24-01</u>		CULTIVAR <u>CHARDONNAY</u>
LOT NO.	<u>541</u>		REMARKS: <u>rec'd 9-22-74°</u>
SERIAL NO.	<u>11-95</u>		<u>triaged for slight</u>
FERMENTER NO.	<u>F-4</u>		<u>botrytis and MOG, no</u>
NET GALLONS	<u>1.814</u>		<u>KM5, 3oz P-enzyme</u>
NET TONS	<u>11.22</u>		<u>6 press loads with</u>
GALLONS PER TON	<u>161.7</u>		<u>rice hulls - juice</u>
pH	<u>3.30</u> T.A.	<u>.915</u>	<u>chilled to 47° in 5-7</u>
BRIX	<u>19.1</u> ALC.	<u>-</u>	<u>for 48 hrs - 5B gals. less</u>
EXT.	<u>-</u>		<u>destroyed on 9-24</u>
AMELIORATION = <u>5</u> % @ <u>22.5</u> ° BRIX = <u>12</u> ALCOHOL <u>1.814</u> GALLONS START = <u>1.909</u> RESULTING TOTAL <u>.95</u> INVERSE OF GALLONS OF PRODUCT AMEL. PERCENTAGE			
<u>1.909</u> TOTAL @ <u>22.5</u> ° BRIX <u>12.054</u> LBS/GAL = <u>3,921</u> TOTAL GALLONS LBS.			
<u>1.814</u> START @ <u>19.1</u> ° BRIX <u>11.720</u> LBS/GAL = <u>3,120</u> START GALLONS LBS.			
<u>95</u> AMELIORATION <u>801</u> ADDITION GALLONS LBS.			
<u>59</u> SUGAR AS <u>59</u> SUGAR AS GALLONS x .074 (GAL/LB) = GALLONS			
<u>36</u> WATER GALLONS			
<u>Pasteur</u> YEAST ADDITION = RATE OF <u>1</u> LBS PER <u>1,000 gals</u>			
REMARKS: <u>dry yeast mixed in 15 gals of water @ 100°</u>			

- (a) General. In producing natural wine from juice having a fixed acid level exceeding .500 gm/100 ml, the winemaker may adjust the fixed acid level by adding ameliorating material (water, sugar, or a combination of both) before, during and after fermentation. The fixed acid level of the juice is determined prior to fermentation and is calculated as tartaric acid for grapes, malic acid for apples, and citric acid for other fruit. Each 20 gallons of ameliorating material added to 1,000 gallons of juice or wine will reduce the fixed acid level of the juice or wine by 0.01 gm/100 ml

(the fixed acid level of the juice or wine may not be less than .500 gm/100 ml after the addition of ameliorating material).

(b) Limitations.

- (1) Amelioration is permitted only at the bonded wine premises where the natural wine is produced.
- (2) The ameliorating material added to juice or wine may not reduce the fixed acid level of the ameliorated juice or wine to less than .500 gm/100 ml.
- (3) Except for wine made exclusively from loganberries, currants, or gooseberries, the volume of ameliorating material added to juice or wine may not exceed 35% of the total volume of ameliorated juice or wine (calculated exclusive of pulp). Where the starting fixed acid level is or exceeds .769 gm/100 ml, a maximum of 538.4 gallons of ameliorating material may be added to each 1,000 gallons of wine or juice.

ACID ADDITIONS

Lower acidities are encountered more frequently in grapes grown in warmer climates, such as central California grapes, than in those grown in more cooler regions of the United States. Tartaric acid additions to increase TA can be calculated using Table 9–1. Stir in all additions, and test for Brix and TA once more to ensure that the desired levels have been achieved. Post additions in the appropriate record.

YEAST INOCULATION

Refer to Chapter 3 and the subsection “Yeast” in the “Materials” section in Chapter 8 to become familiar with media and rates used for juice inoculation. Follow the manufacturer’s instructions for cultured wine yeast preparation, or refer to the discussion of dry yeast pellets in Chapters 3 and 8. Inoculate juice with cultured wine yeast starter, and gently stir. Clean the yeast preparation containers, stirrers, and other associated instruments with scalding water, and drain dry.

Table 9–1 Total acidity table for Tartaric acid additions.

Total acidity (g/100 mL)	Per-gallon addition for 0.600 g/100 mL	Per-gallon addition for 0.700 g/100 mL
0.450	5.7 g	9.5 g
0.500	3.8 g	7.6 g
0.550	1.9 g	5.7 g
0.600	—	3.8 g
0.650	—	1.9 g

At one time winemakers inoculated by simply taking a few gallons of another fermenting white wine, or lees sediments, and adding them straightaway to another lot of juice. This practice is now known to be a potential source for encouraging and perpetuating wild yeast and bacterial infections, as well as other problems.

FERMENTATION

Clean fermenters made of glass or stainless steel are ideal for white wine fermentation, processing, and storage. One variation is to ferment the juice from vinous white grapes in oak barrels to achieve an added “toasty-nutty” flavor complexity.

Fermenters should be filled to about 70% of capacity, which will allow room for foaming to occur without fouling the fermentation lock. Each fermenter should be properly fitted with a clean lock and placed in a cool, dark room or cellar that is freely ventilated. Ideal temperature is in the 55 to 60°F range.

IMPORTANT NOTE: FERMENTATION IS DANGEROUS! IF THE FERMENTER VESSEL IS NOT PROPERLY VENTED WITH A FERMENTATION LOCK, THE CONTAINER MAY EXPLODE. FURTHER, CARBON DIOXIDE GAS GENERATED FROM FERMENTATION CAN BE ASPHYXIATING IN CLOSED, POORLY VENTILATED AREAS. EMPLOYEES, VISITORS, AND ESPECIALLY CHILDREN SHOULD BE CAREFULLY SUPERVISED IF ALLOWED IN LOCATIONS WHERE ACTIVE FERMENTATIONS ARE TAKING PLACE.

It may take a day or two, perhaps several, for fermentation to commence so that regular bubbles can be observed in the lock, along with thin spots of foam forming on the surface. If a delay of more than a week occurs, gently heat the juice up to 65 to 70°F until fermentation starts, and then return it to the cooler temperature. Such delays are generally attributed to deficiencies in free nitrogen and/or poor yeast viability.

No two fermentations are exactly alike. Three fermenters of the same juice may take three different lengths of time to finish fermenting. At 60°F, however, a Balling below –1.5° should be achieved within 3 weeks or so. Exhibit 9–2 shows a White Wine Fermentation Control Form typifying the fermentation process.

STUCK FERMENTATION

A “stuck” fermentation is one that has not run the full course, a fermentation that has stopped even though a good share of the fermentable sugar remains. This is generally caused by having added too much KMS at the crusher and/or having fermented at temperatures that are radically low or high. Conversely, too little KMS may permit certain yeast-inhibiting bacteria to grow in the must. Grapes that have been heavily treated with fungicides in the vineyard can also be a problem source.

Exhibit 9-2 White wine fermentation control form in progress.

FERMENTATION CONTROL RECORD												
TANK NO. <u>F-4</u>		LOT NO. <u>541</u>		VARIETY <u>CHARDONNAY</u>								
DATE	TIME	TEMP.	BALL	ALC.	T.A.	V.A.	C	C	N	T		
9-25	7:00am	55°	22.2	—	.870	—	✓	✓	✓	✓	✓	
9-26	7:30am	57°	21.6	—			✓	✓	✓	✓	✓	
9-27	7:30am	58°	20.0	—	.855	—	✓	✓	✓	✓	✓	
9-28	7:00am	59°	18.0	—			✓	✓	✓	✓	✓	greeny
9-29	7:00am	60°	15.7	—	.833	.012	✓	✓	✓	✓	✓	-stuck
9-30	8:00am	62°	13.1	non	cooling jacket for 2 hrs		✓	✓	✓	✓	✓	
10-1	7:00am	54°	11.5	5.1	.773	.018	✓	✓	✓	✓	✓	
10-2	7:30am	57°	9.5				✓	✓	✓	✓	✓	-stuck
10-3	7:30am	59°	7.2	non	cooling jacket for 2 hrs		✓	✓	✓	✓	✓	
10-4	7:00am	57°	5.5		.743	.018	✓	✓	✓	✓	✓	
10-5	7:30am	59°	4.1				✓	✓	✓	✓	✓	
10-6	7:00am	60°	3.0	9.0	.728	.024	✓	✓	✓	✓	✓	-stuck
10-7	7:00am	61°	1.9	non	cooling jacket for 1/2 hr		✓	✓	✓	✓	✓	
10-8	7:30am	59°	1.0			.024	✓	✓	✓	✓	✓	
10-9	7:00am	59°	0.5	10.9	.720	.024	✓	✓	✓	✓	✓	
10-10	7:30am	59°	0.0				✓	✓	✓	✓	✓	
10-11	7:00am	60°	-0.4	11.3	.713	.024	✓	✓	✓	✓	✓	
10-12	7:30am	60°	-0.7		.713	.024	✓	✓	✓	✓	✓	
10-13	7:00am	60°	-1.1	11.7	.713	.024	✓	✓	✓	✓	✓	
10-13	10:00 am				sampled to TK-5-2							38 gals. Lees destroyed
					work order no 191-93							
REMARKS: <u>No cracked seeds in Lees. TK F-4 needs a new</u> <u>transway gasket</u>												

More recently, blame has been cast upon diseased grapevines, such as those infected with *Phylloxera*, which are unable to take up essential nutrients from the soil.

There is no simple remedy for stuck fermentations. If the problem can be positively diagnosed as excessive sulfur dioxide content, then vigorous agitation may help—at the risk of oxidation. An appropriate temperature adjustment and/or the addition of a modest nitrogen supplement, such as diammonium phosphate (DAP), up to 150 ppm, may also be effective. See 27 CFR, Part 4, Section 24.246 for specific types and limits authorized by the ATF.

FIRST RACKING

As soon as fermentation bubbles have resounded to a very slow repetition, the new white wine should be racked (decanted) off the gross lees (grape particles and dead yeast sediment) into clean storage vessels.

One variant of this process is traditional to the white Burgundy wines made in France, which remain *sur lies*, or “on the lees,” for several months after fermentation is completed. Ideally, this results in the development of complex “cheesy-earthly” flavors. Without proper care and expertise, however, the lees can also provide plenty of growth nutrients for spoilage organisms.

Another variation is to arrest fermentation in order to conserve some predetermined level of residual sweetness. This is usually performed by chilling and then cold storage at about 30°F, along with a coarse filtration. This treatment will slow yeast viability and remove most viable yeast cells.

The new wine should be racked into containers which can be filled with perhaps only a slight amount of air space. This void will allow the final few carbon dioxide bubbles to escape without fouling the fermentation trap which is continued in use after the first racking. It helps to determine which containers will be used beforehand so that the new wine does not require handling over and over again. Continue fermenting at the same 55 to 60°F temperature. Clean and sanitize empty fermenters and racking equipment thoroughly. Post the racking date and resulting gallonage in the appropriate record.

SECOND RACKING

After bubbling has completely stopped (it may be only a day or two after the first racking or as long as perhaps a week or two), repeat the racking process. New white wine should test with only a trace of residual sugar. If so, the tank should be filled completely to expel any air and a loose cap or plug used to cover the second-rack containers instead of the fermentation lock.

An addition of 1 ounce of KMS per 100 gallons should increase the sulfur dioxide content in the young wine by about 20 to 30 ppm. This is recommended to help further protect against spoilage microorganisms and oxidation. Continue storage in a cool, dark, cellarlike environment.

From this point on, throughout the life of the wine, it will be necessary to ensure that it is stored in containers that are completely full. Preventing air contact with the wine reduces its exposure to oxygen, which is a component for oxidative browning and the growth of spoilage organisms.

BLENDING

The general rule of thumb in blending is to associate the different virtues of individual wines into a resulting wine that has a more broad and satisfying appeal. For example, grapes harvested a bit too green may have made a wine that has superb acidity but is shy in flavor. Another lot may have been harvested a bit overripe so that the resulting wine is bountiful in flavor but weak in acidity. Laboratory-scale trials will reveal the best blending ratio.

Wines should be blended before stabilization. Because so many factors are involved with stability, sometimes stable wines blended together form an unstable wine. See Chapter 4 for an in-depth discussion of blending considerations.

Each blend will, of course, give cause for appropriate entries in appropriate records.

FINING AND STABILIZATION

Young white wines (2 to 4 months old) typically have a hazy, translucent appearance, caused principally by proteins and other organic residues in suspension. Most new wines also have excessive potassium and tartrate ions in solution. This eventually crystallizes as potassium bitartrate, or cream of tartar.

Gelatin and Kieselsol, or bentonite clay and Sparkolloid mixtures, are recommended for clarification, or *fining*, of the haze. A typical application is about 1 ounce per 100 gallons of wine. Description and preparation of these materials is found in Chapters 4 and 8 of this book.

After the finings have been added, the wine should be “topped up” to ensure that each container is completely filled, sealed, and allowed to rest for 4 to 6 hours at normal room temperature. Cold stabilization should then commence. This is performed by storing the wine in an insulated chamber at a temperature of 26 to 28°F for at least 3 weeks. This time can be significantly reduced by the addition and frequent agitation of specially prepared potassium bitartrate crystals—the “seeding” of particles that serve as nuclei in the formation of new crystals.

The refrigeration capacity to maintain detartration temperatures is expensive in terms of both equipment and energy. One alternative is allowing ambient winter cold into winery storage cellars, which can work well if controls are in place to counteract severe cold temperatures. Freezing wine can radically distort flavor profiles and, worse, expand to rupture containers. Frozen water pipes are, of course, yet another risk.

It is easy to confuse tartrate crystals with filter media when wine samples are viewed under a light microscope. The preparation of slides with various crystals and filter media can help to develop an expertise in using microscopy as an effective quality control device (see Figures 9–2 through 9–5).

Potassium bitartrate *argols* (crystals) resulting from detartration can easily clog drains, so it is a good idea to remove them into solid waste disposal. Thoroughly clean empty clarification and stabilization containers. Ensure that all clarification agents used are posted in the record, along with the temperature(s) and duration of cold storage for stabilization. Table 9–2 provides a source that may be useful in determining the cause of instability problems.

AGING

There is a wide variation in how oak aging is employed in winemaking. Some white wines, especially the vinous ones described at the beginning of this chapter, can significantly improve from a very carefully controlled regimen of oak aging.

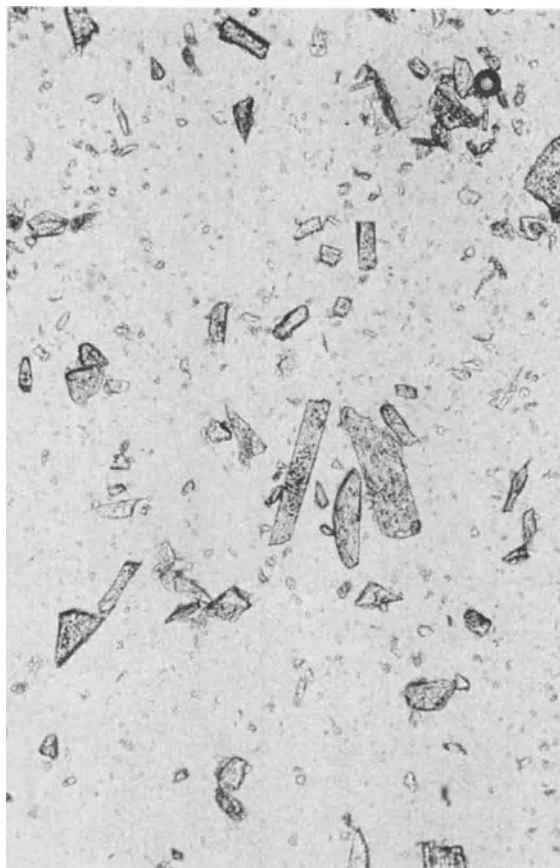


Fig. 9–2 Light microscope image of potassium bitartrate crystals.

The desired effect is a bit of “toasty-vanilla” character, which can be a fine addition to bouquet and flavor. But wines made from the floral and fruity varieties are often better without any oak aging complexity in the bouquet and flavor profile.

Oak aging in commercial wineries is usually performed in expensive barrels having approximately 55 gallons of capacity. Smaller containers such as 15- to 30-gallon kegs can be used effectively but are usually constructed from inferior types and grades of oak. The smaller the keg, the more wood surface area exposure there is per unit volume of capacity, and the faster the wine will respond in aging. Wooden containers require special cleaning and maintenance to be effective. See Chapter 4 for a full discussion of barrel aging.

Oak chips and oak extracts can also be used effectively, provided they are employed properly. Obviously, these do not add the charisma in the cellar as do barrels.

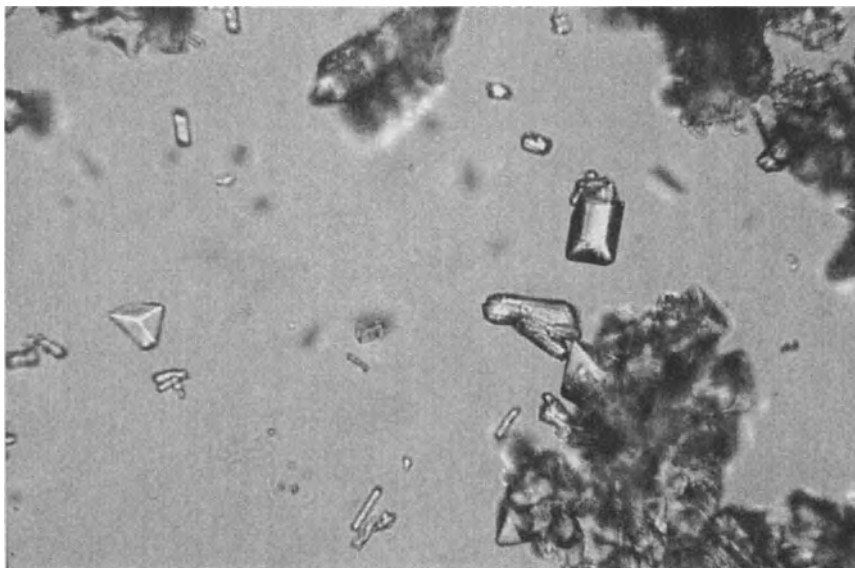


Fig. 9-3 Light microscope image of calcium tartrate.

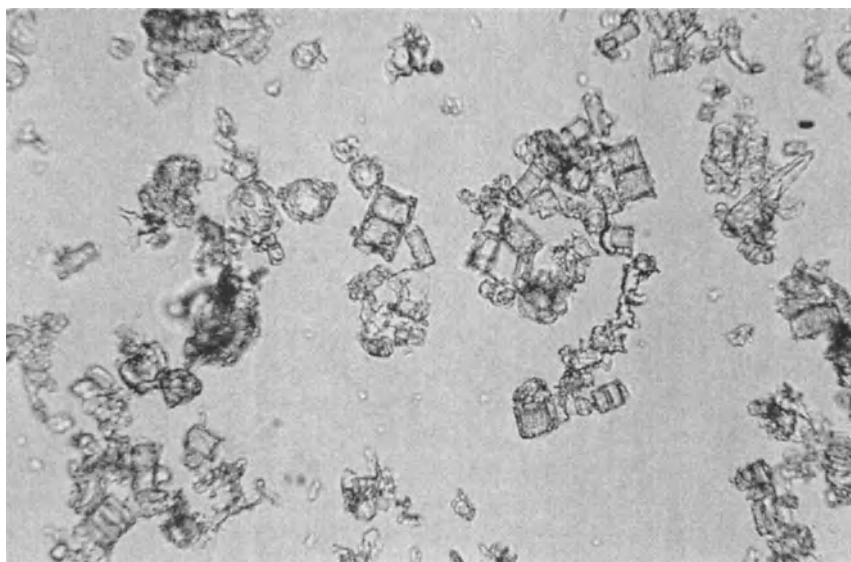


Fig. 9-4 Light microscope image of diatomaceous earth filter media.

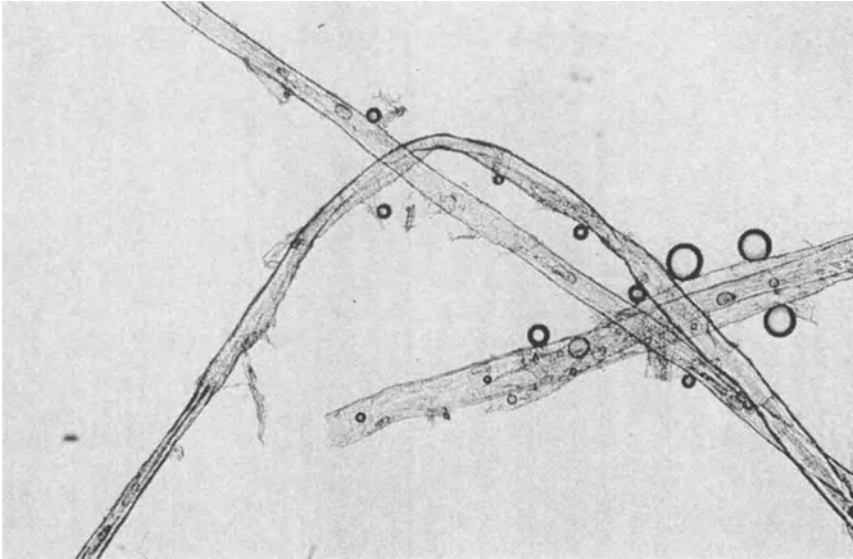


Fig. 9–5 Light microscope image of cellulose pad filter media.

Wine should be evaluated at least once weekly during wood aging to monitor development of bouquet and flavor. As the desired effect appears, evaluate more frequently to prevent the wine from becoming too “oaky.”

Detailed wood aging history is a particularly valuable record to maintain for future reference.

BALANCING AND PRESERVATION

Wines should be balanced and preserved just before bottling.

To balance a wine is to bring harmony to sweetness/acidity ratios. What is perfectly balanced to one person may be out of balance to another. Most avid wine consumers prefer dry wines, especially with food. Some people prefer a touch of sweetness, while still others enjoy wines that are very sweet. Table 9–3 presents some generally recognized levels.

Cane sugar or citric acid additions to achieve the desired balance for each wine should be made just prior to final filtration and bottling. ATF regulates sugar additions in 27 CFR, Part 4, Section 24.179, and acid additions in Section 24.182.

Take small samples and test various laboratory-scale additions first. From this testing, the “ideal” prescription of sugar and/or acidity can be calculated for the entire lot. Stir thoroughly, as sugar and acid additions do not readily dissolve in cold wine.

Table 9-2 Determination, causes, and recovery of wine instability.
Courtesy: Tanner, H., and Vetsch, U. 1956. "How to characterize cloudiness in beverages." Am. J. Enol. 7, 142-149.

Designation of turbidity or separated material	Cause	Prevention	Recovery
Organic copper turbidity	Copper contact of beverages (spray medium, copper piping, poorly silver plated or tinned fittings or equipment).	Avoidance of copper contact. Tinned copper pipes are often defective and should be replaced if possible with stainless steel.	About 90% of the existing copper is eliminated by fermentation. Addition of sulfurous acid is eventually able to reducing copper turbidity to a soluble stage. (Total SO ₂ must not be above 80 mg/L.)
Calcium sulfate sediment	This turbidity occurs in brandy which has been diluted with calcium-containing water.	Avoidance of calcium-containing water for dilution purposes.	Cold storage and filtration, or ionexchange treatment (IR 120)
Winestone sediment	Rapid cooling or higher tartaric acid content, or brief cask storage.	Employment of softened water. Control of tartaric acid content before bottling. At 0°C, about 1.5 g/L tartar is soluble in the beverage (with 10% alcohol by volume)	Cold storage of wines and filtration or do tartaric acid determination and as far as possible on this basis carry out chemical deacidification.
Calcium tartrate sediment	During short observation period after chemical deacidification. Perhaps employment of calcium containing filter sheets.	Wash filter sheets in tartaric acid solution! Observation time before racking off from deacidification sludge should be at least 4 weeks.	Cold storage of beverages and filtration.
Copper sulfate turbidity	Copper containing beverages with simultaneous high total sulfurous acid content (over 120 mg/L).	Prevent copper uptake! Remove copper piping and replace with stainless steel or hose.	Reduce copper concentration by blending: aeration of beverage to bring about disappearance of SO ₂ (insoluble cuprous compounds thereby go over into soluble cupric compounds)

Ferric phosphate turbidity	High iron content (More than 6–8 mg/L)	Prevent iron uptake! Lacquer bare iron parts.	Remove iron by Aferrin fining. Lower the iron content by blending with iron-poor beverage. Gelatin fining in connection with stronger dosing with sulfurous acid.
Pigment-tannin sediment	High pigment and tannin content.	By longer storage of harsh and highly colored beverages in casks, sedimentation in bottles can be largely avoided.	Gelatin fining with 20 g/100 L and dosing with sulfurous acid up to a content of 25–30 mg/L free SO ₂ helps to remove an excess of tannin and leads to stabilized beverage.
Iron-tannin turbidity	High iron and tannin content.	Avoid iron uptake! Lacquer bare iron parts.	Gelatin fining for removal of excess tannin. Blend with iron and tannin-poor beverages. Citric acid addition (50 g/100 L in wine; 200 g/100 L in fruit wine.)
Biological turbidity	Yeast and bacterial activity	Sterile bottling as preventative measure. Sanitary operations!	Tight filtration (EK filter) and dosage with sulfurous acid; pasteurization.
Thermolabile protein turbidity	Unfermented beverages, especially grape juice, contain under certain circumstances thermolabile protein, which, after heating (pasteurization) and cooling lead to turbidity	(a) Pasteurization done twice, filtration between times. (b) Bentonite treatment	Beverages with protein sediment are pasteurized, cooled and filtered. Repeat heating leads to no new turbidity.

Source: Tanner, H., and Vetsch, U. 1956. How to characterize cloudiness in beverages. *Am. J. Enol.* 7, 142–149.

Table 9–3 Typical white table wine sugar–acid balance levels.

Balance level	Degrees balling	Total acidity (g/100 mL)
Tart	–2.0	0.700
Dry	–2.0	0.600
Semidry	–1.0	0.600
Semisweet	0.0	0.600
Sweet	≥2.0	0.600
Insidid	≥2.0	<0.500

Preservation can be achieved with a final addition of KMS, usually at a rate of 1 ounce per 100 gallons for dry wines at a pH of 3.30 or under.

Wines having residual sugar, or added sugar, should be preserved with a KMS addition of at least 2 ounces per 100 gallons and even higher doses for pH levels greater than 3.30. Be certain that the KMS supply used is fresh and active. A failure to halt yeast action can result in secondary fermentation and exploding bottles. As a further safeguard, a dosage of sorbic acid is also recommended. The following formula can be used in calculating the proper addition.

$$\begin{aligned} &\text{mg/L of sorbic acid desired} \times (150/112) \times (3.8) \times \text{gallons of wine} \\ &= \text{mg of sorbic acid required} \end{aligned}$$

Take the following 100-gallon example, in which 350 mg/L of sorbic acid is desired:

$$350 (150/112) (3.8) (100) = 178,125 \text{ mg}$$

Dividing the 178,125-mg result by 1,000 results in 178 g of sorbic acid addition required, or about 6 1/4 oz. The ATF limits sorbic acid additions to a maximum of 300 mg/L in 27 CFR, Part 4, Section 24.246.

Post all sugar/acid, KMS, and sorbic acid components used for balancing and preservation on the appropriate record.

FILTRATION

Brilliantly clear wine has a visual appeal, but filters and filter media are high-cost items.

The most common filter used in small wineries is a cartridge type fed by a small pump. This involves a comparatively small filter investment but a high cost for the cartridge media, which are typically fabricated from various cellulose and other fiber materials. While these materials may be advertised as “inert,” they can impart unwanted flavors into the wine and/or absorb desired flavors from the wine. Consequently, thorough rinsing water, in which 5 g/L each of KMS and citric acid are dissolved, should be pumped through each cartridge before commencing

subsequent wine filtrations. If both white and red wines are to be filtered, filter the white(s) first.

While the savings may be tempting, it is not recommended that cartridges be “cleaned” for using again. Spent cartridges contain heavy concentrations of microorganisms trapped inside the media and should be disposed of immediately. Disinfecting, soaking, and other contrived attempts cannot reclaim cartridges for commercially dependable results.

Once a small winery has grown to produce more gallonage than cartridge filters can handle, it is time to consider a plate-and-frame type that uses cellulose pads for media. These plates are standardized 16" × 16", and setups can exceed 40 pads for larger lots. It is an obvious waste to equip the winery with far greater capacity than foreseeably necessary. Pad media availability is diverse, from several microns of coarse porosity to less than 1 :m nominal, which can remove most yeasts and larger bacteria cells.

While commercial membrane filtrations can remove all yeasts and bacteria from wines, this technology is available only if one can spend many thousands of dollars for equipment and media. For all intents and purposes, filtration is purely cosmetic: repeated filtration does not remove winemaking flaws, but it will diminish bouquet and flavor. See Chapters 4 and 8 for additional information regarding wine filtration. Figure 9-6 illustrates a lab monitor for filtration quality control.

After filtration operations are completed, the filter, tubing, pump, and other equipment used in filtration should be cleaned with scalding hot water and drained dry.

BOTTLING

The bottling operation is usually one of the most festive operations in winemaking, but it should be performed with serious attention to the exceptional care required for sanitary practices. Figure 9-7 shows a schematic of a bottling line operation.

It is recommended that the sections of Chapters 4 and 8 referring to bottles be reviewed before bottling operations begin. New bottles should be rinsed at least once with scalding hot water. High-pressure “jet” spray devices work best. Used bottles should have been cleaned as soon as they were emptied and stored upside down. Under the best of storage conditions, however, used bottles may need scrubbing with a brush to clear out foreign matter. Used bottles are, of course, recyclable, but the expense of cleaning is often more than the cost of new bottles.

New wines will undergo an abrupt change in flavor and overall character due to the rigors of preservation, final filtration, and bottling—a condition referred to as “bottle shock.” This state usually passes in a few weeks, and the wine returns to near its original condition.

Most white wines are best only up to a few years of aging. Some people feel that the older a wine, the better. Not true. While some red wines may improve with

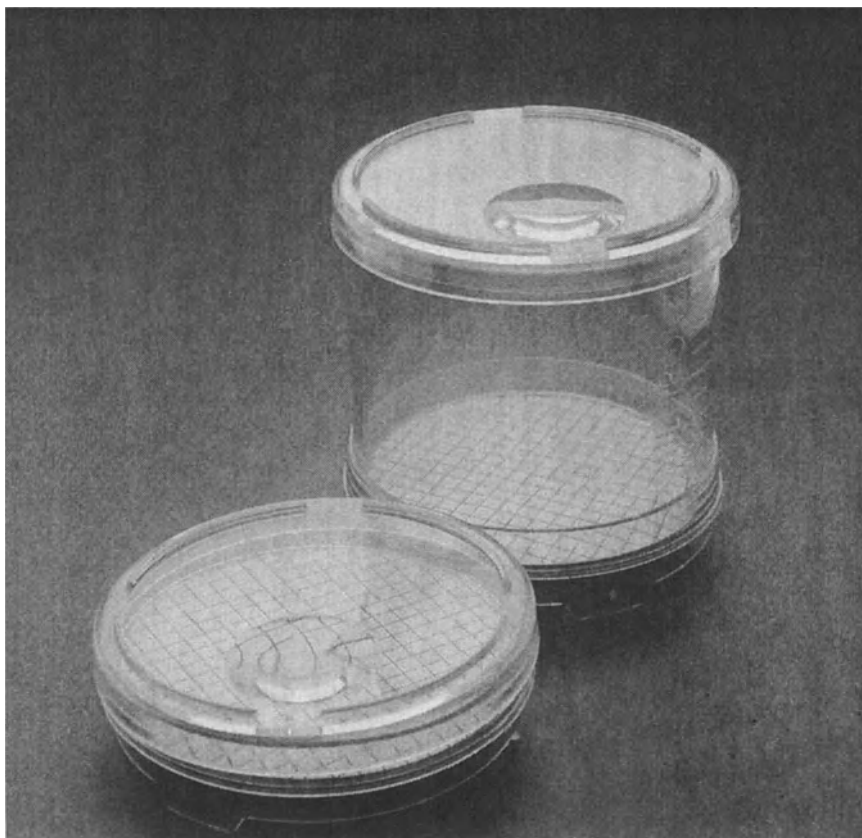


Fig. 9-6 55-Plus beverage monitor.

Courtesy: Millipore Corporation.

several decades or more of aging, they, too, will reach an optimal point in life and then diminish in quality.

White wines should be aged in a cool, dark, cellarlike environment, free from vibration and temperature variations. If space is at a premium, place white and blush wines closest to the floor, where it is generally cooler. Ideal storage temperatures are consistently maintained from 55 to 60°F.

PACKAGING

Some winemakers prefer to apply capsules and labels right after bottling and corking. This can work well as long as the bottles are stored in cases or some other protective container.

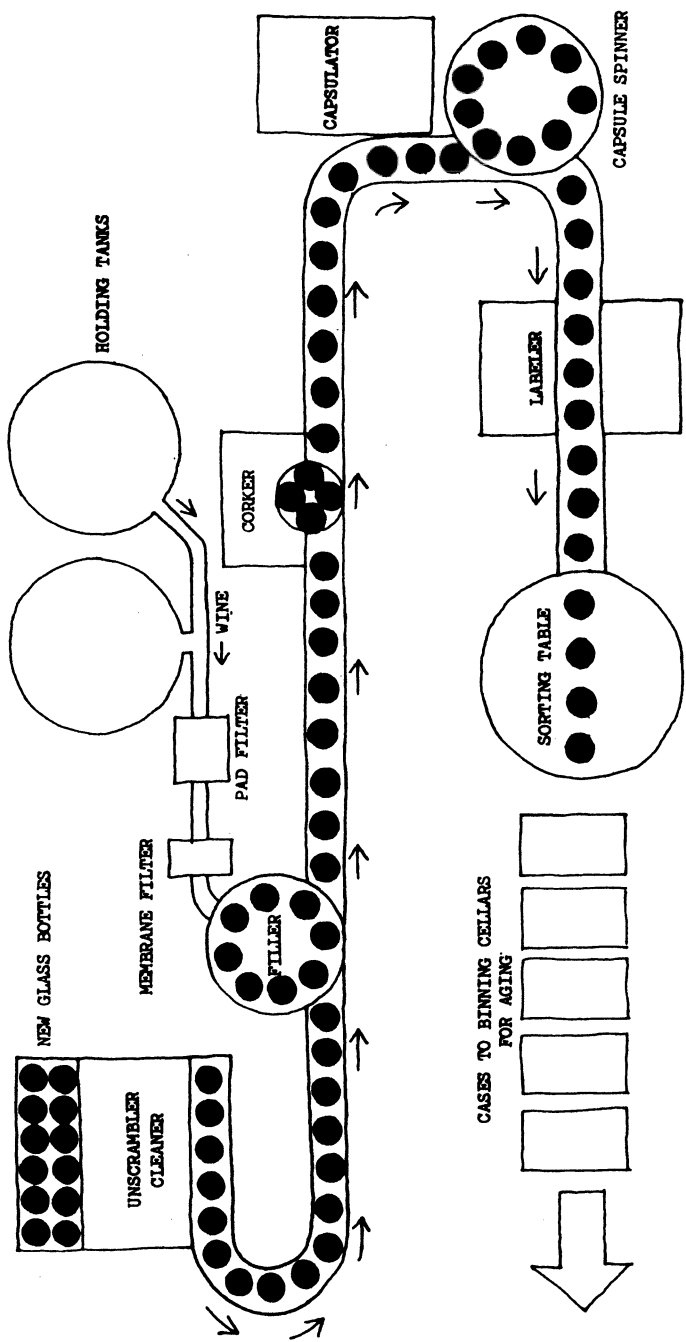


Fig. 9-7 Schematic of bottling line operation.
Courtesy: Sebastiani Vineyards.

Bottles laid to age in cellars, however, can collect cobwebs, dirt, mold, and grime. Cleaning each bottle with warm water will help in the successful application of packaging materials. Hard-water rinsings will generally leave unsightly mineral spots unless each bottle is dried.

Apply glue to fully cover the back of each label with as thin a layer as possible. Identically position labels on bottles, as varying heights of labels create a haphazard impression. Many vintners prefer pressure-sensitive labels produced with an adhesive coating already applied. These can work very effectively in the hands of experienced personnel. Once in contact with a bottle, pressure-sensitive labels are difficult to adjust for position or to remove.

See Chapters 4 and 8 for further information relating to packaging materials and equipment.

Once packaging is applied, bottles can be carefully inserted in cases so as not to scuff the labels. Some vintners choose to wrap bottles in tissue paper before casing. This is obviously an extra expense, and both pros and cons as to consumer preference can be argued.

Cases can then be stored in a cool (60 to 65°F), dry warehouse (Figure 9-8). Be sure that paperboard cartons are stored on a pallet or some other support that will protect them from wet floors.

Fig. 9-8 Wine warehouse.
Courtesy: Bouchard Pere et Fils.



Exhibit 9-3 White wine bulk inventory form in service.

BULK WINE BOOK INVENTORY CARD													
Lot. No.	Type-Variety					Class	Color						
541	CHARDONNAY					-14%	WHITE						
REMARKS:													
Date	Racked From Tank W.O. Gallons			Treatment-Disposition	Racked To Tank W.O. Gallons		PHYS INV	ALC	BALL	T.A.	V.A.	FSO2	
10-13	F-4	191	1814	racking	5-2	850	✓	11.9	-1.9	.683	.024	-	
					5-3	850	✓	12.0	-2.0	.690	.024	-	
					16b1	50	64 gals juice less destroyed						
10-17	to wine acct			analysis	5-2	850	✓		-2.1		.024	26	
	to wine acct			add KMS	5-3	850	✓		-2.0		.024	36	
	to wine acct				16b1	50	✓		-2.1		.030	32	
10-24	5-2	199	850	racking	5-6	1,500		12.0	-2.1	.683	.024	30	
	5-3		850		4 bbls (H10m)	212	18 gals wine less destroyed						
	16b1		50		2 cbys	10							
11-8	5-6	211	1,500	fining	5-2	850	✓	12b hantoms	12b KAT crystals				
	4bbls		212	and	5-3	850	✓	12b hantoms	12b KAT crystals				
	2cbys		10	stabilize	2cbys	10	✓	12gals less destroyed					
11-9					5-2	analysis	850	28°	12.1	-2.1	.668	.030	-
					5-3	analysis	850	29°	12.1	-2.0	.660	.024	8
12-15	5-2	244	850	filter w/	15bbls (H10m)	890	50° ✓	12.1	-2.1	.638	.030	-	
	5-3		850	coarse	14bbls (H10m)	812	51° ✓	12.0	-2.0	.630	.024	-	
	2cbys		10	made	3cbys	15	✓	13gals filter loss					

BULK WINE BOOK INVENTORY CARD											
Lot. No. 541	Type-Variety CHARDONNAY				Class -14%			Color WHITE			
REMARKS: continued from reverse side											
Date	Racked From Tank W.O. Gallons		Treatment- Disposition	Racked To Tank W.O. Gallons		PHYS INV	ALC	BALL	T.A.	V.A.	FSO2
12-22	1cby		top up	4 bbl	SAMPLES	56° ✓				.024	-
1-6-96			top up	4 bbl	SAMPLES	54° ✓				.024	-
1-13			top up	4 bbl	SAMPLES	52° U				.024	-
2-6			top up	4 bbl	SAMPLES	51° ✓				.030	-
2-21			top up	4 bbl	SAMPLES	51° ✓				.024	-
3-5	(1cby used)		top up	4 bbl	SAMPLES	51° U				.030	-
3-20	1cby		top up	4 bbl	SAMPLES	52° ✓				.030	-
4-3			top up	4 bbl	SAMPLES	51° ✓				.024	-
4-17			top up	4 bbl	SAMPLES	52° ✓				.030	-
5-3			top up	4 bbl	SAMPLES	54° ✓				.030	-
5-21	(1cby used)		top up	4 bbl	SAMPLES	56° ✓				.030	-
6-5	15bbls	99.96	870	adjust		58°	12.1	-2.1	.638	.030	-
	14bbls		812	and	added 2 lbs citric acid, 1 1/2 lbs KMS						
	1cby		5	filter	BT-1	54°	12.1	-2.1	.645	.030	48
6-5	BT-1	100	1680	bottle	699cs + 7 bottles	64°	12.1	-2.0	.645	.030	40
6-5	transferred 697 cs and 7 bottles to					cool	good acct				
6-5	this acct closed										

Exhibit 9-4 White wine cased goods inventory form in service.

CASED GOODS BOOK INVENTORY							
Lot No.	Vintage		Type-Variety		Size		
541	1995		CHARDONNAY		750 ML		
Class	Color	Minimum Inventory	Bottle Type	Closure	NAT WARE	Other	
-14%	WHITE		DEAD LEAF GREEN FUNT	13/4"	STERILE		
Date	Cases In	Cases Out	Balance on Hand	Ref No.	Inv.	Remarks	
6-5-96	697cs 7bot		697cs 7bot				
7-10		1bot	697cs 6bot	for analysis	✓	bottle sick	
8-14		1bot	697cs 5bot	for analysis	✓	bottle sick	
9-3		1bot	697cs 4bot	for analysis	✓	bottle sick	
9-4	released	1cs 4bot	696cs	family use			
9-5		10cs	686cs	retail shop		R-94-96	
9-6		50cs	636cs	Royal Dist.		P.O. # 44707	
9-6		40cs	596cs	James Barr		P.O. # JB-991-96	
9-6		40cs	556cs	Castle Whale	✓	P.O. # C-1866-W	
9-10		50cs	506cs	Filmage Dist.		P.O. # 944-6	
9-10		30cs	476cs	Phillips Barr		P.O. # PB-6497	
9-11		40cs	436cs	Morgan Dist.		P.O. # 7833	
9-11		10cs	426cs	retail shop	✓	R-97-96	
9-12		10cs	416cs	O'Brien Dist.		P.O. # 196-96	
9-12		4bot	415cs 8bot			4 bottles broken	
9-12		8bot	415	family use			
9-12		30cs	385	Burns Whale	✓	P.O. # BUA-6447W	

RECORDS

Exhibits 9-3 and 9-4 are examples of a typical white wine processing history using the record format presented in Chapter 8.

CHAPTER 10

RED TABLE WINES

It generally takes more time to bring red wines to the point of bottling than whites. While fermentation should take place in only a week or so and the clarification/stabilization process is relatively simple, the aging of heavy-bodied reds to maturity is more complex.

GRAPE VARIETIES

Cabernet Sauvignon, Chambourcin, Chancellor, Merlot, and Norton make reds that are “vinous” in character (herbal and woody flavors), dense in color, heavy-bodied, and require long terms of aging. Varieties such as Baco Noir, Gamay, Pinot Noir, Marechal Foch, and Zinfandel are typically “berry-cherry” in bouquet and flavor, moderate in densities of color and body, and capable of maturing in just several years of bottle age. Native American reds such as Concord, Isabella, and Ives make “fruity-grapey” wines that are typically bold in every sensory aspect and can often be marketed when just 1 year old.

Most red grapes should be harvested at peak ripeness or just slightly on the overripe side, which would generally be in the pH range of 3.30 to 3.40. This

allows for full color development and will encourage later malo-lactic fermentation, if so desired. The buttery-cheesy flavor component from this bacterial fermentation usually has a negative effect on wines made from the native American red varieties, so these may be harvested just slightly on the green side, in the pH range of 3.20 to 3.30.

As with making white wines, the very first consideration is for sanitation. Reds are fermented in open-top containers at higher temperatures, making the exposure to spoilage organisms an even greater consideration. It is absolutely essential that equipment and containers be thoroughly cleaned with scalding hot water and drained dry before securing from use.

INCREASING RED WINE FRUIT FLAVOR INTENSITY

Close observation of the ripening process each vintage season can provide essential data for anticipating the precise time for harvest in order to achieve the red wine flavor profile desired. Much of this scrutiny can be determined by simply tasting representative berry samples. The flavor identified in grapes should be captured into the wine made from them. Analysis of pH, Brix, and total acidity data should be recorded and correlated with optimal maturity levels for future years.

Immature red grapes in the pH range of 3.10 to 3.20 typically make wine that is rather vegetative in flavor. Generally higher pH levels, in the 3.30 to 3.40 range, are associated with more intense fruit flavor.

One method commonly employed in Europe is the *saignee* process. This involves separating free-run and/or lightly pressed juice from red grape musts immediately after crushing. This initial fraction is usually cool-fermented separately for blush wines or for making blended wines. The drained must is then fermented in the normal red wine manner. This obviously results in a greatly reduced juice portion in the must in which color and other solids are dissolved during extraction by fermentation. Fully ripened grapes are needed for the *saignee* to reduce high concentrations of acidity. Generally shorter on-skin time exposure is made to avert excessive levels of phenol extraction.

Another method of increasing red wine fruit flavor is the *maceration carbonique* process, which is closely identified with the famous “Beaujolais nouveau” wines. Closed fermenters are partially filled with crushed grapes and then filled with whole clusters. Carbon dioxide generated by the fermenting crushed portion is held under a modest pressure in the tank until the desired color and flavor extraction is achieved, and then the entire lot is pressed immediately. Carbonic maceration generates dense purple color extraction and lower concentrations of tannins. Unless this typical “nouveau” character is desired, further blending is needed to achieve good overall balance.

HARVESTING

Hand-harvested grapes are preferred. Broken grape skins resulting from mechanical harvesting at higher pH levels encourage apiculate, or “wild,” spoilage yeasts to grow before a cultured yeast strain can be inoculated. Expect a yield of about 140 to 150 gallons of finished wine from each ton of grapes.

CRUSHING-DESTEMMING

Commence the winemaking process as soon as possible after harvest. Grapes received in the 75 to 80°F temperature range are ideal. Remove leaves and trash along with unripe or spoiled berries in the same triage process as that described for white wines. Record net weight, and rinse grapes (hand harvested only) with cold water. Grapes with higher pH values will respond less effectively from sulfur dioxide additions, as increasing pH reduces molecular sulfur dioxide.

Reds, unlike whites, may benefit from the retention of some portion of the stems in the must because this adds *tannin* (complex phenolic compounds), which may contribute a bit of appealing astringency and extends aging potential. Perhaps up to one half of the stems might be considered for retention in vinous grape musts; less in the berry-cherry flavored varieties; little or none in fruity-grapey reds. Experiments should be conducted to learn optimal stem/must ratios.

Clean the crusher thoroughly with scalding hot water, crush whole bunches of grapes for the stem-retained portion, and destem the balance. Crusher-destemmer devices are discussed in Chapter 8 of this book. After crushing operations are completed, clean the empty grape containers, crusher, and destemmer thoroughly with scalding hot water, and allow to drain dry.

PECTIC ENZYMES

The use of pectolytic enzymes in red grape musts can significantly increase the extraction of phenolic compounds such as anthocyanin pigments and resveratrol. Experimentation is recommended beforehand, as anthocyanin pigmentation can be increased and significant changes in color hue can also result.

Crushing should take place so that the resulting must is deposited directly into a clean fermenting container. Red must fermenters should have a “jacket” around the outer wall in which cold water can be recirculated to lower the rapid temperatures generated by active fermentation (Figure 10–1). Even so, fermentation may take place in only a few days, as yeasts are very active in the ideal elevated temperature environs of red grape musts.

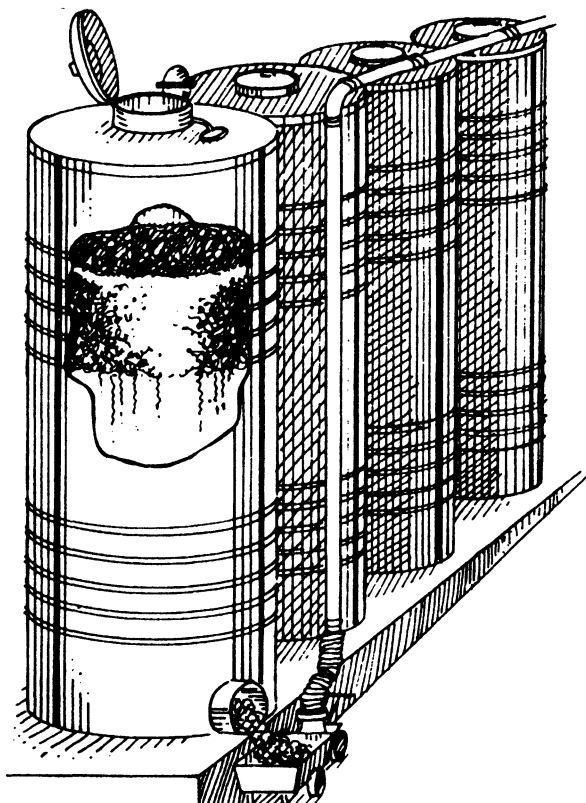


Fig. 10-1 Jacketed red wine fermenting tanks with floating “cap” of skins.
Courtesy: Sebastiani Vineyards.

BRIX (SUGAR) ADJUSTMENTS

Brix in red grape juice typically measures between 14 and 23°. Native *Vitis labrusca* varieties will generally ripen at the lower ends of this range, *Vitis vinifera* at the higher end, and the French-American hybrids at midrange. There are exceptions.

Each degree of Brix will convert to about .535% alcohol during fermentation. Consequently, to achieve 12% alcohol by volume, the red grape much would require approximately 22.5° Brix. A natural Brix of 17.6° Brix would thus need to be adjusted with sufficient sugar to achieve a 4.9° increase in Brix.

As mentioned previously, the addition of sugar is called *chaptalization*. In some cases grapes are excessively high in acidity, so that both sugar and water are added in a process referred to as *amelioration*. Both of these adjustments have historically

been controversial in fine-wine circles. The ATF regulation 27 CFR, Part 4, Section 24.177, relates to Brix adjustments.

The major difference between adding sugar to white juice and adding it to red must is that the winemaker does not have a finite measurement of the juice contained in must that has not yet been pressed. It is thus necessary to estimate conservatively the amount of juice for a sugar addition. Depending upon grape variety, ripeness, and press efficiency, history will usually determine optimal juice potential to be within the range of 160 to 180 gallons per ton. Once the fermented red wine is pressed out, a precise determination can be made for final adjustment if desired.

The same mathematical formulas for calculating sugar and concentrate additions used for white juice in Chapter 9 can be employed for red musts. A well-designed Lot Origination Form will permit vintners to calculate sugar/water additions accurately and simultaneously maintain a good record of changes in gallonage.

ACID ADDITIONS

Total acidity in red musts should ideally measure between .65 and .75 g/100 mL or, roughly, between .65% and .75% acidity. Most red wine styles are constructed upon medium to heavy-bodied consistency. It may be best to attempt no adjustments upon musts having only a slightly high acidity, given the thinning that would result from ameliorating that tartness with water. The calculation and methodology for red wine additions are the same as those provided in Chapter 9 for whites.

Stir in all additions, and test for Brix and total acidity once more to ensure that the desired levels have been achieved. Post additions in the appropriate record.

YEAST INOCULATION

The rationales for selecting and preparing red wine yeast starters are identical to those for white wines as outlined in Chapter 9.

At one time it was rather common practice to utilize yeast sediment remaining in the bottom of red wine fermenters or some actively fermenting red juice as an immediate starter culture for inoculating the next batch of crushed must. Research indicates that neither of these practices is recommended. Wild yeast and/or bacterial infections are easily spread, and significantly more undesired fermentation byproducts are generated by perpetuating yeast cultures.

FERMENTATION

The fermentation philosophy for red wines is much different than for whites. While for white wines it is critical to maintain cooler temperatures and restrict exposure to

grape stem, pulp, and seed particles, the opposite is true for reds. Color and flavor extraction from the red grape skins usually is ideal at 80 to 90°F. Figure 10–2 shows the method of taking the temperature of a red wine fermenter.

As mentioned previously, some portion of grape stems left during the crushing and destemming operation may be preferred by some winemakers in developing added red wine complexity. As will be mentioned below, post-fermentation maceration of red wine in contact with the skins may be another source of desired complexity in the finished product.

IMPORTANT NOTE: CAUTION! DUE TO THE ELEVATED TEMPERATURES AND TUMULTUOUS FERMENTATION OF RED WINE MUSTS, AN INCREASED AMOUNT OF CARBON DIOXIDE GAS WILL BE GENERATED. HEAVY-DUTY VENTILATION SYSTEMS AND AIR MARKS CLOSE AT HAND ARE CRITICAL TO THE PREVENTION OF ASPHYXIATION. UNASSIGNED PERSONNEL AND VISITORS, ESPECIALLY CHILDREN, SHOULD NOT BE ALLOWED IN LOCATIONS WHERE ACTIVE RED WINE FERMENTATIONS ARE TAKING PLACE.

As fermentation commences, grape skins will rise to the top of the must, forming a “cap” over the juice below. So that maximum color is extracted from the skins by the alcohol being generated by fermentation, the cap should be regularly pushed back down into the juice—an operation referred to as “punching.” Punching also serves to discourage aerobic spoilage organisms that may be obtaining oxygen over the cap. The cap should be punched at least three times daily with a clean wooden, plastic, or stainless-steel paddle. Some winemakers prefer to “pump over” rather than punch. This operation is the simple attachment of a pump and hose to the bottom of the fermenter in order to deliver fermenting juice over the top, where it is sprayed evenly across the surface of the cap (Figure 10–3). Like punching, pumping over should be performed at least every 8 hours during fermentation.

Fermenter temperature should be checked at least twice daily and adjusted by running hot or cold water as needed in the jacket around the outside of the fermenter. If there is no jacket, a handy device for this is an old garden hose, which can be punched with small pinholes every inch or so. The hose is plugged at one end and strapped around the upper outside wall of the fermenter. Increase pressure from the faucet slowly until an even flow of cold water covers the tank wall. While the cold water provides some cooling, the evaporation of the water creates the major cooling effect. Fermenting red wine musts under temperature adjustment should be agitated frequently to avoid extreme heat in the center of the vessel.

One of the newest, and most expensive, innovations in red wine fermentation is rotating stainless-steel fermenters (Figure 10–4), which are very efficient in the amount of time and energy needed to gain optimal skin exposure to the fermenting wine—and minimal oxygen exposure to spoilage organisms.

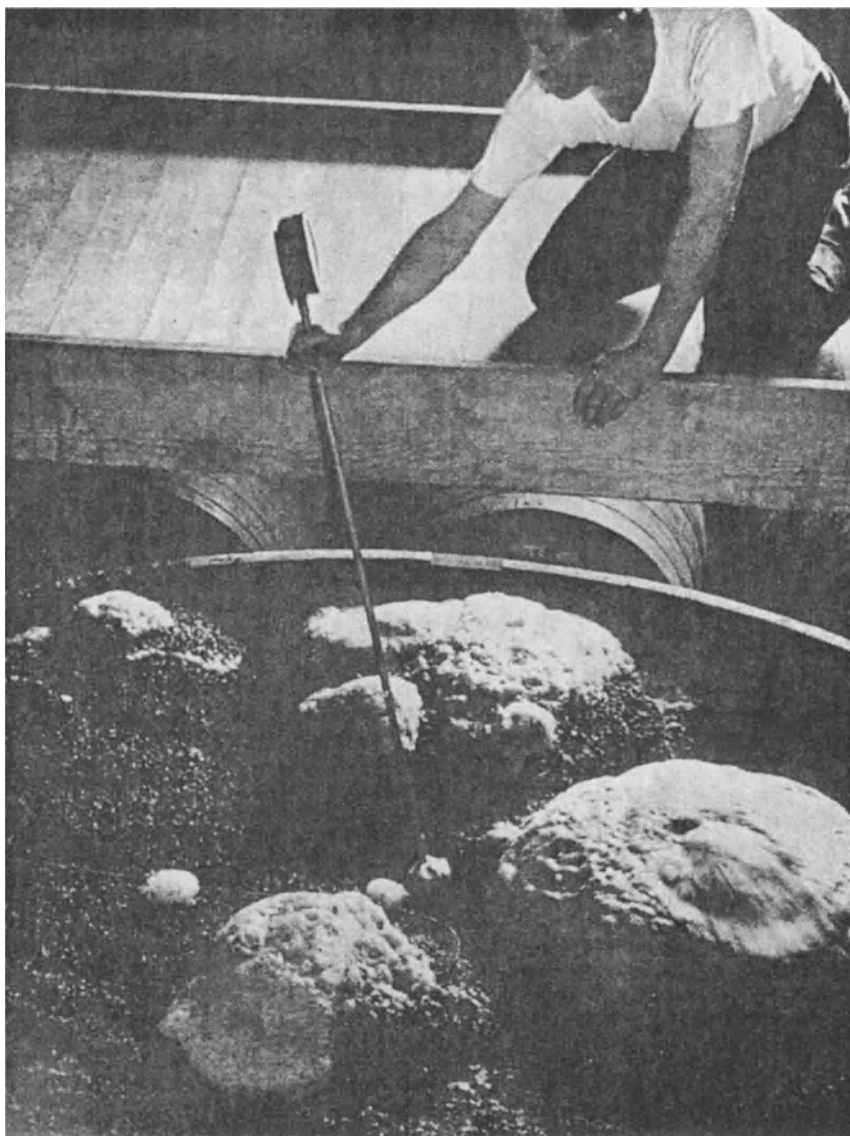


Fig. 10-2 Taking the temperature of a red wine fermenter.
Courtesy: Wine Institute.

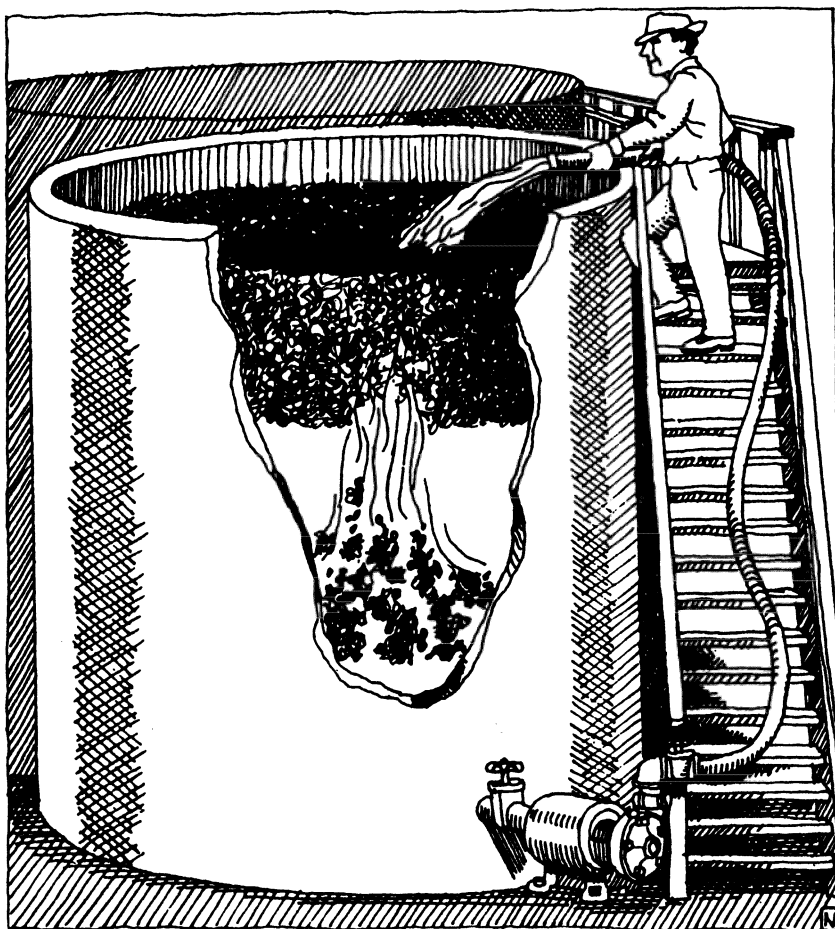


Fig. 10-3 Pumping over the cap.

Courtesy: Sebastiani Vineyards.

Stuck (incomplete) red wine fermentations are rare, and most are due to temperatures having been allowed to exceed 100°F. If the problem can be positively diagnosed as excessive heat, then immediate cooling is an obvious treatment. Light agitation to replenish some oxygen for remaining viable yeast cells may also serve to resume fermentation. The addition of a light dose of food-grade diammonium phosphate (DAP), up to 2 ounces per 100 gallons, will help replenish free nitrogen required by yeasts.

Post the inoculation date, temperature checks, heating/cooling, and other pertinent data in the appropriate record.

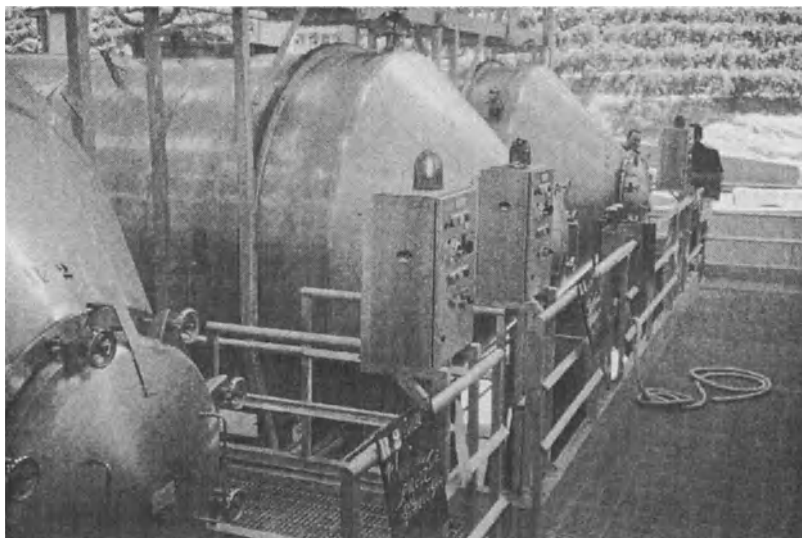


Fig. 10–4 Rotating stainless-steel fermenters.

PRESSING

Several major factors are influential in determining the precise time for pressing new red wine from the must.

One factor is color extraction. Color intensity is primarily a function of pigment development in the berries during a given growing season and the pH at harvest—and is not directly correlated with extended skin contact. Most red pigments are fully extracted by the time Balling reaches 1.0 or so. This level is recommended as a beginning standard for pressing berry-cherry varietal musts.

Yet another factor is limited skin contact, in the range of 10.0 to 5.0 Balling. Pressing fruity-grapey musts at this midway stage of fermentation renders wines with a simpler profile of flavors, uncomplicated by the extraction of tannins and other complex compounds.

MACERATION

Maceration of fermented red wine must simply means holding it in the fermenter for some period of time, usually 10 to 30 days, after fermentation has been completed. This should not be confused with *maceration carbonique* for “nouveau” wines, which is completely separate from red wine must maceration.

Red wines designed for long life, such as the heavily structured Meritage reds, must have sufficient tannin to slow the oxidation process known as aging. One

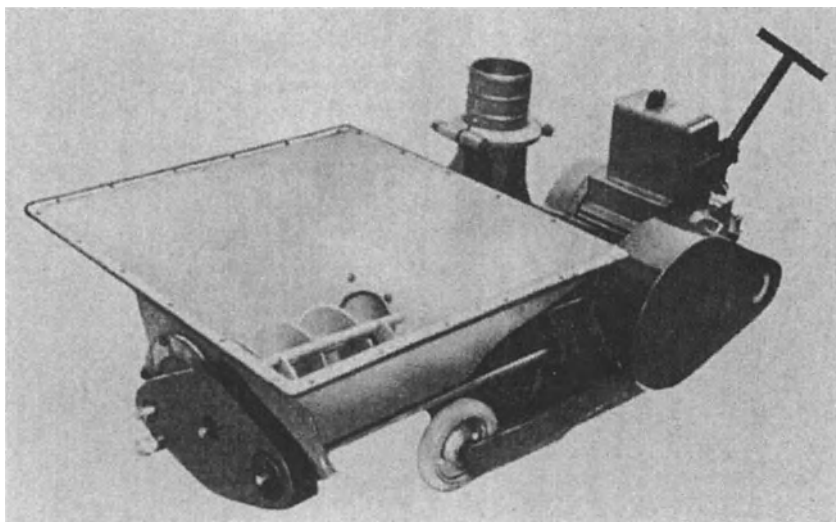


Fig. 10–5 Portable must pump with hopper and auger.

Courtesy: Scott Laboratories.

approach is to determine the amount of tannin that one wishes to be extracted from the skins and seeds. Bordeaux chateaux often keep Cabernet Sauvignon and Merlot musts “on the skins” for up to several weeks of maceration after fermentation has been completed. A close watch needs to be maintained during this time to guard against white “islands” of spoilage bacteria developing on the surface of the wine. Any suspicion of such activity justifies immediate pressing.

Maceration can easily be excessive, primarily because of the chemical makeup of the tannins extracted from the must. This is very complex, as tannins are polyphenolic polymers formed by oxidation and condensation reactions. Tannins from red grapes harvested on the immature side generally tend to form coarse and bitter compounds. Conversely, the more ideal, somewhat astringent tannins, those that complement the aging process, are developed in grapes that have fully matured—and perhaps have even just slightly overripened. Fermented must can be gravity fed to the press from fermenters designed for that purpose. Otherwise, fermenters must should be transferred to the press with a portable must pump such as that illustrated in Figure 10–5.

FREE-RUN AND PRESS WINE

Free-run red wine from the press (Figure 10–6) is usually the best quality. The press wine should be darker in color due to the extraction of pigments from the

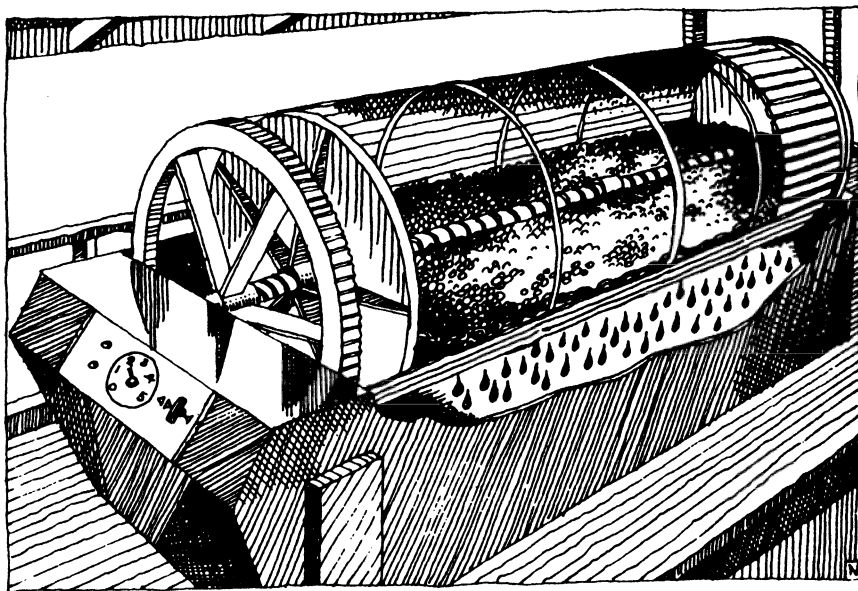


Fig. 10-6 Free-run juice falling from a Vaslin-Type rotating basket press.
Courtesy: Sebastiani Vineyards.

skins under pressure. The free-run and press-run sublots should be held separately for later blending considerations.

If the red wine has fermented to less than -1.0° Balling, it can be placed into clean containers, filled, sealed with fermentation locks, and stored in a dark 55 to 60°F chamber. If it is above -1.0° Balling, the containers should be filled to about 80% to 90% capacity until the -1.0° Balling level is reached, and the wine should then be racked to full containers equipped with fermentation locks. If the estimated gallonage of juice in the red must was reasonably accurate, the level of alcohol developed from fermentation should not vary much from what was calculated. Providing the yeast population is still active, increases in alcohol can be made by a sugar adjustment and calculated by using the ratio: $1.0^{\circ}\text{Brix} = .535\%$ alcohol.

RACKING

Young red wines will generally finish bubbling within a few days after pressing and should indicate no more than a trace of residual sugar using the Dextro-Chek tape. If so, the wine can be racked into a clean vessel filled completely to expel any air, and a loose cap or plug can be used to cover the second-rack containers instead of the fermentation lock.

An addition of 1 ounce of KMS powder per 100 gallons should increase the sulfur dioxide content in the young wine by about 20 to 30 ppm or mg/L. This is recommended to help protect further against spoilage microorganisms and oxidation. Continue storage in a cool, dark, cellarlike environment.

From this point on, throughout the life of the wine, it will be necessary to ensure that it is stored in containers that are completely full. Preventing air contact with the wine reduces its exposure to oxygen that is a component for oxidative browning and the growth of spoilage organisms.

BLENDING

As discussed in Chapter 4, the general rule of thumb in blending is to associate the different virtues of individual wines in a resulting wine that has a more broad and satisfying appeal. A typical blending motive for white wines is to achieve a delicate expression of varietal fruit flavor expressed upon balanced acidity with little tannin astringency. Vinous reds are generally blended to achieve the opposite: bold flavors, heavy body, dry finish and moderate tannin. Berry-cherry reds may be a bit lighter in overall expression, and fruity-grapey reds may approach the white wine blending rationale.

Wines should be blended before stabilization. Because of the many factors involved with stability, sometimes stable wines, when blended together, form an unstable wine.

The new blend will cause a new record to be started. Wines taken for the blend should be deducted from their appropriate records.

MALO-LACTIC FERMENTATION

As mentioned in Chapter 3, the controlled growth of select bacteria species (*Lactobacillus* and *Leuconostoc*) can serve to transform malic acid, which has a rather “apple-like” character, to lactic acid, which has a pronounced “buttery-cheesy” (diacetyl) flavor. Such bacteria are thus known as *malo-lactic*, and their fermentation process also includes a small reduction in total acidity, a small increase in acetic acid, and some carbon dioxide gas. The virtue of malo-lactic fermentation in vinous and berry-cherry red wines is to add an interesting nuance of buttery character. As a rule, fruity-grapey reds are not improved by this treatment.

A viable malo-lactic bacterial culture should be inoculated in a young red wine only when pH is above 3.35 and temperature is in the range of 65 to 70°F. Be forewarned that this condition also invites the growth of acetic bacteria and other spoilage organisms.

Malo-lactic fermentation can be a frustrating process, as the activity of malolactic bacteria can vary significantly and is difficult to anticipate. Since these organisms require only a small amount of oxygen, this fermentation can be conducted in nearly full containers equipped with fermentation locks. Close observation remains critical, however. Inspect several times daily for the emergence of carbon dioxide bubbles in the fermentation lock and the development of buttery-cheesy odor.

FINING AND STABILIZATION

The high level of phenolic compounds in red wines is usually sufficient to reduce protein suspensions. Consequently, young red wines (2–4 months old) are typically “bottle-bright” and have no need for clarification. During barrel aging, red wines usually precipitate unstable tartrates without the use of cold storage. Because of this, many commercial vintners choose not to stabilize red wines in cold storage.

Egg-white fining is a common procedure employed by the finer vintners of red wine. Several egg whites beaten to a froth, not stiff, are gently stirred into 50 gallons of wine, serve to soften the harshness of young wines having an abundance of tannins. Alternative methods of red wine fining can be found in Chapter 8.

BARREL AGING

Oak aging for red wines generally requires a much longer time in wood to achieve the desired “toasty-vanilla” character in bouquet and flavor than oak aging for whites. The vinous red varieties may take up to several years in 55-gallon barrels in a cool cellar environment (Figure 10–7). Berry-cherry varieties may need a year or so, and perhaps no wood aging is best for the fruity-grapey reds. Remember that aging progresses faster in smaller oak containers.

All wooden containers will lose some wine contents, mostly due to assimilation and seepage. Each keg and barrel should be carefully examined at least monthly to “top up” with wine. Generally only a few tablespoons, certainly less than a cup, is needed. This procedure refills the *ullage* (air space) with a similar type of wine so that oxygen is not available for bacterial growth. If a white “film” of bacteria or wild yeast growth appears on the surface, the wine used for filling should be treated with KMS at a rate of about 1 teaspoon per gallon. This should inhibit the microorganisms and allow aging to continue.

Some commercial winemakers prefer to “wet-bung” barrels after filling to reduce the risk of bacterial infection entering barrels through the bunghole. It is a procedure in which barrels are filled with bungs securely driven and secured. Barrels are then rolled to the side so that the bungs are situated at about “two o’clock.” With the barrel in this position, wine constantly covers the bunghole during the



Fig. 10–7 Visitors observing red wine barrel aging.

Courtesy: Sterling Vineyards.

aging period. Subsequent ullage is created in an aseptic atmosphere that resists spoilage.

An in-depth discussion of barrel aging is provided in Chapter 4.

BALANCING AND PRESERVATION

Red wines are rarely finished with any residual sweetness, so red wine sugar/acid balance is not generally an issue unless total acidity is excessively high or low. Red wines at these extremes may be best used in blending.

Dry red wines can be preserved with a final addition of KMS, usually at a rate of 1/2 ounce per 100 gallons for dry wines with a pH at 3.30 or lower.

FILTRATION

Red wines often clarify so well that filtration is not needed or desired. Some degree of sediment is expected in older bottles of red wines anyway—this is why fine crystal wine decanters exist.

Careful racking, fining, and barrel aging regimens can result in perfectly clear red wines. Under these circumstances there may be justification to consider no filtration. This will conserve color and flavor. Should no filtration be decided upon, it is recommended that the label upon this finished wine indicate that the wine was “unfiltered” so that consumers can expect some sedimentation.

Chapters 4, 8, and 9 provide information relating to common filter types, filter media, and filtering techniques.

Low-porosity membrane filtration is not recommended for red wines because extensive color pigmentation may be lost.

BOTTLING AND CORKING

Refer to Chapter 8 to select the proper shape bottles with respect to red wine traditions. Bottling procedure for reds is identical to that for whites, discussed in Chapter 9.

Red wines such as “nouveau” and fruity-grapey types, as well as others designed for consumption in the short term, can be bottled and corked in much way as white wines (Figure 10–8). Heavier reds having greater tannins for longer terms of bottle aging are usually given longer corks, 2 inches or more, to extend sealing potential.

BOTTLE AGING

Greater amounts of color and other phenolic compounds extracted during fermentation are correlated with longer bottle aging times—from just several weeks to several years. Some commercial vintners routinely bottle-age red wines for 1 year prior to market release. The ideal environment is 50 to 60°F, dark, and free of vibration (Figure 10–9).

Bottles can be placed in horizontal tiers or upside down to keep corks moist. If paperboard cases are used, be sure to keep them on a pallet or some other support to avoid wet floors. As time passes, regular sensory evaluations should be made to monitor aging progress, with tasting notes posted in the appropriate record.

PACKAGING

Unless bottled red wines are placed in some type of sealed container for bottle aging, it is difficult to escape at least some sort of dust or grime during that time. As a rule, a buildup of cobwebs, dust, mold, and other matter needs to be washed off with warm water. Hard water can leave unsightly mineral spots unless each bottle is dried.

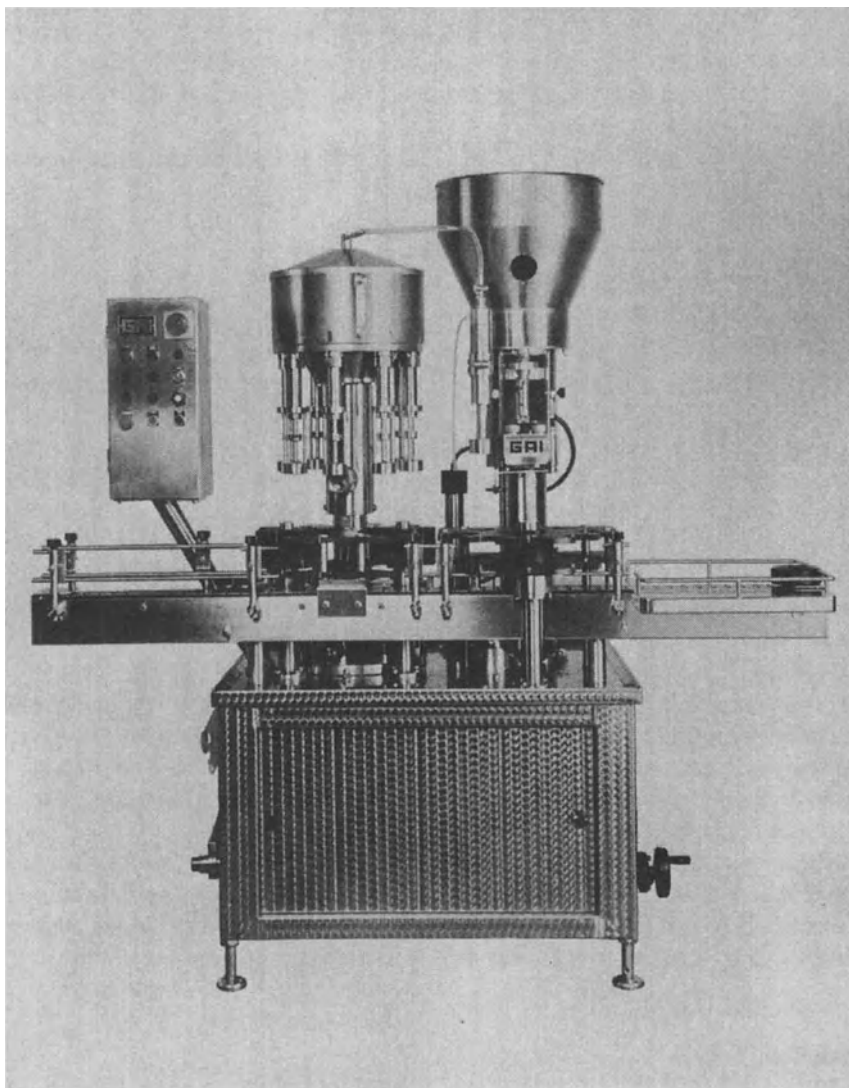


Fig. 10-8 Compact single-unit bottling and corking line.
Courtesy: Prospero Equipment Corporation.

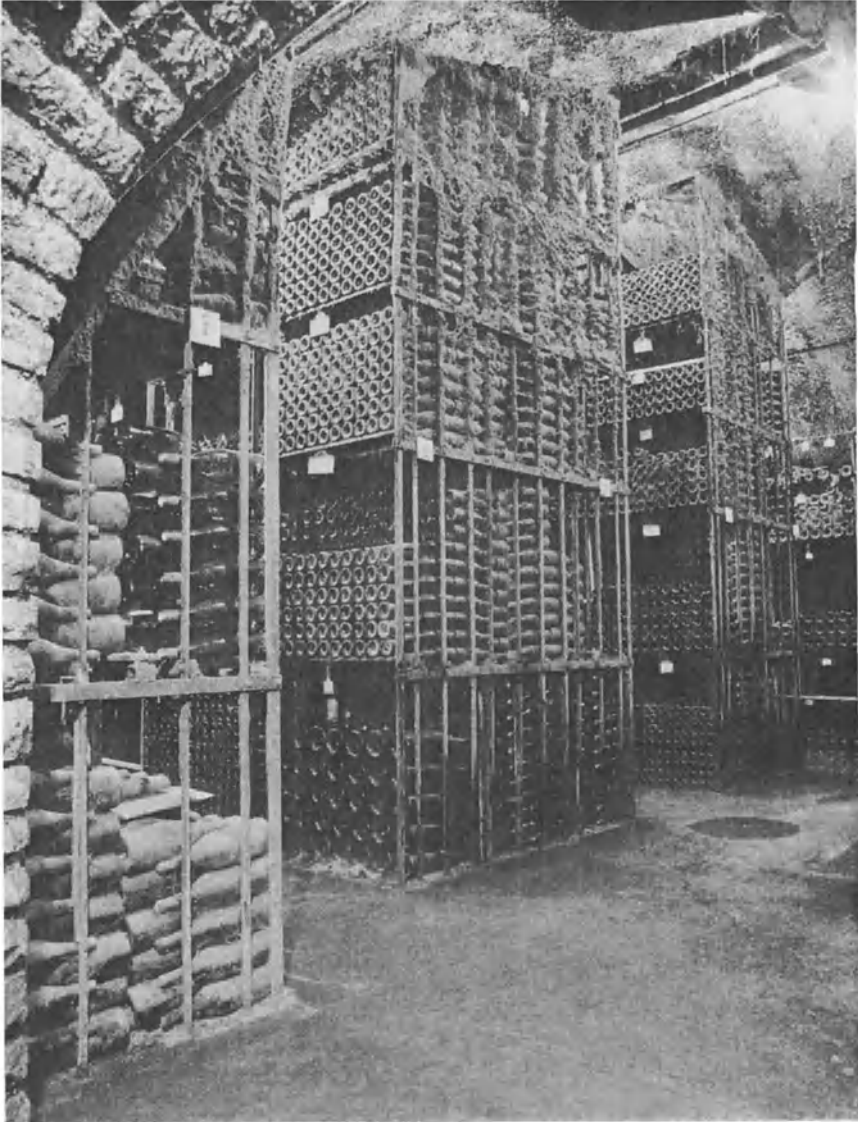


Fig. 10–9 Red wine bottle aging.
Courtesy: Bouchard Pere et Fils.

Exhibit 10-1 Red wine Lot Origination Form in service.

LOT ORIGATION			
DATE	<u>9-14-01</u>	CULTIVAR	<u>CHAMBOURCIN</u>
SERIAL NO.	<u>521</u>	REMARKS:	<u>machine</u>
NET GALLONS	<u>8-95</u>		<u>harvested - ±1% MOG</u>
FERMENTER NO.	<u>F-12</u>		<u>must temp 74°F out</u>
NET TONS	<u>EST. 984</u>		<u>of crusher - added</u>
GALLONS PER TON	<u>5.79</u>		<u>50 ppm KMS during</u>
pH	<u>3.35 T.A. 1.028</u>		<u>crush - yeast added.</u>
BRIX	<u>18.8 ALC.</u>		<u>immediately in F-12</u>
EXT.			
EST. AMELIORATION = <u>10</u> % @ <u>22.0</u> BRIX = <u>11.5</u> ALCOHOL			
<u>EST 984</u>	GALLONS START	<u>= 1093</u>	RESULTING TOTAL
<u>90</u>	INVERSE OF		GALLONS OF PRODUCT
	AMEL. PERCENTAGE		
<u>EST 1,093</u>	TOTAL GALLONS	@ <u>22.0</u> BRIX	<u>(2.004 LBS/GAL) = 2,190</u> TOTAL LBS.
<u>984</u>	START GALLONS	@ <u>18.8</u> BRIX	<u>(1.691 LBS/GAL) = 1,664</u> START LBS.
<u>109</u>	AMELIORATION GALLONS		<u>526</u> ADDITION LBS.
<u>39</u>	SUGAR AS GALLONS	x .074 (GAL/LB) =	<u>39</u> SUGAR AS GALLONS
<u>70</u>	WATER GALLONS		
<u>PDM</u>	YEAST ADDITION = RATE OF	<u>1</u> LBS PER	<u>1,000 gals</u>
REMARKS:	<u>yeast in 4 gals of water @ 105°F</u>		

Information relating to packaging equipment, materials, and techniques is found in Chapters 4, 8, and 9.

Once packaging is applied, bottles can be carefully inserted in cases so as not to scuff the labels. Some vintners choose to wrap bottles in tissue paper before casing. This is obviously an extra expense, and both pros and cons as to consumer preference can be argued.

Cases can then be stored in a cool (60 to 65°F) dry warehouse. As mentioned above, be sure that paperboard cartons are stored on a pallet or some other support that will protect them from wet floors.

RECORDS

Exhibits 10–1 and 10–2 are examples of a typical red wine lot origination format along with a typical fermentation history.

Once young red wines are fully fermented, the bulk wine and cased goods recordkeeping procedures are similar to those illustrated for white wines in Chapter 9.

CHAPTER 11

BLUSH TABLE WINES

Making pink table wines brings together some elements from both white and red winemaking procedures. Consequently, much of this chapter will make reference back to Chapters 9 and 10.

GRAPE VARIETIES

Color hue and intensity are two of the most important criteria in “blush” wines—a delicate pink pigmentation instead of the generally heavier color intensity found in most traditional “rosé” wines.

Both blush and rosé wines require grapes having pigmentation properties that are stable and resist oxidation. Orange hues generally indicate an excess of bound tannins in relationship to the amount of anthocyanin pigmentation in solution. One of the best methods to achieve a true blush color hue is to add a pectolytic enzyme right after crushing and then allow pigmentation to leach out in the must from 24 to 36 hours before inoculation to commence fermentation.

Most of the varieties noted as “vinous” in the “Grape Varieties” section of Chapter 10 are good choices for blush and rosé wine products. Other reds, such as Zinfandel and Concord, can also achieve superior results.

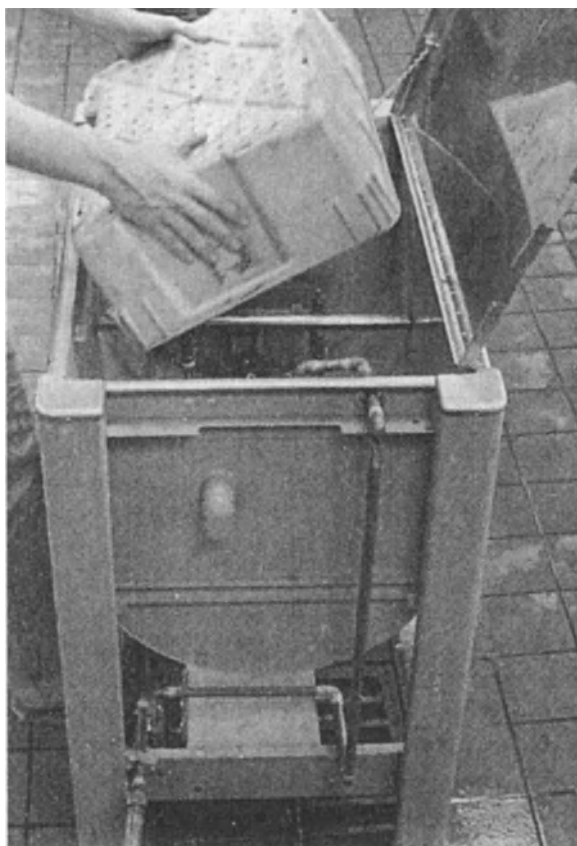
Harvest should take place when grapes are slightly underripened, with the same general quality control parameters as outlined for white table wines in Chapter 9.

CRUSHING-DESTEMMING, SUGAR AND ACIDITY ADJUSTMENTS, AND YEAST INOCULATION

The same general procedures outlined for red table wines in Chapter 10 above can be employed for processing blush and rosé wine. Minimal use of sulfite is recommended to reduce bleaching out pink color hues.

It is imperative that grape containers be thoroughly cleaned after each use to minimize the possibility of contamination. An effective foot-operated custom-made grape lug rinser is illustrated in Figure 11–1.

Fig. 11–1 Plastic grape lug rinser.



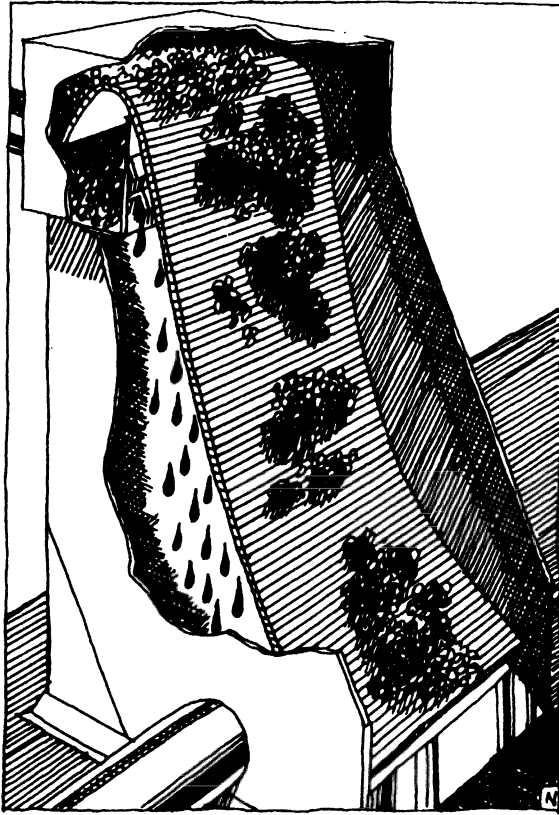


Fig. 11-2 Draining juice from red grapes on a screen.

Courtesy: Sebastiani Vineyards.

For heavily pigmented red grapes, such as Cabernet Sauvignon and Zinfandel, some vintners simply drain the juice from crushed must through a stainless-steel screen prior to fermentation (Figure 11-2). The remaining must is then fermented on the skins for red wines, which will, of course, be much more concentrated in color, flavor, acidity, and tannins—although a bit less strong in alcohol due to much of the sugar having been removed with the loss of free-run juice. This separation process is sometimes referred to as *saignee*.

PRESSING AND FERMENTATION

The principal concern for determining the precise time for pressing is ensuring that the desired amount of color is extracted from the must (Figure 11-3). Allow for a little extra color development, as some pigmentation will be lost in later processing

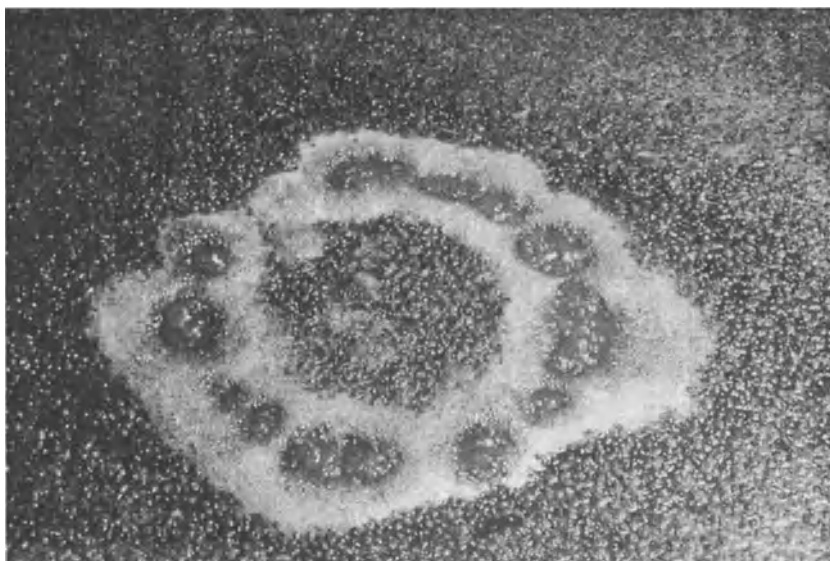


Fig. 11-3 First indication of red must fermentation.

Courtesy: Food and Wines from France.

functions. Some red grape varieties release color more quickly and more intensely than others, requiring close attention to the development of hue and density.

Conduct pressing and fermentation operations in the same manner as discussed for white table wines in Chapter 9. No juice racking should be necessary, but cool fermentation temperatures are particularly important in avoiding development of orange hues due to oxidation.

RACKINGS

These can also be performed in the same procedure as prescribed for white table wines. Blush wines do not generally improve from *sur lies* retention, as the “yeasty” flavor component that results does not typically enhance the fresh fruit flavors desired in most pink wines.

BLENDING

Adjustments in color can be made by appropriate white wine or red wine additions as determined by laboratory-scale blends. The racking of young blush wines should be made to clean, cool-storage jacketed stainless-steel tanks such as those illustrated in Figure 11-4.

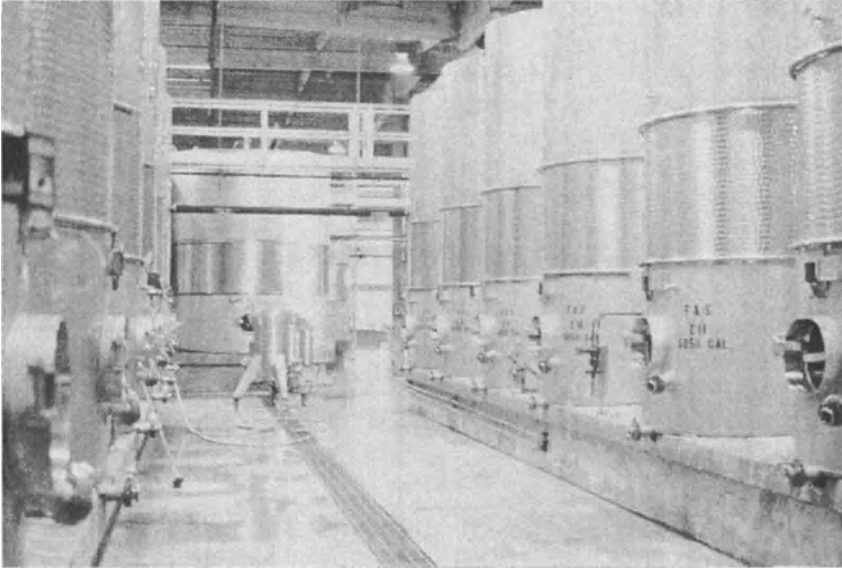


Fig. 11-4 Jacketed stainless-steel fermentation, detartration, storage, and blending Tank Room with Diatomaceous Earth Filter.

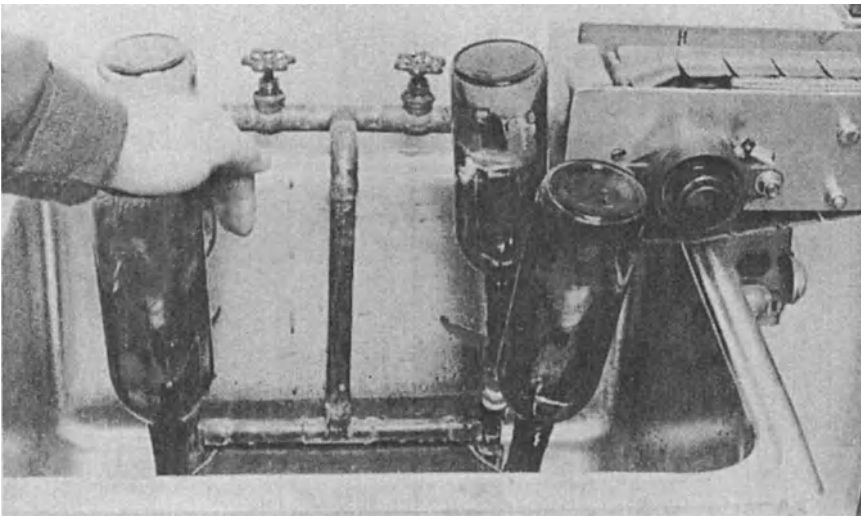


Fig. 11-5 Inverted-bottle spray washer/rinser.

CLARIFICATION AND STABILIZATION

Depending chiefly upon the amount of skin contact during fermentation, blush wines may or may not have sufficient phenolic extraction to reduce suspended proteins. Hazy light blush wines should be clarified and stabilized in the manner of white table wines, while clear darker rosé wines can be treated as reds.

AGING, BALANCING, AND PRESERVATION

Blush wines are usually best when as fresh and young, rarely benefiting from any type of extended aging program. Balance and preserve the same as discussed for white table wines in Chapter 9, ensuring that residually sweet wines are treated with both KMS and sorbic acid.

FILTRATION

Blush wines often create a dilemma regarding filtration. Pink hues in young blush wines are generally dull without the sheen of brilliant clarity. On the other hand, significant color loss can be realized during filtration. This is the primary reason why it was recommended earlier in this chapter (in “Pressing and Fermentation”) to allow a little extra color to develop during fermentation.

Proceed with filtering in the same manner as described for white table wines in Chapter 9.

BOTTLING, CORKING, AND RECORDING

All of these blush and rosé table wine operations can be performed in the same manner as described for whites. An effective custom-made bottle-rinsing device is illustrated in Figure 11-5.

Apart from allowing a few weeks to recover from “bottle shock,” the delicate and elusive character of fresh, uncomplicated pink table wines usually renders bottle aging an unnecessary and undesirable element of processing.

CHAPTER 12

FRUIT AND BERRY WINES

One of the attractive aspects of making fruit and berry wines is that many species ripen during the spring and summer months, well before the vintage season commences for grapes, allowing for more efficiency and variety in the winery facility. The opportunity to add different types of wine to the product portfolio is also an appealing prospect for many vintners.

In developing an understanding of fruit and berry wines, it should be remembered that commercially these products are typically viewed as including everything other than wine made from grapes. Fruit wines offer a far greater array of intense flavors than grapes and can stretch the imagination across a far wider range of possible tastes. Fruits and berries, like grapes, will vary in color, flavor, and acidity profile depending upon the soil and climate in the region where they are grown and the methods of horticulture applied in orchards. Thus, the same care and consideration in the selection of varieties and grades must be given to fruits and berries to achieve top-quality wine results.

While general methods are much the same as for grape musts, one should be extremely careful when applying pressure to fermented berry and fruit musts. Consistency is often very loose, and the must may flow between the press staves without efficient separation of liquids from solids. A stainless-steel or poly colander or mesh nylon bag may be of help. Also, even moderate pressures can extract very

Table 12-1 Relative aroma and acidity table for apple varieties.

Full aroma	Low acidity	High acidity
Delicious	Baldwin	Granny Smith
Golden Delicious	Cortland	Greening
Jonathon	Delicious	Jonathan
McIntosh	McIntosh	Northern Spy
Rome	Rome	Winesap

astringent and bitter phenols from the pits, pulp, seeds, and skins. Taste the press wine frequently to detect excess bitter phenol extraction.

As a rule of thumb, fermentation temperatures and processing regimens for most berry and fruit wines can proceed in the same general manner as suggested for grape wines earlier in this book. In general, use white, red, or blush table wine procedures as appropriate to the color of the berry and fruit wines being made.

While exceptions exist, most berry and fruit wines are rather short-lived and difficult to stabilize—certainly as compared as to red vinous grape wines. Consequently, long-term wood and bottle aging regimens are not recommended.

APPLE WINE

While apples were common in ancient, medieval, and Renaissance Europe, apple trees first came to America from England, quickly becoming a staple in the diet of the New England colonists. By 1776, the apple had made its way westward into the Finger Lakes region of upstate New York, with many plantings made by the Mohawk and Seneca Indians. Today, apples are one of the most widely cultivated fruits across the northern United States and Canada.

Apple wine blending is an essential function in composing the desired product type and style. Varieties of apples vary in their aromatic and flavor character, their sugar content, and their astringency and acidity (Table 12-1).

Most of the characteristics presented in Table 12-1 are of course, subject to variability due to the growing climate and soil, ripeness level, and treatment provided during growing, harvest, and storage. As with grape wine, the variety of apple selected and its state of ripeness and condition are all-important to the level of quality that can be achieved in the final product.

It is generally agreed that the best apple wines have a near colorless or blush hue with brilliant clarity, moderate bouquet intensity, and sugar/acid balance a bit on the tart side. This can be readily achieved by insisting upon fresh fruit harvested slightly on the green side of ripeness. A “triage,” or hand removal of rotten apples, is highly recommended.

It is imperative that pressing immediately follow crushing. One need only recall the last time an apple was eaten to understand how quickly apple pulp will commence to brown from oxidation. Some of the most successful apple winemakers sprinkle from 4 to 6 ounces of malic acid (citric acid is not permitted by 27 CFR, Part 4, Section 24.182c) and KMS evenly over each ton of apples going into the crusher to inhibit oxidation and the growth of spoilage microorganisms. An addition of pectolytic enzyme is also encouraged, in accordance with the manufacturer's recommendations.

Juice temperature should be adjusted to about 55°F as soon as the juice is taken from the press by an immediate inoculation with approximately 1% fermenting white wine yeast starter. Apples generally do not have enough sugar to reach a sufficient alcohol level. Sugar additions can be calculated in the same manner as described in Chapter 9.

Fermentation should be maintained at 55 to 60°F. Rackings, clarification, and the rest of apple wine processing can also be patterned after white table wine production from grapes.

BRAMBLE BERRY WINES*

Most bramble berries (berries that grow on thorny bushes) do not have enough natural sugar to reach a sufficient alcohol level after fermentation. Sugar additions can be calculated in the same manner as described in Chapter 9. Most of the berry pulp will liquify during the enzymatic and fermentation reactions.

Only prime fresh or frozen fully ripened food-grade berries are recommended. Commercial experience has shown repeatedly that poor-quality berries, or those distressed from poor transit and storage, make equally poor wines. Spoiled berries should be culled by triage and the remaining fruit gently crushed so as not to crack seeds, which can release astringent phenolic compounds.

Because bramble berries are well known for their high acid content, developing a berry wine with good balance is a major challenge of utmost importance to the winemaker. Without this balance, most berry wines, particularly raspberry, are excessively acidic.

One of the simplest methods of reducing the acidity level is by amelioration with water and/or sugar—although this can also have the negative effect of producing a wine with light body and thin consistency. Since flavor is the most important single factor upon which the quality of a fruit wine depends, the winemaker should strive to find the best balance of fruit, water, and sugar. The rules for these functions

*Adapted from "All about Raspberries and Gooseberries," by Dan Archibald, *Fruit Winemaking Quarterly* 2, no. 3 (summer, 1992) and "Experiments and Experiences with Raspberry Wine Making," paper presented by Edwin Patterson, Fruit and Berry Wine Symposium Proceedings, Purdue University, Department of Food Science, May 1992.

set forth by the Bureau of Alcohol, Tobacco, and Firearms (ATF) are found in 27 CFR, Part 4, Sections 24.178 through 24.182.

Inoculate with *Pris de Mousse* yeasts, and ferment in much the same manner as for light, fruity red table wines (discussed in Chapter 10). If needed, adding some of the amelioration water in the form of ice can help to control fast-rising fermentation temperatures.

Particular attention should be placed upon monitoring any development of volatile acidity. Berry wines seem to be particularly vulnerable to this type of spoilage. Some producers buy frozen berries to suspend any bacterial action until just before fermentation. In any case, berry wine fermenters should be closed except for fermentation locks, and frequent volatile acidity (VA) analyses should be run in the laboratory.

Crushed berries typically ferment rather quickly, usually in just several days. Temperature should not be allowed to exceed 70°F, and fermentation should achieve at least –1.5° Balling before pressing.

The best results in pressing wine from fermented berry must are generally obtained by gently and slowly pressing the mass in mesh bags. This helps to strain out the seeds and other pulp, which can later cause astringency and/or bitterness. Heavy rates of rice hulls added as a pressing aid, often two or three times that used for other fruits, can help achieve optimal wine extraction from the press.

Some berry winemakers add up to 3 pounds of bentonite per 1,000 gallons of newly pressed wine in order to immediately compact the lees during the final throes of fermentation. This will permit optimal efficiency in the first racking, which should be into storage at 30 to 40°F. Cold storage in combination with a moderate sulfite addition will help to avert malolactic fermentation, to which berry wines are very susceptible. A second racking should also be made in cold storage.

After several months of continual cold storage, the new berry wine may be sufficiently clarified for a coarse filtration. Filtration is frequently difficult and slow, requiring larger filter capacity than for most other wines.

BLUEBERRY WINE*

Blueberries are versatile in that they can make equally good dry red table wines and sweeter dessert wines.

It is recommended that fully ripened, even a bit overripened, blueberries be considered to achieve good color and flavor. Some of the best blueberries for wine are those that commercial packers discard as “floaters” in the washing process.

*Adapted from “Making Better Wines from Blueberries,” by Douglas Welsch, paper presented at the Fruit and Berry Wine Symposium, Purdue University, Department of Food Science, May 1992.

These are overripened and generally perfect for winemaking when carefully sorted. Since only a small percentage of the berries sorted each day are floaters, several trips to the packing house may be required to get a sufficient quantity. They can alternatively be collected and frozen for later delivery.

Blueberries have a delicate skin with tiny seeds in a semisolid pulp. Most crusher-destemmer-must pump machines used for grapes work adequately for crushing blueberries. Frozen berries should be thawed before commencing this operation.

Add 50 to 75 ppm of sulfite to berries in the crusher, and pump must to a holding tank, where pectic enzyme should be added at two to three times the recommended rate for grapes. Allow the crushed berries to set for 2 to 4 days in contact with the skins in order to extract flavor and color into the juice. Then press, settle out juice solids, ameliorate up to 30% of the resulting product, inoculate with active cultured yeast, and ferment as discussed for white table wines in Chapter 9.

Crushed blueberries will generally have a Brix of 7 to 9°, a total acidity of .800 to .900 g/100 mL, and a pH of between 3.10 and 3.30. Brix adjustment up to about 22° is recommended for dry wine—up to 24 or 25° if the resulting wine is to be residually sweet. The amelioration may be as high as 30% with sugar and water to reduce the bold character of the fruit and the astringency from the skins.

One of the most common difficulties encountered in making blueberry wine is in fermentation, as blueberry juice typically lacks a nitrogen source for yeast growth. Diammonium phosphate (DAP) should be added in the full amount of the 8 pounds per 1,000 gallons of juice permitted in the ATF regulation 27 CFR, Part 4, Section 24.246. The recommended yeast strains are Fermivin or Pasteur to best transcend the full term of fermentation. The fermentation may take several weeks to complete, during which time the temperature should be maintained between 55 and 65°F. If the wine is designed to be residually sweet, fermentation can be stopped by chilling to 30°F or so and adding a light treatment of sulfite.

The resulting wine should then be clarified, stabilized, and coarse filtered as for white table wines. Some styles of blueberry wines benefit from brief periods of oak aging.

CHERRY WINE*

As with grape wines, the best cherry wines are made from top-quality fruit, ideally a bit overripened to maximize color and sugar. Brown rot, a common problem in overripened cherries, will ruin cherry wine and must be avoided.

* Adapted from "Making Better Wines from Cherries," by Douglas Welsch, Fruit and Berry Wine Symposium, Department of Food Science, Purdue University, May 1992.

Frozen Montmorency pitted cherries are recommended, as the pits being removed will reduce the almond flavor that results from fermenting in contact with the pits. Most varieties of sweet cherries have a very unstable color.

Cherries should be crushed in a grape crusher-destemmer-must pump machine with the crusher rollers separated so as to minimize cracking of the pits. Sulfite in the 50 to 75 ppm range can be added at the crusher and the must pumped to a red wine fermenting vessel. Inoculate with yeast that accentuates fruit character, such as Epernay, and ferment cool—below 55°F, if possible.

Punch down the cap at least twice daily, and press after 4 to 6 days of skin contact, using gentle pressing action to avoid damaging pits or macerating skins. Skin color is readily extracted during fermentation. Pressing as soon as possible will help to reduce bitter herbaceous skin character and almond flavor from the pits. Be careful if using a bladder or membrane press, as broken pits have sharp edges and may cause some damage. Immediately chaptalize the new wine with sufficient sugar to increase the alcohol to 12% to 13% alcohol by volume. No water additions are recommended, as cherry wine color and flavor are typically rather delicate. Continue fermentation at the cool temperature. The color pigments in young cherry wines tend to brown unless treated with 2 to 3 pounds per 1,000 gallons of polyvinylpyrrolidone (PVPP) during the latter stages of fermentation. Because of high acidity levels in cherry wines, most fermentations are stopped at 0.0°Balling, or even before.

Most cherry wines tend to be protein unstable and should be fined with 4 to 5 pounds of bentonite per 1,000 gallons. In some cases, even more bentonite may be needed. Run a lab trial first to determine the ideal level of bentonite treatment. Excessive use of bentonite diminishes color and flavor in wine. The wine is then finished along the same general practices and procedures as discussed for white table wines in Chapter 9.

Cherry wine does not age well. Hence, it should be made, processed, bottled, and sold well within a 1-year production-marketing cycle.

PEACH WINES

The first major hurdle in the production of peach wines is in the removal of the pit. This is usually too large to go through a grape crusher-destemmer-must pump without damage to both the pits and the machine. The pits must be removed to avoid the astringent and bitter phenolic compounds that are extracted during fermentation. Consequently, access to a special pitting machine is needed, or this arduous task must be performed by hand.

Pitted peaches can be run through a grape crusher with an addition of 50 to 75 ppm of sulfite. The peach must/puree may not be liquid enough to pump with centrifugal pumps—in which case it will need to be transferred by a progressive

cavity pump, or by hand, to a clean fermenter, where an addition of about twice the normal pectic enzyme recommended for grapes is recommended. As peaches are deficient in the essential nitrogen for yeast growth, DAP should also be added at a rate of 4 to 6 pounds per 1,000 gallons—up to the ATF limit of 8 pounds if necessary.

Inoculate with Epernay to accentuate flavor extraction, and ferment cool—at less than 55°F if possible. As fermentation lags below 0.0°Balling, a light sulfite addition can be made and the wine immediately racked to cold storage at about 30°F. It can then be fined with 2 to 3 pounds of bentonite per 1,000 gallons, more if determined necessary by laboratory trials, to achieve protein stability. Continue cold storage until clarity is achieved sufficient to permit at least a coarse filtration.

Balance acidity and sweetness as desired. Citric acid additions are permitted up to a maximum of 5.8 pounds per 1,000 gallons in the ATF regulations (27 CFR, Part 4, Section 24.246). Sweetening additions are regulated in Section 24.179 of the same regulations. The wine is then finished along the same general practices and procedures as discussed for white table wines in Chapter 9.

Peach wine does not age well. Hence, it should be made, processed, bottled, and sold well within a 1-year production-marketing cycle.

STRAWBERRY WINE*

Strawberries are unusual in that they can be frozen and still make superior wine. Whether fresh or frozen, the berries should be fully ripened and unsound berries removed—a tedious triage (sorting and culling) process, but essential, since a small percentage of rotten fruit can have a profound negative effect upon wine flavor.

Fresh strawberries do not need to be crushed before pressing, although frozen berries need to be thawed beforehand. Berries should be carefully placed in a mesh bag to contain the stringy pulp and the tiny seeds while the press slowly extracts the juice and free pulp. Before pressing, a treatment of about 3 g (1/2 teaspoon) of citric acid per 100 pounds of berries, plus an addition of pectolytic enzyme in the manufacturer's recommended dosage, should be made.

Juice from the press should ideally be very pulpy, as the pulp is needed for sufficient flavor extraction during fermentation. Temperature should be adjusted to about 55°F and immediately inoculated with 1% or 2% of actively fermenting yeast starter.

Strawberries, like bramble berries, generally lack enough sugar to reach a sufficient alcohol level. Sugar additions can be calculated in the same manner as described in Chapter 9. Most of the pulp will liquify during the enzymatic and fermentation reactions.

*Adapted from "All about Strawberries," by Dan Archibald, *Fruit Winemaking Quarterly* 4 (July 1994).

Fermentation should be maintained at 55 to 60°F with rackings made every week or so until the new wine is dry of sugar or nearly so. The last racking should be made into a blending tank so that color can be adjusted with natural strawberry concentrate. The color extracted from most varieties of strawberries is unstable and highly susceptible to browning during fermentation. When the desired addition of concentrate for color and sweetness is made and thoroughly mixed in, the wine should be immediately transferred to storage at about 28 to 32°F until sufficient clarity is achieved for filtration and bottling. An additional racking or two may be necessary in cold storage until final clarity.

The rest of strawberry processing can be performed in the same manner as outlined for white table wine production from grapes in Chapter 9.

DRIED FRUIT WINE

The best raisin wines can approach a similarity to some of the “late-harvest” types made from overripened grapes—dark straw-gold wines with a soft apricot-caramel character.

While choices are usually limited, the freshest possible raisins are best, providing the most natural nutrients for yeast growth. Avoid using “golden” raisins, as these are generally processed with high concentrations of sulfur dioxide, which can inhibit fermentation.

The sugar content of most raisins ranges between 70 and 80°Brix and should be diluted with water to about 22°Brix before being inoculated with wine yeast. High sugar concentrations in musts have a dehydrating property that stresses yeast activity.

Precision water addition calculations are difficult due to the presence of skins and other unfermentable solids in the raisins. A typical water addition is about three times the weight of the raisins. For example, 10 pounds of raisins at 75°Brix could be calculated to need 30 pounds of water—or, at 8 pounds per gallon, approximately 3 3/4 gallons. Warm water will help to dissolve sugar solids from the raisins more quickly. Gently knead and mix raisins in the water to help create a raisin “must.”

Measure Brix and total acidity, adjusting each as outlined in Chapter 10 above. Inoculate with wine yeast, and continue with the same general red wine procedure.

Most raisin wines are finished with a bit of residual sweetness, in the typical “late-harvest” style. Be sure that adequate KMS and potassium sorbate are used as a yeasticide to inhibit secondary fermentations.

The following specific ATF regulation applies:

In the production of wine from dried fruit, a quantity of water sufficient to restore the moisture content to that of the fresh fruit may be added. If it is desired not to restore the moisture content of the dried fruit to that of the fresh fruit, or if the moisture content

is not known, sufficient water may be added to reduce the density to 22 degrees Brix. If the dried fruit liquid after restoration is found to be deficient in sugar, sufficient pure dry sugar may be added to increase the total solids content to 25 degrees Brix. After addition of water to dried fruit, the resulting liquid may be ameliorated with either water or sugar, or both, in such total volume as may be necessary to reduce the natural fixed acid level of the mixture to a minimum of 5.0 grams per liter; however, in no event may the volume of the ameliorating material exceed 35 percent of the total volume of the ameliorated juice or wine (calculated exclusive of pulp). Pure dry sugar may be used for sweetening. After complete fermentation or complete fermentation and sweetening, the finished product may not have an alcohol content of more than 14 percent by volume nor may the total solids content exceed 35 degrees Brix. (27 CFR, Part 4, Section 24.202)

MEAD

Mead is wine made from honey. There are many mead “recipes,” and variation far exceeds standardization in the manners and methods employed in its making.

The same general methods can be employed as outlined for white table wines in Chapter 10, commencing with sugar adjustment.

One critical variable is, of course, the sweetness level of the honey to be used—usually in the general range of 60 to 70°Brix. This is difficult to measure directly by hydrometer, and many home meadmakers will add warm water at a rate of twice the volume of honey and then take a Brix test. From this result, a final water adjustment can be calculated to pinpoint the Brix required to achieve the desired potential alcohol.

Honey is highly deficient in natural acidity. Consequently, an addition of about 25 g of citric acid per gallon of diluted honey is recommended. The lack of sufficient yeast nutrients should be supplemented with a dosage of Yeastex 61 in the amount of 6 g per gallon.

Fermentations may be difficult to get started, resulting in the need to consider the use of an unpreserved fruit juice as an active starter medium.

To discourage stuck fermentations, which are common in making mead, fermentation should be conducted in the 65 to 75°F range.

Mead is finished both dry and sweet. Close attention should be given to the following specific ATF regulation for honey wine:

In the production of wine from honey, a quantity of water may be added to facilitate fermentation provided the density of the mixture of honey and water is not reduced below 22 degrees Brix. Hops may be added in quantities not to exceed one pound of each 1,000 pounds of honey. Pure dry sugar or honey may be added for sweetening. After complete fermentation or complete fermentation and sweetening, the wine may not have an alcohol content of more than 14 percent by volume nor may the total solids content exceed 35 degrees Brix. (27 CFR, Part 4, Section 24.203)

CHAPTER 13

MARKETING

Wine marketing encompasses all aspects of the winery at all stages of development. It begins with image development, market research, and development of the marketing plan and marketing budget; continues through the initial sale; and then focuses on consumer retention.

This chapter will guide vintners through the initial stages of developing a marketing strategy and then will discuss the multiple techniques that can be applied to enhance publicity and profitability. These techniques can be divided into six main categories: self-promotion, advertising, public relations, industry relations, customer service, and sales. All are equally important, and all need to be incorporated to ensure the implementation of a successful marketing program.

Be aware that wine marketing is heavily regulated in the United States by both federal and state governments. The federal rules are found in 27 CFR, the U.S. Code of Federal Regulations enforced by the Bureau of Alcohol, Tobacco, and Firearms (ATF). The alcoholic beverage authority operating in each capital city administers state statutes.

GETTING STARTED

A vintner decides to create a new wine. Many steps go into this process. Varietal character will be examined (research), the final grapes will be selected (planning), the wine will be fermented (implementation), and then the wine will be blended and adjusted as needed (revision). An effective marketing strategy is like developing a new wine in that it requires the same steps to be successful: research, planning, implementation, and revision.

Some wines are released immediately, but many wines require aging to develop strong characteristics and flavor. Marketing activities are comparable: they will vary on how quickly they bring business to the winery. Aged wine is considered an investment, a product that will bring future profits. Viewing marketing as an investment too, rather than as a large expenditure, will also enhance winery success.

Additionally, marketing plan development is imperative in the early stages of winery development. Marketing involves every aspect of the winery, from location to packaging to customer service. Creating a marketing strategy early helps link all aspects of the winery and present them to the public in an entertaining and educational manner.

Image Development

Wineries should know who they are, what differentiates them and what makes them unique. . . . Don't be something you're not.

—Cherie Melka, Melka Cellars in California

Focus, consistency, and success all have one thing in common: a direct relation to image development. Developing an image helps a winery determine who they are, why they are unique, and where they expect to go in the future. More important, a focused image portrays that information to consumers. Wineries that lack a defined image often lose direction, spend money inefficiently, and waste precious resources.

Image development involves articulating the winery philosophy—that is, telling the winery story. Start by answering some simple questions:

- Who are the winery owners, and where do they come from?
- What is the winery's signature product?
- When did the winery idea (dream) evolve? Why? How?
- Where are the grapes grown, and what makes them special?
- Why locate the winery here? Why select this theme?
- How will customers feel when they walk in the door or try the product?

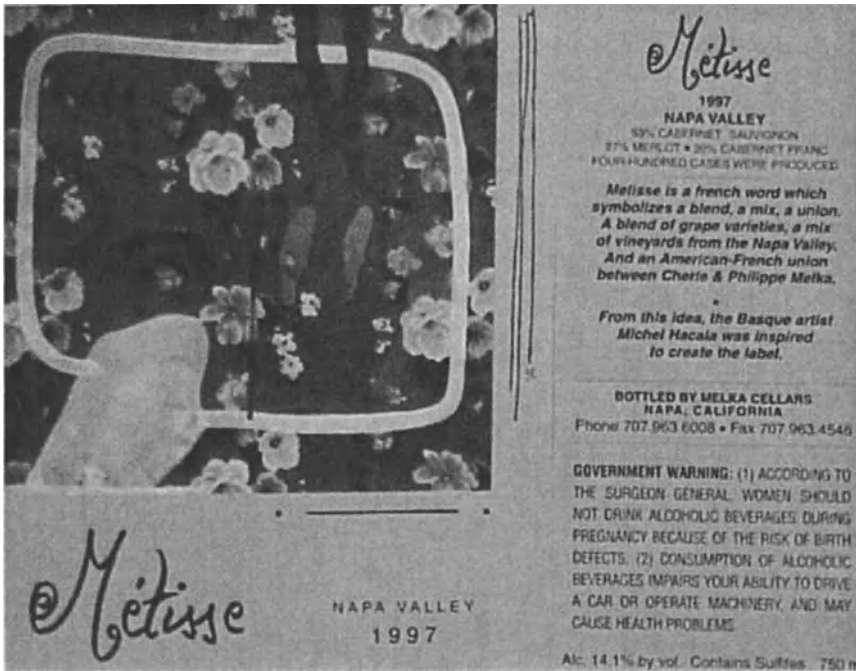


Fig. 13-1 Developing an image: “Metisse” wine.

Melka Cellars in California spent considerable time analyzing their image. The owners wanted to develop a high-quality, fine wine while focusing on differentiation in their backgrounds. Therefore, they named the wine *Metisse*, meaning “union,” and they are consistent in delivering their marketing message: balance in their wine, their life together with their two children, and the cultures in which they were raised (Figure 13-1). As a result, over 75% of their business comes from telling their story. It is beneficial to focus on an image and be consistent in delivering it in every marketing function, from developing the marketing plan to media releases.

Market Research

Market research is the systematic gathering, recording, and analyzing of market data which will help uncover opportunities and reduce risks in decision making.

—William Dillon et al., *Marketing Research in a Marketing Environment*

Success in the wine world is heavily dependent upon selling what people want. Market research identifies who those people are and what type of product they may want. Wine does not fall into the need category; people do not need wine to

survive. Rather, it is a want—something consumers select to enhance their life. Advance market research, done before the winery is developed, will help detect which consumers are most likely to buy a particular product. These core consumers are often referred to as the *target market* or *primary audience*.

Understanding the target market for the winery is very important. Are the core consumers primarily retired empty nesters with an annual income of \$50,000? Or are they Generation Xers, in their late 20s, single, with a large disposable income? Marketing efforts may vary greatly depending on the target audience.

In addition to understanding who the ultimate consumer is, all vintners need to have a firm understanding of the market in which they will exist. Market analysis needs to occur in two areas, the macro-market and the micro-market. The macro-market concerns the overall wine industry or the big picture. What are the national and international wine consumption trends? Why are the trends growing or declining? The micro-market, on the other hand, narrows the overall wine market to focus on specific segments (niches). What are the wine consumption trends in a specific geographic location? What varietals are produced in that area, and how much of each varietal is sold annually? Analyzing the macro- and micro-markets will help a winery lay out plans for selecting the best possible varietals, wines, and image for the proposed location of the winery.

Obtaining accurate market research information can take time, money, and effort. Many market research firms will conduct studies for small business owners. If the budget is lean, search the local library for wine market information and create small consumer focus groups. Additional resources include a state wine association, a local tourism department, or Web sites for the Wine Market Council or Wine Institute. Marketing research is worthwhile; the information can help give direction to the marketing decisions that need to be made in the marketing plan.

Marketing Plan

[The marketing plan] creates an operational “road map” to define a winery’s direction, focus and image.

—Veronica Barclay, Winery Marketing Consultant

Once the basic image is developed, the target audience is defined, and the various markets are researched, it is time to develop the marketing plan. The theory of marketing involves four main variables: product, price, promotion, and place (distribution). The four “P’s” are known as the *marketing mix*. Creative marketing strategies blend the four “P’s” to focus on a specific target market while maximizing profitability and exposure.

Marketing involves unknown risks. There is no guarantee that one promotion will be more effective than another. However, the development of a marketing plan, after thorough market research, helps reduce the unknown aspects of marketing by

highlighting key opportunities. The marketing strategy places a vintner's ultimate goal into perspective, while communicating the marketing vision to employees and other interested parties (i.e., investors).

Many different outlines are available for marketing plans. Each is a little different, but all offer a few important elements that must be included.

- Winery vision: What does the vintner want the winery to be?
- Mission statement: What business is the winery involved in?
- Target market/competitive analysis: Who is the product for?
- Benefit to consumers
- Measurable, attainable objectives
- Strategies for reaching the objectives
- Techniques for implementing the strategies
- Month-by-month action calendar

The winery vision defines what type of winery the vintner wants to own, while the mission statement describes the overall goal for the winery. What is the business focus? To be an ultra-premium contemporary winery, located downtown, creating 15 award-winning *vinifera* wines? Or a winery and farm market offering the best homegrown wine and fresh produce in the town of Nortonville? The mission and vision statements relate directly back to the image development of the business. The style of the winery and products need to be portrayed in both statements.

To make a sale, products need to offer the consumer a benefit that is greater than their cost. Additionally, the marketing strategy should separate product features from benefits. For example, a well-designed wine label is considered a feature, whereas a wine label with suggested food pairings and temperature guidelines offers the consumer a benefit.

Who is the ultimate consumer? This question is one to be answered through market research, generally done before the business or product is developed. Turn the research into a marketing statement that tells what you are offering and to whom: "Triangle Vineyards offers 100% high-quality estate wines to homegrown supporters."

Vintners also need to consider competitors. Finding a niche, or a product area that does not have many competitors, will increase sales and positioning in the marketplace. Figure 13–2 is an example of a niche analysis. A competitive analysis within this particular market has shown that there is currently no winery selling a high-end table wine for a moderate price. Therefore, there is an opportunity for a new winery to fill a niche within that market.

Creating measurable objectives is the core of the marketing plan that combines the vision and mission with the facts obtained from research. Although the business

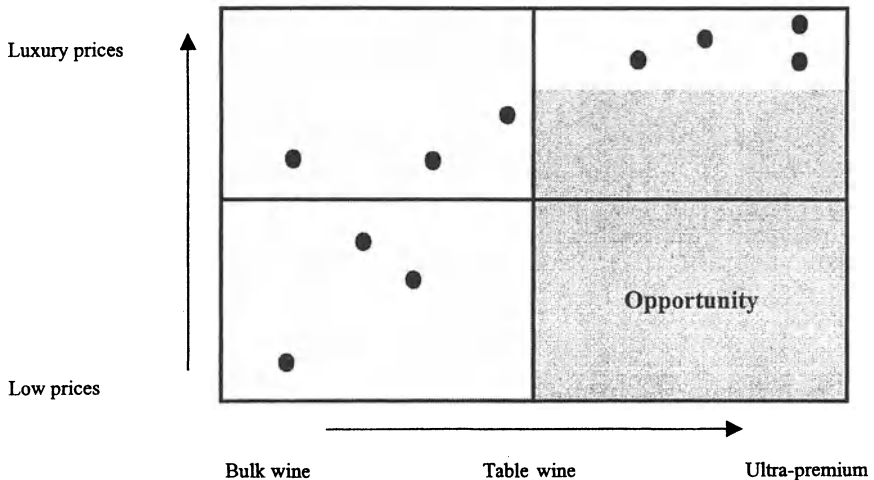


Fig. 13–2 A competitive niche analysis.

may have several goals, three to five main objectives should be selected per year. For example, one objective may be to increase media coverage. Next, strategies or general approaches are developed to reach the objectives. They may include, but are not limited to, pricing and margins, promotion, advertising, and product positioning. Finally, techniques or specific actions are created to execute each of the strategies. Work on combining self-promotion, advertising, public relations, industry relations, customer service, and sales. An effective plan will unite all six elements.

The final piece of an effective marketing plan is the month-by-month action calendar. The calendar puts the plan to work in a concise format that is easy to follow and revise, as shown in Exhibit 13–1. Listing all of the objectives and plotting them by date allows one to schedule around busy times when staffing and financial commitments may be tight. As the marketing plan becomes finalized, it is time to consider the budget that will officially allow the plan to succeed.

Marketing Budget

Prepare your best realistic projections. Triple your expenses and halve your sales numbers. If it still looks positive, go for it.

—Old forecasting rule

Budgeting the marketing objectives for the first time can be difficult. Ideally, marketing budgets should be built around the recently created marketing plan.

Realistically, for new vintners, most marketing plans need to fall within a certain budget. Either way, the marketing budget needs to effectively and efficiently portray the winery's image and products to the target market.

New vintners will need to spend more on marketing budgets than established vintners. *Vineyard and Winery Management* magazine projects that the initial time and money investment should be 35% to 40% (or more) to develop market share. As the customer base and cash flow increase, marketing investments should average 10% to 20%. Calculating projected sales for the first year will help determine the amount available for marketing.

Chart the marketing plan techniques, and determine the appropriate budget allocation for each one. Exhibit 13-2 is a sample marketing budget, based on a budget created by *Vineyard and Winery Management* magazine. Marketing budgets do not need to be high dollar to succeed. Personal time commitment can maximize a marketing plan, often as much as monetary investment. There are multiple ways to receive consumer response without spending a large amount of money, including the Internet, superior customer service, handwritten thank-you notes, and creative special events.

Once the budget is prepared and the marketing plan is put into action, continuous evaluation and adjustment are a necessity. Evaluating past performance allows one to judge the success of what has been done or the likelihood of future success. Measurement comes in many forms, from dollars to cases sold to a percentage increase in winery customers. Someone will need to monitor and measure each objective on the marketing plan and budget. If a particular marketing tactic does not accomplish the overall objective, then the plan needs to be revised.

Marketing strategies are active, not stagnant. A successful program involves continuous measurements and updates. Be patient: some but not all marketing efforts display immediate results. Image development and promotional plans are long-term initiatives; be willing to consider marketing as an investment rather than an expense.

MARKETING TECHNIQUES

A triumphant marketing strategy is one that involves multiple techniques or tactics. After planning the work, it is now time to work the plan. Messages need to be heard and seen in various formats for to be effective. On average, consumers need to hear a product name seven to nine times before they are compelled to purchase. As mentioned above, a vintner needs to adapt techniques that include self-promotion, advertising, public relations, and industry relations. Of equal importance is the customer service offered within the winery that leads to long-term customer relations and enhanced sales.

Exhibit 13–2 Sample Marketing budget.

	Budget \$	Actual \$	Adjustment
Promotional Materials			
Brochure			
Newsletter			
Wine list			
Media kit			
Web site			
Direct Mail			
Database maintenance			
Design/printing/postage			
Letterhead/envelopes			
Point-of-Sale Materials			
Shelf talkers			
Gift items/souvenirs			
Special Events			
Winemaker dinners/theme events			
Media events			
Advertising			
Newspaper/magazine/trade publications			
Radio/TV			
Internet/yellow pages			
Tourism guide/tourism map			
Trade Shows/Competitions			
Entry fee/registration			
Wine donation			
Travel/staffing expense			
Outsource Marketing			
Graphic design			
Web site maintenance			
Photography			
Mail distribution			
Unexpected Opportunities			

Total:

Note: List will vary for each vintner; alter as needed.

Source: Based on marketing budget from *Vineyard and Winery Management*.

Various techniques in each of the main areas will be discussed individually to offer insight and guidance on how to implement an effective marketing program. All of the techniques overlap and blend. They need to be used in conjunction with one another over an extended period of time to achieve their full potential.

Self-Promotion

Brochures, packaging, newsletters, direct mail—all fall under the term *self-promotion*. Instead of being reactive to consumer interest once consumers are

in the winery, self-promotion creates a proactive environment to attract customers. The core of the self-promotion campaign is to tell the winery story over and over again. Additionally, consumers need to understand what benefits they will receive from this proactive approach. What's in it for them?

Total Package

Promoting the winery in the fullest sense includes a discussion on packaging. To start, the winery package includes bottles, labels, case boxes, in-store displays, point-of-sale materials, signage, and letterhead. The category is broad and essential to the overall success of a winery and its product.

Vintners need to keep the winery image in mind when developing all packaging. An attractive and eye-catching package sells. The winery packaging must be consistent, portraying the winery story to the specified target market. Chateau Morrisette, a Virginia winery, created a package that consistently follows its image. Every aspect of the winery is focused on Nicholas, the winery's black dog. Even the newsletter contains dog-related information; Nicholas writes the newsletter (with some human help) and signs it with a large paw print. See Figure 13-3 for an example of Chateau Morrisette's complete packaging.

When designing the marketing plan or budget, the winery needs to consider the packaging requirements needed to meet the plans. Even the name of a wine affects sales. Names should be short and easy to pronounce for consumers. Then graphics

Fig. 13-3 Chateau Morrisette packaging.



should be added to enhance the name, making the brand easily recognizable. Carry the winery image throughout all of the packaging to create an easy-to-remember brand imprint for the consumer.

Brochures/Newsletter

The brochure and newsletter are some of the most popular self-promotion tactics and, if implemented properly, are the most effective for portraying the winery image. A brochure will act as the winery destination guide for consumers. Among other things, it may include the brief winery story, a map with directions, tips, and photographs (Figure 13-4).

Professionalism demands attention in self-promotion programs. Is there a staff member who can turn words into a visual impression or take professional photographs? If not, consider using a portion of the budget to hire a copywriter or photographer. Be realistic about deciding which aspects can be completed in house and which should be outsourced. Consider that a brochure is what consumers take home with them, request over the phone, or pick up at a special event. Take the time to collect samples and analyze the good, the bad, and the ugly. What is visually attractive? Would the brochure entice the consumer to learn more about the winery? Does it tell the winery story in text and photos?

A newsletter keeps consumers informed on winery happenings, new wine releases, recipes, and competition awards. The newsletter should be similar in style

Fig. 13-4 Winery brochures.





Fig. 13-5 Winery newsletters.

to the brochure (Figure 13-5). Remember the importance of following a consistent image that tells the winery story. Newsletters offer an additional benefit: the opportunity to “stay in touch” with customers. Generally newsletters are sent quarterly or biannually through the mail or e-mail.

Who should receive a newsletter? That depends on the main marketing objectives created earlier. If the objective is to maintain contact with frequent buyers, the list should be built from in-house sales reports. However, if the objective is to reach new customers within a particular zip code, a mailing list should be purchased.

Prices vary for brochures and newsletters. A high-quality, full-color brochure with photos and detailed maps may cost up to a dollar each to produce, while a nicely designed, simple brochure with two colors may cost less than 15 cents each, depending on quantity ordered. The same pricing variations are valid for a winery newsletter. To begin, obtain quotes from various graphic designers and printers. The vintner needs to analyze the proposed designs from a consumer point of view. Would the chosen design and print quality make a customer come to the winery door?

As technology moves forward, there are many “do it at home” brochure and newsletter programs. Some are very good, some are not as effective. Be careful by paying close attention to the overall project goal and winery image. Saving a

few dollars by producing a nonprofessional self-promotion project may lead the consumer to someone else's front door.

Video/CD

A picture is worth a thousand words. Visual images drive sales. Draw a picture for the customer. These are all everyday phrases when dealing with marketing. Consumers are attracted to visual objects, things that they can see and relate to. A winery video or CD-ROM will offer current and potential customers a glance at the winery story and daily happenings.

Harvest, bottling, special events, wine tasting, and production tours are available to view year round if a winery video or CD is accessible. Technical presentations allow the off-site consumer to experience the winery "behind the scenes." On site, walking visitors through the production area while showing them the processing steps through video or CD can often enhance winery tours.

Rapid technology is moving toward CD-ROM and DVD use and away from video. Consider each type of medium when deciding on the best fit for the winery. Consider how many years the technology will be usable. As with other marketing projects, prices vary greatly, and professionalism is extremely important. Today's home technology makes amateur production affordable, although it often sends the wrong message to consumers. Wineries can save money by producing a CD-ROM business card (Figure 13-6). The small CD can contain a winery presentation, including photographs and important facts. It is less expensive than video and can offer the same beneficial effect if done properly.

Again, the vintner needs to put him- or herself in the customer's place. What is visually appealing? Does a professional project earn more respect and interest than a homemade presentation? If possible, distribute copies to local visitor bureaus, important media contacts (including television stations), and tour companies. Use the presentation as a marketing resource; let the viewer experience the winery in its entirety.

The Winery Web Site

Busy schedules and limited leisure time make the Internet a great place for consumers to get immediate information. Therefore, a winery Web site is almost essential in today's marketplace. A Web site offers smaller wineries an equal presence to that of the big wineries. It is also a cost-effective way to provide information about and access to a company.

National tourism facts show that a high percentage of travelers are planning their trips on the Internet. In some states, surveys indicate that over 80% of travelers obtain destination information from the Internet. Nationally, the Travel Industry Association of America recently reported that 85 million travelers are online and that 62% consult the Web for travel information. Additionally, nearly one half of Internet users are frequent travelers, taking five or more trips in the last year. That

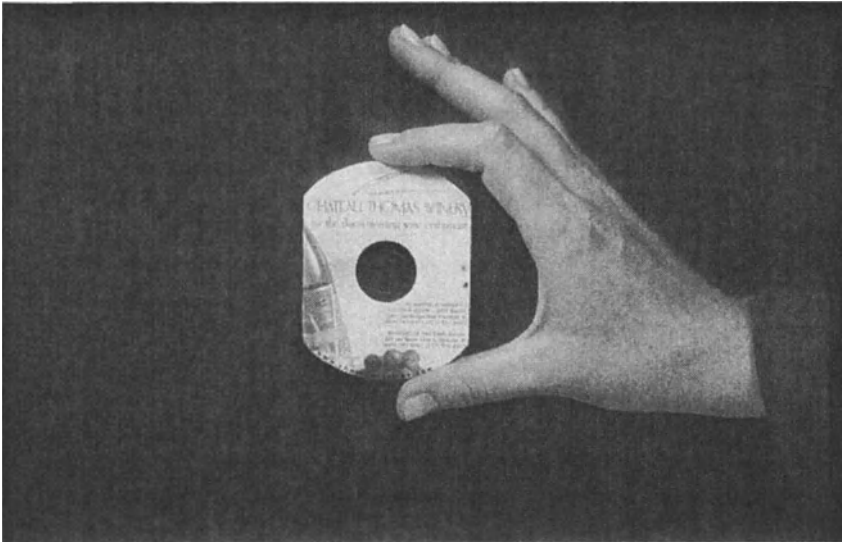


Fig. 13–6 Chateau Thomas business card CD.

means there is a potential audience of over 52 million people planning travel on the Web, travel that could be directed to the wine industry.

Winery Web sites offer a multitude of information, depending on the design. A winery description, product offerings, event calendar, photographs, and maps are all common components of a Web site. However, with technology continually advancing, additional options can enhance the site and drive business to the winery. A winery may consider adding a virtual tour or Web coupons (if legal) for consumers.

Another valuable addition to any winery Web site is media information. Fact sheets, background information, contact lists, and press releases can easily be included for reporters. Media deadlines are tight, and the Internet allows a reporter to quickly gather necessary information for story deadlines.

Consider the creation of a database-backed Web site that allows the site owner to update the contents from any computer at any time, using basic fill-in-the-blank forms. Such a site is easy to update and saves money, since a technical expert is not required to make changes on a regular basis. Another option with a database site is to have Web request forms for brochures, media kits, and so on automatically add requester addresses to your mailing list. Any extra design charge is recuperated through saved labor time by the elimination of rekeying information.

Requesting proposals from multiple Web development companies will help a winery select the best design team for its product and budget. Web site development can range in price from approximately \$1,500 to whatever limit the budget will

allow. Like any other marketing project, money properly invested will be returned many times over. The saying “You get what you pay for” holds true in the world of technology. Expertise is expensive but worth it.

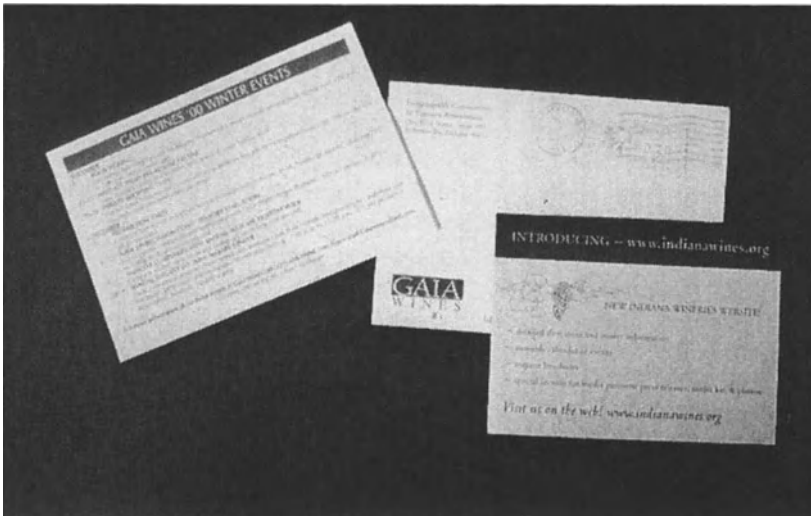
The Internet is an important and powerful tool for any winery. When selecting the designer for the site, be cautious to maintain the winery image. As stated earlier, consistency in marketing efforts definitely includes the winery Web site. Use the site for more than a promotional brochure by incorporating some of the features mentioned above. User-friendliness and interaction are important and increase visitor returns as long as the added features do not add more downloading time for the viewer. Continuous updates are essential; be willing and able to make a commitment to the finished site.

Direct Mail/Database Marketing

Targeting specific consumers with precise marketing efforts is a basic description of direct marketing. When a winery sends a promotional piece directly to a customer's door (direct marketing), it is hoping to get a direct response. Generally, projects are customized for a particular mailing list based on the winery's current target market or desired market.

Direct marketing is delivered in various formats: postcards (Figure 13–7), newsletters, invitations, event announcements, and so forth. Prices are determined on the basis of the style and detail involved in each piece. Money does not have to

Fig. 13–7 Direct marketing postcard.



be abundant to achieve direct marketing success; repetition of simple basic pieces can be sufficient and effective in creating brand awareness.

The key element of a direct marketing campaign is the mailing list or database. Using the proper database will increase the response of the direct mailing. The more defined the list is, the better the response. There are two main types of mailing lists: the winery list and the rented list.

The winery list is composed of people who have already visited the winery or who requested information about the winery in response to various promotions or advertising. A winery guest book is a good place to collect names and information from visitors, and an 800 number or a Web site often handles inquiries for potential visitors.

It is pertinent to maintain an updated winery database. Annual checks for outdated addresses and duplication are important. Sending a response card (Do you want to remain on our list?) at least every 2 years eliminates unwanted mailings and extra postage costs. Another option is to maintain the list based on annual purchases or visits. Linking the mailing list with sales, thereby focusing direct marketing on high-volume and proven consumers, can do this.

The second list type is the rented list. Rented lists are either compiled from phone books and trade journals (*compiled lists*) or developed from customer response studies (*consumer response lists*). Compiled lists can be purchased according to zip code or another broad area and are considered more generalized, whereas consumer response lists contain names of people who have responded to direct mail before, thus increasing the chances that they will respond to the winery mailing. Rented lists can be very detailed. A winery can request, if desired, a targeted list of wine consumers in the Atlanta area with an annual household income of over \$200,000. Rented lists are expensive and rise in price as the level of detail increases. Also, note the word *rented*. Often, the list is allocated for one-time use. There are strict penalties if the lists are reused without permission or payment.

Once the target mailing list is complete, it is time to design the direct mail campaign. General guidelines to follow when creating direct mail are:

- Choose the appropriate direct mailing format based on budget and audience.
- Be consistent; the design of the project needs to match the winery image.
- Keep the mailing simple and easy to read. Include lots of white space and short sentences.
- Pay attention to the 3-second-rule. People often toss something in the trash if they cannot find any perceived value within 3 seconds.
- Be creative to attract the reader's attention. Be friendly and use emotion.
- Refer to a deadline date to gain response. For example, "First 50 people receive a free etched wine glass!"
- Check legal guidelines; they vary by state for direct mailing pieces.

- REPEAT! REPEAT! REPEAT! One direct mailing piece will not be effective. This is a long-term marketing project; seven to nine product exposures may be required before the first purchase.

Advertising

In the marketing profession, the term *advertising* generally refers to paid advertising. As mentioned earlier, consumers need to hear a message seven to nine times before they are compelled to purchase a particular brand. Paid advertising is another way to get the name of the winery in front of the public. It should be used in conjunction with the other main marketing techniques.

The ATF enforces regulations specific to wine advertising. These are found in 27 CFR, Part 4, Subpart G, "Advertising of Wine."

Repetition is important when selling a product. Consumers must recognize products' names and benefits before they will purchase those products. Advertising, when used in conjunction with other promotional items, is a marketing tool that helps drive the consumer to the product. Research shows that wine industry advertising is on the rise (since 1994) and that consumption has followed the increase. As wine advertising dollars increase and/or decrease, so does consumption.

Paid advertising comes in many forms, including print, television, radio, and sponsorship. Advertising should be used to enhance the overall marketing plan. It is often expensive, ranging in price from \$30 per column inch for newspapers to more than \$1 million for a 30-second television advertisement during the Superbowl.

Research is required to make sure that the correct advertising source is selected. One winery may choose to purchase a newspaper advertisement in the local paper to promote a community open house, while another may purchase a national magazine advertisement to highlight a new wine release. Ad selection and placement depends on the winery's overall marketing objectives.

Cooperative Advertising

Decreased advertising costs and strong relationships with other companies are things that all wineries would like to obtain. How would a winery accomplish this? One way is through cooperative advertising. Cooperative advertising allows a winery to team up with another company to enhance its product and advertising dollar.

Think outside the box when considering cooperative advertising programs. Projects can range from wineries forming a holiday shopping spree wine trail with coupon books to a winery and radio station teaming up to jointly promote a "Wine of the Month." Cooperative programs are often created to increase traffic during slow times.



Fig. 13–8 Wente Vineyard's cooperative advertising campaign.

Working with other wineries is one option but not the only one. The options are limitless. Make them fun and easy to obtain, with a long-lasting effect. Some of the most popular and successful cooperative advertising projects involve business collaborations that are not normally considered.

For the past several years, Wente Vineyards in Livermore, California, has teamed up with contrasting and complementary partners in a cooperative concept called “Best of Both Worlds” (Figure 13–8). A complementary program included pairing Wente’s Reserve Cabernet Sauvignon with a box of Godiva Chocolates. An uncommon partnership was formed when Wente’s Pinot Noir was packaged with a famous Burgundian Pinot Noir from Maison Bichot.

While some would consider this odd since it involved winery competitors, Wente considered it an opportunity to showcase two of the leading U.S. and French premium wine brands. As Wente Vineyards demonstrates, creativity can play a large role in cooperative advertising. When a winery is looking for an innovative way to promote its product or advertising dollars are tight, cooperative advertising is a viable option.

Overall, paid advertising and cooperative advertising must fit into the marketing plan. Each is a component of the big picture, placing the winery name in front of consumers to enhance recognition and drive purchases. Research is a necessity for both forms of advertising, increasing the chances for a sound message transfer to consumers.

Signage

Signage is a very effective form of advertising, exposing consumers to the winery brand. Used effectively, it can be an inexpensive way to repeat the winery information to the public, moving them closer to the seven to nine repetitions needed for an initial purchase. Common signage for wineries includes billboards, outdoor signs, winery truck signage, and directional road signs.

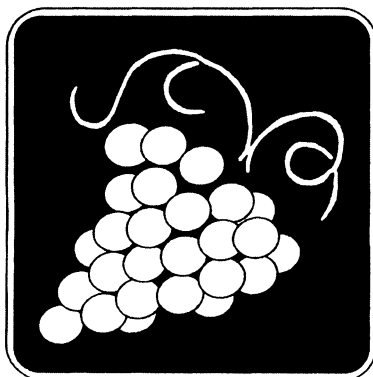
Traditional signage proves effective for many wineries. Billboards located in key positions on surrounding roadways may attract multiple visitors throughout the year. For example, Chateau Thomas Winery, located on a major highway in Indiana, draws in over 70% of its visitors from billboard signage. Billboards are valuable but expensive. Prices range from several hundred dollars per month on small roads to several thousand per month on major roadways. Sharing the billboard with another company can sometimes reduce expenses.

While billboards direct consumers to the correct road or exit, outdoor winery signage notifies visitors that they have arrived at the correct location. Outdoor signage is not so much an advertisement as a landmark. Attractive signage on a winery truck, however, will be seen throughout the region.

Some tourism departments offer directional road signage for wineries. Figure 13-9 is an example of winery directional signage. These programs usually have eligibility requirements and specifications for location, size, and so forth. Requirements may be based on number of visitors, accessibility, restrooms, and hours of operation. To participate, wineries need to show that they meet all requirements.

Billboards, outdoor signs, winery truck signage, and directional road signs are all excellent examples of traditional signs, but the term *signage* can include many more items. What about logo signs at events, window displays, winery T-shirts, and hot air balloons with the winery logo? All of these are also forms of signage, promoting the winery name and image. Depending on where the winery is located,

Fig. 13-9 Winery directional signage.



some forms of signage may be more cost-effective than others. Look back at the winery image, goals, and target audience to determine what type of signage program to implement. Whenever signage is used, make sure it meets local code requirements.

Wine Competitions

The marketing benefit found in wine competitions is this: consumers believe independent evaluation of products. Similar to a high automobile rating by *Consumer Reports*, a wine medal from a reputable wine competition is unbiased and believable. It is not a paid advertisement produced by the winery; rather, it is an independent product analysis (judging) and report (medal/no medal).

Found throughout the world, wine competitions offer vintners the opportunity to have an outside entity evaluate their wine. Competitions often focus on specific wine characteristics. For example, Virginia hosts a regional competition only for Virginia wines. Other competitions may only judge *vinifera* wines or amateur wines. Although, large international competitions will select expert judges to evaluate multiple wine categories, including wine made from *vinifera*, hybrid, or Native American grapes and various fruit.

Entry prices vary for each competition. Average prices range from \$25 to \$125 per wine entry. The most expensive competitions are not necessarily the best. There are other important competition aspects to consider in addition to cost.

Competition reputation and integrity should be examined before agreeing to enter wines. Take time to review the competition rules. Are judges receiving the wines in numbered glasses so they cannot identify entries through bottle shape and color? Are the judges familiar with the types of wine that are entered into the competition, enabling them to make an accurate evaluation? Reviewing the competition design will allow vintners to select competitions that will fairly and accurately judge their wines.

Once a medal is received, promote it! Top awards include trophies and plaques, as shown in Figure 13–10. Many competitions offer bottle stickers that say “Medal Winner” and include the competition name and logo. When wine bottles with medal-winning wine stickers are placed on the shelf, they have instant appeal to the consumer who is not familiar with the product. Use the wine awards to promote the winery to the media and to other industry partners also. Awards are given to well-made, high-quality wine. Consumers, media, and other businesses will be supportive of such products.

Public Relations

Public opinion can have an effect on any product or industry, including wineries. If properly implemented, public relations can build a positive image for the winery and its products. Mark Miller of Brown Miller Communications states, “Public relations is about forging the long-term reputation of the winery—building the



Fig. 13-10 Wine competition trophy.

respect, admiration, and relationships that will support sales, enhance image, and sustain the winery through good times and bad.”

The main focus of public relations is working with the media, maintaining strong contacts and presenting newsworthy information. However, special events and consumer enhancements like wine trails also fall into the category.

Media Relations

Media exposure adds credibility and believability to the winery story. An outside, journalistic view offers a new perspective in an unbiased manner. Longwoods International reports that readers are 75% more likely to get travel information from editorials than from advertising. Since most wineries focus on the travel

industry, this is an important audience to reach. Additionally, media relations are an excellent way to enhance the marketing plan without adding prohibitive costs. Media relations are often promoted heavily because they are essentially free: costs can be limited to media releases and telephone calls.

Media Basics

In work with the media, everything is built on relationships. The relationships are built on providing honest, accurate information in time for the reporters to complete the story. Reporters are under strict deadlines and need to work with reliable sources. Becoming one of those sources is a benefit for any vintner.

How do you become the main source for a reporter?

- Return messages promptly. Be available and easy to reach so that the reporter can meet the story deadline.
- Be an expert on the winery and its products. List other expert contacts relating to the story.
- Provide quotes, statistics, and new facts. Be brief and clear.
- Become a resource for industry-related stories. Subscribe to trade publications and read them.
- Call and offer story ideas that do not directly relate to the winery. Help the reporter find new, interesting story angles.
- Be enthusiastic and helpful.
- Give credit where credit is due. Take time to thank the journalist for his or her story.

Newsworthiness includes breaking news and features. Breaking news consists of “hot topics” like elections and storms; features contain a human interest side that is also known as “soft news.” Additional newsworthy activities include seasonal events, unique angles, or conflictive issues. The wine business is versatile and can fit into various newsworthy activities, including features, business news, wine and food reviews, and travel pages.

Wineries need to be cautious about how stories are presented to the media, since wine, a consumer product, is often viewed as highly commercial. Presenting information in an entertaining and/or educational manner takes the focus off the commercial bias.

Audiences identify with success stories of people achieving the unthinkable, working hard, winning awards, contributing to the community, and so on. Mold stories around similar topics. Instead of writing a story on the winery’s new Chardonnay, add a twist and describe the contest that was held to select the wine name. Create a new story, one that the audience can connect with.

Media Kit

Before the first story idea is sent to the media, the vintner needs to gather pertinent information about the winery and present it in a neat, orderly fashion. This



Fig. 13–11 King Estate Winery (Oregon) media kit.

information is referred to as the media kit. Media kits do not need to be fancy and expensive; the journalist is more concerned about accurate, timely information. File folders or pocket folders with the winery logo or label work well. King Estate Winery in Oregon has an effective media kit that contains useful information simply presented (Figure 13–11).

Some basic information to include in a winery media kit would be

- Background or history sheet on the winery (one page)
- Winery fact sheet containing statistics on production, vineyard acreage, styles of wine offered, etc. (one page)
- Contact/information sheet on winery experts (owner, winemaker, vineyard manager, etc.), including at least two ways to contact each person (one page)
- Visuals (photos, slides, CD-ROM, labels, logo samples, etc.)
- Brochure
- Map and detailed directions
- Recent awards or accolades
- Question-and-answer sheet
- Wine terms (especially for nonwine media)
- Web site address for additional information

This is more than enough to fill a folder, so the vintner must be selective and choose items that best tell the winery story.

Media kits can be delivered in various ways. Meeting a journalist in person is the best way to introduce the winery for the first time. Special luncheons and tours at the winery for individual members of the media are a nice way to begin building a friendly relationship. If a reporter calls the winery directly, mailing a media kit is perfectly acceptable.

Today, it is fairly common to have media kits located on the winery Web site. This provides quick, easy access to the winery background in the case of an approaching deadline. Ensure that facts and figures are updated, easy to download, and simple to understand. Additionally, offer contact information so that reporters can talk to a live voice if they want a direct quote.

News Releases

Who, what, when where, why, and how? These are the traditional press release questions. Recently, a new one was added: Who cares? Answering these seven questions in a clear and concise manner will ultimately lead to an effective news release. Desired length for the answer to the above questions is 30 to 35 words or two sentences. Since reporters receive anywhere from 50 to several thousand press releases each day, brevity improves visibility and readability while getting the message across immediately.

With all of these stories to choose from, journalists select topics that are easy to research. Place contact information (including e-mail, phone, fax, address, etc.) at the top of the release. If possible, list a second contact in case the primary contact is unavailable when the reporter calls.

Write the release using the inverted-pyramid style, putting the most important facts first and saving the winery background for the end. Quotes and anecdotes can fill in the middle. See Figure 13–12 for an example. To reinforce brand identity, include the winery mission statement in every news release. Eliminate grammar and spelling mistakes by carefully reading each sentence from beginning to end. Journalists prefer simple words, short sentences, brief paragraphs, and colorful descriptions. Portraying the sight, smell, and feel of a wine will encourage the audience to read more. A well-written release is less than one page in length, double spaced, with 1-inch margins on all sides.

The traditional press release is changing to fit into current technology. A common way to send releases today is through e-mail, a completely different format from an 8 1/2 × 11-inch piece of paper. Information needs to be condensed and treated as a teaser to get a reporter to the Web site for additional information. On-line releases should be a maximum of 200 words in five short paragraphs. Subject lines should summarize the topic in four to five short words. Keep it short and simple; bullet points are frequently used and encouraged.

It is courteous to contact journalists to find out whether they prefer to receive releases by snail mail, e-mail, or fax. When targeting specific media, take the time to research their work and the publication. Personalizing the release enhances its

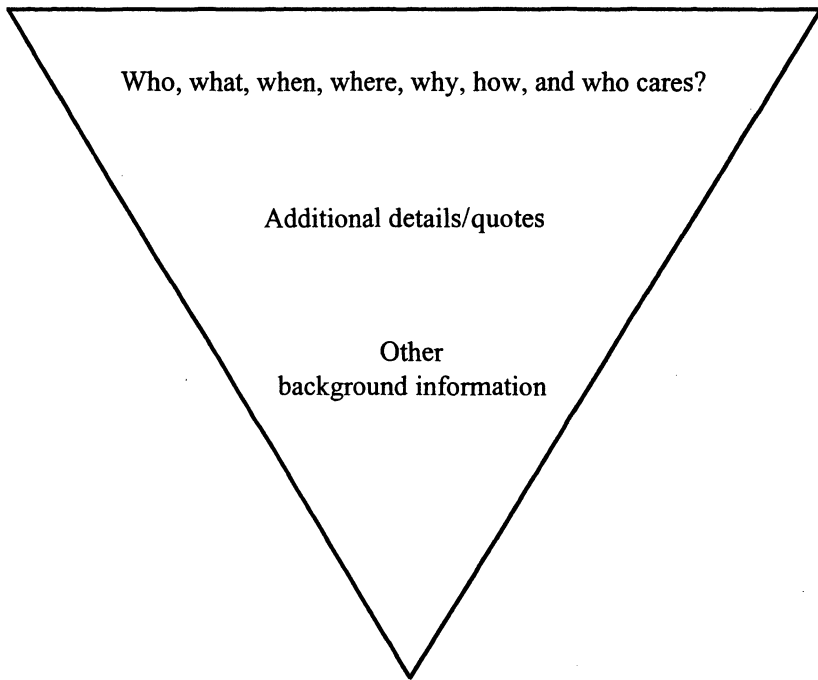


Fig. 13–12 Inverted-pyramid writing style: Easy to read and edit.

attractiveness. Once the release is sent, following up with the journalist to find out if they need additional information is helpful. At any time, it is a good idea to begin the conversation by asking, "Do you have a minute?" Posting the release on the winery Web site will allow additional media and consumers to view it.

Photographs/Visuals

Journalists write visually and also want to present visuals to the audience. There is no better way to describe a product or winery than through visuals. Slides, photographs (Figure 13–13), CD-ROMs, graphs, labels, and logos are all visuals that enhance the winery story. Additionally, they make the story stand out to the journalist, increasing chances for publication.

If using photographs, make sure they are professional quality. Many times the picture will be used directly in print, so high quality is essential. Do not use the same photographs for publicity and advertising. Most reporters do not want to feature visuals that have already been used. Finding out what format to provide the visuals helps the vintner and the journalist. Traditionally, visuals were provided through the mail on slide or print. However, today's technology allows for the immediate transfer of files through e-mail and the Internet.



Fig. 13–13 Professional photo of winery.

Interviewing

When a journalist calls, the winery needs to be prepared for an instant interview. Sometimes the interviews are over the telephone; sometimes they are live on the noon news. Whatever the situation, some basic guidelines will help with the preparation.

To begin, consider the purpose of the interview, and try to imagine the types of questions that will be asked. Work on memorizing key facts and statistics; they need to be accurate and up-to-date. Most reporters cover a wide range of subjects, so keep explanations simple. If appropriate, subtly help rephrase or redirect their questions. For example, to switch a conversation from Merlot to a new Syrah that was recently planted, one might say, “Yes, Merlot is a popular variety. However, we recently planted the region’s first Syrah vines, a new variety that produces a robust, red wine good for any occasion.” The question is redirected in a way that is not offensive but is helpful to the reporter who did not even know about the Syrah.

Vintners should seek help in preparing for interviews. Mock interviews are an excellent way to train and rehearse the two or three main winery points. Additionally, they help prepare for difficult questions like “Why did your new Syrah only win a bronze medal at a recent competition instead of a gold?”

Listen carefully to each question, and take a moment to think about it before answering. Reporters are experts at adding long pauses between questions in an

effort to keep the interviewer talking. Avoid offhand comments by allowing them to break the silence. Nothing is off the record; any statement made can be used. If a question is confusing, let the reporter know by saying, "I'm not sure I understand your question," therefore inviting them to rephrase it. One of the worst responses to make is "No comment." If you are unsure of an answer, an appropriate response would be "I am not sure of that answer, but I'll find out and get back to you."

When interviewing, always tell the truth. Audiences are willing to identify with someone that is honest, even if the truth is not always positive. Honesty is the best plan when dealing with difficult questions and topics.

Wine Trails

In the wine industry, it is often a benefit to have other wineries close by—especially when you are considering a wine trail. Viewing other local wineries as partners instead of competitors helps everyone involved. Wineries that are located close to each other can team up to form a wine trail or passport club.

Wine trails (Figure 13–14) enable wineries to share event and advertising costs, while offering consumers a winery "path" to follow on their visit. Detailed maps in joint brochures guide visitors through the region, encouraging consumers to experience all of the wineries in the area. Wine trail events are very popular and can revolve around holiday themes, shopping themes, or wine-tasting parties.

Passport clubs are similar to wine trails. Cooperative advertising and events still take place, although one extra step is added. At each winery on the trail, the consumer receives a stamp or has the brochure marked with a winery signature. Once the consumer visits a certain number of wineries on the trail, he or she receives special gifts.

Wine trails and passport clubs bring consumers into the wine region. If visitors can travel to multiple wineries in one day or weekend, they are more likely to visit that area. In the wine industry, teaming up with other local vintners often enhances sales instead of hindering them.

Wine Clubs

Everyone, including winery visitors, enjoys being considered a preferred customer or part of a special group. Wineries that offer wine clubs spotlight frequent and dedicated consumers through special events, discounts, and direct mailings. Wine clubs vary by winery from extensive benefits at larger vintners to limited benefits at smaller vintners.

Some clubs offer "members only" dinners or prereleases for new wine. Others offer special club newsletters with winery coupons. Often, there is a yearly fee to join, but membership can be based on yearly consumer purchases or be free to interested visitors. Whatever the costs or benefits, the key to creating a successful wine club is to make the club member feel special.



Fig. 13–14 Wine trail samples.

Special Events

Special events (Figure 13–15) can take on many different forms, from a wine-maker's dinner for 20 preferred customers at the winery to a statewide festival for 100,000 adventure seekers. Some events are planned in house, while others are held in city parks, museums, and exhibition halls. The bottom line for any special event is to create awareness for the winery and its wines.

Therefore, before deciding what type of program to promote or participate in, you need to set an event goal. What does the winery want to accomplish through the event? Increased visitors at the winery? Then an in-house event may be best. What about expanding consumer awareness in a big city? In this situation, a winery may choose to attend a food and wine festival in the state's capital city.

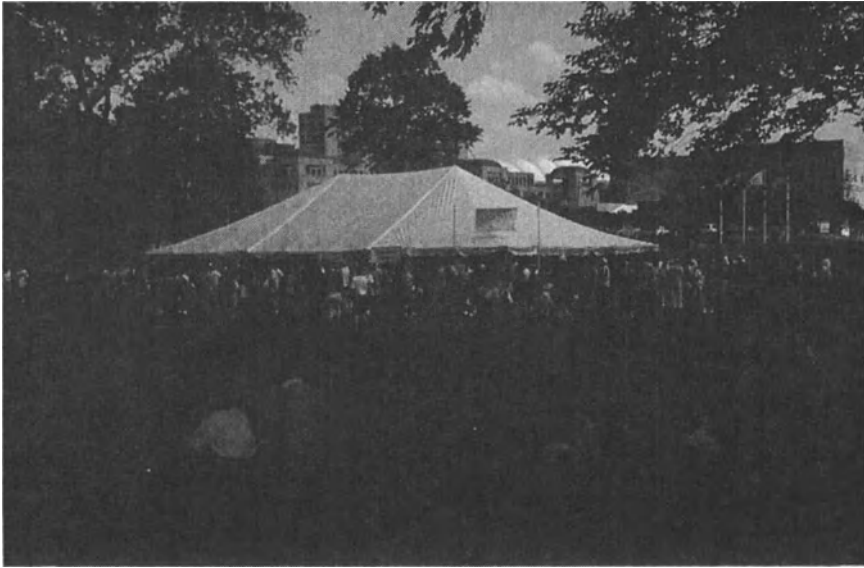


Fig. 13–15 Vintage Indiana wine and food festival.

Any event, no matter how big or small, needs to offer a consumer benefit. The goal may be to promote the awareness of a new wine called Jazzy Red. An event featuring a summer evening of outdoor jazz at the winery introduces guests to this newly released wine. However, the fact that the consumers can relax with friends in a beautiful setting over a glass of fresh-from-the-vineyard wine is the benefit. Focus on the benefit and promote it to draw more guests.

People are very busy with work, family, and friends. Well-designed events encourage consumers to forgo leisure time and attend. They also require a good amount of time and organization. Here are some items to consider when organizing an event.

- Event goal for the winery
- Perceived “want” of the audience or target market
- Event theme
- Time of year, time of day, day of week
- Competing events—check area event calendars
- Staff/volunteer availability and training
- Facilities—can they handle the proposed event?
- Security
- Sponsorship

- Admission price
- Food
- Entertainment—bands, comedians, DJ, etc.
- Licensing/permits—any special permit required? What are the laws for special events involving wineries?
- Additional insurance—any special insurance needed?
- Promotion

The above checklist will provide basic guidelines for planning an event. Expand it to include additional questions and considerations based on each particular event. Some additional thoughts include:

- Offer events that children can attend. Time with family is limited; make sure that all family members are welcome.
- Always have food available. People will stay longer and learn more about the winery.
- Promote drinking in moderation. Designated driver programs are an effective way to make sure everyone gets home safe.
- Create logo gift items with the event information. Provides guests with a take-home memory and the winery with free advertising.
- Charge a fair ticket price. Admission price needs to agree with the event's perceived value. Check other similar events to develop a standard price range.
- Select a time of year when staff are available to assist with planning and implementation. A small winery may not want to plan a large harvest event if all staff members need to be in the vineyards.

Special events organized by an outside group will not take as much time or planning. Select outside events carefully. Make sure the target audience is appropriate and the mission of the event fits one of the winery goals. Review the above checklist to formulate questions for the event committee. It is important to ensure the winery is not liable for items out of its control.

Whether in house or in the community, an effective event notifies the public through a promotional campaign. Begin with posters or announcements in the winery several months in advance to generate interest. Include the event in the winery newsletter and on the Web site. One month out, send a press release to the calendar sections of the paper, area radio, and television. If possible, send a direct mail piece or invitation to the winery mailing list. Distribute additional posters to stores and winery supporters. The week before the event, advertise through radio, television, and newspapers if possible.

The size and target audience for the event will determine the extent of promotion. A small barrel tasting may include only announcements in the winery and a direct mail piece, while the annual October Fest may include the entire campaign mentioned above. Each event needs to be analyzed individually with the aim of helping to achieve the marketing objectives.

Educational Programs

A specific type of special event is educational programming. Educational programs bring small wineries credibility with consumers and media by offering a promotional tool that is less commercial than other marketing tactics.

Adults today are interested in interactive learning sessions that they can share with other adults who have similar interests. Additionally, wine is a popular topic with adults and one that is fun to share. Vintners should take full advantage of the interest in wine by offering educational programs at the winery, community centers, or local colleges. Classes may include topics on wine appreciation, viticulture, or the history of making wine in America. Wine is a broad subject and is present throughout the world, making educational programs fairly easy to research and develop.

Industry Relations/Networking

Teaming up with wine associations, wineries, tourism groups, and other community businesses will help any vintner build a strong network of assistance. Each group can offer support in many areas, including marketing.

Wine Associations

Valuable information and marketing assistance are often found through the state or local wine association. They all have different names and offer different services. Some offer strictly promotional or scientific resources, while others specialize in marketing, enology, and viticulture. Most are created to help develop the wine and grape industry within the area. Whatever the case, contacting the association closest to the winery can help any vintner's marketing plan.

The wineries and associations generally work together to create a win-win situation for all involved. The Lodi-Woodbridge Winegrape Commission is a good example. The wineries, grape growers, and association team up to jointly promote the wine region, as shown in Figure 13–16.

Vintners need to find out how to complement the existing programs within the wine association. Are there existing marketing programs that promote the industry? What do the programs entail, and how does a winery become involved? Ask the association how it can help with winery promotion through a joint wine brochure, tastings, and other special projects. Likewise, offer assistance to the organization through wine donations (if legal) or by providing product and winery information. Building a relationship with the organization will enhance not only the winery but also the entire wine industry for the area. This, in turn, enhances the winery marketing efforts even more through additional promotion and possible cost savings.



Fig. 13-16 Lodi-Woodbridge Wine Commission information.

Wineries

Teaming up with other wineries offers a strong support system for a new vintner. Experience and advice come directly from someone who has been through the same process. Established wineries tend to view new wineries as partners, not competitors. A clustering of wineries in an area invites consumers to visit; they can see multiple wineries in one day and experience the winery region. New wineries should provide a place for other local wineries to place their brochures.

Consumer attraction, educational/production assistance, and general advice are all available from other wineries. Even before the winery is opened, it may be beneficial to make contact with established wineries to build strong industry relations.

Tourism

Many successful small wineries promote themselves as tourism destinations and attractions. Bringing the customer in the winery door to experience the atmosphere is beneficial to awareness and sales. To build this awareness among tourists, wineries should work closely with the department of tourism.

Travel in the United States has increased over 40% in the past 10 years, with most travelers enjoying short trips and weekend travel. A perfect fit for any winery owner! Target winery awareness to American tourists, and share in the over \$400 billion that they spend each year in the United States.

Most tourism departments and convention and visitor bureaus will be more than willing to work with local vintners. Since wineries attract millions of tourists each year, they enhance the state economy and local community—one of the main goals of tourism. Explore the possibilities of joint winery/tourism promotional work for consumers and media. Offer photos, background information, brochures, and other promotional items for tourism use. Volunteer to host tourism-related events at the winery, or donate wine items (if legal) that will be given to special tourism visitors.

In most cases, the opportunities for working with tourism departments are endless and can be very creative. As soon as possible, meet with a local representative to find out how to involve the winery in tourism publications, trip planners, and projects.

Local Business/Community

To develop partnerships in the community, vintners should visit restaurants, hotels, gourmet shops, and other local businesses regularly. Build relationships by inviting the business staff to the winery to enjoy the experience so they will be able to convey the feeling to their customers. Discuss the possibility of working together on projects that will be mutually beneficial to the businesses and the consumer. This may include event sponsorship or co-hosting a wine/food evening, including a hotel stay. Any joint partnership offers consumers enhanced activities and offers participating businesses advertising savings.

Additional community involvement is important for any new business. Joining local organizations or becoming involved in charitable events helps both the winery and community. Allow winery staff to participate in community programs; it will enhance the vintner/staff relationship and the winery/community relationship. In turn, the community will support the winery, forming a strong bond for future partnerships.

Wine donations (if allowed) and speaking engagements are other possible community relations. All of these will help create alliances with local organizations, thereby enhancing the winery's message.

Customer Service

Good customer service consists of making a customer feel as if he or she is the most important person to walk through the winery door all day. The trick is to make every customer feel this way. The result is happy customers who buy wine and return to the winery with friends. When people have a good time, they tell approximately three friends about it. Likewise, if they have a bad time, they tell 10 friends. Providing friendly customer service is a benefit to all involved.

Team Members

One of the most important winery assets is a positive and productive staff. Once the vintner has created a beautiful winery with superb products, it only makes sense to have a high-quality staff to promote the experience. Even if the wine wins multiple awards and the winery is peaceful and relaxing, will consumers return to the winery if they were treated poorly or ignored on their first visit? Probably not.

Developing a motivated team that believes in the winery and its story is very important. Team motivation and positive thinking need to start at the top and work down through the winery staff. If the top management doesn't respect the staff or the winery tasting room, the negative attitude will filter down to the staff very quickly.

To begin creating a winning winery team, here are a few things to consider:

- ***It is possible to educate staff on wine; it is not possible to educate them on friendliness.*** Hire friendly, positive people, then teach them about the winery.
- ***Offer competitive salaries and incentives.*** Everyone appreciates being fairly compensated for his or her work. Follow through on incentives if promised.
- ***Have an open-door policy.*** Create a positive working environment through communication.
- ***Train, train, train.*** Take the time to train employees by walking them around and teaching everyday tasks. Provide a training manual for reference.
- ***Cross-train to offer employees the opportunity to try new tasks in the winery.*** This expands knowledge and helps team members understand the duties of others.
- ***Hold team meetings on a regular basis.*** If team members feel that they contribute to the winery and its success, they will portray it to customers and friends.
- ***Offer flexible scheduling when needed.*** Employees will be more respectful and appreciative if a vintner tries to understand their needs too.
- ***Update the staff.*** The staff deal directly with the public. Keep them updated on recent news in the industry so they can field consumer questions.
- ***Plan team outings.*** Take the staff on private field trips to other wineries or treat them to a team dinner. Reward them for their work and let them know they are appreciated.

Remember, if the staff does well, the winery does well. Some visitors return to the same winery over and over again because the staff are always cheerful and

helpful and understand customer needs. Without a friendly staff that offers good customer service, a winery will not be successful in tasting room sales. For some consumers, one team member's attitude will be the only impression of the winery that they have; every attempt should be made to make it a positive attitude and impression.

Winery Tour and Tasting Room

The instant a visitor walks into the winery, he or she should be welcomed in a warm, friendly manner. The feel of the tasting room has an effect on the level of comfort for the visitors. Overall design and decorations need to coincide with the winery image. It is important that the winery be neat and clean, since first impressions last a lifetime.

Ask new visitors if they are interested in a winery tour, and let them know when the next tour will take place. Guides should be well trained to educate visitors on the winery vision while also answering general wine questions. Tour guides must be friendly people with a pleasing demeanor and must be willing to have open communication with the participants. It helps if tours follow the actual wine process, starting where the grapes are received from harvest and ending in the warehousing area.

Generally, tours last 20 to 30 minutes, with brief stops at each major processing area. If possible, the tour guide should move to the tasting room with the group and provide wine samples (Figure 13–17). Since the visitors already feel comfortable with the guide, the relationship can continue in the tasting room while the guests consider sales.

All wine bottles should be uncorked in front of visitors with a friendly touch. Checking color, clarity, and aroma/bouquet before serving ensures that the wine is in good condition. Small pours (approximately 1 ounce) are sufficient. A small tulip-shaped glass displays table wine in the best manner and is easy to clean and store. Plastic glasses may be used, but they do not always enhance the taste of the wine or the look of it.

During pouring, the neck of the bottle should not touch the edge of the glass. Pointing out one or two characteristics about each wine is enough promotion, since the wine will help sell itself when it is tasted. Between wines, offer additional information on upcoming sales, special events, and recent wine awards.

Dump bowls, for unwanted wine, need to be in clear view. Water and palate cleansers are also a nice addition for visitors. Additionally, a designated driver program is a good idea, both for community relations and for legal protection.

Anything that is free is attractive to customers. For smaller wineries, free parking, free information, free tours, and free samples are typical. It is important, however, that visitors realize the value in these offerings. Therefore, some vintners offer samples for a small fee and include an etched wine glass.

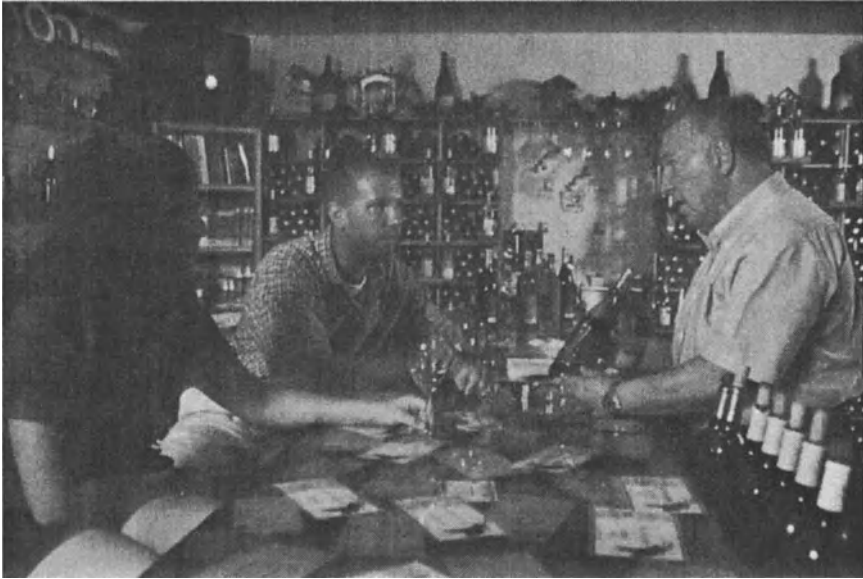


Fig. 13–17 Wine-tasting bar.

Sales

The purpose of all initial marketing efforts and hard work is to complete the sale. However, the sale is not always easy and is definitely not complete when the initial sale is complete. Marketing efforts need to continue after the first sale to bring back repeat sales and develop long-term customers.

Retail Shop

Winery retail shops and gift shops are generally incorporated into the tasting area (Figure 13–18). Some states prohibit the sale of gift items or any nonwine items in the retail facility. Therefore, it is important for all vintners to research their laws before including this revenue source in the marketing plan.

Popular gift items in winery shops include but are not limited to

- Logo items—T-shirts, wine glasses, corkscrews, and cork pullers
- Cookbooks
- Food items—ready to eat at the winery or prepackaged to take home
- Wine carriers and wine racks
- Picnic baskets
- Wine stoppers
- Wine-related jewelry



Fig. 13–18 Winery gift shop.

- Hats, aprons, and other clothing
- Local arts and crafts

The seasonality of some items will correlate to sales demand. For example, wine picnic baskets and ready-to-eat food will be more popular in the summer months, while cookbooks and gift baskets are top sellers around the holidays. Generally, price points remain under \$50. Higher priced items seem to be less popular and have a slow turnover, although having a few of these items is beneficial in case a customer is searching for a special gift.

Just as an unpopular wine needs to be replaced with a new wine that will hopefully sell better, gift items that do not sell need to be replaced. Dead shelf space offers no benefit to anyone and is actually a cost burden, since new merchandise could generate more profit. Different merchandising displays and seasonal decorative changes give the gift shop a new look for frequent visitors and are eye-catching for first-time visitors.

Offer drawings for logo items. These can be shipped to visitor homes and reinforce the winery experience. Demonstrating gift shop items as part of the wine tasting and providing employees with logo shirts, aprons, and hats are free advertising for the winery—especially when the customer leaves the store with a winery logo item.

Personalized Sales

Sales today are becoming more personal. Benefits offered to the customers (10% off case sales) mean much more than wine features (aged in oak). Successful sales involve more listening than talking. Forget the days of pushy salespeople who like to hear themselves talk. Today is the day to listen to customers and give them what they are asking for. Asking open-ended questions will help determine the customer's interests.

It is important for the winery staff to speak at the customer's level. They should be trained to explain the winery products in a simple, nonsnobbish manner. Vintners should attempt to make the winery experience nonthreatening and inviting, thereby encouraging consumers to become familiar with the products and purchase more. One of the major tasting room benefits is that customers can sample the wine before they buy. The better the product explanation and tasting experience, the higher the sales. Additionally, consumers will feel that the promotion is a valuable service rather than a sales pitch.

Wholesale Distribution

When it comes to distributing wine, wineries generally have two options: work with a wholesale distributor or become their own distributor. Before distribution begins, however, check state distribution laws. Alcohol laws and distribution systems vary extensively by state. It is very important to find out what distribution rules need to be followed.

Many states have a few core wine distributors that form the three-tier distribution system of winery to wholesaler to retailer (stores that carry wine and restaurants). Before selecting the correct wholesale distributor, a winery needs to answer some questions:

- Who are the wholesale distributors in the state?
- What do they represent (premium wine, table wine, jug wine, etc.)?
- How do they promote other wineries similar in size and product?
- What discount (off retail price) will they require? Is it negotiable?

The answers to these questions will help select a distributor that will promote the winery product in the best way possible. Generally, wholesalers sell wine to retail outlets at two thirds the winery retail price (33% off). That means the winery needs to sell to the wholesaler at a cost of less than two thirds the retail price. Winery discounts given to wholesalers range from 35% to 50% off winery retail price, depending on negotiation.

Keeping the winery product on the wholesaler/retailer priority list is another consideration. The winery may need to hire a salesperson to attend to wholesale

accounts. The salesman would work with the wholesaler and retailers to continue brand promotion and assist with developing new accounts. To save on distribution dollars, a cooperative salesperson can represent multiple small wineries if they use the same wholesaler.

The second option for distribution is for the winery to create its own wholesale system. Again, since the laws vary by state, research the options before implementing. Direct distribution allows the winery to be in control of all wholesale accounts and to build relationships with retailers through enhanced service. To begin, a winery needs to know where the product will be branded. What class of wine? What price range? After the initial product analysis, the winery can determine what restaurants and retailers fit into the distribution mold.

A major advantage of direct distribution is the ability to sell to the retail market at one third off winery retail price instead of the greater discounts required through the three-tier system. Additionally, the winery is able to offer top-quality service and attention to their retail accounts. However, direct distribution is time intensive and may be expensive.

To offer a full-service direct distribution program, a salesperson is required and generally a delivery person too. Retailers will have individual guidelines and requirements for products and displays. Some retailers may require that the winery provide wine racks, signage, and promotional items; other retailers may say that they only work with wholesale distributors.

Whether you choose a wholesale distributor or working independently, the retailer must be convinced that the winery product is worth accepting and promoting. Each system will require different techniques. Analyzing what best fits the winery and product will lead to a successful and profitable distribution system.

Repeat Sales

Once the initial sale is complete, what is the next step? To maintain contact with consumers and encourage them to purchase again. Repeat sales are very important in the overall success of the winery. Developing long-term, dedicated customers brings a vintner steady business with reliable consumers.

According to a respected marketing source, "Marketing is not over when the sale is made. That's when marketing begins." One winery objective should be to develop repeat sales. This not only results in business year after year but expands business, since those repeat customers give referrals and offer word-of-mouth advertising.

Repeat sales are often a result of maintaining contact and soliciting consumer feedback. Both are important components of the marketing plan. Before sales begin at the winery, there should be a preplanned strategy to follow up with customers. Winery visitors are the prime source of product feedback and satisfaction information. Send a survey to ask what they liked, what they disliked, how they were treated, and their overall impression of the winery and its products.

Also in the survey, thank them for visiting and offer them a special deal if they visit again. Bring them back into the winery, since they may be customers for years to come. It is nice to have a personal touch when contacting important customers, so personal signatures or notes are warmly accepted.

Caring for customers and respecting their feedback can increase referrals, word-of-mouth advertising, and winery profits, all for a small amount of money. Repeat sales depend more on time than on dollars. If a vintner takes the time to maintain contact with valuable customers and invite them back, it will pay off for years to come.

CONCLUSION

Marketing is an ongoing process, one that never ends as long as the winery is in business. Image development begins before the winery location is even selected and continues through every aspect of decisions about the winery and its products. Sometimes marketing is considered most important after the initial sale is made. This is when repeat customers and long-term customer relationships are developed.

For consumers to purchase a product, they need to be exposed to a product seven to nine times in various formats. Effective market research tells a vintner who those potential consumers will be. The marketing plan plots out the strategy and multiple methods that will best reach those consumers. Creating an action calendar and marketing budget puts the plan to work in an affordable manner.

Overall, the marketing tactics and techniques create a multitude of promotional options for wineries that want to sell a product successfully. Some key wine marketing words to remember are *consistency*, *long-term commitment*, *continuous revisions*, and *appreciation* for those involved in all aspects of the winery promotion, including customers, staff, media partners, and other winery owners.

APPENDIX A

SUGGESTED RESOURCES

CATALOGS

Comprehensive catalogs of equipment and supplies for small start-up wineries are offered by the following vendors:

Presque Isle Wine Cellars
9440 Buffalo Road, Northeast, PA 16428
(814) 725-1314

The Compleat Winemaker
955 Vintage Avenue, St. Helena, CA 94574
(707) 963-9681

BUYER'S GUIDES

A complete list of vineyard and winery sources can be found in annual buyer's guides published by the following magazines:

Vineyard and Winery Management
103 Third Street, Watkins Glen, NY 14891
(607) 535-7133

Wines and Vines

1800 Lincoln Avenue, San Rafael, CA 94901
(415) 453-9700

VITICULTURE MATERIAL AND EQUIPMENT

American Nursery
Route 1, Box 8781, Madison, VA 22727
(540) 948-5064 or (707) 942-4100

Bradley Ag Supply, Inc.
917 Bardshar Road, Sandusky, OH 44870
(419) 734-6137

Central Valley Builders Ag Division
1100 Vintage Avenue, St. Helena, CA 94574
(707) 963-3622

Concord Nurseries, Inc.
10175 Mile Block Road, North Collins, NY 14111
(800) 223-2211 or (716) 337-2485

Grafted Grapevine Nursery
2399 Wheat Road, Clifton Springs, NY 14432
(315) 462-3288

Grower's Supply Center
2415 Harford Road, Falston, MD 21047
(410) 931-3111 or (410) 333-2856

Jim's Supply Co., Inc.
P.O. Box 668, Bakersfield, CA 93302
(800) 423-8016 or (661) 324-6514

King Estate Vineyards
80854 Territorial Road, Eugene, OR 97405
(541) 942-9874

Orchard Valley Supply
734 Main Street, Fawn Grove, PA 17321
(717) 382-4612

Sonoma Grapevines, Inc.
P.O. Box 293, Fulton, CA 95439
(707) 542-5510

LABORATORY MATERIALS, EQUIPMENT, AND CONSULTANTS

All-World Scientific
5515 186th Place SW, Lynnwood, WA 98037
(800) 289-6753 or (425) 672-4228

The Compleat Winemaker
955 Vintage Avenue, St. Helena, CA 94574
(707) 963-9681

Lallemand
P.O. Box 5512, Petaluma, CA 94955
(707) 526-9809

Presque Isle Wine Cellars
9440 Buffalo Road, North East, PA 16428
(814) 725-1314

R & D Glass Products and Equipment, Inc.
1808 Harmon Street, Berkeley, CA 94703
(510) 547-6464

Randex Laboratories
70 Dorman Street, #3A, San Francisco, CA 94124
(888) 972-6369 or (415) 643-4460

Scott Laboratories, Inc.
2220 Pine View Way, P.O. Box 4559, Petaluma, CA, 94955
(707) 765-6666

Vinquiry
7795 Bell Road, Windsor, CA 95492
(707) 838-6312

Westover Scientific
P.O. Box 1665, Woodinville, WA 98072
(800) 304-3202

The Wine Lab
477 Walnut Street, Napa, CA 94559
(707) 224-7903

NEW EQUIPMENT

Aftak, Inc.
740 Driving Park, Rochester, NY 14613
(800) 240-5366 and (716) 458-7550

The Compleat Winemaker
955 Vintage Avenue, St. Helena, CA 94574
(707) 963-9681

ConeTech, Inc.
P.O. Box 611, Graton, CA 95444
(707) 829-6200

Criveller Co.
6935 Oakwood Drive, Niagara Falls, ONT, CANADA L2E 6S5
(905) 357-2930

Custom Metalcraft, Inc.
P.O. Box 10587, Springfield, MO 65808
(417) 864-7575

Euro Machines, Inc.
P.O. Box 843, Culpepper, VA 22701
(540) 825-5700

KLM Machines, Inc.
47 W. Steuben Street, Bath, NY 14810
(607) 776-4193

Presque Isle Wine Cellars
9440 Buffalo Road, North East, PA 16428
(814) 725-1314

Prospero Equipment Corp.
123 Castleton Street, Pleasantville, NY 10570
(914) 769-6252

Scott Laboratories, Inc.
P.O. Box 4559, Petaluma, CA 94955
(707) 765-6666

NEW TANKAGE

American Tank Company, Inc.
200 American Way, P.O. Box 340, Windsor, CA 95492
(800) 655-9100

Canton Cooperage Co.
365 S. Woodlawn Avenue, Lebanon, KY 40033
(800) 692-9888

Enerfab

4955 Spring Grove Avenue, Cincinnati, OH 45232
(513) 641-0500

Mendocino Cooperage

12901 East Side Road, Hopland, CA 95449
(707) 744-7425

Mutual Stamping and Manufacturing

P.O. Box 5060, Milford, CT 06460
(800) 735-3933

Paul Mueller Company

P.O. Box 828, Springfield, MO 65801
(800) 683-5537 or (417) 831-3000

PBC International

185 Route 17 North, Mahwah, NJ 07456
(800) 514-2739

Quality Stainless Tanks

510 Caletti Avenue, Windsor, CA 95492
(877) 598-0672 or (707) 837-2721

Santa Rosa Stainless Steel

P.O. Box 518, Santa Rosa, CA 95402
(707) 544-7777

Seguin Moreau, USA

151 Camino Dorado, Napa, CA 94558
(707) 252-3408

USED EQUIPMENT AND TANKAGE

Above Scale Company

2161 Patton Avenue, Napa, CA 94559
(800) 200-9970 or (707) 226-1861

Associated Winery Systems, Inc.

7787 Bell Road, Windsor, CA 95492
(800) 953-3736 or (707) 838-2812

Barrel Builders, Inc.

P.O. Box 268, St. Helena, CA 94574
(707) 942-4291

Euro Machines, Inc.

P.O. Box 843, Culpepper, VA 22701
(540) 825-5700

Idiaz Services, Inc.

4023 Coffey Lane, Santa Rosa, CA 95403

(707) 545-1981

RLS Equipment Co, Inc.

P.O. Box 282, Egg Harbor, NJ 08215

(800) 527-0197 or (609) 965-0074

Redwood Equipment, Inc.

4137 South Moorland Avenuethe, Santa Rosa, CA 95407

(707) 586-1790

Rowlands Sales

Butler Industrial Park, P.O. Box 552, Hazleton, PA 18201

(800) 582-6388 or (717) 455-5813

Transition Equipment Co.

P.O. Box 108, Rio Nido, CA 95471

(707) 869-0396

VBS Industries, Inc.

800 East McGlincey Lane, Campbell, CA 95008

(408) 371-3303

World Cooperage

2557 Napa Valley Corporate Drive, Suite D, Napa, CA 94558

(707) 255-5900

GRAPES, MUST, AND JUICE

Herb Barber and Sons Food Brokers, Inc.

7193 East Main Road, Westfield, NY

(800) 953-3736 or (716) 326-4692

Joseph W. Ciatti Co.

1101 Fifth Avenue, #170, San Rafael, CA 94901

(415) 458-5150

Fulkerson's Winery and Juice Plant

5596 Route 14, Dundee, NY 14837

(607) 243-7883

G. W. Kent, Inc.

3667 Morgan Road, Ann Arbor, MI 48108

(800) 333-4288

Mazza Vineyards

11815 East Lake Road, North East, PA 16428

(800) 796-9463 or (814) 725-8695

Presque Isle Wine Cellars

9440 Buffalo Road, North East, PA 16428

(814) 725-1314

Serra Mission Winery

852 Hanley Industrial Court, St. Louis, MO 63144

(314) 962-4600

Vinotheque, USA

420 Northboro Road, Marlboro, MA 07152

(800) 481-8466

Walker's Fruit Basket/Press House

2860 Route 39, Forestville, NY 14062

(716) 679-1292

Wild Rose Vineyards

1924 Colette, Lodi, CA 95242

(209) 339-0102

COMPUTER SOFTWARE**Agrisystems, Inc.**

7855 South Elizabeth Way, Littleton, CO 80122

(303) 267-0024

Agri-Synthesis, Inc.

P.O. Box 10007, Napa, CA 95481

(707) 258-9300

Beer and Wine Services, Inc.

211 Wapoo, Suite 202, Calistoga, CA 94515

(707) 942-6911

Codeck Manufacturing

P.O. Box 2940, Sausalito, CA 94966

(800) 700-1341

Computer Systems Planning

P.O. Box 1020, South Cle Elum, WA 98943

(509) 674-7086

Ladybug Products, Inc.

P.O. Box 504, Athol, ID 83801

(800) 456-6371

Logix Controls

10518 NE 68th St, Kirkland, WA 98033

(425) 828-4149

Modular Information Systems

3160 Crow Canyon Place, Suite 100, San Ramon, CA 94583
(925) 244-5930

POS Resources, Inc.

3209 Ramos Circle, Sacramento, CA 95827
(916) 369-7990

Smart Tools, Inc.

13010 Boonville Road, Boonville, CA 95415
(707) 462-9734

WINEMAKING MATERIALS**APM Europack**

441 Industrial Way, Benicia, CA 94510
(800) 487-7555 or (707) 745-8060

Barrel Builders, Inc.

P.O. Box 268, St. Helena, CA 94574
(707) 942-4291

California Glass Company

155 98th Avenue, Oakland, CA 94603
(510) 635-7700

Cellulo Company

2949 East Townsend Avenue, Fresno, CA 93721
(209) 485-2692

Demptos Glass Corp.

650 Airpark Road, Suite A, Napa, CA 94558
(707) 224-1000

Epic Products, Inc.

17370 Mt. Hermann Street, Fountain Valley, CA 92708
(800) 548-9791 or (714) 641-8217

Lafitte Cork and Capsule

45 Executive Court, Napa, CA 94558
(800) 343-2675 or (707) 258-2675

Presque Isle Wine Cellars

9440 Buffalo Road, North East, PA 16428
(814) 725-1314

Scott Laboratories, Inc.

P.O. Box 4559, Petaluma, CA 94955
(707) 765-6666

The Wine Lab
477 Walnut Street and, Napa, CA 94559
(707) 224-7903

WINERY TASTING ROOM SUPPLIES

Chris's Stuff
706 Almar Avenue, Pacific Palisades, CA 90272
(800) 218-7085 or (310) 390-0701

Crosby & Baker, Ltd.
999 Main Road, Westport, MA 02790
(800) 999-2440

DC Enterprises
425 Queens Lane, San Jose, CA 95112
(800) 858-8437 or (408) 453-7843

Epic Products, Inc.
17370 Mount Hermann Street, Fountain Valley, CA 92708
(800) 548-9791 or (714) 641-8194

Glass Tech
5695 Southwest Arctic Drive, Beaverton, OR 97005
(503) 646-3987

Grape Impressions
995 Vintage Avenue, Suite 10, St. Helena, CA 94574
(707) 963-1177

Grape Motif Collection
10 Monument Neck Road, Buzzards Bay, MA 02532
(508) 759-9019

Private Reserve
P.O. Box 2841, Napa, CA 94558
(707) 252-4258

Wine Appreciation Guild
360 Swift Avenue, Suite 34, South San Francisco, CA 94080
(800) 231-9463

Wine Glasses
112 Pine Street, San Anselmo, CA 94960
(800) 8060677 or (415) 454-0660

PERIODICALS*American Vineyard*

P.O. Box 626, Clovis, CA 93613
(559) 298-6675

American Wine Society Journal

361 Eldridge Road, Aurora, OH 44202
(330) 562-7188

Fruit Winemaking Quarterly

4330 Gunderson Court, Sebastopol, CA 95472
(707) 823-4711

Practical Winery and Vineyard

15 Grande Paseo, San Rafael, CA 94903
(415) 479-5819

Vineyard and Winery Management

103 3rd Street, P.O. Box 231, Watkins Glen, NY 14891
(607) 535-7133

Wine Business Insider

110 West Napa Street, Sonoma, CA 95476
(707) 939-0822

Wine Country Classifieds

P.O. Box 389, St. Helena, CA 94574
(707) 967-5155

Wine East

620 North Pine Street, Lancaster, PA 17603
(717) 393-0943

Wines and Vines

1800 Lincoln Avenue, San Rafael, CA 94901
(415) 453-9700

Wine Spectator

387 Park Avenue South, New York, NY 10016
(212) 684-4224

APPENDIX B

ANALYTICAL PROCEDURES

MICROSCOPY

The microscope (Figure B-1) is a very delicate precision instrument and should be placed only in the hands of an operator who will accept responsibility for properly using and maintaining it. Given that, the operation of this device can be rather simple and highly rewarding.

Certain general operational instructions should be observed in using any microscope:

1. Moving parts of the microscope should never be forced. They should move freely. If something binds or becomes inoperable, the microscope should be serviced by a qualified person.
2. The lenses in the objectives should never be touched with anything but appropriate cleaning materials. Other items may deposit film or oils or even scratch the precision surfaces.
3. The objectives should never touch slides, cover slips, or the stage.
4. Specimens should be examined first with low-power objective, and then magnification increased as necessary.

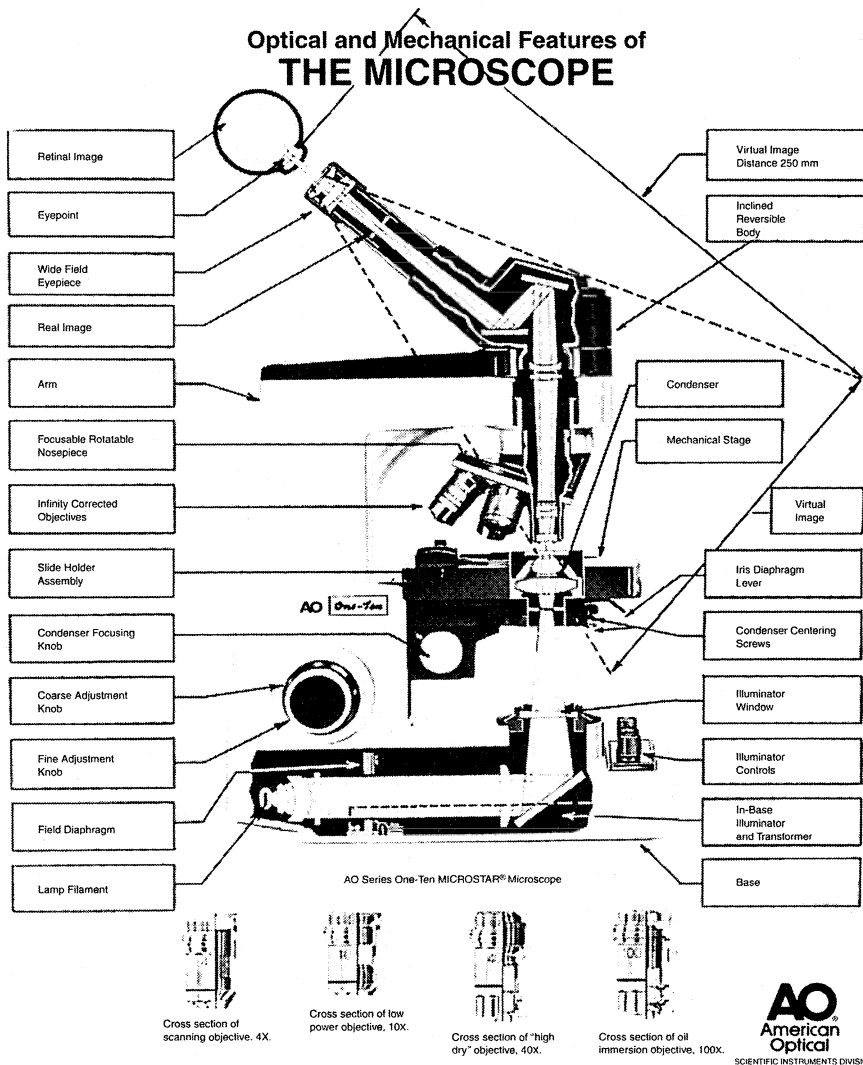


Fig. B-1 Light microscope.
Courtesy: American Optical.

5. The body of the microscope should never be adjusted downward with the coarse knob while the operator is looking through the eyepiece. At eye level, the objective can be brought close to the cover slip and then adjusted for focus upward while one is looking through the eyepiece. This will aid in preventing the objective from touching the slide and perhaps ruining the lens.

6. The microscope should be stored with the low-power (10X) objective in the focusing position.
7. The microscope should be carried only by the arm, ensuring that it is maintained in the upright position.
8. Eye strain can be minimal if both eyes are kept open when viewing through the eyepiece. Squinting causes the eyes to tire quickly and, apart from the discomfort, much can be lost or overlooked in specimen observation.
9. One should become totally familiar with a new or different microscope. It may help to "dummy" with the instrument without a slide in position on the stage to become accustomed to the feel and position of adjustment locations and operation.
10. Securing the microscope from operation should be preceded by a careful dusting and lubrication, as may be instructed by the owner's manual, if available, or as may be advised by a qualified service person. Cover with dust cover jacket and store in a wooden storage box or cupboard.

The following procedure is provided, although if the manufacturer's operating instructions are available, they should take precedence:

1. Open the aperture diaphragm of the condenser.
2. Turn on illuminator or adjust mirror to an external light source.
3. Turn coarse adjustment knob to raise nosepiece and objectives sufficiently for insertion of a glass slide.
4. Place properly prepared specimen slide and cover slip on the stage.
5. Position the low-power objective over the slide, and lower the body with the coarse adjustment knob until the objective is about 1/8 inch from the slide. It is important that the objective does not actually touch the slide to avoid scratching the objective and causing other damage to the instrument. This operation is best accomplished by keeping the eye level with the stage.
6. Slowly elevate the body with the coarse adjustment knob while looking through the eyepiece. Once the image appears in approximate focus, stop adjustment. Adjust only upward with the adjustment knobs when viewing through the eyepiece. Adjust downward only when the eye is level with the stage.
7. Fine-tune focus by adjustment with the fine adjustment knob. Further adjustment may be necessary with the light source to achieve the optimum image.
8. Adjust the diaphragm by moving the iris diaphragm level to the desired opening while viewing the image through the eyepiece.
9. Place the desired object or specimen in the exact center of the image by manipulation of the stage.
10. Turn the coarse adjustment so as to move the body upward, allowing positioning of the 40X objective over the slide. Be careful not to move the slide on the stage, or the centered image will be lost.
11. With the eye at stage level, lower the body with coarse adjustment knob about 1/8 inch from the cover slip.

12. Repeat Step 6.
13. Repeat Step 7.
14. Repeat Step 8.
15. Repeat Step 9.
16. Turn coarse adjustment knob so as to move the body upward, allowing positioning of the 100X oil-immersion objective over the slide. Be careful not to move the slide on the stage, or the centered image will be lost.
17. With the eye at stage level, lower the body until the objective is approximately 1/4 inch from the slide.
18. With extreme patience and care, place one drop immersion oil on the cover slip just below the objective, again being sure not to move the slide.
19. With the eye at stage level, lower the body with the coarse adjustment knob until the lens of the 100X objective comes in contact with the oil. Continue to lower the body very carefully until the objective nearly touches the cover slip. Do not, however, allow the objective to actually come in contact with the slide.
20. Viewing through the eyepiece, slowly elevate the body by turning the fine adjustment knob until a focus is achieved.
21. Make the adjustments of light source and diaphragm that are necessary for the optimum image.
22. When observation is finished, raise the body so that the objective is about 1 inch from the cover slip, and return the 10X objective to the focusing position.
23. The oil-immersion objective should be polished dry with dry lens paper after every use. The other lower power objectives should never be used with the oil-immersion technique. If, however, some oil should come in contact with them, they should be immediately cleaned with lens paper moistened with commercial window cleaner and then polished with dry lens paper.
24. Remove the slide from the stage. If any oil spills or other liquids are on the stage, they should be cleaned with a cheesecloth moistened with commercial window cleaner and then dried with an untreated cheesecloth.

METHYLENE BLUE STAIN PROCEDURE

Microscopic observations of bacteria and yeast can be enhanced by the use of microbial stains. Methylene blue staining is a simple procedure that stains all cells blue when used as a dry mount and has the added advantage of acting as a viability stain when added to a culture and observed as a wet mount.

Apparatus and Reagents

- Methylene blue solution (0.3 g of methylene blue dissolved in 30 mL of 95% ethyl alcohol and 100 mL of citrate buffer, pH 4.6)

- Citrate buffer (Dissolve 2.1 g of NaH₂-citrate and 2.4 g of NaH-citrate in 50 mL of deionized water in a 100-mL volumetric flask. Bring final volume to 100 mL with deionized water.)
- Slide and cover slip

Procedure

Wet Mount

1. Place a drop of methylene blue solution on a glass slide, and mix in a drop of wine or culture.
2. Cover mixture with a cover slip, and observe under microscope immediately. Clear yeast cells are viable, and blue stained cells are dead. The stain is toxic to yeast, and all cells will appear dead (blue) after several minutes. All bacteria will appear dark blue.

Dry Mount

1. Spread a drop of wine or culture on a glass slide, and allow to air-dry.
2. Fix culture to the slide by quickly passing it through a flame several times until just too hot to touch.
3. Flood slide with methylene blue solution, and hold for 1 to 2 minutes before rinsing gently with water.
4. Blot with bibulous paper, and allow to air-dry.
5. Microorganisms will appear blue. This procedure makes yeast and bacteria much easier to differentiate from miscellaneous debris in the wine or culture.

GRAM STAIN PROCEDURE

Bacteria stained by the Gram stain procedure (Figure B-2) exhibit two different staining characteristics: Gram-positive organisms hold the crystal violet stain in the cell wall and appear blue, but Gram-negative organisms are decolorized when rinsed with the alcohol solvent, and they show the color of the red counterstain, safranin. The technique is effective with young cultures, but results vary when the organisms are no longer actively replicating. Gram stain of wine bacteria in wine sediment is not dependable.

Apparatus and Reagents

- Crystal violet solution (10 mL of gentian violet, saturated alcohol solution, and 40 mL of ammonium oxalate 1.0% aqueous solution)
- Gram's iodine solution (1 g of iodine crystals, 2 g of potassium iodide, and 300 mL of distilled or deionized water)
- Safranin solution, saturated aqueous solution
- Bunsen burner

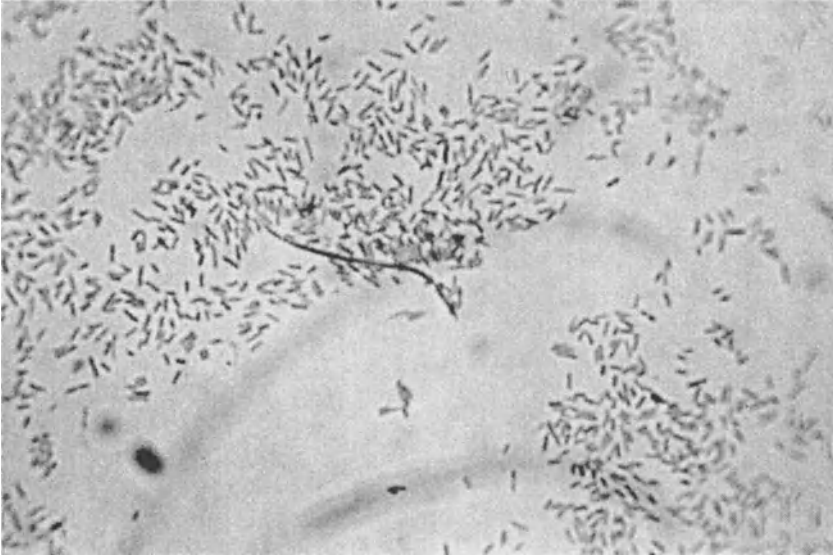


Fig. B-2 Gram stain of bacteria cells.

- Slide and cover slip
- Distilled or deionized water
- 95% alcohol
- Bibulous paper

Procedure

1. Place a drop of the medium or culture on the center of a clean slide, spreading the material over an area of about 1/2 inch squared.
2. Allow the sample material to dry, and then fix by quickly passing the slide over the burner flame several times. Do not allow the slide to become too hot so as to avoid burning the sample.
3. Flood slide with crystal violet, and hold for 30 seconds. Rinse with distilled or deionized water.
4. Cover the sample with Gram's iodine, and hold for 30 seconds. Rinse with distilled or deionized water.
5. Decolorize in 95% alcohol for 20 to 30 seconds. Rinse with distilled or deionized water.
6. Counterstain with safranin, allowing a reaction time of about 10 seconds. Rinse with distilled or deionized water.
7. Dry with bibulous paper.
8. Observe under microscope. Gram-positive organisms will stain purple-black. Gram-negative organisms will stain pink or red.

MALOLACTIC FERMENTATION DETERMINATION BY PAPER CHROMATOGRAPHY*

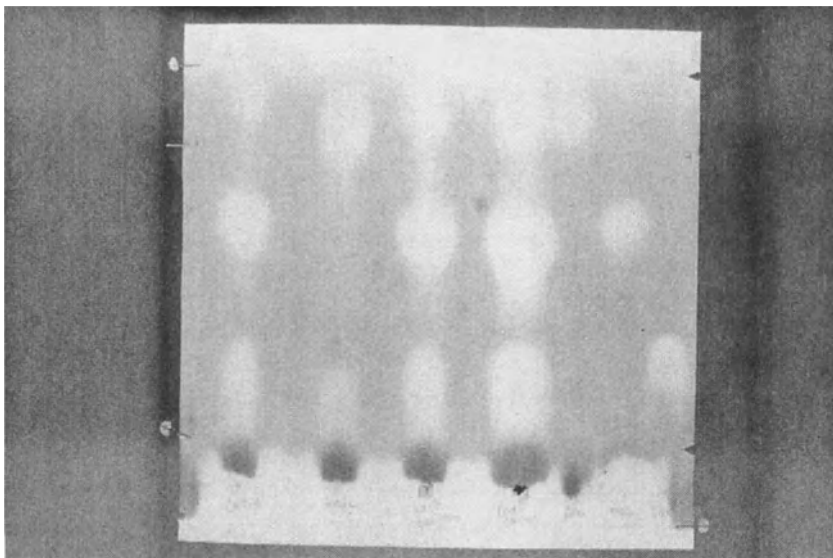
The use of paper chromatogram (Figure B-3) in the detection of malolactic fermentation distinguishes qualitatively between malic acid and lactic acids existing in the wine sample analyzed. The absence of malic acid is an indication that this bacterial fermentation has taken place. The formation of lactic acid in itself is not valid evidence, as this may result from other microbial activity.

This analytical method determines the ratio of distance from a "baseline" to an acid "spot" called the R_f upon each paper chromatogram by the use of standard acid solutions. Once these standards are determined, a comparison can be made with a chromatogram run with wine samples.

Apparatus and Reagents

- Chromatographic-grade filter paper, cut into 20 × 30 cm rectangles
- 1.2 × 75-mm micropipets
- Separatory funnel

Fig. B-3 Paper chromatogram.



* Adapted from R.E. Kunkee. In "Malo-Lactic Fermentation and Winemaking," Part 7 of *Chemistry of Winemaking*, ed. A.D. Webb. Washington, DC: American Chemical Society, 1974.

- 1-gallon wide-mouth glass jars with covers
- Solvent constituents:
 - a. 100 mL of distilled water
 - b. 100 mL of n-butyl alcohol
 - c. 10.7 mL of concentrated formic acid
 - d. 15 mL of 1% water-soluble bromcresol green
- Standard solutions:
 - a. 0.3% tartaric acid
 - b. 0.3% citric acid
 - c. 0.3% malic acid
 - d. 0.3% lactic acid
 - e. 0.3% succinic acid
 - f. 0.3% fumaric acid

Procedure

1. Draw a pencil line (baseline) approximately 2.5 cm parallel to the long edge of the filter paper. Spot the wine sample (or standard) from the micropipet on this line about 2.5 cm apart. Make each spot four times (allowed to dry in between) at a volume of 10 μ L from the micropipet.
2. Make a cylinder from the paper by stapling the short ends, without overlapping.
3. Place solvent constituents in the separatory funnel and mix. After about 20 minutes, draw off and discard the lower aqueous phase.
4. Transfer 70 mL of the upper layer into the wide-mouth jar. Stand the paper cylinder up in the jar such that the spotted edge (baseline) is in the solvent, and cover the jar.
5. The chromatogram should develop in about 6 to 8 hours.
6. Remove chromatogram (now yellow in color), and store in a ventilated area until dry and the formic acid has vaporized, leaving a blue-green background with yellow spots of acid having the following approximate R_f values:
 - tartaric acid 0.28
 - citric acid 0.45
 - malic acid 0.51
 - lactic acid 0.78
 - succinic acid 0.78
 - fumaric acid 0.91
7. Standards and wine samples should be run simultaneously or one immediately following the other.
8. Solvent may be used repeatedly if care is taken to remove any aqueous layer that may have separated after each run.

BOTTLE STERILITY

Select a monitor kit such as the Millipore 55 PLUS, which is designed to enumerate living or viable organisms in wine (Figure B-4). These systems allow for easy use

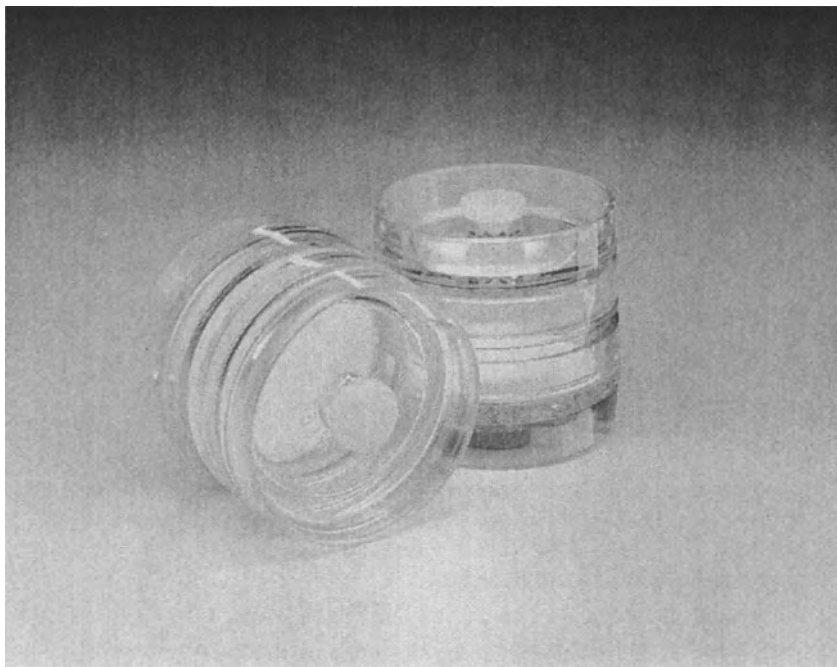


Fig. B-4 Microbiological analysis monitor, 37 mm.

Courtesy: Millipore Corporation, Bedford, MA.

and minimal contamination. Monitors function by filtering a sample of juice or wine, trapping any yeasts or bacteria on the surface of the filter membrane. A nutrient medium is applied to that membrane and then incubated at 75°F for 7 to 10 days. Organisms will grow into “colonies” that can be enumerated and identified.

Carefully wash the exterior of the bottle to be evaluated in order to minimize the potential for contamination. The neck, cork, and cork puller should each be rinsed with 70% alcohol solution. Shake the bottle thoroughly just before opening, and remove the cork gently, keeping the cork puller from forcing particles of cork into the wine.

Extensive directions are provided by the manufacturer for each type of monitor unit. Typically, 25 to 100 mL is poured from the bottle into the funnel and filtered through the membrane. This is followed by filtering sterile water (which can be easily made by boiling several minutes in a microwave and then cooling in a refrigerator) to flush the membrane free of any residual wine. Any wine left in the membrane may contain sufficient sulfur dioxide and/or sorbates to inhibit viable organisms from forming colonies.

Apply the contents of the medium vial through the port at the bottom of the filter. After sealing up the port with its cap, incubate the monitor unit at 65 to 75°F.

Since mold can grow rapidly and obscure the entire filter membrane, observations should be made daily to check for growth.

In general, yeast colonies are smooth, shiny, cream-colored mounds, 2 to 4 mm in diameter, that grow in 2 to 5 days. Bacteria will take 5 to 10 days to form tiny colonies, sometimes nearly transparent and requiring magnification to identify.

The absence of colonies on the filter after 10 days provides confidence, not a guarantee, that the bottled wine is microbiologically stable.

The presence of a few yeast and/or bacteria colonies suggests that the bottled wine should be held for several weeks before being fully released into marketable cased-goods inventory. Additional bottles should be then be taken from various case locations and each run for bottle sterility again. If there is no apparent gas formation, haziness, or other indication of fermentation in the bottle and the bottle sterility test remains the same as before, the wine is probably stable but should undergo a sensory examination at least once per week for the next several weeks or so.

If more than 10 colonies are counted upon the incubated filter, immediate action should be taken before deterioration goes any further. This may mean making the difficult decision to empty all the bottles back into a tank and taking appropriate action and treatments to surmount the situation.

Molds will not grow in the relatively high alcohol levels found in wine. The presence of mold colonies or other nonspoilage organisms indicates contamination from poor technique. Read the directions of the monitor again, and resume analysis from the beginning.

WINERY SANITATION

The first step in any microbiological procedure is to minimize contamination sources from the laboratory itself. The technician should wear a clean lab coat laundered in hot water and hypochlorite in order to cover clothing that has been exposed to countless other organisms, both airborne and waterborne. Hair should be covered and long hair done up. Hands should be washed with soap, then sprayed or rinsed with 70% ethanol solution.

Swab Test

Yeast and mold contamination can be analyzed by the use of swab test kits. If the directions supplied by the swab test manufacturer are followed carefully, there should be accurate estimation of contamination on areas such as tank interiors, valves, corks, and filler tubes. The methodology involves rubbing the area in question with a swab, transferring the swab to a buffer solution to suspend any organisms that have been picked up, then dipping a medium-impregnated filter membrane into the buffer. The filter membrane is shaken to remove excess buffer, then replaced in its sterile housing and incubated, filter surface down, for 5 to 7 days at 75°F.

Each viable organism that lands on the filter will begin to multiply and grow to form colonies. Look for this growth after 48 hours, as mold colonies may grow fuzzy and very large, obscuring the entire filter after a few days and making it impossible to evaluate other growth.

Yeasts will produce visible small, shiny, creamy-white mounds in 3 to 5 days. Molds will begin as very small, rough, gray, green, or black colonies that become larger and fuzzier with time. Most bacteria will not grow on this medium, although occasionally certain species of bacilli will grow to form yeastlike colonies or will appear as a slimy, flat, rapidly spreading white or nearly transparent film on the test membrane (Figure B-5).

Counting the number of colonies of yeast and/or mold provides an estimate of how many living organisms were on the surface tested. If the results of the swab culture show more than several colonies of yeast or mold, a better sanitation program is necessary. Obviously, the absence of yeast or mold growth is an indication, not a guarantee, that the area is free of yeast and mold, and probably bacteria as well.

pH METER

A pH probe consists of a hydrogen ion-sensitive electrode and a reference electrode. Both produce a voltage when in contact with hydrogen ions. The value of that voltage is a linear function of the pH (Figure B-6).

The following general procedure is provided. However, when the manufacturer's operating instructions are available, they should take precedence in the use and care of the pH meter.

Apparatus and Reagents

- pH meter with electrode
- 50-mL poly or glass beakers
- Thermometer
- Distilled-water wash bottle
- pH 4.0 buffer solution

Procedure

1. Turn pH meter on and allow several minutes of "warm-up" in order to become stable.
2. Pour about 20 mL of buffer solution into a clean beaker.
3. Immerse the electrode into the buffer solution, and gently swirl the beaker (be careful not to damage electrode).
4. Ensure that the buffer solution and wine sample(s) are the same temperature.
5. Adjust the temperature control knob of pH meter to the buffer solution temperature, and adjust the reading of pH meter to exactly 4.0.

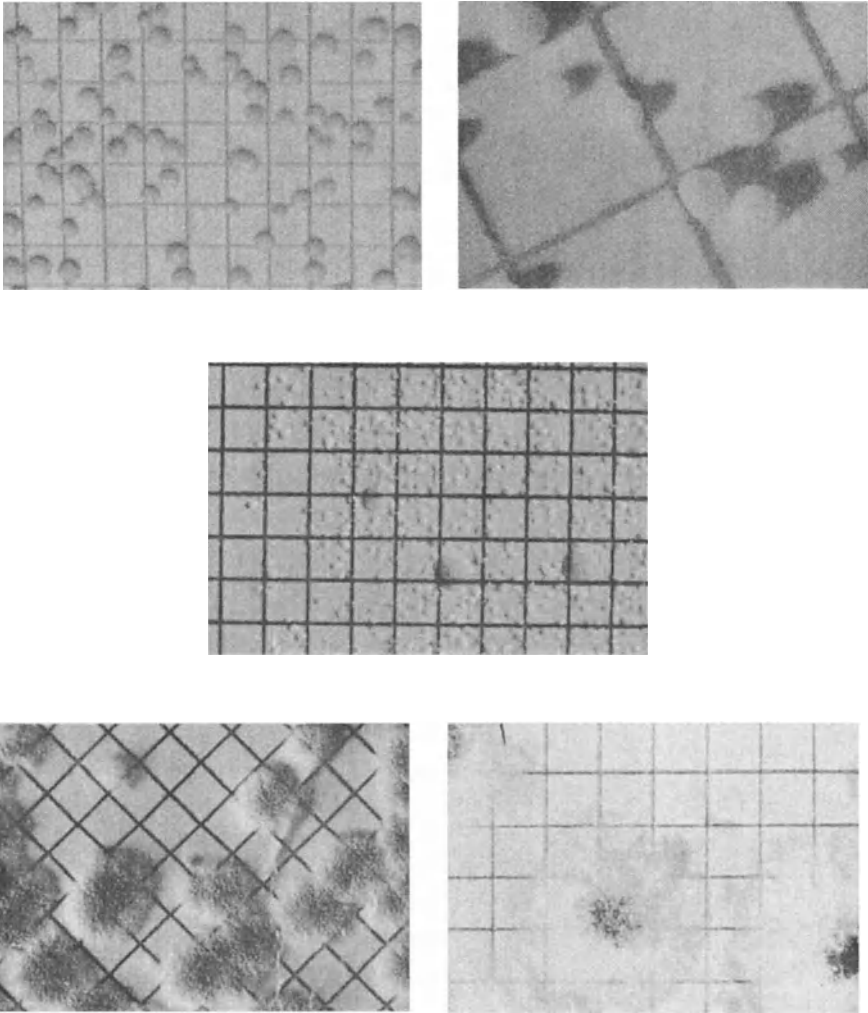


Fig. B-5 Colonies on swab test filter medium.
Courtesy: Millipore Corporation, Bedford, MA.

6. Remove the buffer solution, and rinse the electrode with some of the wine to be analyzed.
7. Pour about 20 mL of the wine sample into a clean beaker.
8. Immerse the electrode into the wine sample, and gently swirl the beaker (be careful not to damage electrode).
9. Take pH reading when stabilized, and record result.

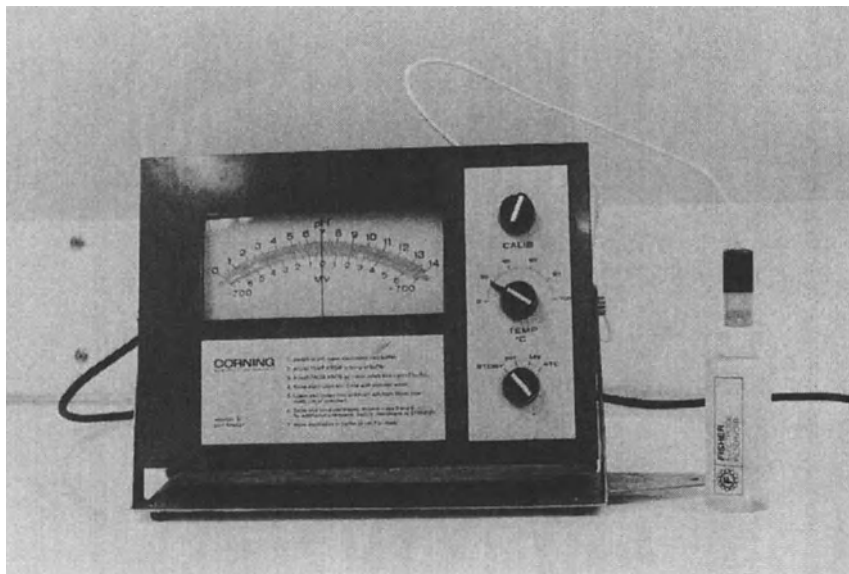


Fig. B-6 pH meter in use.

BRIX-BALLING BY HYDROMETER

Brix is a measurement of dissolved solids in a sample of juice or must being analyzed. If there is any alcohol in the sample, the test is then properly called a Balling. While dissolved solids increase the specific gravity (density) of a liquid, alcohol decreases specific gravity. Consequently, the combination of dissolved solids and alcohol is measured as viscosity ("mouth-feel" or "body") in the Balling test. Hydrometers can be used for both Brix and Balling analyses (Figure B-7).

Apparatus

- Brix-Balling hydrometer in the range of juice or wine to be analyzed
- Hydrometer jar or cylinder
- Thermometer

Procedure

1. If the sample has been taken from a fermenter and contains carbon dioxide gas, the gas should be removed by careful agitation or filtration.
2. Adjust sample temperature to that required as indicated on the stem of the hydrometer.

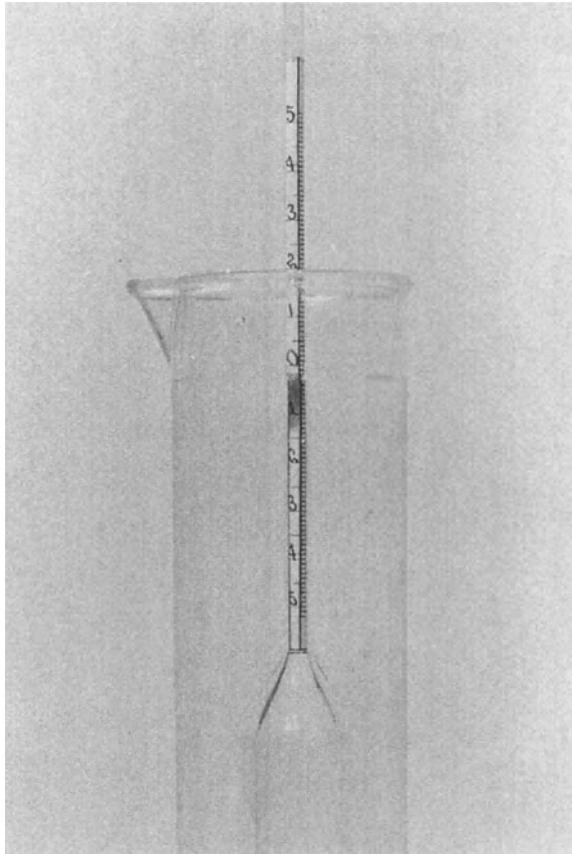


Fig. B-7 Brix-balling hydrometer test.

3. Pour about 50 mL of the sample to be analyzed into the hydrometer jar, swirl to rinse, and discard.
4. Pour the sample into a rinsed hydrometer jar up to a level of about 3 inches from the top.
5. Insert the clean, dry hydrometer carefully, holding the top of the hydrometer stem in a pendulum effect.
6. Spin the hydrometer carefully to free the instrument from the surface tension on the inside of the hydrometer jar.
7. Read the stem directly at the bottom meniscus just before spinning stops, and record result.

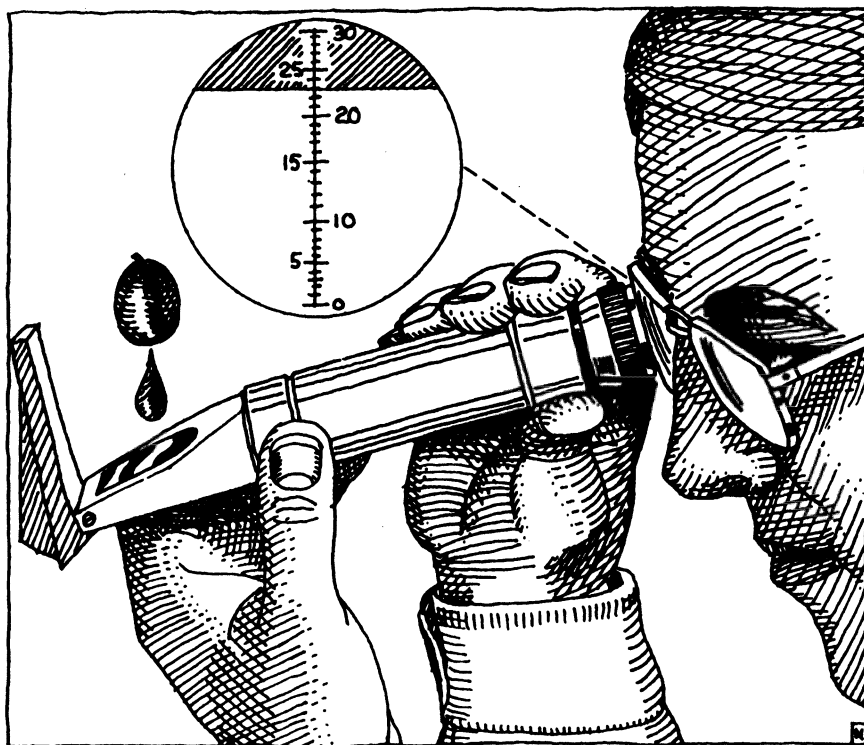


Fig. B-8 Determining brix by refractometer.

Courtesy: Sebastiani Vineyards.

BRIX BY REFRACTOMETER

The rationale for this procedure is to place a drop or two of representative juice sample to a clean refractometer prism. Exposure to a bright light source refracts incident light correlating to dissolved solids concentration, which is measured upon a graduated scale on a screen inside the instrument (Figure B-8). Refractometers should not be used for Balling analysis, as samples containing alcohol will distort the refraction.

Apparatus

- Refractometer
- Bright light source

Procedure

1. Adjust the juice sample to room temperature or to that required by operating instructions of the instrument.
2. Open the prism cover, and rinse the prism surface with several drops of the sample. Gently wipe dry with absorbent lens paper.
3. Apply several drops of the sample again, and close the prism cover. Point the refractometer toward a bright light source, and hold in the same manner as a telescope. Adjust focus, and read the Brix at the light-dark dividing line. Record result.
4. Rinse the refractometer prism surface and prism cover three times with distilled or deionized water, and wipe dry with absorbent lens paper. Avoid scratching the prism surface.

Table B-1 lists the Brix of soluble solids at 20°C.

ALCOHOL BY EBULLIOMETER

This analytical procedure is less accurate with wines that have an alcohol greater than 14% by volume or for wines that have a Balling of more than 0.0° (residual sugar more than 3%).

The ebulliometer (Figure B-9) is used to compare the boiling point of water with the boiling point of a wine sample—the difference being due to the alcohol content in the wine. Because the boiling point of water ranges from about 99 to 101°C from day to day, in response to rising and falling barometric pressure, it is necessary to recheck the boiling point of water in the ebulliometer every hour or so.

Apparatus and Reagents

- Salleron-DuJardin ebulliometer with thermometer and circular slide scale
- 100-mL graduated cylinder
- Distilled or deionized water
- Cold tap water

Procedure

1. Rinse all inside surfaces of the ebulliometer with distilled or deionized water. Drain the valve and close.
2. Fill the upper reflux condenser jacket with cold tap water.
3. Measure 50 mL of distilled or deionized water into a clean 100-mL graduated cylinder, and carefully pour into the lower chamber inlet.
4. Very carefully insert the thermometer into the lower chamber inlet, holding the top of the thermometer in one hand in a pendulum effect and holding the rubber stopper portion in the other hand. Slowly and gently twist the rubber stopper into position for a snug fit. These thermometers are very delicate and break rather easily.

Table B-1 Brix (soluble solids) at 20°C; sucrose (cane sugar) scale.

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
0.1	1.00040	8.34878	.00834	1.000	8.34044
0.2	1.00079	8.35196	.01675	2.007	8.33521
0.3	1.00118	8.35514	.02513	3.011	8.33001
0.4	1.00156	8.35828	.03351	4.015	8.32477
0.5	1.00195	8.36145	.04189	5.020	8.31956
0.6	1.00233	8.36462	.05026	6.024	8.31436
0.7	1.00272	8.36779	.05864	7.028	8.30915
0.8	1.00310	8.37096	.06702	8.031	8.30394
0.9	1.00349	8.37413	.07539	9.035	8.29874
1.0	1.00387	8.37730	.08377	10.039	8.29353
1.1	1.00425	8.38047	.09219	11.047	8.28828
1.2	1.00464	8.38372	.10060	12.055	8.28312
1.3	1.00503	8.38698	.10903	13.065	8.27795
1.4	1.00542	8.39023	.11746	14.075	8.27277
1.5	1.00581	8.39348	.12590	15.086	8.26758
1.6	1.00620	8.39674	.13435	16.099	8.26239
1.7	1.00659	8.39999	.14280	17.111	8.25719
1.8	1.00698	8.40325	.15126	18.125	8.25199
1.9	1.00737	8.40650	.15972	19.139	8.24678
2.0	1.00776	8.40976	.16820	20.155	8.24156
2.1	1.00815	8.41301	.17667	21.170	8.23634
2.2	1.00854	8.41627	.18516	22.187	8.23111
2.3	1.00893	8.41952	.19365	23.205	8.22587
2.4	1.00933	8.42286	.20215	24.223	8.22071
2.5	1.00972	8.42611	.21065	25.242	8.21546
2.6	1.01011	8.42937	.21916	26.262	8.21021
2.7	1.01051	8.43271	.22768	27.282	8.20503
2.8	1.01090	8.43596	.23621	28.305	8.19975
2.9	1.01129	8.43922	.24474	29.327	8.19448
3.0	1.01169	8.44255	.25328	30.350	8.18927
3.1	1.01208	8.44581	.26182	31.373	8.18399
3.2	1.01248	8.44915	.27037	32.398	8.17878
3.3	1.01287	8.45240	.27893	33.423	8.17347
3.4	1.01327	8.45574	.28750	34.450	8.16824
3.5	1.01366	8.45899	.29606	35.476	8.16293
3.6	1.01406	8.46233	.30464	36.504	8.15769
3.7	1.01445	8.46559	.31323	37.534	8.15236
3.8	1.01485	8.46892	.32182	38.563	8.14710
3.9	1.01524	8.47218	.33042	39.593	8.14176
4.0	1.01564	8.47552	.33902	40.624	8.13650
4.1	1.01603	8.47877	.34763	41.656	8.13114
4.2	1.01643	8.48211	.35625	42.689	8.12586
4.3	1.01683	8.48545	.36487	43.722	8.12058
4.4	1.01723	8.48878	.37351	44.757	8.11527
4.5	1.01763	8.49212	.38215	45.792	8.10997
4.6	1.01802	8.49538	.39079	46.827	8.10459
4.7	1.01842	8.49871	.39944	47.864	8.09927
4.8	1.01882	8.50205	.40810	48.902	8.09395
4.9	1.01922	8.50539	.41676	49.939	8.08863
5.0	1.01962	8.50873	.42544	50.980	8.08329
5.1	1.02002	8.51207	.43412	52.020	8.07795
5.2	1.02042	8.51540	.44280	53.060	8.07260
5.3	1.02082	8.51874	.45149	54.101	8.06725
5.4	1.02122	8.52208	.46019	55.144	8.06189
5.5	1.02163	8.52550	.46890	56.187	8.05660
5.6	1.02203	8.52884	.47762	57.232	8.05122
5.7	1.02243	8.53218	.48633	58.276	8.04585
5.8	1.02283	8.53552	.49506	59.322	8.04046
5.9	1.02323	8.53885	.50379	60.368	8.03506

Table B-1 (continued)

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
6.0	1.02364	8.54228	.51254	61.417	8.02974
6.1	1.02404	8.54561	.52128	62.464	8.02433
6.2	1.02444	8.54895	.53003	63.512	8.01892
6.3	1.02485	8.55237	.53880	64.563	8.01357
6.4	1.02525	8.55571	.54757	65.614	8.00814
6.5	1.02566	8.55913	.55634	66.665	8.00279
6.6	1.02606	8.56247	.56512	67.717	7.99735
6.7	1.02646	8.56581	.57391	68.770	7.99190
6.8	1.02687	8.56923	.58271	69.825	7.98652
6.9	1.02727	8.57257	.59151	70.879	7.98106
7.0	1.02768	8.57599	.60032	71.935	7.97567
7.1	1.02808	8.57933	.60913	72.991	7.97020
7.2	1.02849	8.58275	.61796	74.049	7.96479
7.3	1.02890	8.58617	.62679	75.107	7.95938
7.4	1.02931	8.58959	.63563	76.166	7.95396
7.5	1.02972	8.59301	.64448	77.227	7.94853
7.6	1.03012	8.59635	.65332	78.286	7.94303
7.7	1.03053	8.59977	.66218	79.348	7.93759
7.8	1.03094	8.60319	.67105	80.410	7.93214
7.9	1.03135	8.60662	.67992	81.473	7.92670
8.0	1.03176	8.61004	.68880	82.538	7.92124
8.1	1.03216	8.61338	.69768	83.602	7.91570
8.2	1.03257	8.61680	.70658	84.668	7.91022
8.3	1.03298	8.62022	.71548	85.735	7.90474
8.4	1.03339	8.62364	.72439	86.802	7.89925
8.5	1.03380	8.62706	.73330	87.870	7.89376
8.6	1.03421	8.63048	.74222	88.939	7.88826
8.7	1.03462	8.63390	.75115	90.009	7.88275
8.8	1.03503	8.63733	.76009	91.080	7.87724
8.9	1.03544	8.64075	.76903	92.151	7.87172
9.0	1.03585	8.64417	.77798	93.224	7.86619
9.1	1.03626	8.64759	.78693	94.296	7.86066
9.2	1.03667	8.65101	.79589	95.370	7.85512
9.3	1.03708	8.65443	.80486	96.445	7.84957
9.4	1.03750	8.65794	.81385	97.522	7.84409
9.5	1.03791	8.66136	.82283	98.598	7.83853
9.6	1.03832	8.66478	.83182	99.675	7.83296
9.7	1.03874	8.66829	.84082	100.754	7.82747
9.8	1.03915	8.67171	.84983	101.833	7.82188
9.9	1.03956	8.67513	.85884	102.913	7.81629
10.0	1.03998	8.67863	.86786	103.994	7.81077
10.1	1.04039	8.68205	.87689	105.076	7.80516
10.2	1.04081	8.68556	.88593	106.159	7.79963
10.3	1.04122	8.68898	.89496	107.241	7.79402
10.4	1.04164	8.69249	.90402	108.327	7.78847
10.5	1.04205	8.69591	.91307	109.411	7.78284
10.6	1.04247	8.69941	.92214	110.498	7.77727
10.7	1.04288	8.70283	.93120	111.584	7.77163
10.8	1.04330	8.70634	.94028	112.672	7.76606
10.9	1.04371	8.70976	.94936	113.760	7.76040
11.0	1.04413	8.71326	.95846	114.850	7.75480
11.1	1.04454	8.71669	.96755	115.940	7.74914
11.2	1.04496	8.72019	.97666	117.031	7.74353
11.3	1.04538	8.72370	.98578	118.124	7.73792
11.4	1.04580	8.72720	.99490	119.217	7.73230
11.5	1.04622	8.73071	1.00403	120.311	7.72668
11.6	1.04663	8.73413	1.01316	121.405	7.72097
11.7	1.04705	8.73763	1.02230	122.500	7.71533
11.8	1.04747	8.74114	1.03145	123.597	7.70969

(continued)

Table B-1 (continued)

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
11.9	1.04789	8.74464	1.04061	124.694	7.70403
12.0	1.04831	8.74815	1.04978	125.795	7.69837
12.1	1.04873	8.75165	1.05895	126.892	7.69270
12.2	1.04915	8.75516	1.06813	127.992	7.68703
12.3	1.04957	8.75866	1.07732	129.093	7.68134
12.4	1.04999	8.76217	1.08651	130.194	7.67566
12.5	1.05041	8.76567	1.09571	131.297	7.66996
12.6	1.05083	8.76918	1.10492	132.400	7.66426
12.7	1.05125	8.77268	1.11413	133.504	7.65855
12.8	1.05167	8.77619	1.12335	134.609	7.65284
12.9	1.05209	8.77969	1.13258	135.715	7.64711
13.0	1.05251	8.78320	1.14182	136.822	7.64138
13.1	1.05293	8.78670	1.15106	137.929	7.63564
13.2	1.05335	8.79021	1.16031	139.038	7.62990
13.3	1.05378	8.79379	1.16957	140.147	7.62422
13.4	1.05420	8.79730	1.17884	141.258	7.61846
13.5	1.05463	8.80089	1.18812	142.370	7.61277
13.6	1.05505	8.80439	1.19740	143.482	7.60699
13.7	1.05547	8.80790	1.20668	144.594	7.60122
13.8	1.05590	8.81149	1.21599	145.710	7.59550
13.9	1.05632	8.81499	1.22528	146.823	7.58971
14.0	1.05675	8.81858	1.23460	147.940	7.58398
14.1	1.05717	8.82208	1.24391	149.055	7.57817
14.2	1.05760	8.82567	1.25325	150.174	7.57242
14.3	1.05803	8.82926	1.26258	151.292	7.56668
14.4	1.05846	8.83285	1.27193	152.412	7.56092
14.5	1.05889	8.83644	1.28128	153.533	7.55516
14.6	1.05931	8.83994	1.29063	154.654	7.54931
14.7	1.05974	8.84353	1.30000	155.776	7.54353
14.8	1.06017	8.84712	1.30937	156.899	7.53775
14.9	1.06060	8.85071	1.31876	158.044	7.53195
15.0	1.06103	8.85430	1.32815	159.150	7.52615
15.1	1.06146	8.85788	1.33754	160.275	7.52034
15.2	1.06189	8.86147	1.34794	161.401	7.51453
15.3	1.06232	8.86506	1.35635	162.529	7.50871
15.4	1.06275	8.86865	1.36577	163.657	7.50288
15.5	1.06318	8.87224	1.37520	164.787	7.49704
15.6	1.06361	8.87583	1.38463	165.917	7.49120
15.7	1.06404	8.87941	1.39407	167.049	7.48534
15.8	1.06447	8.88300	1.40351	168.180	7.47949
15.9	1.06490	8.88659	1.41297	169.313	7.47362
16.0	1.06534	8.89026	1.42244	170.448	7.46782
16.1	1.06577	8.89385	1.43191	171.583	7.46194
16.2	1.06620	8.89744	1.44139	172.719	7.45605
16.3	1.06663	8.90103	1.45087	173.855	7.45016
16.4	1.06707	8.90470	1.46037	174.993	7.44433
16.5	1.06750	8.90829	1.46987	176.132	7.43842
16.6	1.06793	8.91188	1.47937	177.270	7.43251
16.7	1.06837	8.91555	1.48890	178.412	7.42665
16.8	1.06880	8.91914	1.49842	179.553	7.42072
16.9	1.06923	8.92272	1.50794	180.693	7.41478
17.0	1.06967	8.92640	1.51749	181.838	7.40891
17.1	1.07010	8.92998	1.52703	182.981	7.40295
17.2	1.07054	8.93366	1.53659	184.127	7.39707
17.3	1.07098	8.93733	1.54616	185.273	7.39117
17.4	1.07141	8.94092	1.55572	186.419	7.38520
17.5	1.07185	8.94459	1.56530	187.567	7.37929
17.6	1.07229	8.94826	1.57489	188.716	7.37337
17.7	1.07272	8.95185	1.58448	189.865	7.36737

Table B-1 (continued)

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
17.8	1.07316	8.95552	1.59408	191.015	7.36144
17.9	1.07360	8.95919	1.60370	192.168	7.35549
18.0	1.07404	8.96286	1.61331	193.320	7.34955
18.1	1.07448	8.96654	1.62294	194.474	7.34360
18.2	1.07492	8.97021	1.63258	195.629	7.33763
18.3	1.07536	8.97388	1.64222	196.784	7.33166
18.4	1.07580	8.97755	1.65187	197.940	7.32568
18.5	1.07624	8.98122	1.66153	199.098	7.31969
18.6	1.07668	8.98489	1.67119	200.255	7.31370
18.7	1.07712	8.98857	1.68086	201.414	7.30771
18.8	1.07756	8.99224	1.69054	202.574	7.30170
18.9	1.07800	8.99591	1.70023	203.735	7.29568
19.0	1.07844	8.99958	1.70992	204.896	7.28966
19.1	1.07888	9.00325	1.71962	206.059	7.28363
19.2	1.07932	9.00693	1.72933	207.222	7.27760
19.3	1.07977	9.01068	1.73906	208.388	7.27162
19.4	1.08021	9.01435	1.74878	209.559	7.26557
19.5	1.08066	9.01811	1.75853	210.721	7.25958
19.6	1.08110	9.02178	1.76827	211.888	7.25351
19.7	1.08154	9.02545	1.77801	213.055	7.24744
19.8	1.08199	9.02921	1.78778	214.226	7.24143
19.9	1.08243	9.03288	1.79754	215.396	7.23534
20.0	1.08288	9.03663	1.80733	216.569	7.22930
20.1	1.08332	9.04031	1.81710	217.739	7.22321
20.2	1.08377	9.04406	1.82690	218.914	7.21716
20.3	1.08421	9.04773	1.83669	220.087	7.21104
20.4	1.08466	9.05149	1.84650	221.262	7.20499
20.5	1.08510	9.05516	1.85631	222.438	7.19885
20.6	1.08555	9.05891	1.86614	223.616	7.19277
20.7	1.08599	9.06259	1.87596	224.793	7.18663
20.8	1.08644	9.06634	1.88580	225.972	7.18054
20.9	1.08688	9.07001	1.89563	227.150	7.17438
21.0	1.08733	9.07377	1.90549	228.331	7.16828
21.1	1.08777	9.07744	1.91534	229.511	7.16210
21.2	1.08822	9.08120	1.92521	230.694	7.15599
21.3	1.08867	9.08495	1.93509	231.878	7.14986
21.4	1.08912	9.08871	1.94498	233.063	7.14373
21.5	1.08957	9.09246	1.95488	234.249	7.13758
21.6	1.09002	9.09622	1.96478	235.436	7.13144
21.7	1.09047	9.09997	1.97469	236.623	7.12528
21.8	1.09092	9.10373	1.98461	237.812	7.11912
21.9	1.09137	9.10748	1.99454	239.014	7.11294
22.0	1.09182	9.11124	2.00447	240.192	7.10677
22.1	1.09227	9.11499	2.01441	241.383	7.10058
22.2	1.09272	9.11875	2.02436	242.575	7.09439
22.3	1.09317	9.12250	2.03432	243.768	7.08818
22.4	1.09362	9.12626	2.04428	244.962	7.08198
22.5	1.09408	9.13010	2.05427	246.159	7.07583
22.6	1.09453	9.13385	2.06425	247.355	7.06960
22.7	1.09498	9.13761	2.07424	248.552	7.06337
22.8	1.09543	9.14136	2.08423	249.749	7.05713
22.9	1.09588	9.14512	2.09423	250.947	7.05089
23.0	1.09634	9.14896	2.10426	252.149	7.04470
23.1	1.09679	9.15271	2.11428	253.350	7.03843
23.2	1.09725	9.15655	2.12432	254.553	7.03223
23.3	1.09770	9.16031	2.13435	255.755	7.02596
23.4	1.09816	9.16415	2.14441	256.960	7.01974
23.5	1.09862	9.16798	2.15448	258.167	7.01350
23.6	1.09907	9.17174	2.16453	259.371	7.00721

(continued)

Table B-1 (continued)

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
23.7	1.09953	9.17558	2.17461	260.579	7.00097
23.8	1.09998	9.17933	2.18468	261.786	6.99465
23.9	1.10044	9.18317	2.19478	262.996	6.98839
24.0	1.10090	9.18701	2.20488	264.206	6.98213
24.1	1.10135	9.19085	2.21499	265.418	6.97586
24.2	1.10182	9.19469	2.22511	266.630	6.96958
24.3	1.10228	9.19853	2.23524	267.844	6.96329
24.4	1.10274	9.20237	2.24538	269.059	6.95699
24.5	1.10320	9.20620	2.25552	270.236	6.95068
24.6	1.10366	9.21004	2.26567	271.491	6.94437
24.7	1.10412	9.21388	2.27583	272.708	6.93805
24.8	1.10458	9.21772	2.28599	273.926	6.93173
24.9	1.10504	9.22156	2.29617	275.145	6.92539
25.0	1.10550	9.22540	2.30635	276.365	6.91905
25.1	1.10596	9.22924	2.31654	277.586	6.91270
25.2	1.10642	9.23307	2.32673	278.807	6.90634
25.3	1.10688	9.23691	2.33694	280.031	6.89997
25.4	1.10734	9.24075	2.34715	281.254	6.89360
25.5	1.10781	9.24467	2.35739	282.481	6.88728
25.6	1.10827	9.24851	2.36762	283.707	6.88089
25.7	1.10873	9.25235	2.37785	284.933	6.87450
25.8	1.10919	9.25619	2.38810	286.161	6.86809
25.9	1.10965	9.26003	2.39835	287.389	6.86168
26.0	1.11012	9.26395	2.40863	288.621	6.85532
26.1	1.11058	9.26779	2.41889	289.851	6.84890
26.2	1.11105	9.27171	2.42919	291.085	6.84252
26.3	1.11151	9.27555	2.43947	292.317	6.83608
26.4	1.11198	9.27947	2.44978	293.552	6.82969
26.5	1.11245	9.28340	2.46010	294.789	6.82330
26.6	1.11291	9.28723	2.47040	296.023	6.81683
26.7	1.11338	9.29116	2.48074	297.262	6.81042
26.8	1.11384	9.29499	2.49106	298.499	6.80393
26.9	1.11431	9.29892	2.50141	299.739	6.79751
27.0	1.11478	9.30284	2.51177	300.980	6.79107
27.1	1.11525	9.30676	2.52213	302.222	6.78463
27.2	1.11572	9.31068	2.53250	303.464	6.77818
27.3	1.11619	9.31461	2.54289	304.709	6.77172
27.4	1.11666	9.31853	2.55328	305.954	6.76525
27.5	1.11713	9.32245	2.56367	307.199	6.75878
27.6	1.11760	9.32637	2.57408	308.447	6.75229
27.7	1.11807	9.33029	2.58449	309.694	6.74580
27.8	1.11854	9.33422	2.59491	310.943	6.73931
27.9	1.11901	9.33814	2.60534	312.193	6.73280
28.0	1.11948	9.34206	2.61578	313.444	6.72628
28.1	1.11995	9.34598	2.62622	314.695	6.71976
28.2	1.12042	9.34990	2.63667	315.947	6.71323
28.3	1.12089	9.35383	2.64713	317.200	6.70670
28.4	1.12137	9.35783	2.65762	318.457	6.70021
28.5	1.12184	9.36175	2.66810	319.713	6.69365
28.6	1.12231	9.36568	2.67858	320.969	6.68710
28.7	1.12279	9.36968	2.68910	322.229	6.68058
28.8	1.12326	9.37360	2.69960	323.488	6.67400
28.9	1.12373	9.37753	2.71011	324.747	6.66742
29.0	1.12421	9.38153	2.72064	326.009	6.66089
29.1	1.12468	9.38545	2.73117	327.271	6.65428
29.2	1.12516	9.38946	2.74172	328.535	6.64774
29.3	1.12564	9.39347	2.75229	329.801	6.64118
29.4	1.12611	9.39739	2.76283	331.064	6.63456
29.5	1.12659	9.40139	2.77341	332.332	6.62798

Table B-1 (continued)

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
29.6	1.12707	9.40540	2.78400	333.601	6.62140
29.7	1.12754	9.40932	2.79457	334.868	6.61475
29.8	1.12802	9.41333	2.80517	336.138	6.60816
29.9	1.12850	9.41733	2.81578	337.409	6.60155
30.0	1.12898	9.42134	2.82640	338.682	6.59494
30.1	1.12945	9.42526	2.83700	339.952	6.58826
30.2	1.12993	9.42927	2.84764	341.227	6.58163
30.3	1.13041	9.43327	2.85828	342.502	6.57499
30.4	1.13089	9.43728	2.86893	343.778	6.56835
30.5	1.13137	9.44128	2.87959	345.056	6.56169
30.6	1.13185	9.44529	2.89026	346.334	6.55503
30.7	1.13223	9.44929	2.90093	347.613	6.54836
30.8	1.13281	9.45330	2.91162	348.894	6.54168
30.9	1.13329	9.45731	2.92231	350.175	6.53500
31.0	1.13377	9.46131	2.93301	351.457	6.52830
31.1	1.13425	9.46532	2.94371	352.739	6.52161
31.2	1.13473	9.46932	2.95443	354.023	6.51489
31.3	1.13521	9.47333	2.96515	355.308	6.50818
31.4	1.13570	9.47742	2.97591	356.597	6.50151
31.5	1.13618	9.48142	2.98665	357.884	6.49477
31.6	1.13666	9.48543	2.99740	359.172	6.48803
31.7	1.13715	9.48952	3.00818	360.464	6.48134
31.8	1.13763	9.49352	3.01894	361.754	6.47458
31.9	1.13811	9.49753	3.02971	363.044	6.46782
32.0	1.13860	9.50162	3.04052	364.339	6.46110
32.1	1.13908	9.50562	3.05130	365.631	6.45432
32.2	1.13957	9.50971	3.06213	366.929	6.44758
32.3	1.14005	9.51372	3.07293	368.223	6.44079
32.4	1.14054	9.51781	3.08377	369.522	6.43404
32.5	1.14103	9.52190	3.09462	370.822	6.42728
32.6	1.14151	9.52590	3.10544	372.119	6.42046
32.7	1.14200	9.52999	3.11631	373.421	6.41368
32.8	1.14248	9.53400	3.12715	374.720	6.40685
32.9	1.14297	9.53808	3.13803	376.024	6.40005
33.0	1.14346	9.54217	3.14892	377.329	6.39325
33.1	1.14395	9.54626	3.15981	378.634	6.38645
33.2	1.14444	9.55035	3.17072	379.941	6.37963
33.3	1.14493	9.55444	3.18163	381.248	6.37281
33.4	1.14542	9.55853	3.19255	382.557	6.36598
33.5	1.14591	9.56262	3.20248	383.867	6.35914
33.6	1.14640	9.56671	3.21441	385.176	6.35230
33.7	1.14689	9.57080	3.22536	386.488	6.34544
33.8	1.14738	9.57489	3.23631	387.801	6.33858
33.9	1.14787	9.57898	3.24727	389.114	6.33171
34.0	1.14836	9.58306	3.25824	390.428	6.32482
34.1	1.14885	9.58715	3.26922	391.744	6.31793
34.2	1.14934	9.59124	3.28020	393.060	6.31104
34.3	1.14984	9.59541	3.29123	394.382	6.30418
34.4	1.15033	9.59950	3.30223	395.700	6.29727
34.5	1.15083	9.60368	3.31327	397.023	6.29041
34.6	1.15132	9.60777	3.32429	398.343	6.28348
34.7	1.15181	9.61185	3.33531	399.664	6.27654
34.8	1.15231	9.61603	3.34638	400.990	6.26965
34.9	1.15280	9.62012	3.35742	402.313	6.26270
35.0	1.15330	9.62429	3.36850	403.641	6.25579
35.1	1.15379	9.62838	3.37956	404.966	6.24882
35.2	1.15429	9.63255	3.39066	406.296	6.24189
35.3	1.15479	9.63672	3.40176	407.626	6.23496
35.4	1.15528	9.64081	3.41285	408.955	6.22796

(continued)

Table B-1 (continued)

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
35.5	1.15578	9.64498	3.42397	410.287	6.22101
35.6	1.15628	9.64916	3.43510	411.621	6.21406
35.7	1.15677	9.65325	3.44621	412.952	6.20704
35.8	1.15727	9.65742	3.45736	414.289	6.20006
35.9	1.15777	9.66159	3.46851	415.625	6.19308
36.0	1.15827	9.66576	3.47967	416.962	6.18609
36.1	1.15877	9.66994	3.49085	418.302	6.17909
36.2	1.15927	9.67411	3.50203	419.641	6.17208
36.3	1.15977	9.67828	3.51322	420.982	6.16506
36.4	1.16027	9.68245	3.52441	422.323	6.15804
36.5	1.16078	9.68671	3.53565	423.670	6.15106
36.6	1.16128	9.69088	3.54686	425.013	6.14402
36.7	1.16178	9.69505	3.55808	426.358	6.13697
36.8	1.16228	9.69923	3.56932	427.704	6.12991
36.9	1.16278	9.70340	3.58055	429.050	6.12285
37.0	1.16329	9.70766	3.59183	430.402	6.11583
37.1	1.16379	9.71183	3.60309	431.751	6.10874
37.2	1.16429	9.71600	3.61435	433.100	6.10165
37.3	1.16480	9.72026	3.62566	434.456	6.09460
37.4	1.16530	9.72443	3.63694	435.807	6.08749
37.5	1.16581	9.72868	3.64826	437.164	6.08042
37.6	1.16631	9.73286	3.65956	438.518	6.07330
37.7	1.16681	9.73703	3.67086	439.872	6.06617
37.8	1.16732	9.74129	3.68221	441.232	6.05908
37.9	1.16782	9.74546	3.69353	442.588	6.05193
38.0	1.16833	9.74971	3.70489	443.950	6.04482
38.1	1.16883	9.75389	3.71623	445.308	6.03766
38.2	1.16934	9.75814	3.72761	446.672	6.03053
38.3	1.16985	9.76240	3.73900	448.037	6.02340
38.4	1.17036	9.76665	3.75039	449.402	6.01626
38.5	1.17087	9.77091	3.76180	450.769	6.00911
38.6	1.17138	9.77517	3.77322	452.137	6.00195
38.7	1.17189	9.77942	3.78464	453.506	5.99478
38.8	1.17240	9.78368	3.79607	454.875	5.98761
38.9	1.17291	9.78793	3.80750	456.245	5.98043
39.0	1.17342	9.79219	3.81895	457.617	5.97324
39.1	1.17393	9.79645	3.83041	458.990	5.96604
39.2	1.17444	9.80070	3.84187	460.364	5.95883
39.3	1.17495	9.80496	3.85335	461.739	5.95161
39.4	1.17546	9.80921	3.86483	463.115	5.94438
39.5	1.17598	9.81355	3.87635	464.495	5.93720
39.6	1.17649	9.81781	3.88785	465.873	5.92996
39.7	1.17700	9.82207	3.89936	467.253	5.92271
39.8	1.17751	9.82632	3.91188	468.633	5.91544
39.9	1.17802	9.83058	3.92240	470.013	5.90818
40.0	1.17854	9.83492	3.93397	471.340	5.90095
40.1	1.17905	9.83917	3.94551	472.783	5.89366
40.2	1.17957	9.84351	3.95709	474.170	5.88642
40.3	1.18008	9.84777	3.96865	475.555	5.87912
40.4	1.18060	9.85211	3.98025	476.945	5.87186
40.5	1.18111	9.85636	3.99183	478.333	5.86453
40.6	1.18163	9.86070	4.00344	479.724	5.85726
40.7	1.18214	9.86496	4.01504	481.114	5.84992
40.8	1.18266	9.86930	4.02667	482.508	5.84263
40.9	1.18317	9.87355	4.03828	483.899	5.83527
41.0	1.18369	9.87789	4.04993	485.295	5.82796
41.1	1.18420	9.88215	4.06156	486.689	5.82059
41.2	1.18472	9.88649	4.07323	488.087	5.81326
41.3	1.18524	9.89083	4.08491	489.487	5.80592

Table B-1 (continued)

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
41.4	1.18576	9.89517	4.09660	490.887	5.79857
41.5	1.18628	9.89951	4.10830	492.289	5.79121
41.6	1.18679	9.90376	4.11996	493.687	5.78380
41.7	1.18731	9.90810	4.13168	495.091	5.77642
41.8	1.18783	9.91244	4.14340	496.495	5.76904
41.9	1.18835	9.91678	4.15513	497.901	5.76165
42.0	1.18887	9.92112	4.16687	499.308	5.75425
42.1	1.18939	9.92546	4.17862	500.716	5.74684
42.2	1.18991	9.92980	4.19038	502.125	5.73942
42.3	1.19043	9.93414	4.20214	503.534	5.73200
42.4	1.19096	9.93856	4.21395	504.949	5.72461
42.5	1.19148	9.94290	4.22573	506.361	5.71717
42.6	1.19200	9.94724	4.23752	507.774	5.70972
42.7	1.19253	9.95166	4.24936	509.192	5.70230
42.8	1.19305	9.95600	4.26117	510.607	5.69483
42.9	1.19357	9.96034	4.27299	512.024	5.68735
43.0	1.19410	9.96476	4.28485	513.445	5.67991
43.1	1.19462	9.96910	4.29668	514.863	5.67242
43.2	1.19515	9.97353	4.30856	517.711	5.66497
43.3	1.19568	9.97795	4.32045	519.132	5.65750
43.4	1.19620	9.98229	4.33231	520.559	5.64998
43.5	1.19673	9.98671	4.34422	521.986	5.64249
43.6	1.19726	9.99113	4.35613	523.411	5.63500
43.7	1.19778	9.99547	4.36802	524.842	5.62745
43.8	1.19831	9.99990	4.37996	526.273	5.61994
43.9	1.19884	10.00432	4.39190	527.705	5.61242
44.0	1.19937	10.00874	4.40385	529.126	5.60489
44.1	1.19990	10.01317	4.41581	530.571	5.59736
44.2	1.20043	10.01759	4.42777	532.006	5.58982
44.3	1.20096	10.02201	4.43975	533.442	5.58226
44.4	1.20149	10.02643	4.45173	534.880	5.57470
44.5	1.20202	10.03086	4.46373	536.318	5.56713
44.6	1.20255	10.03528	4.47573	537.758	5.55955
44.7	1.20308	10.03970	4.48775	539.198	5.55195
44.8	1.20361	10.04413	4.49977	540.640	5.54436
44.9	1.20414	10.04855	4.51180	541.722	5.53675
45.0	1.20467	10.05297	4.52384	542.803	5.52913
45.1	1.20520	10.05739	4.53588	543.525	5.52151
45.2	1.20573	10.06182	4.54794	544.971	5.51388
45.3	1.20627	10.06632	4.56004	546.420	5.50628
45.4	1.20680	10.07075	4.57212	547.868	5.49863
45.5	1.20734	10.07525	4.58424	549.320	5.49101
45.6	1.20787	10.07968	4.59633	550.769	5.48335
45.7	1.20840	10.08410	4.60843	552.219	5.47567
45.8	1.20894	10.08860	4.62058	553.675	5.46802
45.9	1.20947	10.09303	4.63270	555.127	5.46033
46.0	1.21001	10.09753	4.64486	556.584	5.45267
46.1	1.21054	10.10196	4.65700	558.039	5.44496
46.2	1.21108	10.10646	4.66918	559.499	5.43728
46.3	1.21162	10.11097	4.68138	560.960	5.42959
46.4	1.21216	10.11548	4.69358	562.422	5.42190
46.5	1.21270	10.11998	4.70579	563.885	5.41419
46.6	1.21323	10.12440	4.71797	565.345	5.40643
46.7	1.21377	10.12891	4.73020	567.028	5.39871
46.8	1.21431	10.13342	4.74244	568.277	5.39098
46.9	1.21485	10.13792	4.75468	569.744	5.38324
47.0	1.21539	10.14243	4.76694	571.213	5.37549
47.1	1.21593	10.14694	4.77921	572.683	5.36773
47.2	1.21647	10.15144	4.79148	574.153	5.35996

(continued)

Table B-1 (continued)

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
47.3	1.21701	10.15595	4.80376	575.625	5.35219
47.4	1.21755	10.16045	4.81605	577.098	5.34440
47.5	1.21809	10.16496	4.82836	578.573	5.33660
47.6	1.21863	10.16947	4.84067	580.049	5.32880
47.7	1.21917	10.17397	4.85298	581.523	5.32099
47.8	1.21971	10.17848	4.86531	583.000	5.31317
47.9	1.22025	10.18299	4.87765	584.479	5.30534
48.0	1.22080	10.18758	4.89004	585.964	5.29754
48.1	1.22134	10.19208	4.90239	587.444	5.28969
48.2	1.22189	10.19667	4.91479	588.929	5.28188
48.3	1.22243	10.20118	4.92717	590.413	5.27401
48.4	1.22298	10.20577	4.93959	591.901	5.26618
48.5	1.22353	10.21036	4.95202	593.391	5.25834
48.6	1.22407	10.21486	4.96442	594.877	5.25044
48.7	1.22462	10.21945	4.97687	596.368	5.24258
48.8	1.22516	10.22396	4.98929	597.857	5.23467
48.9	1.22571	10.22855	5.00176	599.351	5.22679
49.0	1.22626	10.23314	5.01424	600.846	5.21890
49.1	1.22680	10.23765	5.02669	602.338	5.21096
49.2	1.22735	10.24224	5.03918	603.835	5.20406
49.3	1.22790	10.24683	5.05169	605.334	5.19514
49.4	1.22845	10.25142	5.06420	606.833	5.18722
49.5	1.22900	10.25601	5.07562	608.333	5.17929
49.6	1.22955	10.26059	5.08925	609.935	5.17134
49.7	1.23010	10.26518	5.10179	611.337	5.16339
49.8	1.23065	10.26977	5.11435	612.842	5.15542
49.9	1.23120	10.27436	5.12691	614.347	5.14745
50.0	1.23175	10.27895	5.13948	615.854	5.13957
50.1	1.23230	10.28354	5.15205	617.360	5.13149
50.2	1.23285	10.28813	5.16464	618.868	5.12349
50.3	1.23340	10.29272	5.17724	620.378	5.11548
50.4	1.23396	10.29740	5.18989	621.894	5.10751
50.5	1.23451	10.30199	5.20250	623.405	5.09949
50.6	1.23506	10.30658	5.21513	624.919	5.09145
50.7	1.23562	10.31125	5.22780	626.437	5.08345
50.8	1.23617	10.31584	5.24045	627.953	5.07539
50.9	1.23672	10.32043	5.25310	629.468	5.06733
51.0	1.23728	10.32510	5.26580	630.990	5.05930
51.1	1.23783	10.32969	5.27847	632.509	5.05122
51.2	1.23839	10.33436	5.29119	634.033	5.04317
51.3	1.23894	10.33895	5.30388	635.553	5.03507
51.4	1.23950	10.34363	5.31663	637.081	5.02700
51.5	1.24006	10.34830	5.32937	638.608	5.01893
51.6	1.24061	10.35289	5.34209	640.132	5.01080
51.7	1.24117	10.35756	5.35486	641.662	5.00270
51.8	1.24172	10.36215	5.36759	643.188	4.99456
51.9	1.24228	10.36683	5.38038	644.720	4.98645
52.0	1.24284	10.37150	5.39318	646.254	4.97832
52.1	1.24340	10.37617	5.40598	647.788	4.97019
52.2	1.24396	10.38085	5.41880	649.324	4.96205
52.3	1.24452	10.38552	5.43163	650.861	4.95389
52.4	1.24508	10.39019	5.44446	652.399	4.94573
52.5	1.24564	10.39487	5.45731	653.939	4.93756
52.6	1.24620	10.39954	5.47016	655.478	4.92938
52.7	1.24676	10.40421	5.48302	657.019	4.92119
52.8	1.24732	10.40889	5.49589	658.562	4.91300
52.9	1.24788	10.41356	5.50877	660.105	4.90479
53.0	1.24845	10.41832	5.52171	661.655	4.89661
53.1	1.24901	10.42299	5.53461	663.201	4.88838

Table B-1 (continued)

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
53.2	1.24957	10.42766	5.54752	664.748	4.88014
53.3	1.25014	10.43242	5.56048	666.301	4.87194
53.4	1.25070	10.43709	5.57341	667.851	4.86368
53.5	1.25127	10.44185	5.58639	669.406	4.85546
53.6	1.25183	10.44652	5.59933	670.957	4.84719
53.7	1.25239	10.45119	5.61229	672.509	4.82890
53.8	1.25296	10.45595	5.62530	674.068	4.83065
53.9	1.25352	10.46062	5.63827	675.623	4.82235
54.0	1.25409	10.46538	5.65131	677.185	4.81407
54.1	1.25465	10.47005	5.66430	678.742	4.80575
54.2	1.25522	10.47481	5.67735	680.305	4.79746
54.3	1.25579	10.47957	5.69041	681.870	4.78916
54.4	1.25636	10.48432	5.70347	683.435	4.78085
54.5	1.25693	10.48908	5.71655	685.003	4.77253
54.6	1.25749	10.49375	5.72959	686.565	4.76416
54.7	1.25806	10.49851	5.74268	688.134	4.75583
54.8	1.25863	10.50327	5.75579	689.705	4.74748
54.9	1.25920	10.50802	5.76890	691.276	4.73912
55.0	1.25977	10.51278	5.78203	692.849	4.73075
55.1	1.26034	10.51754	5.79516	694.422	4.72238
55.2	1.26091	10.52229	5.80830	695.997	4.71399
55.3	1.26148	10.52705	5.82146	697.574	4.70559
55.4	1.26205	10.53181	5.83462	699.151	4.69719
55.5	1.26263	10.53665	5.84784	700.735	4.68881
55.6	1.26320	10.54140	5.86102	702.314	4.68038
55.7	1.26377	10.54616	5.87421	703.895	4.67195
55.8	1.26434	10.55092	5.88741	705.477	4.66351
55.9	1.26491	10.55567	5.90062	707.059	4.65505
56.0	1.26549	10.56051	5.91389	708.650	4.64662
56.1	1.26606	10.56527	5.92712	710.235	4.63815
56.2	1.26664	10.57011	5.94040	711.826	4.62971
56.3	1.26721	10.57487	5.95365	713.414	4.62122
56.4	1.26779	10.57971	5.96696	715.009	4.61275
56.5	1.26836	10.58446	5.98022	716.598	4.60424
56.6	1.26894	10.58930	5.99354	718.194	4.59576
56.7	1.26951	10.59406	6.00683	719.786	4.58723
56.8	1.27009	10.59890	6.02018	721.386	4.57872
56.9	1.27066	10.60366	6.03348	722.980	4.57018
57.0	1.27124	10.60850	6.04685	724.582	4.56165
57.1	1.27182	10.61334	6.06022	726.184	4.55312
57.2	1.27240	10.61818	6.07360	727.787	4.54458
57.3	1.27298	10.62302	6.08699	729.392	4.53603
57.4	1.27356	10.62786	6.10039	730.998	4.52747
57.5	1.27414	10.63270	6.11380	732.604	4.51890
57.6	1.27472	10.63754	6.12722	734.213	4.51032
57.7	1.27530	10.64238	6.14065	735.822	4.50173
57.8	1.27588	10.64722	6.15409	737.432	4.49313
57.9	1.27646	10.65206	6.16754	739.044	4.48452
58.0	1.27704	10.65690	6.18100	740.657	4.47590
58.1	1.27762	10.66174	6.19447	742.271	4.46727
58.2	1.27820	10.66658	6.20795	743.886	4.45863
58.3	1.27878	10.67142	6.22144	745.503	4.44998
58.4	1.27937	10.67634	6.23498	747.125	4.44136
58.5	1.27995	10.68118	6.24849	748.744	4.43269
58.6	1.28053	10.68602	6.26201	750.364	4.42401
58.7	1.28112	10.69095	6.27559	751.991	4.41536
58.8	1.28170	10.69579	6.28912	753.613	4.40667
58.9	1.28228	10.70063	6.30267	755.236	4.39796
59.0	1.28287	10.70555	6.31627	756.866	4.38928

(continued)

Table B-1 (continued)

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
59.1	1.28345	10.71039	6.32984	758.492	4.38055
59.2	1.28404	10.71531	6.34346	760.124	4.37185
59.3	1.28463	10.72024	6.35710	761.759	4.36314
59.4	1.28521	10.72508	6.37070	763.388	4.35438
59.5	1.28580	10.73000	6.38435	765.024	4.34565
59.6	1.28639	10.73492	6.39801	766.661	4.33691
59.7	1.28697	10.73976	6.41164	768.294	4.32812
59.8	1.28756	10.74469	6.42532	769.933	4.31937
59.9	1.28815	10.74961	6.43902	771.575	4.31059
60.0	1.28874	10.75454	6.45272	773.217	4.30182
60.1	1.28933	10.75946	6.46644	774.861	4.29302
60.2	1.28992	10.76437	6.48016	776.505	4.28422
60.3	1.29051	10.76931	6.49389	778.150	4.27542
60.4	1.29110	10.77423	6.50763	779.796	4.26660
60.5	1.29169	10.77915	6.52139	781.445	4.25776
60.6	1.29228	10.78408	6.53515	783.094	4.24893
60.7	1.29287	10.78900	6.54892	784.744	4.24008
60.8	1.29346	10.79392	6.56270	786.395	4.23122
60.9	1.29405	10.79885	6.57650	788.049	4.22235
61.0	1.29465	10.80385	6.59035	789.708	4.21350
61.1	1.29524	10.80878	6.60416	791.363	4.20462
61.2	1.29584	10.81378	6.61803	793.025	4.19575
61.3	1.29643	10.81871	6.63187	794.684	4.18684
61.4	1.29703	10.82372	6.64576	796.348	4.17796
61.5	1.29762	10.82864	6.65961	798.008	4.16903
61.6	1.29822	10.83365	6.67353	799.676	4.16012
61.7	1.29881	10.83857	6.68740	801.338	4.15117
61.8	1.29941	10.84358	6.70133	803.007	4.14225
61.9	1.30000	10.84850	6.71522	804.671	4.13328
62.0	1.30060	10.85351	6.72918	806.344	4.12433
62.1	1.30119	10.85843	6.74309	808.011	4.11534
62.2	1.30179	10.86344	6.74706	809.685	4.10638
62.3	1.30239	10.86844	6.77104	811.360	4.09740
62.4	1.30299	10.87345	6.78503	813.037	4.08842
62.5	1.30359	10.87846	6.79904	814.715	4.07942
62.6	1.30418	10.88338	6.81300	816.388	4.07038
62.7	1.30478	10.88839	6.82702	818.068	4.06137
62.8	1.30538	10.89340	6.84106	819.751	4.05234
62.9	1.30598	10.89840	6.85509	821.432	4.04331
63.0	1.30658	10.90341	6.86915	823.117	4.03426
63.1	1.30718	10.90842	6.88321	824.801	4.02521
63.2	1.30778	10.91342	6.89728	826.487	4.01614
63.3	1.30838	10.91843	6.91137	828.176	4.00706
63.4	1.30899	10.92352	6.92551	829.870	3.99801
63.5	1.30959	10.92853	6.93962	831.561	3.98891
63.6	1.31019	10.93354	6.95373	833.252	3.97981
63.7	1.31080	10.93863	6.96791	834.951	3.97072
63.8	1.31140	10.94363	6.98204	836.644	3.96159
63.9	1.31200	10.94864	6.99618	838.338	3.95246
64.0	1.31261	10.95373	7.01039	840.041	3.94334
64.1	1.31321	10.95874	7.02455	841.738	3.93419
64.2	1.31382	10.96383	7.03878	843.443	3.92505
64.3	1.31442	10.96883	7.05296	845.142	3.91587
64.4	1.31503	10.97393	7.06721	846.850	3.90672
64.5	1.31564	10.97902	7.08147	848.558	3.89755
64.6	1.31624	10.98402	7.09568	850.261	3.88834
64.7	1.31685	10.98911	7.10995	851.971	3.87916
64.8	1.31745	10.99412	7.12419	853.677	3.86993

Table B-1 (continued)

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
64.9	1.31806	10.99921	7.13849	855.391	3.86072
65.0	1.31867	11.00430	7.15280	857.106	3.85150
65.1	1.31928	11.00939	7.16711	858.820	3.84228
65.2	1.31989	11.01448	7.18144	860.538	3.83304
65.3	1.32050	11.01957	7.19578	862.256	3.82379
65.4	1.32111	11.02466	7.21013	863.975	3.81453
65.5	1.32172	11.02975	7.22449	865.696	3.80526
65.6	1.32233	11.03484	7.23886	867.418	3.79598
65.7	1.32294	11.03993	7.25323	869.140	3.78670
65.8	1.32355	11.04502	7.26762	870.864	3.77740
65.9	1.32416	11.05012	7.28203	872.591	3.76809
66.0	1.32477	11.05521	7.29644	874.318	3.75877
66.1	1.32538	11.06030	7.31086	876.046	3.74944
66.2	1.32600	11.06547	7.32534	877.781	3.74013
66.3	1.32661	11.07056	7.33978	879.511	3.73078
66.4	1.32723	11.07573	7.35428	881.249	3.72145
66.5	1.32784	11.08082	7.36875	882.983	3.71207
66.6	1.32846	11.08600	7.38328	884.724	3.70272
66.7	1.32907	11.09109	7.39776	886.459	3.69333
66.8	1.32969	11.09626	7.41230	888.201	3.68396
66.9	1.33030	11.10135	7.42680	889.939	3.67455
67.0	1.33092	11.10653	7.44138	891.686	3.66515
67.1	1.33153	11.11162	7.45590	893.426	3.65572
67.2	1.33215	11.11679	7.47048	895.173	3.64631
67.3	1.33277	11.12197	7.48509	896.923	3.63688
67.4	1.33338	11.12706	7.49964	898.667	3.62742
67.5	1.33400	11.13223	7.51426	900.419	3.61797
67.6	1.33462	11.13740	7.52888	902.171	3.60852
67.7	1.33523	11.14249	7.54347	903.919	3.59902
67.8	1.33585	11.14767	7.55812	905.674	3.58955
67.9	1.33647	11.15284	7.57278	907.431	3.58006
68.0	1.33709	11.15802	7.58745	909.189	3.57057
68.1	1.33771	11.16319	7.60213	910.948	3.56106
68.2	1.33833	11.16836	7.61682	912.708	3.55154
68.3	1.33895	11.17354	7.63153	914.471	3.54201
68.4	1.33957	11.17871	7.64624	916.234	3.53247
68.5	1.34020	11.18397	7.66102	918.005	3.52295
68.6	1.34082	11.18914	7.67575	919.770	3.51339
68.7	1.34144	11.19432	7.69050	921.537	3.50382
68.8	1.34206	11.19949	7.70525	923.305	3.49424
68.9	1.34268	11.20466	7.72001	925.073	3.48465
69.0	1.34331	11.20992	7.73484	926.850	3.47508
69.1	1.34393	11.21510	7.74963	928.623	3.46547
69.2	1.34456	11.22035	7.76448	930.402	3.45587
69.3	1.34518	11.22553	7.77929	932.177	3.44624
69.4	1.34581	11.23078	7.79416	933.959	3.43662
69.5	1.34643	11.23596	7.80899	935.736	3.42697
69.6	1.34706	11.24122	7.82389	937.521	3.41733
69.7	1.34768	11.24639	7.83873	939.299	3.40766
69.8	1.34831	11.25165	7.85365	941.087	3.39800
69.9	1.34893	11.25682	7.86852	942.869	3.38830
70.0	1.34956	11.26208	7.88346	944.659	3.37862
70.1	1.35019	11.26734	7.89841	946.451	3.36893
70.2	1.35083	11.27268	7.91342	948.249	3.35926
70.3	1.35147	11.27802	7.92845	950.050	3.34957
70.4	1.35210	11.28327	7.94342	951.844	3.33985
70.5	1.35274	11.28862	7.95848	953.649	3.33014
70.6	1.35338	11.29396	7.97354	955.453	3.32042
70.7	1.35401	11.29921	7.98854	957.251	3.31067

(continued)

Table B-1 (continued)

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
70.8	1.35465	11.30455	8.00362	959.058	3.30093
70.9	1.35529	11.30990	8.01872	960.867	3.29118
71.0	1.35593	11.31524	8.03382	962.677	3.28142
71.1	1.35656	11.32049	8.04887	964.480	3.27162
71.2	1.35720	11.32583	8.06399	966.292	3.26184
71.3	1.35784	11.33117	8.07912	968.105	3.25205
71.4	1.35847	11.33643	8.09421	969.913	3.24222
71.5	1.35911	11.34177	8.10937	971.730	3.23240
71.6	1.35975	11.34711	8.12453	973.546	3.22258
71.7	1.36038	11.35237	8.13965	975.358	3.21272
71.8	1.36102	11.35771	8.15484	977.178	3.20287
71.9	1.36166	11.36305	8.17003	978.998	3.19302
72.0	1.36230	11.36839	8.18524	980.821	3.18315
72.1	1.36293	11.37365	8.20040	982.638	3.17325
72.2	1.36357	11.37899	8.21563	984.463	3.16336
72.3	1.36421	11.38433	8.23087	986.289	3.15346
72.4	1.36484	11.38959	8.24606	988.109	3.14353
72.5	1.36548	11.39493	8.26132	989.937	3.13361
72.6	1.36612	11.40027	8.27660	991.768	3.12367
72.7	1.36675	11.40553	8.29182	993.592	3.11371
72.8	1.36739	11.41087	8.30711	995.424	3.10376
72.9	1.36803	11.41621	8.32242	997.259	3.09379
73.0	1.36867	11.42155	8.33773	999.094	3.08382
73.1	1.36930	11.42681	8.35300	1000.923	3.07381
73.2	1.36994	11.43215	8.36833	1002.760	3.06382
73.3	1.37058	11.43749	8.38368	1004.600	3.05381
73.4	1.37121	11.44275	8.39898	1006.433	3.04377
73.5	1.37185	11.44809	8.41435	1008.275	3.03374
73.6	1.37249	11.45343	8.42972	1010.116	3.02371
73.7	1.37312	11.45869	8.44505	1011.953	3.01364
73.8	1.37376	11.46403	8.46045	1013.799	3.00358
73.9	1.37440	11.46937	8.47586	1015.645	2.99351
74.0	1.37504	11.47471	8.49129	1017.494	2.98342
74.1	1.37567	11.47997	8.50666	1019.336	2.97331
74.2	1.37631	11.48531	8.52210	1021.186	2.96321
74.3	1.37695	11.49065	8.53755	1023.038	2.95310
74.4	1.37758	11.49591	8.55296	1024.884	2.94295
74.5	1.37822	11.50125	8.56843	1026.738	2.93282
74.6	1.37886	11.50659	8.58392	1028.594	2.92267
74.7	1.37949	11.51184	8.59934	1030.442	2.91250
74.8	1.38013	11.51718	8.61485	1032.300	2.90233
74.9	1.38077	11.52253	8.63037	1034.160	2.89216
75.0	1.38142	11.52795	8.64596	1036.028	2.88199
75.1	1.38207	11.53337	8.66156	1037.897	2.87181
75.2	1.38273	11.53888	8.67724	1038.776	2.86164
75.3	1.38338	11.54431	8.69287	1041.649	2.85144
75.4	1.38404	11.54981	8.70856	1043.529	2.84125
75.5	1.38470	11.55532	8.72427	1045.412	2.83105
75.6	1.38535	11.56075	8.73993	1047.288	2.82082
75.7	1.38601	11.56625	8.75565	1049.172	2.81060
75.8	1.38666	11.57168	8.77133	1051.051	2.80035
75.9	1.38732	11.57719	8.78709	1052.939	2.79010
76.0	1.38798	11.58269	8.80284	1054.827	2.77985
76.1	1.38863	11.58812	8.81856	1056.710	2.76956
76.2	1.38929	11.59363	8.83435	1058.602	2.75928
76.3	1.38994	11.59905	8.85008	1060.487	2.74897
76.4	1.39060	11.60456	8.86588	1062.381	2.73868
76.5	1.39126	11.61006	8.88170	1064.276	2.72836
76.6	1.39191	11.61549	8.89747	1066.166	2.71802

Table B-1 (continued)

Degrees Brix	Specific Gravity	Lb Total Weight per U.S. Gallon	Lb Solids per U.S. Gallon	G Solids per Liter	Lb Water per U.S. Gallon
76.7	1.39257	11.62100	8.91331	1068.064	2.70769
76.8	1.39322	11.62642	8.92909	1069.955	2.69733
76.9	1.39388	11.63193	8.94495	1071.855	2.68698
77.0	1.39454	11.63744	8.96083	1073.758	2.67661
77.1	1.39519	11.64286	8.97665	1075.654	2.66621
77.2	1.39585	11.64837	8.99254	1077.558	2.65583
77.3	1.39650	11.65379	9.00838	1079.456	2.64541
77.4	1.39716	11.65930	9.02430	1081.364	2.63500
77.5	1.39782	11.66481	9.04023	1083.273	2.62458
77.6	1.39847	11.67023	9.05610	1085.174	2.61413
77.7	1.39913	11.67574	9.07205	1087.086	2.60369
77.8	1.39978	11.68116	9.08794	1088.990	2.59322
77.9	1.40044	11.68667	9.10392	1090.905	2.58275
78.0	1.40110	11.69218	9.11990	1092.819	2.57228
78.1	1.40175	11.69760	9.13583	1094.728	2.56177
78.2	1.40241	11.70311	9.15183	1096.645	2.55128
78.3	1.40306	11.70854	9.16779	1098.558	2.54075
78.4	1.40372	11.71404	9.18381	1100.478	2.53023
78.5	1.40438	11.71955	9.19985	1102.400	2.51970
78.6	1.40503	11.72498	9.21583	1104.314	2.50915
78.7	1.40569	11.73048	9.23189	1106.239	2.49859
78.8	1.40634	11.73591	9.24790	1108.157	2.48801
78.9	1.40700	11.74142	9.26398	1110.084	2.47744
79.0	1.40766	11.74692	9.28007	1112.012	2.46685
79.1	1.40831	11.75235	9.29611	1113.934	2.45624
79.2	1.40897	11.75785	9.31222	1115.865	2.44563
79.3	1.40962	11.76328	9.32828	1117.789	2.43500
79.4	1.41028	11.76879	9.34442	1119.723	2.42437
79.5	1.41094	11.77429	9.36056	1121.657	2.41373
79.6	1.41159	11.77972	9.37666	1123.586	2.40306
79.7	1.41225	11.78523	9.39283	1125.524	2.39240
79.8	1.41290	11.79065	9.40894	1127.454	2.38171
79.9	1.41356	11.79616	9.42513	1129.394	2.37103

- Ignite the ethanol burner, and carefully position under the lower chamber in the proper position.
- Observe the thermometer mercury rising until it stops and holds for 15 to 20 seconds at the same temperature. Remove the ethanol burner, and close carefully to extinguish flame.
- Remove the thermometer carefully in reverse manner to Step 4 above. Hold in vertical position until the mercury drops from the capillary. Dry with a towel carefully, and place upright in a safe place.
- Adjust the circular slide scale to indicate the boiling point temperature of the water.
- Empty the ebulliometer carefully, and rinse inner surfaces with a few milliliters of the wine sample to be analyzed. Drain the instrument, and fill the upper reflux condenser with cold tap water. Ensure that no water goes down the inner tube.

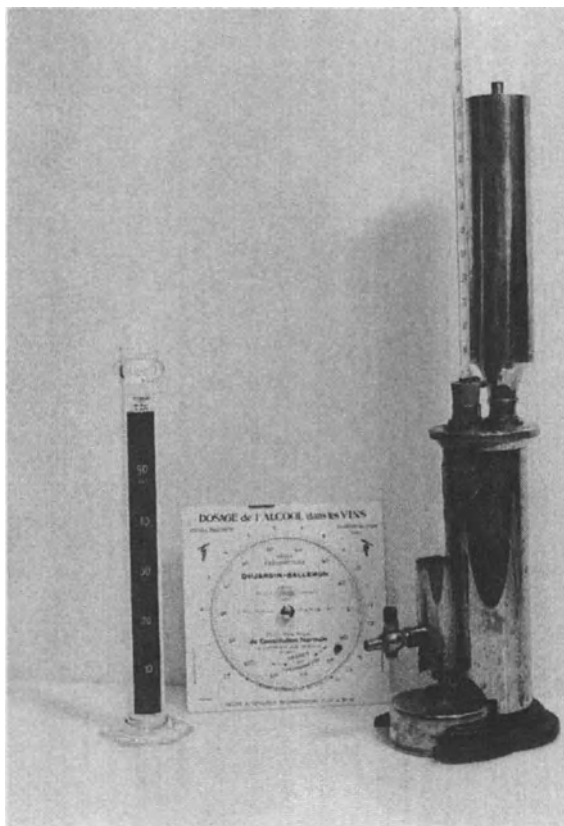


Fig. B-9 Ebulliometer in service.

10. Rinse the 100-mL graduate with a few milliliters of the wine sample to be analyzed. Empty and refill with 50 mL of the wine.
11. Repeat Steps 4, 5, 6, and 7.
12. Compare the reading of the thermometer to corresponding alcohol percentage on the circular slide scale. For example, with water boiling at 99.8°C, and a wine boiling at 91.1°C, the alcohol would be 12.1% by volume.

EXTRACT BY NOMOGRAPH

Extract is virtually the same as Brix, except that the term *extract* implies that the alcohol of the wine has been removed and replaced with water. As this is a rather

tedious process, extract can be determined by carefully using a special graph called a nomograph.

Use of a nomograph requires prior knowledge of the alcohol content and Balling of the wine to be analyzed. The procedure is simply to find the extract by placing a straight-edge, such as a ruler, so as to intersect the known alcohol and Balling levels. For example, using the nomograph in Figure B-10, an alcohol of 12.0% by volume and a Balling of -2.0° results in an extract of 2.0 g/100mL of soluble solids.

TOTAL ACIDITY

This analysis measures the unknown concentration of total wine acidity by neutralizing those acids with a known concentration of a basic (alkaline) reagent. This method is shown in Figure B-11.

Apparatus and Reagents

- 10-mL volumetric pipet
- 250-mL wide-mouth Erlenmeyer flask
- 25- or 50-mL (in 0.1-mL gradations) buret with stopcock
- 1/10 N sodium hydroxide reagent
- 1% phenolphthalein indicator solution
- Distilled or deionized water

Procedure

1. If the sample has been taken from a freshly pressed lot of juice or must, there may be too many suspended solids for the pipet to allow flow through the narrow capillary at the tip. In this case, the juice sample should be carefully centrifuged or filtered through neutral paper. Adjust sample temperature to 68°F or to whatever temperature is indicated upon the 10-mL volumetric pipet.
2. Pipet the sample into a clean 250-mL wide-mouth Erlenmeyer flask, and add five drops of 1% phenolphthalein indicator into the sample.
3. Fill the buret with 1/10 N sodium hydroxide reagent, and slowly drain down to the 0.0 starting point. Ensure that there is no air remaining in the outlet capillary.
4. Titrate carefully drop by drop until the slight pink color from the indicator turns pink (green for red wine) and holds for at least 15 seconds. Note the buret reading for milliliters of 1/10 N sodium hydroxide used.
5. Grams per 100 mL total acidity is found by using the Total Acidity Table (Table B-2). For example, when 9.0 mL of 1/10 N sodium hydroxide is used in titration, the resulting total acidity is .675 g/100 mL.

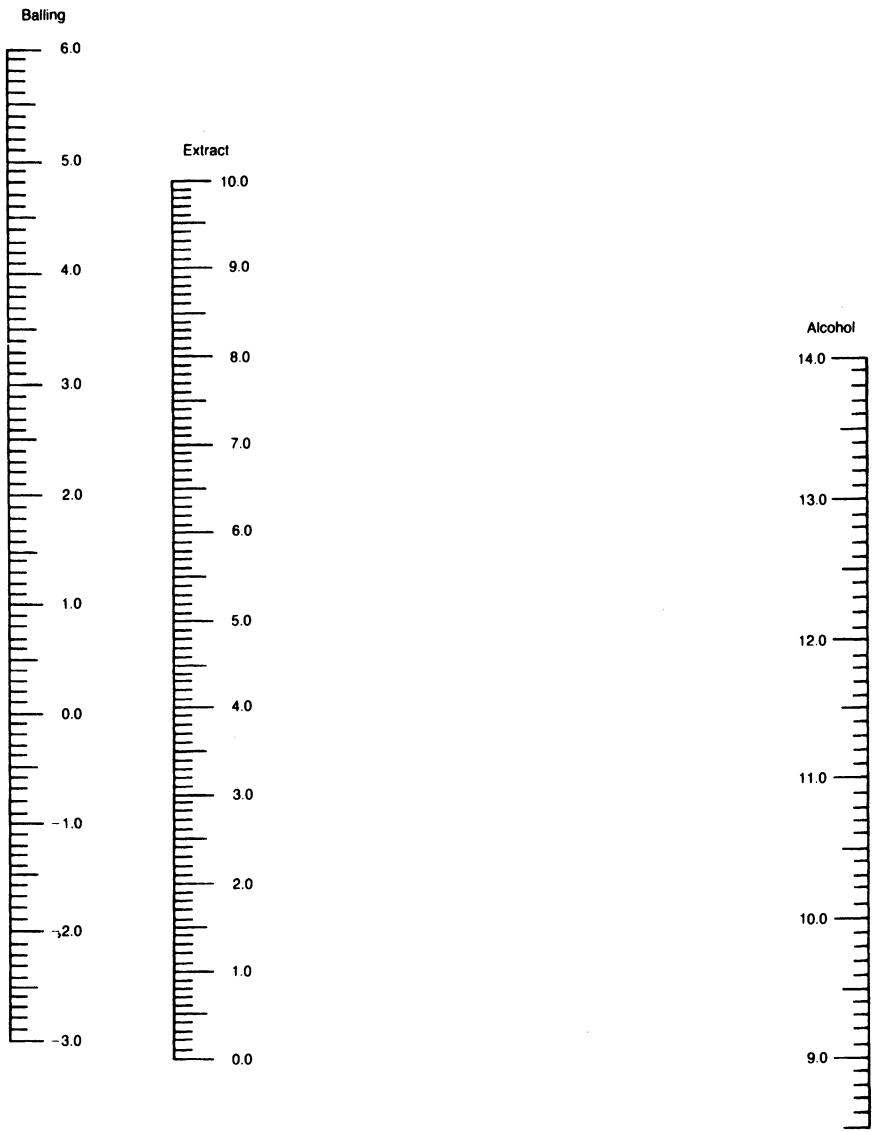


Fig. B-10 Table wine nomograph.
Courtesy: J.M. Vahl, Am. J. Enol. Vitic. 30(3) 1979.

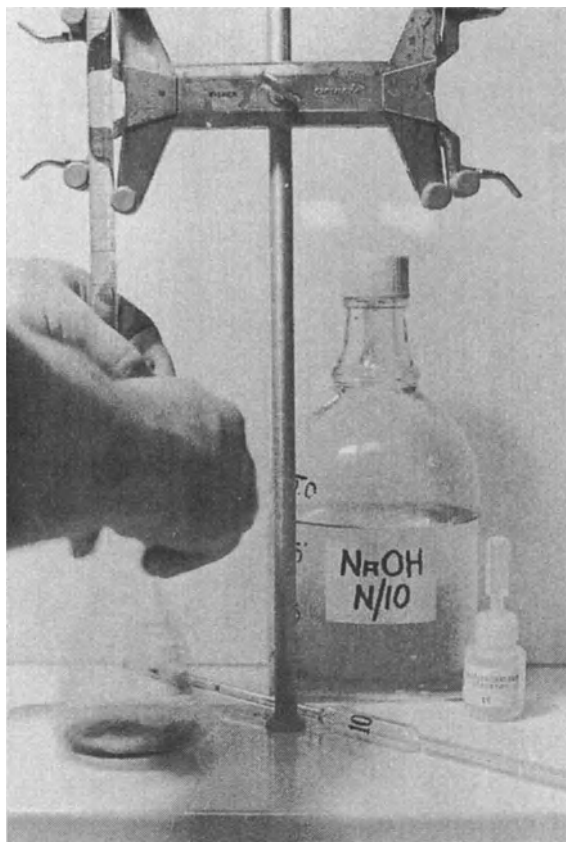


Fig. B-11 Total acidity titration.

VOLATILE ACIDITY BY CASH STILL

This method of determining volatile acidity separates the volatile, or distillable, acids from the fixed acids in the Cash still (Figure B-12). The distilled acids are then titrated separately in the same manner as for a total acidity determination.

Apparatus and Reagents

- Cash volatile acid apparatus, complete with condenser, support clamps, and rods
- 1/4-inch ID Tygon tubing
- 10-mL volumetric pipet

Table B-2 Total acidity table (g/100 mL expressed as tartaric acid).

1/10 N NaOH	g/100 ml T.A.	1/10 N NaOH	g/100 ml T.A.	1/10 N NaOH	g/100 ml T.A.
5.1	.383	9.1	.683	13.1	.983
5.2	.390	9.2	.690	13.2	.990
5.3	.398	9.3	.698	13.3	.998
5.4	.405	9.4	.705	13.4	1.005
5.5	.413	9.5	.713	13.5	1.013
5.6	.420	9.6	.720	13.6	1.020
5.7	.428	9.7	.728	13.7	1.028
5.8	.435	9.8	.735	13.8	1.035
5.9	.443	9.9	.743	13.9	1.043
6.0	.450	10.0	.750	14.0	1.050
6.1	.458	10.1	.758	14.1	1.058
6.2	.465	10.2	.765	14.2	1.065
6.3	.473	10.3	.773	14.3	1.073
6.4	.480	10.4	.780	14.4	1.080
6.5	.488	10.5	.788	14.5	1.088
6.6	.495	10.6	.795	14.6	1.096
6.7	.503	10.7	.803	14.7	1.103
6.8	.510	10.8	.810	14.8	1.110
6.9	.518	10.9	.818	14.9	1.118
7.0	.525	11.0	.825	15.0	1.125
7.1	.533	11.1	.833	15.1	1.133
7.2	.540	11.2	.840	15.2	1.140
7.3	.548	11.3	.848	15.3	1.148
7.4	.555	11.4	.855	15.4	1.155
7.5	.563	11.5	.863	15.5	1.163
7.6	.570	11.6	.870	15.6	1.170
7.7	.578	11.7	.878	15.7	1.178
7.8	.585	11.8	.885	15.8	1.185
7.9	.593	11.9	.893	15.9	1.193
8.0	.600	12.0	.900	16.0	1.200
8.1	.608	12.1	.908	16.1	1.208
8.2	.615	12.2	.915	16.2	1.215
8.3	.623	12.3	.923	16.3	1.223
8.4	.630	12.4	.930	16.4	1.230
8.5	.638	12.5	.938	16.5	1.238
8.6	.645	12.6	.945	16.6	1.245
8.7	.653	12.7	.953	16.7	1.253
8.8	.660	12.8	.960	16.8	1.260
8.9	.668	12.9	.968	16.9	1.268
9.0	.675	13.0	.975	17.0	1.275

- 250-mL Erlenmeyer flask with rubber stopper to fit
- 25- or 50-mL (in 0.1-mL gradations) buret with stopcock
- 1/10 N sodium hydroxide reagent
- 1% phenolphthalein indicator
- Distilled or deionized water
- Continuous cold tap water connected into condenser and out into drain

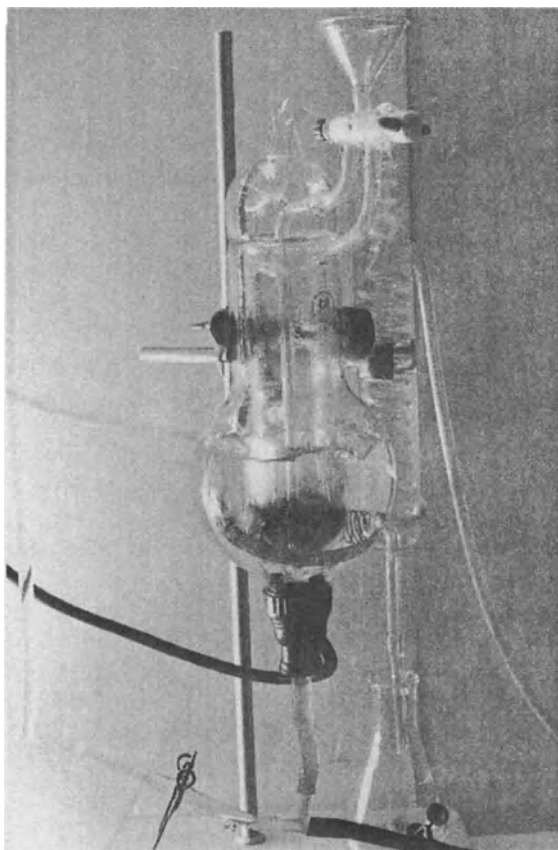


Fig. B-12 Cash volatile acid still in service.

Procedure

1. Turn the stopcock on the Cash still so that the passage runs from the funnel to the inner tube.
2. Adjust the sample temperature to 68°F or to whatever temperature is indicated on the 10-mL volumetric pipet.
3. Fill the lower (outer) pot with distilled water up to a level that is about 1 inch above the heating element.
4. Pipet the sample into the funnel, and drain into the inner tube of the Cash still. Wash the funnel into the inner tube with 10 to 15 mL of distilled water.
5. Turn the stopcock so that the passage goes to the outer tube. Place a clean 250-mL Erlenmeyer flask under the condenser outlet. (It is recommended to place about

- 1 inch of Tygon tubing upon the condenser outlet to avoid chipping of both the outlet and the flask.)
6. Start running cold tap water through the condenser at a moderate rate, and turn on electricity to the heating coil in the lower pot.
 7. Small bubbles should form upon the heating coil. They should be allowed to boil off. Keep the stopcock turned to the outer tube position. When “spitting” commences, turn the stopcock to the horizontal position, closing off both tubes.
 8. Distill over approximately 100 mL of distillate. Be sure that the distillate runs out of the condenser at a temperature not exceeding 80°F. Make several trial runs to determine the ideal cold water flow rate. Increase the condenser water flow as needed, but not so fast to break the condenser from excessive pressure.
 9. After the distillate is collected, turn off electric power to the heating coil, and turn the stopcock to the inner tube passage. Remove the 250-mL Erlenmeyer flask carefully, and close with the rubber stopper.
 10. Open the outlet tube to drain, and add approximately 50 mL of distilled water into the funnel. The Cash apparatus should “cough,” as it cleans itself by the vacuum created by water draining out of the lower pot. Repeat twice more, spacing the “coughing” about equally at the beginning, middle, and end of outer tube drainage. Turn off the condenser water, and allow the apparatus to drain.
 11. Remove the stopper from the distillate flask, and add five drops of phenolphthalein indicator solution. Titrate very carefully with 1/10 N sodium hydroxide solution from buret in the same manner as for the total acidity analysis described above. Note the buret reading for milliliters of 1/10 sodium hydroxide used.
 12. Grams per 100 mL volatile acidity is found by using the Volatile Acidity Table (Table B-3). For example, when 0.5 mL of 1/10 N sodium hydroxide is used in titration, the resulting volatile acidity is .030 g/100 mL.

FREE SULFUR DIOXIDE

Free sulfur dioxide, as the term suggests, is dynamic and can quickly change in concentration. Knowledge of the actual concentration of free sulfur dioxide and the ability to adjust concentrations to a desired level are of immense importance in fine wine production. For this reason, the accurate analysis of sulfur dioxide should be the most important analytical procedure in a winery. Two conditions that contribute to rapidly diminishing free sulfur dioxide in wine are agitation (oxidation) and warmer temperatures (volatilization). Consequently, wine samples taken for free sulfur dioxide determination should be analyzed as soon as possible.

There are inexpensive kits available for instant free sulfur dioxide analysis. They may serve as quick checks but should not be used for finite quantitative determinations. The recommended method for analyzing free and bound sulfur dioxide is the aeration oxidation distillation and titration procedure (Figures B-13 and B-14). This requires specialized glassware and careful technique, but the accuracy of the

Table B-3 Volatile acidity table (g/100 mL expressed as acetic acid) (For use when distilling 10 mL sample in cash V.A. apparatus and titrating using N/10 NaOH).

(Ml) NaOH	G/100 ml VA
.1	.006
.2	.012
.3	.018
.4	.024
.5	.030
.6	.036
.7	.042
.8	.048
.9	.054
1.0	.060
1.1	.066
1.2	.072
1.3	.078
1.4	.084
1.5	.090
1.6	.096
1.7	.102
1.8	.108
1.9	.114
2.0	.120
2.1	.126
2.2	.132
2.3	.138
2.4	.144
2.5	.150
2.6	.156
2.7	.162
2.8	.168
2.9	.174
3.0	.180

results, especially for red wine, justifies the extra effort. The most common procedure for sulfur dioxide analysis has been the Ripper titrametric method (Figure B-15). It is important to note that due to the instability of the iodine solution, requiring frequent calibration, and inaccuracies caused by nonspecific interaction of red wine pigments with iodine, results are often questionable.

It should be fully understood that the pH level in wine has a major influence on the effectiveness of free sulfur dioxide as a preservative. The active portion of free sulfur dioxide is referred to as *molecular sulfur dioxide*, which is correlated to the pH level in every wine. Consequently, the results from every free sulfur dioxide determination should be recalculated using the Approximate Effective Sulfur Dioxide Table (Table B-5). For example, using Table B-5, if the molecular sulfur dioxide desired is 1.4 ppm, then sufficient sulfite will need to be added to reach a free sulfur dioxide level of 37 ppm at a pH of 3.20. To reach the same 1.4 molecular level at a pH of 3.40 will require sufficient sulfite to reach a free sulfur dioxide level of 56 ppm.

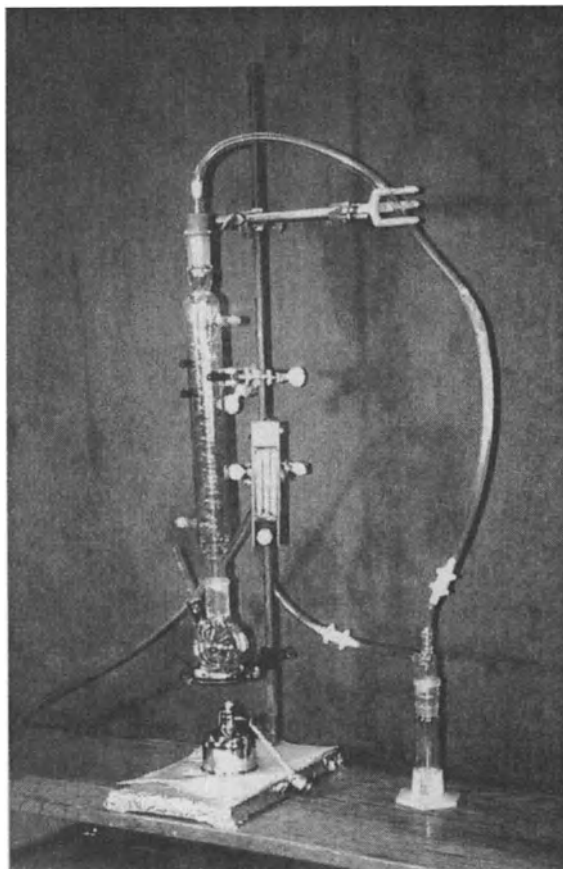


Fig. B-13 Aeration oxidation distillation and titration method for free and total sulfur dioxide.

Aeration Oxidation Distillation and Titration Procedure for Free Sulfur Dioxide*

Apparatus and Reagents

- Glass distillation unit (for free sulfur dioxide, the condenser unit may be omitted)
- Ice bath
- Volumetric pipettes (10- and 20-mL)
- Burette (10-mL)
- 0.01% sodium hydroxide

*From B.W. Zoecklein et al., *Wine Analysis and Production*. Gaithersburg, MD: Aspen Publishers, 1995.

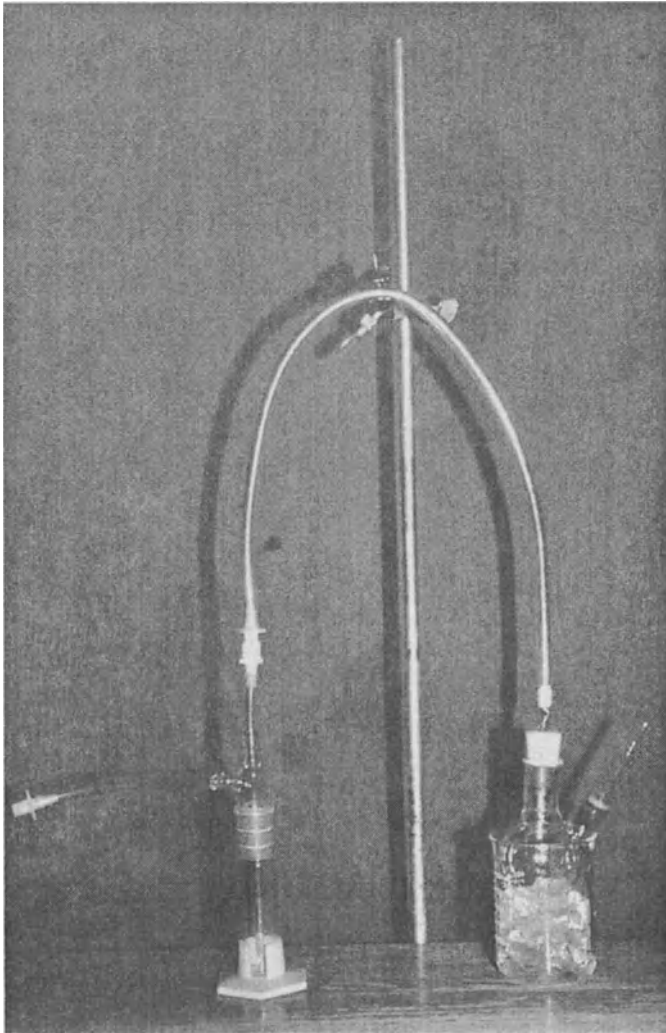


Fig. B-14 Alternative aeration oxidation distillation and Titration Method for free sulfur dioxide only.

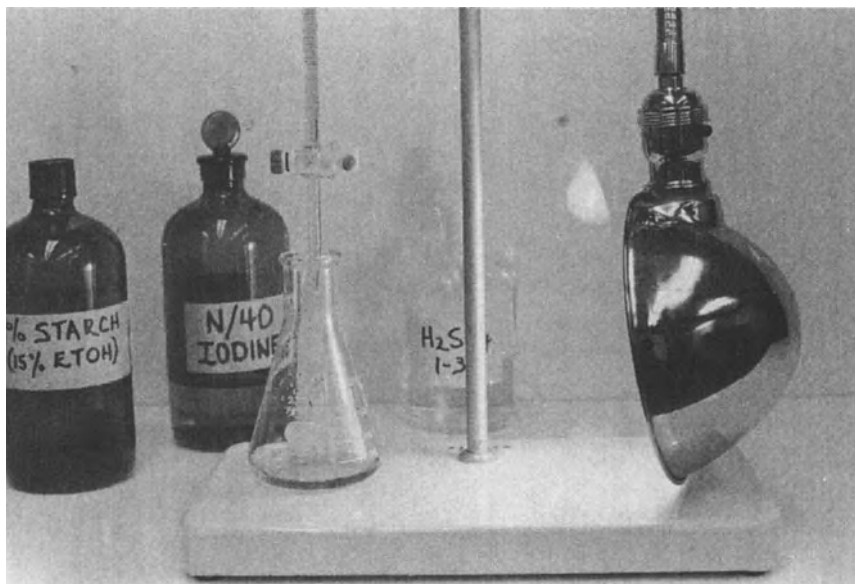


Fig. B-15 Free sulfur dioxide titration by Ripper Method.

- 0.3% hydrogen peroxide (immediately before use, dilute 1 mL of 30% hydrogen peroxide with 99 mL of distilled water)
- Phosphoric acid: 1 + 3 (1 part *o*-phosphoric acid 85% stock, plus 3 parts distilled water)
- Indicator solution (methylene blue/methyl red solution)

Procedure

1. Assemble glassware as directed by manufacturer (Figures B-13 and B-14).
2. Run a vacuum hose between a water aspirator and the side arm of the vacuum adapter. Adjust water flow through the aspirator to produce a moderately vigorous bubbling in the receiving flask.
3. Place an ice bath under the round-bottom flask.
4. Rinse all glassware with deionized water. To the receiving flask, add 10 mL of 0.3% hydrogen peroxide solution and 6 drops of indicator. If the color in the flask is purple, add 0.01N NaOH (usually one drop) to a turquoise-green end point. Connect the receiving flask to the vacuum adapter.
5. Distillation of sample: through the side port, add exactly 20.0 mL of sample to the round-bottom flask, which is immersed in an ice bath. Add 10 mL of 1 + 3 phosphoric acid to the same flask and quickly replace bubbler/stopper.
6. Begin aspirating (bubbles should be vigorous but not enough to go beyond half the volume of the receiving flask,) and continue aspirating the sample for exactly 10 minutes. If free sulfur dioxide was present in the sample, the solution in the receiving flask should become purple.

7. At the end of the 10-minute period, turn off aspiration. Remove the receiving flask. Using deionized water, rinse the inside of the vacuum adapter, and the outside of the bubbler tube connected to it, into the receiving flask.
8. Using a 10-mL burette, titrate the contents of the receiving flask to a turquoise-green end point. Read titration volume to the nearest 0.01 mL. Using the formula below, calculate sulfur dioxide concentration to the nearest ± 1 ppm:

$$\text{SO}_2 \text{ ppm} = \text{mL NaOH} \times N \text{ NaOH} \times 1,600$$

Free Sulfur Dioxide by Ripper Method

Apparatus and Reagents

- 20-mL volumetric pipet
- 250-mL wide-mouth Erlenmeyer flask with fitted stopper
- 25- or 50-mL (in 0.1-mL gradations) buret with stopcock
- 1/40 N iodine reagent
- 1% starch indicator solution
- Bicarbonate of soda
- 25% sulfuric acid
- 25% sodium hydroxide
- Distilled or deionized water

Procedure

1. Adjust sample temperature to 68°F, or to whatever temperature is indicated on the 20-mL volumetric pipet.
2. Pipet the sample into a clean 250-mL wide-mouth Erlenmeyer flask.
3. Add 5 mL of 25% sulfuric acid, then a pinch of bicarbonate of soda and 5 mL of 1% starch indicator solution.
4. Fill the buret with 1/40 N iodine, and drain down to the 0.0 mark. Ensure that no air remains in the outlet capillary.
5. Titrate carefully with 1/40 N iodine from the buret until the color changes to light blue (blue-green in reds) and holds for at least 15 seconds. Note the buret reading for milliliters of 1/40 N iodine used. (Note: Addition of up to 20 mL of distilled or deionized water and/or an appropriately situated light may help with some very dark or turbid samples.)
6. Milligrams per liter (parts per million) of free sulfur dioxide is found by using the Sulfur Dioxide Table (Table B-4). For example, when 1.4 mL of 1/40 N iodine reagent is used in titration, the resulting free sulfur dioxide is 56 mg/L (ppm).

Milligrams per liter (parts per million [ppm]) of free sulfur dioxide is found by using the following formula:

$$\text{ppm sulfur dioxide} = \text{mL iodine} \times N \text{ iodine} \times 1,600$$

Table B-4 Sulphur dioxide table (20 mL sample $\times$ 1/40 N iodine $\times$ 1% starch indicator).

MI Iodine	Ppm SO ₂	MI Iodine	Ppm SO ₂
.1	4	4.1	164
.2	8	4.2	168
.3	12	4.3	172
.4	16	4.4	176
.5	20	4.5	180
.6	24	4.6	184
.7	28	4.7	188
.8	32	4.8	192
.9	36	4.9	196
1.0	40	5.0	200
1.1	44	5.1	204
1.2	48	5.2	208
1.3	52	5.3	212
1.4	56	5.4	216
1.5	60	5.5	220
1.6	64	5.6	224
1.7	68	5.7	228
1.8	72	5.8	232
1.9	76	5.9	236
2.0	80	6.0	240
2.1	84	6.1	244
2.2	88	6.2	248
2.3	92	6.3	252
2.4	96	6.4	256
2.5	100	6.5	260
2.6	104	6.6	264
2.7	108	6.7	268
2.8	112	6.8	272
2.9	116	6.9	276
3.0	120	7.0	280
3.1	124	7.1	284
3.2	128	7.2	288
3.3	132	7.3	292
3.4	136	7.4	296
3.5	140	7.5	300
3.6	144	7.6	304
3.7	148	7.7	308
3.8	152	7.8	312
3.9	156	7.9	316
4.0	160	8.0	320

AERATION OXIDATION DISTILLATION AND TITRATION PROCEDURE FOR TOTAL SULFUR DIOXIDE*

Each wine has some portion of the total amount of sulfur dioxide existing in combination with other wine components. This is called *bound sulfur dioxide*. The

*From B.W. Zoecklein et al., *Wine Analysis and Production*. Gaithersburg, MD: Aspen Publishers, 1995.

Table B-5 Approximate effective sulfur dioxide conversion table.

ppm molecular SO ₂	ppm @ pH 3.10	ppm @ pH 3.20	ppm @ pH 3.30	ppm @ pH 3.40	ppm @ pH 3.50	ppm @ pH 3.60
0.5	10	13	16	20	25	31
0.8	16	21	26	32	30	50
1.1	22	29	36	44	55	69
1.4	28	37	46	56	70	88
1.7	34	45	56	68	85	108
2.0	40	53	66	80	100	126

combination of bound and free sulfur dioxide is called *total sulfur dioxide* and is limited to a maximum of 350 ppm by ATF regulations in 27 CFR, Part 4, Section 4.22.

Apparatus and Reagents

- Glass distillation unit with condenser
- Microburner or alcohol lamp
- Other equipment same as for the free sulfur dioxide method

Procedure

1. Assemble glassware as directed by manufacturer (Figure B-13).
2. Follow procedure as listed above for free sulfur dioxide, replacing ice bath with microburner. The same sample in the round-bottom flask as used above for the free analysis is now used for bound sulfur dioxide. (Alternatively, a new 20-mL sample may be introduced to the cleaned round-bottom flask.) The contents of the round-bottom flask (sample and phosphoric acid) are heated to a boil with the microburner. A few drops of antifoam may be necessary to control excessive foaming during distillation. When boiling commences, aspirate vigorously for exactly 15 minutes.
3. At the end of the 15-minute period, turn off aspiration and burner. Remove the receiving flask. Using deionized water, rinse the inside of the vacuum adapter, and the outside of the bubbler tube connected to it, into the receiving flask.
4. Using a 10-mL burette, titrate the contents of the receiving flask to a turquoise-green end point. Read titration volume to the nearest 0.01 mL. Using the same formula below, calculate bound sulfur dioxide concentration to the nearest ± 1 ppm:

$$\text{SO}_2 \text{ ppm} = \text{mL NaOH} \times N \text{ NaOH} \times 1,600$$

5. To calculate total sulfur dioxide, add the results for free and for bound sulfur dioxide. If a new sample was introduced to perform the bound analysis, those results represent total sulfur dioxide.

TOTAL SULFUR DIOXIDE BY RIPPER METHOD

One of the most popular methods for determining total sulfur dioxide is the following modified Ripper method.

Apparatus and Reagents

These are the same as listed in the section "Free Sulfur Dioxide by Ripper Method."

Procedure

1. Adjust the sample temperature to 68°F, or to whatever temperature is indicated on the 20-mL volumetric pipet.
2. Pipet the sample into a clean 250-mL wide-mouth Erlenmeyer flask, and add 5 mL of 25% sodium hydroxide. Seal with a rubber stopper. Swirl for a few seconds, and set aside for about 15 minutes.
3. Remove the stopper, and add 5 mL of 25% sulfuric acid, a pinch of bicarbonate of soda, and 5 drops of 1% starch solution to sample in flask.
4. Proceed with titration and determination in the same manner as for free sulfur dioxide above.

SENSORY EVALUATION

The single most important element of analysis in quality control is sensory examination and criticism (Figures B-16 and B-17). There is no substitute for building learning and experience in wine judging, which is achieved, of course, through learning proper procedure and continued practice.

Beginning winemakers should sample a large number of positive and negative wine constituents and components in water and/or water-ethanol solutions. This will help in learning to identify these compounds without the masking effects in more complex wine solutions. The ability to commit these to sensory memory will help develop the critical eye, nose, and palate that are essential to long-term success in commercial wine production.

Apparatus and Conditions

- Several dozen identical plain (uncolored) tulip-shape wine glasses
- Solid white table top or counter top
- Mild cheese
- Salt-free crackers
- Noncarbonated bottled water
- Hot water for cleaning
- Deionized water for rinsing

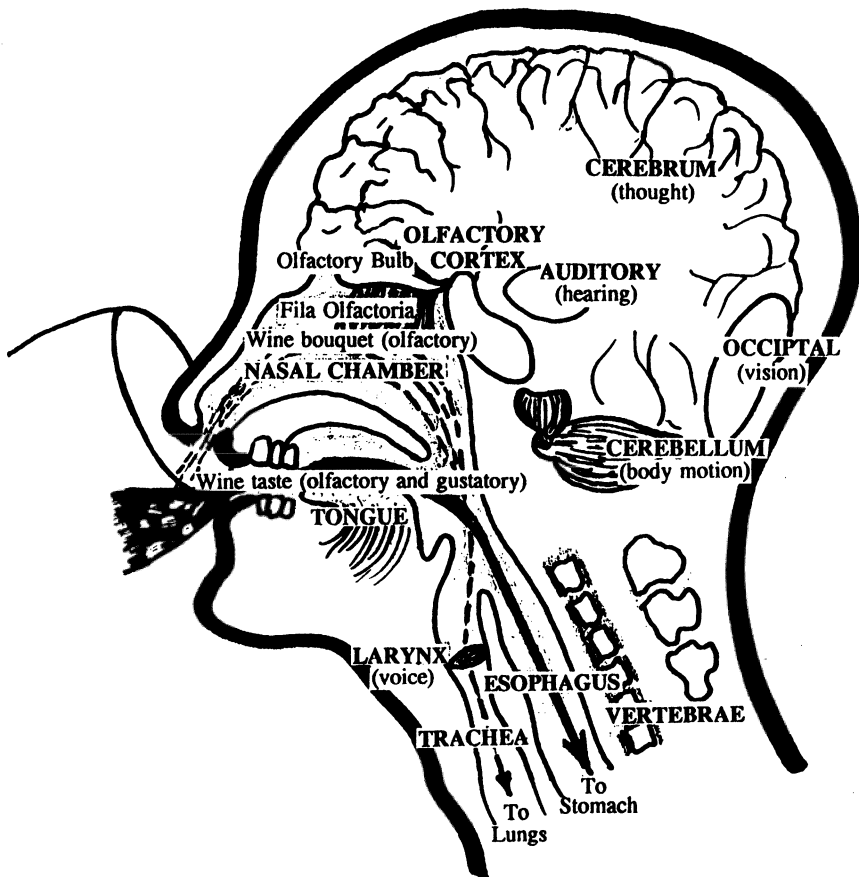


Fig. B-16 Human sensory organs.

- Drain racks for drying
- Cool room temperature
- Noiseless and odorless room maintained at cool temperature and moderate humidity with adjustable incandescent light
- Sufficient pencils and score sheets

Procedure

Visual Mode

The visual mode (Figure B-18) in wine sensory evaluation is divided into two major categories, clarity and color—and, in the case of sparkling wines, effervescence too.

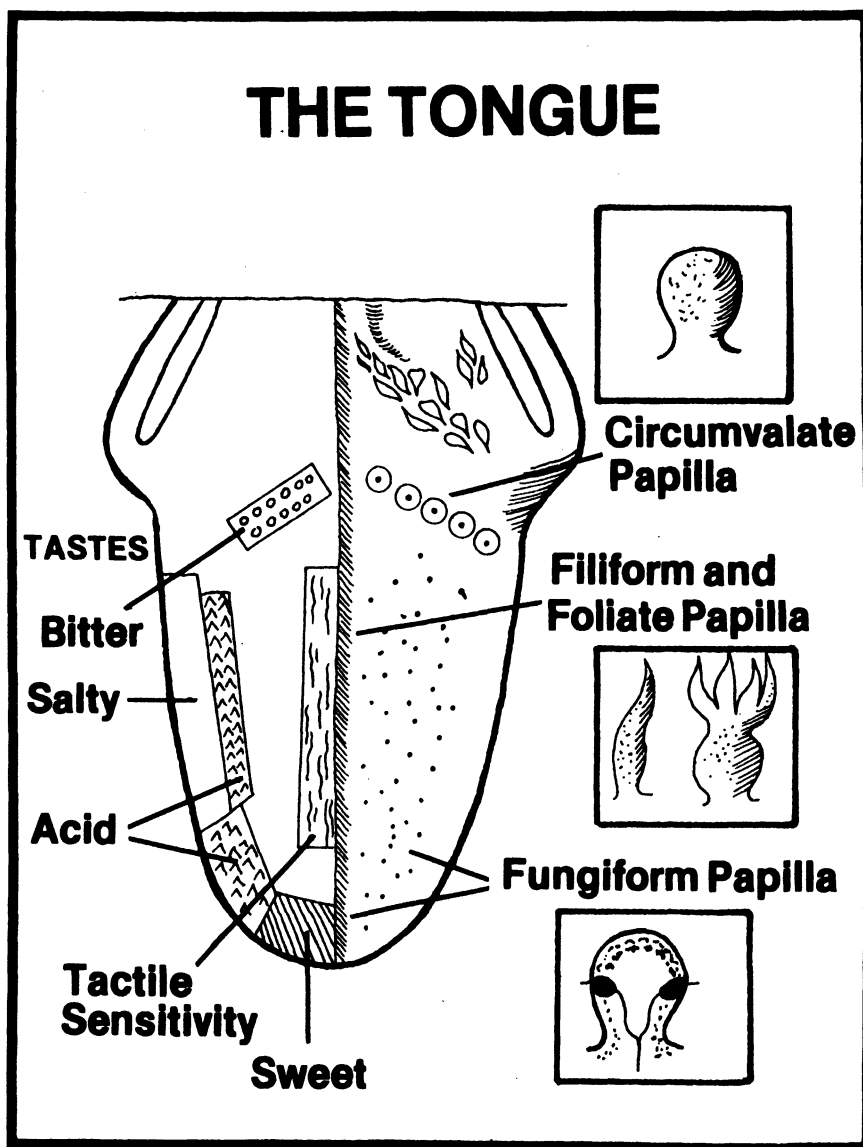


Fig. B-17 Different types of papillae found on the human tongue.

Adapted from Puisais, J. and Chabanon, R.L. 1974. "Initiation into the Art of Wine Tasting," Interpublish, Madison, WI.



Fig. B-18 The visual mode.

Photo by David King, Purdue University Agricultural Communications.

The best procedure in evaluating clarity is to hold the wine glass by the stem or base so that a constant light source behind the glass can filter through the wine. Wine with perfect clarity has no trace of suspended particles or lint. The four generally accepted echelons of wine clarity are (1) brilliant, (2) clear, (3) hazy, and (4) cloudy. To replicate a hazy condition, fill an 8-ounce wine glass half full with water, and add one drop of whole milk. Two or three more drops will render a cloudy mixture.

The first concern for color is its hue, or an identity of specific color value such as “straw-gold” or “ruby-red.” Judgment is then made upon color intensity—the quantity of hue present in the wine. Color evaluations are best made upon a stark white background such as plain paper or a table napkin. By looking downward through the glass, one can perceive variances in both hue and intensity as stationary light passes through different depths of the wine. Obviously, the wine will have the same color hue throughout, but intensity will be more dense in the center of the glass, where the wine is deeper. Conversely, intensity will be lighter around the edge of the glass, where the wine is more shallow. The dilution of intensity in the shallow portion of the glass can reveal nuances of color, such as tones of tawny-brown in older red wines, which are hidden by the heavier density in the center of the glass.

Table B-6 Typical color judgments for various wine types.

Wine type	Color hue	Color intensity
Dry Vermouth	Pale celery	Extremely light
Sauvignon Blanc	Pale straw	Very light
Chardonnay	Straw	Light
Sauternes	Golden	Moderate
Blush	Pink	Light
Anjou Rosé	Rosé	Moderate
Pinot Noir	Crimson red	Light
Merlot	Scarlet red	Moderate
Cabernet Sauvignon	Garnet red	Dark
Port	Ruby red	Dense
Tawny Port	Amber-red	Moderate
Oloroso Sherry	Amber	Dark

Some judges prefer to make this examination by holding the glass at various angles and looking from side to side through the bowl of the glass. Whichever approach is adopted, a judge should remain consistent.

Table B-6 shows some typical color judgments for various wine types.

Olfactory Mode

The olfactory mode is the single most important element in wine judging. It involves the most sensitive human organ, the nose, and is generally the primary factor in deciding upon wine quality. For all these reasons, the entire scope of the olfactory mode comprises the most points on every good cardinal scale of wine judging.

The first concern is that of *bouquet*—often referred to as the “nose” by wine judges. Wine bouquet is perhaps best compared to a bouquet of flowers. Each flower in the bouquet expresses its own characteristic odor that contributes to a complex array of smells overall. In the case of wine, the part of the bouquet that is grape flavor is called the *aroma*. The aroma, combined with odors resulting from the winemaking process, such as malo-lactic fermentation and barrel aging, results in a complex array of smells collectively referred to as bouquet. A typical Chardonnay wine, for example, may have a slight ripe olive and herbal aroma. The nose may also detect a “buttery” flavor resulting from a special malo-lactic bacterial fermentation and a “vanilla” flavor due to oak barrel aging. The combined olive-herbal aroma and buttery-vanilla flavors make up the overall bouquet of this Chardonnay. As mentioned in the previous section, a properly trained human olfactory and temporal system can identify, classify, store, and recall hundreds of different flavors. Some of these are delicious experiences, while others are unpleasant odors resulting

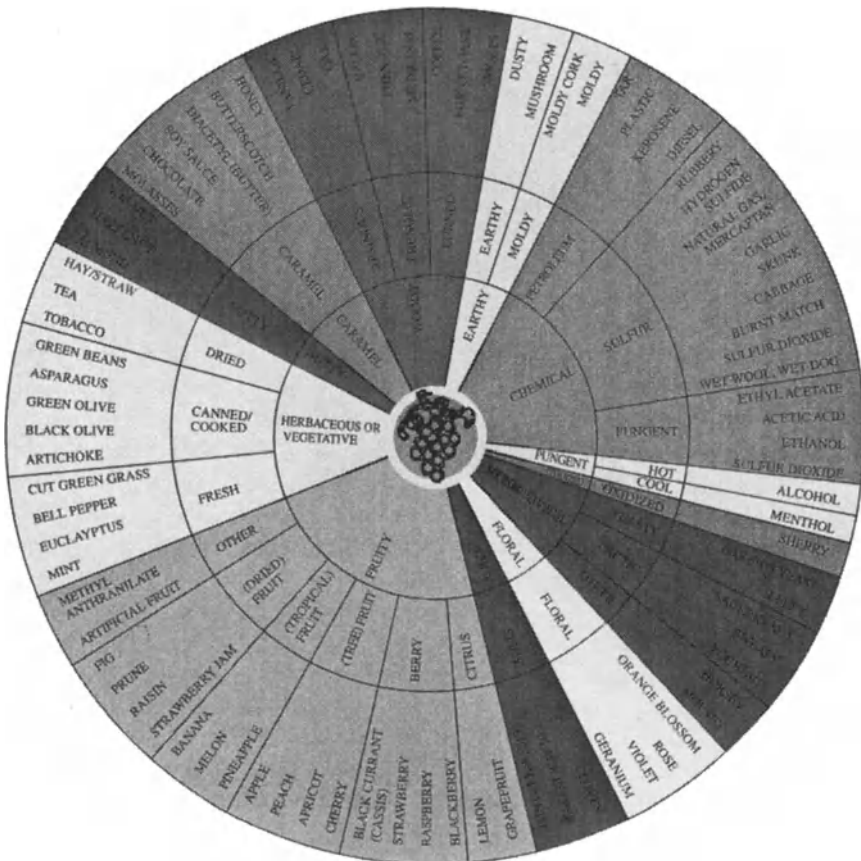


Fig. B-19 The aroma wheel.

Courtesy: Noble, A.C., et al, "Progress Towards a Standardized System of Wine Aroma Terminology," *Am. J. Enol. Vitic.*, Vol. 35, No.2, 1984.

from winemaking problems. An experienced wine judge will be familiar with both.

One system of wine flavor terminology is organized in the form of an "aroma wheel" (Figure B-19). It categorizes primary, secondary, and tertiary levels of flavor definition. While not yet perfect, it does establish a basis for communicating effectively in discussing the many and varied olfactory impressions that exist in the vast world of wine.

In many white grape varieties, we find *primary floral aromas*. Among these are acacia, grapefruit, honeysuckle, linden, peach, pear, rose, peony, and violets. These

flavor values are most often found in grapes that have been harvested slightly before maturity, before the *primary fruit aromas* have formed during full maturation.

Many of the primary floral and fruit aromas exist in the form of higher terpene alcohols such as citronellol, linalol, and geraniol. These are commonly found in Johannisberg Riesling, Gewurztraminer, Vidal Blanc, Vignoles, and most of the Muscat varieties.

Other aromas are identified in the form of esters. Among these are the isoamyl acetate, associated with bananas; ethyl propionate, associated with apples; and methyl anthranilate, found in the native American *Vitis labrusca* cultivars such as Niagara and Concord.

Primary fruit aromas in red grapes are often identified as cherry, black currant or cassis, strawberry, raspberry, and plum—which are often used as flavor descriptives in evaluation of wines made from Chambourcin, Gamay, and Pinot Noir. Many of these are found in ester forms. Strawberry flavor is attributed principally to several butyrate compounds. Raspberry is identified with ethyl caproate.

Another flavor group is called *primary vegetal aromas*. Good examples of these are anise, green pepper, tobacco leaf, and mint. This group is often structured in the form of carbonyl compounds, such as methoxy-isobutyl-pyrazine, which is the principal aroma component associated with bell peppers. These compounds can be very complex, such as the aroma associated with Cabernet Sauvignon, 2-methoxy-3-isobutylpyrazine. Vegetal aromas can also exist as phenols, such as cinnamic acid, found in grass and tobacco. This character is most often desired in very heavy red wines, such as those traditionally made from Cabernet Sauvignon, Chancellor, and Merlot.

Yet another group, *primary wood aromas*, is exemplified by briar, cedar, hazelnut, resin, oak, and eucalyptus. These generally exist in the form of phenolic compounds such as vanillin, which is the aroma component associated with oak. Wood aromas can also be carbonyl compounds, such as benzaldehyde, the aroma associated with bitter almond.

The first phase of olfactory judgment begins with the examination of bouquet and should be made while the wine is in a glass upon the counter top or table top. The judge will be smelling only the vapors that the wine gives off at rest. The nose is inserted in the glass, and several deep sniffs are taken, with a mental note made of each reaction. Sometimes closing the eyes may help concentration (Figure B-20). These primary judgment notes are then recorded.

After a few seconds rest, another examination of bouquet should be made immediately after the wine has been fully swirled around the inside walls of the glass. Grasp the glass stem and gently rotate the bowl. This significantly increases the surface area of wine from which flavor esters can evaporate into the chamber for the nose to receive. Again the nose is inserted in the glass, and several deep sniffs are taken, with a mental note made of each reaction. These secondary judgment notes are also recorded.

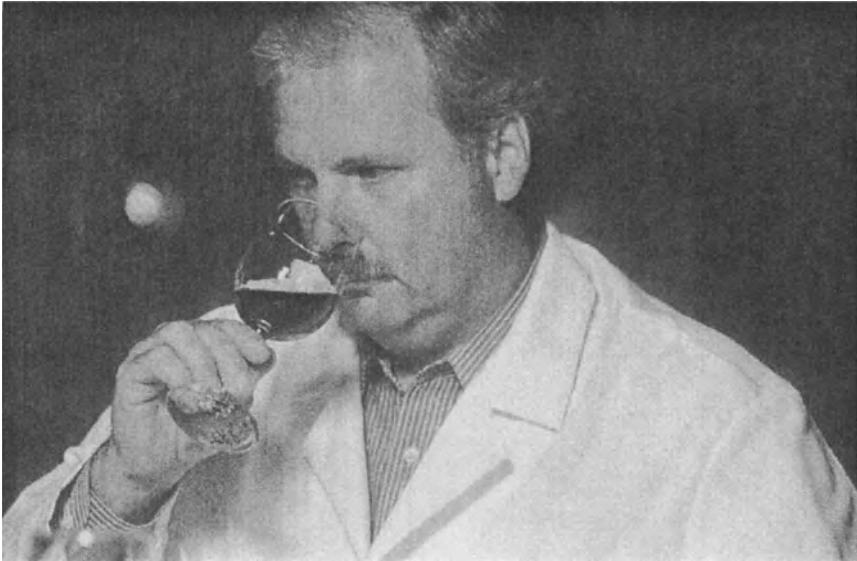


Fig. B-20 The olfactory mode.

Photo by David King, Purdue University Agricultural Communications.

The second phase of olfactory judgment takes place in the mouth, with the taste buds employed as evaluators of wine flavor constituents. The mouth should be freshly rinsed clean with water, and then a small sip of wine should be taken from the glass and rolled around the mouth. The flavors may, or may not, be the same as those identified in the nose. The palate may identify new flavors altogether or the same flavors with different intensity. These primary judgment notes are recorded, and the wine is spit out into the bucket.

After a few seconds rest, during which the taster takes a small nibble of cheese and unsalted cracker, the mouth is rinsed clean again for making the second flavor judgment. This time, a sip of wine is drawn in the mouth along with a little air—almost like whistling in reverse. The taster rolls the wine around in the mouth, drawing in air again to stimulate taste buds and accentuate shy flavors. These secondary judgment impressions are recorded, and the wine is spit into the bucket as before.

Gustatory Mode

The gustatory mode (Figure B-21) evaluates wine on the palate, principally the “feel” of the wine. Conventional wisdom often relates this to four human gustatory senses: acidity, sweetness, bitterness, and saltiness. These senses are supplemented by tactile, or mouth-feel, sensations of touch and temperature on the palate.

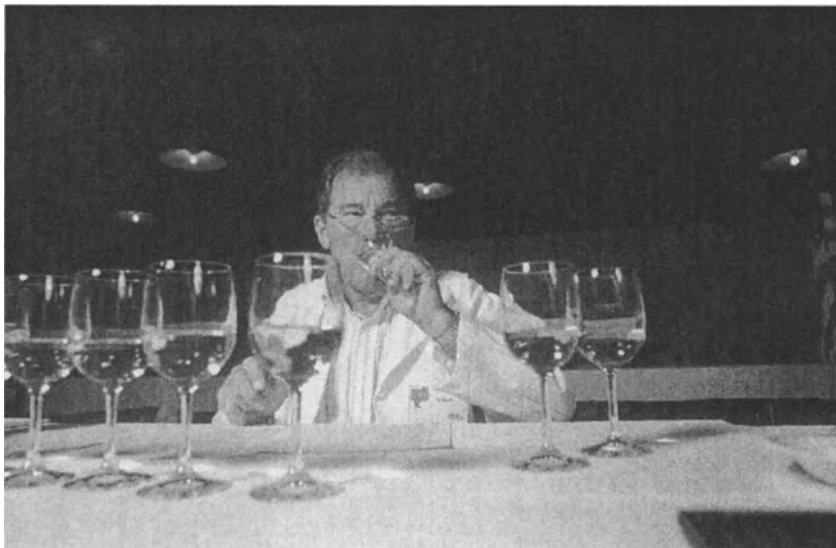


Fig. B-21 The gustatory mode.

Photo by David King, Purdue University Agricultural Communications.

There is comparatively little gustatory sensation in the nose. High alcohol content, carbon dioxide bubbles, and some measures of acidity and phenolic compounds are detectable in the nose and have flavor. Consequently, experienced judges usually make olfactory and gustatory evaluations simultaneously while the wine is on the palate. For beginners, it helps to make gustatory judgments more deliberately until an expertise is developed.

The upper surface of the tongue is constructed of several thousand various types of *papillae*, which are expanded skin protuberances, each containing about 250 taste buds. Each bud delivers taste information directly to the brain via special transmitter nerves.

Total acidity is the sum total of acids present in wine. These consist primarily of tartaric, malic, lactic, and acetic acids. Typically, total acidity in wine is measured between 0.5% and 1.0% of total wine composition. Tartaric acid has very little perceptible flavor. Malic acid is a bit green apple-like in character, lactic acid has a buttery-cheesy flavor, and acetic acid is vinegary.

Acidity is perceived mostly on the sides of the tongue by foliate papilla cells, in various degrees of tartness—with which the term *tartaric acid* is associated. The following is a typical range of acidity expressed in wine judging: (1) insipid, (2) bland, (3) balanced, (4) tart, and (5) harsh.

Sweetness can mask total acidity on the palate. A beginning wine enthusiast should devote some time to studying the interaction of these two gustatory influences. One suggestion is to pour three glasses of the same wine. In one, dissolve

a pinch of ascorbic acid, and in another, dissolve a pinch of sugar. Compare all three to ascertain whether differences can be detected. If not, dissolve another pinch of each and try it again. Continue until the additions are readily obvious on the palate. In a fourth glass, blend together equal parts of the acid- and sugar-treated wines. Compare the resulting blend to both the high-acid wine and the high-sugar wine. This exercise can help illustrate how acidity and sweetness serve to neutralize each other on the palate—and also to gather meaning from the term *sugar-acid balance*. The following is a typical range of sweetness expressed in wine judging: (1) bone dry, (2) dry, (3) semidry, (4) sweet, and (5) cloying.

Glucose, or common corn sugar, is the least sweet of the common sugars. Sucrose, which is cane sugar, is a bit sweeter on the palate, but fructose, or natural fruit sugar, has the greatest gustatory sensation of sweetness.

The bitterness in wines is usually attributed to phenolic compounds extracted mostly from grape seeds. A good example of this gustatory sense can be experienced by biting into a few grape seeds and chewing them for a minute or so before spitting them out.

A related gustatory term is *astringency*—also resulting from phenolic compounds extracted principally from grape skins and stems. Wine judges will often use the term *tannin* to refer to puckery astringency—a rather leathery, aspirin-like sensation on the palate.

Tannins are a special group of phenols that serve to slow the wine aging process by inhibiting oxidation potential. Wine aging is thus wine oxidation. Vintners desiring long-lived red wines purposely conduct fermentations with extended periods of contact with grape skins (and sometimes a portion of stems too) in order to extract greater tannin content. As would be expected, these young red wines will have unpleasant astringent tannins. Over years of aging, however, these tannins will eventually give way to the inevitable oxidation-aging process and will develop an attractive delicate tealike flavor. During that time, all of the other flavor constituency in the wine will also have matured into what has hopefully made the wait worthwhile.

The following is a typical range of bitterness expressed in wine judging: (1) smooth, (2) astringent, (3) coarse, (4) tannic, and (5) harsh.

The tactile phase of gustatory examination has to do with the manner in which a wine “lays” on the palate—more often referred to as “mouth-feel” by food scientists.

A wine judge will consider wine body by whether it feels light and thin, as opposed to full and heavy. Heavy-bodiedness results from dissolved solids in the wine, such as sugars, color pigments, and higher alcohols. Intuitively, sweet wines have more body than dries, but it is the relative degree of body for each individual wine type that is the judgment criterion. A little glycerine mixed to a light dry white wine can create the effect of heavier body.

Higher concentrations of ethyl alcohol in wine can contribute to a slight burning sensation in the nose and on the palate. Mix enough unflavored vodka with water to make a solution of about 13% alcohol—a common level in table wines. If the

Exhibit B-1 Modified 20-point cardinal scale for wine scoring.

PURDUE UNIVERSITY
WINE EVALUATION CHART

NAME: _____

PLACE: _____

DATE: _____

	WINE	PRICE	APPEARANCE 3 MAX	AROMA/ BOUQUET 6 MAX	TASTE 6 MAX	AFTERTASTE 3 MAX	OVERALL 2 MAX	TOTAL 20 MAX
1								
2								
3								
4								
5								
6								
7								
8								
9								
COMMENTS:								

Exhibit B-2 Rationale for the modified 20-point cardinal scale for wine scoring.

APPEARANCE (the visual mode - what one is seeing in a wine)

The color and clarity of a wine can be an indication of its character - of potential qualities and flaws. Darker colors (both reds and whites) generally indicate heavier, more full-bodied wines. Rosé wines should be pink, perhaps just a bit of orange tint to older selections. All wines should be transparent - younger whites brilliant, while older selections may exhibit a very slight haziness, and very old reds often "throw" a noticeable sediment.

- 3 - Excellent
 - 2 - Good
 - 1 - Poor
 - 0 - Objectable
- brilliant with outstanding characteristic color
 - clear with characteristic color
 - excessive haziness and/or uncharacteristic color
 - cloudiness and/or very poor color

AROMA AND BOUQUET (the olfactory mode - what one is smelling in a wine)

This second step in wine evaluation is directed towards aroma (fruit flavor) and bouquet (fruit flavor plus added odors from vinification). Swirl the wine inside the glass to expand the surface area from which flavor esters can evaporate. Take several large whiffs. Some wines may be herbaceous, while others are truly - experience serving to distinguish which should be which. In every case the result should be balanced (not either neutral or overpowering) and, most importantly, pleasant. This mode is often called the "nose" in organoleptic testing.

- 6 - Extraordinary
 - 5 - Excellent
 - 4 - Good
 - 3 - Fair
 - 2 - Poor
 - 1 - Unacceptable
 - 0 - Objectable
- outstanding in character with exceptional balance in bouquet constituency
 - characteristic and well-balanced
 - distinguishable and adequately balanced
 - somewhat neutral and/or slightly unbalanced
 - undistinguishable and detectable "off" odors
 - obvious "off" odors and unbalanced
 - offensive odors and very unbalanced

TASTE (the gustation mode - what one is tasting in a wine)

The human tongue can detect variances in acidity, bitterness, and sweetness, levels in wine - typical values for each wine type are learned with experience. Other portions of the mouth are anatomically connected to the olfactory lobe and are, therefore, sensitive to specific flavors and essences associated with the "nose" of each wine - again judged authoritatively through continued practice. Take a small sip and "wash" it around the mouth so that all surfaces are given a chance to experience the wine - "whistling" in some air to activate taste bud activity. Generally, after ten to fifteen seconds of examination, the wine is spit out.

- 6 - Extraordinary
 - 5 - Excellent
 - 4 - Good
 - 3 - Fair
 - 2 - Poor
 - 1 - Unacceptable
 - 0 - Objectable
- outstanding in character with exceptional balance of acidity and sweetness
 - characteristic and well-balanced
 - distinguishable and adequately balanced
 - somewhat neutral and/or slightly unbalanced
 - undistinguishable and detectable "off" flavors and/or unbalanced
 - obvious "off" flavors and unbalanced
 - offensive flavors and very unbalanced

AFTERTASTE (the taste and flavor values that linger following completion of the gustatory mode)

Take a second, perhaps smaller, sip of the wine and swallow - endeavoring to judge how long the taste and flavor constituents remain detectable in the mouth. Lighter wines may linger only momentarily, while heavier wines can last for perhaps a dozen seconds or more.

- 3 - Excellent
 - 2 - Good
 - 1 - Poor
 - 0 - Objectable
- outstandingly pleasant aftertaste, lasting beyond normal duration of time
 - pleasant aftertaste, lasting a normal duration of time
 - little or no distinguishable aftertaste
 - unpleasant aftertaste

OVERALL IMPRESSION (the true sense of type and price/value judgement one determines in a wine)

The total quality level achieved by a wine in comparison to other wines of the same variety or type, and in regard to the monetary price, is addressed in this mode. This not a "hedge factor" used indiscriminately to give or take points in response to subjective notions.

- 2 - Excellent
 - 1 - Good
 - 0 - Poor
- outstanding example of the variety or type and/or an exceptional value for the money
 - representative of the variety or type and/or an acceptable value for the money
 - uncharacteristic of the variety or type and/or an unacceptable value for the money

SCORES

- 18 - 20
 - 15 - 17
 - 12 - 14
 - 9 - 11
 - 0 - 8
- truly great wines, unusually superior attributes orchestrated in a faultless character
 - excellent wines, perhaps faulted only by a tryness of one or another attributes
 - good wines, with perhaps only one or two detectable faults
 - poor wines, with at least two obvious faults
 - objectionable wines, having many faults

vodka is 80 proof, it is about 40% alcohol by volume. Mixing two parts of water with one part of vodka results in a little over 13% alcohol by volume.

Smell and taste the plain water side by side with the ethyl alcohol solution at room temperature. The ethanol mixture should have no flavor, but there should be a gustatory sensation—perhaps a slight burning in the nostrils and on the tongue. Then chill both the water and the ethanol solution to refrigerator temperature and repeat the evaluation. The result should significantly mute the gustatory effect of the alcohol—a term called *closed*. Then heat both solutions up to lukewarm temperature and try the experiment again. This should have a profound burning effect—a term called *hot* or *strong* by wine judges.

Aftertaste relates to how long both the olfactory and gustatory effects remain on the palate after the wine has been spit out. The common judgment terminology for aftertaste is simply *short*, *lingering*, or *long*. As a rule of thumb, lengthy aftertaste is a virtue. Some very high-quality wines are, however, also delicate and shy—precluding any measure of lengthy aftertaste.

Proceed with judging the gustatory mode in precisely the same manner as for tasting the olfactory mode. After a few seconds' rest, take a small bite of cheese and unsalted cracker. Rinse out the mouth with water and take a sip of wine. Roll it around the mouth to obtain the full effects of acidity, sweetness, and bitterness. Spit out the wine, and record all impressions. While recording, make a judgment as to the length of the aftertaste.

Remember that this entire discussion has to do with wine judging and not wine drinking. When wine is swallowed, an additional set of olfactory and gustatory impressions is made in the back of the mouth and in the pharyngeal passage of the throat. For obvious reasons, wine judges cannot swallow every wine presented them in a whole day of wine competition.

Overall Impression

The overall impression is used to rate the combined visual, olfactory, and gustatory impressions of each wine.

More often than not, judgment of overall impression is relegated to a rather “fudge factor” role. A given wine might possibly score only moderately when scrutinized mode by mode, yet all the modes might fit together very nicely—in which case a judge would be generous in scoring overall impression. The opposite can hold as well.

There are many cardinal (numerical) scales for the sensory scoring of wines (Exhibits B-1 and B-2). One of the most commonly used is the Modified Davis 20-Point Scale, modeled after the original devised by the University of California, Davis, years ago.

APPENDIX C

CHARTS, TABLES, AND CONVERSION TABLES

Table C-1 Tons of grapes per acre.

Lb Grapes per Vine	(Acre = 202 ft × 202 ft = 20 rows at 200 ft spaced 10 ft apart)					
	Vines at 5 ft (800 Vines)	Vines at 6 ft (667 Vines)	Vines at 7 ft (571 Vines)	Vines at 8 ft (500 Vines)	Vines at 15 ft (267 Vines)	Vines at 20 ft (200 Vines)
2	.80	.67	.57	.50	.27	.20
4	1.60	1.33	1.14	1.00	.54	.40
6	2.40	2.00	1.71	1.50	.81	.60
8	3.20	2.67	2.28	2.00	1.08	.80
10	4.00	3.34	2.85	2.50	1.35	1.00
12	4.80	4.00	3.42	3.00	1.62	1.20
14	5.60	4.67	3.99	3.50	1.89	1.40
16	6.40	5.34	4.56	4.00	2.16	1.60
18	7.20	6.00	5.13	4.50	2.43	1.80
20	8.00	6.67	5.70	5.00	2.70	2.00
22	8.80	7.34	6.27	5.50	2.97	2.20
24	9.60	8.00	6.84	6.00	3.24	2.40
26	10.40	8.67	7.41	6.50	3.51	2.60
28	11.20	9.34	7.98	7.00	3.78	2.80
30	12.00	10.01	8.55	7.50	4.05	3.00
32	12.80	10.67	9.12	8.00	4.32	3.20
34	13.60	11.34	9.69	8.50	4.59	3.40
36	14.40	12.01	10.26	9.00	4.86	3.60
38	15.20	12.67	10.83	9.50	5.13	3.80
40	16.00	13.34	11.40	10.00	5.40	4.00
42	16.80	14.07	11.97	10.50	5.67	4.20
44	17.60	14.74	12.54	11.00	5.94	4.40
46	18.40	15.41	13.11	11.50	6.21	4.60
48	19.20	16.08	13.68	12.00	6.48	4.80
50	20.00	16.75	14.25	12.50	6.75	5.00
60	24.00	20.04	17.10	15.00	8.10	6.00
70	28.00	23.38	19.95	17.50	9.45	7.00
80	32.00	26.72	22.80	20.00	10.80	8.00
90	36.00	30.06	25.65	22.50	12.15	9.00
100	40.00	33.40	28.50	25.00	13.50	10.00
110	44.00	36.74	31.35	27.50	14.85	11.00
120	48.00	40.08	34.20	30.00	16.20	12.00
130	52.00	43.42	37.05	32.50	17.55	13.00
140	56.00	46.76	39.90	35.00	18.90	14.00
150	60.00	50.10	42.75	37.50	20.25	15.00

160	53.44	45.60	40.00	21.60	16.00
170	56.78	48.45	42.50	22.95	17.00
180	60.12	51.30	45.00	24.30	18.00
190		54.15	47.50	25.65	19.00
200		57.00	50.00	27.00	20.00
210		59.85	52.50	28.35	21.00
220			55.00	29.70	22.00
230			57.50	31.05	23.00
240			60.00	32.40	24.00
250				33.75	25.00
260				35.10	26.00
270				36.45	27.00
280				37.80	28.00
290				39.15	29.00
300				40.50	30.00

Table C-2 Traditional bulk wine vessels.

Vessel	Origin	Capacity in	
		U.S. Gal.	Liters
Aroba	Spain	3-5	12-18
Aum	Germany	42.27	160.00
Baril	Lisbon	4.42	16.74
Baril	Malaga	7.93	30.00
Barile	Rome	15.41	58.34
Barrel	U.S.	31.50	119.24
Barril	Madeira	4.08	15.44
Barrique	Algeria	58.12	220.00
Barrique	Beaujolais	57.07	216.00
Barrique	Bordeaux	59.45	225.00
Barrique	Champagne	52.84	200.00
Barrique	Côte d'Or	60.24	228.00
Barrique	Côtes du Rhone	59.45	225.00
Barrique	Macon	56.80	215.00
Barrique	Yonne	65.52	248.00
Bocoy	Spain	162.00	613.17
Botte (small)	Sardinia	11.77	44.54
Botte (large)	Sardinia	132.10	500.00
Brente	Switzerland	13.21	50.00
Butt	Spain	132.00	499.62
Butte (Buttig)	Hungary	3.59	13.6
Demi-queue	Burgundy	60.24	228.00
Double-Aum	Germany	84.54	320.00
Dreiling	Vienna	358.78	1358.00
Eimer	Switzerland	9.91	37.50
Eimer	Vienna	14.95	56.58
Eimer	Württemberg	77.65	293.92
Fuder	Mosel	253.63	960.00
Fuder	Germany (Prussian)	217.81	824.42
Fuder	Württemberg	465.94	1763.57
Gallon	England	1.20	5.54
Gallon	U.S.	1.00	3.79
Halb-Fuder	Mosel	126.82	480.00
Halb-Stuck	Rhine and Palatinate	169.09	640.00
Hogshead	England	66.00	249.81
Octavilla	England and Spain	16.50	62.45
Ohm	Alsace	13.21	50.00
Ohm	Baden	39.63	150.00
Ohm	Bavaria	33.82	128.00
Ohm	Saar	38.04	144.00
Ohm	Switzerland	10.57	40.00
Oka	Balkans	.34	1.28
Oxhoft	Hamburg	59.71	226.00
Piece	Burgundy	60.24	228.00
Piece	Champagne	52.84	200.00
Piece	Saumur	58.12	220.00
Piece	Vouvray	66.05	250.00
Pipe	England	126.02	477.00
Pipe	Lisbon	132.63	502.00
Pipe	Madeira	109.91	416.00
Pipe	Oporto	141.35	535.00
Pipe	Tarragona	134.74	510.00
Pipe	Valencia	19.92	75.39
Punchon	England	84.00	317.94
Queue	Burgundy	120.48	456.00
Stuck	Rhineland	317.04	1200.00
Tierce	England	42.00	158.97
Tonneau	Bordeaux	237.78	900.00
Tun	England	252.05	954.00
Vedro	Russia	3.27	12.39

Table C-3 Alcohol measurements.

Specific Gravity ² 20°C 4°C	Percent by Volume at 20°C	Percent by Weight	G per 100 ml
0.99823	0.00	0.00	0.00
0.99675	1.00	0.79	0.79
0.99528	2.00	1.59	1.58
0.99384	3.00	2.38	2.37
0.99243	4.00	3.18	3.16
0.99106	5.00	3.98	3.95
0.98973	6.00	4.78	4.74
0.98845	7.00	5.59	5.53
0.98718	8.00	6.40	6.32
0.98596	9.00	7.20	7.10
0.98476	10.00	8.02	7.89
0.98416	11.00	8.83	8.68
0.98296	11.50	9.23	9.08
0.98238	12.00	9.64	9.47
0.98180	12.50	10.05	9.87
0.98122	13.00	10.40	10.26
0.98066	13.50	10.86	10.66
0.98009	14.00	11.28	11.05
0.97953	14.50	11.68	11.44
0.97897	15.00	12.09	11.84
0.97841	15.50	12.50	12.23
0.97786	16.00	12.92	12.63
0.97732	16.50	13.33	13.02
0.97678	17.00	13.74	13.42
0.97624	17.50	14.15	13.81
0.97570	18.00	14.56	14.21
0.97517	18.50	14.97	14.60
0.97464	19.00	15.39	15.00
0.97412	19.50	15.80	15.39
0.97359	20.00	16.21	15.79
0.97306	20.50	16.63	16.18
0.97252	21.00	17.04	16.58
0.97199	21.50	17.46	16.97
0.97145	22.00	17.88	17.37
0.97091	22.50	18.29	17.76
0.97036	23.00	18.71	18.16
0.96980	23.50	19.13	18.55
0.96925	24.00	19.55	18.94
0.96869	24.50	19.96	19.34
0.96812	25.00	20.38	19.73
0.96755	25.50	20.80	20.13
0.96699	26.00	21.22	20.52
0.96641	26.50	21.64	20.92
0.96583	27.00	22.07	21.31
0.96525	27.50	22.49	21.71
0.96465	28.00	22.91	22.10
0.96406	28.50	23.33	22.50
0.96346	29.00	23.76	22.89
0.96285	29.50	24.18	23.29
0.96224	30.00	24.61	23.68
0.96163	30.50	25.04	24.08
0.96100	31.00	25.46	24.47
0.96036	31.50	25.89	24.86
0.95972	32.00	26.32	25.26

(continued)

Table C-3 (continued)

Specific Gravity ² 20°C 4°C	Percent by Volume at 20°C	Percent by Weight	G per 100 ml
0.95906	32.50	26.75	25.64
0.95839	33.00	27.18	26.05
0.95771	33.50	27.61	26.44
0.95703	34.00	28.04	26.84
0.95634	34.50	28.48	27.23
0.95563	35.00	28.91	27.63
0.95492	35.50	29.34	28.02
0.95419	36.00	29.78	28.42
0.95346	36.50	30.22	28.81
0.95272	37.00	30.66	29.21
0.95196	37.50	31.09	29.60
0.95120	38.00	31.53	29.99
0.95043	38.50	31.97	30.39
0.94964	39.00	32.42	30.79
0.94885	39.50	32.86	31.18
0.94805	40.00	33.30	31.57
0.94724	40.50	33.75	31.97
0.94643	41.00	34.19	32.36
0.94560	41.50	34.64	32.76
0.94477	42.00	35.09	33.15
0.94393	42.50	35.54	33.55
0.94308	43.00	35.99	33.94
0.94222	43.50	36.44	34.34
0.94135	44.00	36.89	34.73
0.94046	44.50	37.35	35.13
0.93957	45.00	37.80	35.52
0.93867	45.50	38.26	35.92
0.93776	46.00	38.72	36.31
0.93684	46.50	39.18	36.70
0.93591	47.00	39.64	37.10
0.93498	47.50	40.10	37.49
0.93404	48.00	40.56	37.89
0.93308	48.50	41.03	38.29
0.93213	49.00	41.49	38.68
0.93116	49.50	41.96	39.07
0.93017	50.00	42.43	39.47

¹ Calculated by U.S. Bureau of Standards.² This table is valid for specific gravity determinations at temperatures between an upper limit of 20°C and a lower limit of 4°C.

Table C-4 Calculation of partially filled horizontal tanks (cylindrical, straight-sided tanks only, for which total capacity of tank is known).**Procedure:**

- (1) Determine inside diameter of tank.
- (2) Determine depth of wine in tank.
- (3) Compute percent depth of wine (line 2 less line 1).
- (4) From table below find percent total volume that corresponds to this percent depth
- (5) Multiply percent found by total volume = amount of wine in tank

Percent Depth Wine	Percent Volume	Percent Depth Wine	Percent Volume	Percent Depth Wine	Percent Volume
99	99.6	64	67.8	29	23.8
98	99.2	63	66.6	28	22.7
97	98.8	62	65.4	27	21.6
96	98.4	61	64.2	26	20.5
95	98.0	60	63.0	25	19.4
94	97.4	59	61.6	24	18.3
93	96.8	58	60.2	23	17.2
92	96.2	57	58.8	22	16.1
91	95.6	56	57.4	21	15.0
90	95.0	55	56.0	20	14.0
89	94.2	54	54.8	19	13.0
88	93.4	53	53.6	18	12.0
87	92.6	52	52.4	17	11.0
86	91.8	51	51.2	16	10.0
85	91.0	50	50.0	15	9.0
84	90.0	49	48.8	14	8.2
83	89.0	48	47.6	13	7.4
82	88.0	47	46.4	12	6.6
81	87.0	46	45.2	11	5.8
80	86.0	45	44.0	10	5.0
79	85.0	44	42.6	9	4.4
78	83.9	43	41.2	8	3.8
77	82.8	42	39.8	7	3.2
76	81.7	41	38.4	6	2.6
75	80.5	40	37.0	5	2.0
74	79.5	39	35.8	4	1.6
73	78.4	38	34.6	3	1.2
72	77.3	37	33.4	2	.8
71	76.2	36	32.2	1	.4
70	75.0	35	31.0	0	.0
69	73.8	34	29.8		
68	72.6	33	28.6		
67	71.4	32	27.4		
66	70.2	31	26.2		
65	69.0	30	25.0		

Table C-5 Wine tank capacities (for cylindrical straight-sided tanks only, measured at the waist¹).

Calculation of total capacity in U.S. gal.:

Diam. (in.) $\times$ diam. (in.) $\times$ height (in.) $\times$ 0.0034 = U.S. gal capacity of tank.

Example:

72 in. diam. $\times$ 72 in. diam $\times$ 96 in. height (or length) $\times$ 0.0034 = 1692.1 U.S. gal. capacity of tank.

Calculation of U.S. gal. per in.:

Total U.S. gal. capacity divided by total number of in. = U.S. gal. per in.

Example:

1692.1 U.S. gal. capacity divided by 96 in. = 17.626 U.S. gal. per in.

Calculation of total capacity in liters:

Total U.S. gal. capacity $\times$ 3.785 = liters capacity of tank.

Example 1692.1 U.S. gal. capacity $\times$ 3.785 = 6404.6 liters capacity of tank.

Calculation of liters per cm:

U.S. gal. per in. $\times$ 1.49016 = liters per cm.

Example:

17.626 gal. per in. $\times$ 1.49016 = 26.266 liters per cm.

More Examples:

Diameter In. I.D.	Length In. I.D.	Capacity Gal. U.S.	Gal. per In. U.S.	Capacity Liters	Liters per Cm
36	36	158.6	4.41	600.3	6.44
36	48	211.5	4.41	800.5	6.44
36	60	264.4	4.41	1000.8	6.44
48	48	376.0	7.83	1423.2	11.67
48	60	470.0	7.83	1779.0	11.67
48	72	564.0	7.83	2134.7	11.67
60	60	734.4	12.24	2779.7	18.24
60	72	881.3	12.24	3335.7	18.24
60	84	1028.2	12.24	3891.7	18.24
60	96	1175.0	12.24	4447.4	18.24
72	72	1269.0	17.63	4803.2	26.27
72	84	1480.6	17.63	5604.1	26.27
72	96	1692.1	17.63	6404.6	26.27
72	108	1903.6	17.63	7205.1	26.27
72	120	2115.1	17.63	8005.7	26.27
84	84	2015.2	23.99	7627.5	35.75
84	96	2303.1	23.99	8717.2	35.75
84	108	2591.0	23.99	9806.9	35.75
84	120	2878.8	23.99	10,896.3	35.75
84	132	3166.7	23.99	11,986.0	35.75
84	144	3454.6	23.99	13,075.7	35.75
96	96	3008.1	31.33	11,385.7	46.69
96	108	3384.1	31.33	12,810.0	46.69
96	120	3760.1	31.33	14,232.0	46.69
96	132	4136.1	31.33	15,655.1	46.69
96	144	4512.2	31.33	17,078.7	46.69

¹ Whether the tank is horizontal or vertical, the diameter must be measured at the center of the tank between the two parallel, flat ends.

Table C-6 Vat capacity.

(Per 1 Ft of Depth—Inside Measure)

Diameter (Ft) (In.)		Area (Ft)	U.S. Gal.
1	0	.7854	5.8735
1	2	1.0690	7.9944
1	4	1.3962	10.4413
1	6	1.7671	13.2150
1	8	2.1816	16.3148
1	10	2.6398	19.7414
2	0	3.1416	23.4940
2	2	3.6869	27.5720
2	4	4.2760	32.6976
2	6	4.9087	36.7092
2	8	5.5850	41.7668
2	10	6.3049	47.1505
3	0	7.0686	52.8618
3	2	7.8757	58.8976
3	4	8.7265	65.2602
3	6	9.6211	73.1504
3	8	10.5591	78.9652
3	10	11.5409	86.3074
4	0	12.5664	93.9754
4	2	13.6353	101.9701
4	4	14.7479	110.2907
4	6	15.9043	118.9386
4	8	17.1041	127.9112
4	10	18.3476	137.2105
5	0	19.6350	146.8384
5	2	20.9656	156.7891
5	4	22.3400	167.0674
5	6	23.7583	177.6740
5	8	25.2199	188.6045
5	10	26.7251	199.8610
6	0	28.2744	211.4472
6	6	33.1831	248.1564
7	0	38.4846	287.8230
7	6	44.1787	330.3859
8	0	50.2656	375.9062
8	6	56.7451	424.3625
9	0	63.6174	475.7563
9	6	70.8823	530.0861
10	0	78.5400	587.3534
10	6	86.5903	647.5568
11	0	95.0034	710.6977
11	6	103.8691	776.7746
12	0	113.0976	848.1890

Table C-7 Example of tank chart for a straight-sided horizontal tank¹.

Tank No. S-41		Gal. per In.		Wet In.	
In.	Gal.	In.	Gal.	In.	Gal.
1	10	37	1855	73	3896
2	23	38	1920	74	3928
3	41	39	1987	75	3955
4	66	40	2052	76	3980
5	91	41	2118	77	3998
6	120	42	2183	78	4008
7	153	43	2248		
8	189	44	2314		
9	228	45	2379		
10	271	46	2445		
11	317	47	2508		
1 ft	362	4 ft	2573		
13	409	49	2638		
14	459	50	2703		
15	512	51	2765		
16	568	52	2827		
17	622	53	2888		
18	677	54	2949		
19	733	55	3010		
20	791	56	3070		
21	842	57	3130		
22	903	58	3188		
23	962	59	3245		
2 ft	1023	5 ft	3300		
25	1084	61	3356		
26	1147	62	3411		
27	1209	63	3463		
28	1271	64	3515		
29	1336	65	3565		
30	1399	66	3612		
31	1464	67	3658		
32	1528	68	3703		
33	1593	69	3745		
34	1659	70	3786		
35	1724	71	3825		
3 ft	1789	6 ft	3862		

¹ Tank has an inside diameter of 122.41 in. at the waist (39 in.).

Table C-8 Example of tank chart for a straight-sided vertical tank.

Tank No. S-56 Liters per Cm		Measured at Center of Tank Wet Cm			
Cm	Liters	Cm	Liters	Cm	Liters
0	0	59	1040.64	118	2081.28
1	17.64	60	1058.28	119	2098.92
2	35.28	61	1075.92	120	2116.56
3	52.91	62	1093.56	121	2134.20
4	70.55	63	1111.19	122	2151.84
5	88.19	64	1128.83	123	2169.47
6	105.83	65	1146.47	124	2187.11
7	123.47	66	1164.11	125	2204.75
8	141.10	67	1181.75	126	2222.39
9	158.74	68	1199.38	127	2240.03
10	176.38	69	1217.02	128	2257.66
11	194.02	70	1234.66	129	2275.30
12	211.66	71	1252.30	130	2292.94
13	229.29	72	1269.94	131	2310.58
14	246.93	73	1287.57	132	2328.22
15	264.57	74	1305.21	133	2345.85
16	282.21	75	1322.85	134	2363.49
17	299.85	76	1340.49	135	2381.13
18	317.48	77	1358.13	136	2398.77
19	335.12	78	1375.76	137	2416.41
20	352.76	79	1393.40	138	2434.04
21	370.40	80	1411.04	139	2451.68
22	388.04	81	1428.68	140	2469.32
23	405.67	82	1446.32	141	2486.96
24	423.31	83	1463.95	142	2504.60
25	440.95	84	1481.59	143	2522.23
26	458.59	85	1499.23	144	2539.87
27	476.23	86	1516.87	145	2557.51
28	493.86	87	1534.51	146	2575.15
29	511.50	88	1552.14	147	2592.79
30	529.14	89	1569.78	148	2610.42
31	546.78	90	1587.42	149	2628.06
32	564.42	91	1605.06	150	2645.70
33	582.05	92	1622.70	151	2663.34
34	599.69	93	1640.33	152	2680.98
35	617.33	94	1657.97	153	2698.61
36	634.97	95	1675.61	154	2716.25
37	652.61	96	1693.25	155	2733.89
38	670.24	97	1710.89	156	2751.53
39	687.88	98	1728.52	157	2769.17
40	705.52	99	1746.16	158	2786.80
41	723.16	100	1763.80	159	2804.44
42	740.80	101	1781.44	160	2822.08
43	758.43	102	1799.08	161	2839.72
44	776.07	103	1816.71	162	2857.36
45	793.71	104	1834.35	163	2874.99
46	811.35	105	1851.99	164	2892.63
47	828.99	106	1869.63	165	2910.27
48	846.62	107	1887.27	166	2927.91
49	864.26	108	1904.90	167	2945.55
50	881.90	109	1922.54	168	2963.18
51	899.54	110	1940.18	169	2980.82
52	917.18	111	1957.82	170	2998.46
53	934.81	112	1975.46	171	3016.10
54	952.45	113	1993.09	172	3033.74
55	970.09	114	2010.73	173	3051.37
56	987.73	115	2028.37	174	3069.01
57	1005.37	116	2046.01	175	3086.65
58	1023.00	117	2063.65	176	3104.29

(continued)

Table C-8 (continued)

Cm	Tank No. S-56 Liters per Cm		Measured at Center of Tank Wet Cm		
	Liters	Cm	Liters	Cm	Liters
177	3121.93	210	3703.98	242	4268.40
178	3139.56	211	3721.62	243	4286.03
179	3157.20	212	3739.26	244	4303.67
180	3174.84	213	3756.89	245	4321.31
181	3192.48	214	3774.53	246	4338.95
182	3210.12	215	3792.17	247	4356.59
183	3227.75	216	3809.81	248	4374.22
184	3245.39	217	3827.45	249	4391.86
185	3263.03	218	3845.08	250	4409.50
186	3280.67	219	3862.72	251	4427.14
187	3298.31	220	3880.36	252	4444.78
188	3315.94	221	3898.00	253	4462.41
189	3333.58	222	3915.64	254	4480.05
190	3351.22	223	3933.27	255	4497.69
191	3368.86	224	3950.91	256	4515.33
192	3386.50	225	3968.56	257	4532.97
193	3404.13	226	3986.19	258	4550.60
194	3421.77	227	4003.83	259	4568.24
195	3439.41	228	4021.46	260	4585.88
196	3457.05	229	4039.10	261	4603.52
197	3474.69	230	4056.74	262	4621.16
198	3492.32	231	4074.38	263	4638.79
199	3509.96	232	4092.02	264	4656.43
200	3527.60	233	4109.65	265	4674.07
201	3545.24	234	4127.29	266	4691.71
202	3562.88	235	4144.93	267	4709.35
203	3580.51	236	4162.57	268	4726.98
204	3598.15	237	4180.21	269	4744.62
205	3615.80	238	4197.84	270	4762.26
206	3633.43	239	4215.48	271	4779.90
207	3651.07	240	4233.12	272	4797.54
208	3668.70	241	4250.76	273	4815.70
209	3686.34				

¹ Tank has an inside diameter of 149.86 cm. at the waist (137 cm.).

Table C-9 Periodic table of the elements.

PERIODIC TABLE

Table of Radioactive Isotopes

PERIOD	GROUP	IA	IIA	IIIB	IVB	VB	VIB	VIIB	VIII
1		1.00797 1H Hydrogen							
2		6.939 3Li Lithium	9.0122 4Be Beryllium						
3		22.9898 11Na Sodium	24.312 12Mg Magnesium						
4		39.102 19K Potassium	40.08 20Ca Calcium	44.956 21Sc Scandium	47.90 22Ti Titanium	50.942 23V Vanadium	51.996 24Cr Chromium	54.938 25Mn Manganese	55.847 26Fe Iron
5		85.47 37Rb Rubidium	87.62 38Sr Strontium	88.905 39Y Yttrium	91.22 40Zr Zirconium	92.906 41Nb Niobium	95.94 42Mo Molybdenum	(99) 43Tc Technetium	101.07 44Ru Ruthenium
6		132.905 55Cs Cesium	137.34 56Ba Barium	138.91 57La Lanthanum	178.49 72Hf Hafnium	180.948 73Ta Tantalum	183.85 74W Wolfram	186.2 75Re Rhenium	190.2 76Os Osmium
7		(223) 87Fr Francium	(226) 88Ra Radium	(227) 89Ac Actinium	★	★	★	★	★
					58.140.12 Ce Cerium	140.907 59Pr Praseodymium	144.24 60Nd Neodymium	(147) 61Pm Promethium	150.35 62Sm Samarium
					90.232.038 Th Thorium	91.231 91Pa Protactinium	238.04 92U Uranium	(237) 93Np Neptunium	(242) 94Pu Plutonium

ATOMIC WEIGHT (2)

OXIDATION STATES (Bold most stable)

BOILING POINT, °C

MELTING POINT, °C

DENSITY (g/ml) (4)

NAME

SYMBOL (1)

ELECTRON STRUCTURE

KEY

30 65.37
Zn
Zinc

NOTES:

(1) Black - solid.
Red - gas.
Blue - liquid.
Outline - synthetically prepared.

(2) Based upon carbon - 12. () indicates most stable or best known isotope.

(3) Proposed; not officially accepted

(4) Values for gaseous elements are for liquids at the boiling point.

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OF THE ELEMENTS

α	alpha particle	$\bar{\nu}$	anti-neutrino
β^-	beta particle	SF	spontaneous fission
β^+	positron	γ	gamma ray
K	K-electron capture	e^-	internal electron conversion

2 4.0026
-268.9
-269.7
0.126 He

Courtesy of E.H. Sargent & Co.

Table C-10 Table of periodic properties of the elements.

TABLE OF PERIODIC PROPERTIES

PERIOD

GROUP

IA

IIA

IIIB

IVB

VB

VIB

VII B

VIII

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Table C-10 (continued)

OF THE ELEMENTS

[illegible]

NOTES: • Element sublimes

- (1) For representative oxides (higher valence) of group. Oxide is acidic if color is red, basic if color is blue and amphoteric if both colors are shown. Intensity of color indicates relative strength.
- (2)  Cubic, face centered,  cubic, body centered,  diamond,  cubic,  hexagonal,  rhombohedral,  tetragonal,  orthorhombic,  monoclinic.
- (3) Proposed; not officially accepted (4) At room temperature. (5) At boiling point.
- (6) From 0° to 20°C. (7) Ionic [crystal] radii for coordination number 6
- (8) Metallic radii for coordination number of 12

Courtesy of E.H. Sargent & Co.

Table C-11 Basic conversions: units of area.

1 square inch	645.2 square millimeters
1 square inch	6.452 square centimeters
1 square inch	.000645 square meters
1 square inch	.006944 square feet
1 square inch	.000772 square yards
1 square foot	92,903 square millimeters
1 square foot	929.034 square centimeters
1 square foot	.092903 square meters
1 square foot	144 square inches
1 square foot	0.11111 square yards
1 square yard	8361.31 square centimeters
1 square yard	.836131 square meters
1 square yard	1296 square inches
1 square yard	9 square feet
1 square rod	30.25 square feet
1 acre	40.47 ares
1 acre	0.4047 hectares
1 acre	4480 square yards
1 square-mile	2.58998 kilometers
1 square mile	640 acres
1 square centimeter	100 square millimeters
1 square centimeter	0.1550 square inches
1 square centimeter	.001076 square feet
1 square centimeter	.00012 square yards
1 square centimeter	.00010 square meters
1 square meter	10000 square centimeters
1 square meter	1550 square inches
1 square meter	10.7639 square feet
1 square meter	1.19598 square yards
1 square kilometer	247.1 acres
1 square kilometer	0.3861 square miles

Table C-12 Area conversion formulas.

To Convert	into	Multiply by
square inches	square centimeters	6.451600
square feet	square meters	0.092903
square yards	square meters	0.836127
acres	square meters	4046.856422
square miles	square kilometers	2.589980
square centimeters	square inches	0.155000
square meters	square feet	10.763910
square meters	square yards	1.195990
square kilometers	square miles	0.386102

Table C-13 Square inches into square centimeters.

Sq. In	0	1	2	3	4	5	6	7	8	9
	Sq. Cm									
0	—	6.45	12.90	19.36	25.81	32.26	38.71	45.16	51.61	58.06
10	64.52	70.97	77.42	83.87	90.32	96.77	103.23	109.68	116.13	122.58
20	129.03	135.48	141.94	148.39	154.84	161.29	167.74	174.19	180.65	187.10
30	193.55	200.00	206.45	212.90	219.35	225.81	232.26	238.71	245.16	251.61
40	258.06	264.52	270.97	277.42	283.87	290.32	296.77	303.23	309.68	316.13
50	322.58	329.03	335.48	341.94	348.39	354.84	361.29	367.74	374.19	380.64
60	387.10	393.55	400.00	406.45	412.90	419.35	425.81	432.26	438.71	445.16
70	451.61	458.06	464.52	470.97	477.42	483.87	490.32	496.77	503.23	509.68
80	516.13	522.58	529.03	535.48	541.93	548.39	554.84	561.29	567.74	574.19
90	580.64	587.10	593.55	600.00	606.45	612.90	619.35	625.81	632.26	638.71
100	645.16	651.61	658.06	664.52	670.97	677.42	683.87	690.32	696.77	703.22

Table C-14 Square centimeters into square inches.

Sq. cm	0	1	2	3	4	5	6	7	8	9
	Sq. In									
0	—	0.155	0.310	0.465	0.620	0.775	0.930	1.085	1.240	1.395
10	1.550	1.705	1.860	2.015	2.170	2.325	2.480	2.635	2.790	2.945
20	3.100	3.255	3.410	3.565	3.720	3.875	4.030	4.185	4.340	4.495
30	4.650	4.805	4.960	5.115	5.270	5.425	5.580	5.735	5.890	6.045
40	6.200	6.355	6.510	6.665	6.820	6.975	7.130	7.285	7.440	7.595
50	7.750	7.905	8.060	8.215	8.370	8.525	8.680	8.835	8.990	9.145
60	9.300	9.455	9.610	9.765	9.920	10.075	10.230	10.385	10.540	10.695
70	10.850	11.005	11.160	11.315	11.470	11.625	11.780	11.935	12.090	12.245
80	12.400	12.555	12.710	12.865	13.020	13.175	13.330	13.485	13.640	13.795
90	13.950	14.105	14.260	14.415	14.570	14.725	14.880	15.035	15.190	15.345
100	15.500	15.655	15.810	15.965	16.120	16.275	16.430	16.585	16.740	16.895

Table C-15 Square feet into square meters.

Sq. Ft	0	1	2	3	4	5	6	7	8	9
	Sq. M									
0	—	0.0929	0.1858	0.2787	0.3716	0.4645	0.5574	0.6503	0.7432	0.8361
10	0.9290	1.0219	1.1148	1.2077	1.3006	1.3936	1.4865	1.5794	1.6723	1.7652
20	1.8581	1.9510	2.0439	2.1368	2.2297	2.3226	2.4155	2.5084	2.6013	2.6942
30	2.7871	2.8800	2.9729	3.0658	3.1587	3.2516	3.3445	3.4374	3.5303	3.6232
40	3.7161	3.8090	3.9019	3.9948	4.0877	4.1806	4.2735	4.3664	4.4594	4.5523
50	4.6452	4.7381	4.8310	4.9239	5.0168	5.1097	5.2026	5.2955	5.3884	5.4813
60	5.5742	5.6671	5.7600	5.8529	5.9458	6.0387	6.1316	6.2245	6.3174	6.4103
70	6.5032	6.5961	6.6890	6.7819	6.8748	6.9677	7.0606	7.1535	7.2464	7.3393
80	7.4322	7.5252	7.6181	7.7110	7.8039	7.8968	7.9897	8.0826	8.1755	8.2684
90	8.3613	8.4542	8.5471	8.6400	8.7329	8.8258	8.9187	9.0116	9.1045	9.1974
100	9.2903	9.3832	9.4761	9.5690	9.6619	9.7548	9.8477	9.9406	10.0335	10.1264

Table C-16 Square meters into square feet.

Sq. M	0	1	2	3	4	5	6	7	8	9
	Sq. Ft									
0	—	10.76	21.53	32.29	43.06	53.82	64.58	75.35	86.11	96.88
10	107.64	118.40	129.17	139.93	150.70	161.46	172.22	182.99	193.75	204.51
20	215.28	226.04	236.81	247.57	258.33	269.10	279.86	290.63	301.39	312.15
30	322.92	333.68	344.45	355.21	365.97	376.74	387.50	398.27	409.03	419.79
40	430.56	441.32	452.08	462.85	473.61	484.38	495.14	505.90	516.67	527.43
50	538.20	548.96	559.72	570.49	581.25	592.02	602.78	613.54	624.31	635.07
60	645.84	656.60	667.36	678.13	688.89	699.65	710.42	721.18	731.95	742.71
70	753.47	764.24	775.00	785.77	796.53	807.29	818.06	828.82	839.59	850.35
80	861.11	871.88	882.64	893.41	904.17	914.93	925.70	936.46	947.22	957.99
90	968.75	979.52	990.28	1001.04	1011.81	1022.57	1033.34	1044.10	1054.86	1065.63
100	1076.39	1087.15	1097.92	1108.68	1119.45	1130.21	1140.97	1151.74	1162.50	1173.27

Table C-17 Acres to hectares.

[illegible]**Table C-18 Hectares to acres.**[illegible]

Table C-19 Basic conversions: units of capacity and volume.

1 cubic inch	16.38706 cubic centimeters
1 cubic inch	.01638706 liters
1 cubic inch	0.5541 fluid ounces
1 cubic inch	.017316 U.S. quarts
1 cubic inch	.004329 U.S. gallons
1 cubic foot	0.02832 cubic meters
1 cubic foot	1728 cubic inches
1 cubic foot	28.317 liters
1 cubic foot	.03704 cubic yards
1 cubic foot	7.48052 U.S. gallons
1 cubic yard	764,559 cubic centimeters
1 cubic yard	46,656 cubic inches
1 cubic yard	27 cubic feet
1 cubic yard	0.7646 cubic meters
1 fluid ounce	8 fluid drams
1 fluid ounce	.03125 U.S. quarts
1 fluid ounce	.007812 U.S. gallons
1 fluid ounce	.29573 liters
1 U.S. quart	256 fluid drams
1 U.S. quart	32 fluid ounces
1 U.S. quart	.94633 liters
1 U.S. gallon	3.78533 liters
1 U.S. gallon	268.8 cubic inches
1 cubic centimeter	0.061 cubic inches
1 milliliter	0.03381 fluid ounces
1 milliliter	0.001057 U.S. quarts
1 milliliter	0.000264 U.S. gallons
1 cubic decimeter	1 liter
1 liter	61.023 cubic inches
1 liter	0.0353 cubic feet
1 liter	1.0567 U.S. quarts
1 liter	0.2642 U.S. gallons
1 liter	33.8147 fluid ounces

Table C-20 Capacity and volume conversion formulas.

To Convert	into	Multiply by
U.S. pints	liters	0.4732
U.S. quarts	liters	0.9463
U.S. gallons	liters	3.78533
U.S. gallons	cubic meters	0.00455
cubic inches	cubic centimeters	16.387064
cubic feet	liters	28.316847
cubic yards	cubic meters	0.764555
liters	U.S. pints	2.1134
liters	U.S. quarts	1.0567
liters	U.S. gallons	0.2642

Table C-21 Dry measure.

1 pint	0.5506 liters
1 quart	1.1012 liters
1 peck	8.8096 liters
1 bushel	35.2383 liters
1 liter	1.8162 pints
1 liter	.9081 quarts
1 liter	.1135 pecks
1 liter	.0284 bushels

Table C-22 Liquid measure.

1 dash	6 drops—1⅓ milliliters
1 teaspoon	⅓ fluid ounces—4 milliliters
1 tablespoon	½ fluid ounces—15 milliliters
1 pony	1 fluid ounce—28⅓ milliliters
1 jigger	1½ fluid ounces—42½ milliliters
1 miniature	3.4 fluid ounces—100 milliliters
1 nip (split)	6.3 fluid ounces—187 milliliters
1 half-bottle	12.7 fluid ounces—375 milliliters
1 pint	16 fluid ounces—378½ milliliters
1 bottle	25.4 fluid ounces—750 milliliters
1 fifth	25.6 fluid ounces—757 milliliters
1 quart	32 fluid ounces—946⅓ milliliters
1 liter	33.8 fluid ounces—1000 milliliters
1½ liters	50.7 fluid ounces—1500 milliliters
3 liters	101.4 fluid ounces—3000 milliliters
1 gallon	128.0 fluid ounces—3785 milliliters

Table C-23 U.S. gallons into liters.

Gal.	0	1	2	3	4	5	6	7	8	9
0	—	3.785	7.571	11.356	15.141	18.927	22.712	26.497	30.283	34.068
10	37.853	41.639	45.424	49.209	52.995	56.780	50.565	64.351	68.136	71.921
20	75.707	79.492	83.277	87.063	90.848	94.633	98.419	102.204	105.989	109.775
30	113.560	117.345	121.131	124.916	128.701	132.487	136.272	140.057	143.843	147.628
40	151.413	155.199	158.984	162.769	166.555	170.340	174.125	177.911	181.696	185.481
50	189.267	193.052	196.837	200.622	204.408	208.193	211.978	215.764	219.549	223.334
60	227.120	230.905	234.690	238.476	242.261	246.046	249.832	253.617	257.402	261.188
70	264.973	268.758	272.544	276.329	280.114	283.900	287.685	291.470	295.256	299.041
80	302.826	306.612	310.397	314.182	317.968	321.753	325.538	329.324	333.109	336.894
90	340.680	344.465	348.250	352.036	355.821	359.606	363.392	367.177	370.962	374.748
100	378.533	382.318	386.104	389.889	393.674	397.460	401.245	405.030	408.816	412.601

Table C-24 Liters into U.S. gallons.

Liters	0	1	2	3	4	5	6	7	8	9
0	—	.265	.528	.793	1.057	1.321	1.585	1.849	2.114	2.378
10	2.642	2.906	3.170	3.435	3.699	3.963	4.227	4.491	4.756	5.020
20	5.284	5.548	5.812	6.077	6.341	6.605	6.869	7.133	7.398	7.662
30	7.926	8.190	8.454	8.719	8.983	9.247	9.511	9.775	10.040	10.304
40	10.568	10.832	11.096	11.361	11.625	11.889	12.153	12.417	12.682	12.946
50	13.210	13.474	13.738	14.003	14.267	14.531	14.795	15.059	15.324	15.588
60	15.852	16.116	16.380	16.645	16.909	17.173	17.437	17.701	17.966	18.230
70	18.494	18.758	19.022	19.287	19.551	19.815	20.079	20.343	20.608	20.872
80	21.136	21.400	21.664	21.929	22.193	22.457	22.721	22.985	23.250	23.514
90	23.778	24.042	24.306	24.571	24.835	25.099	25.363	25.627	25.892	26.156
100	26.420	26.684	26.948	27.213	27.477	27.741	28.005	28.269	28.534	28.798

Table C-25 Cubic inches into cubic centimeters.

Cu. In.	0	1	2	3	4	5	6	7	8	9
	Cu. Cm									
0	—	16.39	32.77	49.16	65.55	81.94	98.32	114.71	131.10	147.48
10	163.87	180.26	196.64	213.03	229.42	245.81	262.19	278.58	294.97	311.35
20	327.74	344.13	360.51	376.90	393.29	409.68	426.06	442.45	458.84	475.22
30	491.61	508.00	524.38	540.77	557.16	573.55	589.93	606.32	622.71	639.09
40	655.48	671.87	688.25	704.64	721.03	737.42	753.80	770.19	786.58	802.96
50	819.35	835.74	852.12	868.51	884.90	901.29	917.67	934.06	950.45	966.83
60	983.22	999.61	1016.0	1032.4	1048.8	1065.2	1081.5	1097.9	1114.3	1130.7
70	1147.1	1163.5	1179.9	1196.3	1212.6	1229.0	1245.4	1261.8	1278.2	1294.6
80	1311.0	1327.4	1343.7	1360.1	1376.5	1392.9	1409.3	1425.7	1442.1	1458.5
90	1474.8	1491.2	1507.6	1524.0	1540.4	1556.8	1573.2	1589.6	1605.9	1622.3
100	1638.7	1655.1	1671.5	1687.9	1704.3	1720.7	1737.1	1753.4	1769.8	1786.2

Table C-26 Cubic centimeters into cubic inches.

Cu. Cm	0	1	2	3	4	5	6	7	8	9
	Cu. In									
0	—	0.0610	0.1221	0.1831	0.2441	0.3051	0.3661	0.4272	0.4882	0.5492
10	0.6102	0.6713	0.7323	0.7933	0.8543	0.9154	0.9764	1.0374	1.0984	1.1595
20	1.2205	1.2815	1.3425	1.4036	1.4646	1.5256	1.5866	1.6477	1.7087	1.7697
30	1.8307	1.8917	1.9528	2.0138	2.0748	2.1358	2.1969	2.2579	2.3189	2.3799
40	2.4410	2.5020	2.5630	2.6240	2.6850	2.7461	2.8071	2.8681	2.9291	2.9902
50	3.0512	3.1122	3.1732	3.2343	3.2953	3.3563	3.4173	3.4784	3.5394	3.6004
60	3.6614	3.7225	3.7835	3.8445	3.9055	3.9665	4.0276	4.0886	4.1496	4.2106
70	4.2717	4.3327	4.3937	4.4548	4.5158	4.5768	4.6378	4.6988	4.7599	4.8209
80	4.8819	4.9429	5.0040	5.0650	5.1260	5.1870	5.2480	5.3091	5.3701	5.4311
90	5.4921	5.5532	5.6142	5.6752	5.7362	5.7973	5.8583	5.9193	5.9803	6.0414
100	6.1024	6.1634	6.2244	6.2854	6.3465	6.4075	6.4685	6.5295	6.5906	6.6516

Table C-27 Cubic feet into liters.

Cu. Ft.	0	1	2	3	4	5	6	7	8	9
	Liters									
0	—	28.32	56.63	84.95	113.26	141.58	169.90	198.21	226.53	254.84
10	283.16	311.48	339.79	368.11	396.42	424.74	453.06	481.37	509.69	538.01
20	566.32	594.64	622.95	651.27	679.59	707.90	736.22	764.53	792.85	821.17
30	849.48	877.80	906.11	934.43	962.75	991.06	1019.4	1047.7	1076.0	1104.3
40	1132.6	1161.0	1189.3	1217.6	1245.9	1274.2	1302.5	1330.9	1359.2	1387.5
50	1415.8	1444.1	1472.4	1500.8	1529.1	1557.4	1585.7	1614.0	1642.3	1670.7
60	1699.0	1727.3	1755.6	1783.9	1812.2	1840.5	1868.9	1897.2	1925.5	1953.8
70	1982.1	2010.4	2038.8	2067.1	2095.4	2123.7	2152.0	2180.3	2208.7	2237.0
80	2265.3	2293.6	2321.9	2350.2	2378.6	2406.9	2435.2	2463.5	2491.8	2520.1
90	2548.4	2576.8	2605.1	2633.4	2661.7	2690.0	2718.3	2746.7	2775.0	2803.3
100	2831.6	2859.9	2888.2	2916.6	2944.9	2973.2	3001.5	3029.8	3058.1	3086.5

Table C-28 Liters into cubic feet.

Liters	0	1	2	3	4	5	6	7	8	9
	Cu. Ft.									
0	—	0.0353	0.0706	0.1060	0.1413	0.1766	0.2119	0.2472	0.2825	0.3178
10	0.3532	0.3885	0.4238	0.4591	0.4944	0.5297	0.5651	0.6004	0.6357	0.6710
20	0.7063	0.7416	0.7769	0.8123	0.8476	0.8829	0.9182	0.9535	0.9888	1.0242
30	1.0595	1.0948	1.1301	1.1654	1.2007	1.2361	1.2714	1.3067	1.3420	1.3773
40	1.4126	1.4479	1.4833	1.5186	1.5539	1.5892	1.6245	1.6598	1.6952	1.7305
50	1.7658	1.8011	1.8364	1.8717	1.9071	1.9424	1.9777	2.0130	2.0483	2.0836
60	2.1189	2.1543	2.1896	2.2249	2.2602	2.2955	2.3308	2.3662	2.4015	2.4368
70	2.4721	2.5074	2.5427	2.5780	2.6134	2.6487	2.6840	2.7193	2.7546	2.7899
80	2.8253	2.8606	2.8959	2.9312	2.9665	3.0018	3.0372	3.0725	3.1078	3.1431
90	3.1784	3.2137	3.2490	3.2844	3.3197	3.3550	3.3903	3.4256	3.4609	3.4963
100	3.5315	3.5669	3.6021	3.6375	3.6728	3.7081	3.7434	3.7787	3.8140	3.8493

Table C-29 Cubic feet into cubic meters.

Cu. Ft.	0	1	2	3	4	5	6	7	8	9
	Cu. M									
0	—	0.0283	0.0566	0.0850	0.1133	0.1416	0.1689	0.1982	0.2265	0.2549
10	0.2832	0.3115	0.3398	0.3681	0.3964	0.4248	0.4531	0.4814	0.5097	0.5380
20	0.5663	0.5947	0.6230	0.6513	0.6796	0.7079	0.7352	0.7646	0.7929	0.8212
30	0.8495	0.8778	0.9061	0.9345	0.9628	0.9911	1.0194	1.0477	1.6760	1.1044
40	1.1327	1.1610	1.1893	1.2176	1.2459	1.2743	1.3026	1.3309	1.3592	1.3875
50	1.4158	1.4442	1.4725	1.5008	1.5291	1.5574	1.5857	1.6141	1.6424	1.6707
60	1.6990	1.7273	1.7556	1.7840	1.8123	1.8406	1.8689	1.8972	1.9256	1.9539
70	1.9822	2.0105	2.0388	2.0671	2.0955	2.1238	2.1521	2.1804	2.2087	2.2370
80	2.2654	2.2937	2.3220	2.3503	2.3786	2.4069	2.4353	2.4636	2.4919	2.5202
90	2.5485	2.5768	2.6052	2.6335	2.6618	2.6901	2.7184	2.7467	2.7751	2.8034
100	2.8317	2.8600	2.8883	2.9166	2.9450	2.9733	3.0016	3.0299	3.0582	3.0865

Table C-30 Cubic meters into cubic feet.

Cu. M	0	1	2	3	4	5	6	7	8	9
	Cu. Ft									
0	—	35.3	70.6	105.9	141.3	176.6	211.9	247.2	282.5	317.8
10	353.1	388.5	423.8	459.1	494.4	529.7	565.0	600.3	635.7	671.0
20	706.3	741.6	776.9	812.2	847.6	882.9	918.2	953.5	988.8	1024.1
30	1059.4	1094.8	1130.1	1165.4	1200.7	1236.0	1271.3	1306.6	1342.0	1377.3
40	1412.6	1447.9	1483.2	1518.5	1553.9	1589.2	1624.5	1659.8	1695.1	1730.4
50	1765.7	1801.1	1836.4	1871.7	1907.0	1942.3	1977.6	2012.9	2048.3	2083.6
60	2118.9	2154.2	2189.5	2224.8	2260.1	2295.5	2330.8	2366.1	2401.4	2436.7
70	2472.0	2507.3	2542.7	2578.0	2613.3	2648.6	2683.9	2719.2	2754.5	2789.9
80	2825.2	2860.5	2895.8	2931.1	2966.4	3001.8	3037.1	3072.4	3107.7	3143.0
90	3178.3	3213.6	3249.0	3284.3	3319.6	3354.9	3390.2	3425.5	3460.8	3496.2
100	3531.5	3566.8	3602.1	3637.4	3672.7	3708.0	3743.4	3778.7	3814.0	3849.3

Table C-31 Basic conversions: units of length.

1 inch	25.4 millimeters
1 inch	2.54 centimeters
1 inch	0.0254 meters
1 inch	0.08333 feet
1 inch	0.02777 yards
1 foot	304.8 millimeters
1 foot	30.48 centimeters
1 foot	0.3048 meters
1 foot	12 inches
1 foot	0.3333 yards
1 yard	91.44 centimeters
1 yard	0.9144 meters
1 yard	36 inches
1 yard	3 feet
1 mile	1609.34 meters
1 mile	63,360 inches
1 mile	5,280 feet
1 mile	1760 yards
1 rod	16.5 feet
1 furlong	40 rods
8 furlongs	1 mile
1 league	3 miles
1 knot	6,085 feet
1 knot	1.1526 miles
1 millimeter	.03937 inches
1 millimeter	1000 micrometers (µms)
1 centimeter	0.3937 inches
1 centimeter	0.03281 feet
1 centimeter	0.01094 yards
1 centimeter	0.01 meters
1 meter	39.37 inches
1 meter	3.2808 feet
1 meter	1.0936 yards
1 meter	100 centimeters
1 kilometer	0.62137 miles

Table C-32 Length conversion formulas.

To Convert	into	Multiply by
inches	millimeters	25.4
inches	centimeters	2.54
feet	meters	0.304800
yards	meters	0.914400
miles	kilometers	1.609344
centimeters	inches	0.393701
meters	feet	3.280840
meters	yards	1.093613
kilometers	miles	0.6213712

Table C-33 Millimeters into inches.

Mm	In.	Mm	In.
1	.03937	19	.74803
2	.07874	20	.78740
3	.11811	21	.82677
4	.15743	22	.86614
5	.19685	23	.90551
6	.23622	24	.94488
7	.27559	25	.98425
8	.31496	25.4	1.0
9	.35433	38.1	1.5
10	.39370	50.0	1.968
11	.43307	50.8	2.0
12	.47244	75.0	2.953
13	.51181	76.2	3.0
14	.55118	100.0	3.937
15	.59055	101.6	4.0
16	.62992	127.0	5.0
17	.66929	152.4	6.0
18	.70866		

Table C-34 Length equivalents.

Fraction In.	Decimal In.	Mm
1/64	.015625	0.397
1/32	.03125	0.794
3/64	.046875	1.191
1/16	.0625	1.588
5/64	.078125	1.984
3/32	.09375	2.381
1/8	.125	3.175
3/16	.1875	4.763
1/4	.25	6.35
5/16	.3125	7.938
3/8	.375	9.525
3/16	.4375	11.113
1/2	.50	12.70
9/16	.5625	14.288
5/8	.625	15.875
11/16	.6875	17.463
3/4	.75	19.05
13/16	.8125	20.638
7/8	.875	22.225
15/16	.9375	23.813
1	1.0	25.40

Table C-35 Hundredths of a millimeter into inches.

Mm	In.	Mm	In.	Mm	In.	Mm	In.	Mm	In.
0.01	0.0004	0.21	0.0083	0.41	0.0161	0.61	0.0240	0.81	0.0319
0.02	0.0008	0.22	0.0087	0.42	0.0165	0.62	0.0244	0.82	0.0323
0.03	0.0012	0.23	0.0091	0.43	0.0169	0.63	0.0248	0.83	0.0327
0.04	0.0016	0.24	0.0094	0.44	0.0172	0.64	0.0252	0.84	0.0331
0.05	0.0020	0.25	0.0098	0.45	0.0177	0.65	0.0256	0.85	0.0335
0.06	0.0024	0.26	0.0102	0.46	0.0181	0.66	0.0260	0.86	0.0339
0.07	0.0028	0.27	0.0106	0.47	0.0185	0.67	0.0264	0.87	0.0343
0.08	0.0032	0.28	0.0110	0.48	0.0189	0.68	0.0268	0.88	0.0346
0.09	0.0035	0.29	0.0114	0.49	0.0193	0.69	0.0272	0.89	0.0350
0.10	0.0039	0.30	0.0118	0.50	0.0197	0.70	0.0276	0.90	0.0354
0.11	0.0043	0.31	0.0122	0.51	0.0201	0.71	0.0280	0.91	0.0358
0.12	0.0047	0.32	0.0126	0.52	0.0205	0.72	0.0283	0.92	0.0362
0.13	0.0051	0.33	0.0130	0.53	0.0209	0.73	0.0287	0.93	0.0366
0.14	0.0055	0.34	0.0134	0.54	0.0213	0.74	0.0291	0.94	0.0370
0.15	0.0059	0.35	0.0138	0.55	0.0217	0.75	0.0295	0.95	0.0374
0.16	0.0063	0.36	0.0142	0.56	0.0220	0.76	0.0299	0.96	0.0378
0.17	0.0067	0.37	0.0146	0.57	0.0224	0.77	0.0303	0.97	0.0382
0.18	0.0071	0.38	0.0150	0.58	0.0228	0.78	0.0307	0.98	0.0386
0.19	0.0075	0.39	0.0154	0.59	0.0232	0.79	0.0311	0.99	0.0390
0.20	0.0079	0.40	0.0157	0.60	0.0236	0.80	0.0315	1.00	0.0394

Table C-36 Inches into millimeters.

In.	Mm	In.	Mm	In.	Mm
1/64	0.3969	51/64	20.2406	2-5/32	54.7688
1/32	0.7938	13/16	20.5375	2-3/16	55.5625
3/64	1.1906	53/64	21.0344	2-9/32	56.3563
1/16	1.5875	27/32	21.4313	2-1/4	57.1500
5/64	1.9844	55/64	21.8281	2-9/32	57.9438
3/32	2.3813	7/8	22.2250	2-5/16	58.7375
7/64	2.7781	57/64	22.6219	2-11/32	59.5313
1/8	3.1750	29/32	23.0188	2-3/8	60.3250
9/64	3.5719	59/64	23.4156	2-13/32	61.1188
5/32	3.9688	15/16	23.8125	2-7/16	61.9125
11/64	4.3656	61/64	24.2094	2-15/32	62.7063
3/16	4.7625	31/32	24.6063	2-1/2	63.5000
13/64	5.1594	63/64	25.0031	2-17/32	64.2938
7/32	5.5563	1	25.4000	2-9/16	65.0875
15/64	5.9531	1-1/32	26.1938	2-19/32	65.8813
1/4	6.3500	1-1/16	26.9875	2-5/8	66.6750
17/64	6.7469	1-3/32	27.7813	2-21/32	67.4688
9/32	7.1438	1-1/6	28.5750	2-11/16	68.2625
19/64	7.5406	1-5/32	29.3688	2-23/32	69.0563
5/16	7.9375	1-3/16	30.1625	2-3/4	69.8500
21/64	8.3344	1-7/32	30.9563	2-25/32	70.6438
11/32	8.7313	1-1/4	31.7500	2-13/16	71.4375
23/64	9.1281	1-9/32	32.5438	2-27/32	72.2313
3/8	9.5250	1-5/16	33.3375	2-7/8	73.0250
25/64	9.9219	1-11/32	34.1313	2-29/32	73.8188
13/32	10.3188	1-3/8	34.9250	2-15/16	74.6125
27/64	10.7156	1-13/32	35.7188	2-31/32	75.4063
7/16	11.1125	1-7/16	36.5125	3	76.2000
29/64	11.5094	1-15/32	37.3063	3-1/32	76.9938
15/32	11.9063	1-1/2	38.1000	3-1/16	77.7875
31/64	12.3031	1-17/32	38.8938	3-3/32	78.5813
1/2	12.7000	1-9/16	39.6875	3-1/8	79.3750
33/64	13.0969	1-19/32	40.4813	3-5/32	80.1688
17/32	13.4938	1-5/8	41.2750	3-3/16	80.9625
35/64	13.8906	1-21/32	42.0688	3-7/32	81.7563
9/16	14.2875	1-11/16	42.8625	3-1/4	82.5500
37/64	14.6844	1-23/32	43.6563	3-9/32	83.3438
19/32	15.0813	1-3/4	44.4500	3-5/16	84.1375
39/64	15.4781	1-25/32	45.2438	3-11/32	84.9313
5/8	15.8750	1-13/16	46.0375	3-3/8	85.7250
41/64	16.2719	1-27/32	46.8313	3-13/32	86.5188
21/32	16.6688	1-7/8	47.6250	3-7/16	87.3125
43/64	17.0656	1-29/32	48.4188	3-15/32	88.1063
11/16	17.4625	1-15/16	49.2125	3-1/2	88.9000
45/64	17.8594	1-31/32	50.0063	3-17/32	89.6938
23/32	18.2563	2	50.8000	3-9/16	90.4875
47/64	18.6531	2-1/32	51.5938	3-19/32	91.2813
3/4	19.0550	2-1/16	52.3875	3-5/8	92.0750
49/64	19.4469	2-3/32	53.1813	3-21/32	92.8688
25/32	19.8438	2-1/8	53.9750	3-11/16	93.6625

Table C-36 (continued)

In.	Mm	In.	Mm	In.	Mm
3-23/32	94.4563	5-9/32	134.144	7-11/16	195.262
3-3/4	95.2500	5-5/16	134.938	7-3/4	196.850
3-25/32	96.0438	5-11/32	135.731	7-13/16	198.438
3-13/16	96.8375	5-3/8	136.525	7-7/8	200.025
3-27/32	97.6313	5-13/32	137.319	7-15/16	201.612
3-7/8	98.4250	5-7/16	138.112	8	203.200
3-29/32	99.2188	5-15/32	138.906	8-1/16	204.783
3-15/16	100.012	5-1/2	139.700	8-1/8	206.375
3-31/32	100.806	5-17/32	140.494	8-3/16	207.962
4	101.600	5-9/16	141.288	8-1/4	209.550
4-1/32	102.394	5-19/32	142.081	8-5/16	211.138
4-1/16	103.188	5-5/8	142.875	8-3/8	212.725
4-3/32	103.981	5-21/32	143.669	8-7/16	214.312
4-1/8	104.775	5-11/16	144.462	8-1/2	215.900
4-5/32	105.569	5-23/32	145.256	8-9/16	217.488
4-3/16	106.362	5-3/4	146.050	8-6/8	219.075
4-7/32	107.156	5-25/32	146.844	8-11/16	220.662
4-1/4	107.950	5-13/16	147.638	8-3/4	222.250
4-9/32	108.744	5-27/32	148.431	8-13/16	223.838
4-5/16	109.538	5-7/8	149.225	8-7/8	225.425
4-11/32	110.331	5-29/32	150.019	8-15/16	227.012
4-3/8	111.125	5-15/16	150.812	9	228.600
4-13/32	111.919	5-31/32	151.606	9-1/16	230.188
4-7/16	112.712	6	152.400	9-1/8	231.775
4-15/32	113.506	6-1/16	153.194	9-3/16	233.362
4-1/2	114.300	6-1/8	153.988	9-1/4	234.950
4-17/32	115.094	6-3/16	154.782	9-5/16	236.538
4-9/16	115.888	6-1/4	155.575	9-3/8	238.125
4-19/32	116.681	6-5/16	156.369	9-7/16	239.712
4-5/8	117.475	6-3/8	157.162	9-1/2	241.300
4-21/32	118.269	6-7/16	157.956	9-9/16	242.888
4-11/16	119.062	6-1/2	158.750	9-5/8	244.475
4-23/32	119.856	6-9/16	159.544	9-11/16	246.062
4-3/4	120.650	6-5/8	160.338	9-3/4	247.650
4-25/32	121.444	6-11/16	161.132	9-13/16	249.238
4-13/18	122.238	6-3/4	161.925	9-7/8	250.825
4-27/32	123.031	6-13/16	162.719	9-15/16	252.412
4-7/8	123.825	6-7/8	163.512	10	254.000
4-29/32	124.619	6-15/16	164.306	10-1/16	255.588
4-15/16	125.412	7	165.100	10-1/8	257.175
4-31/32	126.206	7-1/16	165.894	10-3/16	258.762
5	127.000	7-1/8	166.688	10-1/4	260.350
5-1/32	127.794	7-3/16	167.482	10-5/16	261.938
5-1/16	128.588	7-1/4	168.275	10-3/8	263.525
5-3/32	129.381	7-5/16	169.069	10-7/16	265.112
5-1/8	130.175	7-3/8	169.862	10-1/2	266.700
5-5/32	130.969	7-7/16	170.656	10-9/16	268.288
5-3/16	131.762	7-1/2	171.450	10-5/8	269.875
5-7/32	132.556	7-9/16	172.244	10-11/16	271.462
5-1/4	133.350	7-5/8	173.038	10-3/4	273.050

(continued)

Table C-36 (continued)

In.	Mm	In.	Mm	In.	Mm
10-13/16	274.638	13	330.200	28	711.200
10-7/8	276.225	14	355.600	29	736.600
10-15/16	277.812	15	381.000	30	762.000
11	279.400	16	406.400	31	787.400
11-1/16	280.988	17	431.800	32	812.800
11-1/8	282.575	18	457.200	33	838.200
11-3/16	284.162	19	482.600	34	863.600
11-1/4	285.750	20	508.000	35	889.000
11-5/16	287.338	21	533.400	36	914.400
11-3/8	288.925	22	558.800	37	939.800
11-7/16	290.512	23	584.200	38	965.200
11-1/2	292.100	24	609.600	39	990.600
11-9/16	293.688	25	635.000	40	1016.00
11-5/8	295.275	26	660.400	41	1041.40
11-11/16	296.862	27	685.800	42	1066.80
11-3/4	298.450				
11-13/16	300.038				
11-7/8	301.625				
11-15/16	303.212				
12	304.800				

Table C-37 Millimeters into inches.

Mm	In.	Mm	In.	Mm	In.	Mm	In.
1	0.0394	51	2.0079	101	3.9764	151	5.9449
2	0.0787	52	2.0472	102	4.0158	152	5.9843
3	0.1181	53	2.0866	103	4.0551	153	6.0236
4	0.1575	54	2.1260	104	4.0945	154	6.0630
5	0.1969	55	2.1654	105	4.1339	155	6.1024
6	0.2362	56	2.2047	106	4.1732	156	6.1417
7	0.2756	57	2.2441	107	4.2126	157	6.1811
8	0.3150	58	2.2835	108	4.2520	158	6.2205
9	0.3543	59	2.3228	109	4.2913	159	6.2599
10	0.3937	60	2.3622	110	4.3307	160	6.2992
11	0.4331	61	2.4016	111	4.3701	161	6.3386
12	0.4724	62	2.4409	112	4.4095	162	6.3780
13	0.5118	63	2.4803	113	4.4488	163	6.4173
14	0.5512	64	2.5197	114	4.4882	164	6.4567
15	0.5906	65	2.5591	115	4.5276	165	6.4961
16	0.6299	66	2.5984	116	4.5669	166	6.5354
17	0.6693	67	2.6378	117	4.6063	167	6.5748
18	0.7087	68	2.6772	118	4.6457	168	6.6142
19	0.7480	69	2.7165	119	4.6850	169	6.6535
20	0.7874	70	2.7559	120	4.7244	170	6.6929
21	0.8268	71	2.7953	121	4.7638	171	6.7323
22	0.8661	72	2.8347	122	4.8032	172	6.7717
23	0.9055	73	2.8740	123	4.8425	173	6.8110
24	0.9449	74	2.9134	124	4.8819	174	6.8504
25	0.9843	75	2.9528	125	4.9213	175	6.8898
26	1.0236	76	2.9921	126	4.9606	176	6.9291
27	1.0630	77	3.0315	127	5.0000	177	6.9685
28	1.1024	78	3.0709	128	5.0394	178	7.0079
29	1.1417	79	3.1102	129	5.0787	179	7.0472
30	1.1811	80	3.1496	130	5.1181	180	7.0866
31	1.2205	81	3.1890	131	5.1575	181	7.1260
32	1.2598	82	3.2284	132	5.1969	182	7.1654
33	1.2992	83	3.2677	133	5.2362	183	7.2047
34	1.3386	84	3.3071	134	5.2756	184	7.2441
35	1.3780	85	3.3465	135	5.3150	185	7.2835
36	1.4173	86	3.3858	136	5.3543	186	7.3228
37	1.4567	87	3.4252	137	5.3937	187	7.3622
38	1.4961	88	3.4646	138	5.4331	188	7.4016
39	1.5354	89	3.5039	139	5.4724	189	7.4409
40	1.5748	90	3.5433	140	5.5118	190	7.4803
41	1.6142	91	3.5827	141	5.5512	191	7.5197
42	1.6535	92	3.6221	142	5.5906	192	7.5591
43	1.6929	93	3.6614	143	5.6299	193	7.5984
44	1.7323	94	3.7008	144	5.6693	194	7.6378
45	1.7717	95	3.7402	145	5.7087	195	7.6772
46	1.8110	96	3.7795	146	5.7480	196	7.7165
47	1.8504	97	3.8189	147	5.7874	197	7.7559
48	1.8898	98	3.8583	148	5.8268	198	7.7953
49	1.9291	99	3.8976	149	5.8661	199	7.8347
50	1.9685	100	3.9370	150	5.9055	200	7.8740

(continued)

Table C-37 (continued)

Mm	In.	Mm	In.	Mm	In.	Mm	In.
201	7.9134	251	9.8819	301	11.8504	351	13.8189
202	7.9528	252	9.9213	302	11.8898	352	13.8583
203	7.9921	253	9.9606	303	11.9291	353	13.8976
204	8.0315	254	10.0000	304	11.9686	354	13.9370
205	8.0709	255	10.0393	305	12.0079	355	13.9764
206	8.1102	256	10.0787	306	12.0472	356	14.0157
207	8.1496	257	10.1181	307	12.0866	357	14.0551
208	8.1890	258	10.1575	308	12.1260	358	14.0945
209	8.2284	259	10.1969	309	12.1654	359	14.1339
210	8.2677	260	10.2362	310	12.2047	360	14.1732
211	8.3071	261	10.2756	311	12.2441	361	14.2126
212	8.3465	262	10.3150	312	12.2835	362	14.2520
213	8.3858	263	10.3543	313	12.3228	363	14.2913
214	8.4252	264	10.3937	314	12.3622	364	14.3307
215	8.4646	265	10.4331	315	12.4016	365	14.3701
216	8.5039	266	10.4724	316	12.4409	366	14.4094
217	8.5433	267	10.5118	317	12.4803	367	14.4488
218	8.5827	268	10.5512	318	12.5197	368	14.4882
219	8.6221	269	10.5906	319	12.5591	369	14.5276
220	8.6614	270	10.6299	320	12.5984	370	14.5669
221	8.7008	271	10.6693	321	12.6378	371	14.6063
222	8.7402	272	10.7087	322	12.6772	372	14.6457
223	8.7795	273	10.7480	323	12.7165	373	14.6850
224	8.8189	274	10.7874	324	12.7559	374	14.7244
225	8.8583	275	10.8268	325	12.7953	375	14.7638
226	8.8976	276	10.8661	326	12.8346	376	14.8031
227	8.9370	277	10.9055	327	12.8740	377	14.8425
228	8.9764	278	10.9449	328	12.9134	378	14.8819
229	9.0158	279	10.9843	329	12.9528	379	14.9213
230	9.0551	280	11.0236	330	12.9921	380	14.9606
231	9.0945	281	11.0630	331	13.0315	381	15.0000
232	9.1339	282	11.1024	332	13.0709	382	15.0394
233	9.1732	283	11.1417	333	13.1102	383	15.0787
234	9.2126	284	11.1811	334	13.1496	384	15.1181
235	9.2520	285	11.2205	335	13.1890	385	15.1575
236	9.2913	286	11.2598	336	13.2283	386	15.1969
237	9.3307	287	11.2992	337	13.2677	387	15.2362
238	9.3701	288	11.3386	338	13.3071	388	15.2756
239	9.4095	289	11.3780	339	13.3465	389	15.3150
240	9.4488	290	11.4173	340	13.3858	390	15.3543
241	9.4882	291	11.4567	341	13.4252	391	15.3937
242	9.5276	292	11.4961	342	13.4646	392	15.4331
243	9.5669	293	11.5354	343	13.5039	393	15.4724
244	9.6063	294	11.5748	344	13.5433	394	15.5118
245	9.6457	295	11.6142	345	13.5827	395	15.5512
246	9.6850	296	11.6535	346	13.6220	396	15.5906
247	9.7244	297	11.6929	347	13.6614	397	15.6299
248	9.7638	298	11.7323	348	13.7008	398	15.6693
249	9.8031	299	11.7717	349	13.7402	399	15.7087
250	9.8425	300	11.8110	350	13.7795	400	15.7480

Table C-37 (continued)

Mm	In.	Mm	In.	Mm	In.	Mm	In.
401	15.7874	451	17.7559	501	19.7244	551	21.6929
402	15.8268	452	17.7963	502	19.7638	552	21.7323
403	15.8661	453	17.8346	503	19.8031	553	21.7717
404	15.9055	454	17.8740	504	19.8425	554	21.8110
405	15.9449	455	17.9134	505	19.8819	555	21.8504
406	15.9832	456	17.9528	506	19.9213	556	21.8898
407	16.0236	457	17.9921	507	19.9606	557	21.9291
408	16.0630	458	18.0315	508	20.0000	558	21.9685
409	16.1024	459	18.0709	509	20.0394	559	22.0079
410	16.1417	460	18.1102	510	20.0787	560	22.0472
411	16.1811	461	18.1496	511	20.1181	561	22.0866
412	16.2205	462	18.1890	512	20.1575	562	22.1260
413	16.2598	463	18.2283	513	20.1969	563	22.1654
414	16.2992	464	18.2677	514	20.2362	564	22.2047
415	16.3386	465	18.3071	515	20.2756	565	22.2441
416	16.3780	466	18.3465	516	20.3150	566	22.2835
417	16.4173	467	18.3858	517	20.3543	567	22.3228
418	16.4567	468	18.4252	518	20.3937	568	22.3622
419	16.4961	469	18.4646	519	20.4331	569	22.4016
420	16.5354	470	18.5039	520	20.4724	570	22.4409
421	16.5748	471	18.5433	521	20.5118	571	22.4803
422	16.6142	472	18.5827	522	20.5512	572	22.5197
423	16.6535	473	18.6220	523	20.5906	573	22.5591
424	16.6929	474	18.6614	524	20.6299	574	22.5984
425	16.7323	475	18.7008	525	20.6693	575	22.6378
426	16.7716	476	18.7402	526	20.7087	576	22.6772
427	16.8110	477	18.7795	527	20.7480	577	22.7165
428	16.8504	478	18.8189	528	20.7874	578	22.7559
429	16.8898	479	18.8583	529	20.8268	579	22.7953
430	16.9291	480	18.8976	530	20.8661	580	22.8346
431	16.9685	481	18.9370	531	20.9055	581	22.8740
432	17.0079	482	18.9764	532	20.9449	582	22.9134
433	17.0472	483	19.0157	533	20.9843	583	22.9528
434	17.0866	484	19.0551	534	21.0236	584	22.9921
435	17.1260	485	19.0945	535	21.0630	585	23.0315
436	17.1654	486	19.1339	536	21.1024	586	23.0709
437	17.2047	487	19.1732	537	21.1417	587	23.1102
438	17.2441	488	19.2126	538	21.1811	588	23.1496
439	17.2835	489	19.2520	539	21.2205	589	23.1890
440	17.3228	490	19.2813	540	21.2598	590	23.2283
441	17.3622	491	19.3307	541	21.2992	591	23.2677
442	17.4016	492	19.3701	542	21.3386	592	23.3071
443	17.4409	493	19.4094	543	21.3780	593	23.3465
444	17.4803	494	19.4488	544	21.4173	594	23.3858
445	17.5197	495	19.4882	545	21.4567	595	23.4252
446	17.5591	496	19.5276	546	21.4961	596	23.4646
447	17.5984	497	19.5669	547	21.5354	597	23.5039
448	17.6378	498	19.6063	548	21.5743	598	23.5433
449	17.6772	499	19.6457	549	21.6142	599	23.5827
450	17.7165	500	19.6850	550	21.6535	600	23.6220

(continued)

Table C-37 (continued)

Mm	In.	Mm	In.	Mm	In.	Mm	In.
601	23.6614	651	25.6299	701	27.5984	751	29.5669
602	23.7008	652	25.6693	702	27.6378	752	29.6063
603	23.7402	653	25.7087	703	27.6772	753	29.6457
604	23.7795	654	25.7480	704	27.7165	754	29.6850
605	23.8189	655	25.7874	705	27.7559	755	29.7244
606	23.8583	656	25.8268	706	27.7953	756	29.7638
607	23.8976	657	25.8661	707	27.8346	757	29.8031
608	23.9370	658	25.9055	708	27.8740	758	29.8425
609	23.9764	659	25.9449	709	27.9134	759	29.8819
610	24.0157	660	25.9843	710	27.9528	760	29.9213
611	24.0551	661	26.0236	711	27.9921	761	29.9606
612	24.0945	662	26.0630	712	28.0315	762	30.0000
613	24.1339	663	26.1024	713	28.0709	763	30.0394
614	24.1732	664	26.1417	714	28.1102	764	30.0787
615	24.2126	665	26.1811	715	28.1496	765	30.1181
616	24.2520	666	26.2205	716	28.1890	766	30.1575
617	24.2913	667	26.2598	717	28.2283	767	30.1969
618	24.3307	668	26.2992	718	28.2677	768	30.2362
619	24.3701	669	26.3386	719	28.3071	769	30.2756
620	24.4094	670	26.3780	720	28.3465	770	30.3150
621	24.4488	671	26.4173	721	28.3858	771	30.3543
622	24.4882	672	26.4567	722	28.4252	772	30.3937
623	24.5276	673	26.4961	723	28.4646	773	30.4331
624	24.5669	674	26.5354	724	28.5039	774	30.4724
625	24.6063	675	26.5748	725	28.5433	775	30.5118
626	24.6457	676	26.6142	726	28.5827	776	30.5512
627	24.6850	677	26.6535	727	28.6220	777	30.5906
628	24.7244	678	26.6929	728	28.6614	778	30.6299
629	24.7638	679	26.7323	729	28.7008	779	30.6693
630	24.8031	680	26.7717	730	28.7402	780	30.7087
631	24.8425	681	26.8110	731	28.7795	781	30.7480
632	24.8819	682	26.8504	732	28.8189	782	30.7874
633	24.9213	683	26.8898	733	28.8583	783	30.8268
634	24.9606	684	26.9291	734	28.8976	784	30.8661
635	25.0000	685	26.9685	735	28.9370	785	30.9055
636	25.0394	686	27.0079	736	28.9764	786	30.9449
637	25.0787	687	27.0472	737	29.0157	787	30.9843
638	25.1181	688	27.0866	738	29.0551	788	31.0236
639	25.1575	689	27.1260	739	29.0945	789	31.0630
640	25.1969	690	27.1654	740	29.1339	790	31.1024
641	25.2362	691	27.2047	741	29.1732	791	31.1417
642	25.2756	692	27.2441	742	29.2126	792	31.1811
643	25.3150	693	27.2835	743	29.2520	793	31.2205
644	25.3543	694	27.3228	744	29.2913	794	31.2598
645	25.3937	695	27.3622	745	29.3307	795	31.2992
646	25.4331	696	27.4016	746	29.3701	796	31.3386
647	25.4724	697	27.4409	747	29.4094	797	31.3780
648	25.5118	698	27.4803	748	29.4488	798	31.4173
649	25.5512	699	27.5197	749	29.4882	799	31.4567
650	25.5906	700	27.5591	750	29.5276	800	31.4961

Table C-37 (continued)

Mm	In.	Mm	In.	Mm	In.	Mm	In.
801	31.5354	851	33.5039	901	35.4724	951	37.4409
802	31.5748	852	33.5433	902	35.5118	952	37.4803
803	31.6142	853	33.5827	903	35.5512	953	37.5197
804	31.6535	854	33.6220	904	35.5906	954	37.5591
805	31.6929	855	33.6614	905	35.6299	955	37.5984
806	31.7323	856	33.7008	906	35.6693	956	37.6378
807	31.7717	857	33.7402	907	35.7087	957	37.6772
808	31.8110	858	33.7795	908	35.7480	958	37.7165
809	31.8504	859	33.8189	909	35.7874	959	37.7559
810	31.8898	860	33.8583	910	35.8268	960	37.7953
811	31.9291	861	33.8976	911	35.8661	961	37.8346
812	31.9685	862	33.9370	912	35.9055	962	37.8740
813	32.0079	863	33.9764	913	35.9449	963	37.9134
814	32.0472	864	34.0157	914	35.9843	964	37.9528
815	32.0866	865	34.0551	915	36.0236	965	37.9921
816	32.1260	866	34.0945	916	36.0630	966	38.0315
817	32.1654	867	34.1339	917	36.1024	967	38.0709
818	32.2047	868	34.1732	918	36.1417	968	38.1102
819	32.2441	869	34.2126	919	36.1811	969	38.1496
820	32.2835	870	34.2520	920	36.2205	970	38.1890
821	32.3228	871	34.2913	921	36.2598	971	38.2283
822	32.3622	872	34.3307	922	36.2992	972	38.2677
823	32.4016	873	34.3701	923	36.3386	973	38.3071
824	32.4409	874	34.4094	924	36.3780	974	38.3465
825	32.4803	875	34.4488	925	36.4173	975	38.3858
826	32.5197	876	34.4882	926	36.4567	976	38.4252
827	32.5591	877	34.5276	927	36.4961	977	38.4646
828	32.5984	878	34.5670	928	36.5354	978	38.5039
829	32.6378	879	34.6063	929	36.5748	979	38.5433
830	32.6772	880	34.6457	930	36.6142	980	38.5827
831	32.7165	881	34.6850	931	36.6535	981	38.6220
832	32.7559	882	34.7244	932	36.6929	982	38.6614
833	32.7953	883	34.7638	933	36.7323	983	38.7008
834	32.8346	884	34.8031	934	36.7717	984	38.7402
835	32.8740	885	34.8425	935	36.8110	985	38.7795
836	32.9134	886	34.8819	936	36.8504	986	38.8189
837	32.9528	887	34.9213	937	36.8898	987	38.8583
838	32.9921	888	34.9606	938	36.9291	988	38.8976
839	33.0315	889	35.0000	939	36.9685	989	38.9370
840	33.0709	890	35.0394	940	37.0079	990	38.9764
841	33.1102	891	35.0787	941	37.0472	991	39.0157
842	33.1496	892	35.1181	942	37.0866	992	39.0551
843	33.1890	893	35.1575	943	37.1260	993	39.0945
844	33.2283	894	35.1969	944	37.1654	994	39.1339
845	33.2677	895	35.2362	945	37.2047	995	39.1732
846	33.3071	896	35.2756	946	37.2441	996	39.2126
847	33.3465	897	35.3150	947	37.2835	997	39.2520
848	33.3858	898	35.3543	948	37.3228	998	39.2913
849	33.4252	899	35.3937	949	37.3622	999	39.3307
850	33.4646	900	35.4331	950	37.4016	1000	39.3701

(continued)

Table C-38 Decimals of an inch into millimeters.

In.	Mm	In.	Mm	In.	Mm	In.	Mm
0.001	0.025	0.200	5.08	0.470	11.94	0.740	18.80
0.002	0.051	0.210	5.33	0.480	12.19	0.750	19.05
0.003	0.076	0.220	5.59	0.490	12.45	0.760	19.30
0.004	0.102	0.230	5.84	0.500	12.70	0.770	19.56
0.005	0.127	0.240	6.10	0.510	12.95	0.780	19.81
0.006	0.152	0.250	6.35	0.520	13.21	0.790	20.07
0.007	0.178	0.260	6.60	0.530	13.46	0.800	20.32
0.008	0.203	0.270	6.86	0.540	13.72	0.810	20.57
0.009	0.229	0.280	7.11	0.550	13.97	0.820	20.83
0.010	0.254	0.290	7.37	0.560	14.22	0.830	21.08
0.020	0.508	0.300	7.62	0.570	14.48	0.840	21.34
0.030	0.762	0.310	7.87	0.580	14.73	0.850	21.59
0.040	1.016	0.320	8.13	0.590	14.99	0.860	21.84
0.050	1.270	0.330	8.38	0.600	15.24	0.870	22.10
0.060	1.524	0.340	8.65	0.610	15.49	0.880	22.35
0.070	1.778	0.350	8.89	0.620	15.75	0.890	22.61
0.080	2.032	0.360	9.14	0.630	16.00	0.900	22.86
0.090	2.286	0.370	9.40	0.640	16.26	0.910	23.11
0.100	2.540	0.380	9.65	0.650	16.51	0.920	23.37
0.110	2.794	0.390	9.91	0.660	16.76	0.930	23.62
0.120	3.048	0.400	10.16	0.670	17.02	0.940	23.88
0.130	3.302	0.410	10.41	0.680	17.27	0.950	24.13
0.140	3.56	0.420	10.67	0.690	17.53	0.960	24.38
0.150	3.81	0.430	10.92	0.700	17.78	0.970	24.64
0.160	4.06	0.440	11.18	0.710	18.03	0.980	24.89
0.170	4.32	0.450	11.43	0.720	18.29	0.990	25.15
0.180	4.57	0.460	11.68	0.730	18.54	1.000	25.40
0.190	4.83						

Table C-39 Inches into centimeters.

In.	0	1	2	3	4	5	6	7	8	9
	Cm	Cm	Cm	Cm	Cm	Cm	Cm	Cm	Cm	Cm
0		2.54	5.08	7.62	10.16	12.70	15.24	17.78	20.32	22.86
10	25.40	27.94	30.48	33.02	35.56	38.10	40.64	43.18	45.72	48.26
20	50.80	53.34	55.88	58.42	60.96	63.50	66.04	68.58	71.12	73.66
30	76.20	78.74	81.28	83.82	86.36	88.90	91.44	93.98	96.52	99.06
40	101.60	104.14	106.68	109.22	111.76	114.30	116.84	119.38	121.92	124.46
50	127.00	129.54	132.08	134.62	137.16	139.70	142.24	144.78	147.32	149.86
60	152.40	154.94	157.48	160.02	162.56	165.10	167.64	170.18	172.72	175.26
70	177.80	180.34	182.88	185.42	187.96	190.50	193.04	195.58	198.12	200.66
80	203.20	205.74	208.28	210.82	213.36	215.90	218.44	220.98	223.52	226.06
90	228.60	231.14	233.68	236.22	238.76	241.30	243.84	246.38	248.92	251.46
100	254.00	256.54	259.08	261.62	264.16	266.70	269.24	271.78	274.32	276.86

Table C-40 Centimeters into inches.

Cm	0	1	2	3	4	5	6	7	8	9
	In.	In.	In.	In.	In.	In.	In.	In.	In.	In.
0		0.394	0.787	1.181	1.575	1.969	2.362	2.756	3.150	3.543
10	3.937	4.331	4.724	5.118	5.512	5.906	6.299	6.693	7.087	7.480
20	7.874	8.268	8.661	9.055	9.449	9.843	10.236	10.630	11.024	11.417
30	11.811	12.205	12.598	12.992	13.386	13.780	14.173	14.567	14.961	15.354
40	15.748	16.142	16.535	16.929	17.323	17.717	18.110	18.504	18.898	19.291
50	19.685	20.079	20.472	20.866	21.260	21.654	22.047	22.441	22.835	23.228
60	23.622	24.016	24.409	24.803	25.197	25.591	25.984	26.378	26.772	27.164
70	27.559	27.953	28.346	28.740	29.134	29.528	29.921	30.315	30.709	31.102
80	31.496	31.890	32.283	32.677	33.071	33.465	33.858	34.252	34.646	35.039
90	35.433	35.827	26.220	36.614	37.008	37.402	37.795	38.189	38.583	38.976
100	39.370	39.764	40.157	40.551	40.945	41.339	41.732	42.126	42.520	42.913

Table C-41 Feet into meters.

Ft	0	1	2	3	4	5	6	7	8	9
	M	M	M	M	M	M	M	M	M	M
0	—	0.305	0.610	0.914	1.219	1.524	1.829	2.134	2.438	2.743
10	3.048	3.353	3.658	3.962	4.267	4.572	4.877	5.182	5.486	5.791
20	6.096	6.401	6.706	7.010	7.315	7.620	7.925	8.230	8.534	8.839
30	9.144	9.449	9.754	10.058	10.363	10.668	10.973	11.278	11.582	11.887
40	12.192	12.497	12.802	13.106	13.411	13.716	14.021	14.326	14.630	14.935
50	15.240	15.545	15.850	16.154	16.459	16.764	17.069	17.374	17.678	17.983
60	18.288	18.593	18.898	19.202	19.507	19.812	20.117	20.422	20.726	21.031
70	21.336	21.641	21.946	22.250	22.555	22.860	23.165	23.470	23.774	24.079
80	24.384	24.689	24.994	25.298	25.603	25.908	26.213	26.518	26.822	27.127
90	27.432	27.737	28.042	28.346	28.651	28.956	29.261	29.566	29.870	30.175
100	30.480	30.785	31.090	31.394	31.699	32.004	32.309	32.614	32.918	33.223

Table C-42 Meters into feet.

M	0	1	2	3	4	5	6	7	8	9
	Ft	Ft	Ft	Ft	Ft	Ft	Ft	Ft	Ft	Ft
0	—	3.281	6.562	9.842	13.123	16.404	19.685	22.966	26.247	29.528
10	32.808	36.089	39.370	42.661	45.932	49.212	52.493	55.774	59.055	62.336
20	65.617	68.897	72.178	75.459	78.740	82.021	85.302	88.582	91.863	95.144
30	98.425	101.71	104.99	108.27	111.55	114.83	118.11	121.39	124.67	127.95
40	131.23	134.51	137.79	141.08	144.36	147.64	150.92	154.20	157.48	160.76
50	164.04	167.32	170.60	173.88	177.16	180.45	183.73	187.01	190.29	193.57
60	196.85	200.13	203.41	206.69	209.97	213.25	216.53	219.82	223.10	226.38
70	229.66	232.94	236.22	239.50	242.78	246.06	249.34	252.62	255.90	259.19
80	262.47	265.75	269.03	272.31	275.59	278.87	282.15	285.43	288.71	291.99
90	295.27	298.56	301.84	305.12	308.40	311.68	314.96	318.24	321.52	324.80
100	328.08	331.36	334.64	337.93	341.21	344.49	347.77	351.05	354.33	357.61

Table C-43 Miles into kilometers.

Miles	0	1	2	3	4	5	6	7	8	9
	Km	Km	Km	Km	Km	Km	Km	Km	Km	Km
0	—	1.609	3.219	4.828	6.437	8.047	9.656	11.265	12.875	14.484
10	16.093	17.703	19.312	20.922	22.531	24.140	25.750	27.359	28.968	30.578
20	32.187	33.796	35.406	37.015	38.624	40.234	41.843	43.452	45.062	46.671
30	48.280	49.890	51.499	53.108	54.718	56.327	57.936	59.546	61.155	62.764
40	64.374	65.983	67.592	69.202	70.811	72.421	74.030	75.639	77.249	78.858
50	80.467	82.077	83.686	85.295	86.905	88.514	90.123	91.733	93.342	94.951
60	96.561	98.170	99.779	101.39	103.00	104.61	106.22	107.83	109.44	111.05
70	112.65	114.26	115.87	117.48	119.09	120.70	122.31	123.92	125.53	127.14
80	128.75	130.36	131.97	133.58	135.19	136.79	138.40	140.01	141.62	143.23
90	144.84	146.45	148.06	149.67	151.28	152.89	154.50	156.11	157.72	159.33
100	160.93	162.54	164.15	165.76	167.37	168.98	170.59	172.20	173.81	175.42

Table C-44 Kilometers into miles.

Km	0	1	2	3	4	5	6	7	8	9
	Miles	Miles	Miles	Miles	Miles	Miles	Miles	Miles	Miles	Miles
0	—	0.621	1.243	1.864	2.486	3.107	3.728	4.350	4.971	5.592
10	6.214	6.835	7.457	8.078	8.699	9.321	9.942	10.563	11.185	11.806
20	12.427	13.049	13.670	14.292	14.913	15.534	16.156	16.777	17.398	18.020
30	18.641	19.263	19.884	20.505	21.127	21.748	22.369	22.991	23.612	24.234
40	24.855	25.476	26.098	26.719	27.340	27.962	28.583	29.204	29.826	30.447
50	31.069	31.690	32.311	32.933	33.554	34.175	34.797	35.418	36.040	36.661
60	37.282	37.904	38.525	39.146	39.768	40.389	41.011	41.632	42.253	42.875
70	43.496	44.117	44.739	45.370	45.982	46.603	47.224	47.846	48.467	49.088
80	49.710	50.331	50.952	51.574	52.195	52.817	53.438	54.059	54.681	55.302
90	55.923	56.545	57.166	57.788	58.409	59.030	59.652	60.273	60.894	61.516
100	62.137	62.759	63.380	64.001	64.623	65.244	65.865	66.487	67.108	67.730

Table C-45 Units of power.

1 watt	0.73756 foot pound per second
1 foot pound per second	1.35582 watts
1 watt	0.056884 BTU per minute
1 BTU per minute	17.580 watts
1 watt	0.001341 U.S. horsepower
1 U.S. horsepower	745.7 watts
1 watt	0.01433 kilogram-calorie per minute
1 kilogram-calorie per minute	69.767 watts
1 watt	1×10^7 ergs per second
1 lumen	0.001496 watt

Table C-46 Pounds per square inch into kilograms per square centimeter.

Lb per Sq. In.	0	1	2	3	4	5	6	7	8	9
	Kg per Sq. Cm									
0	—	0.0703	0.1406	0.2109	0.2812	0.3515	0.4218	0.4922	0.5625	0.6328
10	0.7031	0.7734	0.8437	0.9140	0.9843	1.0546	1.1249	1.1952	1.2655	1.3358
20	1.4061	1.4765	1.5468	1.6171	1.6874	1.7577	1.8280	1.8983	1.9686	2.0389
30	2.1092	2.1795	2.2498	2.3201	2.3904	2.4607	2.5311	2.6014	2.6717	2.7420
40	2.8123	2.8826	2.9529	3.0232	3.0935	3.1638	3.2341	3.3044	3.3747	3.4450
50	3.5154	3.5857	3.6560	3.7263	3.7966	3.8669	3.9372	4.0075	4.0778	4.1481
60	4.2184	4.2887	4.3590	4.4293	4.4997	4.5700	4.6403	4.7106	4.7809	4.8512
70	4.9215	4.9918	5.0621	5.1324	5.2027	5.2730	5.3433	5.4136	5.4839	5.5543
80	5.6246	5.6949	5.7652	5.8355	5.9058	5.9761	6.0464	6.1167	6.1870	6.2573
90	6.3276	6.3980	6.4682	6.5386	6.6089	6.6792	6.7495	6.8198	6.8901	6.9604
100	7.0307	7.1010	7.1713	7.2416	7.3120	7.3822	7.4525	7.5228	7.5932	7.6635

Table C-47 Kilograms per square centimeter into pounds per square inch.

Kg per Sq. Cm	0	1	2	3	4	5	6	7	8	9
	Lb per Sq. In.									
0	—	14.22	28.45	42.67	56.89	71.12	85.34	99.56	113.79	128.01
10	142.23	156.46	170.68	184.90	199.13	213.35	227.57	241.80	256.02	270.24
20	284.47	298.69	312.91	327.14	341.36	355.58	369.81	384.03	398.25	412.48
30	426.70	440.92	455.15	469.37	483.59	497.82	512.04	526.26	540.49	554.71
40	568.93	583.16	597.38	611.60	625.83	640.05	654.27	668.50	682.72	696.94
50	711.17	725.39	739.61	753.84	768.06	782.28	796.51	810.73	824.95	839.18
60	853.40	867.62	881.85	896.07	910.29	924.52	938.74	952.96	967.19	981.41
70	995.63	1009.9	1024.1	1038.3	1052.5	1066.8	1081.0	1095.2	1109.4	1123.6
80	1137.9	1152.1	1166.3	1180.5	1194.8	1209.0	1223.2	1237.4	1251.7	1265.9
90	1280.1	1294.3	1308.6	1322.8	1337.0	1351.2	1365.4	1379.7	1393.9	1408.1
100	1422.3	1436.6	1450.8	1465.0	1479.2	1493.4	1507.7	1521.9	1536.1	1550.3

Table C-48 Pounds per square foot into kilograms per square meter.

Lb per Sq. Ft	Kg per Sq. M									
	0	1	2	3	4	5	6	7	8	9
0	...	4.88	9.77	14.65	19.53	24.41	29.30	34.18	39.06	43.94
10	48.82	53.70	58.59	63.47	68.35	73.23	78.12	83.00	87.88	92.76
20	97.65	102.53	107.42	112.30	117.18	122.06	126.95	131.83	136.71	141.59
30	146.47	151.35	156.24	161.12	166.00	170.88	175.77	180.65	185.53	190.41
40	195.30	200.18	205.07	209.95	214.83	219.71	224.60	229.48	234.36	239.24
50	244.12	249.00	253.89	258.77	263.65	268.53	273.42	278.30	283.18	288.06
60	292.95	297.83	302.72	307.60	312.48	317.36	322.25	327.13	332.01	336.89
70	341.77	346.65	351.54	356.42	361.30	366.18	371.07	375.95	380.83	385.71
80	390.59	395.47	400.36	405.24	410.12	415.00	419.89	424.77	429.65	434.53
90	439.43	444.30	449.19	454.07	458.95	463.83	468.72	473.60	478.48	483.36
100	488.24	493.12	498.01	502.89	507.77	512.65	517.54	522.42	527.30	532.18

Table C-49 Kilograms per square meter into pounds per square foot.

Kg per Sq. M	Lb per Sq.Ft									
	0	1	2	3	4	5	6	7	8	9
0	0	0.2048	0.4096	0.6144	0.8193	1.0241	1.2289	1.4337	1.6385	1.8433
10	2.0481	2.2530	2.4578	2.6626	2.8674	3.0722	3.2771	3.4819	3.6867	3.8915
20	4.0963	4.3011	4.5060	4.7108	4.9156	5.1204	5.3252	5.5300	5.7349	5.9397
30	6.1445	6.3493	6.5541	6.7589	6.9638	7.1686	7.3734	7.5782	7.7830	7.9878
40	8.1927	8.3975	8.6023	8.8071	9.0119	9.2167	9.4215	9.6262	9.8310	10.036
50	10.241	10.446	10.650	10.855	11.060	11.265	11.470	11.675	11.879	12.084
60	12.289	12.494	12.698	12.903	13.108	13.313	13.518	13.723	13.927	14.132
70	14.337	14.542	14.747	14.952	15.156	15.361	15.566	15.771	15.976	16.181
80	16.385	16.590	16.795	17.000	17.205	17.409	17.614	17.819	18.024	18.229
90	18.434	18.638	18.843	19.048	19.253	19.458	19.662	19.867	20.072	20.277
100	20.482	20.686	20.891	21.096	21.301	21.506	21.711	21.915	22.120	22.325

Table C-50 Water heads and equivalent pressures.

(weight of water at 62.4 lb per cubic ft)		
Head in Feet	Pressure in Lb Sq. In.	Pressure in Lb Sq. Ft
5	2.17	312
10	4.33	624
15	6.50	936
20	8.66	1,248
30	12.99	1,872
40	17.32	2,496
50	21.65	3,120
60	25.98	3,744
70	30.31	4,368
80	34.64	4,992
90	38.97	5,616
100	43.30	6,240
125	54.13	7,800
150	64.95	9,360
175	75.78	10,920
200	86.60	12,480

Table C-51 Inches vacuum into feet suction.

In. Vacuum	Ft Suction
.5	.56
1.0	1.13
1.5	1.70
2.0	2.27
2.5	2.84
3.0	3.41
4.0	4.54
5.0	5.67
6.0	6.80
7.0	7.94
8.0	9.07
9.0	10.21
10.0	11.34
15.0	17.01
20.0	22.68
25.0	28.35
30.0	34.02

Table C-52 Friction of water in pipes.

(approximate loss of heat in ft due to friction per 100 ft of pipeline)				
U.S. Gal. per Min.	1 In. I.D.	1½ In. I.D.	2 In. I.D.	4 In. I.D.
10	11.7	1.4	0.5	
20	42.0	5.2	1.8	
30	89.0	11.0	3.8	
40	152.0	18.8	6.6	0.2
50		28.4	9.9	0.3
70		53.0	18.4	0.6
100		102.0	35.8	1.2
125			54.0	1.9
150			76.0	2.6
175			102.0	3.4
200			129.0	4.4
250				6.7

Table C-53 Friction in fittings.

Type of Fitting	(reduced to equivalent feet of pipeline loss)			
	1 In. I.D.	1½ In. I.D.	2 In. I.D.	4 In. I.D.
90° Elbow	2.8	4.3	5.5	11.0
45° Elbow	1.3	2.0	2.6	5.0
Tee Side Outlet	5.6	9.1	12.0	22.0
Close Return Bend	6.3	10.2	13.0	24.0
Gate Valve	.6	.9	1.2	2.3
Check Valve	10.5	15.8	21.1	42.3
Ball Valve (ported)	27.0	43.0	55.0	115.0

Examples:

A 2-in. inside diameter pipeline loses 29.7 ft of head pumping at 50 gal. per min for 300 ft of straight length. (9.9 per hundred ft $\times$ 3 hundreds)

If the pipeline in the above example also contained, say, three 90° elbows, three 45° elbows, two close return bends, two check valves and one ported ball valve, we could add to the loss of 29.7 ft of head as follows:

Three 90° elbows	= 3 $\times$ 5.5	= 16.5 additional pipeline feet of friction
Three 45° elbows	= 3 $\times$ 2.6	= 7.8 additional pipeline feet of friction
Two close return ends	= 2 $\times$ 13.0	= 26.0 additional pipeline feet of friction
Two check valves	= 2 $\times$ 21.1	= 42.2 additional pipeline feet of friction
One ported ball valve	= 1 $\times$ 55.0	= 55.0 additional pipeline feet of friction
Total fittings		= 147.5 additional pipeline feet of friction

Then 1.475 (hundreds of feet) $\times$ 9.9 (per 100 at 2 in. I.D.) = 14.6 additional ft of head loss

Grand total pipeline (300 straight ft plus 11 fittings) = 44.3 ft of head loss

44.3 ft of head = approximately 19.2 lb pressure per sq. in.

Table C-54 Temperature conversions.

°F	°C	°F	°C	°F	°C	°F	°C
-40.	-40.	2.	-16.67	44.60	7.	87.	30.56
-39.	-39.44	3.	-16.11	45.	7.22	87.80	31
-38.20	-39.	3.20	-16.	46.	7.78	88.	31.11
-38.	-38.89	4.	-15.56	46.40	8.	89.	31.67
-37.	-38.33	5.	-15.	47.	8.33	89.60	32.
-36.40	-38.	6.	-14.44	48.	8.89	90.	32.22
-36.	-37.78	6.80	-14.	48.20	9.	91.	32.78
-35.	-37.22	7.	-13.89	49.	9.44	91.40	33.
-34.60	-37.	8.	-13.33	50.	10.	92.	33.33
-34.	-36.67	8.60	-13.	51.	10.56	93.	33.89
-33.	-36.11	9.	-12.78	51.80	11.	93.20	34.
-32.80	-36.	10.	-12.22	52.	11.11	94.	34.44
-32.	-35.56	10.40	-12.	53.	11.67	95.	35.
-31.	-35.	11.	-11.67	53.60	12.	96.	35.56
-30.	-34.44	12.	-11.11	54.	12.22	96.80	36.
-29.20	-34.	12.20	-11.	55.	12.78	97.	36.11
-29.	-33.89	13.	-10.56	55.40	13.	98.	36.67
-28.	-33.33	14.	-10.	56.	13.33	98.60	37.
-27.40	-33.	15.	-9.44	57.	13.89	99.	37.22
-27.	-32.78	15.80	-9	57.20	14.	100.	37.38
-26.	-32.22	16.	-8.89	58.	14.44	100.40	38.
-25.60	-32.	17.	-8.33	59.	15.	101.	38.33
-25.	-31.67	17.60	-8.	60.	15.56	102.	38.89
-24.	-31.11	18.	-7.78	60.80	16.	102.20	39.
-23.80	-31.	19.	-7.22	61.	16.11	103.	39.44
-23.	-30.56	19.40	-7.	62.	16.67	104.	40.
-22.	-30.	20.	-6.67	62.60	17.	105.	40.56
-21.	-29.44	21.	-6.11	63.	17.22	105.80	41.
-20.20	-29.	21.20	-6.	64.	17.78	106.	41.11
-20.	-28.89	22.	-5.56	64.40	18.	107.	41.67
-19.	-28.33	23.	-5.	65.	18.33	107.60	42.
-18.40	-28.	24.	-4.44	66.	18.89	108.	42.22
-18.	-27.78	24.80	-4.	66.20	19.	109.	42.78
-17.	-27.22	25.	-3.89	67.	19.44	109.40	43.
-16.60	-27.	26.	-3.33	68.	20.	110.	43.33
-16.	-26.67	26.60	-3.	69.	20.56	111.	43.89
-15.	-26.11	27.	-2.78	69.80	21.	111.20	44.
-14.80	-26.	28.	-2.22	70.	21.11	112.	44.44
-14.	-25.56	28.40	-2.	71.	21.67	113.	45.
-13.	-25.	29.	-1.67	71.60	22.	114.	45.56
-12.	-24.44	30.	-1.11	72.	22.22	114.80	46.
-11.20	-24.	30.20	-1.	73.	22.78	115.	46.11
-11.	-23.89	31.	-0.56	73.40	23.	116.	46.67
-10.	-23.33	32.	0.	74.	23.33	116.60	47.
-9.40	-23.	33.	0.56	75.	23.89	117.	47.22
-9.	-22.78	33.80	1.	75.20	24.	118.	47.78
-8.	-22.22	34.	1.11	76.	24.44	118.40	48.
-7.60	-22.	35.	1.67	77.	25.	119.	48.33
-7.	-21.67	35.60	2.	78.	25.56	120.	48.89
-6.	-21.11	36.	2.22	78.80	26.	120.20	49.
-5.80	-21.	37.	2.78	79.	26.11	121.	49.44
-5.	-20.56	37.40	3.	80.	26.67	122.	50.
-4.	-20.	38.	3.33	80.60	27.	123.	50.56
-3.	-19.44	39.	3.89	81.	27.22	123.80	51.
-2.20	-19	39.20	4.	82.	27.78	124.	51.11
-2.	-18.89	40.	4.44	82.40	28.	125.	51.67
-1.	-18.33	41.	5.	83.	28.33	125.60	52.
-0.40	-18.	42.	5.56	84.	28.89	126.	52.22
0.	-17.78	42.80	6.00	84.20	29.	127.	52.78
1.	-17.22	43.	6.11	85.	29.44	127.40	53.
1.40	-17.	44.	6.67	86.	30.	128.	53.33

(continued)

Table C-54 (continued)

°F	°C	°F	°C	°F	°C	°F	°C
129.	53.89	159.	70.56	188.60	87.	217.40	103.
129.20	54.	159.80	71.	189.	87.22	218.	103.33
130.	54.44	160.	71.11	190.	87.78	219.	103.89
131.	55.	161.	71.67	190.40	88.	219.20	104.
132.	55.56	161.60	72.	191.	88.33	220.	104.44
132.80	56.	162.	72.22	192.	88.89	221.	105.
133.	56.11	163.	72.78	192.20	89.	222.	105.56
134.	56.67	163.40	73.	193.	89.44	222.80	106.
134.60	57.	164.	73.33	194.	90.	223.	106.11
135.	57.22	165.	73.89	195.	90.56	224.	106.67
136.	57.78	165.20	74.	195.80	91.	224.60	107.
136.40	58.	166.	74.44	196.	91.11	225.	107.22
137.	58.33	167.	75.	197.	91.67	226.	107.78
138.	58.89	168.	75.56	197.60	92.	226.40	108.
138.20	59.	168.80	76.	198.	92.22	227.	108.33
139.	59.44	169.	76.11	199.	92.78	228.	108.89
140.	60.	170.	76.67	199.40	93.	228.20	109.
141.	60.56	170.60	77.	200.	93.33	229.	109.44
141.80	61.	171.	77.22	201.	93.89	230.	110.
142.	61.11	172.	77.78	201.20	94.	231.	110.56
143.	61.67	172.40	78.	202.	94.44	231.80	111.
143.60	62.	173.	78.33	203.	95	232.	111.11
144.	62.22	174.	78.89	204.	95.56	233.	111.67
145.	62.78	174.20	79.	204.80	96.	233.60	112.
145.40	63.	175.	79.44	205.	96.11	234.	112.22
146.	63.33	176.	80.	206.	96.67	235.	112.78
147.	63.89	177.	80.56	206.60	97.	235.40	113.
147.20	64.	177.80	81.	207.	97.22	236.	113.33
148.	64.44	178.	81.11	208.	97.78	237.	113.89
149.	65.	179.	81.67	208.40	98.	237.20	114.
150.	65.56	179.60	82.	209.	98.33	238.	114.44
150.80	66.	180.	82.22	210.	98.89	239.	114
151.	66.11	181.	82.78	210.20	99.	240.	115.56
152.	66.67	181.40	83.	211.	99.44	240.80	116.
152.60	67.	182.	83.33	212.	100.	241.	116.11
153.	67.22	183.	83.89	213.	100.56	242.	116.67
154.	67.78	183.20	84.	213.80	101.	242.60	117.
154.40	68.	184.	84.44	214.	101.11	243.	117.32
155.	68.33	185.	85.	215.	101.67	244.	117.78
156.	68.89	186.	85.56	215.60	102.	244.40	118.
156.20	69.	186.80	86.	216.	102.22	245.	118.33
157.	69.44	187.	86.11	217.	102.78	246.	118.89
158.	70.	188.	86.67				

Table C-55 Basic conversions: units of weight.

1 grain	.0029 ounces
1 grain	.000143 pounds
1 grain	.064799 grams
1 dram	60 grains
1 dram	.1371 ounces
1 dram	.008571 pounds
1 dram	3.88794 grams
1 ounce	437.5 grains
1 ounce	28.3495 grams
1 ounce	.0625 pounds
1 pound	453.592 grams
1 pound	16 ounces
1 ton	2000 pounds
1 ton	907.18581 kilograms
1 gram	.03527 ounces
1 gram	.002205 pounds
1 gram	15.432 grains
1 gram	.001 milligram
1 kilogram	1000 grams
1 kilogram	35.2739 ounces
1 kilogram	2.20462 pounds

Table C-56 Weight conversion formulas.

To Convert	Into	Multiply By
grains	grams	0.0648
drams	grams	1.7718
ounces	grams	28.3495
pounds	grams	435.5924
tons	kilograms	907.18581
grams	grains	15.4324
grams	drams	0.5644
grams	ounces	0.0353
kilograms	pounds	2.2046

Table C-57 Weight fractions.

Oz	Fraction of a Lb	Decimal of a Lb	G	Oz	Fraction of a Lb	Decimal of a Lb	G
1/4	1/64	.0156	7.09	8-1/4	33/65	.5156	233.88
1/2	1/32	.0313	14.17	8-1/2	17/32	.5313	240.97
3/4	3/64	.0469	21.26	8-3/4	35/64	.5469	248.06
1	1/16	.0625	28.35	9	9/16	.5625	255.15
1-1/4	5/64	.0781	35.44	9-1/4	37/64	.5781	262.23
1-1/2	3/32	.0938	42.52	9-1/2	19/32	.5938	269.30
1-3/4	7/64	.1094	49.61	9-3/4	39/64	.6094	276.41
2	1/8	.125	56.70	10	5/8	.625	283.50
2-1/4	9/64	.1406	63.79	10-1/4	41/64	.6406	290.58
2-1/2	5/32	.1563	70.87	10-1/2	21/32	.6563	297.67
2-3/4	11/64	.1719	77.96	10-3/4	43/64	.6719	304.76
3	3/16	.1875	85.05	11	11/16	.6875	311.84
3-1/4	13/64	.2031	92.14	11-1/4	45/64	.7031	318.93
3-1/2	7/32	.2188	99.22	11-1/2	23/32	.7188	326.02
3-3/4	15/64	.2344	106.31	11-3/4	47/64	.7344	333.11
4	1/4	.250	113.40	12	3/4	.750	340.19
4-1/4	17/64	.2656	120.49	12-1/4	49/64	.7656	347.28
4-1/2	9/32	.2813	127.57	12-1/2	25/32	.7813	354.37
4-3/4	19/64	.2969	134.66	12-3/4	51/64	.7969	361.46
5	5/16	.3125	141.75	13	13/16	.8125	368.54
5-1/4	21/64	.3281	148.84	13-1/4	53/64	.8281	375.63
5-1/2	11/32	.3438	155.92	13-1/2	27/32	.8438	382.72
5-3/4	23/64	.3594	163.01	13-3/4	55/64	.8594	389.81
6	3/8	.375	170.10	14	7/8	.875	396.89
6-1/4	25/64	.3906	177.18	14-1/4	57/64	.8906	403.93
6-1/2	13/32	.4063	184.27	14-1/2	29/32	.9063	411.07
6-3/4	27/64	.4219	191.36	14-3/4	59/64	.9219	418.16
7	7/16	.4375	198.45	15	15/16	.9375	425.24
7-1/4	29/64	.4531	205.53	15-1/4	61/64	.9531	432.33
7-1/2	15/32	.4688	212.62	15-1/2	31/32	.9688	439.42
7-3/4	31/64	.4844	219.71	15-3/4	63/64	.9844	446.50
8	1/2	.500	226.80	16	1	1.000	453.59

Table C-58 Ounces into grams.

Oz	0	1	2	3	4	5	6	7	8	9
					G					
0	—	28.350	56.699	85.049	113.40	141.75	170.10	198.45	226.80	255.15
10	283.50	311.84	340.19	368.54	396.89	425.24	453.59	481.94	510.29	538.64
20	566.99	595.34	623.69	652.04	680.39	708.74	737.09	765.44	793.79	822.14
30	850.49	878.84	907.19	935.53	963.88	992.23	1020.6	1048.9	1077.3	1105.6
40	1134.0	1162.3	1190.7	1219.0	1247.4	1275.7	1304.1	1332.4	1360.8	1389.1
50	1417.5	1445.8	1474.2	1502.5	1530.9	1559.2	1587.6	1615.9	1644.3	1672.6
60	1701.0	1729.3	1757.7	1786.0	1814.4	1842.7	1871.1	1899.4	1927.8	1956.1
70	1984.5	2012.8	2041.2	2069.5	2097.9	2126.2	2154.6	2182.9	2211.3	2239.6
80	2268.0	2296.3	2324.7	2353.0	2381.4	2409.7	2438.1	2466.4	2494.8	2523.1
90	2551.5	2579.8	2608.2	2636.5	2664.9	2693.2	2721.6	2749.9	2778.3	2806.6
100	2835.0	2863.3	2891.6	2920.0	2948.3	2976.7	3005.0	3033.4	3061.7	3090.1

Table C-59 Grams into ounces.

G	0	1	2	3	4	5	6	7	8	9
					Oz					
0	—	0.035274	0.070548	0.10582	0.14110	0.17637	0.21164	0.24692	0.28219	0.31747
10	0.35274	0.38801	0.42329	0.45856	0.49384	0.52911	0.56438	0.59966	0.63493	0.67021
20	0.70548	0.74075	0.77603	0.81130	0.84658	0.88185	0.91712	0.95240	0.98767	1.0229
30	1.0582	1.0935	1.1288	1.1640	1.1993	1.2346	1.2699	1.3051	1.3404	1.3757
40	1.4110	1.4462	1.4815	1.5168	1.5521	1.5873	1.6226	1.6579	1.6932	1.7284
50	1.7637	1.7990	1.8348	1.8695	1.9048	1.9401	1.9753	2.0106	2.0459	2.0812
60	2.1164	2.1517	2.1870	2.2223	2.2575	2.2928	2.3281	2.3634	2.3986	2.4339
70	2.4692	2.5045	2.5397	2.5750	2.6103	2.6456	2.6808	2.7161	2.7514	2.7866
80	2.8219	2.8572	2.8925	2.9277	2.9630	2.9983	3.0336	3.0688	3.1041	3.1394
90	3.1747	3.2099	3.2452	3.2805	3.3158	3.3510	3.3863	3.4216	3.4569	3.4921
100	3.5274	3.5627	3.5979	3.6332	3.6685	3.7038	3.7390	3.7743	3.8096	3.8449

Table C-60 Pounds into kilograms.

Lb	0	1	2	3	4	5	6	7	8	9
					Kg					
0	—	0.454	0.907	1.361	1.814	2.268	2.722	3.175	3.629	4.082
10	4.536	4.990	5.443	5.897	6.350	6.804	7.257	7.711	8.165	8.618
20	9.072	9.525	9.979	10.433	10.886	11.340	11.793	12.247	12.701	13.154
30	13.608	14.061	14.515	14.969	15.422	15.876	16.329	16.783	17.237	17.690
40	18.144	18.597	19.051	19.504	19.958	20.412	20.865	21.319	21.772	22.226
50	22.680	23.133	23.587	24.040	24.494	24.948	25.401	25.855	26.308	26.762
60	27.216	27.669	28.123	28.576	29.030	29.484	29.937	30.391	30.844	31.298
70	31.752	32.205	32.659	33.112	33.566	34.019	34.473	34.927	35.380	35.834
80	36.287	36.741	37.195	37.648	38.102	38.555	39.009	39.463	39.916	40.370
90	40.823	41.277	41.731	42.184	42.638	43.091	43.545	43.999	44.452	44.906
100	45.359	45.813	46.266	46.720	47.174	47.627	48.081	48.534	48.988	49.442

Table C-61 Kilograms into pounds.

Kg	0	1	2	3	4	5	6	7	8	9
					Lb					
0	—	2.205	4.409	6.614	8.819	11.023	13.228	15.432	17.637	19.842
10	22.046	24.251	26.456	28.660	30.865	33.069	35.274	37.479	39.683	41.888
20	44.093	46.297	48.502	50.706	52.911	55.116	57.320	59.525	61.729	63.934
30	66.139	68.343	70.548	72.753	74.957	77.162	79.366	81.571	83.776	85.980
40	88.185	90.390	92.594	94.799	97.003	99.208	101.41	103.62	105.82	108.03
50	110.23	112.44	114.64	116.85	119.05	121.25	123.46	125.66	127.87	130.07
60	132.28	134.48	136.69	138.89	141.10	143.30	145.51	147.71	149.91	152.12
70	154.32	156.53	158.73	160.94	163.14	165.35	167.55	169.76	171.96	174.17
80	176.37	178.57	180.78	182.98	185.19	187.39	189.60	191.80	194.01	196.21
90	198.42	200.62	202.83	205.03	207.24	209.44	211.64	213.85	216.05	218.26
100	220.46	222.67	224.87	227.08	229.28	231.49	233.69	235.90	238.10	240.30

APPENDIX D

GLOSSARY

Amabile Italian term referring to sweetness

Amaro Italian term referring to bitterness

Amelioration dilution—the addition of water and/or sugar to juice, must, or wine

Anaerobic not requiring oxygen—a term describing microorganisms that do not require free oxygen in order to grow, such as wine yeasts

Anthocyanins phenolic color pigments found in the skins of grapes

AOC acronym for the French “Appellation d’Origine Contrôlée,” the official regulatory system for wine origin and quality control

Aperitif appetizer wine—wine containing added flavoring from herbs, roots, seeds, and spices, such as vermouth

Appearance sensory term relating to the entire visual aspect of wine evaluation, principally color and clarity

Appellation geographic source—the specific location where a given wine was grown

Aroma fruit fragrance—that portion of wine bouquet contributed by the fruit

Astringency an aspirin-like sensory response on the palate, generally due to tannins in wine

ATF acronym for the U.S. Bureau of Alcohol, Tobacco, and Firearms

Austere sensory term relating to wines that have excessive acidity and/or astringency

- Bacteria** one-celled microorganisms that do not contain chlorophyll, such as bacilli (which are rod-shaped) and cocci (which are spherical)
- Balance** sensory term relating to the ratio of dryness and acidity
- Balling** analytical measurement of wine viscosity
- Barrique** wine barrel containing approximately 60 U.S. gallons
- Baume** scale for measuring dissolved solids
- Bentonite** a montmorillonite clay compound used as a clarification agent in wine processing
- Bianco** Italian term relating to white wine
- Big** sensory term used in reference to wines that exhibit an abundance of positive qualities
- Bite** same as **austere**
- Blanc de blancs** French term relating to white wines made from white grapes
- Blanc de noirs** French term relating to white wines made from black grapes
- Bland** sensory term used to describe wines deficient in acidity and/or tannin
- Blush** very light pink wines, generally made from a brief fermentation of red must just prior to pressing
- Body** sensory evaluation of wine viscosity; relates to lightness or heaviness of mouth-feel
- Bonded** an ATF term relating to a premises upon which wines may be legally stored prior to the payment of excise taxes
- Botrytis cinerea** same as **noble mold**
- Bottle fermentation** secondary fermentation of wine in special bottles designed for high pressures; captures natural carbon dioxide to make wine effervesce, or “sparkle”; the traditional champagne bottle fermentation method is called *traditional method*
- Bottle shock** a temporary reduction in wines’ bouquet and flavor due to addition of preservatives, final filtration, and bottling; sometimes called “bottle sickness”; recovery generally appears within a few weeks
- Bouquet** sensory term relating to the entire fragrance of a wine; composed chiefly of fruit aroma, fermentation flavors from yeast and/or bacteria, wood essences, and other components
- Brandy** the distillate or “spirits” of wine
- Breathing** the practice of uncorking a wine bottle well in advance of serving to allow the wine to expel headspace gases
- Brilliant** sensory term relating to wines that have flawless clarity
- Brix** analytical measurement of total dissolved solids in juice or must, most of which are sugars
- Bulk process** same as **Charmat**
- Bung** stopper, usually made of wood or glass, that is used to seal a keg, barrel, cask, or some other bulk wine storage vessel
- Bunghole** opening at the top of a keg, barrel, cask, or some other bulk wine storage vessel in which a bung is inserted
- Cantina** Italian for “cellar”

- Capsule** the decorative finishing closure over a cork or cap on a wine bottle, usually made of plastic or nonlead metal alloys
- Carbohydrate** chemical term relating to a food energy compound, such as sugar and ethyl alcohol, composed of carbon, hydrogen, and oxygen
- Carbon dioxide** the tasteless and odorless gas produced by fermentation
- Carbonic maceration** method by which whole grapes are fermented in a closed vessel, during which the carbon dioxide gas generated from the fermentation permeates the grapes to release greater intensity of color and flavor
- Carboy** a narrow-mouthed glass container, usually 3 or 5 gallons in capacity
- Casa vinicola** Italian for “winery”
- Cask** wooden vessel for bulk wine storage and aging, usually made of some species of white oak and containing at least 200 U.S. gallons
- Caskiness** sensory term relating to wines that may have been aged or stored in casks that were not cleaned or treated properly beforehand; much the same as **tankiness**
- Cassé** sensory term to describe wine haziness, generally due to high levels of copper and/or iron
- Cava** Spanish for “sparkling wine”
- Cave** French for “cellar”
- Cépage** French for “variety” or “cultivar”
- Chai** French for “wine cellar”; a term most often identifying Bordeaux wine cellars
- Chambrier** French for adjusting wine temperature to that in a cool room, or about 65°F
- Chaptalization** the addition of sugar to must or juice before fermentation into wine
- Character** sensory term describing the entire profile of a wine, including color, clarity, bouquet, balance, body, aftertaste, and overall impression
- Charmat** the bulk, or tank, method of making ordinary sparkling wines
- Château** French for “castle” or “mansion”; the wine estate, comprising vineyards and chai, in France, usually in Bordeaux
- Chiaretto** Italian for “light red”
- Claret** term used by the English for red wines grown in the Bordeaux region of France
- Clarify** to make a wine brilliantly clear; same as “fining”
- Classified growth** a particular vineyard, domaine, estate, or other geographical locale that is officially recognized by a national government as a controlled appellation of wine origin
- Clean** sensory term relating to a wine having no musty or other “off” elements in character
- Clos** French for “wall,” usually around a vineyard or estate
- Cloudy** sensory term relating to a wine exhibiting heavy suspended solids
- Cloying** sensory term for “excessively sweet”
- Coarse** sensory term relating to a wine having abnormally crude elements of character, especially acidity and/or tannins
- Colloid** the suspension material, usually protein, that causes wines to be cloudy
- Color** relates to both the hue and density of wine pigments

- Complex** sensory term relating to an interrelationship of extensive bouquet and/or flavor components
- Concentrate** dehydrated grape juice, usually at a Brix of 60° or more, which is used for sweetening juice, musts, or wines deficient in natural sugar; sometimes reconstituted with water and fermented to make low-grade wines
- Cooked** sensory term used to describe a “baked” character or an oxidized bouquet and flavor
- Cooperage** cellar term relating to wooden containers, but sometimes loosely used to also include vessels constructed of glass, stainless steel, and other materials
- Cooperative** winery owned by a group of member grape growers and/or vintners
- Cork** bark of the cork oak, *Quercus suber*, largely grown in Portugal and Spain, from which wine bottle closures are made
- Corkiness** sensory term relating to the mold *Penicillium expansum*, which manifests a foul mildewlike compound called trichloroanisole (TCA) in wine bouquet
- Cradle** device in which old bottles with sediment are placed for cork removal; often used for holding bottles while being poured
- Cream of tartar** potassium bitartrate crystals precipitated by new wines, especially in colder storage temperatures
- Cremant** sparkling wine that has a reduced level of effervescence; not to be confused with Cramant, which is an important winegrowing commune in the Champagne region of France
- Cru** French for a specified “growth” or “vineyard”
- Crust** sediment that has collected and solidified on the inside surface of a bottle, often in association with old Port wines
- Cuvée** a French enology term relating to a special blend, such as one designed for secondary fermentation into sparkling wine
- Cultivar** viticultural term relating to a manmade hybrid; set apart from the term **variety**, which usually refers to natural progeny
- Decant** operation of delicately transferring wine from a bottle to a decanter so as to separate any sediments that have formed in the bottle during aging
- Dégorgement** the technique of removing the plug of sediment from a bottle of sparkling wine after the remuage, or “riddling”
- Delicate** sensory term used in reference to wine with subtle bouquet and flavor
- Demi-sec** French for “half dry” or “near dry”
- Density** sensory term relating to the amount of certain character constituents, usually color
- Dessert wine** a sweet wine generally consumed with, or instead of, dessert; made from sweet table wine that has been “fortified” up to about 20% alcohol content with brandy
- Dinner wine** same as table wine
- Dirty** sensory term used in reference to wines tainted with musty odors and/or flavor
- DOC** acronym for the Italian “Denominazione di Origine Controllata,” the official regulatory system for wine quality and origin control

Dolce Italian for “sweet”

Domaine French for “vineyard estate” or “wine estate”; most commonly used in the Burgundy region of France

Dosage addition of wine to sparkling wines directly after degorgement in order to replace the expelled “ice plug”

Doux French for “sweet”

Dry sensory term used in reference to an absence of sweetness

Dull sensory term used in reference to uninteresting wines

Earthy sensory term used to describe soil flavors such as “chalky,” “flinty,” or “stony”—but not “dirty”

Edelfaule German for “noble mold”

Effervescent sensory term for “sparkling”—the discharge of carbon dioxide bubbles in wine

Enology the scientific study of making wine

Enzymes complex organic compounds that trigger chemical reactions

Estate-bottled label term referring to grapes having been grown on the estate of the vintner or vineyards under estate control

Esters complex organic acid-alcohol flavor compounds

Ethyl alcohol essential wine compound produced from fermentation of grape juice or must—ethanol

Expressive sensory term relating to wine that has an abundance of character; the opposite of flat

Extract analytical term referring to total dissolved solids in wine; the same as Brix in juice or must

Facultative microorganisms that can grow either with or without an oxygen source

Fattoria Italian for “vintner”

Fermentation usually relates to the transformation of sugar to alcohol, carbon dioxide, and energy in response to enzymes produced by the growth of yeasts; bacterial fermentations also exist in which malic acid is transformed to lactic acid or in which ethyl alcohol is transformed to acetic acid

Fermentation lock cellar device used on fermenting vessels that allows carbon dioxide gas to escape but does not allow air to enter

Filtration cellar treatment in which wine is pressured through media in which suspended solids or microorganisms are removed

Fining cellar treatment in which agents such as bentonite are used to clarify wine

Finish sensory term relating to the last impression of a wine just before, during, and after swallowing, including aftertaste

Firm sensory term relating to wines that have ample acidity and/or tannins

Flat sensory term relating to wines that have a deficiency of character; the opposite of expressive

Flavor sensory term relating to olfactory values of taste such as “berry,” “earthy,” “vegetative,” “fruity,” or “yeasty”

- Flinty** sensory term relating to “stony” or “rocklike” flavors
- Flor** the surface-growing yeasts that synthesize acetaldehyde, the “nutlike” flavor compound common to sherry wines
- Floral** sensory term relating to a “flowery” or “blossom” value in bouquet
- Flowers** cellar term for white film on the surface of wine that may indicate the growth of spoilage bacteria or wild yeasts
- Fortified wine** wine that has been blended with brandy to increase percentage of alcohol
- Foxiness** sensory term loosely used to describe the aroma of wines made from native *Vitis labrusca* grapes such as Concord and Niagara
- Free run** juice or wine that flows from a press without the exertion of pressure
- Frizzante** Italian for *petillant* or “lightly sparkling”
- Fruity** sensory term relating to a “fruitlike” value in bouquet and/or flavor
- Full** sensory term relating to a heavy density of one or another element of wine character
- Fumé** sensory term relating to wines having a pronounced floral bouquet, usually in reference to Sauvignon Blanc white wines
- Generic** label term relating to wine geography or type, such as Burgundy (wine grown in the Burgundy region of France) or vermouth (wine type in which herb and spice essences are added)
- Green** sensory term relating to wines made from premature grapes or wines not fully developed in aging
- Gross lees** cellar term referring to the large amount of initial sedimentation that precipitates before and during fermentation
- Harsh** sensory term relating to wine that is excessively high in total acidity and/or tannin
- Heady** sensory term relating to table wines with a high alcohol content; the same as **strong**
- Heavy** sensory term relating to high density of one or another character constituents, usually body
- Hock** English term for German white table wines
- Hogshead** small wine cask
- Hue** sensory term relating to the value of color, such as amber, straw-gold, brick-red, crimson, ruby, etc.
- Hybrid** viticultural term relating to the sexual crossing of one grape variety with another in order to create a new cultivar, such as the French-American hybrids Seyval Blanc and Chambourcin
- Hydrometer** analytical device that responds to the specific gravity, or relative density, of a liquid; may be used to measure Brix and Balling
- Indicator** analytical compound that changes color when a solution has been neutralized, such as phenolphthalein, which is used to indicate when a total acidity titration is completed
- Jeroboam** large wine bottle, usually with a capacity of 3 liters
- Keg** very small wooden wine vessel, usually with a capacity of less than 30 U.S. gallons

- KMS** acronym for potassium meta-bisulfite, used as a preservative in winemaking
- Labrusca** viticultural term relating to *Vitis labrusca*, the native grapes of the northeastern United States, such as Concord and Niagara
- Lactic acid** a “cheese-like” flavored compound produced from malic acid in a malolactic fermentation by special bacteria
- Late harvest** wines made from grapes purposely left past peak ripeness in order to dehydrate berries, which concentrates sugar and flavor
- Lees** production term relating to wine sedimentation from fermentation, clarification, or instability of acid salts, color pigments, and other constituents
- Legs** sensory term relating to the reflux condensation of volatile wine components inside the glass; long “legs” are widely believed to be the mark of superior wines, but there is no correlation, as condensation rates are determined by the relative temperatures of the wine and the glass
- Light** sensory term relating to low density of one or another character constituent, usually color or body
- Long** sensory term used in reference to wine flavors that linger in the mouth; extended aftertaste
- Maderization** the oxidation of ethyl alcohol and acetic acid into aldehydes; the process employed in the making of Madeira, Sherry, and Marsala wines
- Magnum** a double bottle containing 1.5 L
- Malolactic bacteria** bacteria that transform malic acid to lactic acid, resulting in a “buttery” (diacetyl) flavor component
- Marque** French for “mark,” usually in reference to “Grand Marque” Champagnes, the highest quality level
- Methuselah** very large wine bottle, usually containing 6 L, often used in the Bordeaux and Champagne regions of France
- Micron (:** filtration term for one-millionth of a meter, or about 0.00004 inch; most yeasts are removed at about 1.0: of filter porosity and most bacteria at about .5:
- Millesime** French for “vintage”
- Moelleux** French for “mellow”
- Moldy** sensory term referring to an “off” value of bouquet and flavor, usually given to wines that have been made from moldy grapes and/or from wines stored in tankage that has harbored mold
- Monopole** French for “monopoly”; generally refers to an entire cru under a single ownership
- Mousseux** French for “foaming”; a term used to describe wines that are effervescent, or sparkling
- Must** cellar term relating to crushed grapes
- Musty** same as **moldy**
- Mutage** the process of adding brandy to fermenting juice or must in order to arrest fermentation and retain remaining grape sugar as residual sweetness in the resulting wine; often used in reference to Port wines

Negociant French for “negotiant”—bulk wine buyer who ages, blends, processes, bottles, and markets wines

Nero Italian for “dark red”

Noble mold viticultural term relating to *Botrytis cinerea*, a mold that permeates grape skins, allowing berry water to evaporate and remaining color, sugar, and flavor components to be concentrated

Nose sensory term relating to the evaluation of aroma and bouquet

Nouveau French for “new”; generally relates to young red wines, such as Beaujolais nouveau, which are made and marketed soon after each vintage

Nutty sensory term relating to the acetaldehyde “nutlike” flavor characteristic in Madeira and sherry wines

Oenology British for **enology**

Oily sensory term used in reference to wines that have a “fatty” consistency on the palate

Organoleptic having to do with sensory evaluation

Oxidation chemical term relating to the reaction of constituents with oxygen, such as the aging of wine or the browning of color

Pasteurization cellar term relating to the inhibition of spoilage microorganisms by the controlled application of heat, generally between 160 and 180°F for less than a minute; rarely used in modern winemaking

Petillant sensory term relating to wines that are just slightly effervescent

pH analytical term relating to the entire range of the most intense acidity at pH 0.0 to the most intense base at pH 14.0, with neutrality at 7.0; often used as one of several indicators of optimal grape ripeness

Phenols complex organic compounds extracted from skins, seeds, and stems in grapes and from oak storage vessels; see **tannin** and **tannic**

Phylloxera the grapevine root louse, more precisely known as *Phylloxera vastatrix*

Pierce’s disease a bacterial disease that blocks the passage of fluids in a grapevine

Piquant French for “tart”

Pomace cellar term referring to the seeds and skins remaining in the press

Pop wine wine containing added flavoring(s) from other berries, melons, and fruits

Potassium bitartrate same as **cream of tartar**

ppm abbreviation for the chemical measurement expression “parts per million,” which is roughly equivalent to “milligrams per liter”

Press juice or Press wine cellar terms relating to the juice or wine extracted under pressure; usually more dense in color and excessively astringent from higher levels of phenols

Puncheon British wine vessel measure of about 70 U.S. gallons

Punt indentation in the bottom of a wine bottle, originally intended to provide added strength to the container

Racking cellar term relating to the decanting operation in separating juice or wine from lees

- Rancio** sensory term relating to the bouquet of some maderized wines; has no connection with “rancid”
- Recolte** French for “harvest”
- Rheoboam** a large wine bottle, about 4.5 L in capacity, often used in the Bordeaux and Champagne regions of France
- Remuage** the process of placing unfinished bottle-fermented sparkling wines into a table or rack in order to “work” the yeast sediment into the neck of the bottle prior to degorgement—the same as “riddling”
- Rich** sensory term having the same general meaning as **heavy**
- Rose** pink wines, generally made from a brief fermentation of red grape must just prior to pressing; typically more pigmented than “blush” wines
- Rotten eggs** sensory term relating to wines that have degenerated because of the formation of hydrogen sulfide (H_2S)
- Round** sensory term relating to wines that have a good balance and harmony of character and constituency
- Salamanazar** very large wine bottle, containing 9 L, now rarely found in use commercially
- Sec** French for “dry”
- Secco** Italian for “dry”
- Sekt** German for “sparkling wine”
- Shy** sensory term relating to wines which have low intensities of bouquet and flavor
- Soft** sensory term relating to wines having lower intensities of total acidity and astringency
- Sorbic acid** preservative used to inhibit yeast growth, ineffective against bacteria
- Sour** sensory term referring to spoilage; often misused in reference to wines having higher intensities of total acidity or astringency
- Stemmy** sensory term referring to wines that express a rather astringent vegetative character often related to excessive contact with the grape stems during fermentation
- Strong** sensory term referring to table wines expressing a burn of alcohol in the bouquet
- Sulfite** cellar term generally referring to KMS, or potassium meta-bisulfite, a source of sulfur dioxide used as a preservative and antioxidant in wines
- Sur lies** cellar term referring to holding new wines on the fermentation lees in order to promote malo-lactic fermentation and extract complex flavors
- Tankiness** same as **caskiness**
- Tannic** sensory term relating to phenolic astringency in wines with a rather “leatherlike” flavor
- Tannin** a group of phenolic compounds found in grape seeds, skins, and stems, as well as in wood used for aging; tannins are antioxidants and thus serve to slow the wine aging process
- Tart** sensory term relating to wines having higher intensities of total acidity
- Tartaric acid** principal organic acid in grapes; combines with potassium to form potassium bitartrate salts, or cream of tartar

- Tastevin** a small shallow tasting cup, usually made of silver, typically related to the Burgundy region of France
- Tawny** sensory term referring to amber-red color, usually as a result of wines aged for long periods of time
- Teinturier** a red grape that has particularly high density in color, often used in blending to enhance color-deficient wines
- Tenuta** Italian for “estate”
- Tirage** cellar term referring to the laying of bottles horizontally, tier upon tier
- Tonneau** wine cask with a capacity of about 238 U.S. gallons, traditional to the Bordeaux region in France
- Total acidity (TA)** the aggregate of all acids measured in a given wine, the sum of fixed and volatile acidities
- Triage** the removal of defective clusters and/or berries, usually on a moving belt inspection line
- Ullage** air space in a bottle or cellar container resulting from assimilation, evaporation, and seepage
- Varietal wine** a wine labeled for the variety of vine from which it was predominantly made
- Variety** a individual natural strain of vine recognized with a group of other closely related strains: for example, Chardonnay, Trebbiano, Johannisberg Riesling, Pinot Noir, Cabernet Sauvignon, and Nebbiolo, are each a variety of the *Vitis* genus and *vinifera* species
- Vendange** French for “grape harvest”
- Vendemmia** Italian for “grape harvest”
- Veraison** viticultural term referring to the point when grape berries start to ripen, softening and changing color
- Vert** French for “green”; often used in reference to wines made from underripened grapes, or for tart young white wines
- Vigne** French for “vine”
- Vignoble** French for “vineyard”
- Vin** French for “wine”
- Vino** Italian for “wine”
- Vintage** label term relating to the year in which the grapes were grown; sometimes used in reference to a particularly good year of growth
- Vintner** a wine producer
- Viticulture** the part of horticulture that has to do with the cultivation of grape vines
- Vitis vinifera*** viticultural term referring to the genus and species of “Old World” vines—the native wine-grape vines of Europe such as Chardonnay and Cabernet Sauvignon
- Volatile acidity (VA)** analytical term referring to the acidity produced by spoilage microorganisms

Woody sensory term referring to wine that has the bouquet and/or taste of excessive wood aging

Yeast single-cell plant organisms that produce enzymes that convert simple sugars into ethyl alcohol, carbon dioxide gas, and energy

Yeasty sensory term referring to wine that has the bouquet and/or taste of excessive exposure to yeast in *sur lies*

Zymase the enzymes produced by wine yeasts that convert sugars to ethyl alcohol, carbon dioxide gas, and heat energy during fermentation

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